CHAPTER 7
PNEUDRAULIC SYSTEM

CHAPTER OVERVIEW

SECTION 7-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

7-1-1 HYDRAULIC SYSTEMS DESCRIPTION AND DATA.

7-1-1.1 Hydraulic Systems Description.

The Hydraulic Systems provide between 3050 ± 3100 pounds per square inch (psi) of hydraulic pressure to operate the primary servos, tail rotor servos, pilot-assist servos, and Auxiliary Power Unit (APU) starter motor. There are three Hydraulic Systems:

a. No. 1 (First Stage) Hydraulic System

b. No. 2 (Second Stage) Hydraulic System

c. Backup Hydraulic System

The major components of these Hydraulic Systems are:

a. Three Hydraulic Pump Modules

b. Two Transfer Modules

c. Utility Module

d. Pilot-Assist Module

e. Three Primary Servos

f. Tail Rotor Servo

g. Four Pilot-Assist Servos

h. APU Accumulator

i. APU Hand Pump

j. Refill Hand Pump

Most of these components are grouped together on the upper deck in front of the main transmission. These servos are connected to the hydraulic modules through manifolds and self-sealing couplings.

The three Hydraulic Systems have pressure switches that illuminate appropriate capsules on the caution/advisory panel when pressure loss is detected (Figure 7-1-1-2, Figure 7-1-1-3, and Figure 7-1-1-4).

A leak detection/isolation feature is built into the Hydraulic System using pressure switches on the pump modules, check valves and shutoff valves in the transfer modules, and electronic logic modules. When a pressure switch senses a pressure loss in a Hydraulic System, the logic module will shut off the required valve or valves to isolate the leak and turn on the backup hydraulic pump.

7-1-1.2 No. 1 Hydraulic System.

The No. 1 Hydraulic System supplies hydraulic pressure from the No. 1 pump module to the No. 1 transfer module. From the transfer module, pressure is supplied to the first stage of the primary servos (lateral, forward, and aft) and the first stage of the tail rotor servo.

7-1-1.3 No. 2 Hydraulic System.

The No. 2 Hydraulic System supplies pressure from the No. 2 pump module to the No. 2 transfer module. From the transfer module, pressure is supplied to the second stage of the primary servos (lateral, forward, and aft) and the pilot-assist module. From the pilot-assist module, pressure is supplied to the pilot-assist servos (collective boost, yaw boost, yaw Stability Augmentation Set (SAS) actuator, roll SAS actuator, pitch SAS actuator, and pitch/trim). The pitch/trim servo is supplied pressure at a reduced rate of 1000 psi by means of a pressure regulating valve.

7-1-1.4 Backup Hydraulic System.

The Backup Hydraulic System supplies hydraulic pressure for ground checks, rescue hoist operation, backup for first and second stage hydraulic pressure, second stage pressure of the tail rotor servo if first stage pressure is lost, and recharging the APU accumulator. The BACKUP HYD PUMP switch (marked OFF, ON, and AUTO) on the upper console is placed ON during ground checks (APU running) or OFF if the APU is running and hydraulic pressure is not needed. For flight, the switch is placed to AUTO. If hydraulic pressure drops below 2000 psi in the First and/or Second Stage System, the Backup Hydraulic System automatically picks up the load regardless of switch position during flight. After the APU has started, the backup pump recharges the APU accumulator regardless of the position of the BACKUP HYD PUMP switch. The BACK-UP PUMP ON capsule on the caution/advisory panel goes on any time the pump is running. If, during flight, hydraulic pressure drops below 2000 psi in the First and/or Second Stage System, the Backup Hydraulic System automatically picks up the load regardless of BACKUP HYD PUMP switch position.

7-1-1.5 Hydraulic Pump Modules.

The hydraulic pump modules are combination hydraulic pumps and reservoirs (Figure 7-1-1-3). The No. 1, No. 2, and backup pump modules are identical and interchangeable. The No. 1 pump module is mounted on and driven by the left accessory transmission module. The No. 2 pump module is mounted on and driven by the right accessory transmission module. The backup pump module is mounted on and driven by an AC electric motor.

The reservoir part of each pump module has a fluid level indicator window marked EMPTY, REFILL, and FULL. The position of the piston indicator stripe viewed through the sight glass of the pump module compared to the fluid identification and level indicator plate indicates amount of hydraulic fluid in the pumps reservoir. The pressure relief valve and bleed valve protect the pump from high pressure in the return System.

The pump has two valves: a high pressure relief valve and a bleed relief valve; two filters: a pressure filter and return filter. A red indicator button on each filter housing will pop out when differential pressure across the filter reaches 70 ± 10 pounds per square inch differential (psid). The indicator button can only be reset from inside the filter housing when the filter element is replaced. The return filter has a bypass valve that opens when return pressure reaches 100 ± 10 psid. The pressure filter has no bypass.

Each pump has three check valves: one at the external ground coupling, one at the pressure side, and one at the return side.

A low-level switch, mounted on top of each pump module, senses reservoir fluid quantity for that Hydraulic System. When the piston in the pump module reaches the REFILL mark, the piston closes the low-level switch which lights the #1 RSVR LOW, #2 RSVR LOW, or BACK-UP RSVR LOW caution capsule in the caution/advisory panel. In the First and Second Stage Systems, activation of the low-level switch will also signal the Leak Detection Isolation System of a leak.

A depressurization valve in the backup pump module allows the motor to get up to rated speed before a load is applied. When the backup pump motor is turned on, the depressurization valve in the backup pump module destrokes the output pressure of the pump to 700 psi. This valve is held open by the logic module in the right relay panel for four seconds when either the APU generator or external power is supplying power, or for ½ second when the helicopter generators are supplying power. After the pump motor is started, the valve closes, allowing the pump to develop between 3050 ± 50 psi output pressure.

7-1-1.6 Primary Servo Shutoff Switch.

These switches, on both collective stick grips, are marked SVO 1ST STG and 2ND STG. If either stage of any primary servo jams or if pressure is lost, that stage may be shut off. The #1 and #2 PRI SERVO PRESS capsules on the caution/advisory panel show which stage has jammed or lost pressure. The Hydraulic Systems are electrically interlocked through the opposite Hydraulic System's servo pressure switch to prevent both Hydraulic Systems from being shut off at the same time. As an example, when the SVO OFF switch is placed to 1ST STG, second stage pressure must be above 2350 pounds per square inch gaged (psig) before the first stage shutoff valve closes, and the #1 PRI SERVO PRESS capsule goes on. The tail rotor servo is not affected. With the switch in 2ND STG off, only the #2 PRI SERVO PRESS capsule goes on. The pilot-assist servos are not affected.

7-1-1.7 Hydraulic Leak Test Switch.

The HYD LEAK TEST switch, on the upper console, verifies that the #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules and their associated logic circuits in logic modules U1 and U2 (right and left relay panels) will function properly when the associated fluid level switch closes to the refill position caused by the reservoir fluid level dropping to an unacceptable level. Placing the switch in TEST lights the #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules and closes the first stage and second stage tail rotor shut-off valves and pilot assist shut-off valve. When these valves close, the associated capsules go on. Placing the switch in RESET and then back to NORM sets the Hydraulic System back to its normal state. The leak detection isolation feature is shown in Figure 7-1-1-4.

7-1-1.8 APU Accumulator.

The accumulator supplies 3050 ± 50 psig hydraulic charge to the APU starter motor (Figure 7-1-1-5 and 7-1-1-6). Hydraulic fluid in the accumulator compresses a charge of nitrogen. If the back-up pump fails to recharge the accumulator, it may be manually charged by pumping the APU hand pump. The hand pump is on the rear cabin ceiling. A tape indicator assembly on the accumulator shows the percent of the pressure charge in the accumulator. A pressure gauge shows the pressure of the nitrogen precharge. The tape will indicate zero (0) when the hydraulic charge has been released. However, the pressure gauge will indicate the nitrogen precharge pressure, which at 21°C shall be 1450 psig.

7-1-1.9 Winterization Kit.

The Winterization Kit consists of an additional accumulator, identical to the accumulator already installed, and time delay relay for the APU starter motor. The kit provides easier cold weather starting between -31.6°C and 18.3°C. The time delay relay allows the backup hydraulic pump to charge the accumulators for an additional 90 seconds, making the total run time of 180 seconds when the kit is installed.

Both accumulators are mounted on the APU start valve (Figure 7-1-1-6) and are serviced with nitrogen at a common servicing point. If either accumulator malfunctions, the other accumulator will continue to operate normally. The Winterization Kit accumulator is precharged with nitrogen to 1450 psig at 21°C and then pressurized hydraulically to 2900 ± 200 psig. When starting the APU, both accumulators are discharged at the same time. The Winterization Kit relay ensures that the backup pump will run long enough for both accumulators to be recharged hydraulically.

7-1-1.10 Hydraulic Refill Hand Pump and Selector Valve.

The hydraulic refill hand pump on the upper deck, in front of the No. 2 AC generator, is used to refill the pump module reservoirs and the rotor brake system (Figure 7-1-1-4). The refill pump has a total capacity of 1.3 quarts of hydraulic fluid, a replaceable 15-micron filter, and a selector valve. A sight gauge window on the side of the pump shows fluid level. When the fluid level is even with the line through the sight gauge window, a one-quart can of fluid can be added to the pump. The selector valve has four numbered positions: port 1 for the No. 1 pump reservoir, port 2 for the No. 2 pump reservoir, port 3 for the backup pump reservoir, and port 4 for the rotor brake reservoir. The reservoir can be easily refilled by turning the selector valve handle to the desired port, holding the handle down, and cranking the pump handle. This is continued until the indicator on the reservoir indicates FULL.

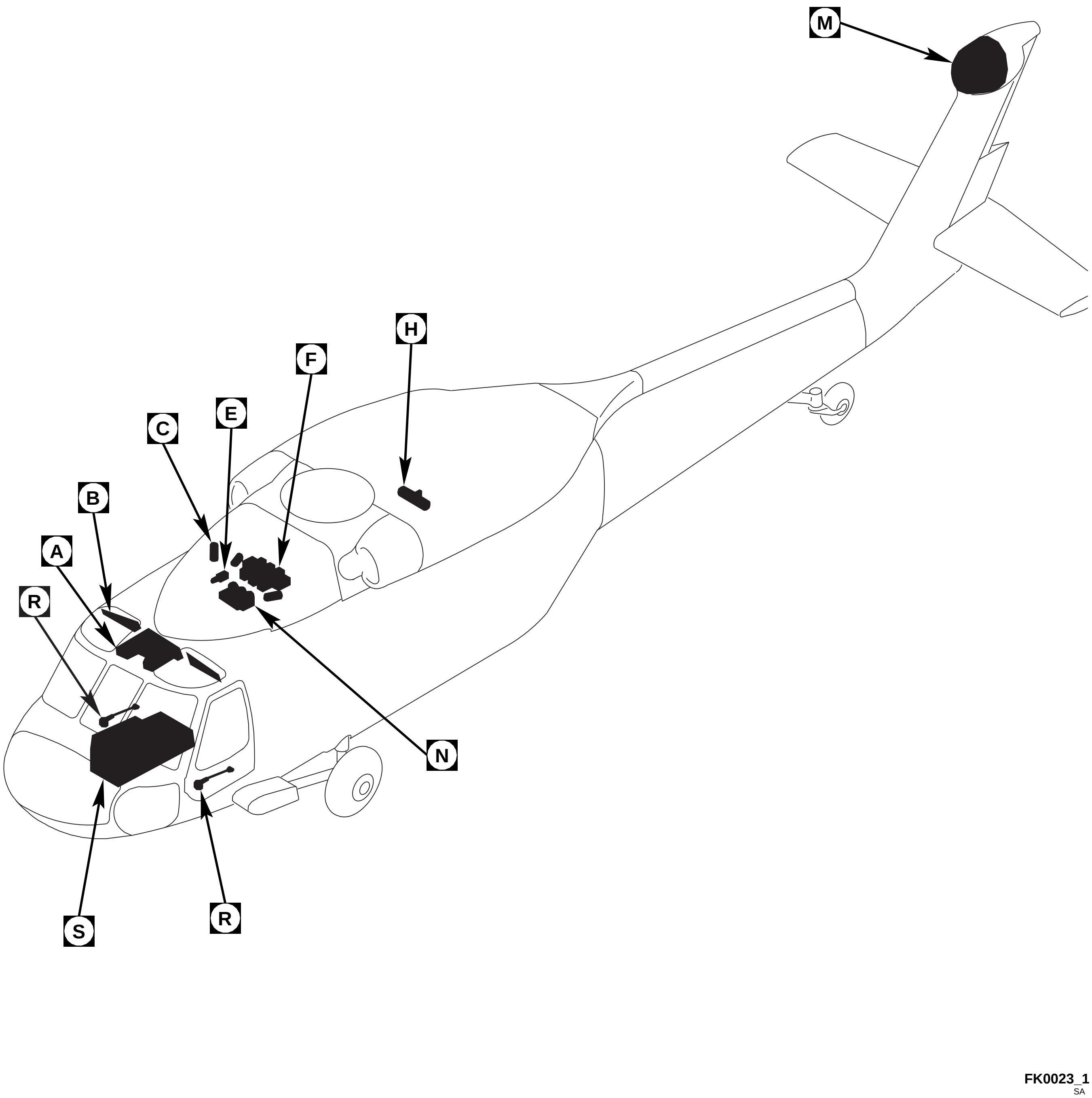

FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 11)

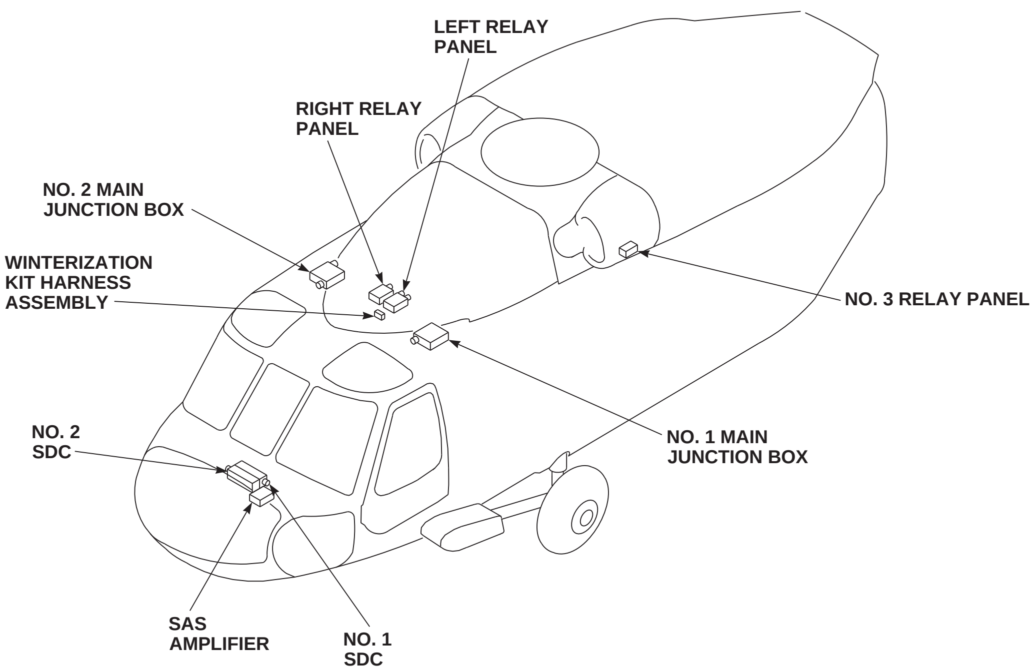

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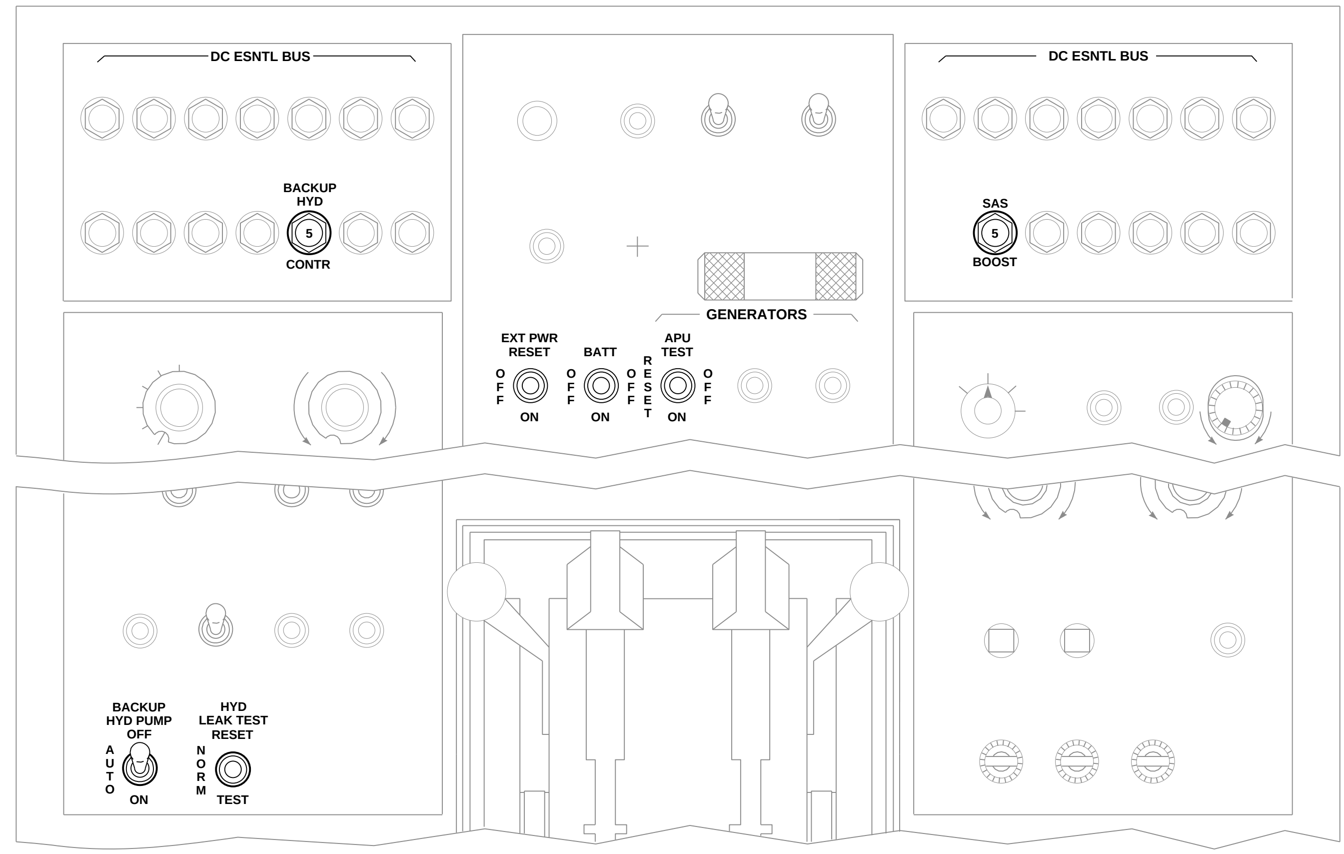
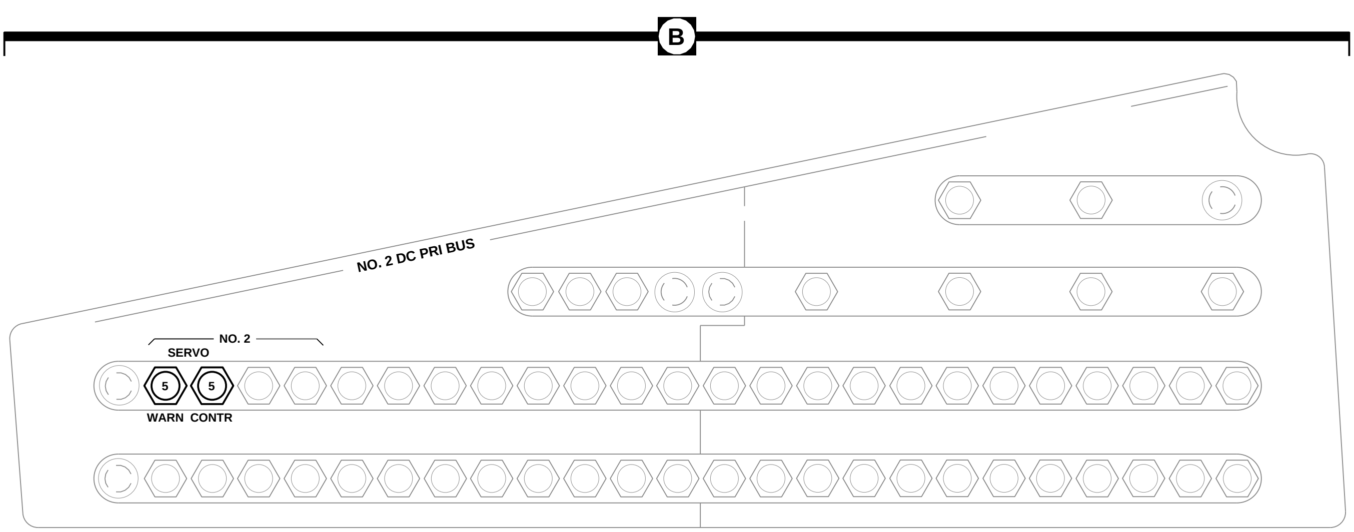
FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 11)



UPPER CONSOLE



PILOT'S CIRCUIT BREAKER PANEL

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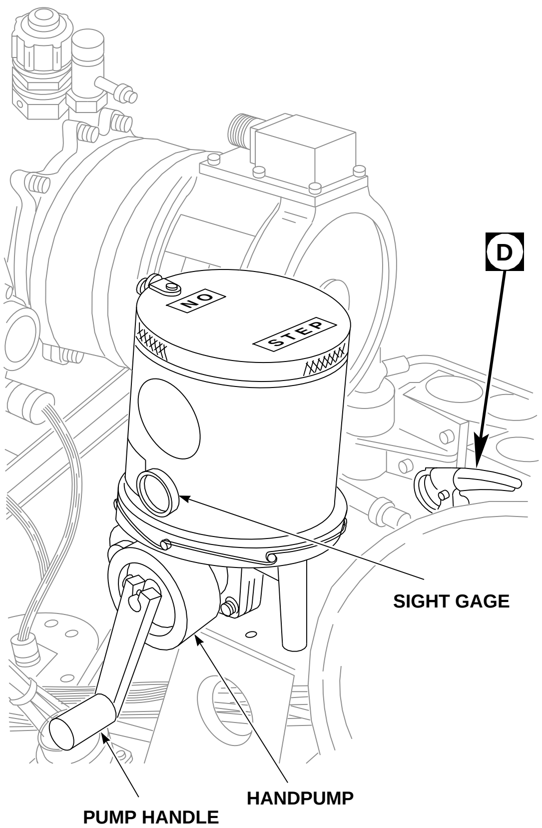
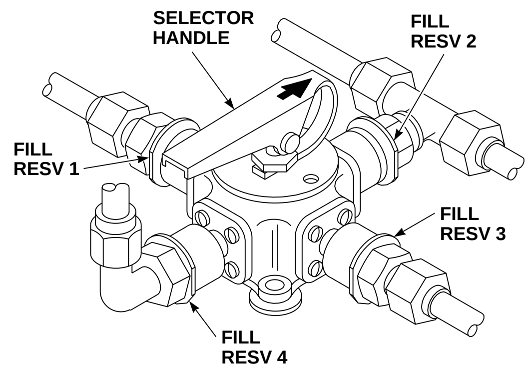
FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 11)



HYDRAULIC REFILL PUMP
AND HYDRAULIC SELECTOR VALVE

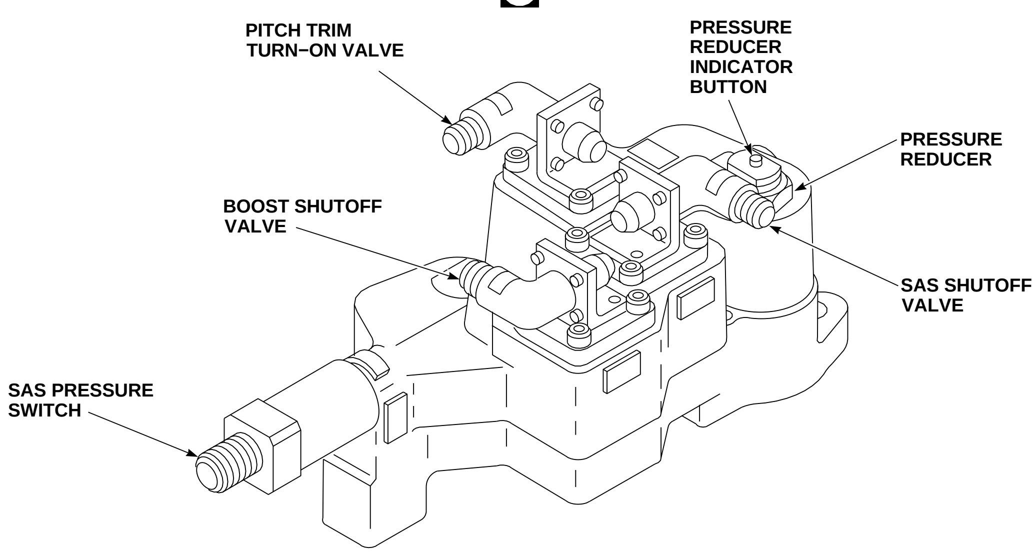
C



SELECTOR VALVE

D

E



PILOT ASSIST MODULE

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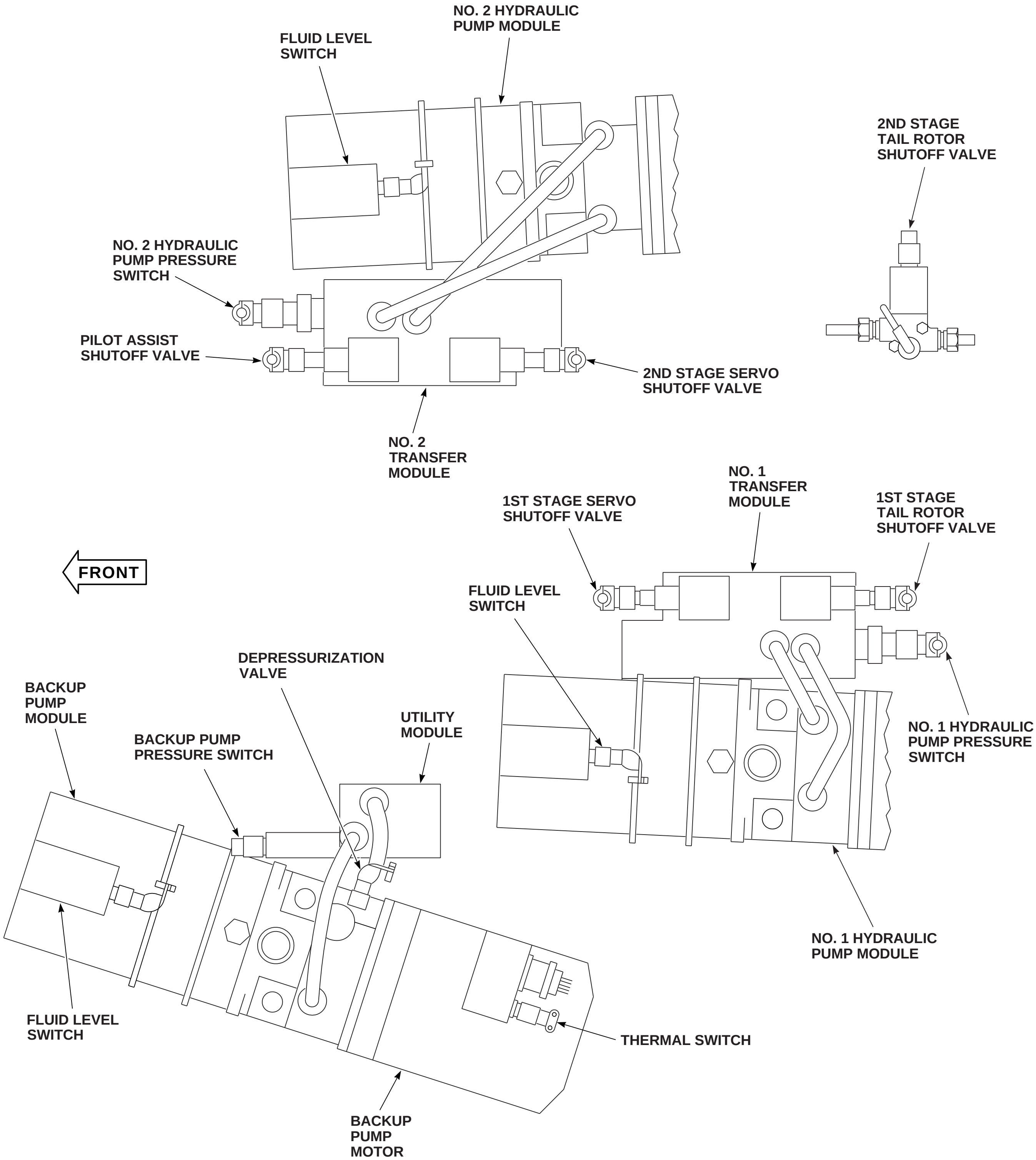
FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 11)

F



MAIN ROTOR PYLON HYDRAULIC COMPONENTS

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FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 5 OF 11)

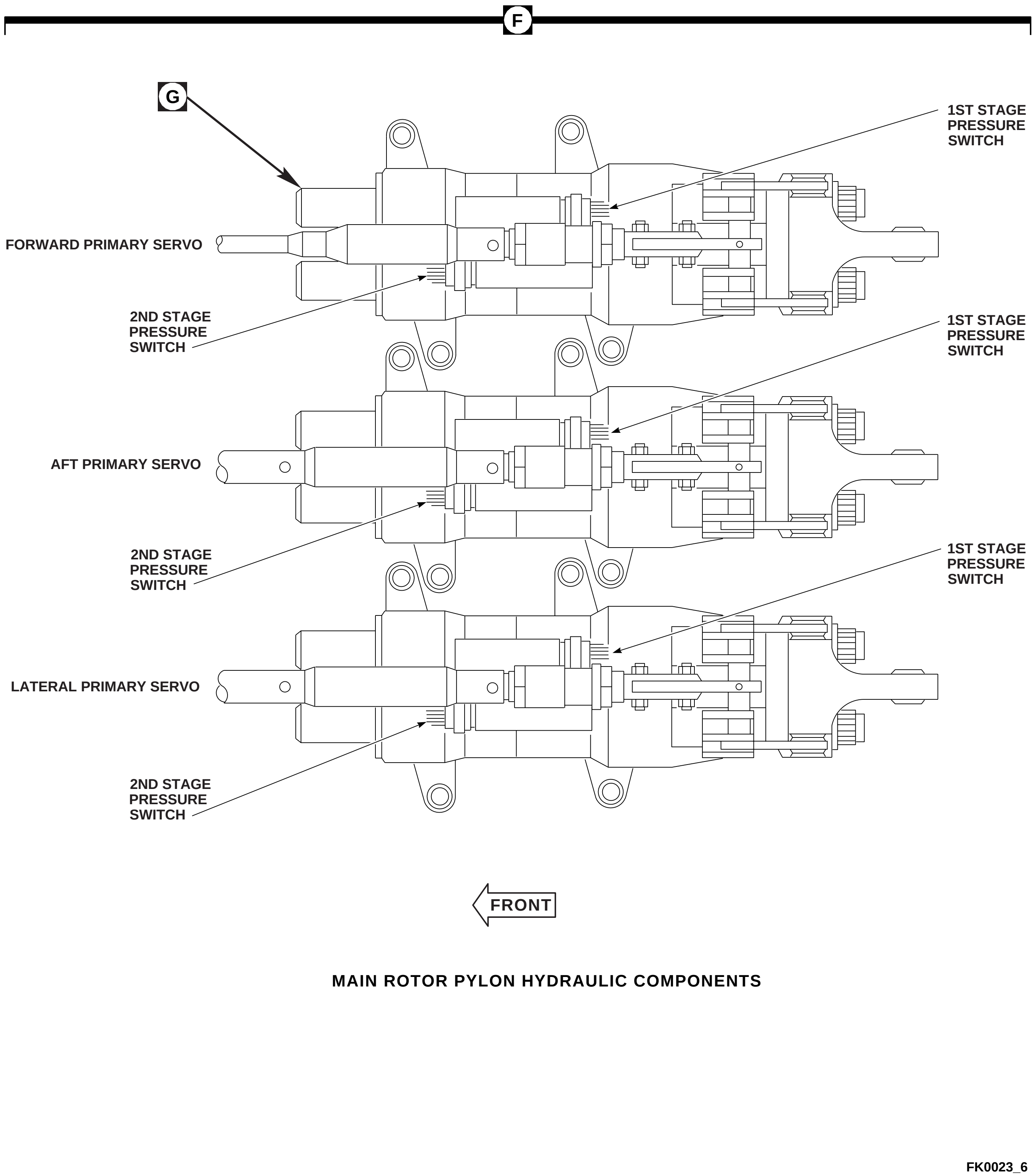


FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 6 OF 11)

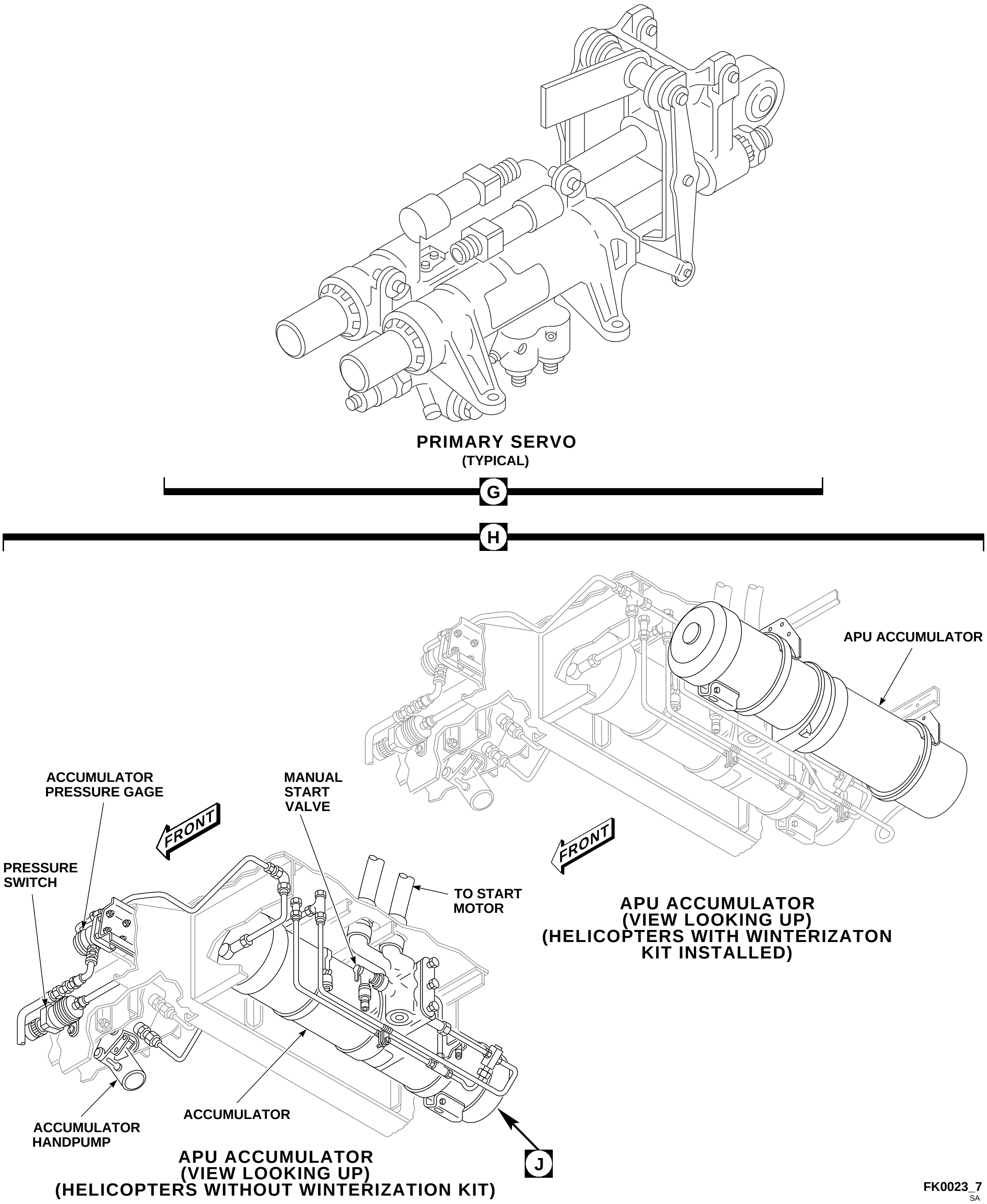


FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 7 OF 11)

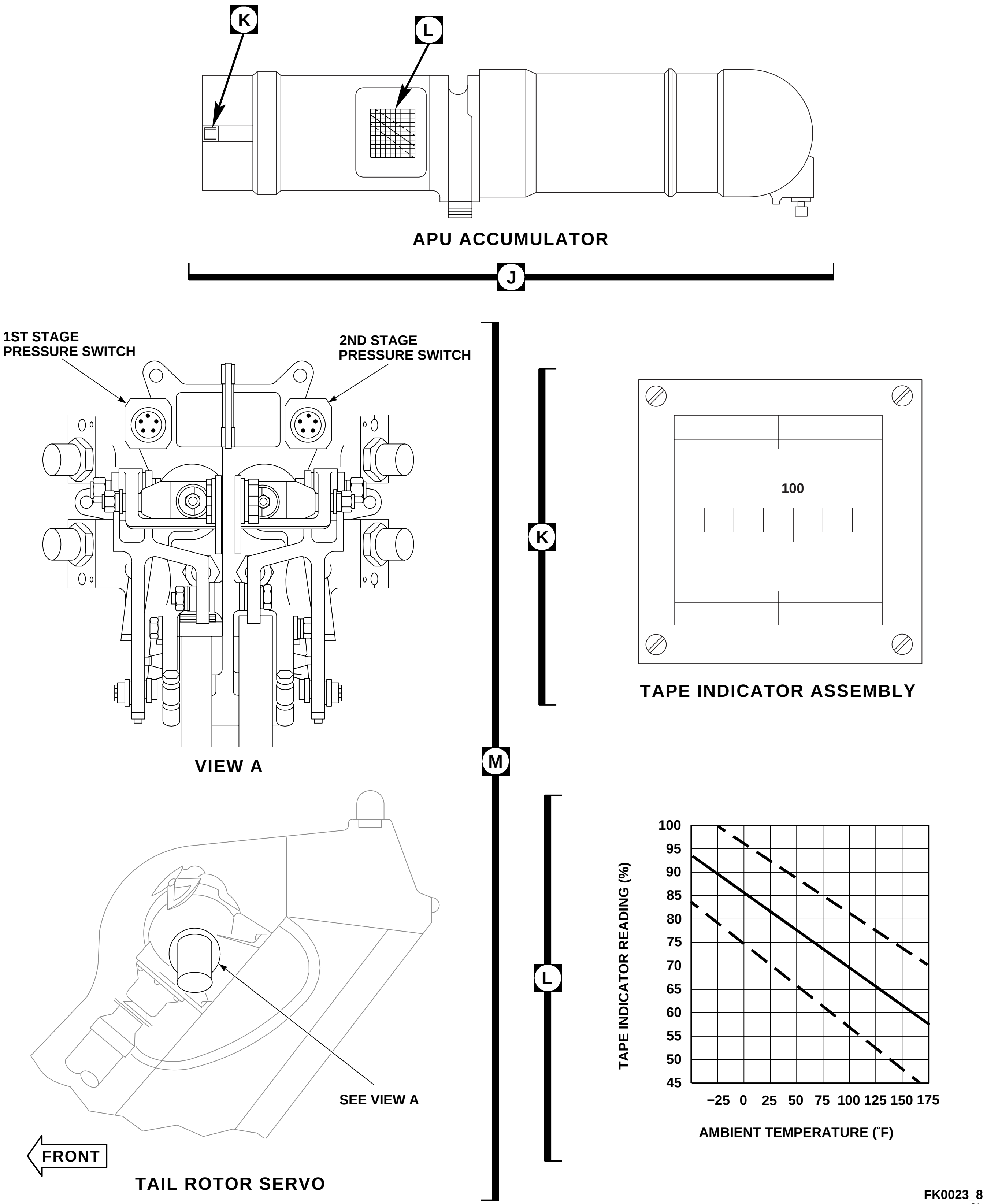
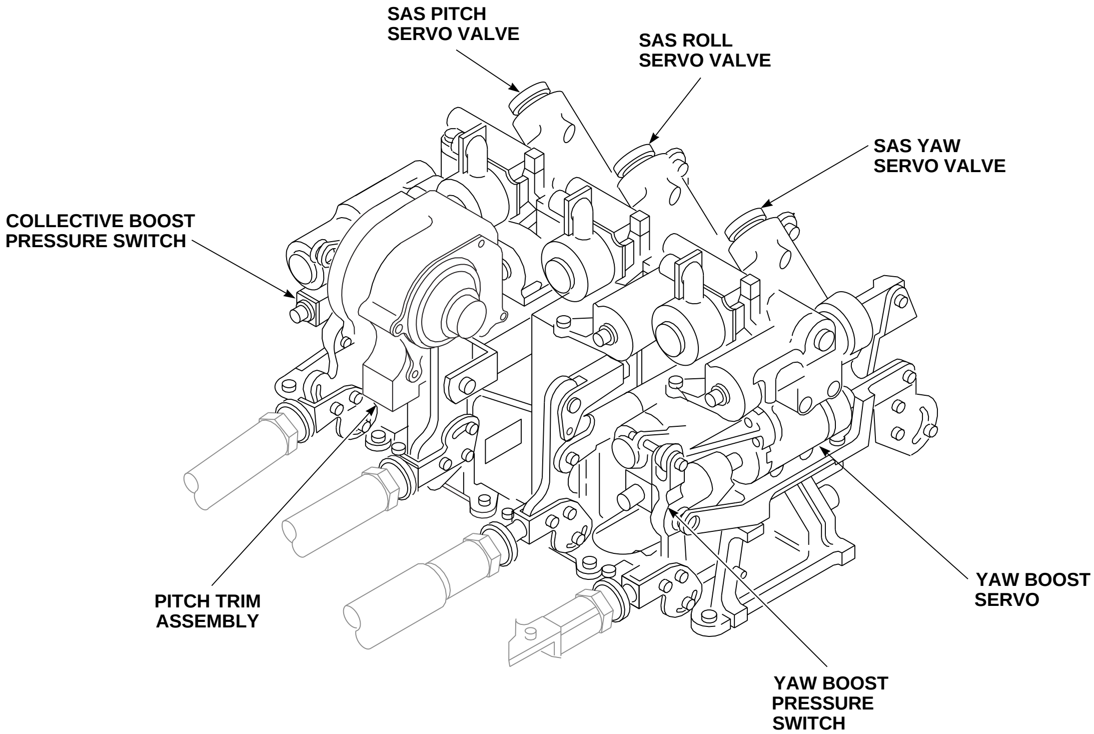
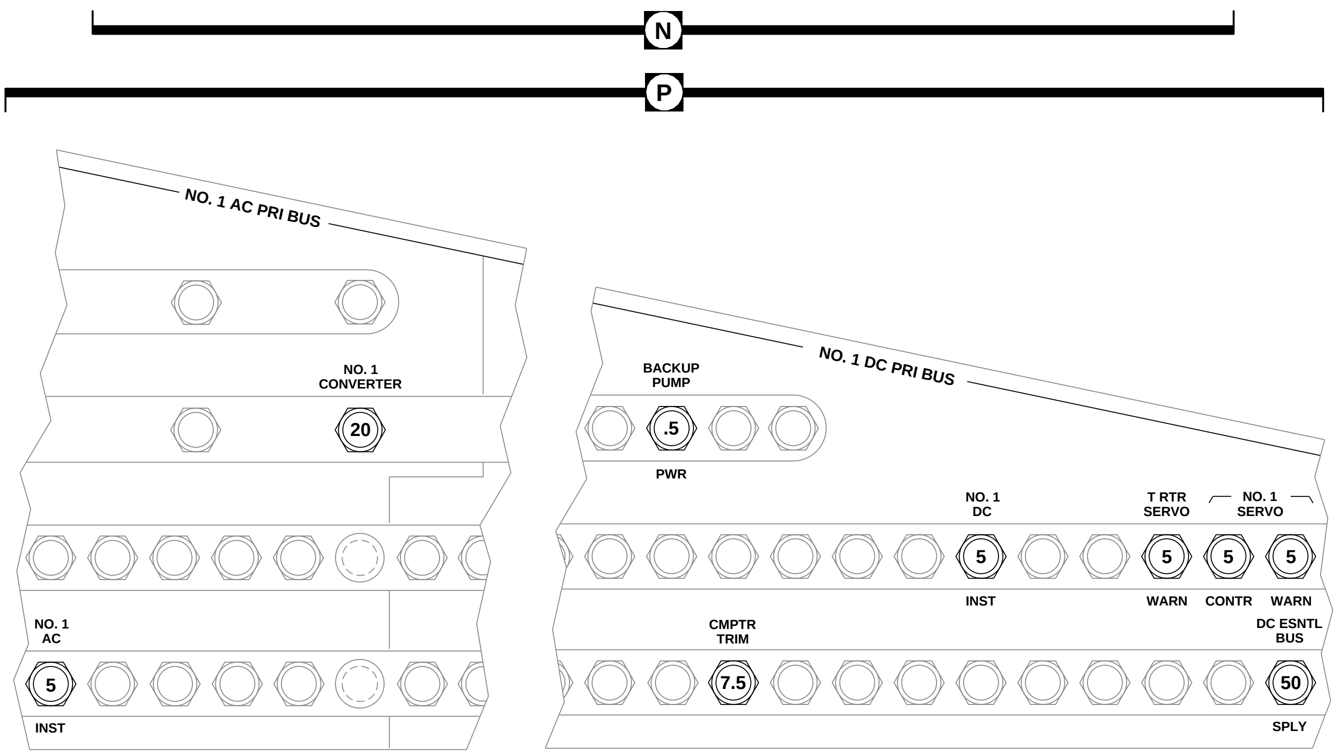


FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 8 OF 11)



PILOT ASSIST SERVOS



COPILOT’S CIRCUIT BREAKER PANEL

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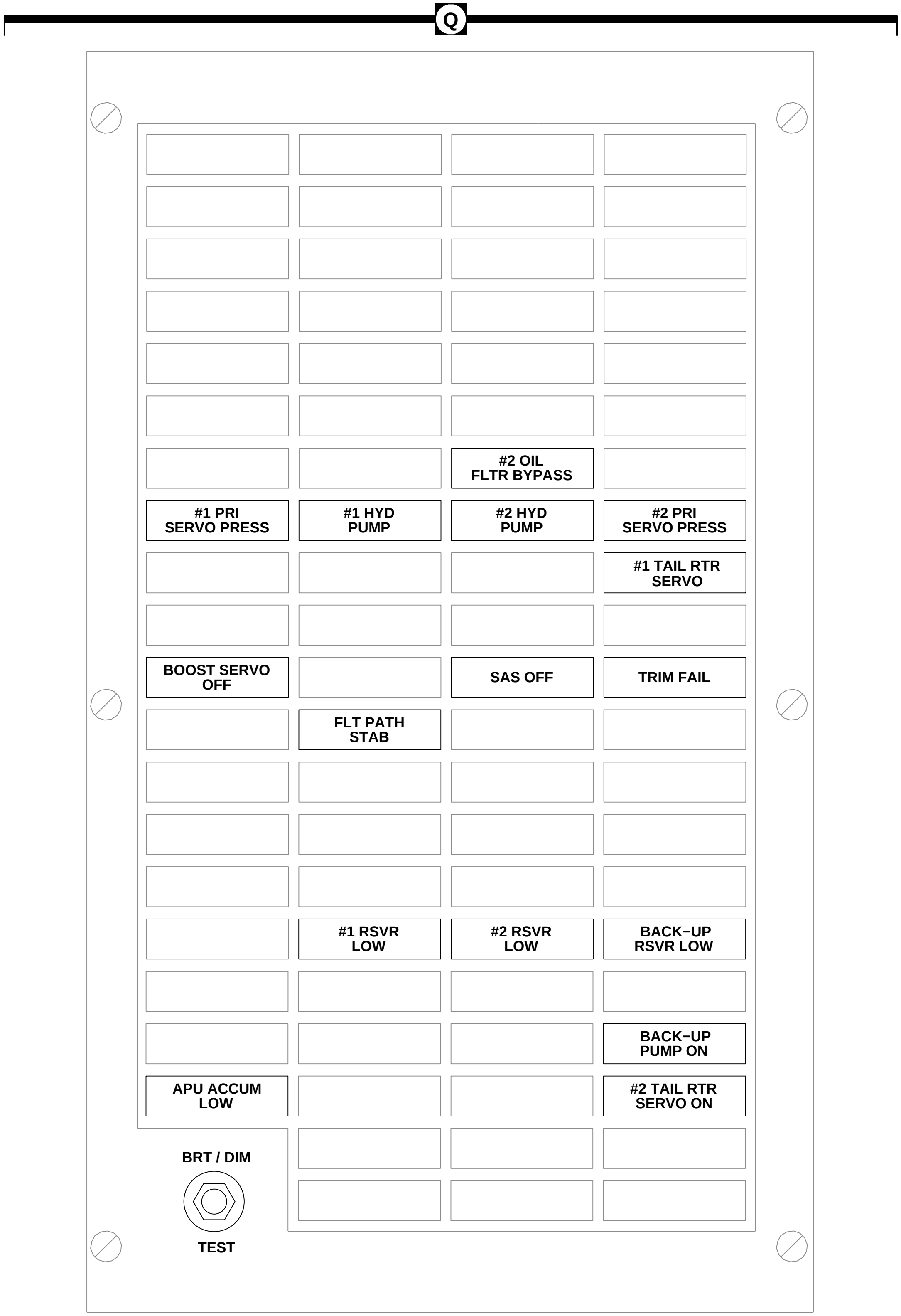
FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 9 OF 11)



CAUTION/ADVISORY PANEL

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FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 10 OF 11)

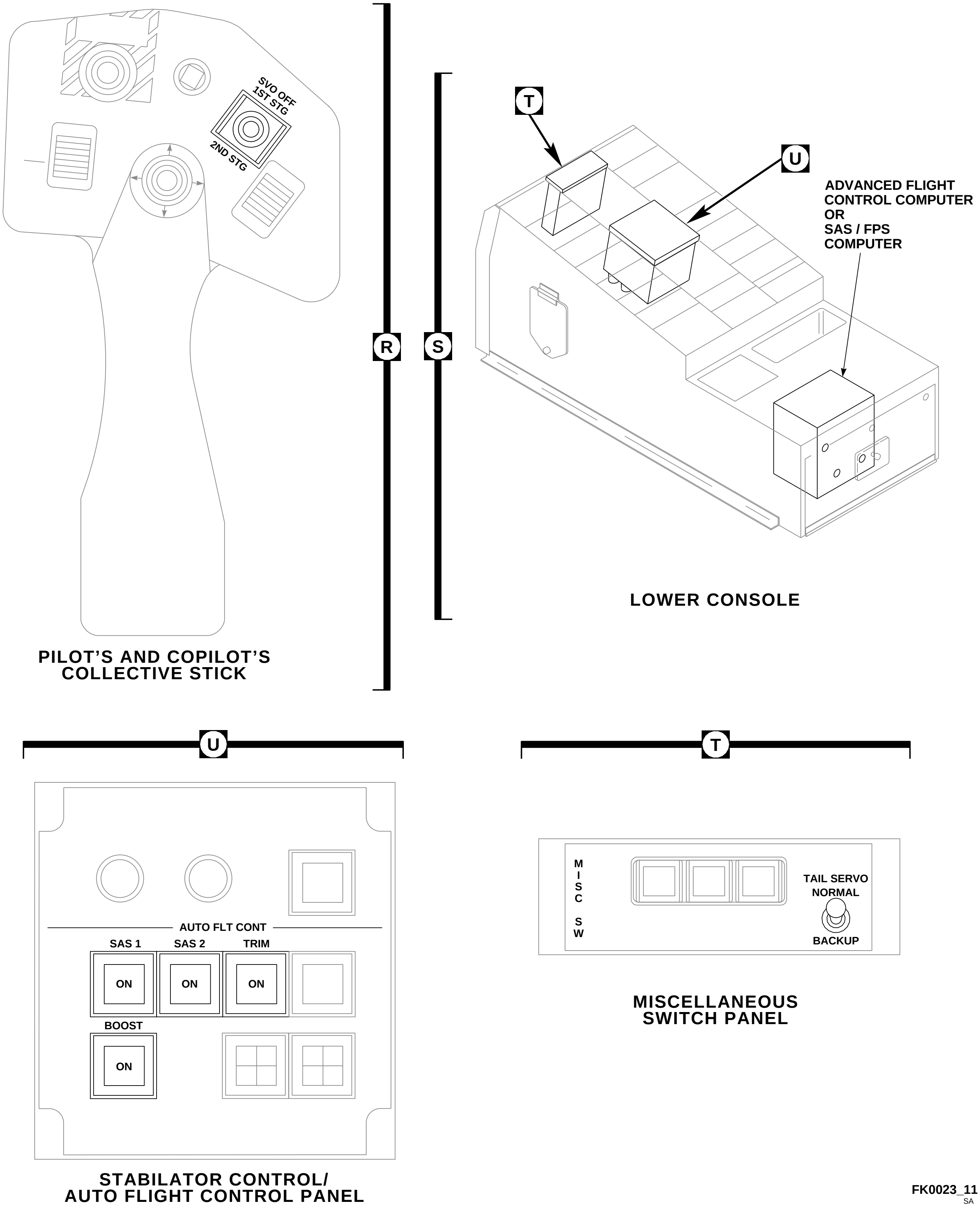


FIGURE 7-1-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 11 OF 11)

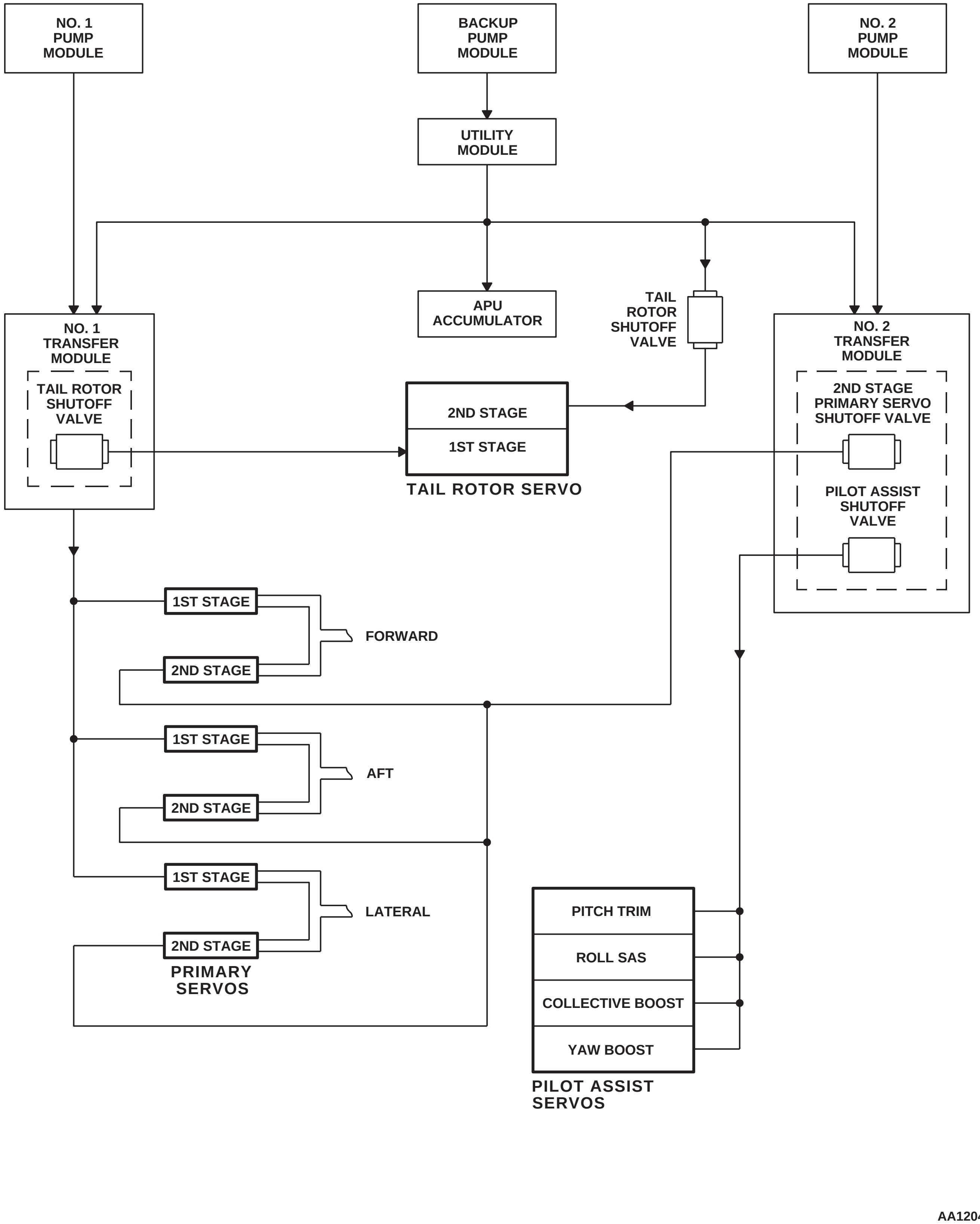


FIGURE 7-1-1-2. HYDRAULIC SYSTEMS SIMPLIFIED BLOCK DIAGRAM

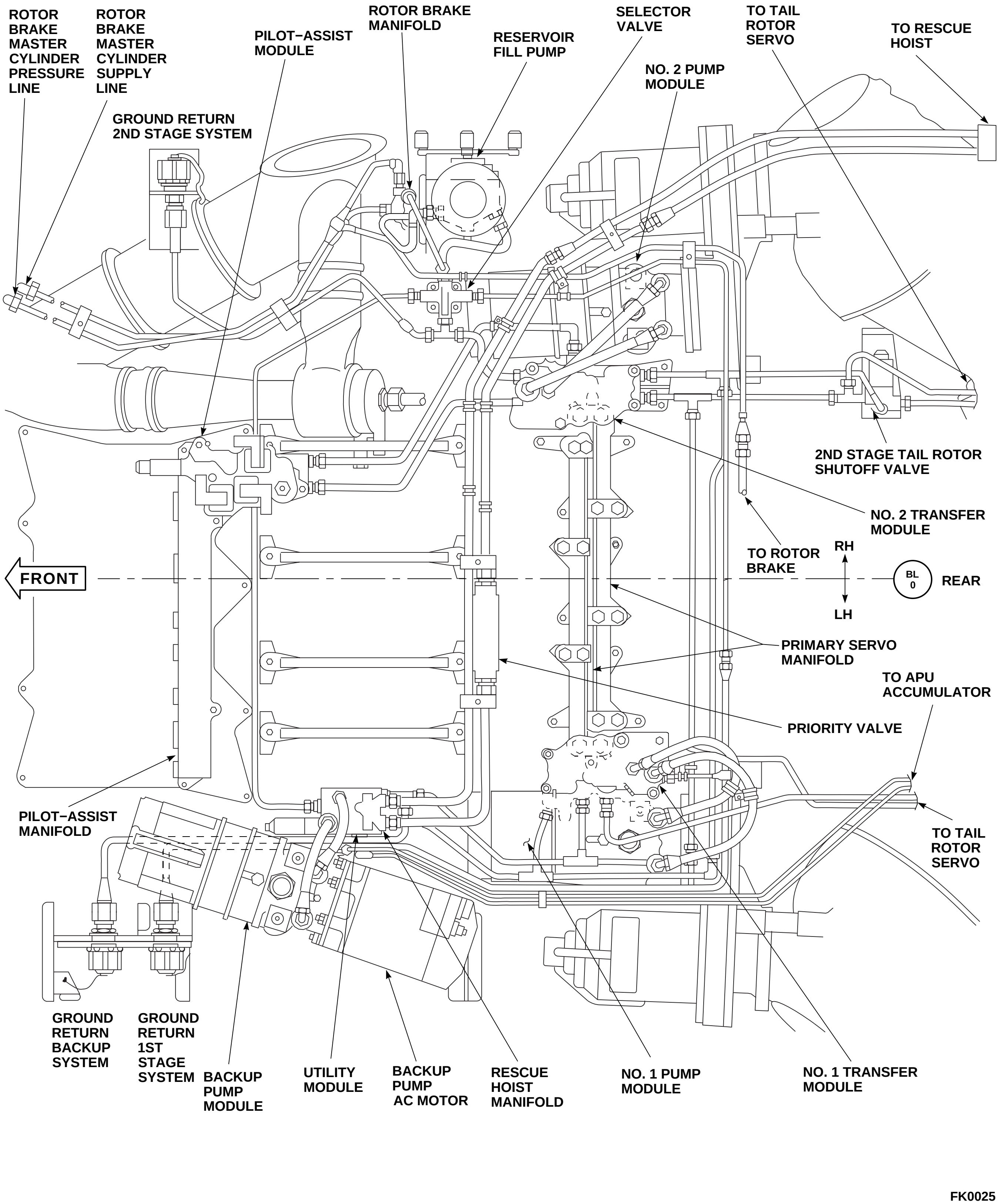
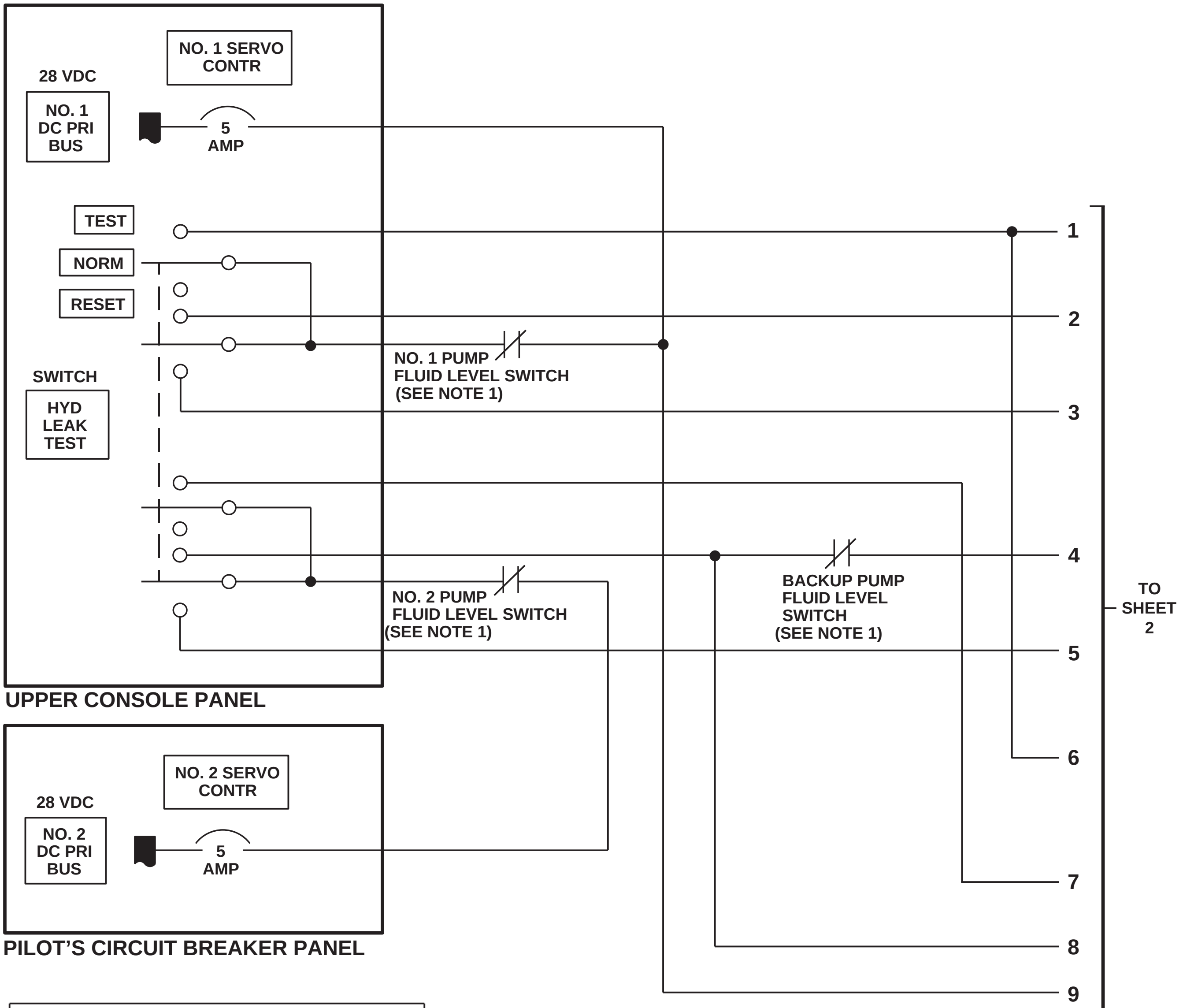


FIGURE 7-1-1-3. MAIN ROTOR PYLON HYDRAULIC COMPONENT LOCATION



NOTES

1. FLUID LEVEL SWITCH CONTACTS ARE SHOWN WITH PUMP MODULE RESERVOIRS PROPERLY SERVICED.
2. WHEN THE 1ST STAGE TAIL ROTOR VALVE CLOSES, THE BACKUP PUMP WILL OPERATE, CAUSING THE BACKUP PUMP ON CAPSULE ON THE CAUTION / ADVISORY PANEL TO GO ON. THEN THE 2ND STAGE TAIL ROTOR SHUTOFF VALVE WILL OPEN (DE−ENERGIZE), CAUSING THE #2 TAIL RTR SERVO ON CAPSULE ON THE CAUTION / ADVISORY PANEL TO GO ON.
3. WHEN THE PILOT ASSIST SHUTOFF VALVE IS CLOSED, THESE CAPSULES SHOULD GO ON: SAS OFF , BOOST SERVO OFF , TRIM FAIL AND FLT PATH STAB (IF ENGAGED).

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FIGURE 7-1-1-4. HYDRAULIC LEAK DETECTION ISOLATION BLOCK DIAGRAM (SHEET 1 OF 2)

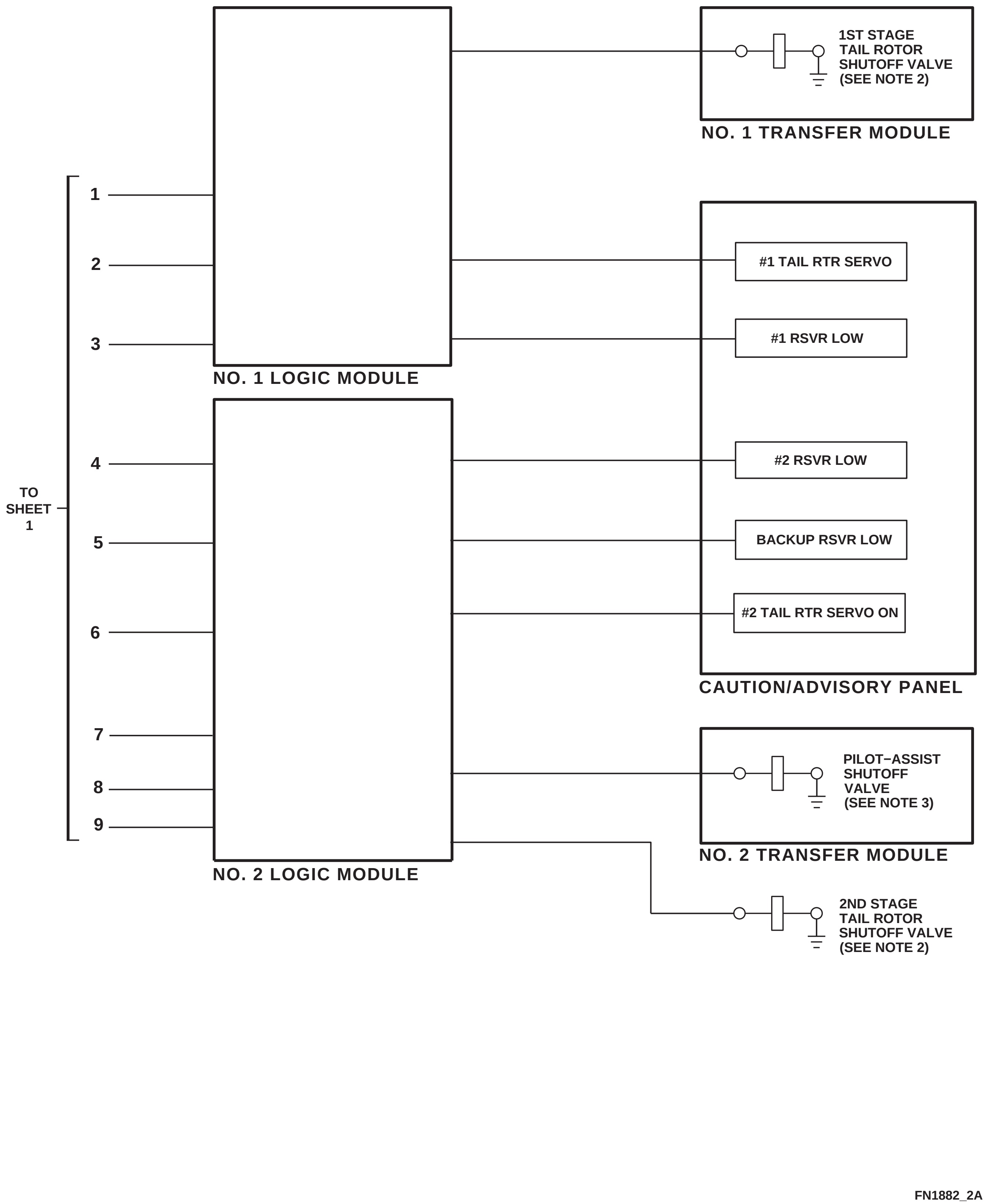


FIGURE 7-1-1-4. HYDRAULIC LEAK DETECTION ISOLATION BLOCK DIAGRAM (SHEET 2 OF 2)

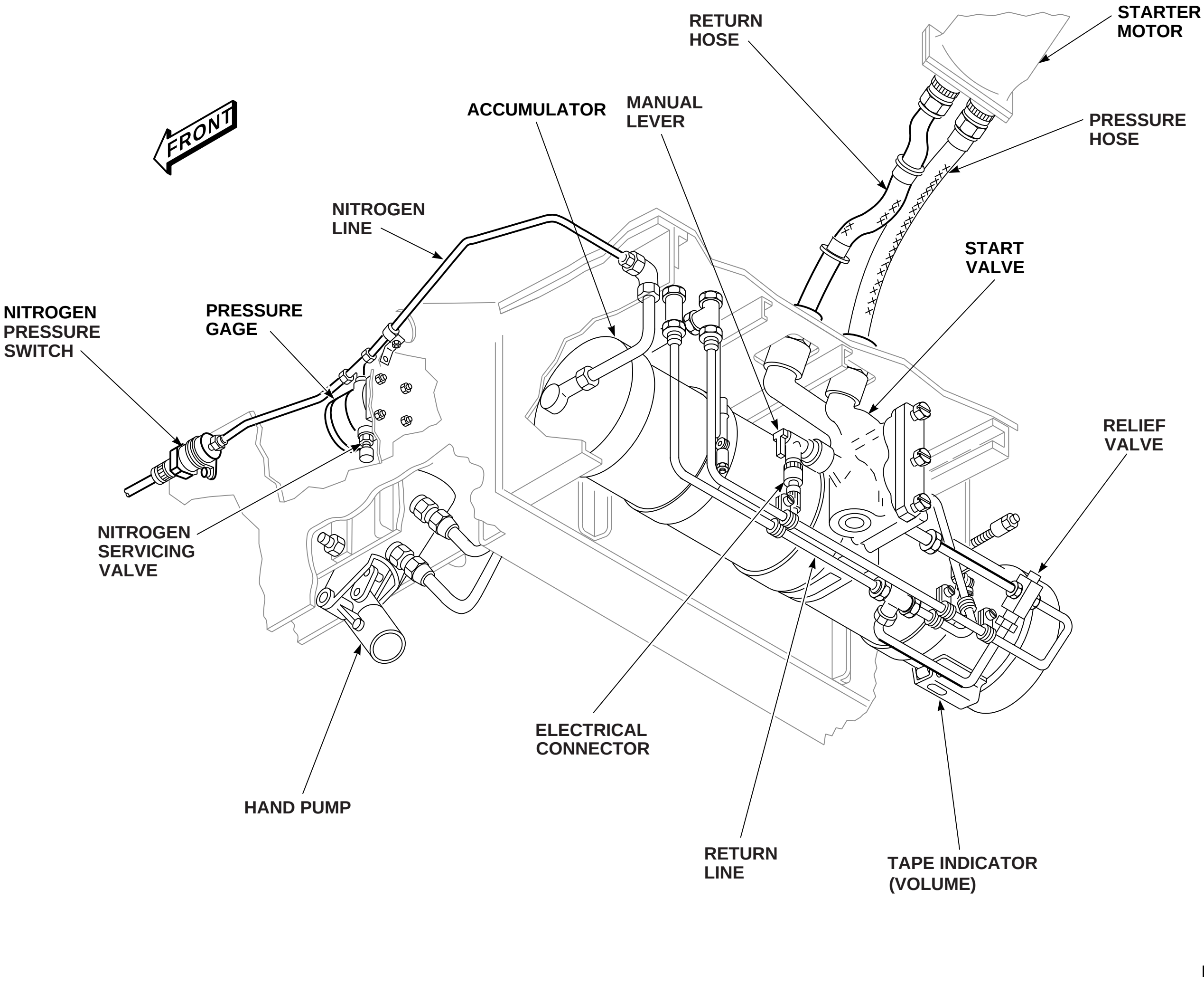


FIGURE 7-1-1-5. APU ACCUMULATOR

7-1-2 HYDRAULIC SERVOS, ACTUATORS, AND MODULES DESCRIPTION AND DATA.

7-1-2.1 Transfer Modules.

The No. 1 and No. 2 transfer modules connect hydraulic pressure from the pump modules to the flight control servos. Each module consist of:

a. Shutoff Valves

b. Pressure Switches

c. Check Valves

d. Shuttle Valves

e. Restrictor

7-1-2.2 No. 1 Transfer Module.

The No. 1 Transfer Module consists:

a. Transfer Valve

b. Pressure Switch

c. First Stage Primary Servo Shutoff Valve

d. First Stage Tail Rotor Shutoff Valve

e. Restrictor

f. Check Valves

The transfer valve is spring-loaded to the open or normal position. If first stage hydraulic pressure is lost, the valve automatically transfers backup pump pressure to the First Stage System.

The first stage primary shutoff valve lets the pilot or copilot, by use of the SVO OFF switch on the collective sticks, shut off first stage pressure to the primary servos.

The pressure switch lights the #1 HYD PUMP caution capsule on the caution/advisory panel when pressure drops to 2,000 psi and also sends a signal to a logic module that pressure is lost in the First Stage Hydraulic System.

The restrictor allows fluid to circulate for cooling under no-flow conditions. If a fluid leak develops past the transfer module, the check valves prevent fluid loss on the return side of the transfer module.

7-1-2.3 No. 2 Transfer Module.

The No. 2 Transfer Module is like the No. 1 module except that it supplies second stage pressure.

The No. 2 Transfer Module consists:

a. Transfer Valve

b. Pressure Switch

c. Pilot-Assist Shutoff Valve

d. Second Stage Tail Primary Servo Shutoff Valve

e. Restrictor

f. Check Valves

The transfer valve is spring-loaded to the open or normal position. If second stage hydraulic pressure is lost, the valve automatically transfers backup pump pressure to the Second Stage System.

The pilot-assist shutoff valve turns off pressure to the pilot-assist module.

The second stage primary servo shutoff valve, controlled by the SVO OFF switch on the collective sticks, turns off pressure to the second stage of the primary servos.

The pressure switch turns on the #2 HYD PUMP caution capsule on the caution/advisory panel when Second Stage System pressure is below 2,000 psi and also sends a signal to a logic module that pressure is lost in the Second Stage System.

The restrictor allows fluid to circulate for cooling under no-flow conditions. If a fluid leak develops past the transfer module, the check valves prevent fluid loss on the return side of the transfer module.

7-1-2.4 Utility Module.

The Utility Module connects hydraulic pressure from the backup pump to:

a. No. 1 Transfer Module

b. No. 2 Transfer Module

c. Second Stage of The Tail Rotor Servo

d. Rescue Hoist

e. APU Accumulator

A pressure switch on the module senses the backup pump operating and turns on the BACK-UP PUMP ON advisory capsule on the caution/advisory panel.

If the flow rate through the module to the APU accumulator goes over 1 ½ gallons per minute (gpm), a velocity fuse shuts off flow.

7-1-2.5 Primary Servos.

There are three interchangeable primary servos (Figure 7-1-2-1):

a. Forward Servo

b. Aft Servo

c. Lateral Servo

The servos provide a power boost to the main rotor flight controls. Servos also reduce feedback forces from the main rotor head.

Each servo has two independent stages (first stage and second stage). Each stage has an independent piston, valve housing, and hydraulic supply. The input linkage is common. The servos are interchangeable.

The primary servo manifold connects the servos to the No. 1 and No. 2 transfer modules.

Each stage of the servo has a jam simulation button. When pressed, the jam simulation button displaces the spool valve sleeve, causing the # 1 or # 2 PRI SERVO PRESS caution light on the caution/advisory panel to go on.

Each stage of a primary servo has a ballistic tolerant feature built in so that if a projectile should damage one stage, that stage will be inoperative, but will not stop the other stage from operating properly.

7-1-2.6 Pilot-Assist Servos and Module.

The Pilot-Assist Servo and Module reduce pilot work load by providing:

a. Control Boost

b. Stick Trimming

c. Stability Augmentation

d. Control Inputs From Automatic Flight Control Set (AFCS)

7-1-2.7 Collective Boost Servo.

The Collective Boost Servo reduces stick and flight control friction (Figure 7-1-2-2). The servo is controlled by a BOOST pushbutton switch, on the STABILATOR CONTROL/AUTO FLT CONT panel.

The collective boost servo has a jam simulation button. When pressed, the pushbutton switch displaces the spool valve sleeve and causes the BOOST SERVO OFF capsule on the caution/advisory panel to go on.

7-1-2.8 Yaw Boost Servo.

The Yaw Boost Servo reduces stick and flight control friction (Figure 7-1-2-3). The yaw boost servo is the same as the collective boost except for the addition of a Stability Augmentation Set (SAS) actuator, which provides rate dampening.

The servo is controlled by BOOST pushbutton switch, on the STABILATOR CONTROL/AUTO FLT CONT panel.

The yaw boost servo has a jam simulation button. When pressed, the button displaces the spool valve sleeve and causes the BOOST SERVO OFF capsule on the caution/advisory panel to go on.

7-1-2.9 Roll SAS Actuator.

The Roll SAS Actuator is a Dynamic Rate Stabilization System that gives rate dampening for the helicopter in the roll axis (Figure 7-1-2-4). When engaged, the helicopter cockpit controls do not move. The actuator is controlled by SAS 1 and SAS 2 pushbutton switches on the STABILATOR CONTROL/AUTO FLT CONT panel.

7-1-2.10 Pitch Trim Actuator Assembly.

The Pitch Trim Actuator Assembly controls the longitudinal axis and the attitude of the helicopter (Figure 7-1-2-5). The actuator is controlled by the SAS 1, SAS 2, TRIM, and FPS pushbutton switches on the STABILATOR CONTROL/AUTO FLT CONT panel. Trim maintains a position of the cyclic stick in the longitudinal axis.

7-1-2.11 Pilot-Assist Module.

The Pilot-Assist Module consists of:

a. Thermal Relief Valve

b. Pressure Reducer

c. SAS Shutoff Valve

d. Boost Shutoff Valve

e. Pitch Trim Turn-On Valve

f. Pressure Switch

g. Self-Sealing Quick-Disconnect Couplings

The Thermal Relief Valve protects the module from damage due to thermal expansion of hydraulic fluid kept in the module during storage. The thermal relief valve has no function when the module is installed on the helicopter.

The Pressure Reducer reduces Hydraulic System hydraulic pressure from 3,000 to 1,000 psi for pitch/trim servo operation. It has a built-in relief valve to protect the pitch trim servo from adverse Hydraulic System pressure. If the pressure reducer fails, the relief valve goes into bypass, and a visual indicator, on the pressure reducer, pops. The indicator will remain visible until manually reset.

The SAS shutoff valve turns off system pressure to the SAS actuators.

The boost shutoff valve turns off system pressure to the collective and yaw boost servos.

The pitch trim turn-on valve turns on system pressure to the pitch trim servo.

The pressure switch on the module turns on the SAS OFF capsule on the caution/advisory panel when pressure drops below limits.

The module also has self-sealing, quick-disconnect couplings on all input and output ports for ease of maintenance.

7-1-2.12 Tail Rotor Servo.

The Tail Rotor Servo is located on the tail gear box. It furnishes a power boost to the tail rotor flight controls (Figure 7-1-2-6). The servo has two independent stages, first and second. The stages of the servo are controlled by the TAIL SERVO switch located on the miscellaneous switch panel on the lower console. A cooling restrictor is installed for the No. 1 pump module. Normally only the first stage of the servo is pressurized.

7-1-2.13 Logic Modules.

Two logic modules, one in the left relay panel and the other in the right relay panel, are used to control the operation of the Hydraulic Systems. The logic modules continually monitor the operation of the Hydraulic Systems by inputs received from pressure switches, fluid level switches on the pump modules, and inputs received from control switches in the Hydraulic System. The outputs of the logic modules will turn on capsules on the caution/advisory panel notifying the pilot of a failure, turn off a valve due to a Hydraulic System malfunction, or command the backup pump to operate.

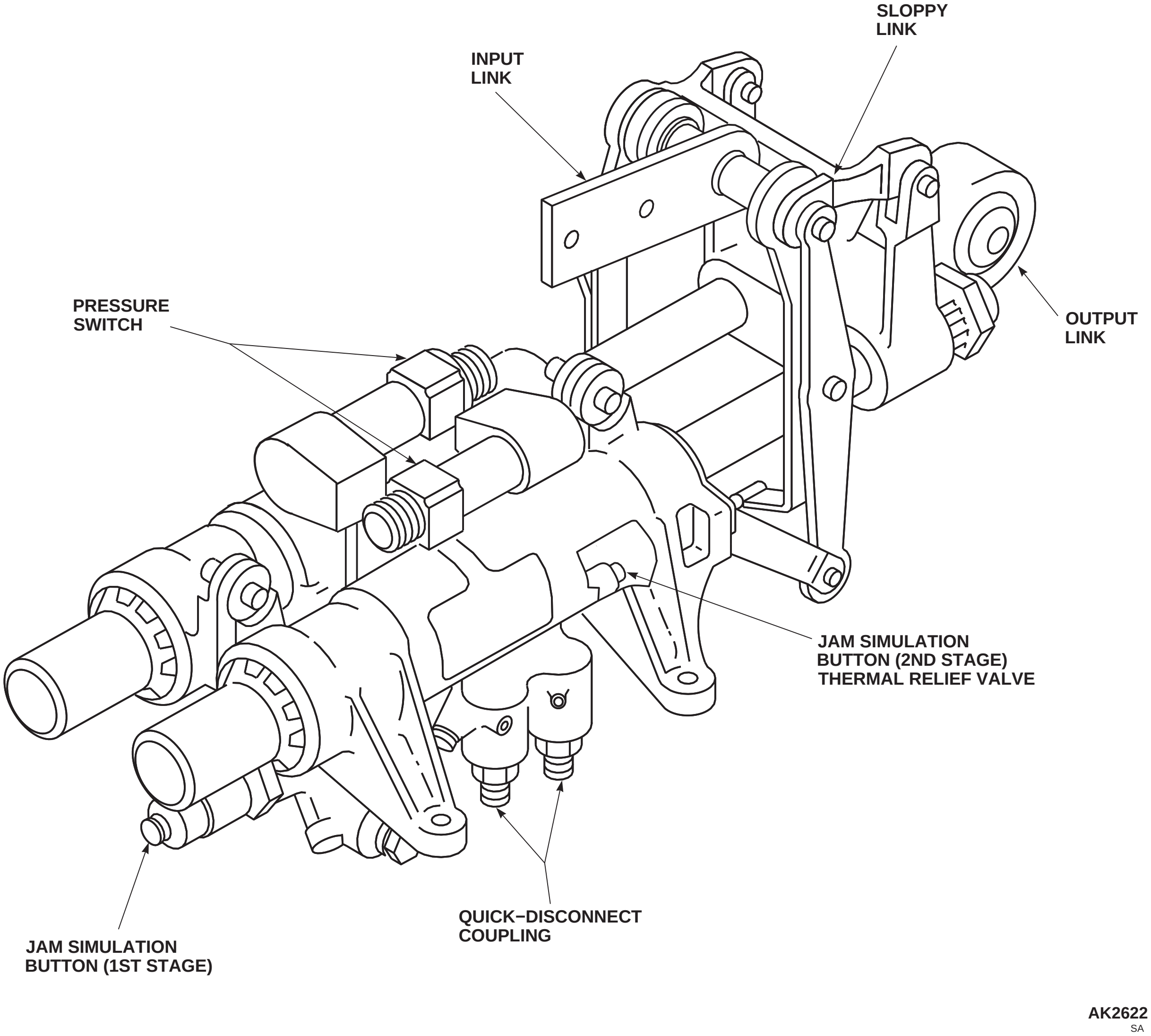


FIGURE 7-1-2-1. PRIMARY SERVO

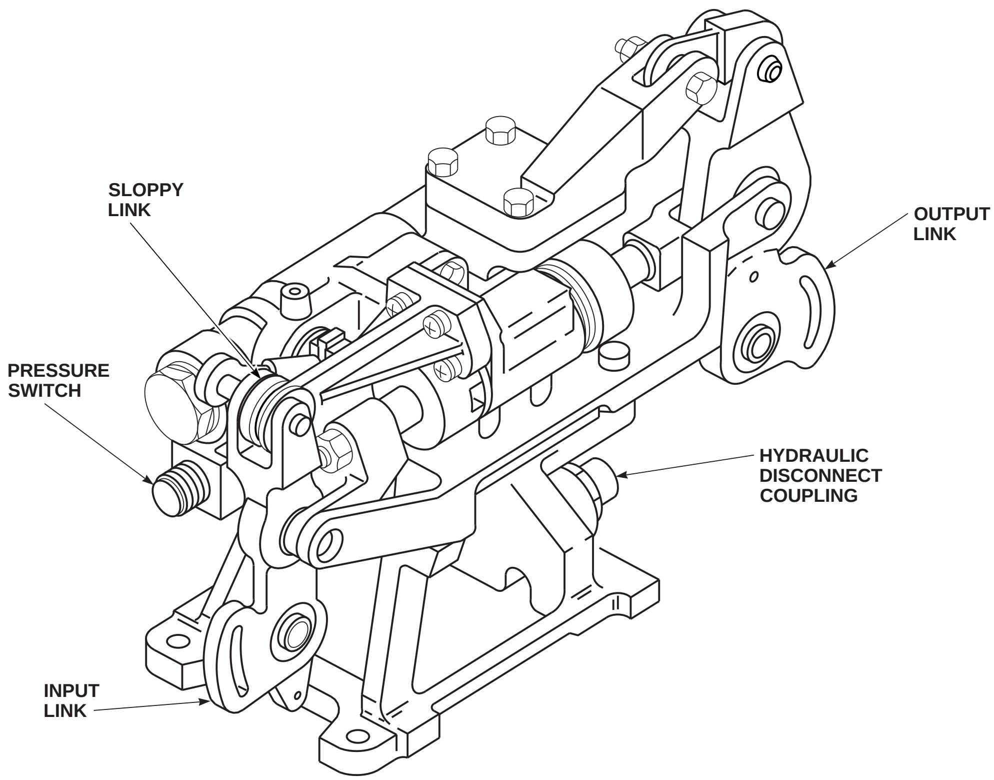


FIGURE 7-1-2-2. COLLECTIVE BOOST SERVO

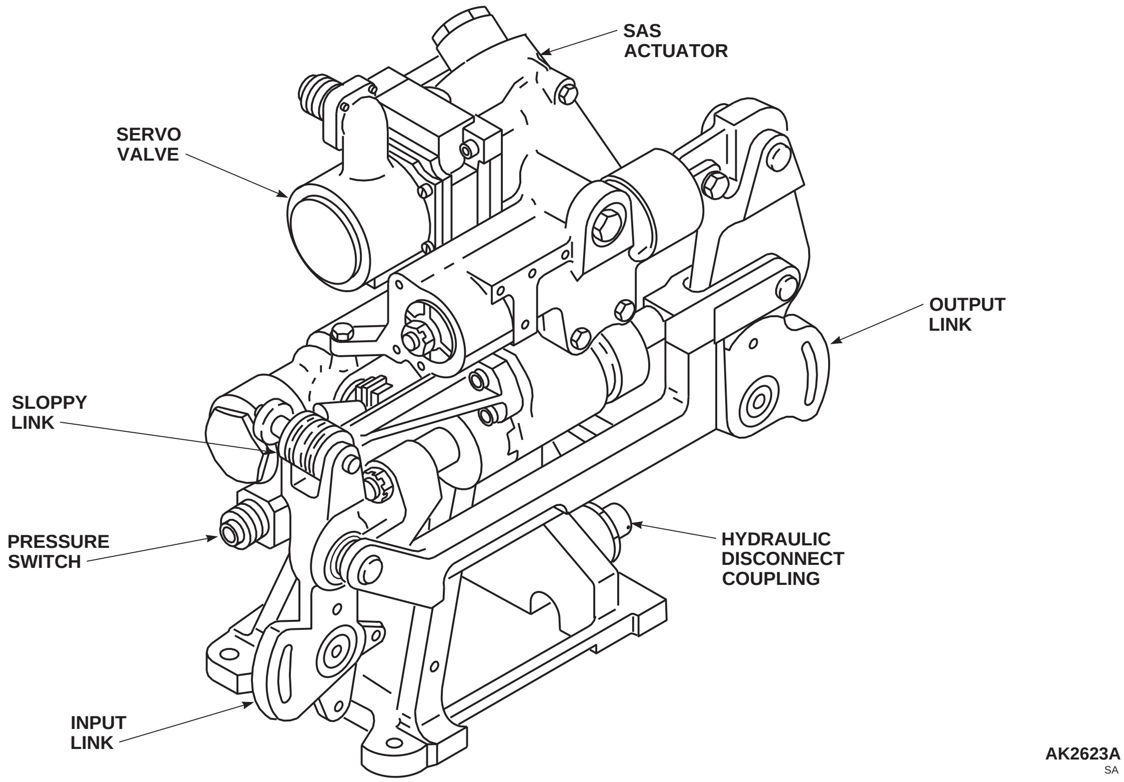


FIGURE 7-1-2-3. YAW BOOST SERVO

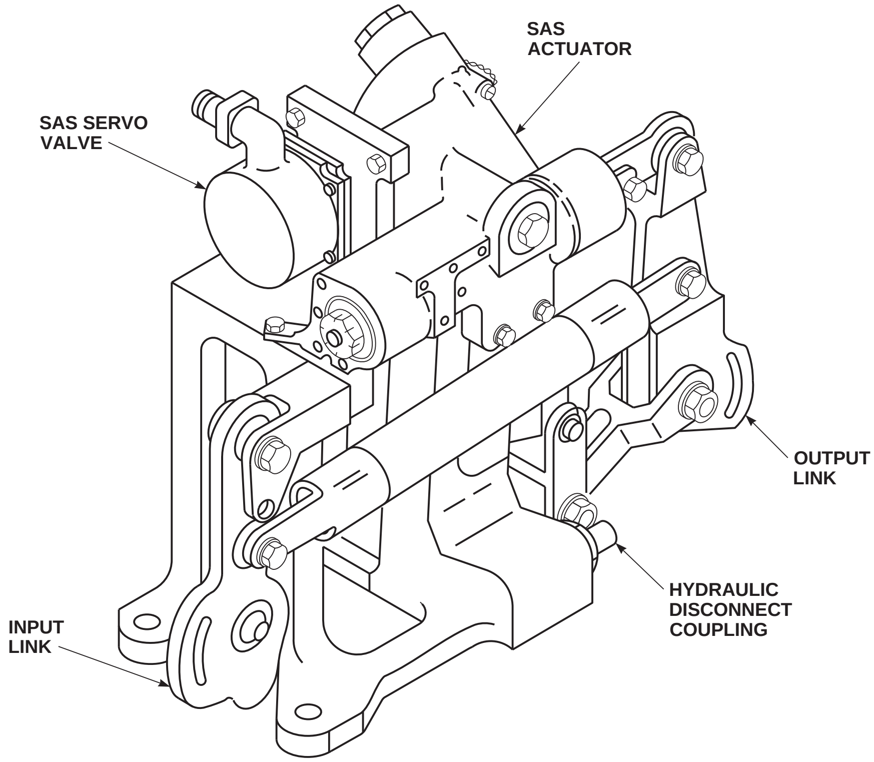


FIGURE 7-1-2-4. ROLL SAS ACTUATOR

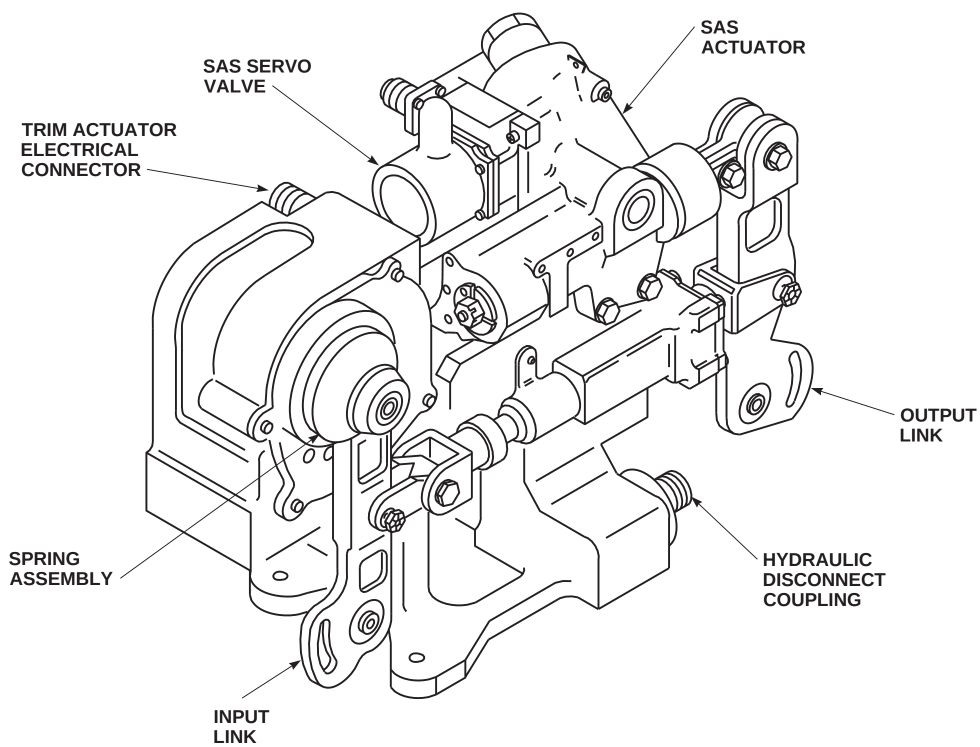


FIGURE 7-1-2-5. PITCH TRIM ACTUATOR ASSEMBLY

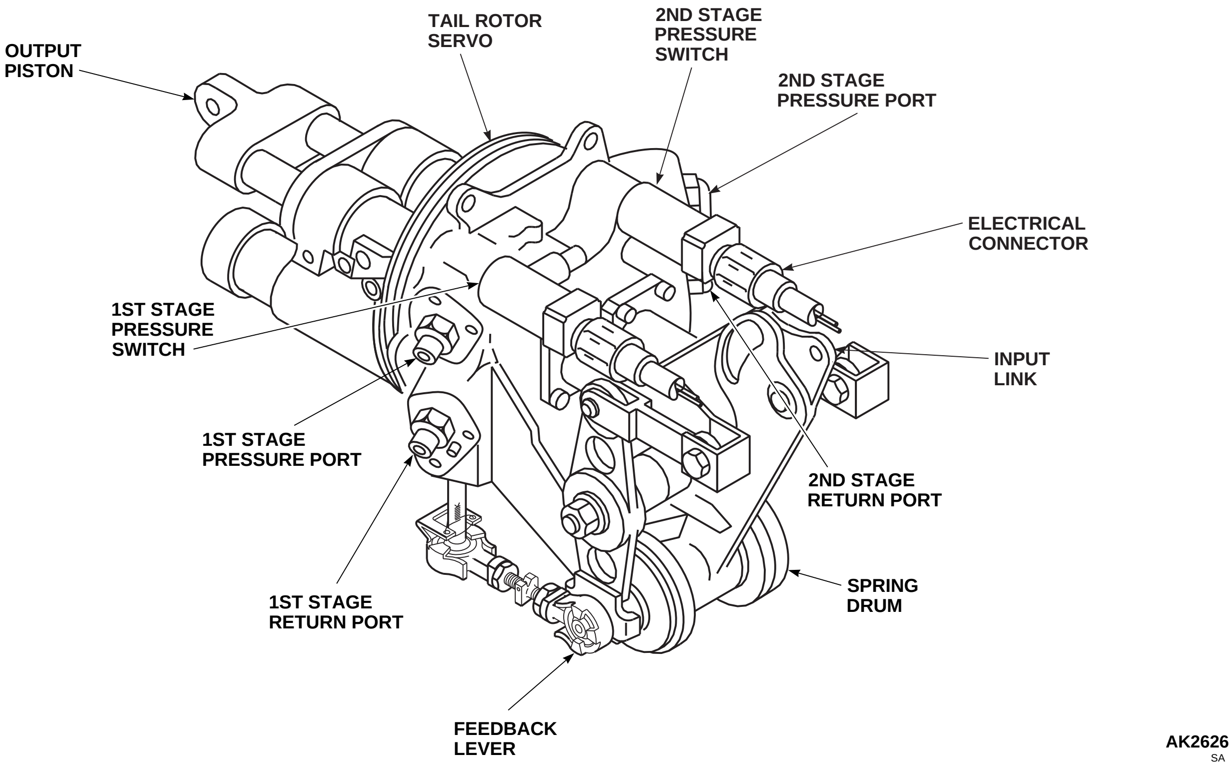


FIGURE 7-1-2-6. TAIL ROTOR SERVO

SECTION 7-2

TROUBLESHOOTING

SECTION OVERVIEW

7-2-1 HYDRAULIC SYSTEMS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Drag Beam Wedge, Figure H-167, Appendix H
- Electrical Repairer Toolkit
- Hydraulic Cart, 05-7040-1200
- Insulated Jumper Wire (2-inch long, 2 required)
- Stopwatch

References

- Appendix H
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- Appropriate Flight Manual

Equipment Conditions

- No. 1 Pump Module Reservoir, Properly Serviced
- No. 2 Pump Module Reservoir, Properly Serviced
- Backup Pump Module Reservoir, Properly Serviced
- APU Accumulator, Properly Serviced
- Winterization Kit Accumulator (If Installed), Properly Serviced
- External Electrical Power Available and Auxiliary Power Unit (APU) Operational

CAUTION

To prevent overheating backup hydraulic pump, use this chart for limits when running backup pump.

FAT C°	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
32° max	Unlimited	-
33° to 38°	24	72
39° to 52°	16	48

CAUTION

If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (NO. 1 DC PRI BUS - copilot's circuit breaker panel).

NOTE

Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.

NOTE

To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.

NOTE

Connecting External Hydraulic Power To Helicopter

Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective reservoir low capsule on the caution/advisory panel to go on.

NOTE

With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel shall be on:

#1 HYD PUMP
#2 HYD PUMP
#1 PRI SERVO PRESS
#2 PRI SERVO PRESS
#1 TAIL RTR SERVO
SAS OFF
BOOST SERVO OFF
#1 RSVR LOW
#2 RSVR LOW

Ignore TRIM FAIL and FLT PATH STAB capsules.
These capsules on the caution/advisory panel shall be off:

BACK-UP PUMP ON
BACK-UP RSVR LOW
#2 TAIL RTR SERVO ON
APU ACCUM LOW

Special Instructions

An individual procedure is to be done after a particular component change. All procedures are to be done if a complete checkout of the hydraulic system is required.

The hydraulic system operational/troubleshooting procedure is subdivided as follows:

(1) Backup Pump Operation - General Information

(2) Switch And Circuit Breaker Setup

(3) APU Accumulator - Recharge Using APU

(4) APU Accumulator - Recharge Using External Power

(5) Hydraulic Leak Test

(6) No. 1 Hydraulic System

(7) No. 2 Hydraulic System

(8) Backup Hydraulic System

(9) Primary Servo Pressure Switch Check

(10) Shutdown

7-2-1.1 Backup Pump Operation - General Information.

Whenever the backup pump is turned on, either by placing the BACKUP HYD PUMP switch S31 to ON or by simulating a failure, the depressurization valve in the backup pump module is energized, allowing the pump to go into bypass (low pressure output) while the pump motor is accelerating to rated speed.

If the APU generator is supplying helicopter power, there is a four second delay after the pump is turned on. If a main generator is supplying helicopter power, there is a 0.5 second delay after the pump is turned on, the depressurization valve is deenergized and the backup pump strokes up to rated pressure (3000 psig), causing the BACK-UP PUMP ON capsule on the caution/advisory panel to go on.

Whenever the backup pump is turned on by a simulated failure (when BACKUP HYD PUMP switch S31 is placed to AUTO), the backup pump will continue to operate for approximately 90 seconds, (180 seconds when winterization kit is installed), even if the failure input is removed.

7-2-1.2 Switch And Circuit Breaker Setup.

a. Make sure switches are placed as indicated:

SWITCH	LOCATION	POSITION
TAIL SERVO	Lower Console	NORMAL
BACKUP HYD PUMP	Upper Console	OFF
HYD LEAK TEST	Upper Console	NORM
SVO OFF	Pilot's and Copilot's Collective Stick Grip	Centered

b. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

7-2-1.3 APU Accumulator - Recharge Using APU.

a. Perform Switch And Circuit Breaker Setup, if not already done.

b. Place upper console BATT switch to ON and start APU (Appropriate Flight Manual).

c. Place upper console GENERATORS APU switch to ON.

RESULT:

(1) APU ACCUM LOW capsule shall go on.

(2) After four seconds, backup pump shall operate.

(3) BACK-UP PUMP ON capsule shall go on.

(4) After 90 seconds, (180 seconds when winterization kit is installed), backup pump shall shut down.

(5) BACK-UP PUMP ON capsule shall go off.

(6) APU ACCUM LOW capsule shall go off.

(7) APU accumulator piston position indicator (tape) shall indicate from 63% to 85% (from 126% to 170%, the sum of the two accumulator tapes when the winterization kit is installed).

(8) APU accumulator pressure gauge shall indicate from 2700 to 3100 psig.

- If results (1) and (2) are not as specified, replace accumulator pressure switch (PARA 7-4-58).
- If result (1) is not as specified, go to TABLE NO. 7-2-1-1.
- If results (2) and (3) are not as specified, go to TABLE NO. 7-2-1-2.
- If result (2) occurs, but result (3) is not as specified, go to TABLE NO. 7-2-1-3.
- If results (4), (5), and (6) are not as specified, go to TABLE NO. 7-2-1-4.
- If result (4) is not as specified, but result (6) occurs, replace right relay panel (PARA 9-4-2).

- If result (4) is not as specified, check if BACKUP PUMP PWR circuit breaker (NO. 1 DC PRI BUS - copilot's circuit breaker panel) or BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) has popped.

  (a) If either one has popped, troubleshoot System Circuit Breakers (PARA 9-2-3).

  (b) If circuit breakers are engaged, go to TABLE NO. 7-2-1-5.

- If APU accumulator tape indicates 0 and APU accumulator pressure gauge reads 1450 psig, check backup pump module lines for leakage.

  (a) If leaks are found, repair/replace components as required.

- If result (7) is not as specified, service accumulator(s) (PARA 1-3-10).

- If accumulator(s) does not hold pressure, replace accumulator(s) (PARA 7-4-52).

- If trouble remains, disconnect P423 from backup pump and check for 115 VAC between pins A, B, or C, and gnd.

  (a) If 115 VAC is present at any pin, replace relay K19, NO. 3 HYD PWR PUMP relay (PARA 9-4-47).

d. Place upper console GENERATORS APU switch to OFF/RESET.

e. Shut down APU (Appropriate Flight Manual).

f. Place BATT switch to OFF.

g. If no other check is to be done, go to Shutdown.

7-2-1.4 APU Accumulator - Recharge Using External Power.

a. Perform Switch And Circuit Breaker Setup, if not already done.

b. Connect external electrical power to helicopter.

c. Turn on electrical power (PARA 1-6-16).

d. Pull up manual lever of APU start valve solenoid for 30 seconds to discharge APU accumulator, and then push lever down.

NOTE

APU start motor will spin and APU compressor will rotate until hydraulic pressure is discharged.

RESULT:

(1) APU ACCUM LOW capsule shall go on.

(2) After four seconds, backup pump shall operate.

(3) BACK-UP PUMP ON capsule goes on.

(4) After 90 seconds, (180 seconds, if winterization kit is installed), backup pump shall shut down.

(5) BACK-UP PUMP ON capsule shall go off.

(6) APU ACCUM LOW capsule shall go off.

(7) APU accumulator position indicator (tape) shall indicate from 63% to 85% (from 126% to 170%, the sum of the two accumulator tapes when the winterization kit is installed).

(8) APU accumulator pressure gauge shall indicate from 2700 to 3100 psig.

- If results (1) and (2) are not as specified, replace accumulator pressure switch (PARA 7-4-58).

- If result (1) is not as specified, go to TABLE NO. 7-2-1-1.

- If results (2) and (3) are not as specified, go to TABLE NO. 7-2-1-2.

- If result (2) occurs, but result (3) is not as specified, go to TABLE NO. 7-2-1-3.
- If results (4), (5), and (6) are not as specified, go to TABLE NO. 7-2-1-4.
- If result (4) is not as specified, but result (6) occurs, replace right relay panel (PARA 9-4-2).
- If result (4) is not as specified, check if BACKUP PUMP PWR circuit breaker (NO. 1 DC PRI BUS - copilot's circuit breaker panel) or BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) has popped.
  - (a) If either one has popped, troubleshoot System Circuit Breakers (PARA 9-2-3).
  - (b) If circuit breakers are engaged, go to TABLE NO. 7-2-1-5.
- If APU accumulator tape indicates 0 and APU accumulator pressure gauge reads 1450 psig, check backup pump module lines for leakage.
  - (a) If leaks are found, repair/replace components as required.
- If result (7) is not as specified, service accumulator(s) (PARA 1-3-10).
- If accumulator(s) does not hold pressure, replace accumulator(s) (PARA 7-4-52).
- If trouble remains, disconnect P423 from backup pump and check for 115 VAC between pins A, B, or C and gnd.
  - (a) If 115 VAC is present at any pin, replace relay K19, NO. 3 HYD PWR PUMP relay (PARA 9-4-47).

e. If no other check is to be done, go to Shutdown.

7-2-1.5 Hydraulic Leak Test.

a. Perform APU Accumulator - Recharge Using APU, if not already done.

b. If necessary, service No. 1, No. 2, and backup pump modules until reservoirs are full (indicator in green area of decal) (PARA 1-3-8).

c. Pull out BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel).

d. Turn on electrical power (PARA 1-6-16).

e. Place upper console HYD LEAK TEST switch to TEST.

RESULT: #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules shall go on.

- If result is not as specified, place and hold caution/advisory panel BRT/DIM-TEST switch to TEST.
  - (a) If capsules do not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
  - (b) If any capsule does not go on, go to TABLE NO. 7-2-1-6.

NOTE

The #1 HYD PUMP, #2 HYD PUMP, #1 PRI SERVO PRESS, or #2 PRI SERVO PRESS capsules may flicker when the HYD LEAK TEST switch is moved from TEST to RESET and then to NORM. This is considered normal.

f. Place upper console HYD LEAK TEST switch to RESET and then to NORM.

RESULT:

(1) #1 RSVR LOW capsule shall go off.

(2) #2 RSVR LOW and BACK-UP RSVR LOW capsules shall go off.

- If result (1) is not as specified, replace left relay panel (PARA 9-4-2).

- If result (2) is not as specified, replace right relay panel (PARA 9-4-2).

g. Push in BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) and place upper console BACKUP HYD PUMP switch S31 to AUTO.

h. Place upper console HYD LEAK TEST switch to TEST.

RESULT:

(1) #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules shall go on.

(2) Backup hydraulic pump shall run.

(3) BACK-UP PUMP ON capsule shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-6.

- If result (2) is not as specified, check for 28 VDC between P244-H and gnd.

  (a) If voltage is as specified, replace right relay panel (PARA 9-4-2).

  (b) If voltage is not as specified, repair/replace wiring (Appendix F).

- If result (3) is not as specified, go to TABLE NO. 7-2-1-3.

i. Place miscellaneous switch panel TAIL SERVO switch to BACKUP.

RESULT:

(1) #1 TAIL RTR SERVO capsule shall go on.

(2) #2 TAIL RTR SERVO ON capsule shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-7.

- If result (2) is not as specified, go to TABLE NO. 7-2-1-8.

j. Place miscellaneous switch panel TAIL SERVO switch to NORMAL.

RESULT:

(1) #1 TAIL RTR SERVO capsule shall go off.

(2) #2 TAIL RTR SERVO ON capsule shall go off.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-9.

- If result (2) is not as specified, go to TABLE NO. 7-2-1-10.

k. Place upper console HYD LEAK TEST switch to RESET and then back to NORM, and place BACKUP HYD PUMP switch S31 to OFF.

RESULT:

(1) Backup pump shall shut down.

(2) BACK-UP PUMP ON capsule shall go off.

(3) #1 RSVR LOW, #2 RSVR LOW, BACK-UP RSVR LOW capsules shall go off.

- If result (1) is not as specified, check for 28 VDC between P244-D and gnd with HYD LEAK TEST switch in RESET position.

  (a) If voltage is not as specified, repair/replace wiring between P244-D and S25-7 (Appendix F).

- If result (2) is not as specified, check continuity between J445-1 and J445-3.

(a) If continuity is present, replace backup pump pressure switch (PARA 7-4-39).

l. Insert drag beam wedge in left drag beam switch to simulate flight condition.

m. Place upper console HYD LEAK TEST switch to TEST.

RESULT:

(1) #1 RSVR LOW capsule shall not go on.

(2) #2 RSVR LOW capsule shall not go on.

(3) BACK-UP RSVR LOW capsule shall go on.

- If result (1) is not as specified, check No. 1 pump module sight glass window.

  (a) If indicator is in red area of decal, service pump reservoir (PARA 1-3-8).

  (b) If indicator is in green area of decal, replace left relay panel (PARA 9-4-2).

- If result (2) is not as specified, check No. 2 pump module sight glass window.

  (a) If indicator is in red area of decal, service the respective pump reservoir (PARA 1-3-8).

  (b) If indicator is in green area, replace right relay panel (PARA 9-4-2).

n. Place upper console HYD LEAK TEST switch to RESET and then to NORM, and remove drag beam wedge.

o. Pull out BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) and place BACKUP HYD PUMP switch S31 to OFF.

p. Connect external hydraulic power to No. 1, No. 2, and backup pump modules, but do not pressurize (PARA 1-6-15).

q. Increase hydraulic power to No. 1, No. 2, and Backup Hydraulic Systems to between 3000 and 3100 psig.

RESULT: Hydraulic fluid shall not leak from No. 1, No. 2, or backup pump module reservoir relief valve.

- If result is not as specified, replace pump module (PARA 7-4-12 or PARA 7-4-13).

r. Decrease hydraulic pressure to No. 1, No. 2, and backup pump to 0 psig.

s. Place upper console BATT switch ON.

RESULT: EXT PWR CONNECTED, #1 CONV, #2 CONV, and AC ESS BUS OFF capsules on the caution/advisory panel shall go off.

t. Push in BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel).

u. Make sure switches listed in CHART NO. 7-2-1-1. are as indicated and caution/advisory capsules are on.

CHART NO. 7-2-1-1. Switch Positions.

SWITCH	LOCATION	POSITION	CAPSULES
BOOST	stabilator control/ auto flight control panel	ON	BOOST SERVO OFF
SAS 1 and SAS 2	stabilator control/ auto flight control panel	ON	SAS OFF

TRIM	stabilator control/ auto flight control panel	off	
Rotary	SAS/FPS Computer	GND (non-AFCC)	
SVO OFF	collective grip	CENTERED	#1 PRI SERVO PRESS #2 PRI SERVO PRESS #1 HYD PUMP #2 HYD PUMP
TAIL SERVO	miscellaneous switch panel	NORMAL	#1 TAIL RTR SERVO
BACKUP HYD PUMP	upper console	OFF	#1 RSVR LOW #2 RSVR LOW

v. Slowly increase hydraulic power to No. 1, No. 2, and Backup Hydraulic Systems to between 3000 and 3100 psig with a 6.5 gpm flow.

RESULT: The following capsules on the caution/advisory panel shall not go on:

#1 PRI SERVO PRESS
#2 PRI SERVO PRESS
BOOST SERVO OFF
SAS OFF
TRIM FAIL
#1 TAIL RTR SERVO

NOTE

The helicopter must be powered on and the NO. 1 SERVO CONTR and NO. 2 SERVO CONTR circuit breakers pushed in for at least one hour prior to doing the following steps. If an output transistor in the hydraulic logic module in the left or right relay panel has an excessive voltage leak, it may cause failure of the Backup Hydraulic System and loss of the second stage tail rotor. The procedure in the following steps is intended to verify operation of the logic modules and Backup Hydraulic System. If transistors in the logic modules are leaking output voltage, the leak may only appear after an extended period of operation.

w. Pull out NO. 2 DC INST circuit breaker (NO. 2 DC PRI BUS - pilot's circuit breaker panel).

x. Place upper console BACKUP HYD PUMP switch S31 to AUTO.

NOTE

The stopwatch is used to measure the operating time of the backup pump from when it goes on in step y. until it goes off in step aa.

y. Simultaneously start stopwatch and place upper console HYD LEAK TEST switch to TEST.

RESULT:

(1) The following capsules on the caution/advisory panel shall go on:

SAS OFF
BOOST SERVO OFF
#1 RSVR LOW
#2 RSVR LOW
BACKUP RSVR LOW
#1 TAIL RTR SERVO
#2 TAIL RTR SERVO ON
BACKUP PUMP ON

(2) Electric motor of backup pump shall start.

- If BACKUP PUMP ON capsule does not go on or #2 TAIL RTR SERVO ON capsule flickers or does not go on, replace right relay panel (PARA 9-4-2).

z. Place upper console HYD LEAK TEST switch to NORM.

RESULT: There shall be no change in caution/advisory panel capsules.

aa. Momentarily place upper console HYD LEAK TEST switch to RESET and then back to NORM.

RESULT:

(1) The following capsules on the caution/advisory panel shall go off immediately:

SAS OFF
BOOST SERVO OFF
#1 RSVR LOW
#2 RSVR LOW
BACKUP RSVR LOW
#1 TAIL RTR SERVO
#2 TAIL RTR SERVO ON
BACKUP PUMP ON

(2) The BACKUP PUMP ON capsule shall go off and the pump motor shall stop after approximately 90 seconds (180 seconds when winterization kit is installed).

ab. Place upper console BACKUP HYD PUMP switch S31 to OFF.

ac. Push in NO. 2 DC INST circuit breaker (NO. 2 DC PRI BUS - pilot's circuit breaker panel).

ad. Decrease hydraulic pressure to No. 1, No. 2, and Backup Hydraulic Systems to 0 psig.

ae. Disconnect external hydraulic power from backup pump.

af. If no other check is to be done, go to Shutdown.

7-2-1.6 No. 1 Hydraulic System.

a. Perform Hydraulic Leak Test, if not already done.

b. Slowly increase No. 2 hydraulic pressure to between 3000 and 3100 psig, make sure MISC SW panel TAIL SERVO switch is in NORMAL, and place upper console EXT PWR switch first to RESET and then to ON.

RESULT:

(1) #1 HYD PUMP capsule on caution/advisory panel shall go on.

(2) #1 PRI SERVO PRESS capsule on caution/advisory panel shall go on.

(3) #1 TAIL RTR SERVO capsule on caution/advisory panel shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-11.
- If result (2) is not as specified, go to TABLE NO. 7-2-1-12.
- If result (3) is not as specified, go to TABLE NO. 7-2-1-13.

c. Slowly increase No. 1 hydraulic pressure to between 3000 and 3100 psig.

NOTE

The #1 HYD PUMP and #2 PRI SERVO PRESS capsules may flicker when #1 TAIL RTR SERVO capsule goes off (No. 1 tail rotor servo pressurized). This is considered normal.

RESULT:

(1) #1 HYD PUMP capsule shall go off.

(2) #1 PRI SERVO PRESS capsule shall go off.

(3) #1 TAIL RTR SERVO capsule shall go off.

- If result (1) is not as specified, check for popped pressure filter.

  (a) If filter is popped, replace filter (PARA 7-4-14).

  (b) If filter is not popped, replace pump pressure switch (PARA 7-4-27).

- If result (2) is not as specified, go to TABLE NO. 7-2-1-14.

- If result (3) is not as specified, go to TABLE NO. 7-2-1-9.

d. Place either pilot's or copilot's collective stick SVO OFF switch to 1ST STG.

RESULT: #1 PRI SERVO PRESS capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-15.

e. Place pilot's or copilot's collective stick SVO OFF switch to center position.

RESULT: #1 PRI SERVO PRESS capsule shall go off.

- If capsule does not go off, perform the following steps:

  (a) Interchange first and second stage servo pressure switches (PARA 11-4-88).

  (b) Retest No. 1 Hydraulic System.

  (c) If trouble remains, replace malfunctioning pressure switch (PARA 11-4-88).

  (d) If trouble still remains, replace No. 1 primary servo manifold (PARA 7-4-25).

f. Place miscellaneous switch panel TAIL SERVO switch to BACKUP.

RESULT: #1 TAIL RTR SERVO capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-7.

g. Decrease No. 1 and No. 2 hydraulic pressure to zero.

h. If no other check is to be done, go to Shutdown.

7-2-1.7 No. 2 Hydraulic System.

a. Perform Hydraulic Leak Test, if not already done.

b. Slowly increase No. 1 hydraulic pressure to between 3000 and 3100 psig and place upper console EXT PWR switch first to RESET, and then to ON.

RESULT:

(1) #2 HYD PUMP capsule on caution/advisory panel shall go on.

(2) #2 PRI SERVO PRESS capsule on caution/advisory panel shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-16.

- If result (2) is not as specified, go to TABLE NO. 7-2-1-17.

c. Check pilot-assist module pressure reducer for popped indicator button.

RESULT: Pressure reducer indicator button shall not be popped.

- If result is not as specified, go to TABLE NO. 7-2-1-18.

d. Slowly increase No. 2 hydraulic pressure to between 3000 and 3100 psig.

RESULT:

(1) #2 HYD PUMP capsule shall go off.

(2) #2 PRI SERVO PRESS capsule shall go off.

- If result (1) is not as specified, check for popped pressure filter.
  - (a) If filter is popped, replace filter (PARA 7-4-14).
  - (b) If filter is not popped, replace pump pressure switch (PARA 7-4-32).
- If result (2) is not as specified, go to TABLE NO. 7-2-1-19.

e. Place either pilot's or copilot's collective stick SVO OFF switch to 2ND STG.

RESULT: #2 PRI SERVO PRESS capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-20.

f. Place either pilot's or copilot's collective stick SVO OFF switch to center position.

RESULT: #2 PRI SERVO PRESS capsule shall go off.

- If capsule does not go off, perform the following steps:
  - (a) Interchange first and second stage servo pressure switches (PARA 11-4-88).
  - (b) Retest No. 2 Hydraulic System.
  - (c) If trouble remains, replace malfunctioning pressure switch (PARA 11-4-88).
  - (d) If trouble still remains, replace No. 2 primary servo manifold (PARA 7-4-30).

NOTE

- The following steps check out the pilot-assist servos.
- When both SAS 1 and SAS 2 are disengaged, the MASTER CAUTION capsule may go on when the TRIM switch on the stabilator control/auto flight control panel or either of the TRIM REL switches on the cyclic sticks is pressed. If this occurs, press to reset the MASTER CAUTION capsule.

g. Make sure all pilot-assist servos BOOST, SAS 1, SAS 2, TRIM, and FPS switches on the stabilator control/auto flight control panel are disengaged.

RESULT:

(1) BOOST, SAS 1, SAS 2, TRIM, and FPS legends on stabilator control/auto flight control panel shall be off.

(2) SAS OFF and BOOST SERVO OFF capsules on caution/advisory panel shall be on.

(3) TRIM FAIL and FLT PATH STAB caution advisories shall be off.

- If results (1) and (2) are not as specified, troubleshoot digital AFCS (TM 11-70-23).
- If result (3) is not as specified, simultaneously press and release both fail advisory reset pushbuttons on the stabilator control/auto flight control panel to reset.

h. Press face of stabilator control/auto flight control panel TRIM switch to ON.

RESULT:

(1) TRIM button light shall go on.

(2) TRIM FAIL and FLT PATH STAB caution advisories shall be off.

- If result (1) is not as specified, troubleshoot digital AFCS (TM 11-70-23).

- If result (2) is not as specified, simultaneously press and release both fail advisory reset pushbuttons on the stabilator control/auto flight control panel to reset.

i. Move cyclic stick back and forth.

RESULT: Stick shall be hard to move.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

NOTE

Do not try to oppose stick movement. If hand is held on stick, follow stick movement.

j. Move cyclic stick STICK TRIM switch FWD and AFT.

RESULT:

(1) TRIM FAIL capsule shall not go on.

(2) Stick shall move slowly by itself.

- If results (1) or (2) are not as specified, troubleshoot digital AFCS (TM 11-70-23).

k. Turn boost servo on and then off, by pressing stabilator control/auto flight control panel BOOST switch.

RESULT: BOOST SERVO OFF capsule shall go off and then on.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

NOTE

The #2 HYD PUMP, #2 PRI SERVO OFF, and BOOST SERVO OFF capsules may flicker when either SAS 1 or SAS 2 is turned on after both have been off. This is considered normal.

l. Turn SAS actuators on, and then off, by pressing stabilator control/auto flight control panel SAS 1 and SAS 2 switches, one at a time. Observe SAS OFF capsule.

RESULT: SAS OFF capsule shall go off when either switch is engaged (lit), and shall go on only if both switches are disengaged (not lit).

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

m. Slowly decrease hydraulic pressure to zero.

n. Place upper console EXT PWR switch to OFF.

o. If no other check is to be done, go to Shutdown.

7-2-1.8 Backup Hydraulic System.

a. Perform Hydraulic Leak Test, if not already done.

b. Place upper console BACKUP HYD PUMP switch S31 to OFF and miscellaneous switch panel TAIL SERVO switch to BACKUP.

c. Make sure BACKUP PUMP PWR circuit breaker (NO. 1 DC PRI BUS - copilot’s circuit breaker panel) is pushed in and BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot’s side upper console circuit breaker panel) is pulled out.

d. Place upper console EXT PWR switch first to RESET and then to ON.

RESULT: #1 HYD PUMP AND #2 HYD PUMP capsules shall go on.

- If result is not as specified, perform No. 1 Hydraulic System or the No. 2 Hydraulic System check, as required.

e. Check pilot-assist module pressure reducer for popped indicator button.

RESULT: Pressure reducer indicator shall not be popped.

- If result is not as specified, go to TABLE NO. 7-2-1-18.

f. Pressurize hydraulic cart's Backup Hydraulic System to between 3000 and 3100 psig.

RESULT:

(1) #1 PRI SERVO PRESS capsule shall go off.

(2) #1 TAIL RTR SERVO capsule shall go on.

(3) #2 PRI SERVO PRESS capsule shall go off.

(4) #2 TAIL RTR SERVO ON capsule shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-14.
- If result (2) is not as specified, go to TABLE NO. 7-2-1-7.
- If result (3) is not as specified, go to TABLE NO. 7-2-1-19.
- If result (4) is not as specified, go to TABLE NO. 7-2-1-8.

g. Decrease hydraulic pressure to approximately 1000 psig, and push in BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel).

h. Place upper console BACKUP HYD PUMP switch S31 to ON.

RESULT: After 4 seconds, backup pump shall operate.

- If result is not as specified, go to TABLE NO. 7-2-1-21.

i. Place upper console BACKUP HYD PUMP switch S31 to AUTO.

RESULT: Backup pump continues to operate.

- If result is not as specified, replace BACKUP HYD PUMP switch S31 (PARA 9-4-5).

j. Place upper console BACKUP HYD PUMP switch S31 to OFF.

RESULT: Backup pump shall shut down.

- If result is not as specified, go to TABLE NO. 7-2-1-22.

k. Insert drag beam wedge in left drag beam switch to simulate flight condition.

RESULT:

(1) Backup pump shall operate.

(2) BACK-UP PUMP ON capsule shall go on.

- If result (1) is not as specified, go to TABLE NO. 7-2-1-2.
- If result (2) is not as specified, go to TABLE NO. 7-2-1-3.

l. Remove drag beam wedge.

m. If no other check is to be done, go to Shutdown.

7-2-1.9 Primary Servo Pressure Switch Check.

a. Slowly decrease hydraulic pressure to 0 psig, if not already done.

b. Connect external electrical power.

c. Turn on electrical power.

d. Disconnect 1st stage pressure switch connectors P424, P425, and P426.

e. Connect P424 to lateral primary servo 1st stage pressure switch.

RESULT: #1 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-23.

f. Disconnect P424 from lateral primary servo 1st stage pressure switch.

g. Connect P425 to aft primary servo 1st stage pressure switch.

RESULT: #1 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-23.

h. Disconnect P425 from aft primary servo 1st stage pressure switch.

i. Connect P426 to forward primary servo 1st stage pressure switch.

RESULT: #1 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-23.

j. Disconnect 2nd stage pressure switch connectors P427, P428, and P429.

k. Connect P427 to lateral primary servo 2nd stage pressure switch.

RESULT: #2 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-24.

l. Disconnect P427 from lateral primary servo 2nd stage pressure switch.

m. Connect P428 to aft primary servo 2nd stage pressure switch.

RESULT: #2 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-24.

n. Disconnect P428 from aft primary servo 2nd stage pressure switch.

o. Connect P429 to forward primary servo 2nd stage pressure switch.

RESULT: #2 PRI SERVO PRESS caution capsule shall go on.

- If result is not as specified, go to TABLE NO. 7-2-1-24.

p. Reconnect all primary servo 1st and 2nd stage pressure switch connectors.

q. If no other check is to be done, go to Shutdown.

7-2-1.10 Shutdown.

a. Turn off hydraulic power (PARA 1-6-15).

b. Turn off electrical power (PARA 1-6-16).

TABLE NO. 7-2-1-1. APU ACCUM LOW Capsule Does Not Go On.

TEST OR INSPECTION CORRECTIVE ACTION
1. Check helicopter configuration. Step 1. **EMEP** , go to **2.** Step 2. **W/O EMEP** , go to **4.**

TABLE NO. 7-2-1-1. APU ACCUM LOW Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

EMEP **2. Remove and check pin filter adapter (PFA) P118 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **8.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P118-C and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **8.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P302-2 and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P118-C and P302-1.

Step 1. If continuity is present, replace APU accumulator pressure switch (PARA 7-4-58). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. Check for 28 VDC between terminal 1 (BACKUP HYD CONTR circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P302-2 (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 7-2-1-2. Backup Pump Does Not Operate.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Place BACKUP HYD PUMP switch S31 to ON, and check that pump operates.

TABLE NO. 7-2-1-2. Backup Pump Does Not Operate. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If pump operates, go to **2.**
Step 2. If pump does not operate, go to **TABLE NO. 7-2-1-22.**

2. Place BACKUP HYD PUMP switch S31 to OFF. Go to 3.

3. Check for 28 VDC between P244-B and gnd.

Step 1. If voltage is as specified, go to **4.**
Step 2. If voltage is not as specified, repair/replace wiring between P244-B and P302-1 (Appendix F). Go to **9.**

4. Check helicopter configuration.

Step 1. **WINTER** , go to **5.**
Step 2. **W/O WINTER** go to **7.**

WINTER **5. Check for 28 VDC between P244B-B and gnd.**

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **6.**

WINTER **6. Check continuity between J244A-B and P244B-B.**

Step 1. If continuity is present, go to **7.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **9.**

7. Check for 28 VDC between terminal 3 (BACKUP HYD PUMP switch S31) and gnd.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **9.**
Step 2. If voltage is not as specified, go to **8.**

8. Check continuity between terminal 3 (BACKUP HYD PUMP switch S31) and P241-L.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

TABLE NO. 7-2-1-2. Backup Pump Does Not Operate. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Procedure completed.

TABLE NO. 7-2-1-3. Backup Pump Operates But BACK-UP PUMP ON Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P118 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **8.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P118-R and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **8.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P445-3 and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P118-R and P445-1.

Step 1. If continuity is present, replace utility module backup pump pressure switch (PARA 7-4-39). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. Check for 28 VDC between terminal 1 (NO. 2 SERVO WARN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd.

TABLE NO. 7-2-1-3. Backup Pump Operates But BACK-UP PUMP ON Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P445-3 (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 7-2-1-4. Backup Pump Does Not Shut Off And APU ACCUM LOW Capsule Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check APU accumulator pressure gauge for proper charge.

Step 1. If charge is correct, go to **2.**
Step 2. If charge is not correct, service APU accumulator (PARA 1-3-10). Go to **4.**

2. Check helicopter configuration.

Step 1. WINTER , go to **3.**
Step 2. W/O WINTER , replace APU accumulator pressure switch (PARA 7-4-58). Go to **4.**

WINTER **3. Check continuity between J302-1 and J302-3.**

Step 1. If continuity is present, replace APU accumulator pressure switch (PARA 7-4-58). Go to **4.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **4.**

4. Procedure completed.

TABLE NO. 7-2-1-5. Backup Pump Operates Then Shuts Down.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Allow pump to cool down. See equipment conditions. Go to 2.

2. Place drag beam wedge in left drag beam switch to simulate weight off wheels. Go to 3.

3. Check that backup pump starts and continues to run.

Step 1. If backup pump starts and continues to run, remove drag beam wedge and replace motor (PARA 9-4-44). Go to **4.**
Step 2. If backup pump does not start, remove drag beam wedge and replace K19 NO. 3 HYD PWR PUMP relay (PARA 9-4-47). Go to **4.**

4. Procedure completed.

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does #1 RSVR LOW capsule remain off?

Step 1. If capsule remains off, go to **2.**
Step 2. If capsule goes on, go to **16.**

2. Check sight glass window on No. 1 hydraulic pump for full indication.

Step 1. If sight glass indication is full, go to **3.**
Step 2. If sight glass does not indicate full, service pump reservoir (PARA 1-3-8). Go to **40.**

3. Check helicopter configuration.

Step 1. EMEP , go to **4.**
Step 2. W/O EMEP , go to **6.**

EMEP **4. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **5.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **40.**

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

EMEP **5. Install PFA (PARA 9-4-98). Go to 6.**

6. Check for 28 VDC between P117-32 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **40.**
Step 2. If voltage is not as specified, go to **7.**

7. Check for 28 VDC between P902-t and gnd.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **9.**

8. Check continuity between P117-32 and P902-y.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

9. Check for 28 VDC between HYD LEAK TEST switch terminal 6 and gnd.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 6 and P902-t (Appendix F). Go to **40.**
Step 2. If voltage is not as specified, go to **10.**

10. Check for 28 VDC between HYD LEAK TEST switch terminal 5 and gnd.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **40.**
Step 2. If voltage is not as specified, go to **11.**

11. Check for 28 VDC between P447-3 and gnd.

Step 1. If voltage is as specified, go to **12.**
Step 2. If voltage is not as specified, go to **13.**

12. Check continuity between HYD LEAK TEST switch terminal 5 and P447-2.

Step 1. If continuity is present, replace pump (PARA 7-4-12). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

13. Check helicopter configuration.

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. WINTER , go to **14.**
Step 2. W/O WINTER , go to **15.**

WINTER **14. Check continuity between P244B-X and P447-3.**

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **40.**
Step 3. If trouble remains, go to **15.**

15. Check continuity between P244-X and P447-3.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

16. Check that #2 RSVR LOW capsule remains off.

Step 1. If capsule remains off, go to **17.**
Step 2. If capsule does not remain off, go to **28.**

17. Check sight glass window on No. 2 hydraulic pump for full indication.

Step 1. If pump indication is full, go to **18.**
Step 2. If pump indication is not full, service pump reservoir (PARA 1-3-8). Go to **40.**

18. Check for 28 VDC between P117-48 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **40.**
Step 2. If voltage is not as specified, go to **19.**

19. Check for 28 VDC between P900-a and gnd.

Step 1. If voltage is as specified, go to **20.**
Step 2. If voltage is not as specified, go to **23.**

20. Check helicopter configuration.

Step 1. WINTER , go to **21.**

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. **W/O WINTER** , go to **22.**

WINTER **21. Check continuity between P117-48 and P244B-g.**

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **40.**
Step 3. If trouble remains, go to **22.**

22. Check continuity between P117-48 and P244-g.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

23. Check for 28 VDC between terminal 12 (HYD LEAK TEST switch) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 12 and P900-a (Appendix F). Go to **40.**
Step 2. If voltage is not as specified, go to **24.**

24. Check for 28 VDC between terminal 11 (HYD LEAK TEST switch) and gnd.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **40.**
Step 2. If voltage is not as specified, go to **25.**

25. Check for 28 VDC between P432-3 and gnd.

Step 1. If voltage is as specified, go to **26.**
Step 2. If voltage is not as specified, go to **27.**

26. Check continuity between terminal 11 (HYD LEAK TEST switch) and P432-2.

Step 1. If continuity is present, replace pump (PARA 7-4-12). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

27. Check for 28 VDC between terminal 1 (NO. 2 SERVO CONTR circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P432-3 (Appendix F). Go to **40.**

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **40.**

28. BACK-UP RSVR LOW capsule does not go on. Check sight glass window on back up hydraulic pump for full indication.

Step 1. If pump shows full indication, go to **29.**
Step 2. If pump does not show full indication, service pump reservoir (PARA 1-3-8). Go to **40.**

29. Check for 28 VDC between P117-64 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **40.**
Step 2. If voltage is not as specified, go to **30.**

30. Check helicopter configuration.

Step 1. WINTER , go to **31.**
Step 2. W/O WINTER , go to **33.**

WINTER **31. Check continuity between P117-64 and P244B-f.**

Step 1. If continuity is present, go to **32.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **40.**

WINTER **32. Check for 28 VDC between P244B-f and gnd.**

Step 1. If voltage is as specified, go to **34.**
Step 2. If voltage is not as specified, go to **35.**

33. Check continuity between P117-64 and P244-f.

Step 1. If continuity is present, replace right relay panel (PARA 9-4-2). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

34. Check for 28 VDC between P442-2 and gnd.

Step 1. If voltage is as specified, go to **35.**
Step 2. If voltage is not as specified, go to **38.**

TABLE NO. 7-2-1-6. Caution/Advisory Capsules Do Not Go On. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION

35. Check helicopter configuration.

Step 1. WINTER , go to **36.**
Step 2. W/O WINTER , go to **37.**

WINTER **36. Check continuity between P244B-e and P442-3.**

Step 1. If continuity is present, replace pump (PARA 7-4-12). Go to **40.**
Step 2. If continuity is not present, replace winterization kit relay (PARA 16-6-2). Go to **40.**
Step 3. If trouble remains, go to **37.**

37. Check continuity between P244-e and P442-3.

Step 1. If continuity is present, replace pump (PARA 7-4-12). Go to **40.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **40.**

38. Check for 28 VDC between terminal 9 (HYD LEAK TEST switch) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 9 and P442-2 (Appendix F). Go to **40.**
Step 2. If voltage is not as specified, go to **39.**

39. Check for 28 VDC between terminal 8 (HYD LEAK TEST switch) and gnd.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **40.**
Step 2. If voltage is not as specified, repair/replace wiring between terminal 8 and P432-2 (Appendix F). Go to **40.**

40. Procedure completed.

TABLE NO. 7-2-1-7. #1 TAIL RTR SERVO Capsule Does Not Go On When TAIL SERVO Switch Is Placed To BACKUP.

TEST OR INSPECTION CORRECTIVE ACTION

1. Check for 28 VDC between P902-v and gnd.

TABLE NO. 7-2-1-7. #1 TAIL RTR SERVO Capsule Does Not Go On When TAIL SERVO Switch Is Placed To BACKUP. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to **4.**
Step 2. If voltage is not as specified, go to **2.**

2. Check for 28 VDC between P135-S and gnd.

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, repair/replace wiring between P135-S and terminal 1 (NO. 1 SERVO CONTR circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) (Appendix F). Go to **4.**

3. Check continuity between P135-T and P902-v.

Step 1. If continuity is present, replace miscellaneous switch panel (PARA 9-4-7). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **4.**

4. Procedure completed.

TABLE NO. 7-2-1-8. #2 TAIL RTR SERVO ON Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. **EMEP** , go to **2.**
Step 2. **W/O EMEP** , go to **4.**

EMEP **2. Remove and check PFA P118 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **8.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P118-S and gnd.

TABLE NO. 7-2-1-8. #2 TAIL RTR SERVO ON Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **8.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P609-1 and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P118-S and P609-3.

Step 1. If continuity is present, replace 2nd stage pressure switch (PARA 11-4-114). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. Check for 28 VDC between terminal 1 (TRTR SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P609-1 (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 7-2-1-9. #1 TAIL RTR SERVO Capsule Remains On When No. 1 Pump Module Is Pressurized.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect P610. Go to 2.

2. Move pilot's or copilot's pedals.

Step 1. If pedals move normally, replace 1st stage pressure switch (PARA 11-4-114). Go to **5.**
Step 2. If pedals do not move normally, go to **3.**

3. Grasp 1st stage pressure line at tail rotor servo. Go to 4.

TABLE NO. 7-2-1-9. #1 TAIL RTR SERVO Capsule Remains On When No. 1 Pump Module Is Pressurized. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

4. **Cycle TAIL SERVO switch from NORMAL to BACKUP several times and check that a jump is felt in pressure line.**

   Step 1. If jump is felt, replace tail rotor servo (PARA 11-4-113). Go to **5.**
   Step 2. If jump is not felt, replace No. 1 transfer module (PARA 7-4-26). Go to **5.**

5. **Procedure completed.**

TABLE NO. 7-2-1-10. #2 TAIL RTR SERVO ON Capsule Remains On When No. 1 Pump Module Is Pressurized.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Disconnect P609. Go to 2.**

2. **Move pilot's or copilot's pedals.**

   Step 1. If pedals move normally, replace 1st stage pressure switch (PARA 11-4-114). Go to **5.**
   Step 2. If pedals do not move normally, go to **3.**

3. **Grasp 2nd stage pressure line at tail rotor servo. Go to 4.**

4. **Cycle TAIL SERVO switch from NORMAL to BACKUP several times and check that a jump is felt in pressure line.**

   Step 1. If jump is felt, replace tail rotor servo (PARA 11-4-113). Go to **5.**
   Step 2. If jump is not felt, replace No. 2 transfer module (PARA 7-4-31). Go to **5.**

5. **Procedure completed.**

TABLE NO. 7-2-1-11. #1 HYD PUMP Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 VDC between P117-24 and gnd.**

   Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **8.**
   Step 2. If voltage is not as specified, go to **2.**

2. **Check for 28 VDC between P421-6 and gnd.**

   Step 1. If voltage is as specified, go to **3.**
   Step 2. If voltage is not as specified, go to **7.**

3. **Check helicopter configuration.**

   Step 1. EMEP , go to **4.**
   Step 2. W/O EMEP , go to **6.**

EMEP 4. **Remove and check PFA P117 (PARA 9-4-98).**

   Step 1. If PFA is good, go to **5.**
   Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **8.**

EMEP 5. **Install PFA (PARA 9-4-98). Go to 6.**

6. **Check continuity between P117-24 and P421-1.**

   Step 1. If continuity is present, replace hydraulic pump pressure switch (PARA 7-4-27). Go to **8.**
   Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. **Check for 28 VDC between terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and gnd.**

   Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P421-6 (Appendix F). Go to **8.**
   Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. **Procedure completed.**

TABLE NO. 7-2-1-12. #1 PRI SERVO PRESS Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **7.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P117-8 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **7.**
Step 2. If voltage is not as specified, go to **5.**

5. Check continuity between:
P117-8 and P424-1
P117-8 and P425-1
P117-8 and P426-1

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **7.**

6. Check continuity between:
J424-1 and J424-6
J425-1 and J425-6
J426-1 and J426-6

Step 1. If continuity is present, repair/replace wiring as required between the following (Appendix F). Go to **7.**
Terminal 1 (NO. 1 SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and P424-6
Terminal 1 (NO. 1 SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and P425-6
Terminal 1 (NO. 1 SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and P426-6
Step 2. If continuity is not present, replace 1st stage pressure switch as required (PARA 11-4-88). Go to **7.**

TABLE NO. 7-2-1-12. #1 PRI SERVO PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

7. Procedure completed.

TABLE NO. 7-2-1-13. #1 TAIL RTR SERVO Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **9.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P117-57 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **9.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P243-M and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P117-57 and P902-z.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

7. Check for 28 VDC between P610-7 and gnd.

Step 1. If voltage is as specified, go to **8.**

TABLE NO. 7-2-1-13. #1 TAIL RTR SERVO Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If voltage is not as specified, repair/replace wiring between P610-7 and terminal 1 (NO. 1 SERVO CONTR circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) (Appendix F). Go to **9.**

8. Check continuity between P243-M and P610-4.

Step 1. If continuity is present, replace 1st stage pressure switch (PARA 11-4-114). Go to **9.**

Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **9.**

9. Procedure completed.

TABLE NO. 7-2-1-14. #1 PRI SERVO PRESS Capsule Remains On When No. 1 System Is Pressurized.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect P424.

Step 1. If capsule goes off, replace lateral primary servo 1st stage pressure switch (PARA 11-4-88). Go to **3.**

Step 2. If capsule does not go off, go to **2.**

2. Disconnect P425.

Step 1. If capsule goes off, replace aft primary servo 1st stage pressure switch (PARA 11-4-88). Go to **3.**

Step 2. If capsule does not go off, replace forward primary servo 1st stage pressure switch (PARA 11-4-88). Go to **3.**

3. Procedure completed.

TABLE NO. 7-2-1-15. #1 PRI SERVO PRESS Capsule Does Not Go On When 1st Stage Is Shut Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. If pilot's SVO OFF switch was used, place copilot's SVO OFF switch to 1ST STG. If copilot's SVO OFF switch was used, place pilot's SVO OFF switch to 1ST STG. Check that capsule goes on.

Step 1. If capsule goes on, troubleshoot copilot's or pilot's collective stick as required (PARA 11-2-2). Go to **14.**
Step 2. If capsule does not go on, go to **2.**

2. Check for 28 VDC between P417-1 and P417-2.

Step 1. If voltage is as specified, replace No. 1 transfer module (PARA 7-4-26). Go to **14.**
Step 2. If voltage is not as specified, go to **3.**

3. Check for 28 VDC between P417-1 and gnd.

Step 1. If voltage is as specified, repair/replace wiring between P417-2 and gnd (Appendix F). Go to **14.**
Step 2. If voltage is not as specified, go to **4.**

4. Check for 28 VDC between P429-9 and gnd.

Step 1. If voltage is as specified, go to **5.**
Step 2. If voltage is not as specified, go to **6.**

5. Check continuity between P417-1 and P429-4.

Step 1. If continuity is present, replace forward primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **14.**

6. Check for 28 VDC between P428-9 and gnd.

Step 1. If voltage is as specified, go to **7.**
Step 2. If voltage is not as specified, go to **8.**

7. Check continuity between P428-4 and P429-9.

Step 1. If continuity is present, replace aft primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **14.**

TABLE NO. 7-2-1-15. #1 PRI SERVO PRESS Capsule Does Not Go On When 1st Stage Is Shut Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Check for 28 VDC between P427-9 and gnd.

Step 1. If voltage is as specified, go to **9.**
Step 2. If voltage is not as specified, go to **10.**

9. Check continuity between P427-4 and P428-9.

Step 1. If continuity is present, replace lateral primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **14.**

10. With pilot's SVO OFF switch at 1ST STG, check for 28 VDC between:
J103-f and gnd
J104-f and gnd

Step 1. If voltage is as specified, go to **11.**
Step 2. If voltage is not as specified, repair/replace wiring as required between the following (Appendix F). Go to **14.**
Terminal 1 (NO. 1 SERVO CONTR circuit breaker - copilot's circuit breaker panel) and J103-f
Terminal 1 (NO. 1 SERVO CONTR circuit breaker - copilot's circuit breaker panel) and J104-f

11. Check helicopter configuration.

Step 1. **70894 – SUBQ** , go to **12.**
Step 2. **70584–70893** , go to **13.**

70894 – SUBQ

12. Check continuity between:
J103-J and P427-9
J104-J and P427-9

Step 1. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (PARA 11-2-2). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **14.**

70584–70893

13. Check continuity between:
J103-a and P427-9
J104-a and P427-9

TABLE NO. 7-2-1-15. #1 PRI SERVO PRESS Capsule Does Not Go On When 1st Stage Is Shut Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (PARA 11-2-2). Go to **14.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **14.**

14. Procedure completed.

TABLE NO. 7-2-1-16. #2 HYD PUMP Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **8.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P117-40 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **8.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P420-6 and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P117-40 and P420-1.

TABLE NO. 7-2-1-16. #2 HYD PUMP Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, replace No. 2 hydraulic pump pressure switch (PARA 7-4-32). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

7. Check for 28 VDC between terminal 1 (NO. 2 SERVO WARN circuit breaker - pilot's circuit breaker panel) and gnd.

Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P420-6 (Appendix F). Go to **8.**
Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to **8.**

8. Procedure completed.

TABLE NO. 7-2-1-17. #2 PRI SERVO PRESS Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check helicopter configuration.

Step 1. EMEP , go to **2.**
Step 2. W/O EMEP , go to **4.**

EMEP **2. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **3.**
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **7.**

EMEP **3. Install PFA (PARA 9-4-98). Go to 4.**

4. Check for 28 VDC between P117-56 and gnd.

Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **7.**
Step 2. If voltage is not as specified, go to **5.**

5. Check continuity between:

TABLE NO. 7-2-1-17. #2 PRI SERVO PRESS Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

P117-56 and P427-1
P117-56 and P428-1
P117-56 and P429-1

Step 1. If continuity is present, go to **6.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **7.**

6. Check continuity between:
J427-1 and J427-6
J428-1 and J428-6
J429-1 and J429-6

Step 1. If continuity is present, repair/replace wiring as required between the following (Appendix F). Go to **7.**
Terminal 1 (NO. 2 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P427-6
Terminal 1 (NO. 2 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P428-6
Terminal 1 (NO. 2 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P429-6
Step 2. If continuity is not present, replace 2nd stage pressure switch as required (PARA 11-4-88). Go to **7.**

7. Procedure completed.

TABLE NO. 7-2-1-18. Pilot-Assist Module Pressure Reducer Indicator Button Is Popped.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Turn on Backup Hydraulic System to reset button.

Step 1. If button resets, go to **2.**
Step 2. If button does not reset, replace pilot-assist module (PARA 7-4-34). Go to **4.**

2. Do not engage/disengage solenoids or run No. 1 or No. 2 Hydraulic Systems. Go to 3.

TABLE NO. 7-2-1-18. Pilot-Assist Module Pressure Reducer Indicator Button Is Popped. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

3. **Run Backup Hydraulic System for 2 minutes. Go to 5.**

4. **If button is popped, replace pilot-assist module (PARA 7-4-34). Go to 5.**

5. **Procedure completed.**

TABLE NO. 7-2-1-19. #2 PRI SERVO PRESS Capsule Remains On When No. 2 System Is Pressurized.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Disconnect P427.**

   Step 1. If capsule goes off, replace lateral primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **3.**
   Step 2. If capsule does not go off, go to **2.**

2. **Disconnect P428.**

   Step 1. If capsule goes off, replace forward primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **3.**
   Step 2. If capsule does not go off, replace aft primary servo 2nd stage pressure switch (PARA 11-4-88). Go to **3.**

3. **Procedure completed.**

TABLE NO. 7-2-1-20. #2 PRI SERVO PRESS Capsule Does Not Go On When 2nd Stage Is Shut Off.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **If pilot's SVO OFF switch was used, place copilot's SVO OFF switch to 2ND STG. If copilot's SVO OFF switch was used, place pilot's SVO OFF switch to 2ND STG. Go to 2.**

TABLE NO. 7-2-1-20. #2 PRI SERVO PRESS Capsule Does Not Go On When 2nd Stage Is Shut Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check that #2 PRI SERVO PRESS capsule goes on.

Step 1. If capsule goes on, troubleshoot pilot's or copilot's collective stick as required (PARA 11-2-2). Go to **13.**
Step 2. If capsule does not go on, go to **3.**

3. Check for 28 VDC between P416-1 and P416-2.

Step 1. If voltage is as specified, replace No. 2 transfer module (PARA 7-4-31). Go to **13.**
Step 2. If voltage is not as specified, go to **4.**

4. Check for 28 VDC between P416-1 and gnd.

Step 1. If voltage is as specified, repair/replace wiring between P416-2 and gnd (Appendix F). Go to **13.**
Step 2. If voltage is not as specified, go to **5.**

5. Check for 28 VDC between P426-9 and gnd.

Step 1. If voltage is as specified, go to **6.**
Step 2. If voltage is not as specified, go to **7.**

6. Check continuity between P416-1 and P426-4.

Step 1. If continuity is present, replace forward primary servo 1st stage pressure switch (PARA 11-4-88). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

7. Check for 28 VDC between P425-9 and gnd.

Step 1. If voltage is as specified, go to **8.**
Step 2. If voltage is not as specified, go to **9.**

8. Check continuity between P425-4 and P426-9.

Step 1. If continuity is present, replace aft primary servo 1st stage pressure switch (PARA 11-4-88). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

TABLE NO. 7-2-1-20. #2 PRI SERVO PRESS Capsule Does Not Go On When 2nd Stage Is Shut Off. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Check for 28 VDC between P424-9 and gnd.

Step 1. If voltage is as specified, go to **10.**
Step 2. If voltage is not as specified, go to **11.**

10. Check continuity between P424-4 and P425-9.

Step 1. If continuity is present, replace lateral primary servo 1st stage pressure switch (PARA 11-4-88). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **13.**

11. Check for 28 VDC between J103-c or J104-c and gnd.

Step 1. If continuity is present, go to **12.**
Step 2. If continuity is not present, repair/replace wiring as required between the following (Appendix F). Go to **13.**
Terminal 1 (NO. 2 SERVO CONTR circuit breaker - pilot's circuit breaker panel) and J103-c
Terminal 1 (NO. 2 SERVO CONTR circuit breaker - pilot's circuit breaker panel) and J104-c

12. Check continuity between J103-d or J104-d and P424-9.

Step 1. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (PARA 11-2-2). Go to **13.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **13.**

13. Procedure completed.

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Place drag-beam wedge in left drag beam switch to simulate weight off wheels. Go to 2.

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

2. Check that backup pump runs.

Step 1. If backup pump runs, go to **3.**
Step 2. If backup pump does not run, go to **6.**

3. Remove drag-beam wedge. Go to 4.

4. Check continuity between terminal X2 (NO. 3 HYD PWR PUMP RELAY) and P422-2.

Step 1. If continuity is present, go to **5.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

5. Check continuity between P422-1 and gnd.

Step 1. If continuity is present, replace backup pump module (PARA 7-4-13). Go to **32.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

6. Remove locally-made wedge. Go to 7.

7. Check for 115 VAC between:
P423-A and P423-D
P423-B and P423-D
P423-C and P423-D

Step 1. If voltage is as specified, replace backup pump module (PARA 7-4-13). Go to **32.**
Step 2. If voltage is not as specified, go to **8.**

8. Check continuity between P423-D and gnd.

Step 1. If continuity is present, go to **9.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

9. Check for 115 VAC between:
Terminal A2 (NO. 3 HYD PWR PUMP RELAY) and gnd
Terminal B2 (NO. 3 HYD PWR PUMP RELAY) and gnd
Terminal C2 (NO. 3 HYD PWR PUMP RELAY) and gnd

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between relay and P423 as required (Appendix F). Go to **32.**
Step 2. If voltage is not as specified, go to **10.**

10. Check for 115 VAC between:

Terminal A1 (NO. 3 HYD PWR PUMP RELAY) and gnd
Terminal B1 (NO. 3 HYD PWR PUMP RELAY) and gnd
Terminal C1 (NO. 3 HYD PWR PUMP RELAY) and gnd

Step 1. If voltage is as specified, go to **11.**
Step 2. If voltage is not as specified, go to **30.**

11. Check for 28 VDC between terminals X1 and X2 (NO. 3 HYD PWR PUMP RELAY).

Step 1. If voltage is as specified, replace relay K19 (PARA 9-4-47). Go to **32.**
Step 2. If voltage is not as specified, go to **12.**

12. Check for 28 VDC between terminal X1 (NO. 3 HYD PWR PUMP RELAY) and gnd.

Step 1. If voltage is as specified, go to **13.**
Step 2. If voltage is not as specified, go to **17.**

13. Check for 28 VDC between P902-CC and gnd.

Step 1. If voltage is as specified, go to **14.**
Step 2. If voltage is not as specified, repair/replace wiring between P902-CC and terminal 1 (BACKUP HYD CONTR circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel) (Appendix F). Go to **32.**

14. Check continuity between P902-BB and terminal X2 (NO. 3 HYD PWR PUMP RELAY).

Step 1. If continuity is present, go to **15.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

15. Check continuity between:
J163-A and gnd
J163-C and P242-L

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If continuity is present, go to **16.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

16. Check continuity between P163-A and P163-C.

Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to **32.**
Step 2. If continuity is not present, replace left drag beam switch (PARA 3-4-6). Go to **32.**

17. Check helicopter configuration.

Step 1. WINTER , go to **18.**
Step 2. W/O WINTER , go to **20.**

WINTER **18. Check for 28 VDC between P244B-C and gnd.**

Step 1. If voltage is as specified, go to **19.**
Step 2. If voltage is not as specified, replace winterization kit harness (PARA 16-6-2). Go to **32.**
Step 3. If trouble remains, go to **20.**

WINTER **19. Check continuity between P244B-A and terminal X1 (NO. 3 HYD PWR PUMP RELAY).**

Step 1. If continuity is present, go to **22.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **32.**
Step 3. If trouble remains, go to **21.**

20. Check for 28 VDC between P244-C and gnd.

Step 1. If voltage is as specified, go to **21.**
Step 2. If voltage is not as specified, go to **27.**

21. Check continuity between P244-A and terminal X1 (NO. 3 HYD PWR PUMP RELAY).

Step 1. If continuity is present, go to **22.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

22. Check for 28 VDC between P241-p and gnd.

Step 1. If voltage is as specified, replace right relay panel (PARA 9-4-2). Go to **32.**
Step 2. If voltage is not as specified, go to **23.**

23. Check for 28 VDC between:
P212-E and gnd
P206-E and gnd
P204-E and gnd

Step 1. If voltage is as specified, go to **24.**
Step 2. If voltage is not as specified, go to **25.**

24. Check continuity between:
P204-F and P241-p
P206-F and P241-p
P212-F and P241-p

Step 1. If continuity is present, troubleshoot DC Electrical System (PARA 9-2-2). Go to **32.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **32.**

25. Check for 28 VDC between terminal 3 (BACKUP HYD PUMP switch S31) and gnd.

Step 1. If voltage is as specified, repair/replace wiring as required between the following (Appendix F). Go to **32.**
Terminal 3 (BACKUP HYD PUMP switch S31) and P204-E
Terminal 3 (BACKUP HYD PUMP switch S31) and P206-E
Terminal 3 (BACKUP HYD PUMP switch S31) and P212-E
Step 2. If voltage is not as specified, go to **26.**

26. Check for 28 VDC between terminal 2 (BACKUP HYD PUMP switch S31) and gnd.

Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to **32.**
Step 2. If voltage is not as specified, repair/replace wiring between terminal 1 (BACKUP HYD CONTR circuit breaker, DC ESNTL BUS - copilot's circuit breaker panel) and terminal 2 (Appendix F). Go to **32.**

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

27. Check helicopter configuration.

Step 1. WINTER , go to **28.**
Step 2. W/O WINTER , go to **29.**

WINTER

28. Check continuity between P244B-C and terminal 1 (BACKUP HYD CONTR circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel).

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **32.**
Step 2. If continuity is not present, replace winterization kit harness (PARA 16-6-2). Go to **32.**
Step 3. If trouble remains, go to **29.**

29. Check continuity between P244-C and terminal 1 (BACKUP HYD CONTR circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel).

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **32.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **32.**

30. Check for 115 VAC between the following (HYD PWR PUMP circuit breaker, DC - copilot's circuit breaker panel):
Terminal A2 and gnd
Terminal B2 and gnd
Terminal C2 and gnd

Step 1. If voltage is as specified, repair/replace wiring between circuit breaker and relay as required (Appendix F). Go to **32.**
Step 2. If voltage is not as specified, go to **31.**

31. Check for 115 VAC between the following (HYD PWR PUMP circuit breaker, NO. 1 GEN CNTOR - No. 1 junction box):
Terminal A1 and gnd
Terminal B1 and gnd
Terminal C1 and gnd

Step 1. If voltage is as specified, replace circuit breaker (PARA 9-4-1). Go to **32.**
Step 2. If voltage is not as specified, troubleshoot AC Electrical System (PARA 9-2-1). Go to **32.**

TABLE NO. 7-2-1-21. Backup Pump Fails To Run With BACKUP HYD PUMP switch S31 Placed On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

32. Procedure completed.

TABLE NO. 7-2-1-22. Backup Pump Runs Continuously With No Fault Capsule Lit On Caution/ Advisory Panel.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between:

J420-4 and J420-7 (CONTINUITY)
J420-4 and J420-9 (OPEN)
J421-1 and J421-6 (CONTINUITY)
J421-5 and J421-6 (OPEN)

Step 1. If continuity is present, replace BACKUP HYD PUMP switch S31 (PARA 9-4-5). Go to **2.**

Step 2. If continuity is not present, replace applicable hydraulic pump pressure switch (PARA 7-4-27 or PARA 7-4-32). Go to **2.**

2. Procedure completed.

TABLE NO. 7-2-1-23. #1 PRI SERVO PRESS Capsule Does Not Go On With No Hydraulic Pressure.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 VDC between:

P424-6 and gnd
P425-6 and gnd
P426-6 and gnd

Step 1. If voltage is as specified, go to **3.**

Step 2. If voltage is not as specified, go to **2.**

2. Check continuity between:

TABLE NO. 7-2-1-23. #1 PRI SERVO PRESS Capsule Does Not Go On With No Hydraulic Pressure. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P424-6
terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P425-6
terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P426-6

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **4.**

3. **Check continuity between:**
   P117-8 (lateral servo) and P424-1
   P117-8 (aft servo) and P425-1
   P117-8 (forward servo) and P426-1

Step 1. If continuity is present, replace malfunctioning 1st stage pressure switch as required (PARA 11-4-88). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **4.**

4. **Procedure completed.**

TABLE NO. 7-2-1-24. #2 PRI SERVO PRESS Capsule Does Not Go On With No Hydraulic Pressure.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check for 28 VDC between:**
   P427-6 and gnd
   P428-6 and gnd
   P429-6 and gnd

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, go to **2.**

2. **Check continuity between:**

TABLE NO. 7-2-1-24. #2 PRI SERVO PRESS Capsule Does Not Go On With No Hydraulic Pressure. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P427-6
terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P428-6
terminal 1 (NO. 1 SERVO WARN circuit breaker - copilot's circuit breaker panel) and P429-6

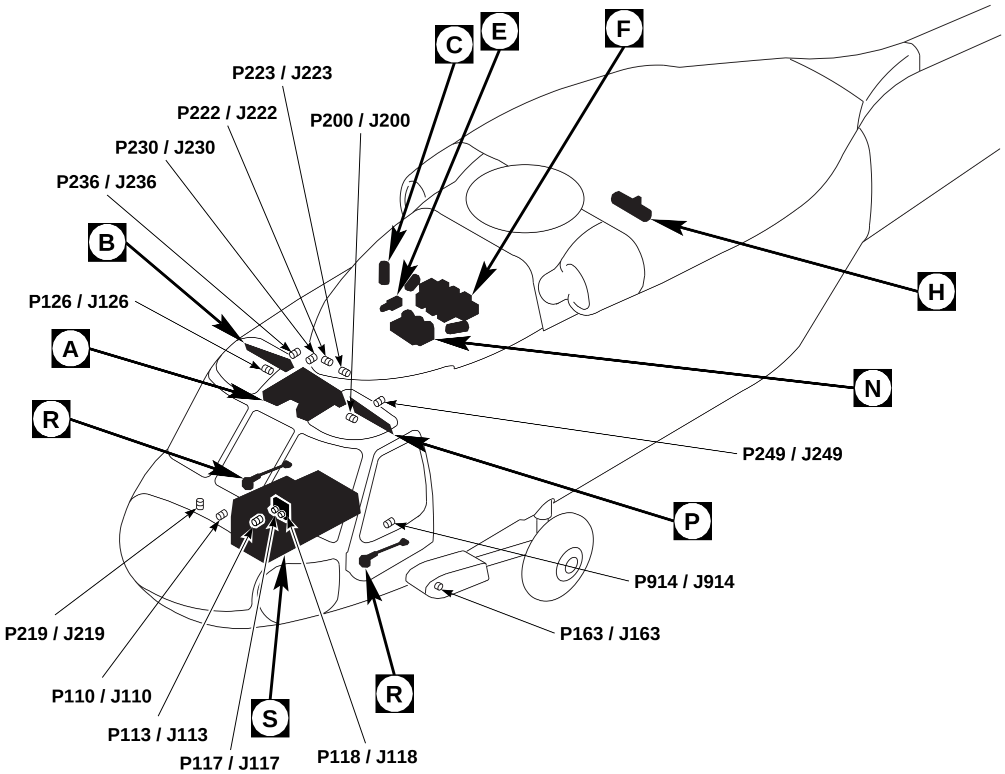
Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **4.**

3. Check continuity between:

P117-56 (lateral servo) and P427-1
P117-56 (aft servo) and P428-1
P117-56 (forward servo) and P429-1

Step 1. If continuity is present, replace malfunctioning 2nd stage pressure switch as required (PARA 11-4-88). Go to **4.**
Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to **4.**

4. Procedure completed.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P3R / J3R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P4R / J4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P103 / J103	COCKPIT TUB, BL 32 LH, STA 247
P104 / J104	COCKPIT TUB, BL 15 RH, STA 247
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	COCKPIT, BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P126 / J126	COCKPIT CEILING, BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P135 / J135	MISCELLANEOUS SWITCH PANEL
P137 / J1	NO. 2 SDC
P138 / J1	NO. 1 SDC
P142 / J142	COCKPIT, BL 3 RH, STA 210
P163 / J163	LEFT DRAG BEAM SWITCH
P200 / J200	COCKPIT CEILING, BL 13 LH, STA 246

NOTES
1. W/O RTR BRAKE
2. RTR BRAKE
3. WINTER
4. SAS / FPS OR AFCC

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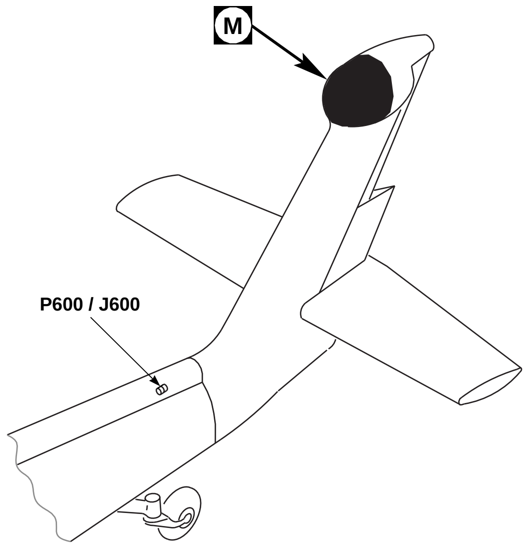
SA

FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 13)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P202 / J202	NO. 2 JUNCTION BOX
P203 / J203	NO. 2 JUNCTION BOX
P210 / J210	NO. 1 JUNCTION BOX
P217 / J217	CABIN CEILING, BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 242
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P224 / J224	JUNCTION BOX, MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23.0 RH
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P244 / J244A P244B / J244	WINTERIZATION KIT HARNESS ASSEMBLY (SEE NOTE 3)

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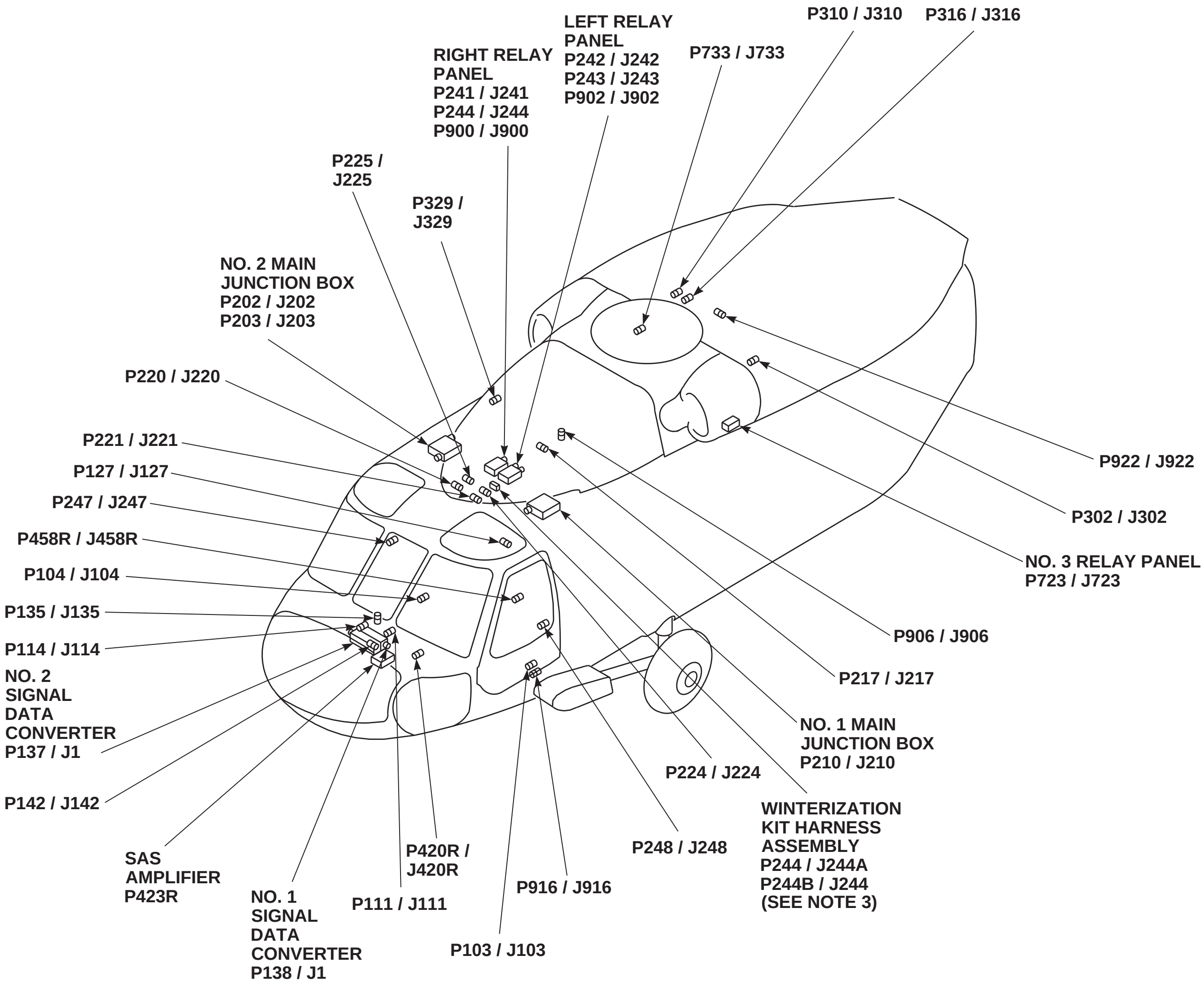
FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 13)



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL 23.0 LH
P302 / J302	APU ACCUMULATOR PRESSURE SWITCH
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P316 / J316	CABIN CEILING, BL 17 RH, STA 387
P329 / J329	MAIN ROTOR BLADE DE−ICE JUNCTION BOX
P416 / J416	2ND STAGE SERVO SHUTOFF VALVE
P417 / J417	1ST STAGE SERVO SHUTOFF VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P420 / J420	NO. 2 HYDRAULIC PUMP PRESSURE SWITCH
P420R / J420R	COCKPIT, BL 0, STA 191.5
P421 / J421	NO. 1 HYDRAULIC PUMP PRESSURE SWITCH
P422 / J422	BACKUP PUMP MOTOR THERMAL SWITCH
P422R	SAS / FPS COMPUTER
P423 / J423	BACKUP PUMP MOTOR
P423R	SAS AMPLIFIER
P424 / J424	1ST STAGE PRESSURE SWITCH, LATERAL PRIMARY SERVO
P425 / J425	1ST STAGE PRESSURE SWITCH, AFT PRIMARY SERVO
P426 / J426	1ST STAGE PRESSURE SWITCH, FORWARD PRIMARY SERVO

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FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 13)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P427 / J427	2ND STAGE PRESSURE SWITCH, LATERAL PRIMARY SERVO
P428 / J428	2ND STAGE PRESSURE SWITCH, AFT PRIMARY SERVO
P428R / J11	SAS / FPS COMPUTER
P429 / J429	2ND STAGE PRESSURE SWITCH, FORWARD PRIMARY SERVO
P432 / J432	NO. 2 HYDRAULIC PUMP MODULE, FLUID LEVEL SWITCH
P433	2ND STAGE TAIL ROTOR SHUTOFF VALVE
P441 / J441	1ST STAGE TAIL ROTOR SHUTOFF VALVE
P442 / J442	BACKUP PUMP MODULE FLUID LEVEL SWITCH
P445 / J445	BACKUP PUMP PRESSURE SWITCH
P446 / J446	DEPRESSURIZATON VALVE
P447 / J447	NO.1 HYDRAULIC PUMP MODULE, FLUID LEVEL SWITCH
P448	PILOT ASSIST SHUTOFF VALVE
P458R / J458R	TOP DECK, BL 4 LH, STA 262
P460R / J460R	PITCH TRIM SAS MODULE
P461R / J461R	ROLL SAS MODULE
P462R / J462R	YAW SAS MODULE
P463R / J463R	COLLECTIVE BOOST SERVO PRESSURE SWITCH
P464R / J464R	PITCH TRIM ACTUATOR ASSEMBLY
P466R / J466R	PITCH TRIM TURN−ON VALVE
P467R / J467R	SAS PRESSURE SWITCH
P468R / J468R	SAS SHUTOFF VALVE
P469R / J469R	BOOST SHUTOFF VALVE
P470R / J470R	YAW BOOST SERVO PRESSURE SWITCH
P600 / J600	TAIL CONE, BL 0, STA 650
P609 / J609	TAIL ROTOR SERVO, 2ND STAGE PRESSURE SWITCH
P610 / J610	TAIL ROTOR SERVO, 1ST STAGE PRESSURE SWITCH
P723 / J723	NO. 3 RELAY PANEL
P733 / J733	BL 24.71, STA 360, WL 265.30
P900 / J900	RH RELAY PANEL ASSEMBLY
P902 / J902	LH RELAY PANEL ASSEMBLY
P906 / J906	CABIN CEILING, BL 0, STA 284
P914 / J914	COCKPIT, BL 24 LH, STA 247
P916 / J916	CABIN TUB, BL 25 RH, STA 247
P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380

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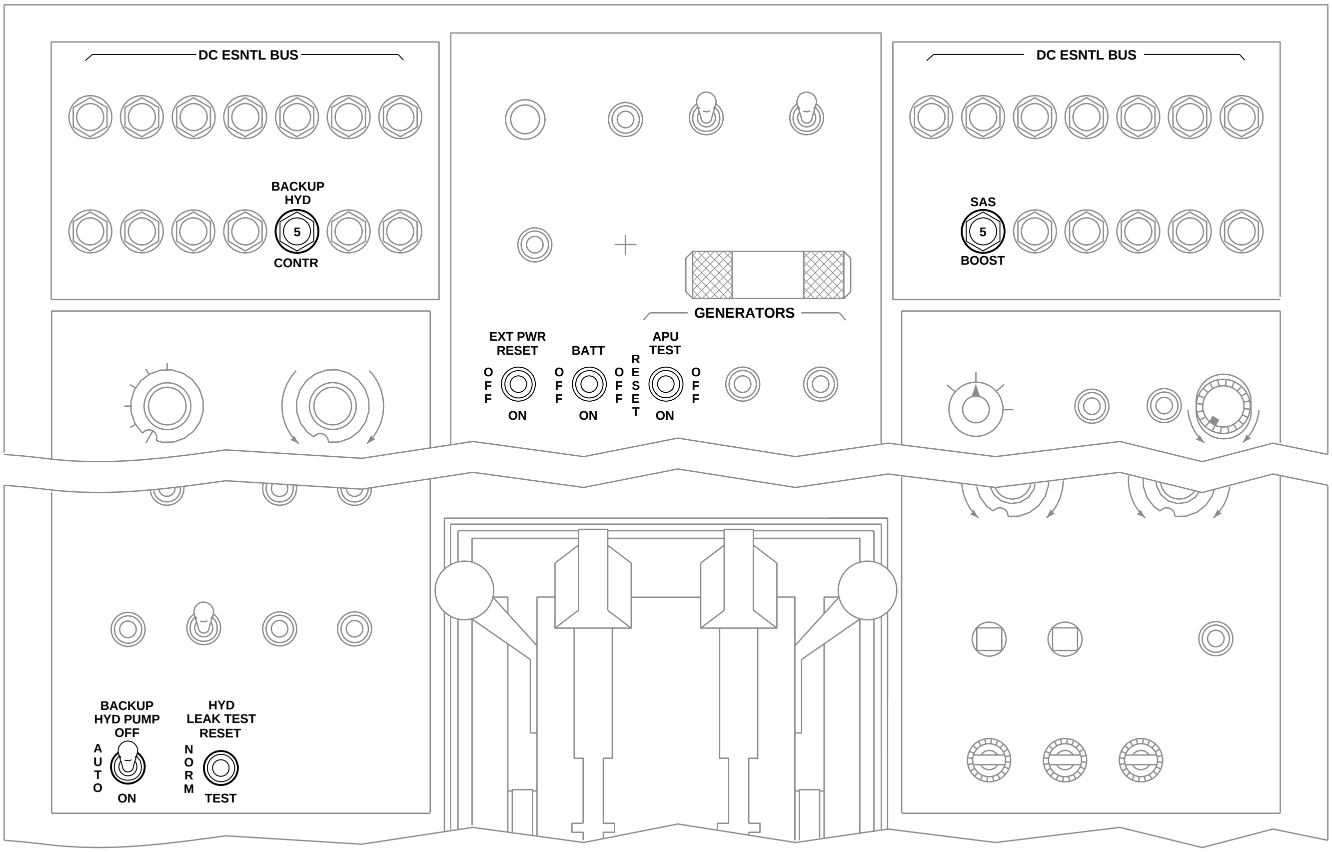
SA

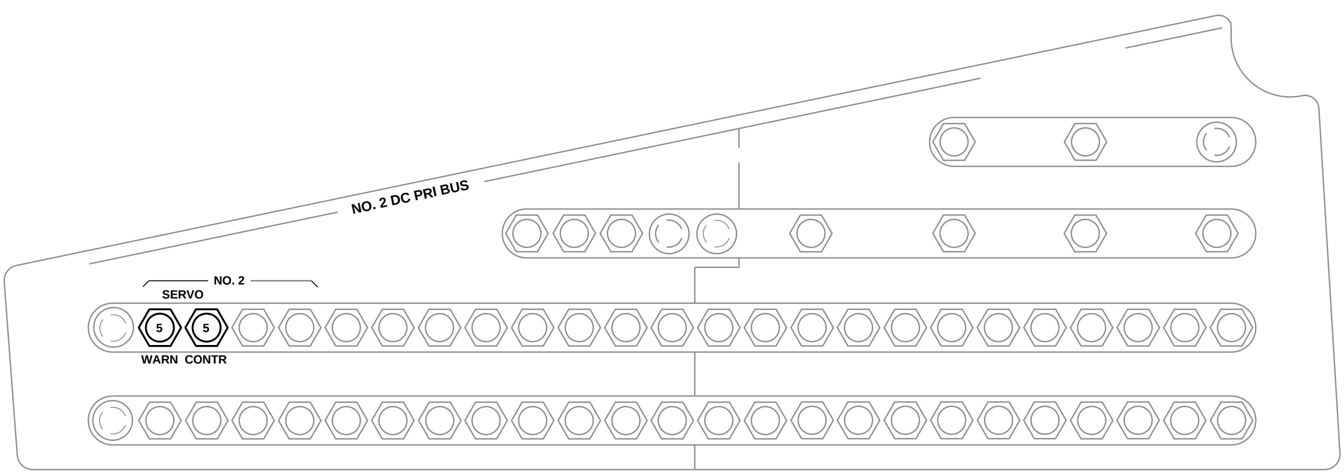
FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 13)



UPPER CONSOLE

A

B



PILOT'S CIRCUIT BREAKER PANEL

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SA

FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 5 OF 13)

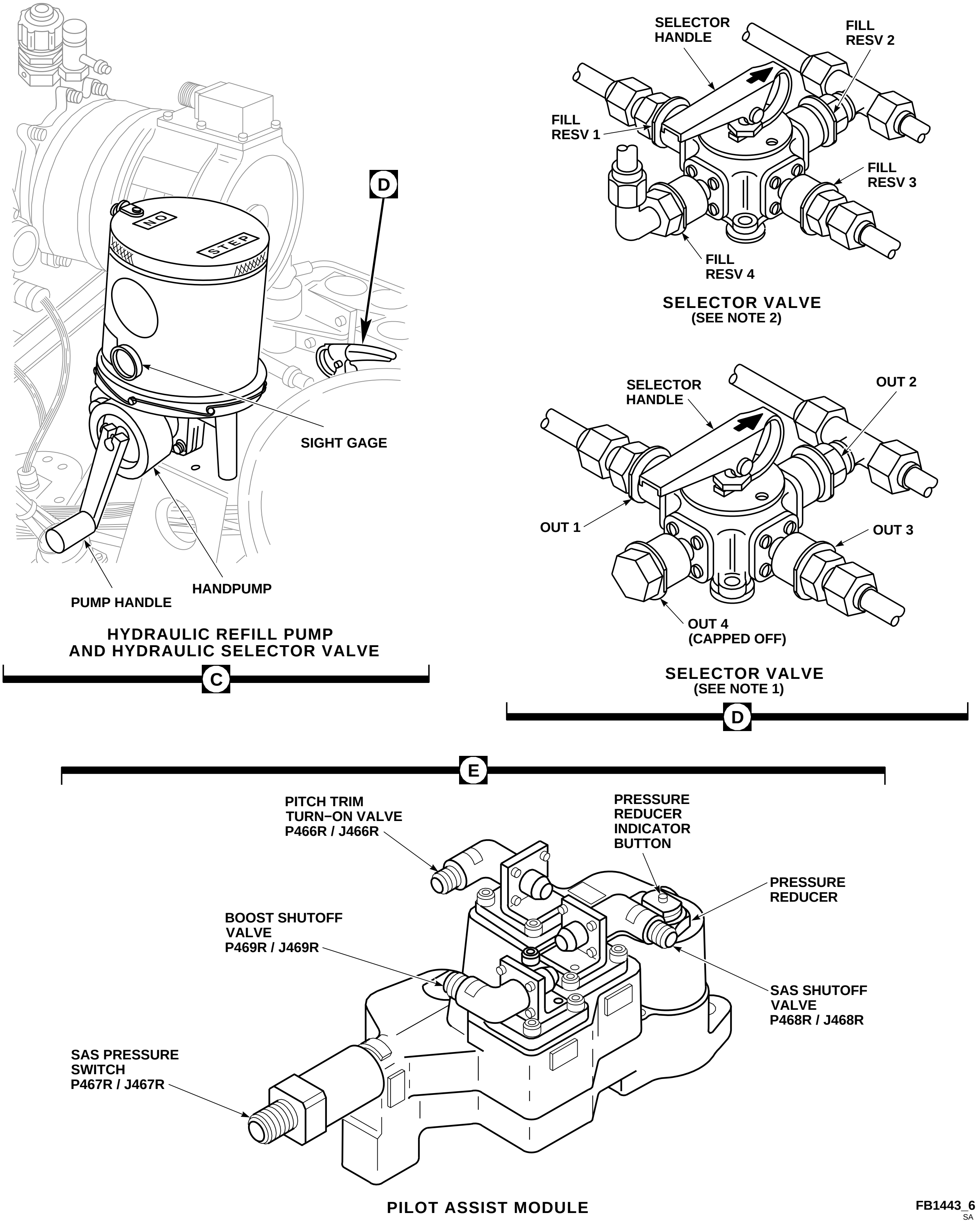
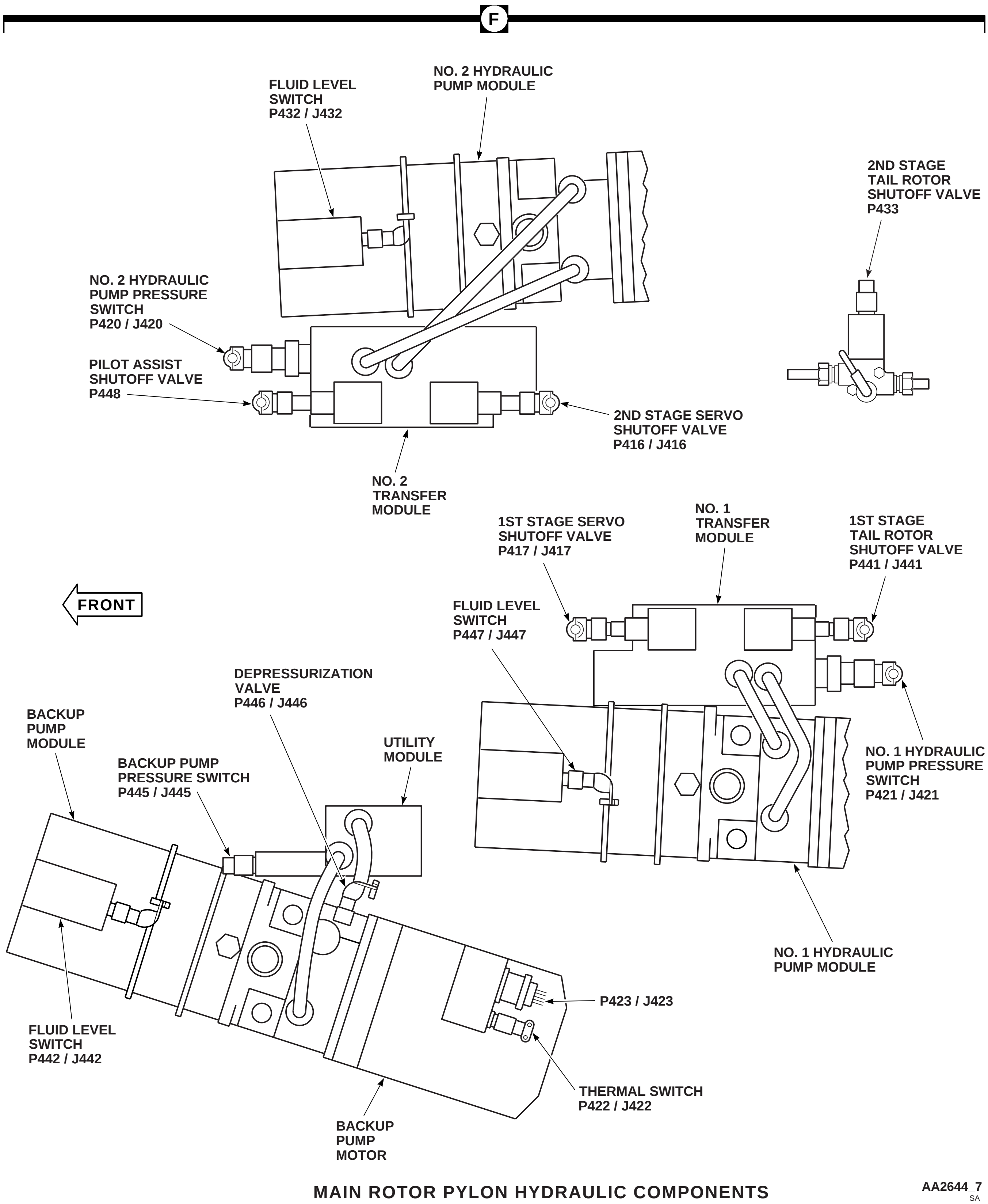


FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 6 OF 13)



MAIN ROTOR PYLON HYDRAULIC COMPONENTS

FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 7 OF 13)

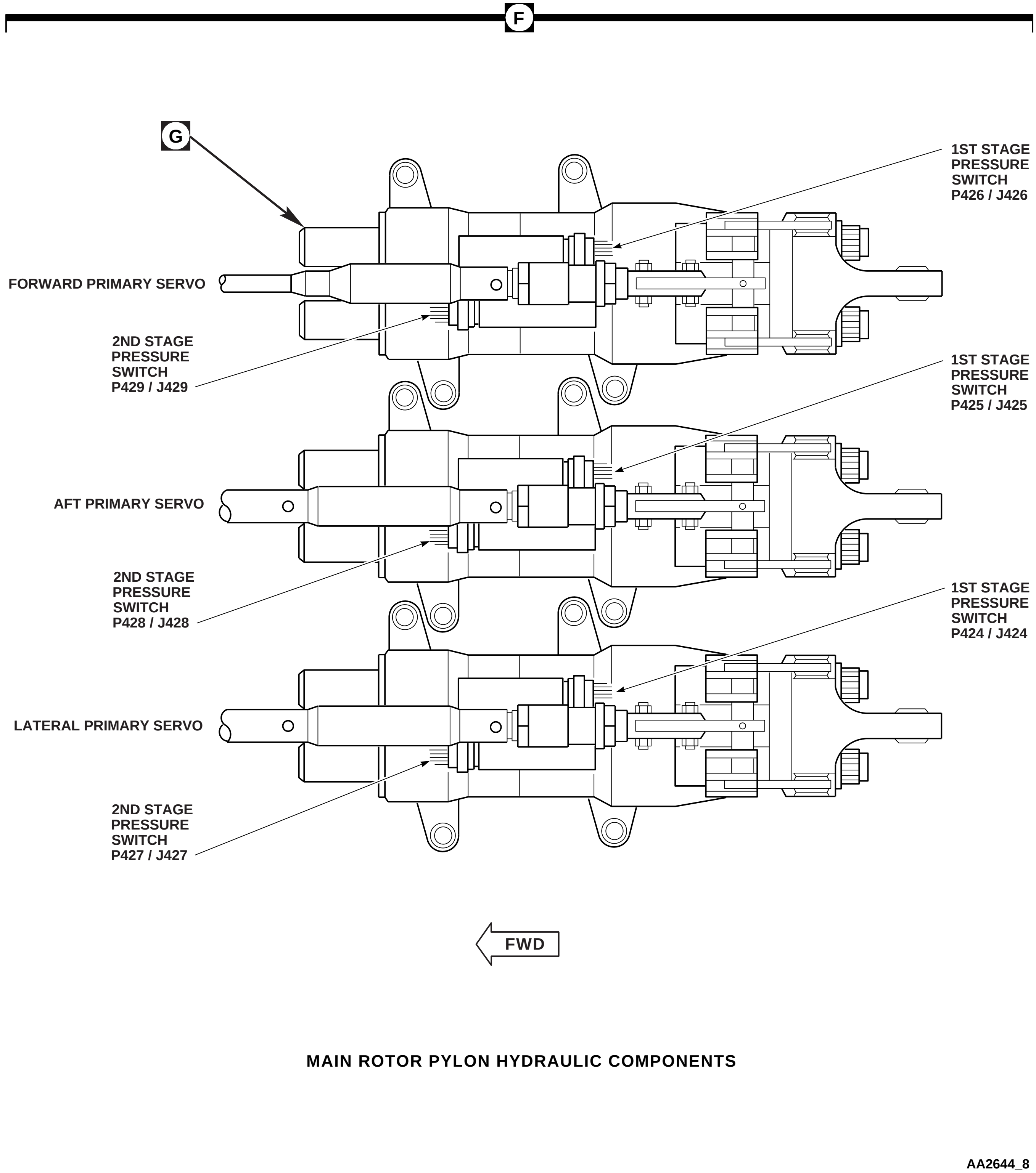


FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 8 OF 13)

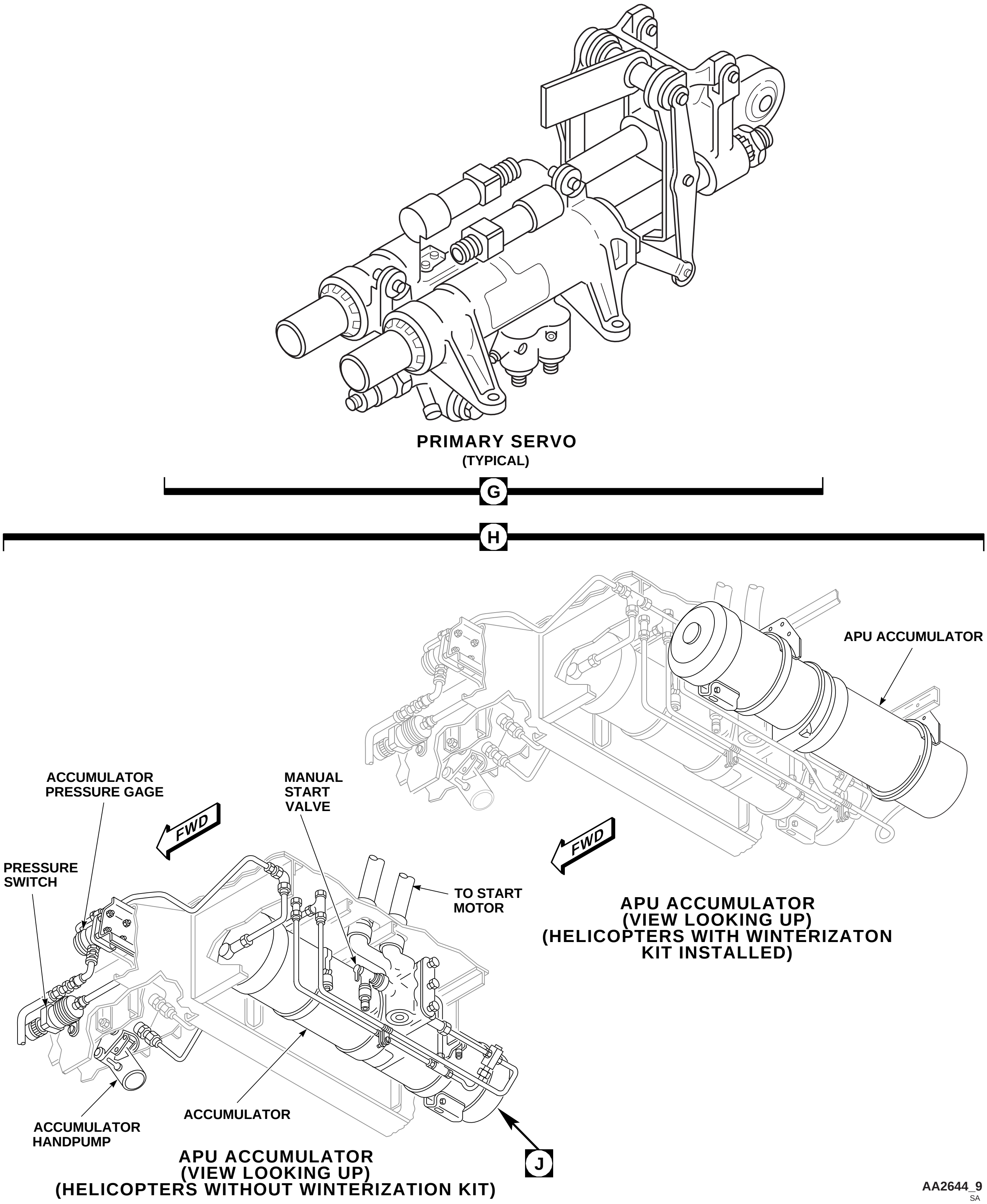


FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 9 OF 13)

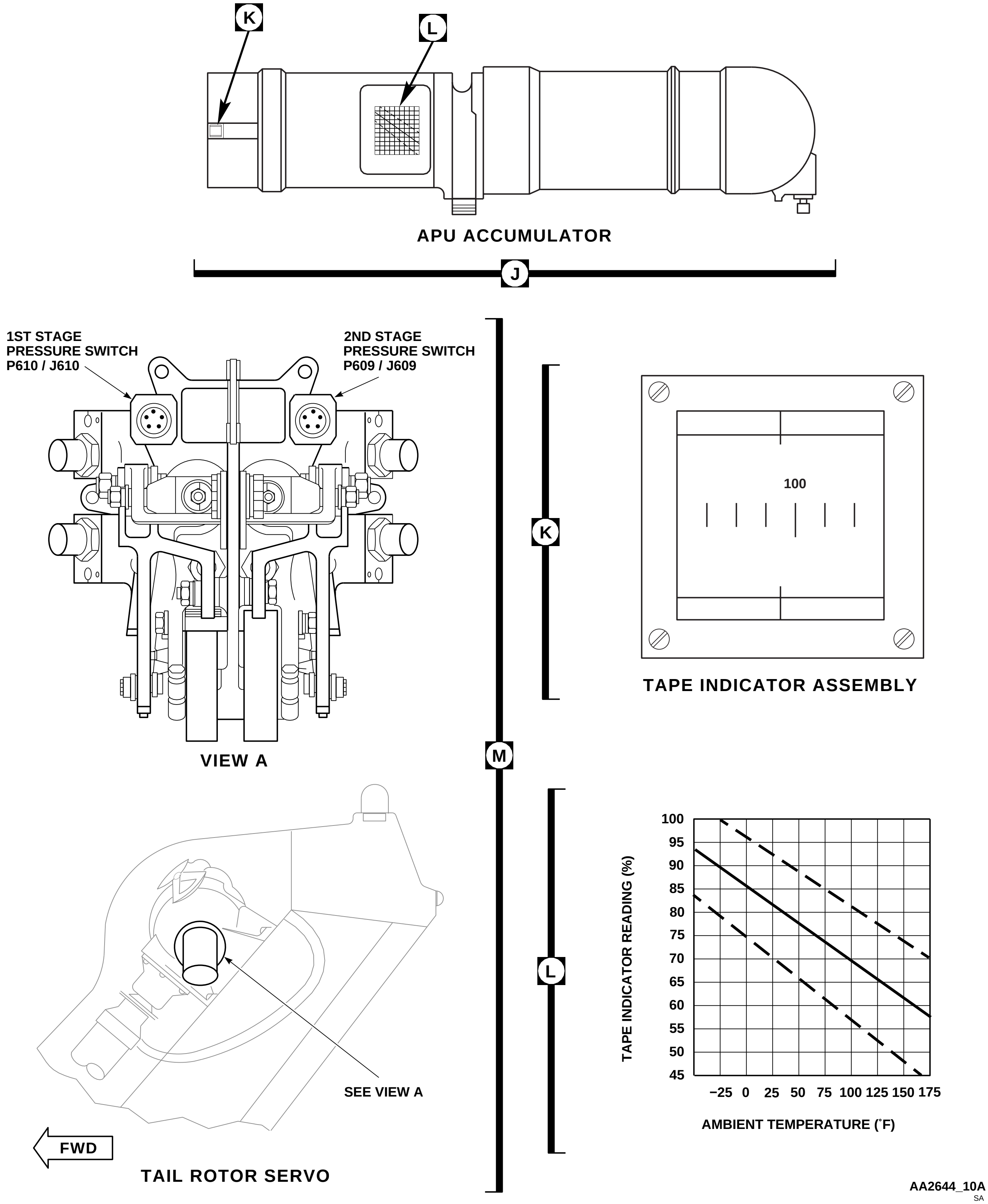
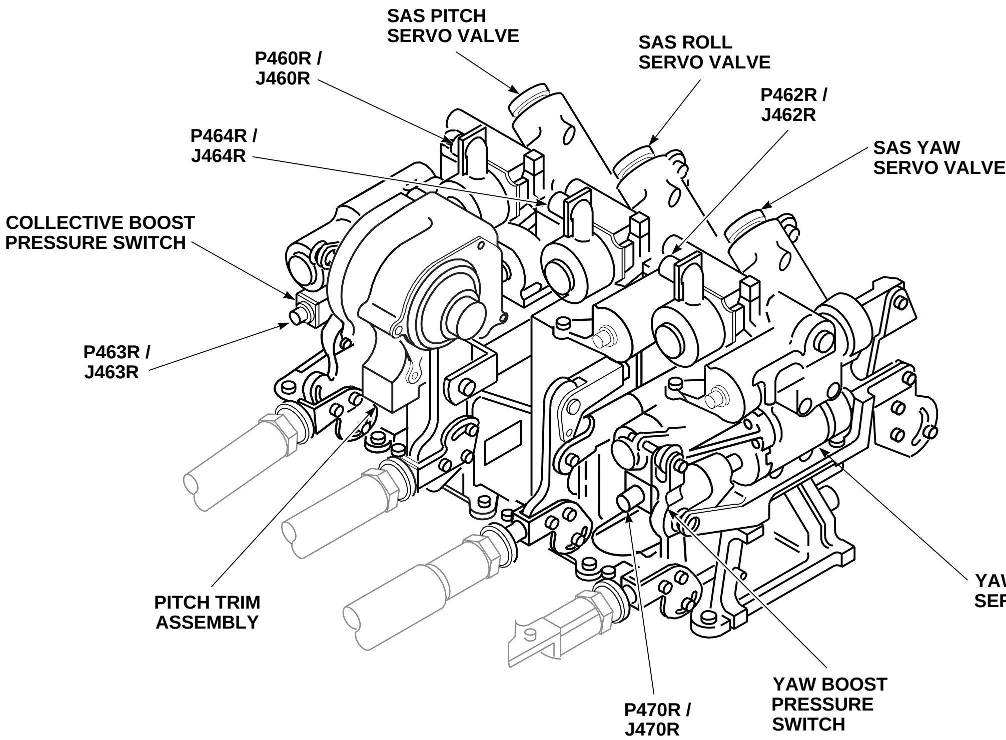


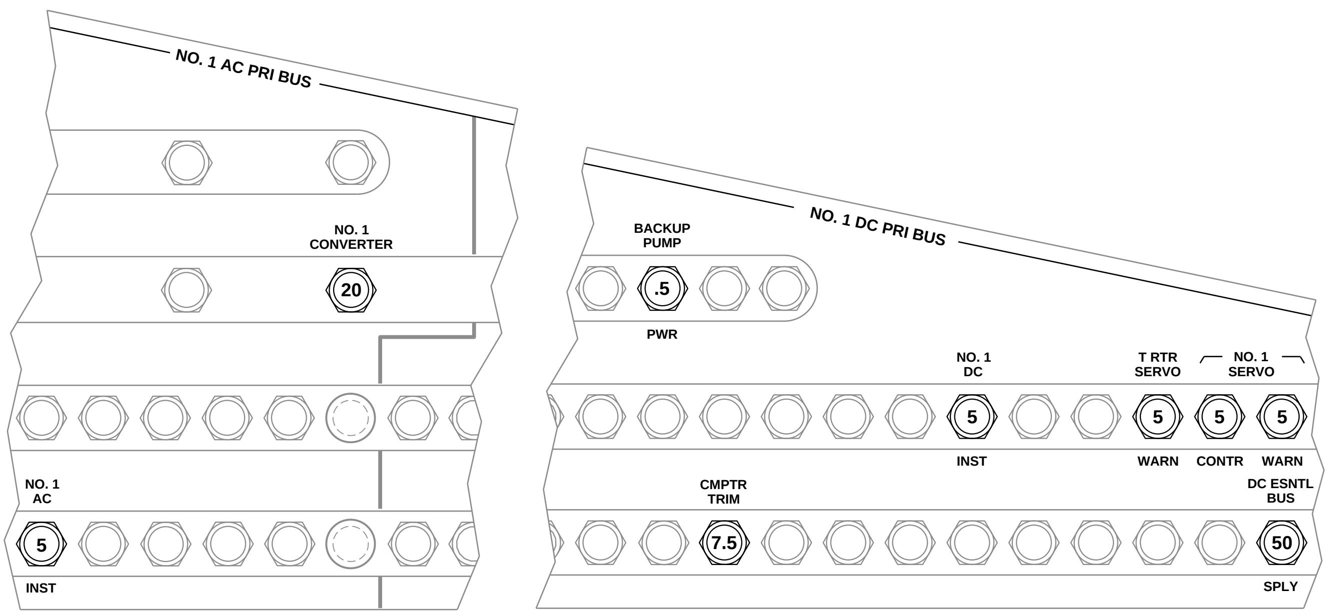
FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 10 OF 13)



PILOT ASSIST SERVOS

N

P



COPILOT'S CIRCUIT BREAKER PANEL

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FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 11 OF 13)

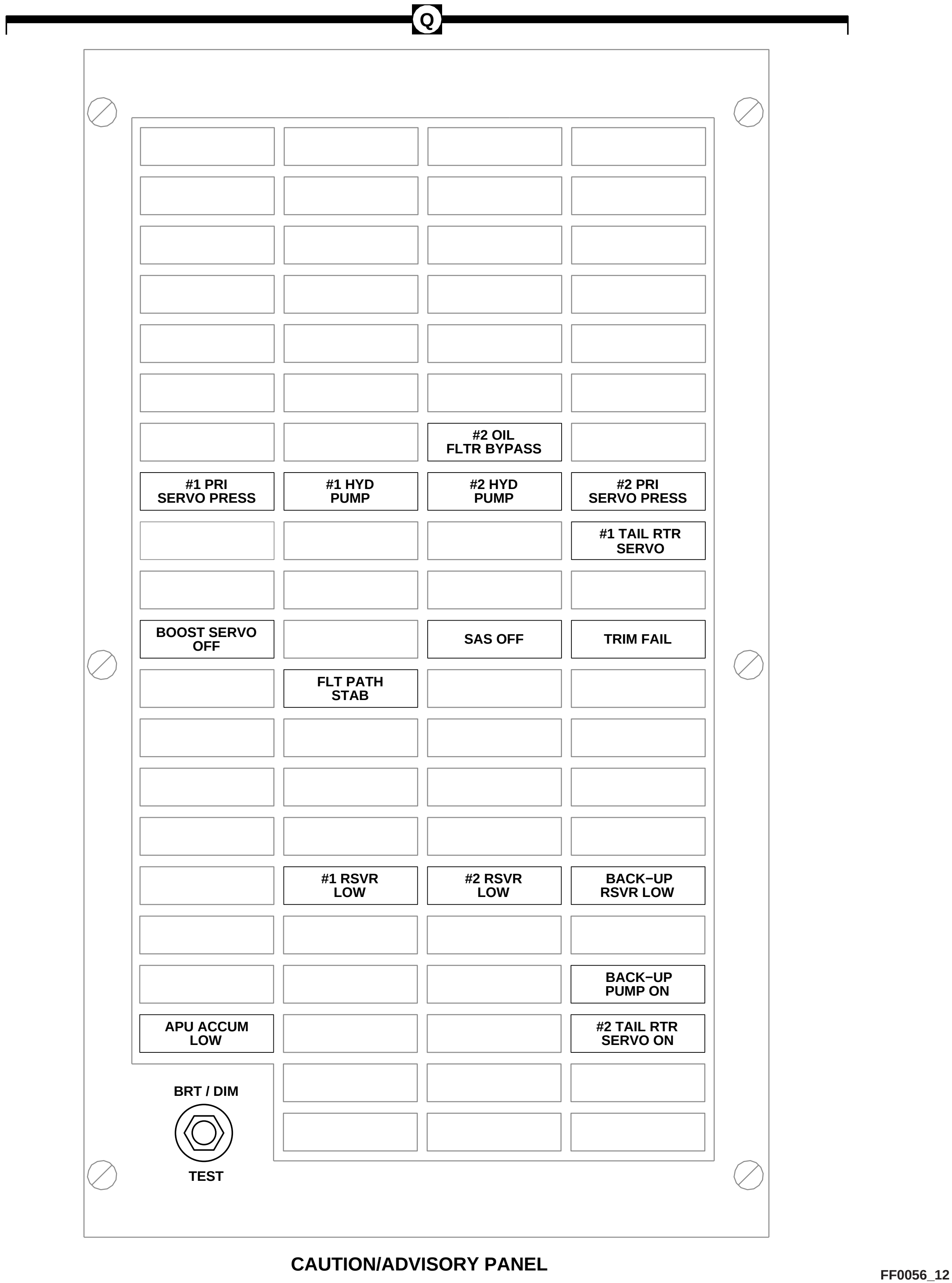


FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 12 OF 13)

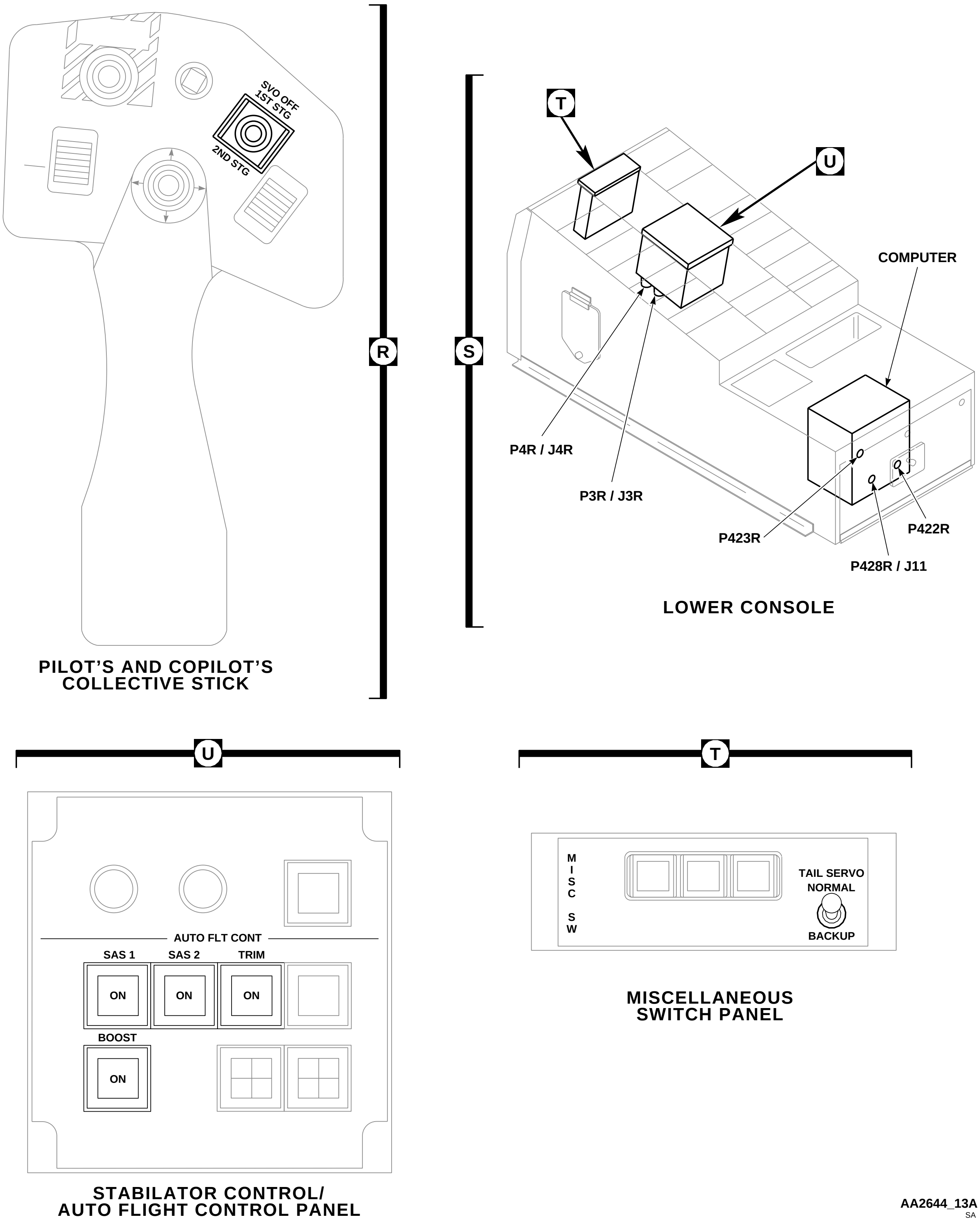


FIGURE 7-2-1-1. HYDRAULIC SYSTEMS LOCATION DIAGRAM (SHEET 13 OF 13)

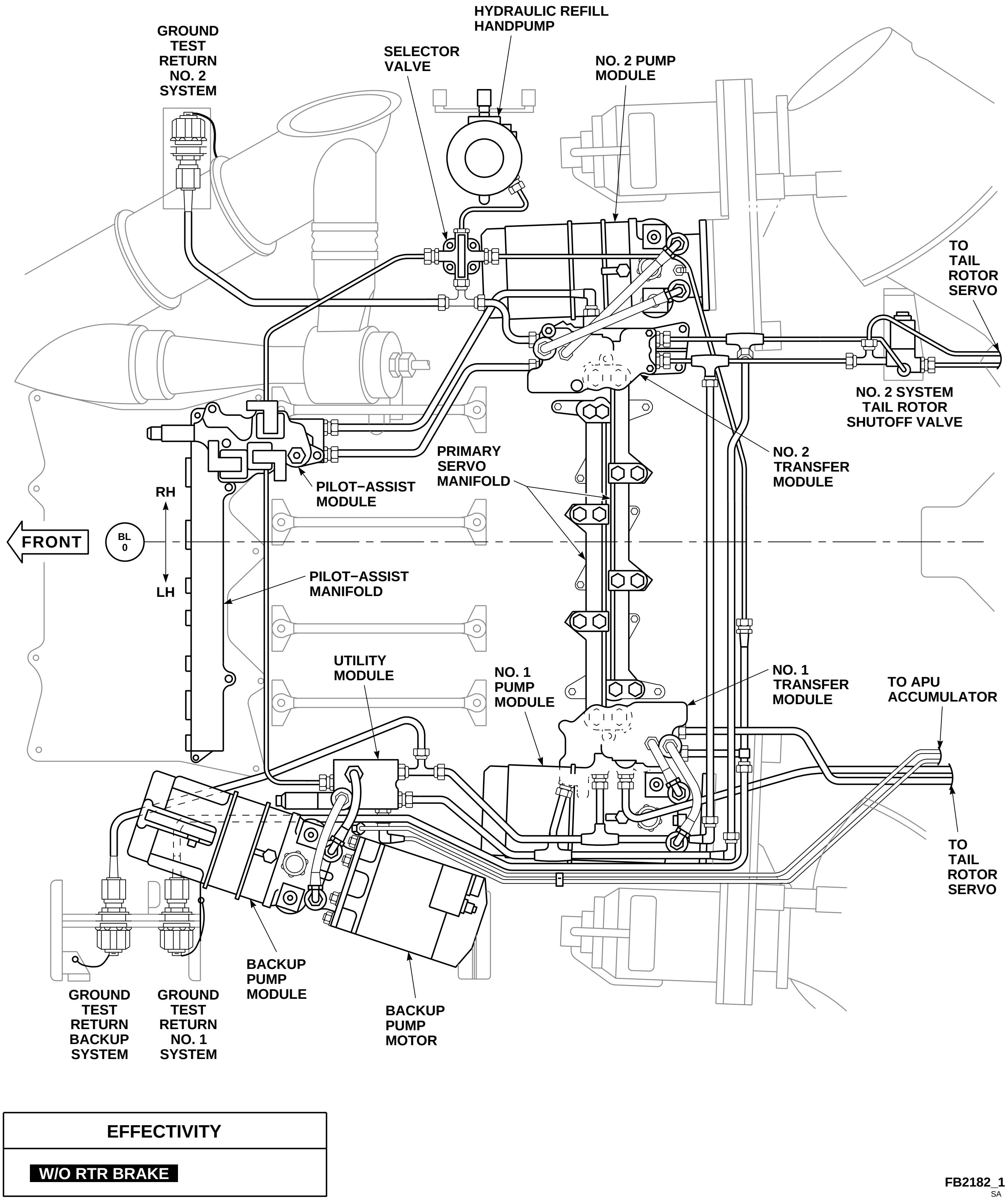


FIGURE 7-2-1-2. MAIN ROTOR PYLON HYDRAULIC COMPONENT LOCATION (SHEET 1 OF 2)

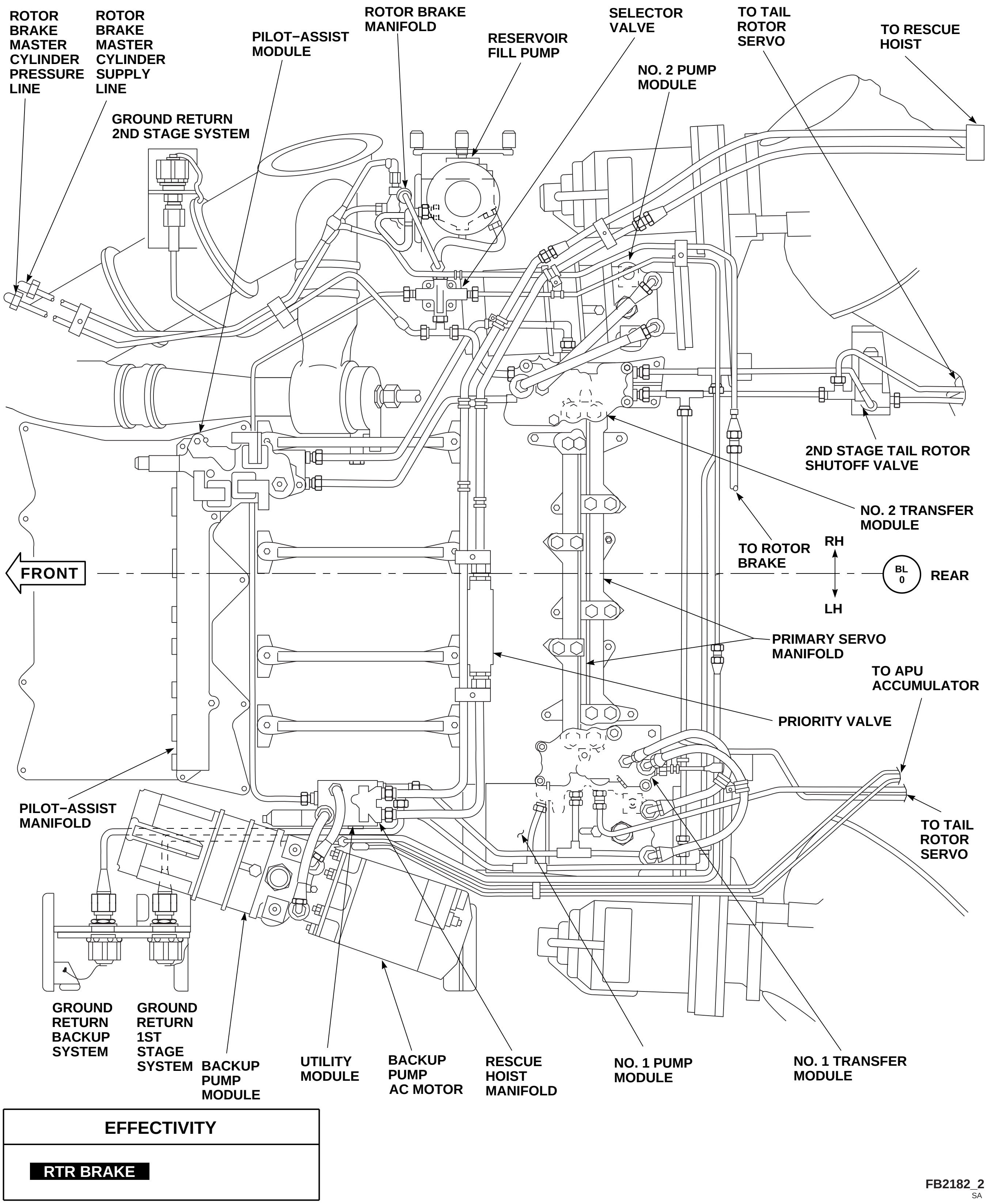
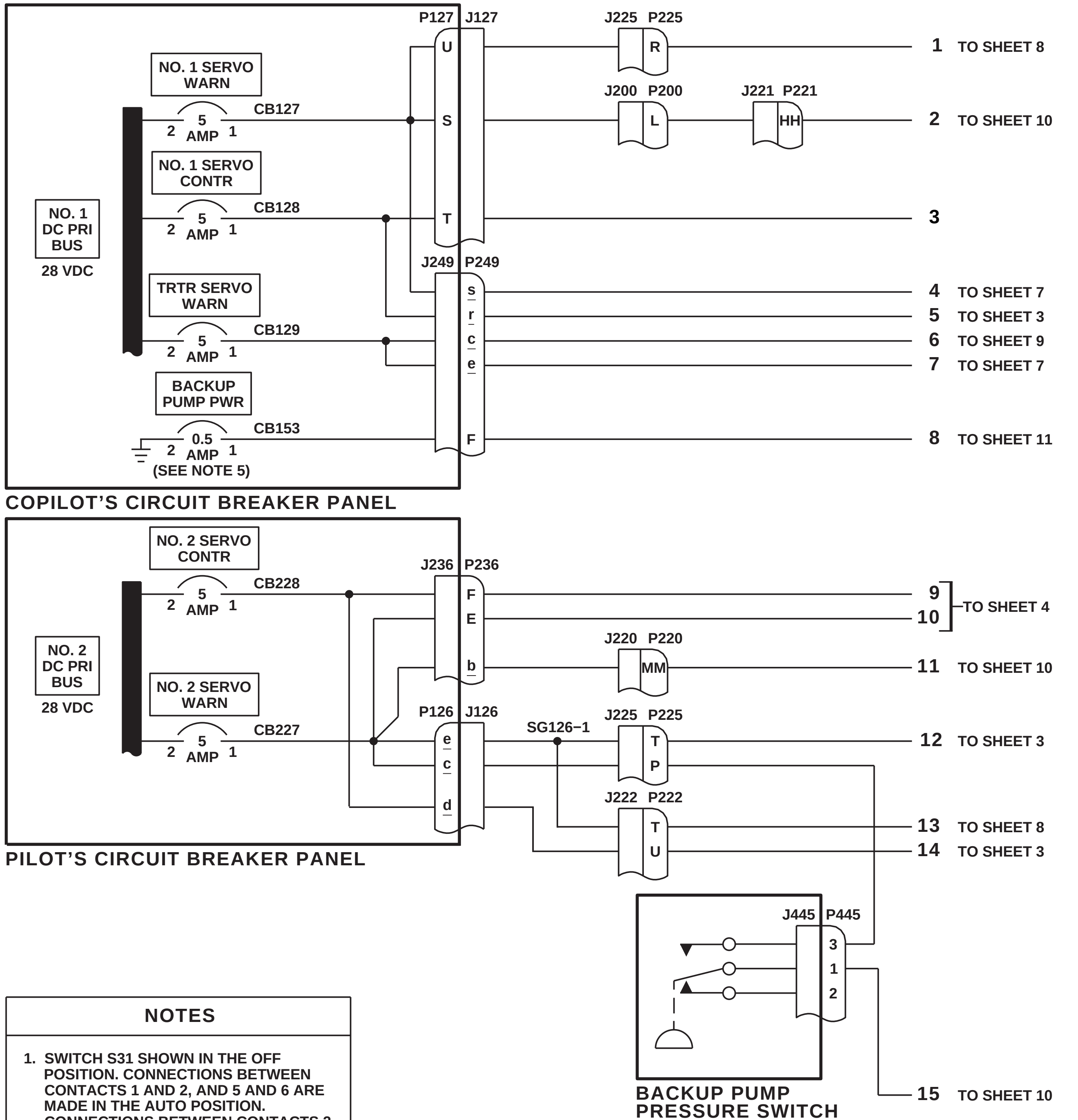


FIGURE 7-2-1-2. MAIN ROTOR PYLON HYDRAULIC COMPONENT LOCATION (SHEET 2 OF 2)



NOTES

1. SWITCH S31 SHOWN IN THE OFF POSITION. CONNECTIONS BETWEEN CONTACTS 1 AND 2, AND 5 AND 6 ARE MADE IN THE AUTO POSITION. CONNECTIONS BETWEEN CONTACTS 2 AND 3, AND 5 AND 6 ARE MADE WITH S31 ON.
2. **WINTER**
3. SWITCH SHOWN WITH WEIGHT ON WHEELS.
4. THIS VALVE (WHEN ENERGIZED) ALLOWS THE BACKUP PUMP TO BE IN BYPASS UNTIL THE MOTOR IS AT RATED SPEED.
5. THE 0.5 AMP CIRCUIT BREAKER FUNCTIONS AS A REMOTE INDICATOR WHEN CB404 TRIPS, AND IS USED TO RESET CB404.
6. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.
7. **70583 − 70893**
8. SAS / FPS OR AFCC.

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FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 1 OF 13)

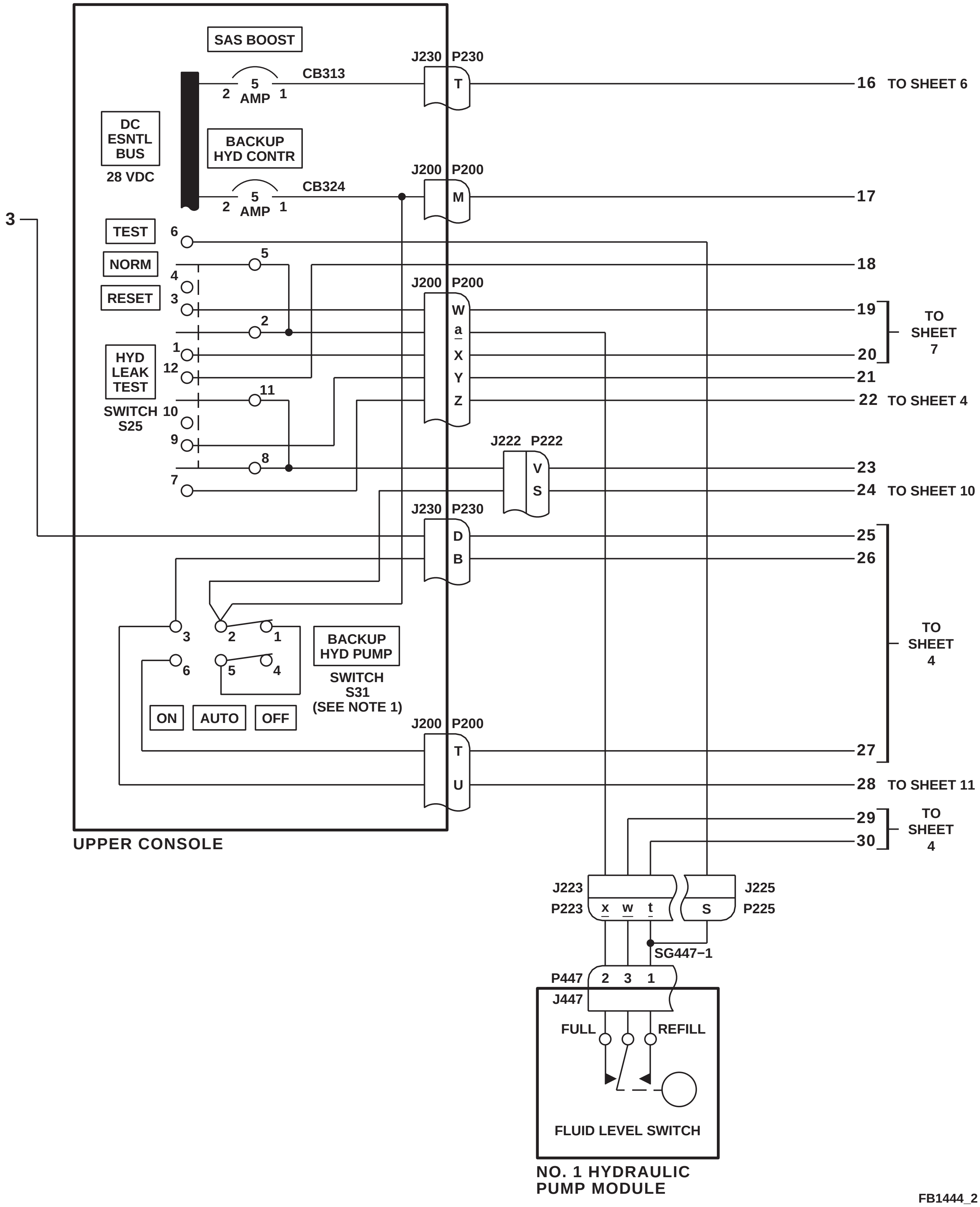


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 2 OF 13)

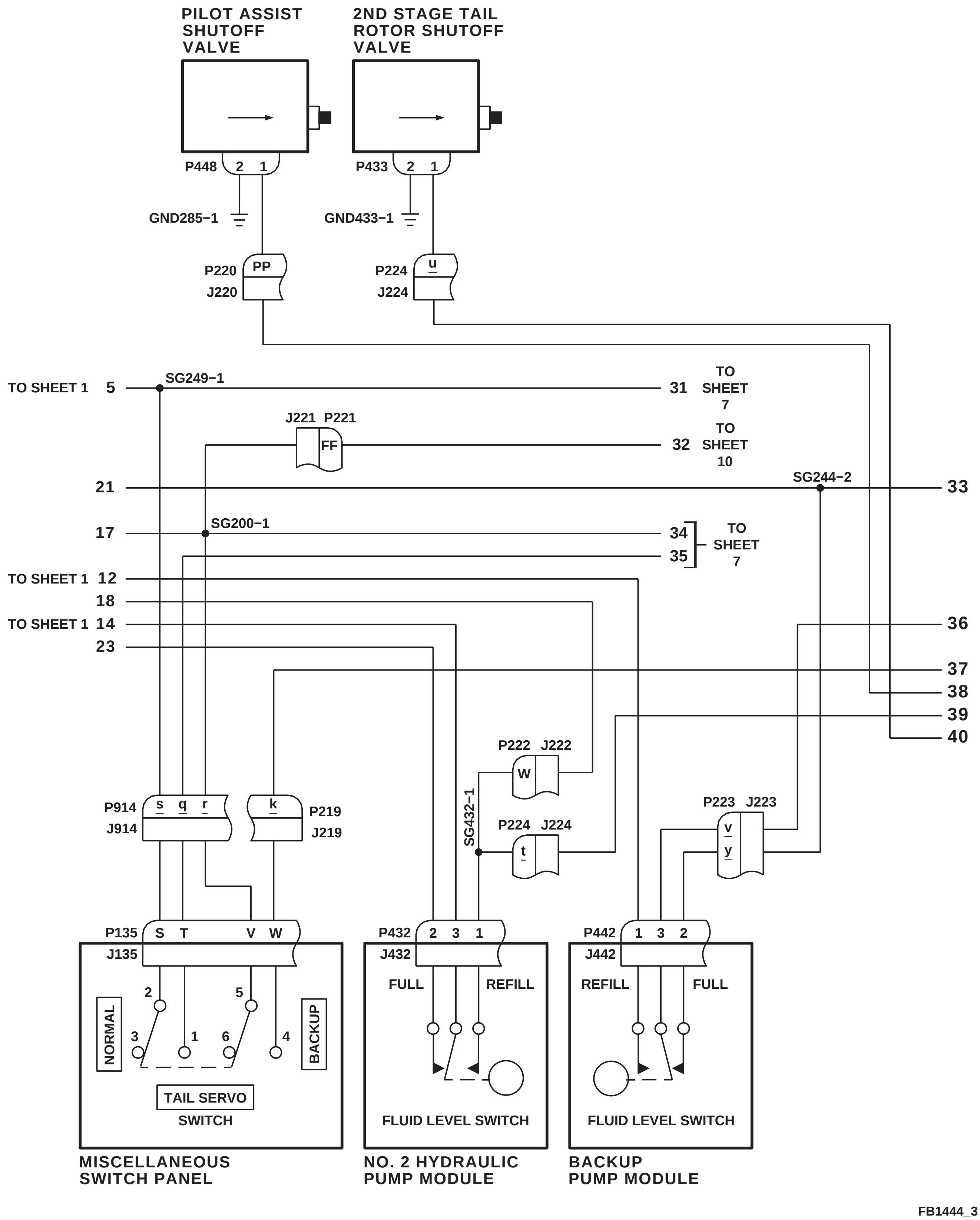


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 3 OF 13)

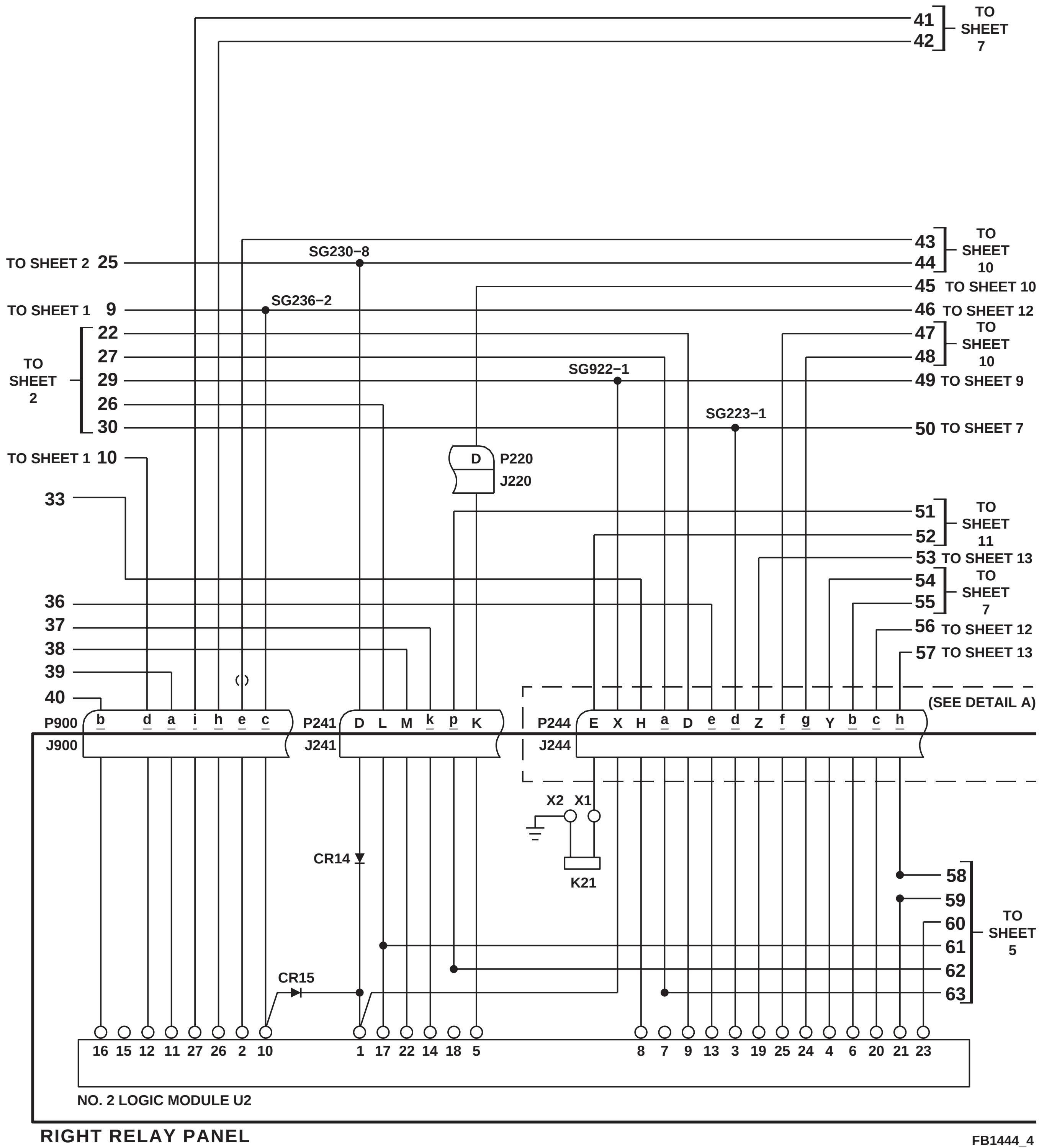
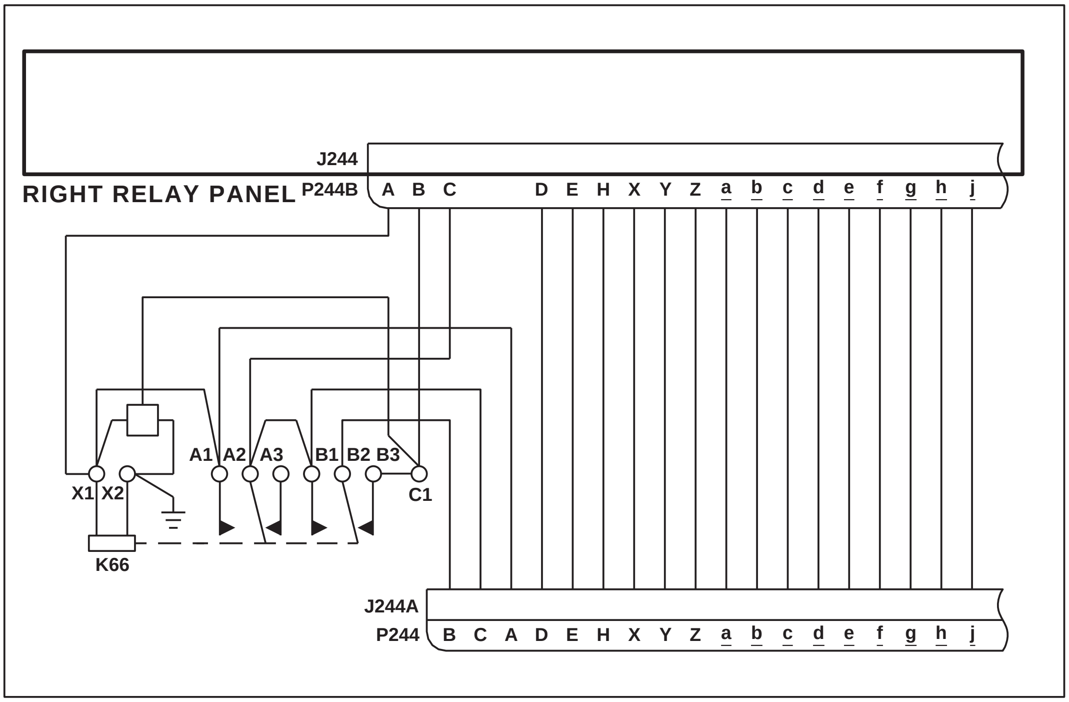
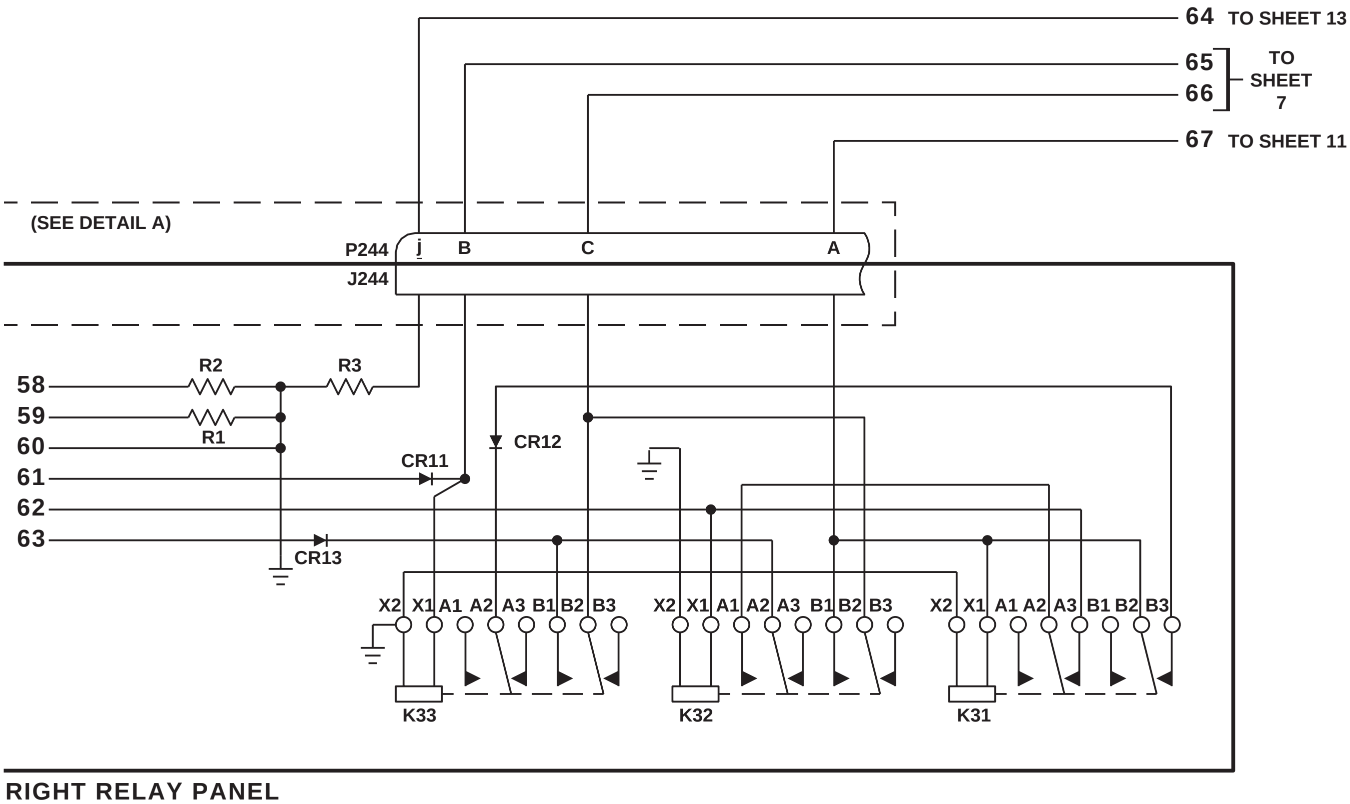


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 4 OF 13)



DETAIL A
(SEE NOTE 2)



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SA

FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 5 OF 13)

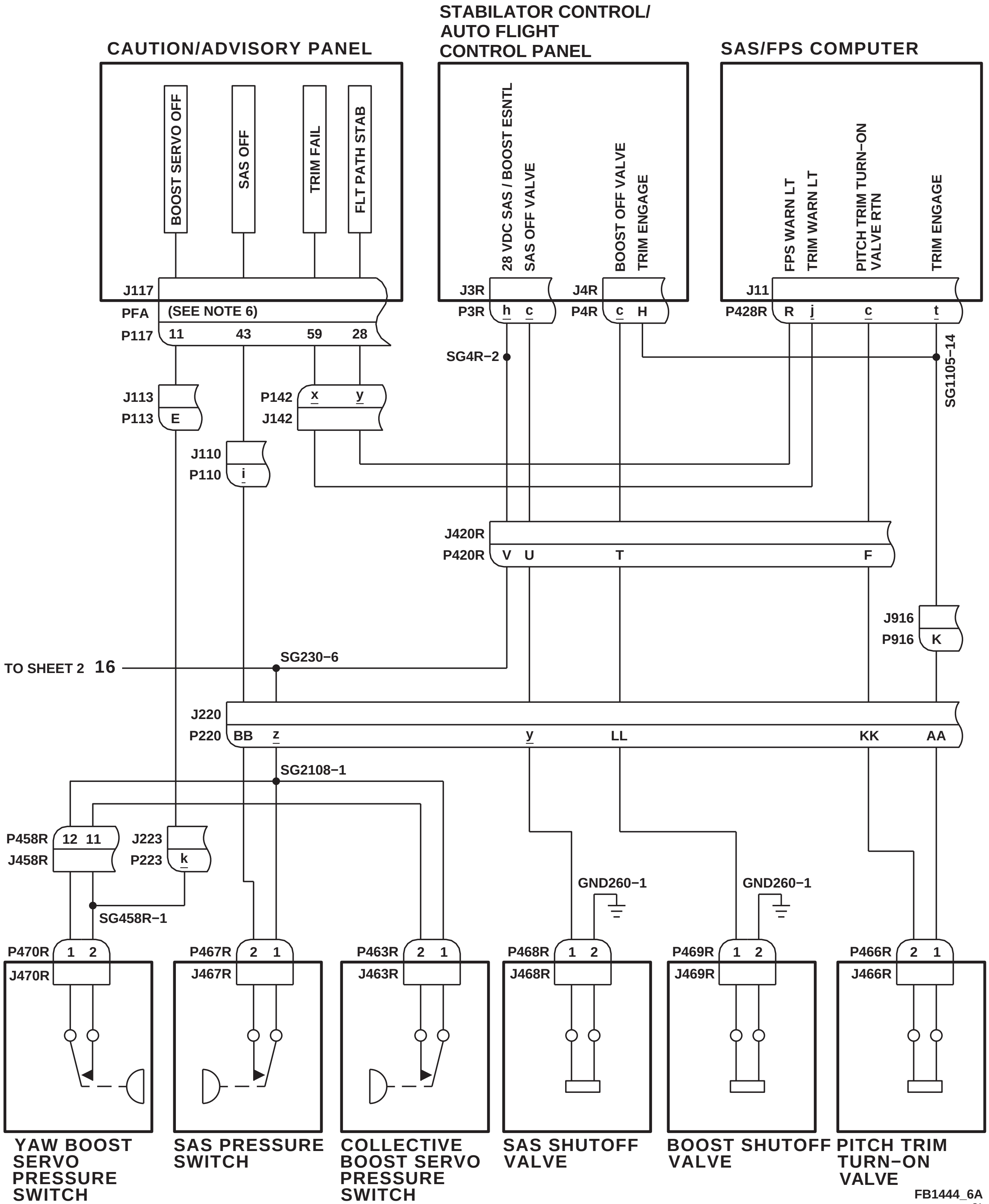


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 6 OF 13)

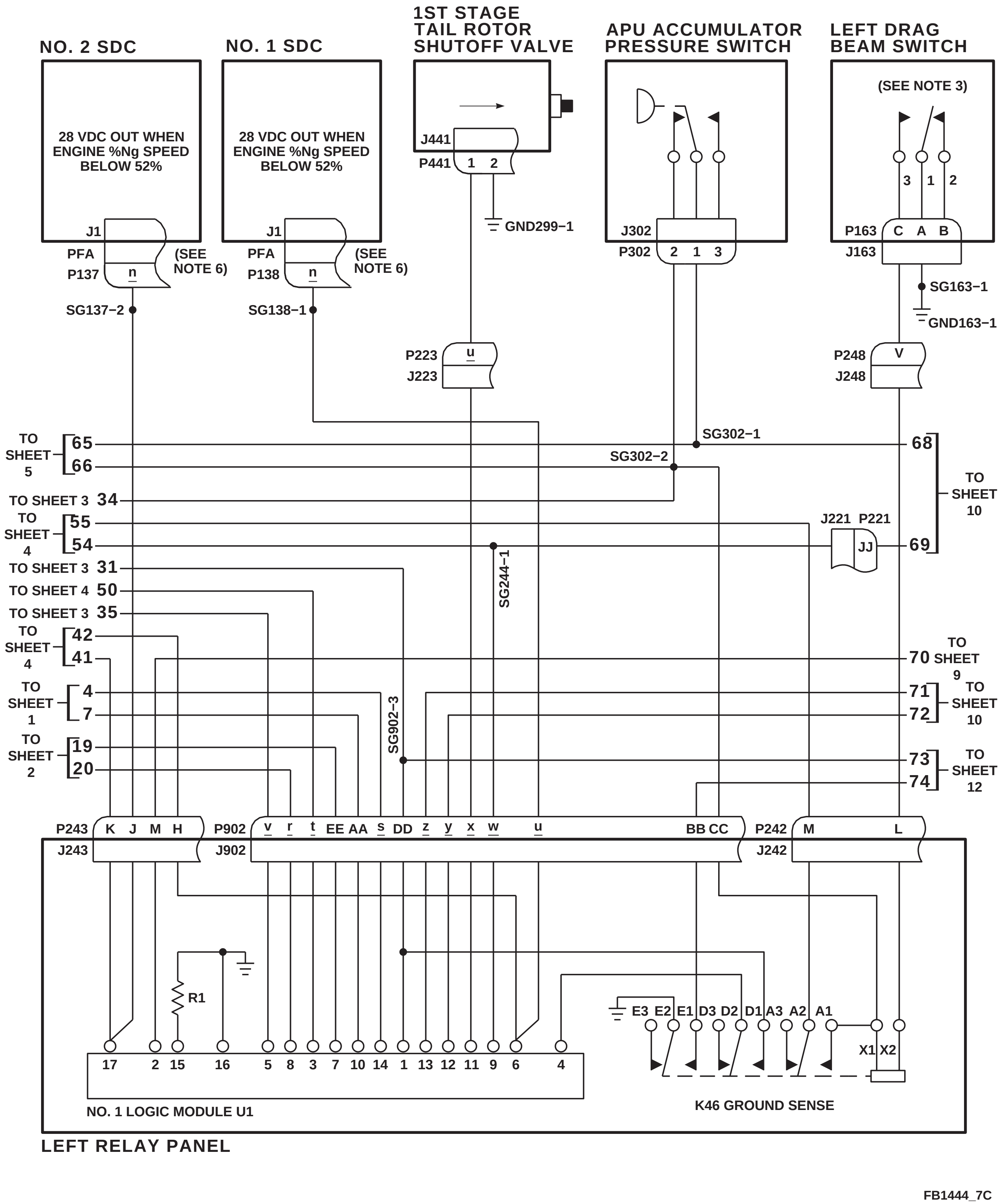


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 7 OF 13)

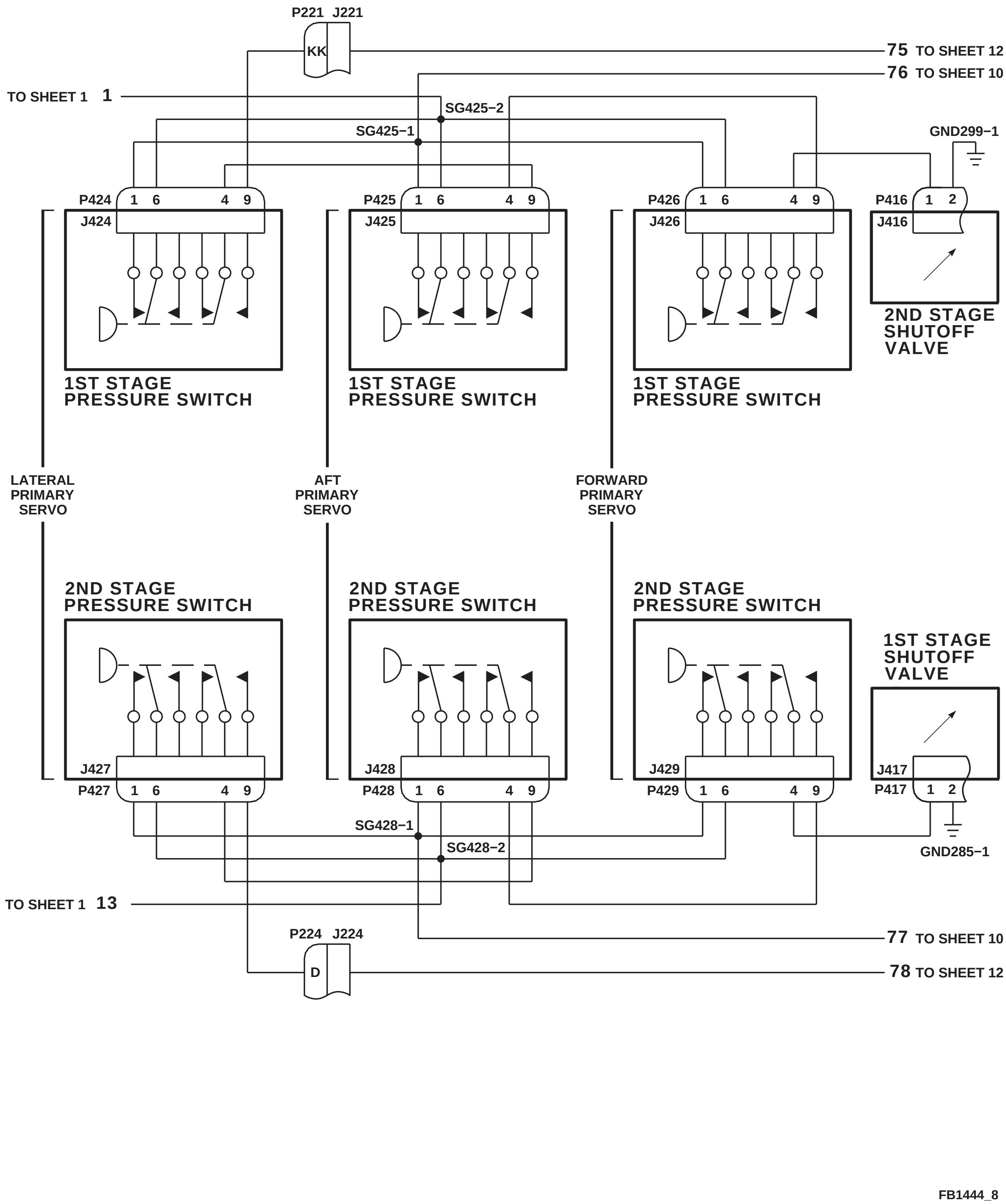


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 8 OF 13)

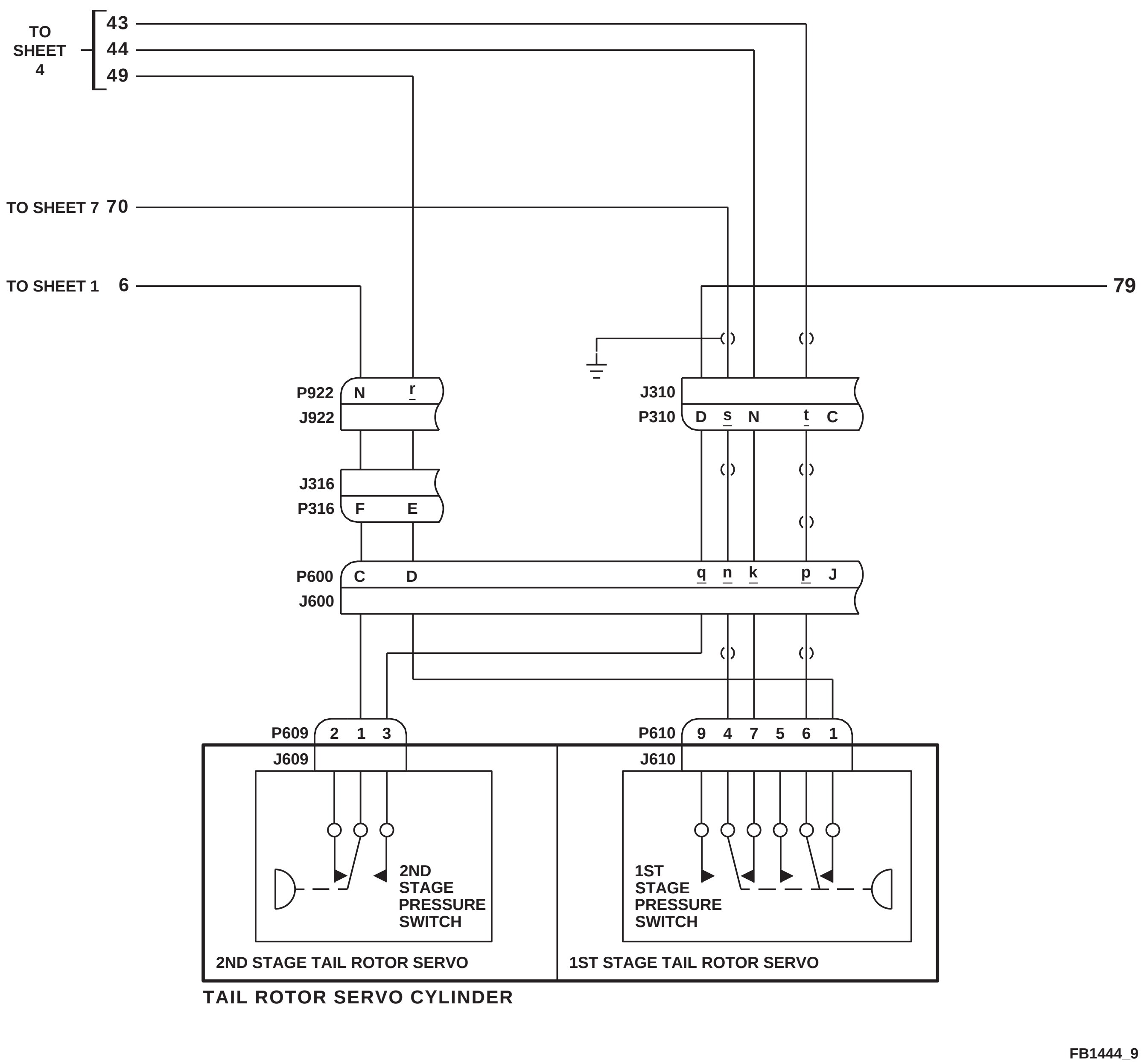


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 9 OF 13)

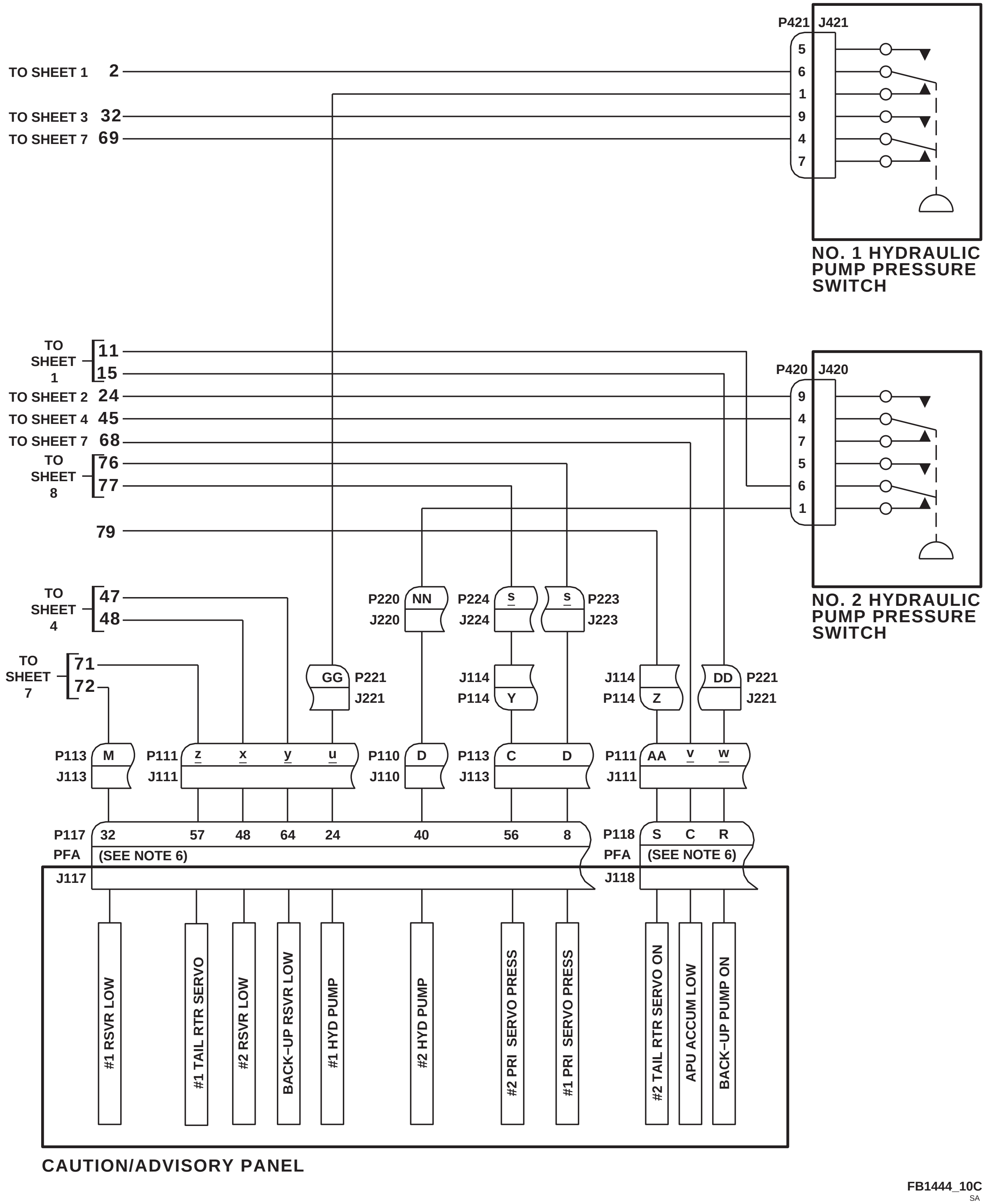


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 10 OF 13)

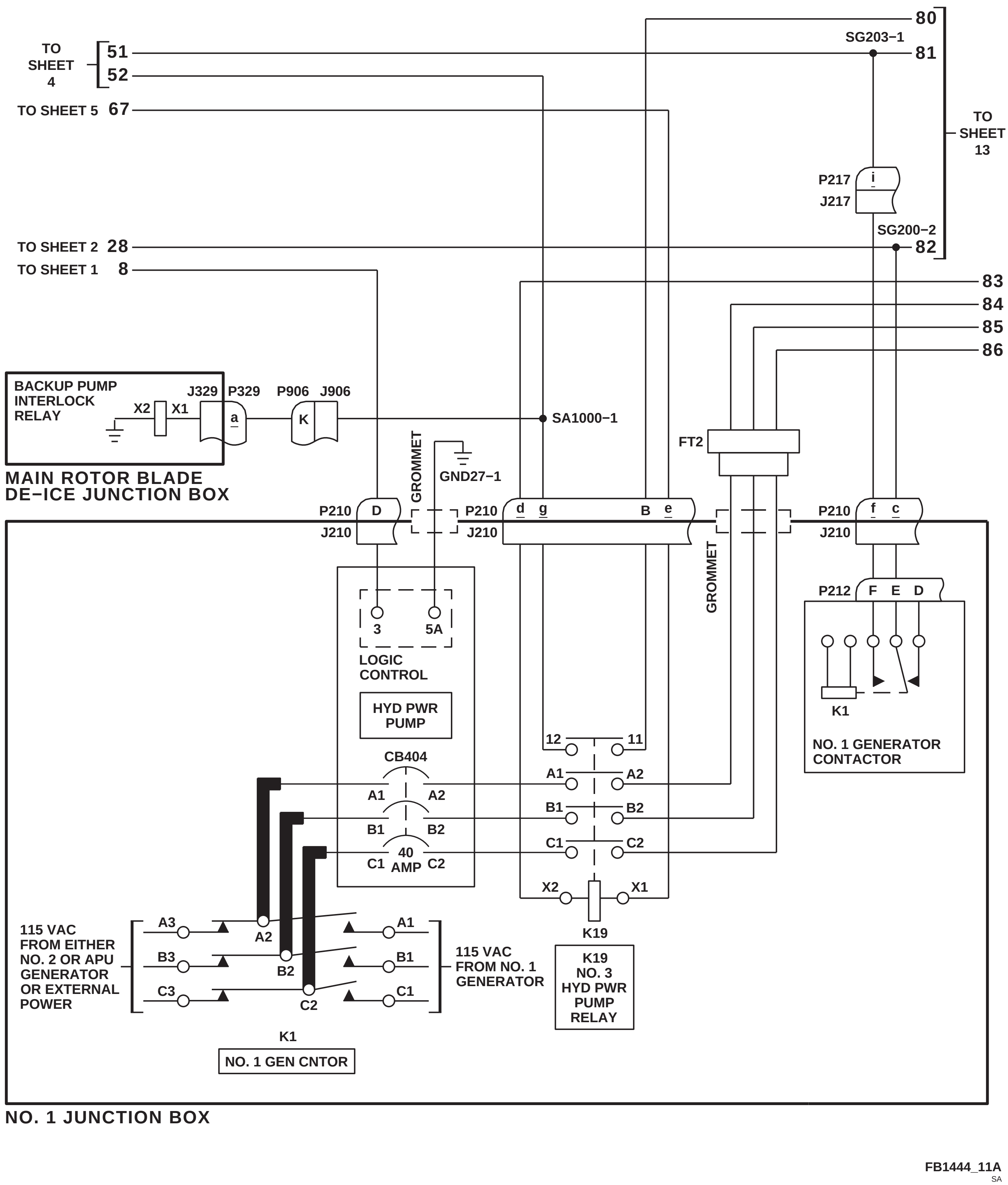


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 11 OF 13)

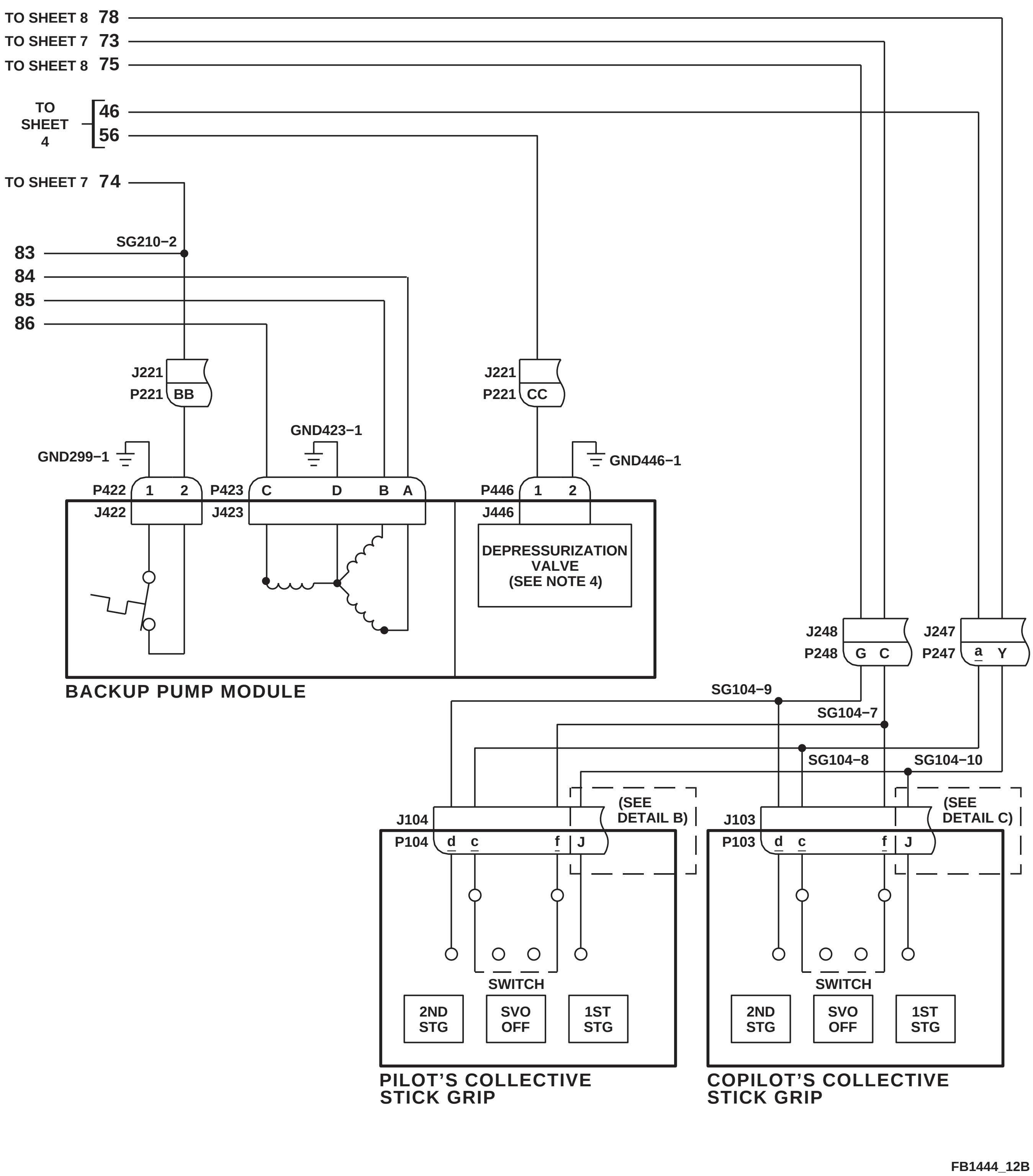


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 12 OF 13)

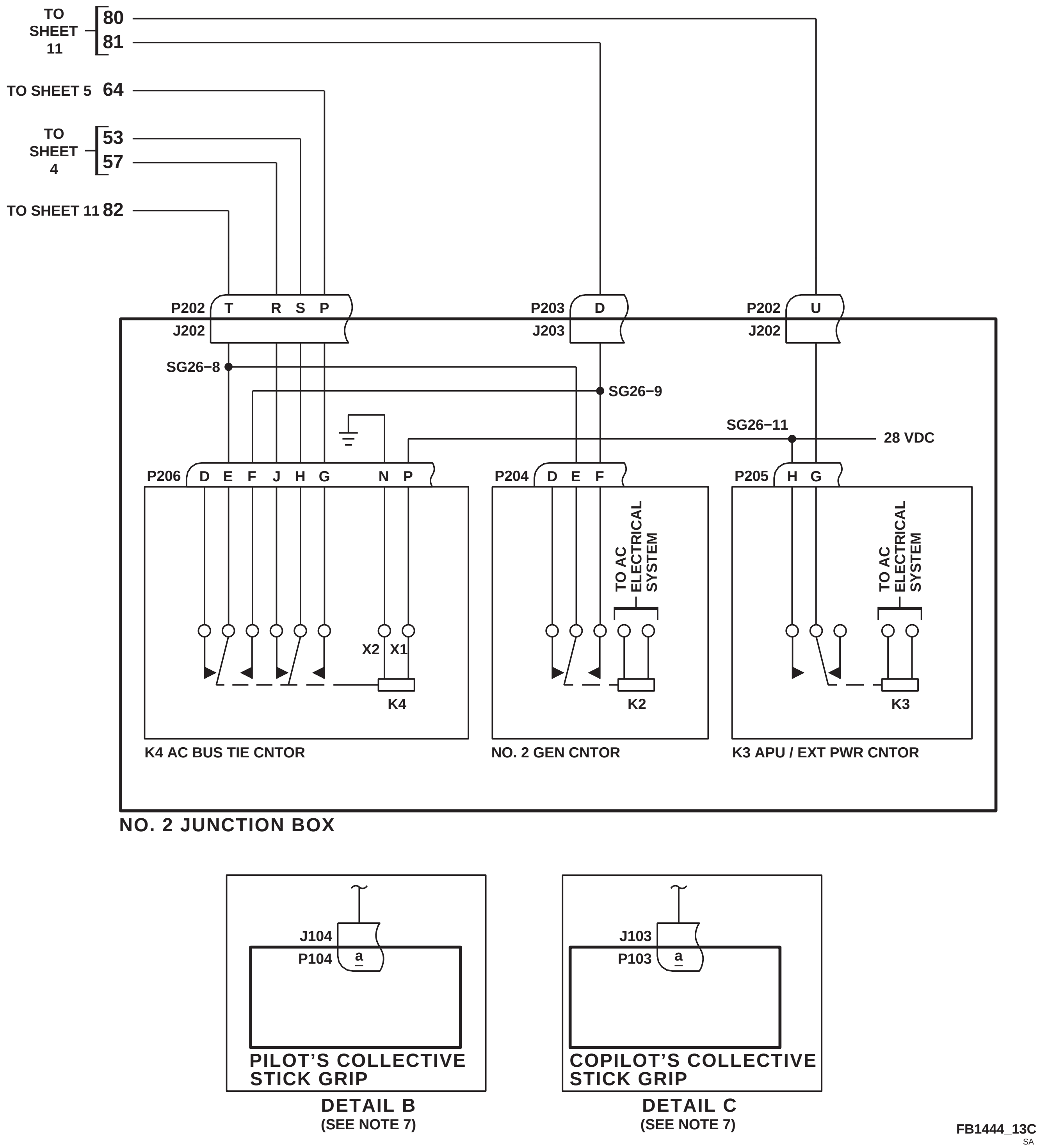


FIGURE 7-2-1-3. HYDRAULIC SYSTEMS SCHEMATIC DIAGRAM (SHEET 13 OF 13)

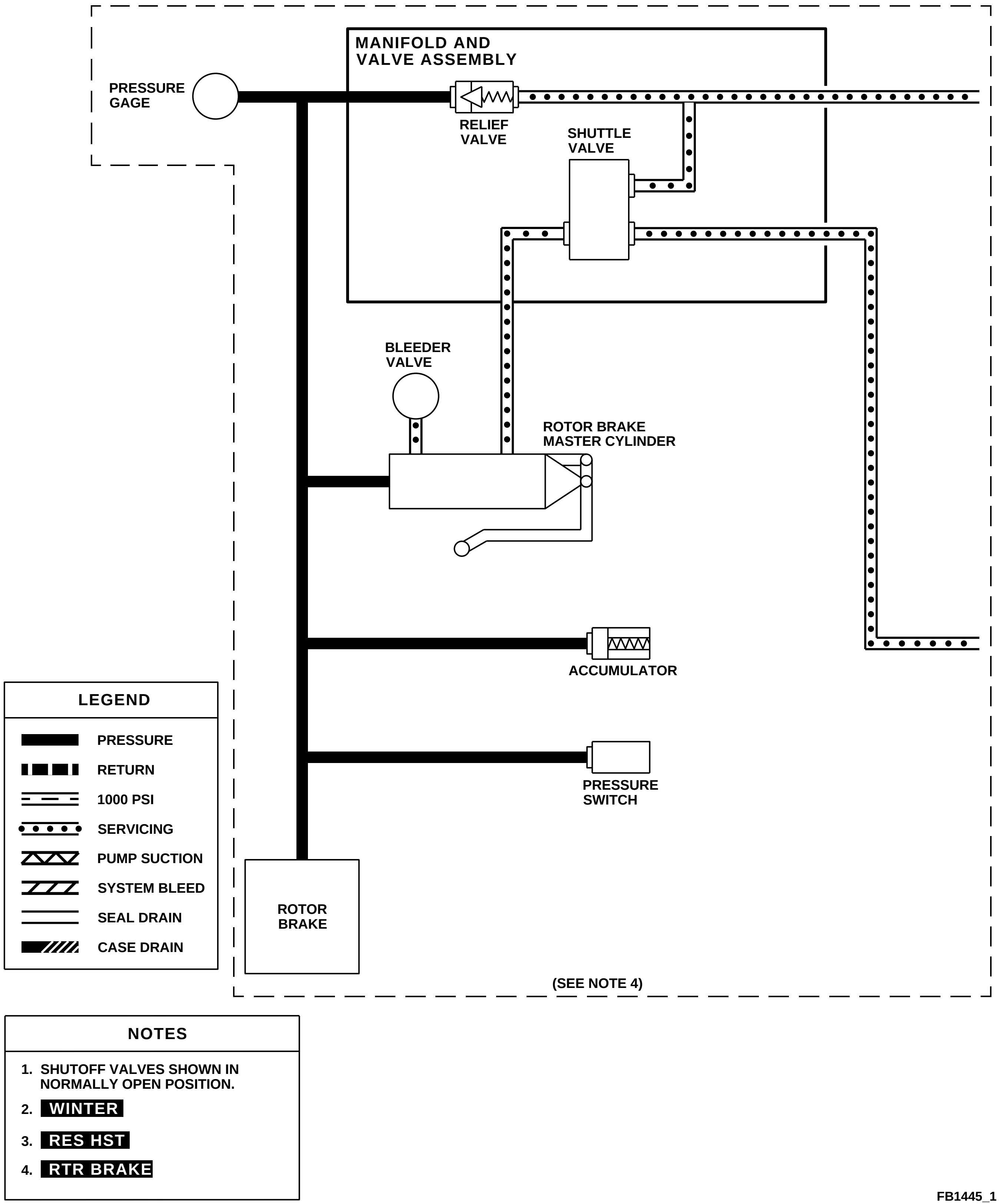


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 1 OF 10)

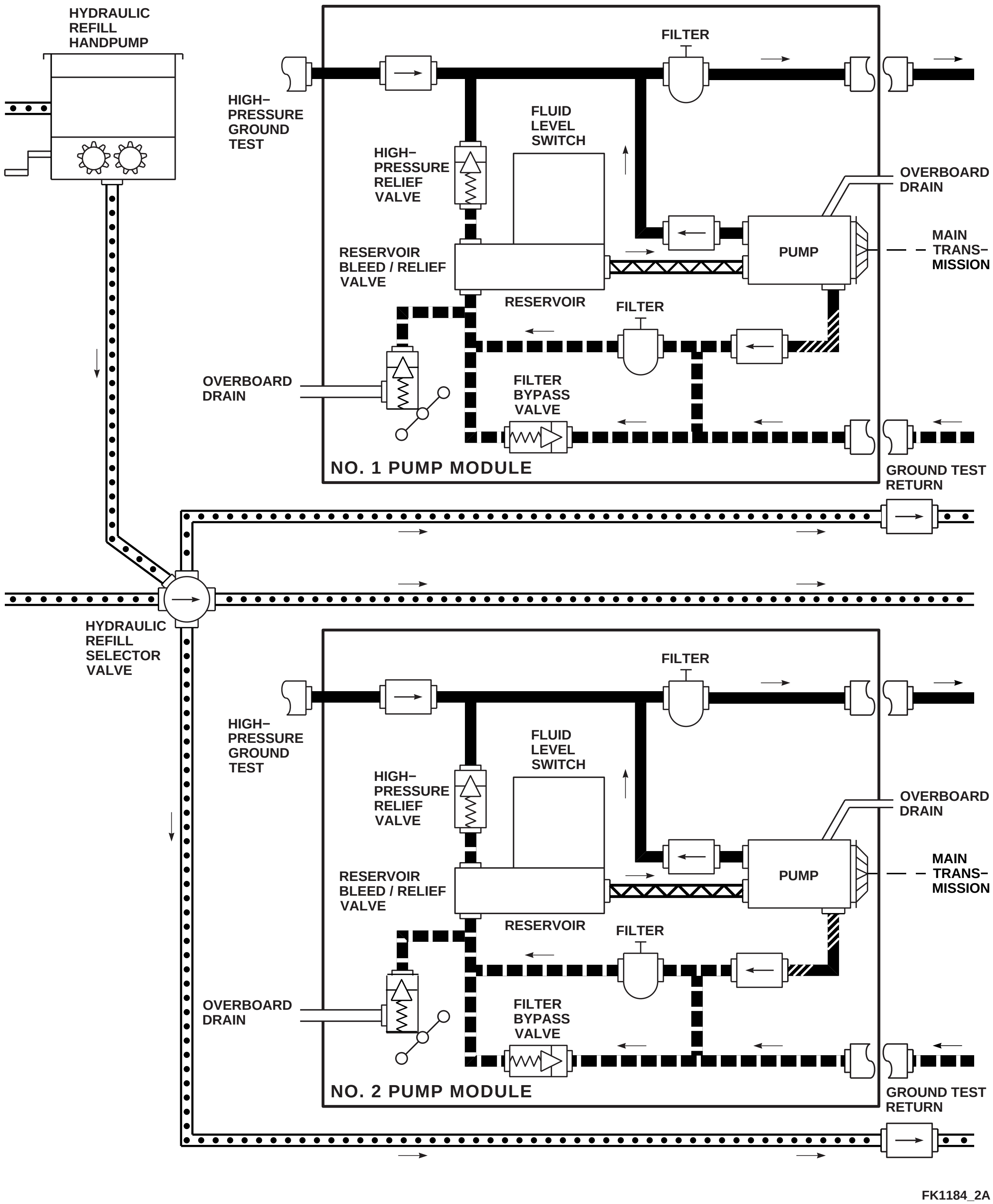


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 2 OF 10)

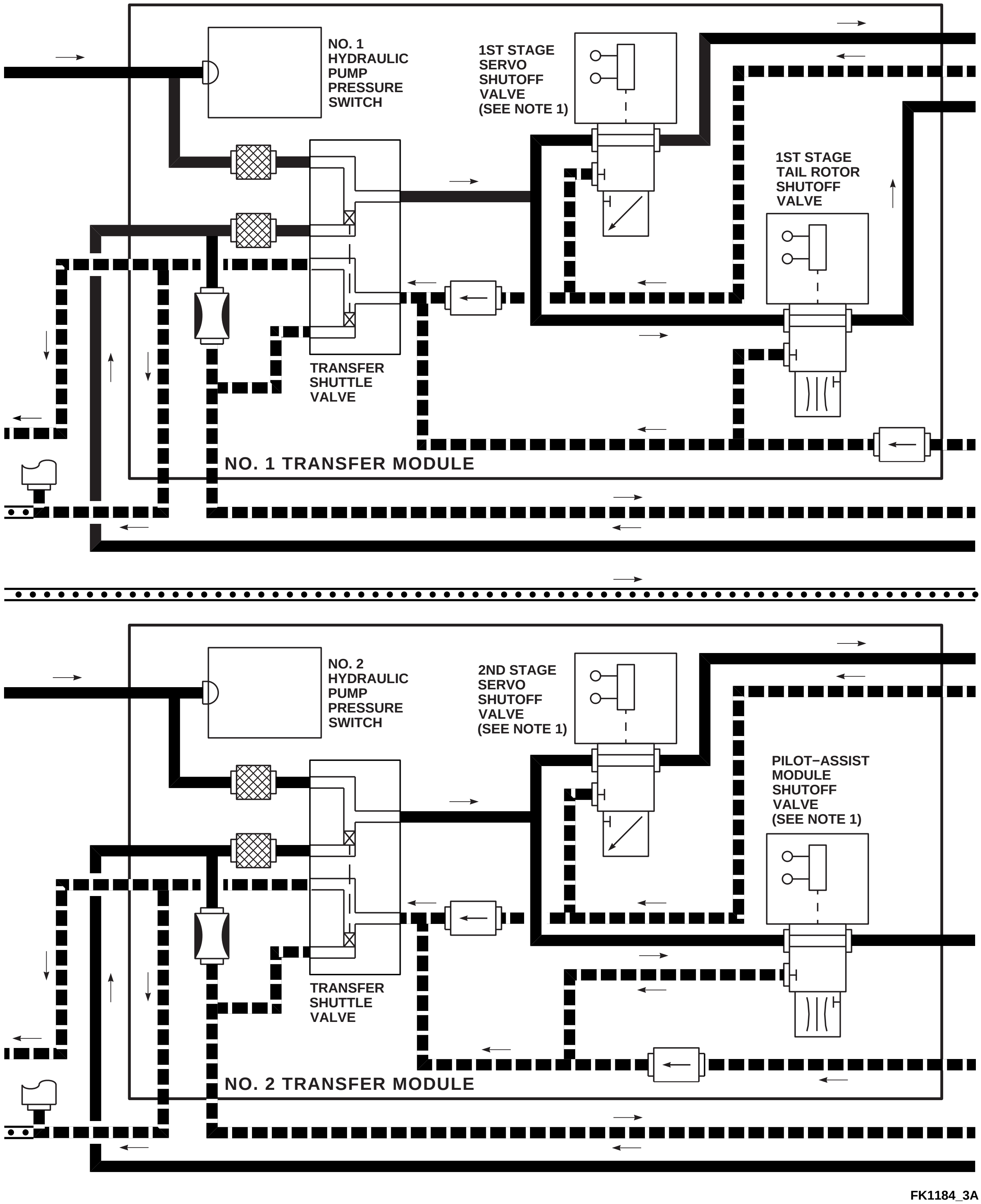


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 3 OF 10)

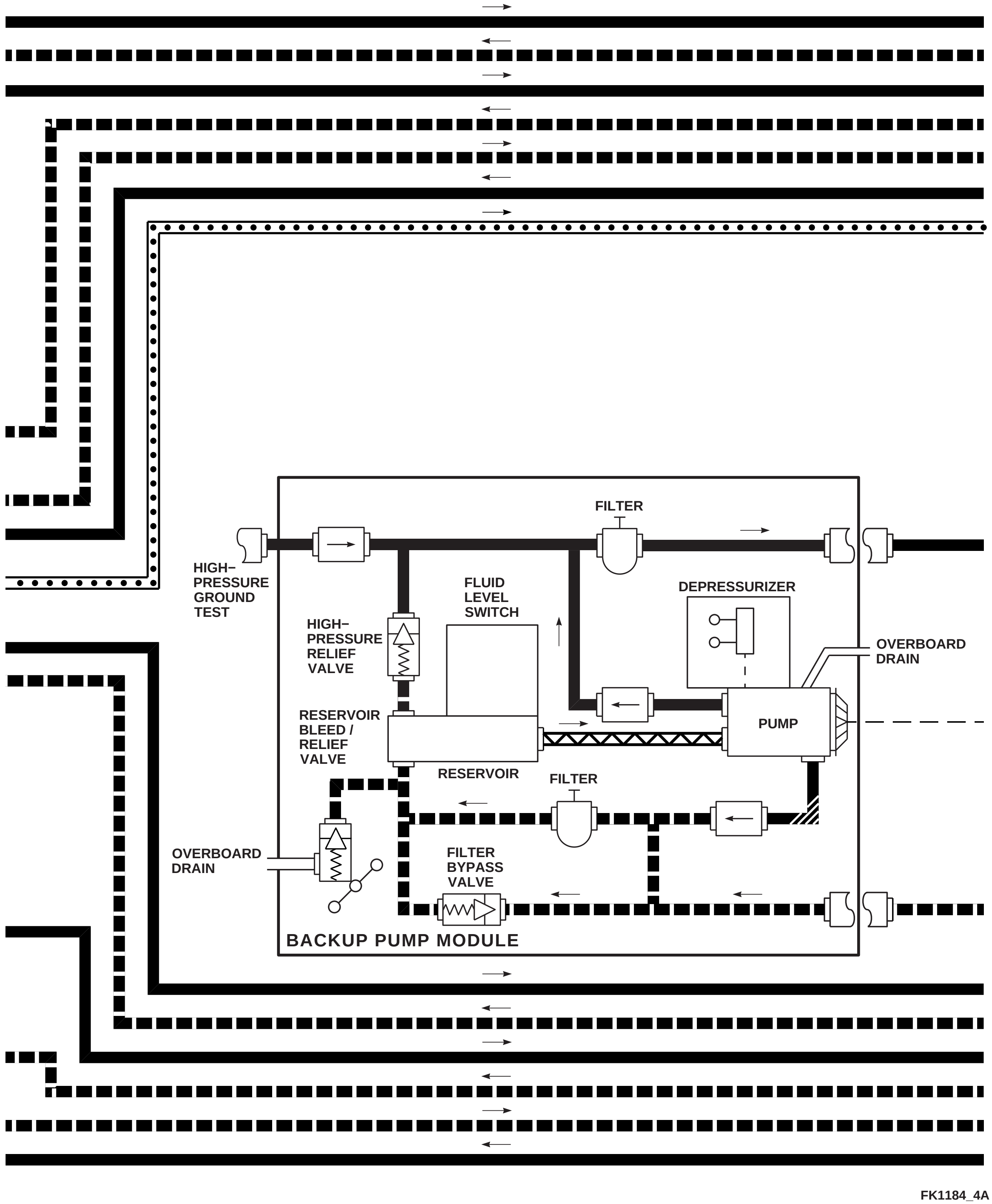


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 4 OF 10)

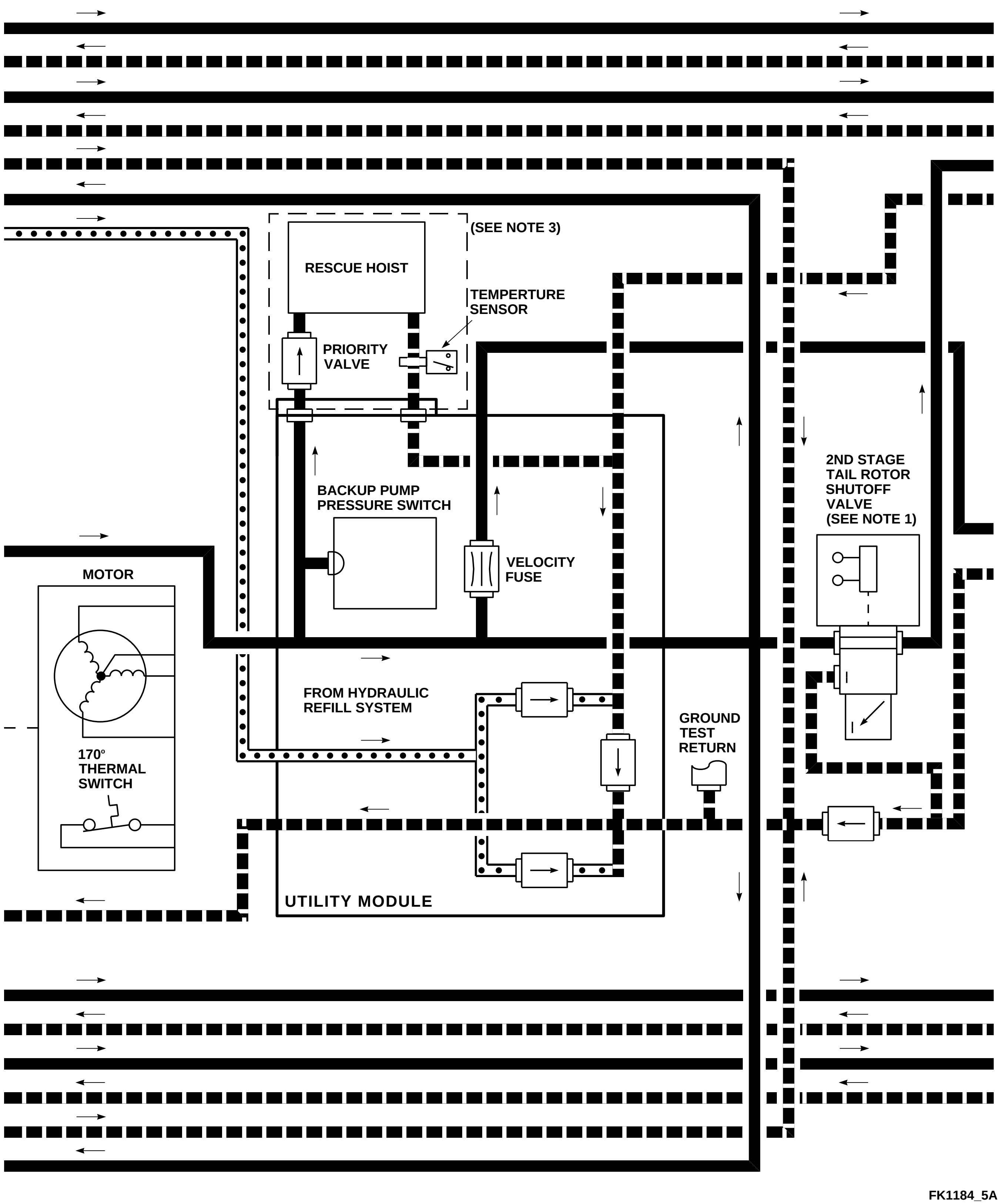


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 5 OF 10)

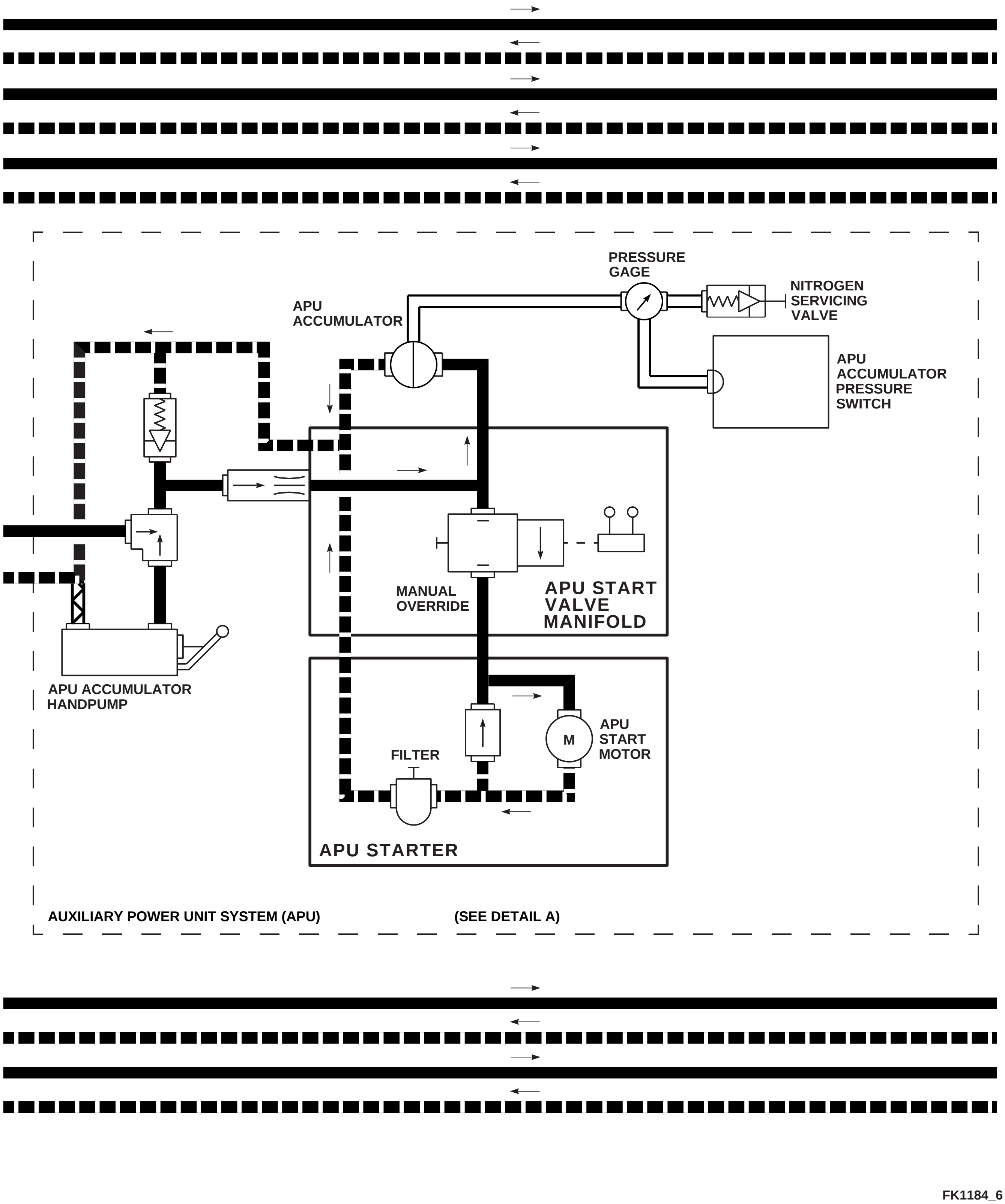


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 6 OF 10)

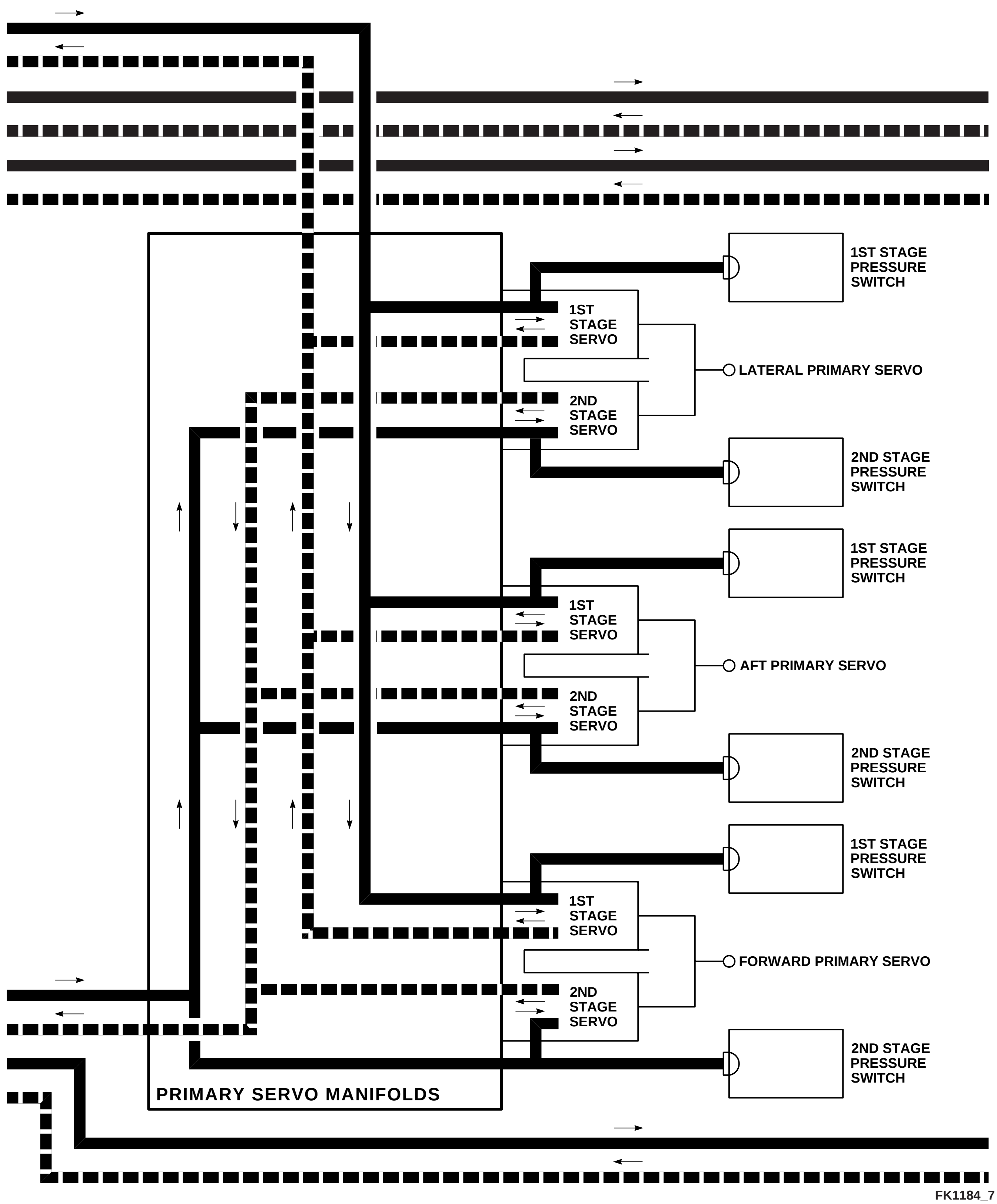


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 7 OF 10)

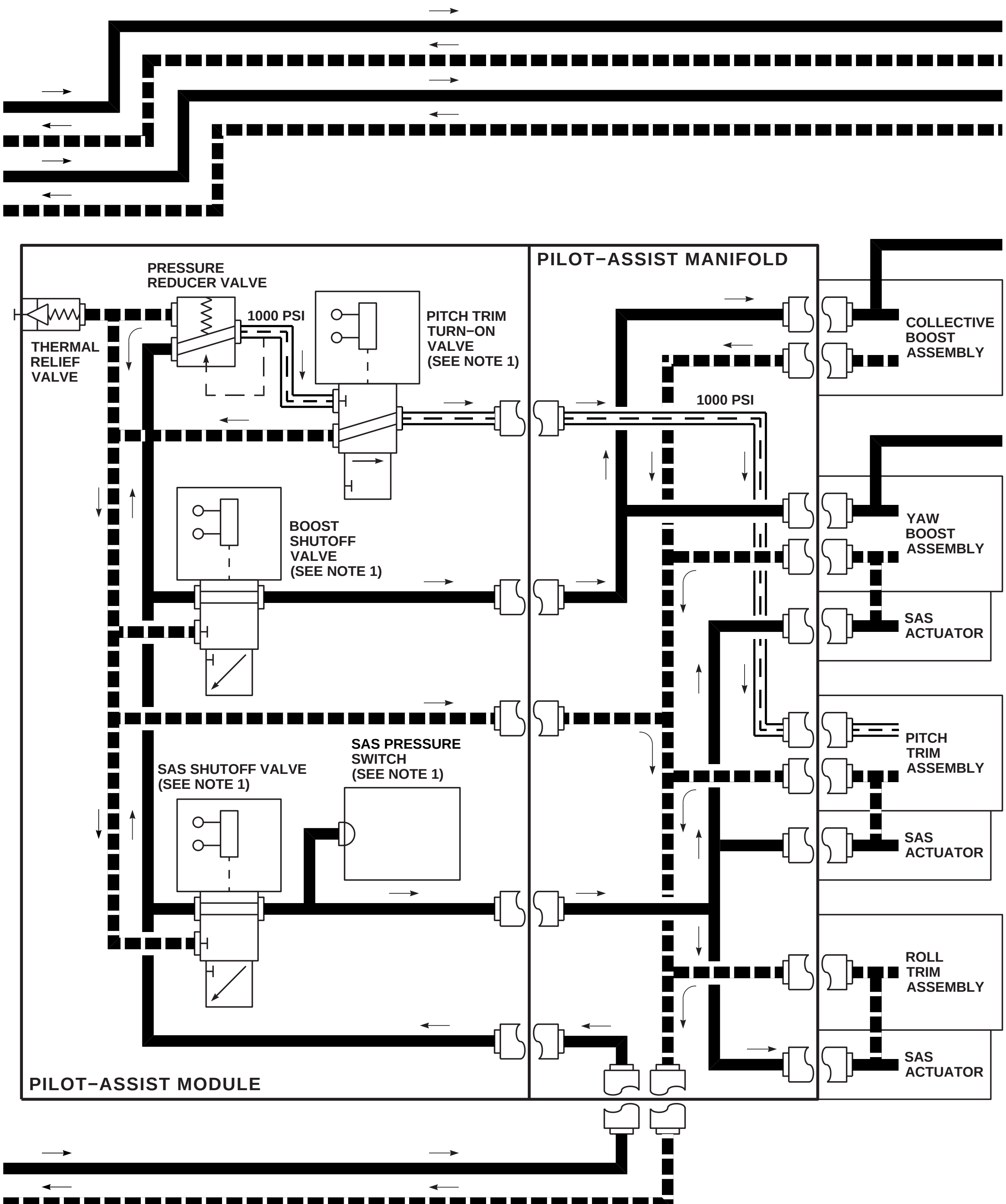


FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 8 OF 10)

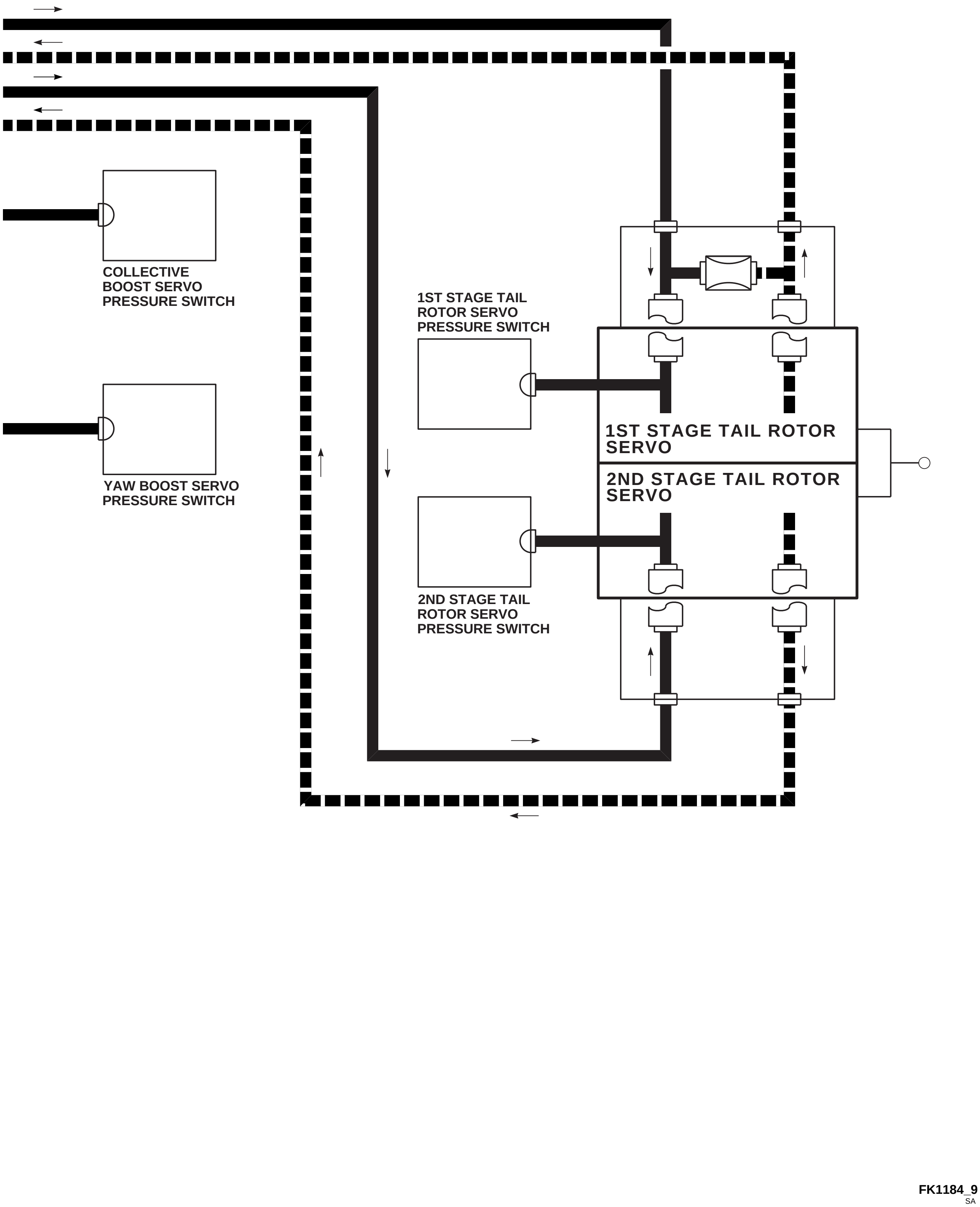
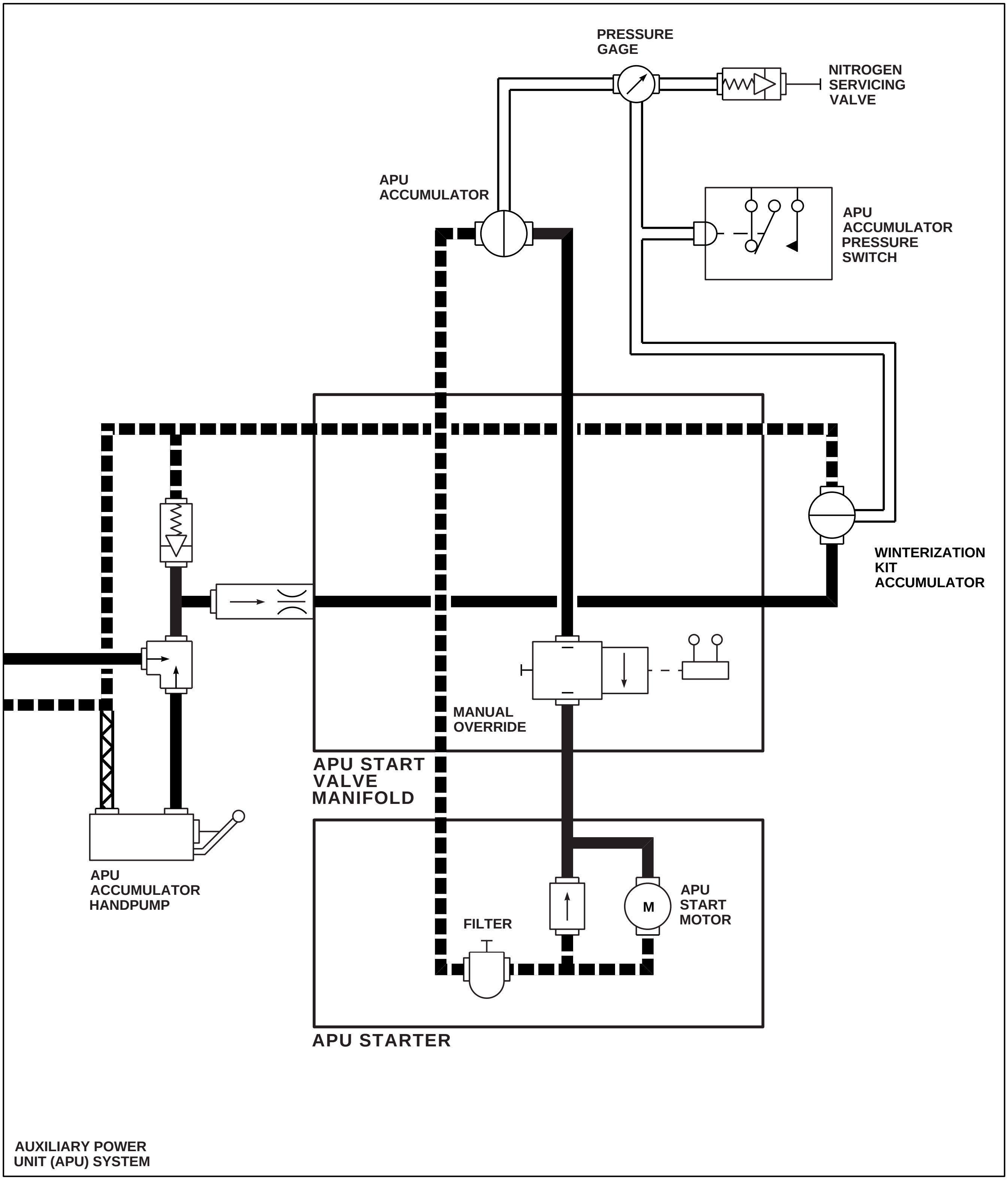


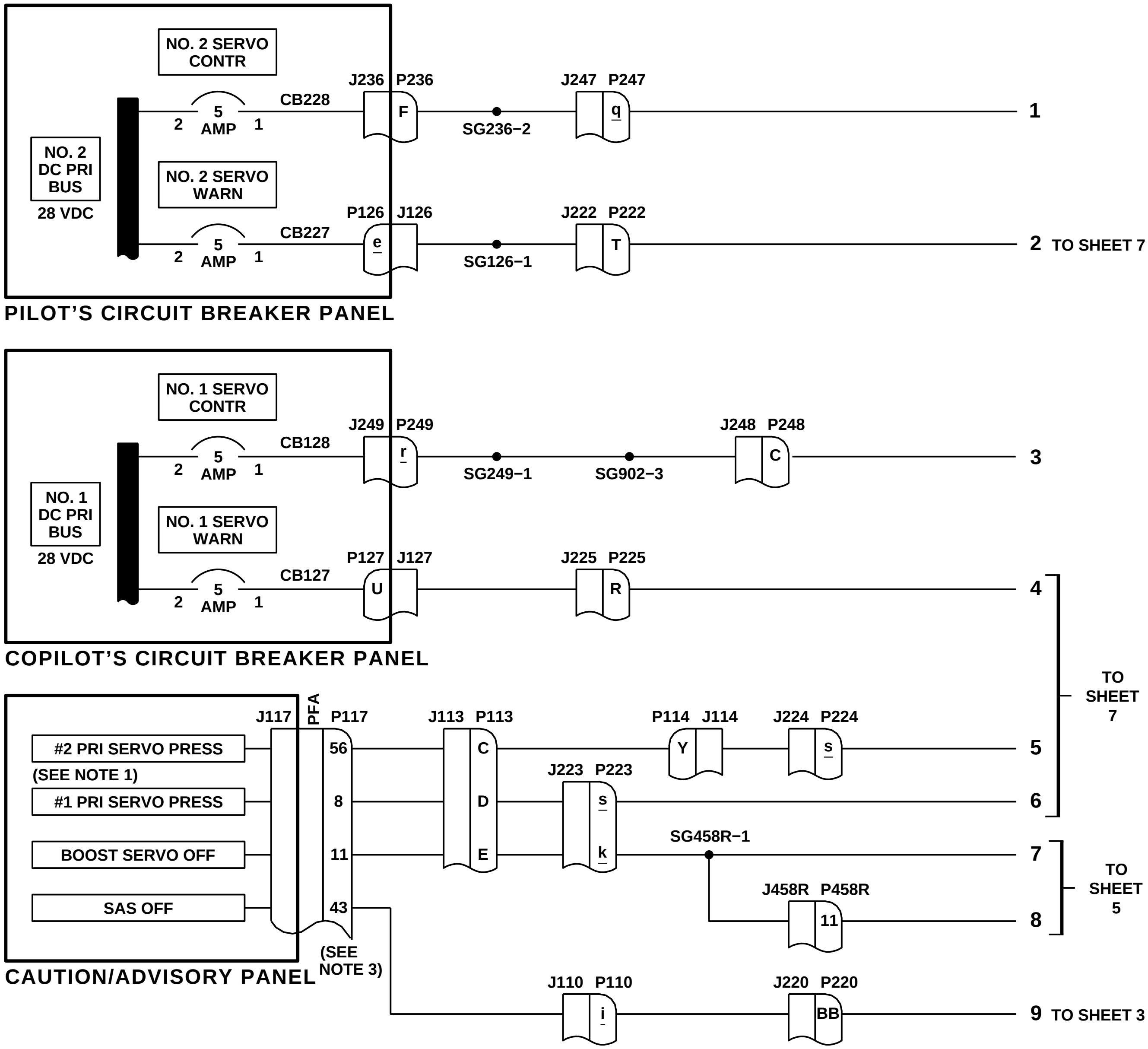
FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 9 OF 10)



DETAIL A
(SEE NOTE 2)

AA2647_8
SA

FIGURE 7-2-1-4. HYDRAULIC SYSTEMS POWER SCHEMATIC (SHEET 10 OF 10)



NOTES
1. #1 AND #2 PRI SERVO CAUTION CAPSULES GO ON WHEN RESPECTIVE HYDRAULIC STAGE IS OFF.
2. **70583−70893** PIN J IS IDENTIFIED AS PIN a ON PILOT'S AND COPILOT'S COLLECTIVE STICK CONNECTORS.
3. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

LEGEND
PRESSURE (1000 PSI)
PRESSURE (3050 PSI)
RETURN
ELECTRICAL
MECHANICAL

FB1446_1B
SA

FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 1 OF 8)

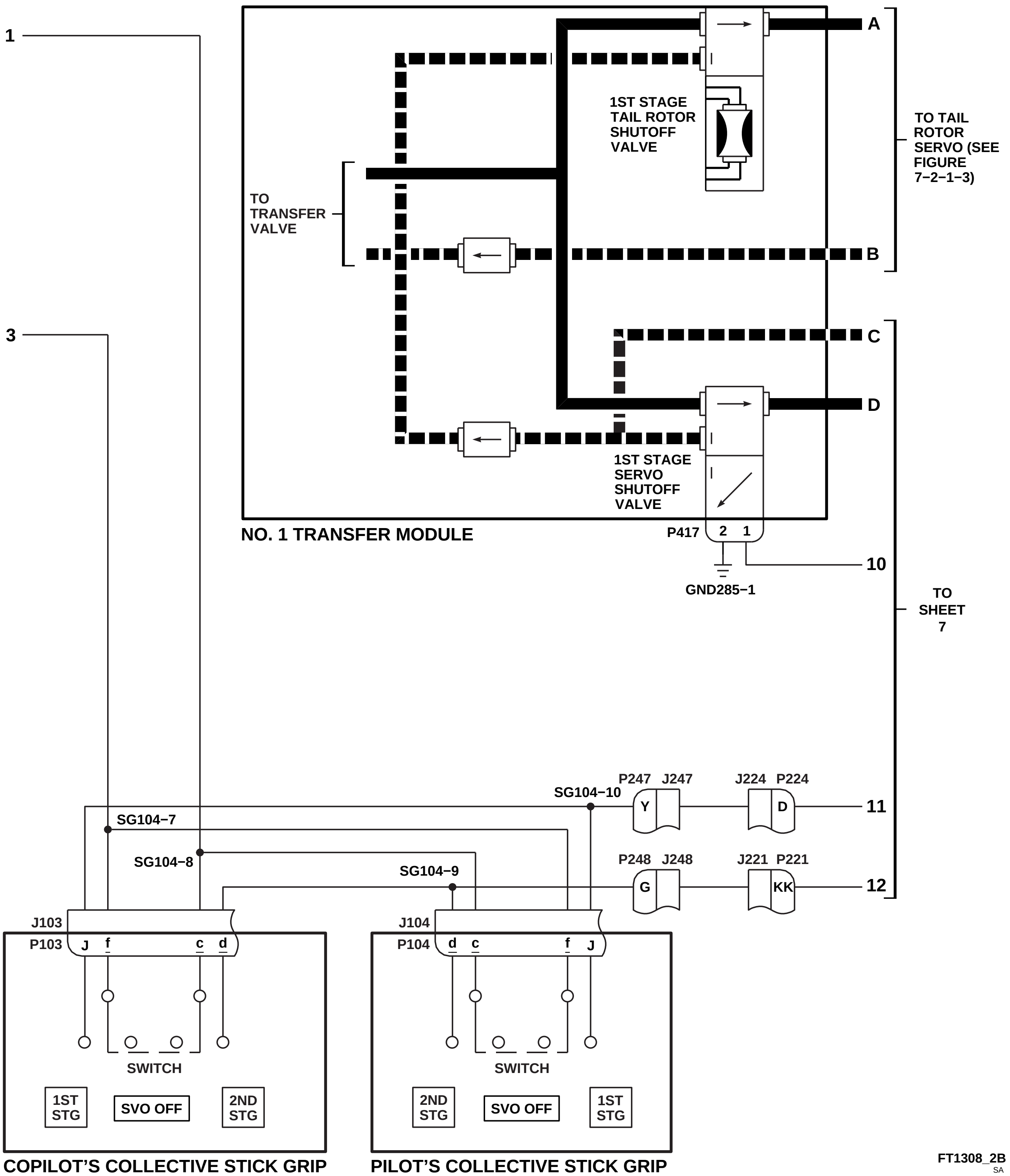


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 2 OF 8)

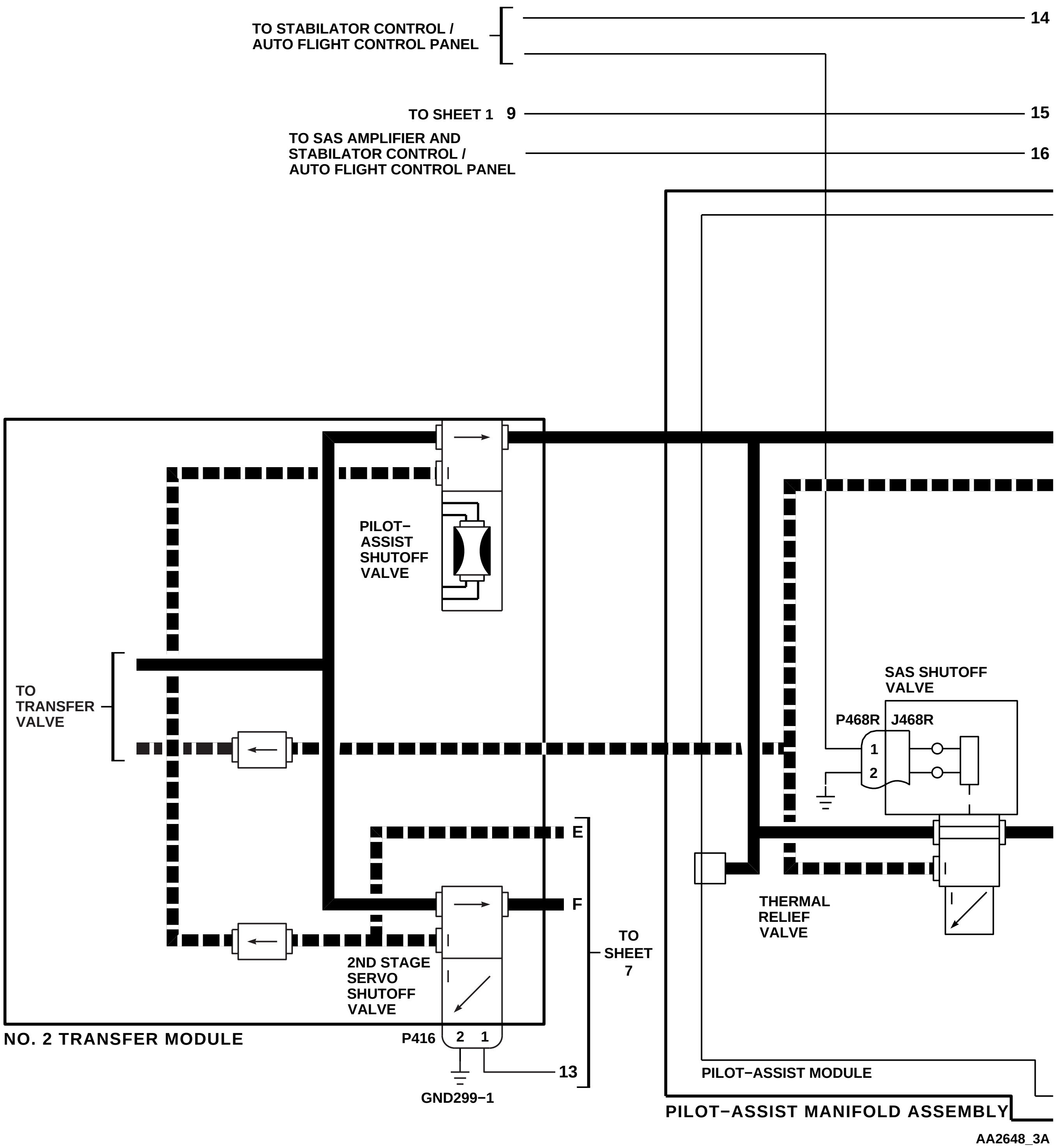


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 3 OF 8)

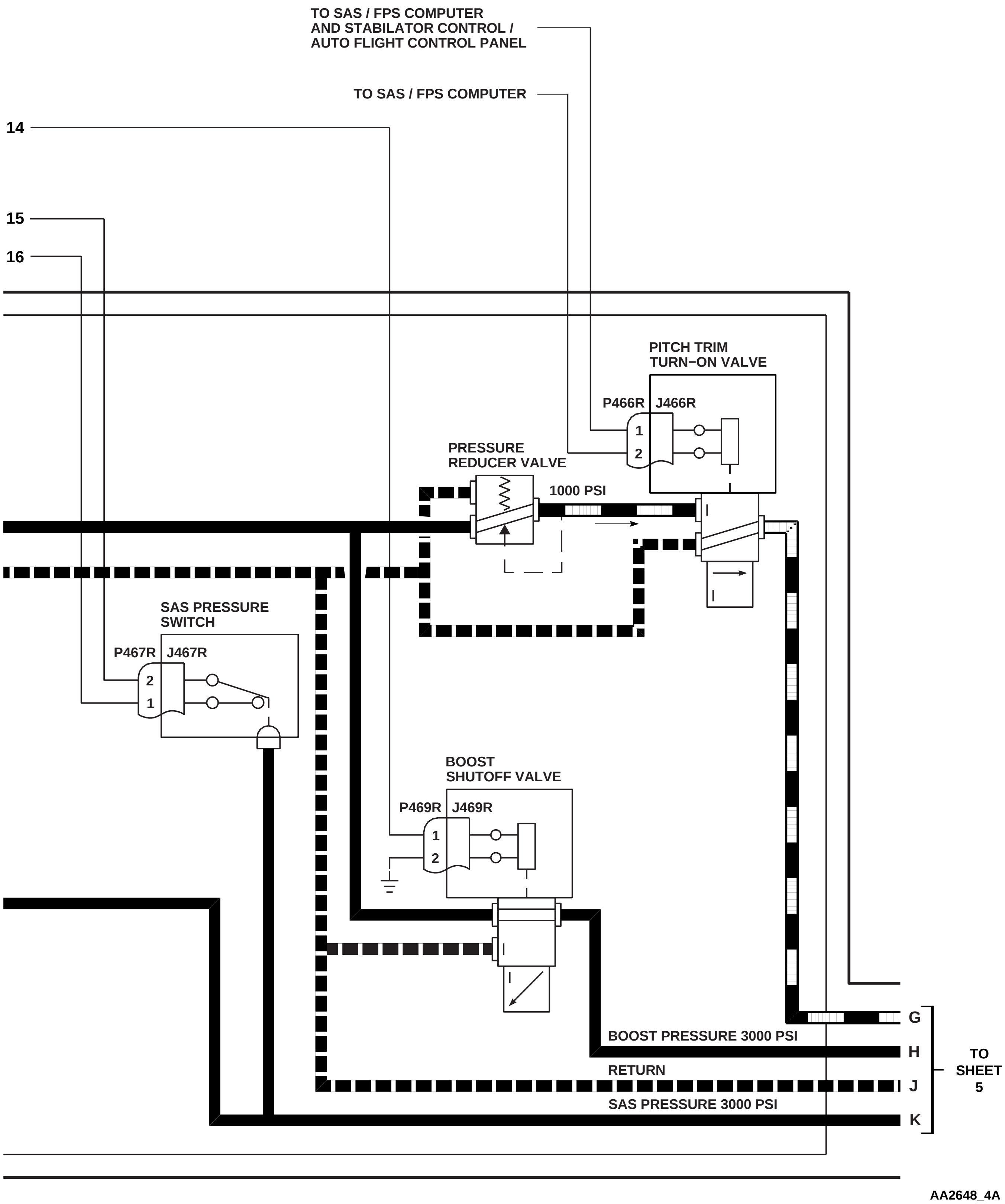


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 4 OF 8)

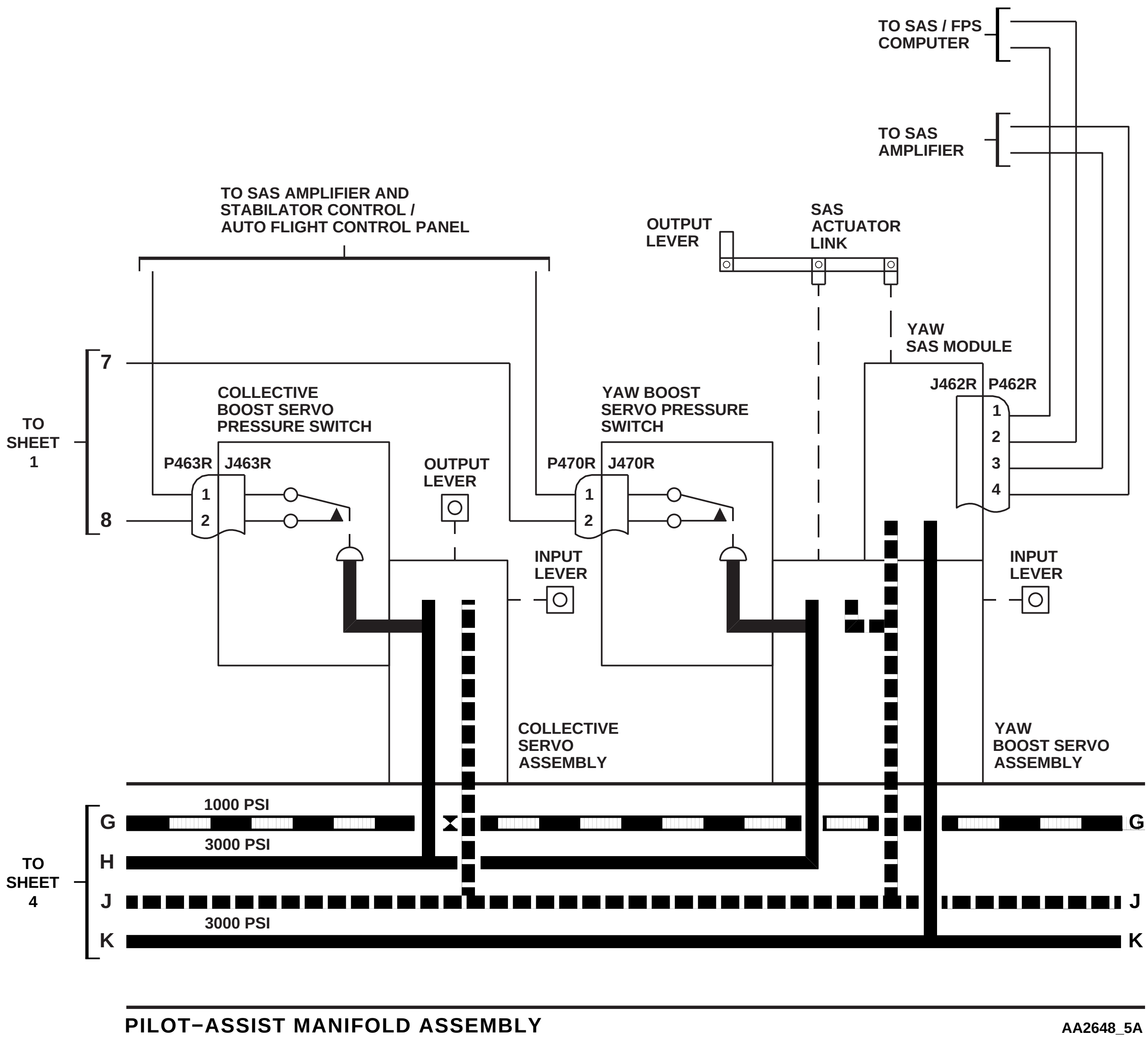


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 5 OF 8)

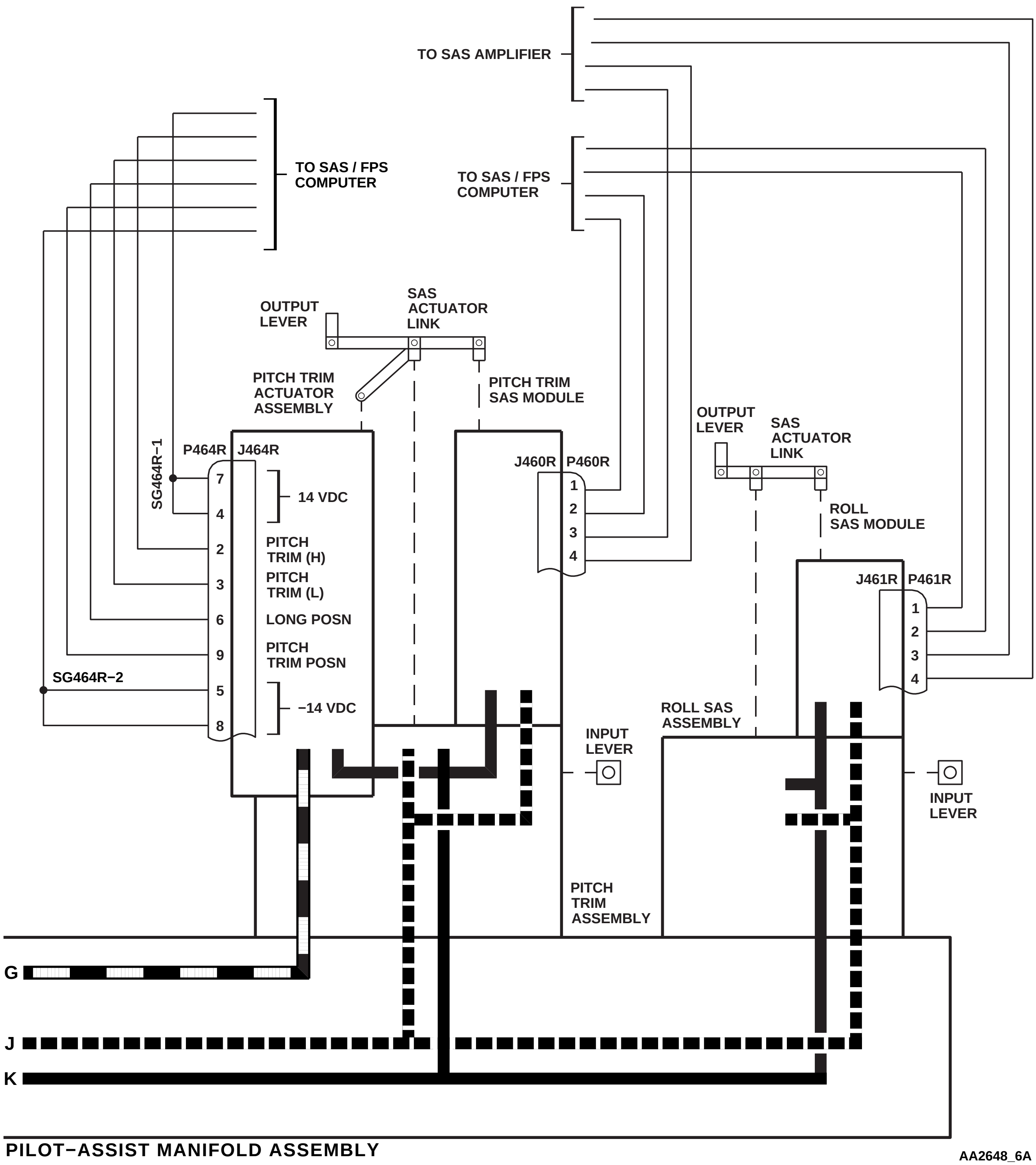


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 6 OF 8)

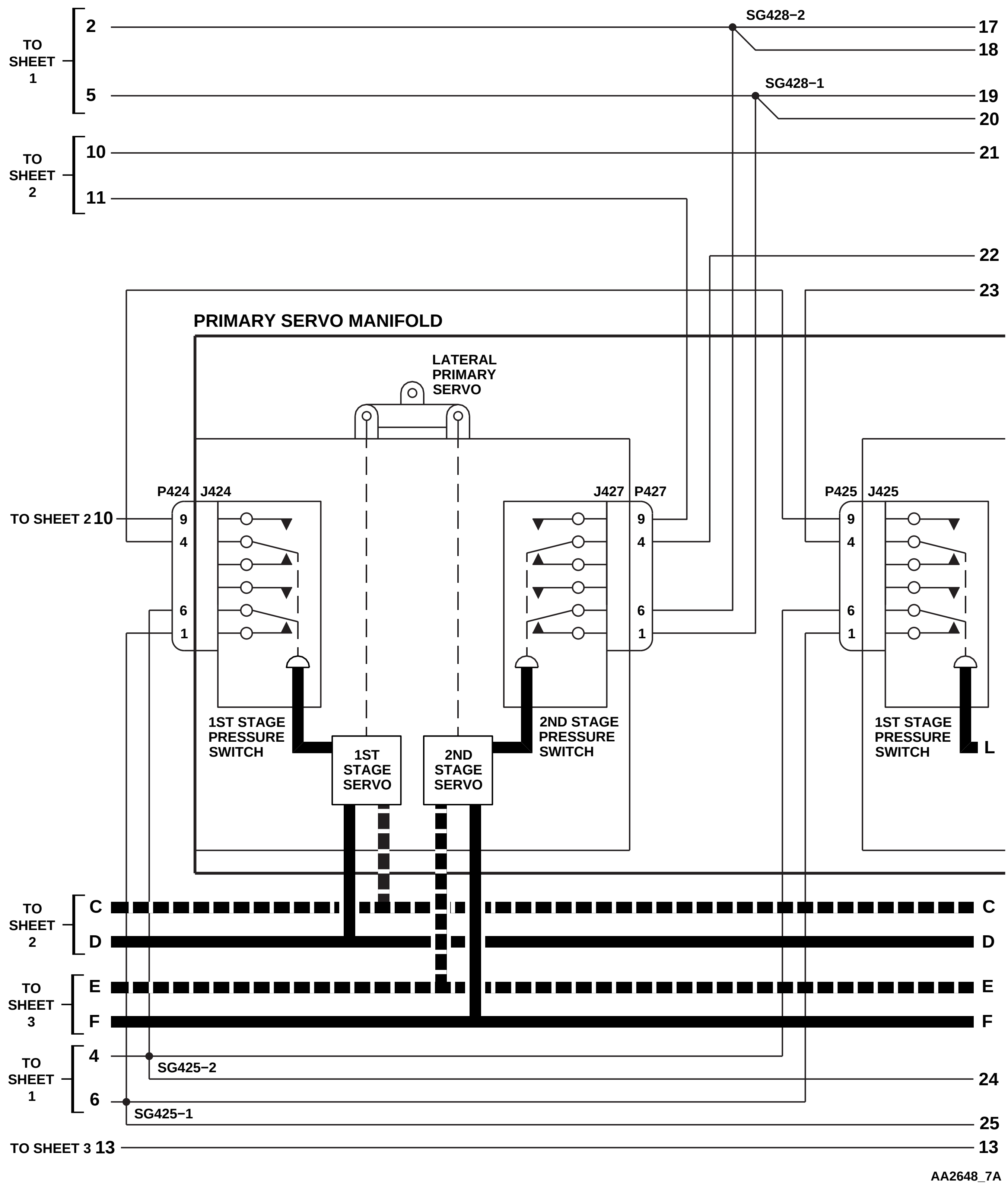


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 7 OF 8)

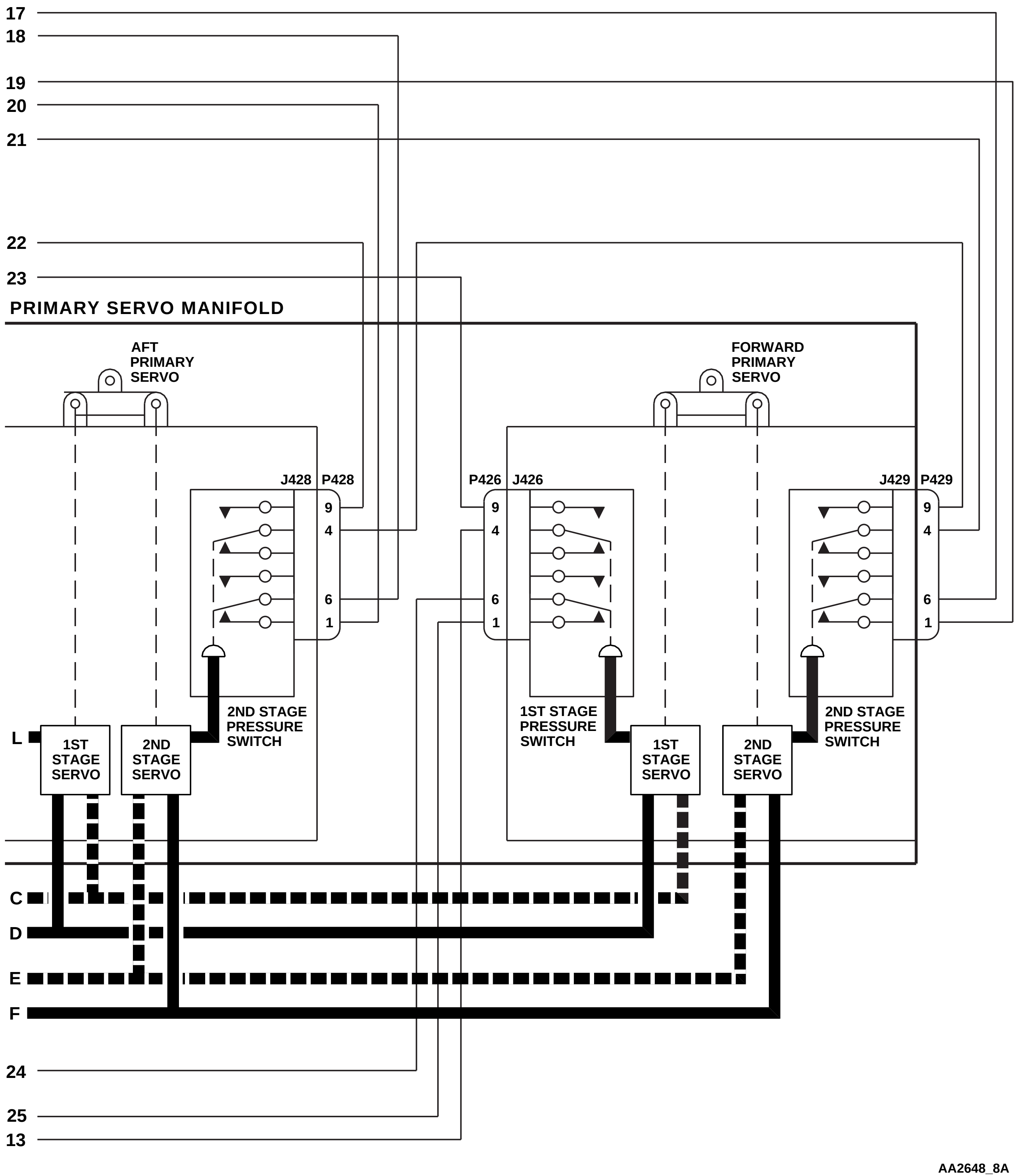


FIGURE 7-2-1-5. HYDRAULIC SYSTEMS PRIMARY AND PILOT ASSIST SERVOS SCHEMATIC DIAGRAM (SHEET 8 OF 8)

SECTION 7-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

7-3-1 HYDRAULIC SYSTEM SEAL LEAKAGE INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Blotter
- Chalk
- Cloth, Clean
- Cloth, White
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-3-8

7-3-1.1 Hydraulic System Seal Leakage Criteria.

NOTE

Leakage may exceed rates given in Table 7-3-1-1 during periods of extremely cold weather. This is normal and is not cause for removal of item from service. Exposure of seals to cold weather causes seals to shrink, allowing fluid to leak past seals. During periods of extremely cold weather, helicopter should be warmed up and controls operated until fluid is warm. After fluid is warmed, leakage rates of Table 7-3-1-1 should be applied.

a. Prior to initiating leakage checks, systems should be activated, components operated a number of times, then any hydraulic fluid should be wiped off components with machinery towel, Item 211, Appendix D.

b. Direct observation: Where fluid dropping from a component may be directly observed, measure leakage as follows:

NOTE

Do not wipe up leakage from surfaces below component(s).

(1) Pressurize and cycle component(s) until a drop falls free.

(2) Continue operating component(s) until another drop falls.

(3) Compare results with leakage rates shown in Table 7-3-1-1.

c. Indirect observation: In some cases, leakage measurement becomes a problem because location of the component does not permit direct observation. Proceed as follows:

(1) If there is a flat surface below component, measure leakage as follows:

(a) Wipe surface clean with machinery towel, Item 211, Appendix D, and place a drop of hydraulic fluid on it.

(b) When drop stabilizes, note size of area by outlining it with chalk and wiping it clean of hydraulic fluid.

(c) Pressurize and cycle suspected component, correlating area of wetted surface with area previously occupied by one drop.

(d) Compare results with leakage rates shown in Table 7-3-1-1.

(2) If there is no flat surface below component, measure leakage as follows:

(a) Locally make a pre-marked panel, using procedure in step c. (1) (a) and step c. (1) (b).

(b) Pressurize and cycle suspected component, correlating area of wetted surface with area previously occupied by one drop.

(c) Compare results with leakage rates shown in Table 7-3-1-1.

(d) Remove pre-marked panel after completion of inspection.

d. For tests requiring long periods of time, and where fluid can drop, measure leakage as follows:

NOTE

Do not use a solvent when wiping up leakage from surfaces below component(s).

(a) Wipe surface clean and dry with machinery towel, Item 211, Appendix D.

(b) Secure or place a clean blotter or white cloth below suspected leak.

(c) Operate system, or allow to remain idle, for required period of time.

(d) Examine blotter or cloth. Compare results with leakage rates shown in Table 7-3-1-1.

e. Hydraulic fluid leakage for components not listed in Table 7-3-1-1 is defined as follows:

(1) Allowable leakage: Fluid leakage is such that quantity lost does not affect component, system, or helicopter operation, and time to complete corrective maintenance is not warranted.

(2) Excessive leakage: Fluid leakage is such that hydraulic reservoir levels may be dangerously lowered or depleted during normal operation, a fire hazard may be created, or airworthiness of helicopter may be otherwise effected.

f. A primary servo leakage rate which exceeds the amount in the Leakage Rate column of Table 7-3-1-1, but is less than or equal to the amount in the Preflight Leakage Rate with Inspection and Servicing column may remain in service, providing that a mandatory preflight inspection and servicing of the three hydraulic pump modules is performed. The preflight inspection and servicing is required whether one or all three primary servos exceed basic leakage rates. If required, service hydraulic pump modules (PARA 1-3-8).

g. If excessive leakage is suspected from a primary servo, perform the following:

(1) Using clean cloth, wipe clean the top deck area and all three primary servos.

(2) Using machinery towel, Item 211, Appendix D, dampened with hydraulic fluid, wipe clean piston rods.

(3) Immediately following next flight, inspect the top deck area with leakage rates shown in Table 7-3-1-1.

h. In order to determine the actual leakage rate of a pitch trim actuator rotary shaft seal or a roll trim actuator rotary shaft seal, perform the following:

(1) Remove protective dust cap from rotary shaft of the pitch/roll trim actuator and clean cap thoroughly.

(2) Wipe off any remaining hydraulic fluid from the rotary shaft seal area of the pitch/roll trim actuator.

(3) Install the protective dust cap on the rotary shaft of the pitch/roll trim actuator.

(4) Fly helicopter on a normal flight.

(5) Immediately following flight, carefully remove protective dust cap to retain any hydraulic fluid which has leaked past rotary shaft seal.

(6) Slowly and carefully allow hydraulic fluid to drip into another container while counting the number of drops.

(7) Refer to Table 7-3-1-1 for allowable leakage rates.

i. Table Table 7-3-1-1 defines allowable static and dynamic hydraulic system seal leakage rates.

Table 7-3-1-1. Hydraulic System Seal Leakage Rates

Component (See Note 1)	Function	Type (See Note 2)	Leakage Rate	Preflight Leakage Rate with Inspection and Servicing
All Servos	End cap	S	None allowed	
	Thermal relief valve	S-D	1 drop per day (However, during shutoff of boost or trim, the relief valve may vent small quantities of fluid. This is normal, provided fluid does not leak continuously)	2 drops per day (primary servos only) (See Note 3)
Primary Servos	Each piston rod seal	D	Post Flight Inspection, a puddle of 2-inches in diameter or less on top deck, or top deck surface wetting of any size with no puddle (See Notes 3 and 4).	1 drop in 5 full cycles (See Note 3)

Table 7-3-1-1. Hydraulic System Seal Leakage Rates (Cont)

Component (See Note 1)	Function	Type (See Note 2)	Leakage Rate	Preflight Leakage Rate with Inspection and Servicing
	Vent hole (beneath bypass valve)	S	2 drops per day	
	Rotary shaft input	D	1 drop in 25 cycles	
		S-D	1 drop per hour	
Collective and Yaw Boost Servos	Each piston rod seal	D	1 drop in 50 full-stroke cycle	
		S-D	1 drop per hour	
	Rotary shaft input	D	1 drop in 25 cycles	
		S-D	1 drop per hour	
Tail Rotor Servo	Each external piston rod seal	D	1 drop in 25 full-stroke cycles	
		S-D	1 drop per hour	
Pitch Trim Assembly	Each piston rod seal	D	1 drop in 25 cycles	
		S-D	1 drop per hour	
	Actuator rotary shaft seal (spring as-sembly) (See Note 5)	D	3 drops for flights up to 1 ½ hours, or 7 drops for flights of 1 ½ to 3 hours, or 11 drops for flights of 3 to 4 ½ hours	

Table 7-3-1-1. Hydraulic System Seal Leakage Rates (Cont)

Component (See Note 1)	Function	Type (See Note 2)	Leakage Rate	Preflight Leakage Rate with Inspection and Servicing
Roll Trim Assembly	Actuator rotary shaft seal (spring assembly) (See Note 5)	D	3 drops for flights up to 1 ½ hours, or 7 drops for flights of 1 ½ to 3 hours, or 11 drops for flights of 3 to 4 ½ hours	
Pilot-Assist Module	Thermal relief valve	S-D	1 drop per day (However, during shutoff of boost or trim, the relief valve may vent small quantities of fluid. This is normal, provided fluid does not leak continuously)	
SAS Actuator	Each external piston rod seal	S-D	1 drop per day	
Pressure Switches	Each vent hole	S-D	1 drop per day	
Hydraulic Pump Modules & Hydraulic Pump Motors	Shaft seal drain	D	4 drops per minute	
		S-D	1 drop per hour	
	Housing (mating surfaces)	S	2 drops per day	
	Reservoir vent	D	1 drop in 5 cycles	
		S-D	2 drops per hour	
	Reservoir bleed relief valve	S	1 drop per hour	

Table 7-3-1-1. Hydraulic System Seal Leakage Rates (Cont)

Component (See Note 1)	Function	Type (See Note 2)	Leakage Rate	Preflight Leakage Rate with Inspection and Servicing
Disconnects	Pump quick-disconnects	S	1 drop per day	2 drops per day 1 drop per flight hour
	¼-inch SS-65 couplings	S S-D	1 drop per day 1 drop in 5 flight hours	
	½-inch SS-65 couplings	S S-D	1 drop per day 1 drop in 3 flight hours	
Solenoid Valves	Valve interface with manifold	S	1 drop per day	
Fittings	Flareless	S	None allowed	
	Permaswaged	S	None allowed	
APU Start Valve	Manual input stem	S-D	1 drop per day	
	Transfer tubes	S	1 drop per day	
APU Accumulator	Reservoir leak-age	S S-D	1 drop per day 1 drop in 5 cycles	
	Cylinder seal	S	None allowed	
APU Starter Motor	Shaft seal drain	S-D	2 drops per hour	
APU Accumulator Handpump	Pump seal	S-D	2 drops per day	
Shock Struts	Shock strut seal	S	2 drops per day	
	Strut internal seals	S	Max 200 ml leakage when servicing	
Ground Cart Test Connections	Quick-Disconnects	S	1 drop in 6 minutes	

Table 7-3-1-1. Hydraulic System Seal Leakage Rates (Cont)

Component (See Note 1)	Function	Type (See Note 2)	Leakage Rate	Preflight Leakage Rate with Inspection and Servicing
Rescue Hoist	Fittings	S	None allowed	
	Fittings	D	None allowed	
	Pump & Motor Shaft Seals	D	4 drops per minute	
	Pump & Motor Shaft Seals	S-D	1 drop per hour	
	Motor	D	25 ml in 15 seconds	
	Proof Pressure Test		None allowed (No external leakage from components, manifolds, tubes, or fit-tings at elevated testing pressures)	

NOTES

1. Refer to PARA 7-3-1.1, step e.
2. Codes:
   D = Dynamic
   S = Static
   S-D = Static leakage through a dynamic seal
3. Refer to PARA 7-3-1.1, step f.
4. Refer to PARA 7-3-1.1, step g.
5. Refer to PARA 7-3-1.1, step h.
6. 20 drops = 1 ml

7-3-2 PNEUMATIC SYSTEM INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 7-4-5
- PARA 7-4-6
- PARA 7-4-7
- PARA 7-4-8
- PARA 7-4-60

7-3-2.1 Inspect Pneumatic System Tubes.

NOTE

Inspection of pneumatic system tubes in transition, oil cooler, APU, and engine areas is the same.

a. Inspect tubes for dents. **DENTS UP TO 25% OF TUBE DIAMETER ARE ACCEPTABLE.** If dents exceed limit, replace tube (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, or PARA 7-4-8).

b. Inspect tubes for nicks and scratches. **NONE ALLOWED.** If nicks or scratches are found, replace tubes (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, or PARA 7-4-8).

c. Inspect all welds on tubes for cracks. **NONE ALLOWED.** If cracks are found, replace tubes (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, or PARA 7-4-8).

d. Inspect rubber couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** If coupling is too stiff to slide over tube, or if chafed or torn, replace rubber couplings (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, or PARA 7-4-8).

e. Inspect pneumatic tubes, check valves, and connections for signs of leaking or damage. **NONE ALLOWED.** If leaking or damaged components are found, replace components (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, PARA 7-4-8, or PARA 7-4-60).

7-3-3 HYDRAULIC PUMP MODULE INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Identification Marker, Item 156, Appendix D
- Temp Indicator Card, Item 323, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 7-3-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-14
- PARA 7-4-21
- PARA 7-4-51
- PARA 7-4-65
- PARA 7-5-4

7-3-3.1 Inspect Hydraulic Pump Modules.

NOTE

Inspection of No. 1, No. 2, and backup hydraulic pump modules is the same.

a. Check hydraulic pump modules for leaks (PARA 7-3-1).

b. Check temperature indicators for condition and security. Make sure indicators are sealed against hydraulic fluid, oils, and solvents. If indicator is loose or damaged, replace indicator (PARA 7-4-21).

c. Check temperature indicators on both sides of hydraulic pump module. If all dots are black, up to and including 270°F (132°C) on identification marker, Item 156, Appendix D, or 275°F (135°C) on temp indicator card, Item 323, Appendix D, perform the following:

(1) Inspect bleed valve drain line on hydraulic pump module. If drain line shows signs of melting, replace drain line and hydraulic pump module (PARA 7-4-12, PARA 7-4-13, and PARA 7-4-51).

(2) Drain a small amount of hydraulic fluid from hydraulic pump module/relief valve. Check hydraulic fluid for discoloration. If hydraulic fluid is red in color, service hydraulic pump module as required (PARA 1-3-8). If hydraulic fluid is brownish, drain entire reservoir. Flush hydraulic fluid from affected system and service (PARA 1-3-8 and PARA 7-4-65).

(3) Check hydraulic system components for leakage (PARA 7-3-1). If leakage is excessive, replace seals or components as required.

d. Check the red indicator buttons located on each filter of the three hydraulic pump modules. If an indicator button is extended, change the hydraulic pump module filter element below the extended indicator button (PARA 7-4-14).

e. Check closure for condition and security. If damaged or loose, replace closure (PARA 7-5-4).

SECTION 7-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

7-4-1 ENGINE STARTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Braid, Textile, Item 81, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-7
- PARA 1-6-16
- PARA 1-7-20
- PARA 4-2-8
- PARA 7-4-8

7-4-1.1 Replace Engine Starter.

NOTE

- No. 1 engine starter is on inner side near firewall. No. 2 engine starter is on outer side.
- Replacement of No. 1 and No. 2 engine starter is the same, except as noted.

7-4-1.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open engine cowling.

c. Disconnect electrical connector, P807 (No. 1 engine), or P806 (No. 2 engine), from engine starter electrical connector (Figure 7-4-1-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove upper pneumatic tube from engine starter and firewall (PARA 7-4-8).

f. Loosen engine starter mounting nuts only enough to be able to turn engine starter, and remove engine starter from engine.

g. Remove packings from grooves on engine starter and engine starter splined shaft. Discard packings.

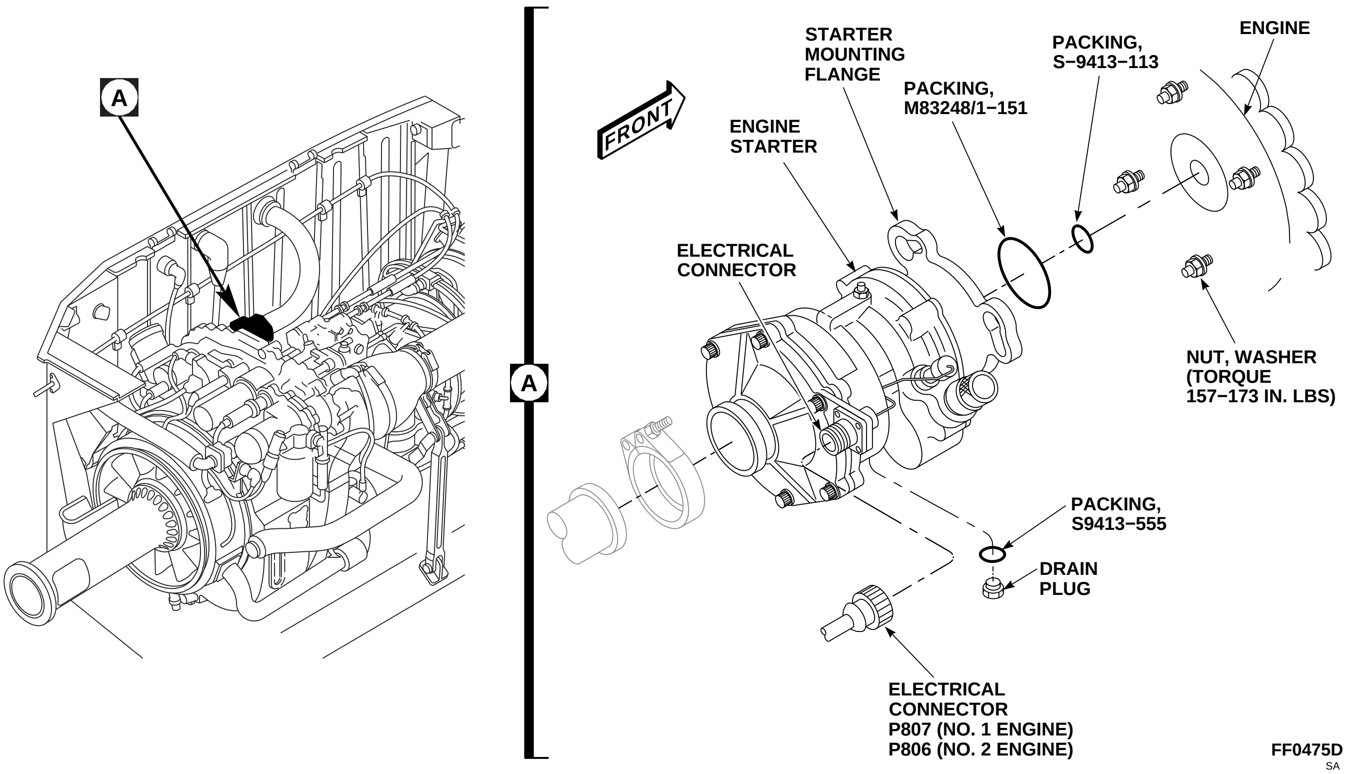


FIGURE 7-4-1-1. REPLACE ENGINE STARTER

7-4-1.1.2 Remove Inlet Housing.

a. If necessary, remove engine starter (PARA 7-4-1.1.1).

b. Position engine starter in a secure upright position.

CAUTION

Damage to engine starter will occur if interior parts are disturbed. Do not remove or allow internal parts of engine starter to be disturbed.

c. Remove screw and nut from electrical connector bracket (Figure 7-4-1-2).

NOTE

Inlet housing, 3501183-1, does not have packing installed.

d. Remove bolts, washers, inlet housing, and electrical connector bracket from engine starter. If removing inlet housing, 3501183-2, remove and discard packing.

7-4-1.1.3 Install Inlet Housing.

NOTE

Inlet housing, 3501183-1, does not require packing.

a. On inlet housing, 3501183-2, lubricate packing with petrolatum, Item 234, Appendix D, and install in machined groove in inlet housing (Figure 7-4-1-2).

b. Position inlet housing on engine starter.

c. Position electrical connector bracket on inlet housing.

NOTE

Electrical connector bracket is secured to inlet housing with bolt and washer, and screw and nut.

d. Secure inlet housing and electrical connector bracket to engine starter with bolts and wash-

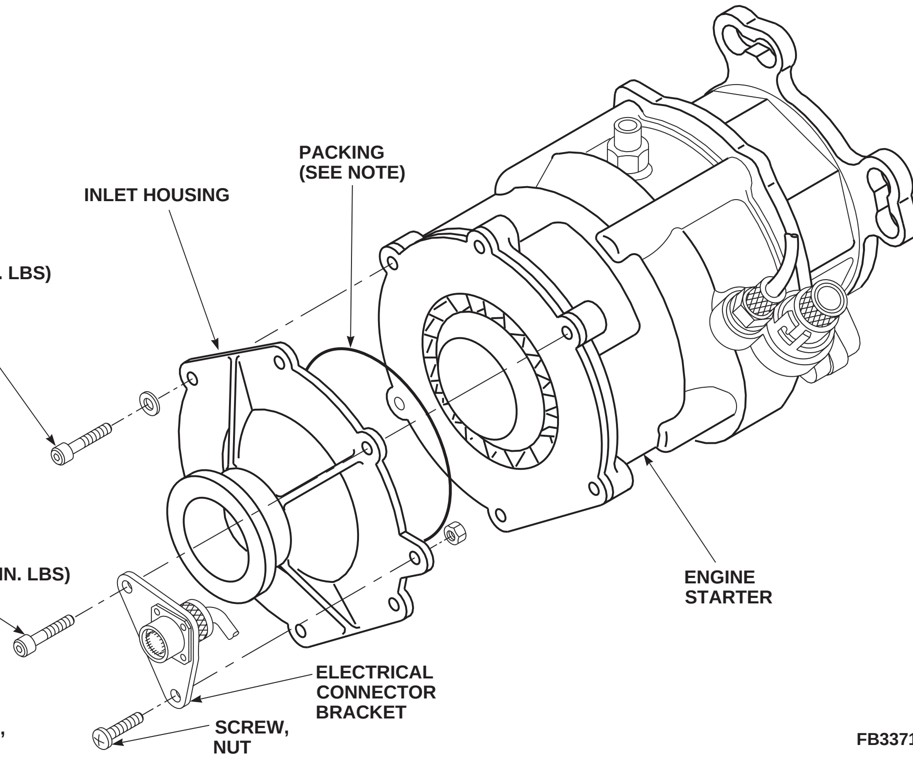


FIGURE 7-4-1-2. REPLACE INLET HOUSING

ers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

e. Secure electrical connector bracket to inlet housing with screw and nut.

f. If necessary, install engine starter (PARA 7-4-1.1.6).

7-4-1.1.4 Remove Transducer.

a. If necessary, remove engine starter (PARA 7-4-1.1.1).

b. Cut textile braid securing wiring to engine starter (Figure 7-4-1-3).

c. Remove bolt, washer, screw, and nut securing electrical connector bracket to inlet housing.

d. Remove transducer, packing, and shims (if installed) from engine starter. If removed, retain shims. Discard packing.

7-4-1.1.5 Install Transducer.

NOTE

Length must be measured without packing installed on transducer.

a. If installing new transducer, measure length of transducer (Figure 7-4-1-3, Detail A). **LENGTH TO BE 0.489 - 0.491-INCH.** Proceed as follows:

(1) If length is 0.489 - 0.491-inch, no shimming is required.

(2) If length is greater than 0.491-inch, add shims as required to achieve required measurement.

(3) If length is less than 0.489-inch, replace transducer.

b. Install packing on transducer (Figure 7-4-1-3).

c. Install transducer in engine starter. **TORQUE TRANSDUCER TO 30 - 50 INCH-POUNDS.**

d. Install electrical connector bracket to inlet housing with screw, bolt, washer, and nut. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**

e. Secure wiring to engine starter housing with textile braid, Item 81, Appendix D, wrapped and tied.

f. If necessary, install engine starter (PARA 7-4-1.1.6).

7-4-1.1.6 Install.

a. If installing replacement engine starter, perform the following (Figure 7-4-1-1, Detail A):

(1) Remove drain plug from engine starter and drain all oil from engine starter. Remove and discard packing.

(2) Lubricate packing, S9413-555, with lubricating oil, Item 205, Item 206, or Item 208, Appendix D.

(3) Install packing on drain plug.

(4) Install drain plug on engine starter.

(5) Service engine starter (PARA 1-3-7).

b. Turn off all electrical power (PARA 1-6-16).

c. Lubricate packing, S-9413-113, with lubricating oil, Item 205, Item 206, or Item 208, Appendix D, and install in groove on engine starter splined shaft.

d. Lubricate packing, M83248/1-151, with petrolatum, Item 234, Appendix D, and install in groove on engine starter mounting flange.

e. Install engine starter on engine. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

f. Install upper pneumatic tube between engine starter and firewall (PARA 7-4-8).

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connectors (PARA 1-7-20).

i. Connect electrical connector, P807 (No. 1 engine), or P806 (No. 2 engine), to engine starter electrical connector.

j. Make sure area is clean and free of foreign material.

k. Perform operational check of engine start and ignition system (PARA 4-2-8).

l. Visually check engine starter for evidence of oil leakage. **NONE ALLOWED.**

m. Close engine cowling.

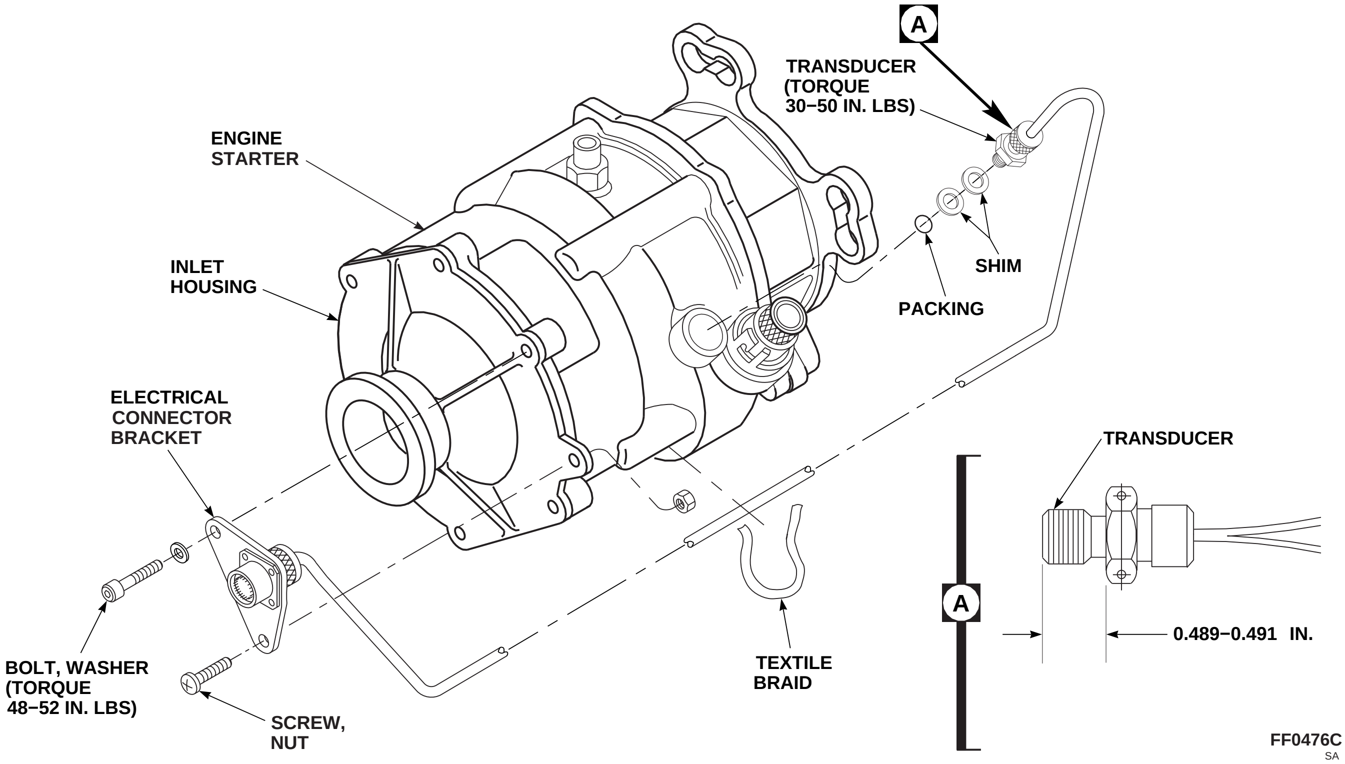


FIGURE 7-4-1-3. REPLACE TRANSDUCER

7-4-2 ENGINE STARTER CONTROL VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Rubber Mallet
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-113

7-4-2.1 Replace Engine Starter Control Valve.

NOTE

Replacement of No. 1 and No. 2 engine starter control valves is the same, except as noted.

7-4-2.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove oil cooler compartment access doors (PARA 2-4-113).

c. Disconnect electrical connector P440 (No. 1 engine), or P450 (No. 2 engine), from starter control valve electrical connector (Figure 7-4-2-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Disconnect clamps securing pneumatic tubes to starter control valve. Remove starter control valve from helicopter.

7-4-2.1.2 Disassemble.

a. Remove screws, washers, and rim clenching clamps securing solenoid valve to starter control valve (Figure 7-4-2-1, Sheet 2, Detail B).

b. Carefully remove solenoid valve from starter control valve, and remove spring retainer, spring, washer, and filter.

c. Remove and discard packings.

7-4-2.1.3 Clean.

a. Clean filter using dry-cleaning solvent, Item 124, Appendix D.

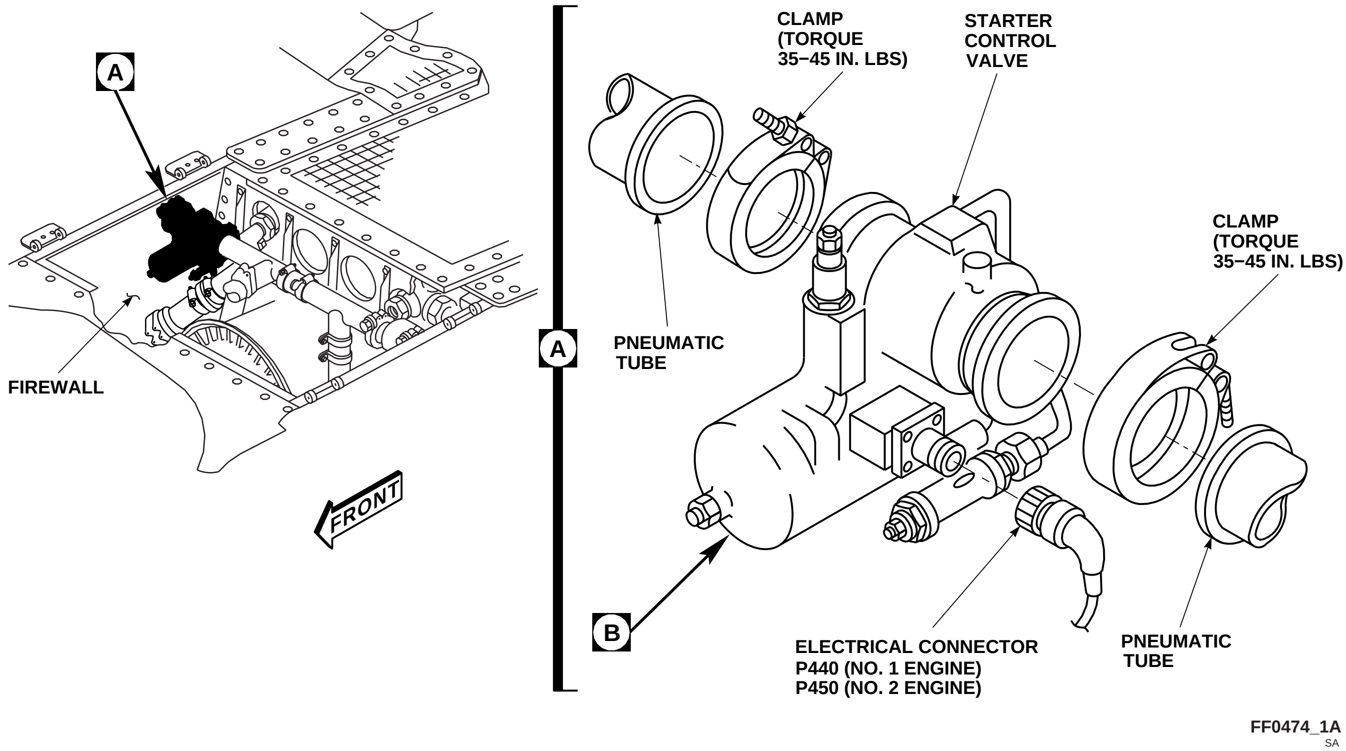


FIGURE 7-4-2-1. REPLACE ENGINE STARTER CONTROL VALVE (SHEET 1 OF 2)

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

b. When cleaning is completed, blow dry filter element and solenoid assembly air holes using compressed air (30 psi max.) to remove any material and solvent.

7-4-2.1.4 Assemble.

a. Lubricate packings with petrolatum, Item 234, Appendix D.

b. Install packings on solenoid valve (Figure 7-4-2-1, Sheet 2, Detail B).

c. Install filter, washer, spring, and spring retainer in starter control valve.

d. Secure solenoid valve to starter control valve body with screws, washers and rim clenching clamps.

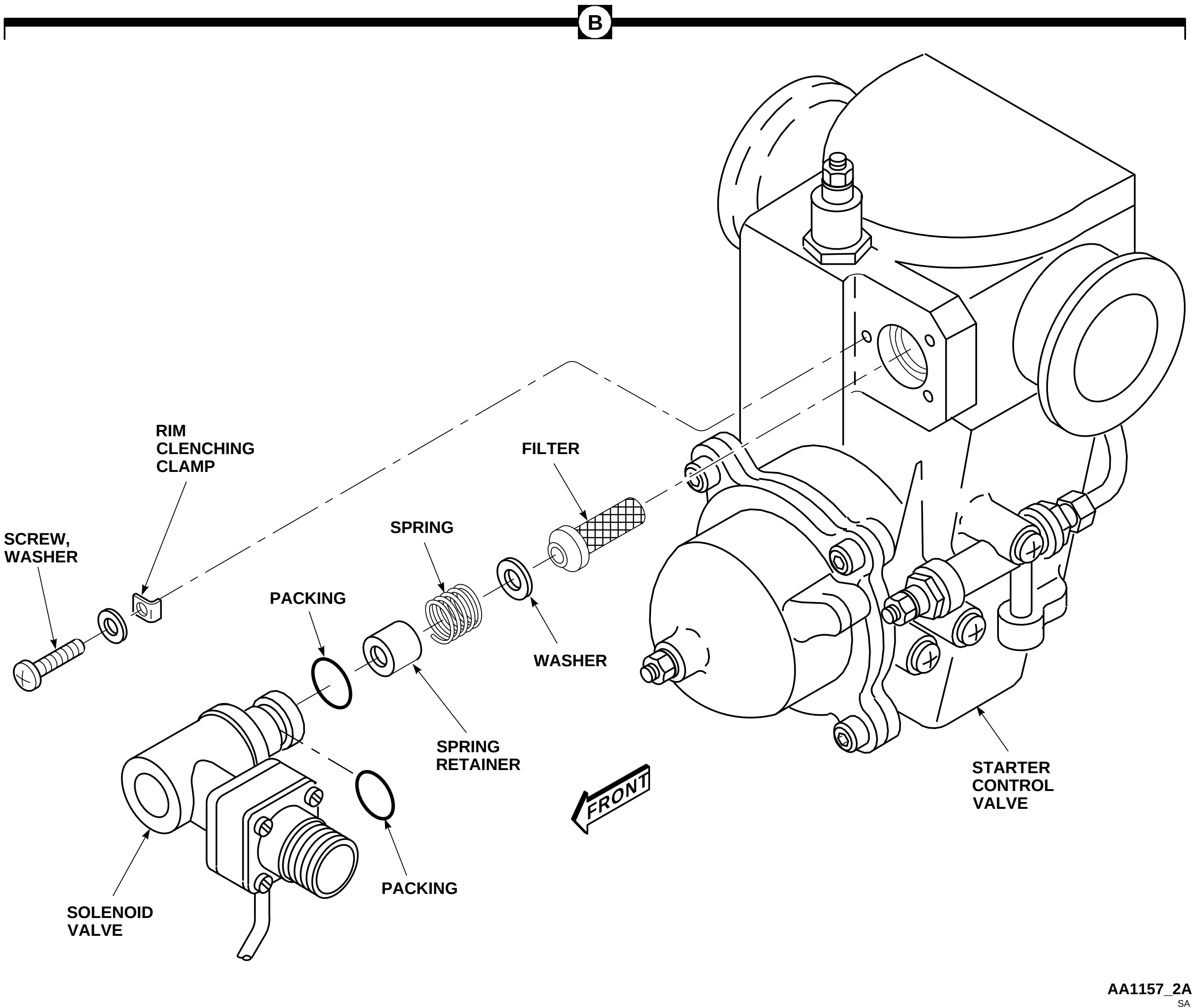


FIGURE 7-4-2-1. REPLACE ENGINE STARTER CONTROL VALVE (SHEET 2 OF 2)

7-4-2.1.5 Install.

CAUTION

- Damage to oil cooler compartment access doors will result if clamp bolts are positioned incorrectly. Make sure clamp bolts are parallel with oil cooler compartment access doors.

- Damage to starter control valve can result if improperly installed. Make sure there is adequate clearance between starter control valve and oil cooler inlet line elbow.

NOTE

Make sure clamp bolts are parallel with oil cooler compartment doors.

a. Place starter control valve between pneumatic tubes with directional flow arrow pointing toward engine. Connect pneumatic tubes to starter control valve with clamps (Figure 7-4-2-1, Sheet 1, Detail A). **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

b. Tap around clamp with rubber mallet.

c. **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

d. Continue tapping and torquing clamp until torque stabilizes.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P440 (No. 1 engine), or P450 (No. 2 engine), to electrical connector on starter control valve.

h. Make sure area is clean and free of foreign material.

i. Install oil cooler compartment access doors (PARA 2-4-113).

j. Motor starter and check for air leaks along pneumatic lines, valves, and at starter. Motor starter to at least 24% Ng speed.

7-4-3 BLEED-AIR SHUTOFF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Rubber Mallet
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vacuum

Materials/Parts

- Adhesive, Item 53, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 193, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-113
- PARA 4-2-8
- PARA 7-5-1

7-4-3.1 Replace Bleed-Air Shutoff Valve.

NOTE

Replacement of No. 1 and No. 2 engine bleed-air shutoff valves is the same, except as noted.

7-4-3.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove oil cooler compartment access doors (PARA 2-4-113).

c. Disconnect electrical connector, P434 (No. 1 engine), or P435 (No. 2 engine), from bleed-air shutoff valve (Figure 7-4-3-1, Sheet 1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Disconnect clamps securing pneumatic tubes to bleed-air shutoff valve. Remove bleed-air shutoff valve from helicopter.

7-4-3.1.2 Disassemble.

a. Remove screws and washers securing manifold assembly to housing. If necessary, tap lightly with rubber mallet to remove manifold from housing (Figure 7-4-3-1, Sheet 2, Detail B).

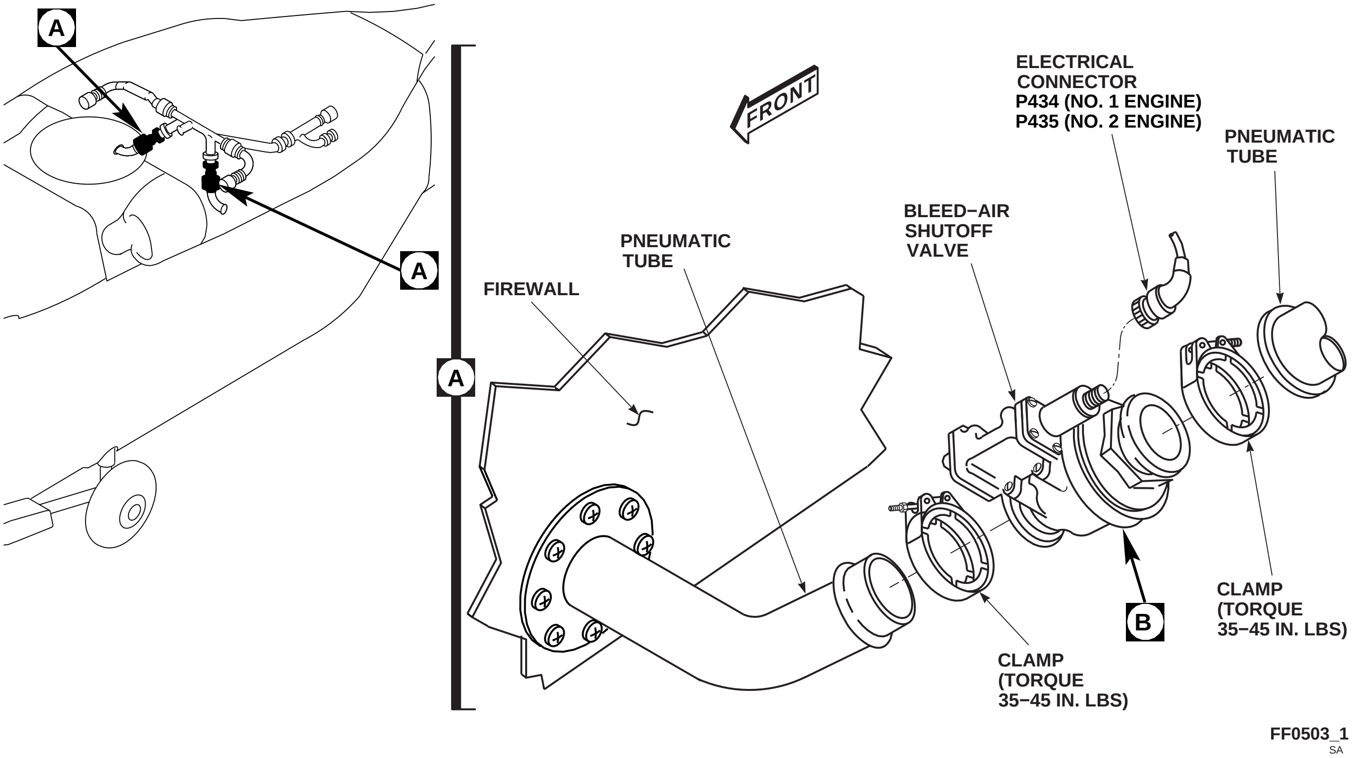


FIGURE 7-4-3-1. REPLACE BLEED-AIR SHUTOFF VALVE (SHEET 1 OF 2)

b. Remove flange from housing.

7-4-3.1.3 Clean.

a. While spraying generous amounts of dry-cleaning solvent, Item 124, Appendix D, clean valve assembly inside of housing by gently pushing sleeve in and out until valve is clean of sand and carbon-like black residue (Figure 7-4-3-1, Sheet 2, Detail B).

b. Using dry-cleaning solvent, Item 124, Appendix D, clean and flush all ports and passages of housing and manifold. Flush ambient port with dry-cleaning solvent. Drain solvent from port after flushing.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

c. Using clean, dry, compressed air (50 psi max.), blow dry valve assembly.

d. If further cleaning of bleed-air shutoff valve is required, clean per PARA 7-5-1.

7-4-3.1.4 Assemble.

NOTE

Do not allow adhesive within ⅛-inch of any air passage.

a. Apply adhesive, Item 53, Appendix D, to threads of flange and install in housing (Figure 7-4-3-1, Sheet 2, Detail B). **TORQUE FLANGE TO 200 - 236 INCH-POUNDS.**

b. Apply adhesive, Item 53, Appendix D, to mating surface of housing and manifold assembly.

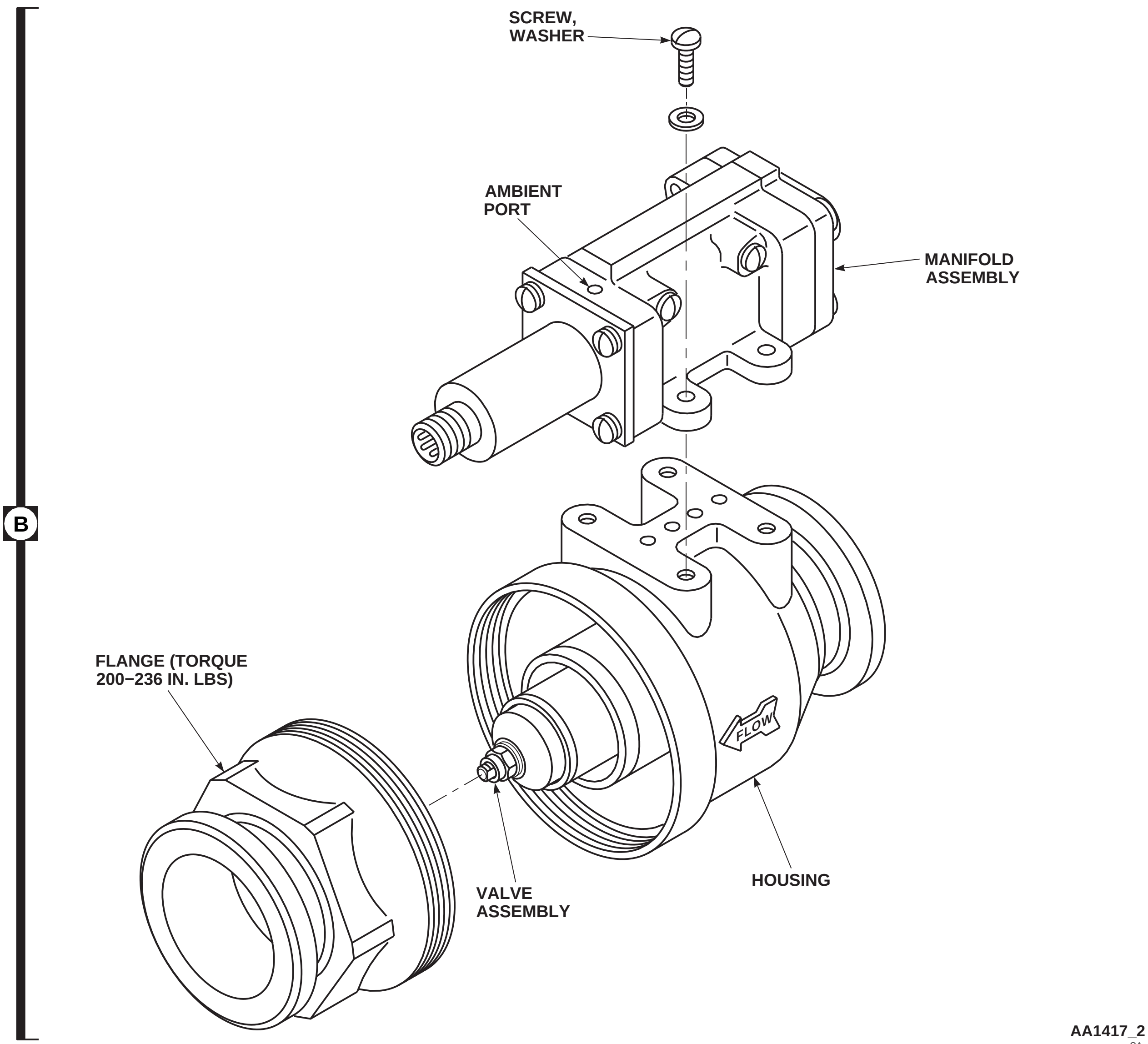


FIGURE 7-4-3-1. REPLACE BLEED-AIR SHUTOFF VALVE (SHEET 2 OF 2)

c. Install manifold assembly on housing with screws and washers.

d. Using lockwire, Item 193, Appendix D, lockwire screws.

7-4-3.1.5 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Clean and vacuum any debris from pneumatic tubes and clamps.

c. Place bleed-air shutoff valve between pneumatic tubes with directional arrow pointing away from firewall. Connect pneumatic tubes to bleed-air shutoff valve with clamps (Figure 7-4-3-1, Sheet 1, Detail A). **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

d. Tap around clamp with rubber mallet. **RE-TORQUE CLAMP TO 35 - 45 INCH-POUNDS.** Repeat until torque stabilizes.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P434 (No. 1 engine), or P435 (No. 2 engine), to electrical connector on bleed-air shutoff valve.

h. Make sure area is clean and free of foreign material.

i. Perform operational check of engine start and ignition system to make sure check valve is not leaking (PARA 4-2-8).

j. Install oil cooler compartment access doors (PARA 2-4-113).

k. Perform maintenance test flight (Appropriate MTF).

7-4-4 APU CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Rubber Mallet
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16
- PARA 4-2-8
- PARA 7-3-2
- PARA 7-4-5
- PARA 7-4-6
- PARA 7-4-7
- PARA 7-4-8

7-4-4.1 Replace APU Check Valve.

7-4-4.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open APU access doors.

c. Loosen clamps on coupling and slide clamps on pneumatic tube. Remove coupling (Figure 7-4-4-1, Detail A).

d. Disconnect clamp holding APU check valve to pneumatic tube. Remove APU check valve from helicopter.

7-4-4.1.2. Inspect.

a. Inspect APU check valve for damaged butterfly valves. **NONE ALLOWED.** Replace APU check valve if butterfly valves are damaged. If parts of valves are missing, parts must be located and removed from pneumatic tubes (PARA 7-4-5, PARA 7-4-6, PARA 7-4-7, and PARA 7-4-8).

b. Inspect pneumatic tubes and couplings (PARA 7-3-2).

7-4-4.1.3. Install.

a. Place APU check valve between pneumatic tubes with directional arrow pointing towards front of helicopter (Figure 7-4-4-1, Detail A). Connect pneumatic tube to APU check valve with coupling. **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

b. Position coupling between check valve and pneumatic tube. Slide clamps onto valve and coupling next to beads on check valve. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

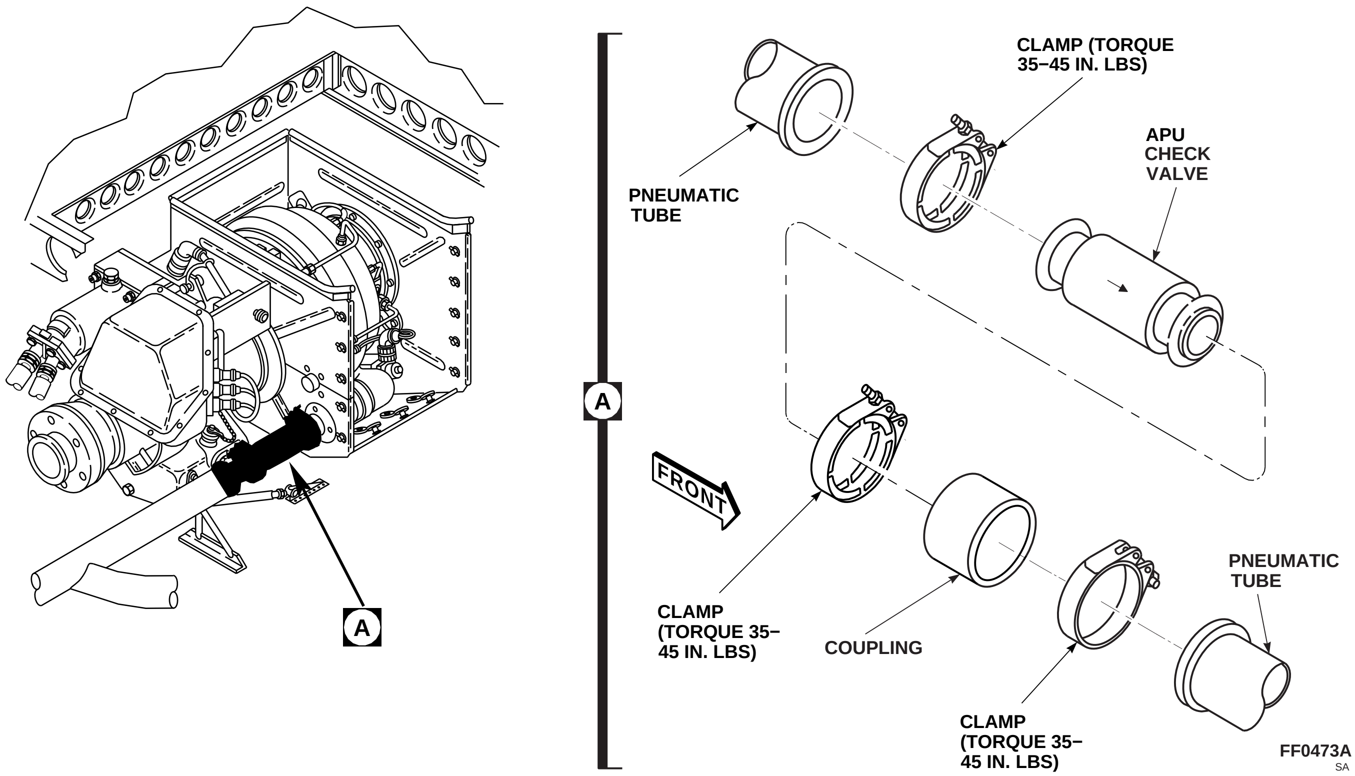


FIGURE 7-4-4-1. REPLACE APU CHECK VALVE

c. Tap around clamps with rubber mallet. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.** Repeat until torque stabilizes.

d. Make sure area is clean and free of foreign material.

e. Perform operational check of engine start and ignition system to make sure APU check valve is not leaking (PARA 4-2-8).

f. Close APU access doors.

7-4-5 TRANSITION SECTION PNEUMATIC TUBES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- File
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-45
- PARA 2-4-69
- PARA 7-3-2
- PARA 7-4-9

7-4-5.1 Replace Transition Section Pneumatic Tubes.

7-4-5.1.1 Remove.

a. Remove nipple-check valve (PARA 7-4-9).

b. Remove left side rear passenger's seats (PARA 2-4-45).

c. Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).

d. Remove cargo netting from stowage compartment.

e. Inside transition section, loosen clamps on couplings, and remove middle and lower pneumatic tubes and gasket (Figure 7-4-5-1, Detail A). Discard gasket.

7-4-5.1.2 Inspect.

Inspect pneumatic tubes and couplings (PARA 7-3-2).

7-4-5.1.3 Install.

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.12-inch (Figure 7-4-5-1, Detail B). If necessary, metal may be removed from pneumatic tube end using file to achieve this gap. Do not remove any metal from bead on pneumatic tubes.

a. If necessary, file end of pneumatic tube(s) to achieve 0.12-inch gap.

b. Inside transition section, connect middle pneumatic tube to upper pneumatic tube with coupling and clamps (Figure 7-4-5-1, Detail A and Detail B). Position clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

c. Connect lower pneumatic tube to middle pneumatic tube with coupling and clamps. Position clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

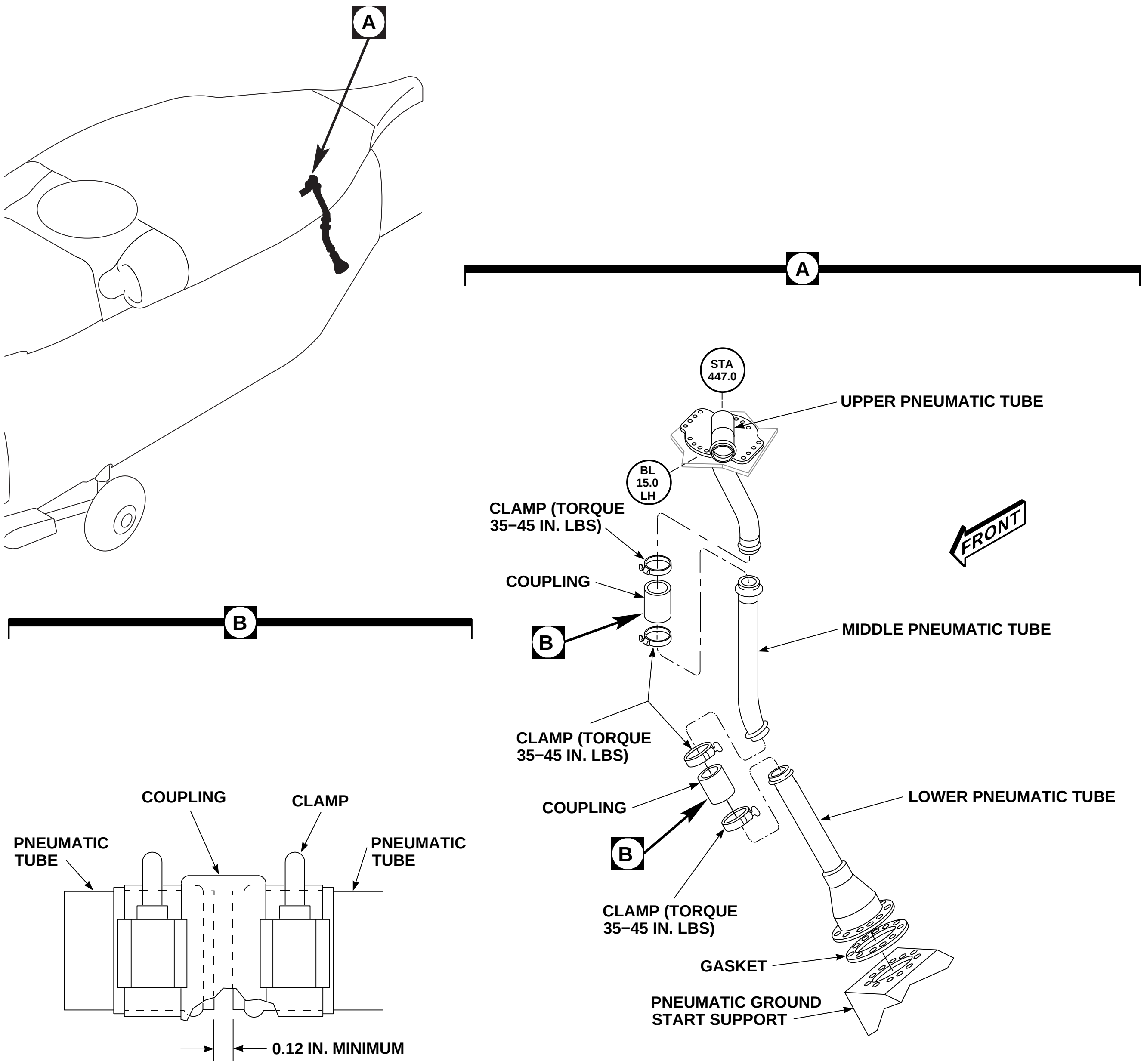


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FIGURE 7-4-5-1. REPLACE TRANSITION SECTION PNEUMATIC TUBES

d. Position gasket between lower pneumatic tube and pneumatic ground start support.

e. Install nipple-check valve (PARA 7-4-9).

f. Make sure area is clean and free of foreign material.

g. Install cargo netting.

h. Install soundproofing panel (PARA 2-4-69).

i. Install rear passenger's seats (PARA 2-4-45).

7-4-6 OIL COOLER COMPARTMENT PNEUMATIC TUBES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- File
- Rubber Mallet
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-113
- PARA 4-2-8
- PARA 7-3-2

7-4-6.1 Replace Oil Cooler Compartment Pneumatic Tubes.

7-4-6.1.1. Remove.

a. Remove oil cooler compartment access doors (PARA 2-4-113).

b. Unclamp pneumatic tubes and couplings from bleed-air shutoff valves, engine starter control valves, and pneumatic tube (Figure 7-4-6-1, Detail A).

c. Loosen clamp on coupling of rear bleed-air tube connection.

d. Loosen clamps on coupling of APU compartment front pneumatic tube and remove tubes from compartment.

7-4-6.1.2. Inspect.

Inspect pneumatic tubes and couplings (PARA 7-3-2).

7-4-6.1.3. Install.

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.12-inch (Figure 7-4-6-1, Detail B). If necessary, metal may be removed from pneumatic tube end using file to achieve this gap. Do not remove any metal from bead on pneumatic tubes.

a. If necessary, file end of pneumatic tube(s) to achieve 0.12-inch gap.

b. Install pneumatic tube between engine starter control valves with clamps (Figure 7-4-6-1, Detail A). **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

c. Connect pneumatic tube sections with coupling and clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

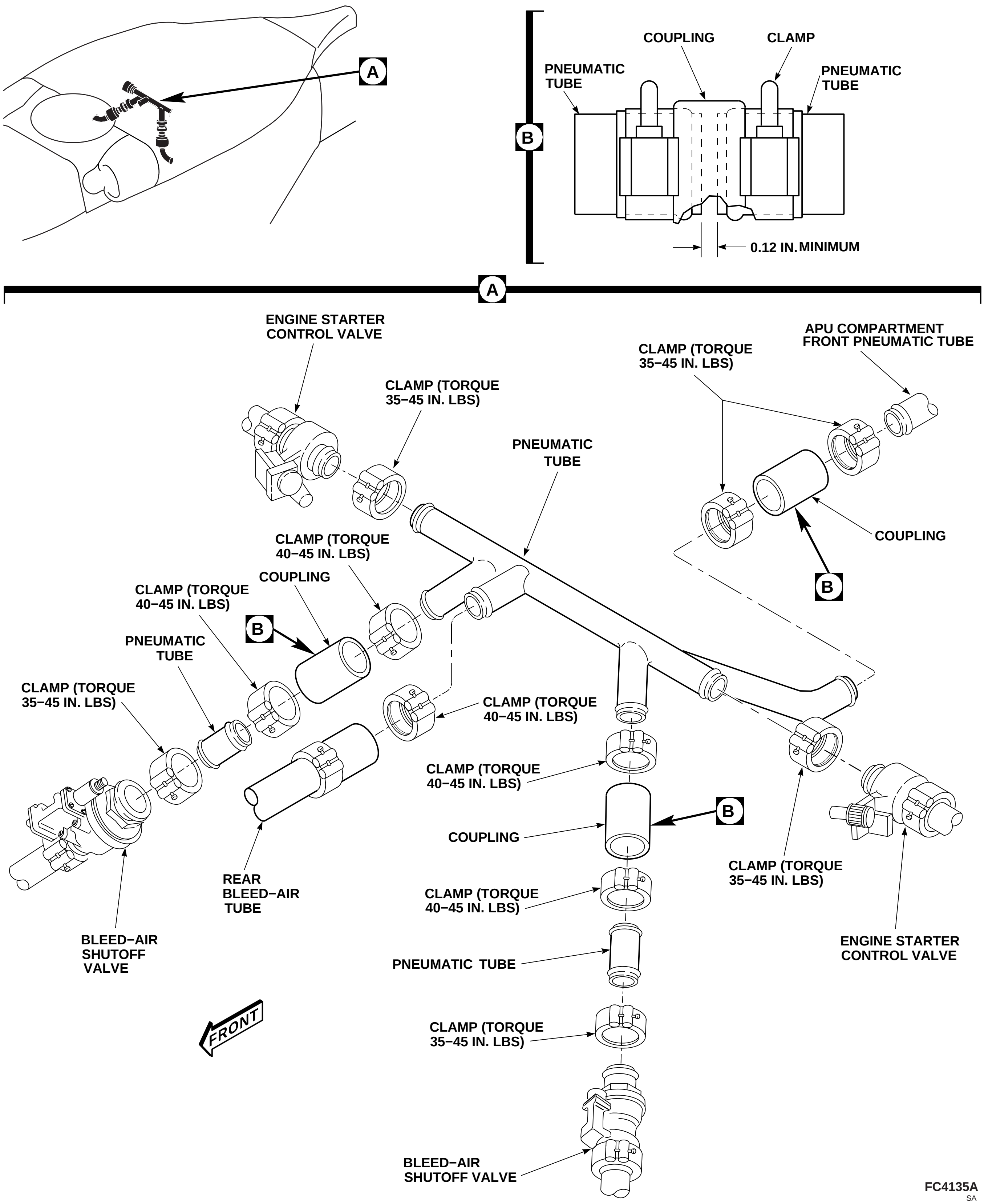


FIGURE 7-4-6-1. REPLACE OIL COOLER COMPARTMENT PNEUMATIC TUBES

d. Connect pneumatic tube sections to bleed-air shutoff valves with clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

e. Connect rear bleed-air tube to pneumatic tube with coupling and clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

f. Connect pneumatic tube to APU compartment front pneumatic tube with coupling and clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

g. Tap around clamps with rubber mallet. Retorque clamps per Figure 7-4-6-1, Detail A. Repeat until torque stabilizes.

h. Make sure area is clean and free of foreign material.

i. Perform operational check (PARA 4-2-8). Check for air leaks.

j. Install oil cooler compartment access doors (PARA 2-4-113).

7-4-7 APU COMPARTMENT PNEUMATIC TUBES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- File
- Rivet Gun
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-108
- PARA 7-3-2
- PARA 7-4-4

7-4-7.1 Replace APU Compartment Pneumatic Tubes.

7-4-7.1.1 Remove.

a. Remove APU access door (PARA 2-4-108).

b. Remove APU check valve (PARA 7-4-4).

c. Remove screws, washers, and nuts securing clamps to brackets (Figure 7-4-7-1, Detail A).

d. Loosen clamps on couplings and remove front and rear pneumatic tubes from compartment.

e. Remove rivets securing upper pneumatic tube to airframe and remove tube.

7-4-7.1.2 Inspect.

a. Inspect pneumatic tubes and couplings (PARA 7-3-2).

b. Inspect APU check valve (PARA 7-4-4).

7-4-7.1.3 Install.

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.12-inch (Figure 7-4-7-1, Detail B). If necessary, metal may be removed from pneumatic tube end using file to achieve this gap. Do not remove any metal from bead on pneumatic tubes.

a. If necessary, file end of pneumatic tube(s) to achieve 0.12-inch gap.

b. Position upper pneumatic tube on airframe with mounting holes lined up. Install rivets using rivet gun (Figure 7-4-7-1, Detail A).

c. Install APU check valve (PARA 7-4-4).

d. Connect rear pneumatic tube to upper pneumatic tube with coupling and clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

e. Connect front pneumatic tube to rear pneumatic tube and oil cooler compartment pneumatic tube with couplings and clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

f. Install clamps on pneumatic tubes and secure to brackets with screws, washers, and nuts.

g. Make sure area is clean and free of foreign material.

h. Install APU access door (PARA 2-4-108).

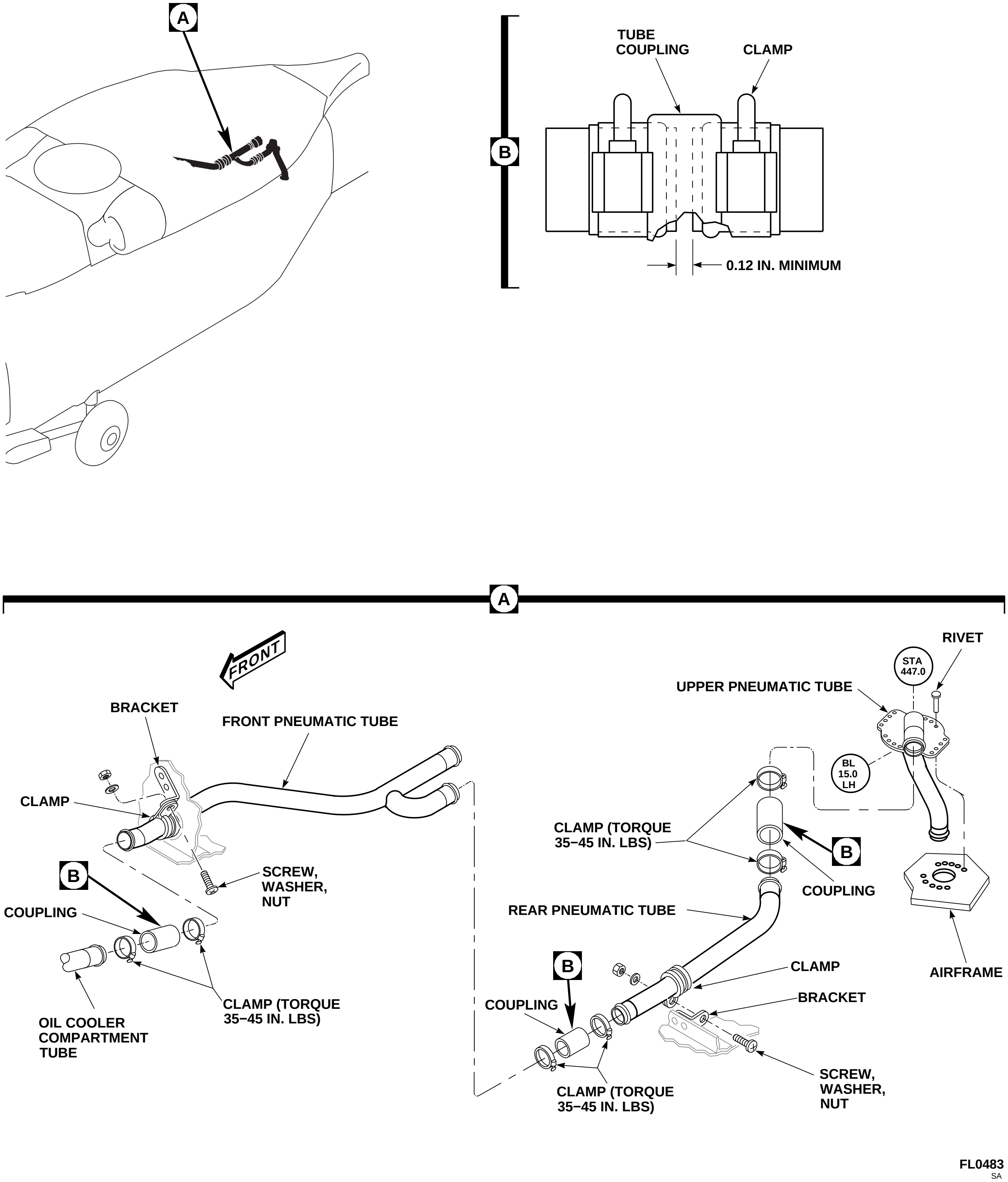


FIGURE 7-4-7-1. REPLACE APU COMPARTMENT PNEUMATIC TUBES

7-4-8 ENGINE COMPARTMENT PNEUMATIC TUBES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- File
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-113
- PARA 7-3-2
- PARA 7-4-2

7-4-8.1 Replace Engine Compartment Pneumatic Tubes.

NOTE

There are two upper and two lower pneumatic tubes. Replacement for left and right, upper and lower pneumatic tubes is the same.

7-4-8.1.1. Remove.

a. Open engine cowling.

b. Remove oil cooler compartment access doors (PARA 2-4-113).

c. Unclamp pneumatic tube and coupling from engine starter and firewall tube (Figure 7-4-8-1, Detail A).

d. Remove engine starter control valve (PARA 7-4-2).

e. Remove screws, washers, plates, seal, and firewall tube from firewall.

f. Unclamp lower pneumatic tube from bleed-air shutoff valve and engine bleed-air tube.

g. Remove screws, washers, and nuts and remove lower pneumatic tube from firewall.

7-4-8.1.2. Inspect.

Inspect pneumatic tubes and couplings (PARA 7-3-2).

7-4-8.1.3. Install.

NOTE

Minimum gap between pneumatic tube coupling connection is 0.12-inch (Figure 7-4-8-1, Detail B). If necessary, metal may be removed from pneumatic tube end using file to achieve this gap. Do not remove any metal from bead on pneumatic tube.

a. If necessary, file end of pneumatic tube(s) to achieve 0.12-inch gap.

b. Install lower pneumatic tube in firewall with screws, washers, and nuts (Figure 7-4-8-1, Detail A).

c. Clamp lower pneumatic tube to bleed-air shutoff valve and engine bleed-air tube with

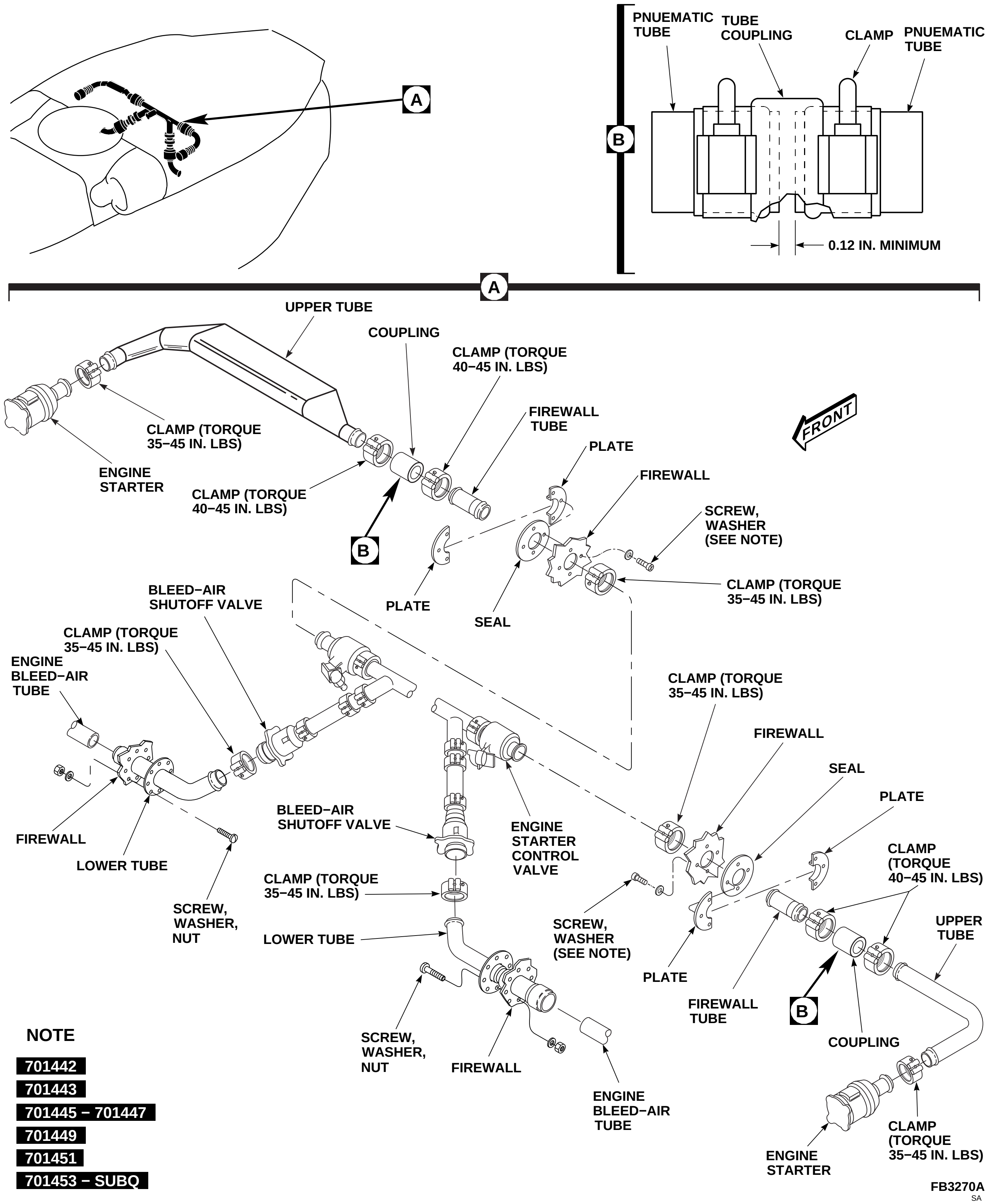


FIGURE 7-4-8-1. REPLACE ENGINE COMPARTMENT PNEUMATIC TUBES clamps. Place clamps next to bead on each pneumatic tube. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**

NOTE

701442–SUBQ Washers are installed on firewall tube, seal, and plate assemblies.

d. Install firewall tube in firewall with seals, plates, screws, and washers.

e. Install engine starter control valve (PARA 7-4-2).

f. Clamp upper pneumatic tube to engine starter. **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

g. Connect upper pneumatic tube to firewall tube with coupling and clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

h. Make sure area is clean and free of foreign material.

i. Install oil cooler compartment access doors (PARA 2-4-113).

j. Close engine cowling.

7-4-9 NIPPLE-CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Grease Pencil
- Torque Wrench, 30 - 150 in. lbs

7-4-9.1 Replace Nipple-Check Valve.

7-4-9.1.1. Remove.

a. Open ground start access panel.

b. Using grease pencil, mark flange on nipple-check valve and pneumatic ground start support for proper alignment at installation (Figure 7-4-9-1, Detail A).

c. Remove bolts, washers, nipple-check valve, and gasket from pneumatic ground start support and lower pneumatic tube. Discard gasket.

7-4-9.1.2. Install.

a. Place gasket on nipple-check valve flange and line up gasket and flange against pneumatic ground start support (Figure 7-4-9-1, Detail A).

b. Secure gasket and pneumatic ground start support to lower pneumatic tube with bolts and washers. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**

c. Make sure area is clean and free of foreign material.

d. Close ground start access panel.

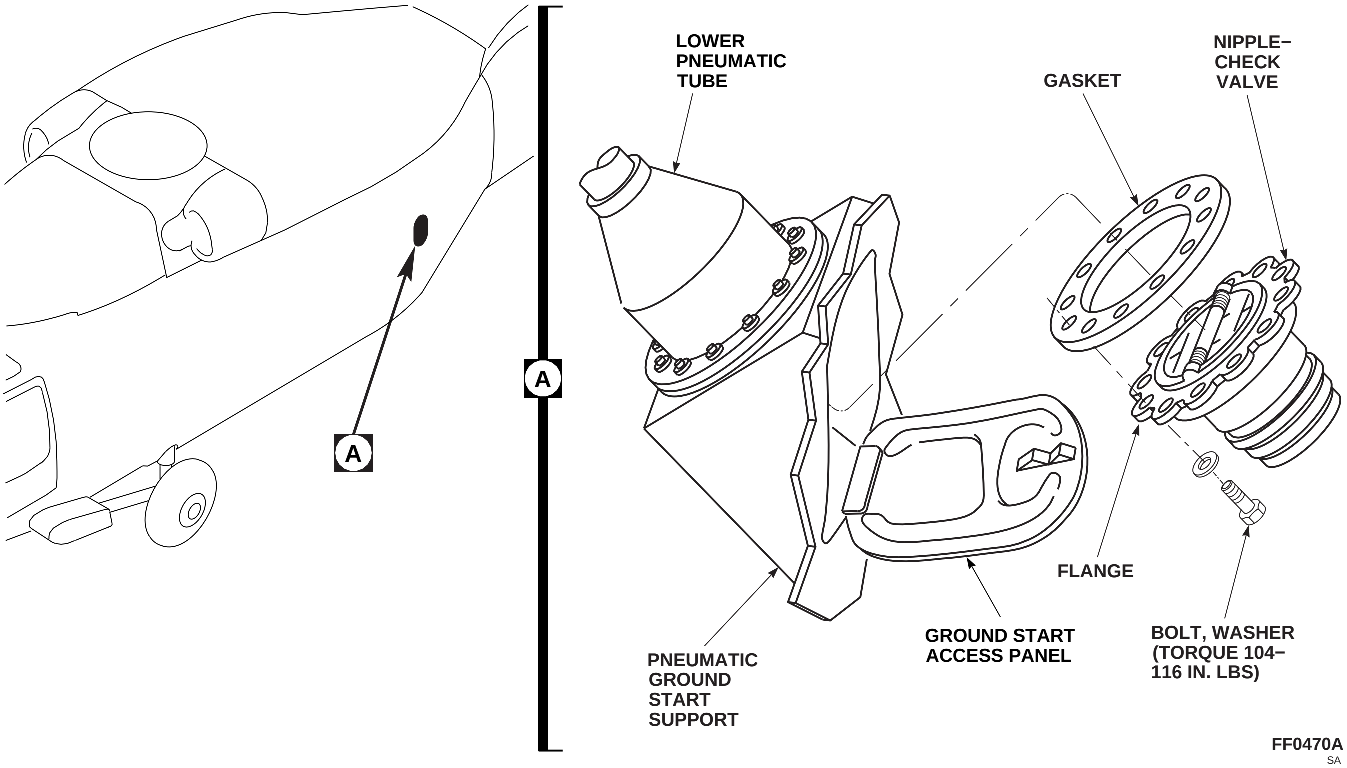


FIGURE 7-4-9-1. REPLACE NIPPLE-CHECK VALVE

7-4-10 STARTER SPEED SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 4-2-8

7-4-10.1 Replace Starter Speed Switch.

7-4-10.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panel from left cabin ceiling (PARA 2-4-69).

c. Disconnect electrical connector, P314, from starter speed switch (Figure 7-4-10-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove screws, lockwashers, and washers and remove starter speed switch from cabin ceiling.

7-4-10.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place starter speed switch on cabin ceiling and secure with screws, lockwashers, and washers (Figure 7-4-10-1, Detail A).

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P314, to starter speed switch.

f. Make sure area is clean and free of foreign material.

g. Perform operational check of engine start and ignition system (PARA 4-2-8).

h. Install soundproofing panel (PARA 2-4-69).

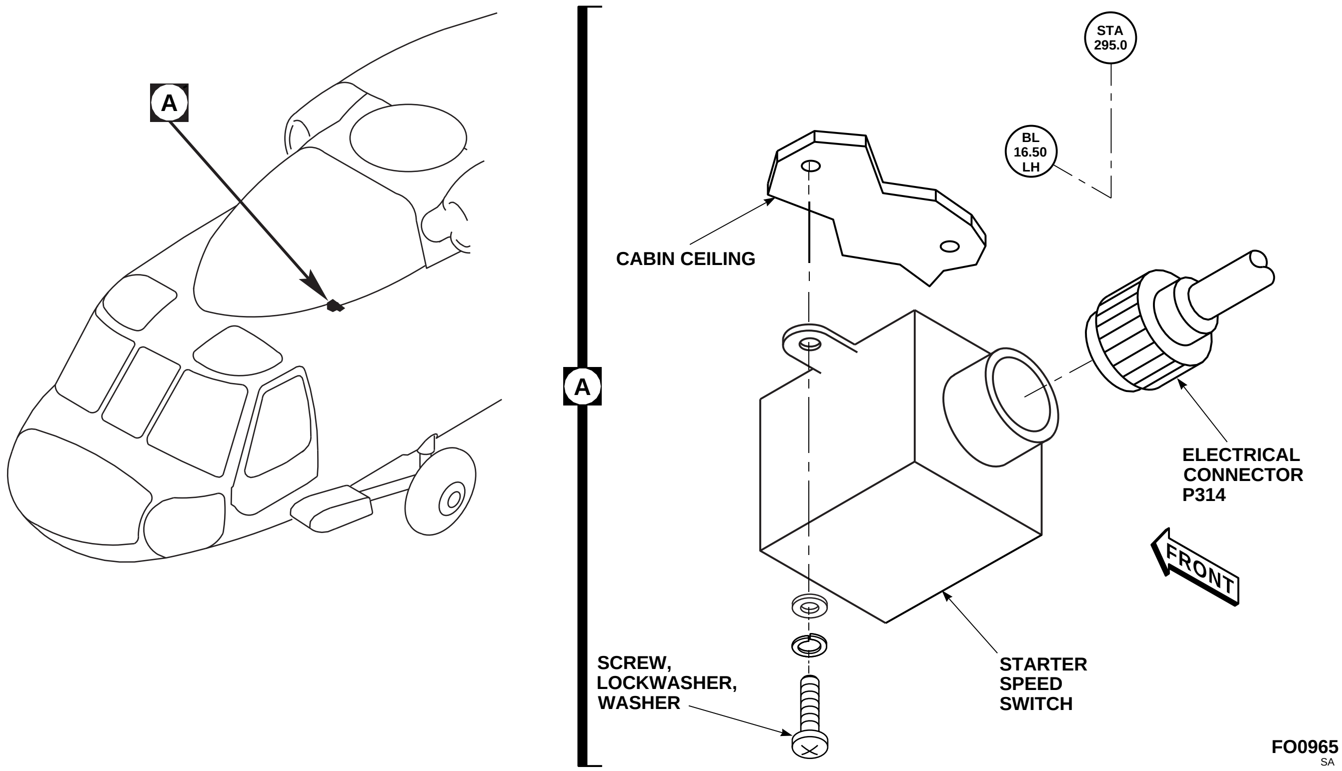


FIGURE 7-4-10-1. REPLACE STARTER SPEED SWITCH

7-4-11 CONVERSION OF HYDRAULIC FLUIDS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Hydraulic Test Stand (Servicing Unit)

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Tiedown Strap, Item 338, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16

7-4-11.1 General.

a. Conversion of hydraulic fluid, Item 155, Appendix D, to hydraulic fluid, Item 154, Appendix D, is required during cold weather operations of -25°F (-32°C) or below. During cold weather, hydraulic fluid, Item 155, Appendix D, becomes extremely viscous and causes sluggish operation in control systems.

b. Conversion of hydraulic fluid, Item 154, Appendix D, to hydraulic fluid, Item 155, Appendix D, is required when helicopter is operating above -25°F (-32°C). A flashpoint check of hydraulic fluid is recommended after converting back to hydraulic fluid, Item 155, Appendix D. Failure to achieve proper flashpoint is not cause to ground helicopter. System operation, plus normal servicing, should dilute any remaining fluid after following conversion procedure in PARA 7-4-11.3.

7-4-11.2 Convert MIL-PRF-83282 to MIL-PRF-5606.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Turn selector valve handle to OUT 3. Hold selector valve handle down and crank refill handpump until no hydraulic fluid is left in handpump.

d. Position 2-quart container under hydraulic drain line near left main landing gear.

e. Turn and press each bleed relief valve on three hydraulic pump modules and drain pump module reservoirs. Discard hydraulic fluid from container under drain line.

f. Fill refill handpump to top with hydraulic fluid, Item 154, Appendix D. Turn selector valve to OUT 1. Hold selector valve handle down and crank refill handpump until No. 1 pump module reservoir piston moves about ½-inch. Turn selector valve to OUT 2 and repeat procedure for No. 2 pump module. Turn selector valve to OUT 3 and repeat procedure for backup pump module.

g. Turn and press each bleed relief valve on the three pump modules and drain pump module reservoirs. Discard hydraulic fluid from container under drain line.

NOTE

Make sure servicing unit is serviced with hydraulic fluid, Item 154, Appendix D.

h. Connect external hydraulic power to backup pump module (PARA 1-6-15).

i. Connect line to backup system external return disconnect, and run line into drum. Do not connect return line to servicing unit.

CAUTION

Damage to antiflap cam and antiflap stop attaching bolt will result if cam is not secured in flight position when cycling flight controls with main rotor blades removed. Secure antiflap cam in flight position prior to cycling flight controls.

j. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and secure them to spindle horn with tiedown straps, Item 338, Appendix D.

k. Make sure electrical power is still off (PARA 1-6-16), and servicing unit is connected to backup hydraulic system. Increase servicing unit hydraulic pressure to 3,000 psi.

l. Remove APU exhaust plug (PARA 1-6-5).

m. Remove ignition connector from APU.

n. Manually operate APU start valve to motor APU and release hydraulic charge.

o. Recharge APU accumulator (PARA 1-3-10).

p. After APU accumulator has fully recharged, manually operate APU start valve to motor APU and release hydraulic charge a second time.

q. Install APU exhaust plug (PARA 1-6-5).

r. Install ignition connector on APU.

s. Slowly cycle flight controls, stop-to-stop, in this sequence:

(1) Perform the following combination six times:

(a) Low collective to high collective.

(b) Forward cyclic to aft cyclic.

(c) Right pedal to left pedal.

(d) Left lateral to right lateral.

(2) Position collective in midposition. Cycle left pedal to right pedal twice.

t. Turn hydraulic pressure off.

u. Disconnect servicing unit from backup pump module and connect to No. 1 pump module.

v. Connect servicing unit return line to No. 1 system external return disconnect.

w. Increase servicing unit hydraulic pressure to 3,000 psi.

x. Slowly cycle cyclic controls from left lateral to right lateral three times, to purge pump module pressure line and No. 1 transfer module.

y. Turn hydraulic pressure off.

z. Disconnect servicing unit from No. 1 pump module and connect to No. 2 pump module.

aa. Connect servicing unit return line to No. 2 system external return disconnect.

ab. Increase servicing unit hydraulic pressure to 3,000 psi.

ac. Slowly cycle cyclic controls from left lateral to right lateral three times to purge pump module pressure line and No. 2 transfer module.

ad. Turn hydraulic pressure off.

ae. Disconnect external hydraulic power (PARA 1-6-15).

af. Service all three hydraulic pump modules with hydraulic fluid, Item 154, Appendix D (PARA 1-3-8).

ag. If antiflap cam was secured in flight position with tiedown straps, cut and remove tiedown straps.

NOTE

When one indicator tape is off scale above 100%, other indicator should read between 20% and 60%.

ah. Check indicator tapes on accumulator. Add tape readings together. Sum should be between 140% to 160%.

ai. Close main rotor pylon sliding cover.

7-4-11.3 Convert MIL-PRF-5606 to MIL-PRF-83282.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Turn selector valve handle to OUT 3. Hold selector valve handle down and crank refill handpump until no hydraulic fluid is left in handpump.

d. Position 2-quart container under hydraulic drain line near left main landing gear.

e. Turn and press each bleed relief valve on three hydraulic pump modules and drain pump module reservoirs. Discard hydraulic fluid from container under drain line.

f. Fill refill handpump to top with hydraulic fluid, Item 155, Appendix D, and perform the following:

(1) Turn selector valve to OUT 1.

(2) Hold selector valve handle down and crank refill handpump until No. 1 pump module reservoir piston moves about ½-inch.

(3) Turn selector valve to OUT 2 and repeat procedure for No. 2 pump module.

(4) Turn selector valve to OUT 3 and repeat procedure for backup pump module.

g. Turn and press each of bleed relief valves on three pump modules and drain pump module reservoirs. Discard hydraulic fluid from container under drain line.

NOTE

Make sure servicing unit is serviced with hydraulic fluid, Item 155, Appendix D.

h. Connect external hydraulic power to backup pump module (PARA 1-6-15).

i. Connect line to backup system external return disconnect and run line into drum. Do not connect return line to servicing unit.

CAUTION

Damage to antiflap cam and antiflap stop attaching bolt will result if cam is not secured in flight position when cycling flight controls with main rotor blades removed. Secure antiflap cam in flight position prior to cycling flight controls.

j. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and secure them to spindle horn with tiedown straps, Item 338, Appendix D.

k. Make sure electrical power is still off (PARA 1-6-16), and servicing unit is connected to backup hydraulic system. Increase servicing unit hydraulic pressure to 3,000 psi.

l. Remove APU exhaust plug (PARA 1-6-5).

m. Remove ignition connector from APU.

n. Manually operate APU start valve to motor APU and release hydraulic charge.

o. Recharge APU accumulator (PARA 1-3-10).

p. After APU accumulator has fully recharged, manually operate APU start valve to motor APU and release hydraulic charge a second time.

q. Install APU exhaust plug (PARA 1-6-5).

r. Install ignition connector on APU.

s. Slowly cycle flight controls, stop-to-stop, in the following sequence:

(1) Perform the following combination six times:

(a) Low collective to high collective.

(b) Forward cyclic to aft cyclic.

(c) Right pedal to left pedal.

(d) Left lateral to right lateral.

(2) Position collective in midposition. Cycle left pedal to right pedal twice.

t. Turn hydraulic pressure off.

u. Disconnect servicing unit from backup pump module and connect to No. 1 pump module.

v. Connect servicing unit return line to No. 1 system external return disconnect.

w. Increase servicing unit hydraulic pressure to 3,000 psi.

x. Slowly cycle cyclic controls from left lateral to right lateral three times to purge pump module pressure line and No. 1 transfer module.

y. Turn hydraulic pressure off.

z. Disconnect servicing unit from No. 1 pump module and connect to No. 2 pump module.

aa. Connect servicing unit return line to No. 2 system external return disconnect.

ab. Increase servicing unit hydraulic pressure to 3,000 psi.

ac. Slowly cycle cyclic controls from left lateral to right lateral three times to purge pump module pressure line and No. 2 transfer module.

ad. Disconnect external hydraulic power (PARA 1-6-15).

ae. Service all three hydraulic pump module reservoirs with hydraulic fluid, Item 155, Appendix D (PARA 1-3-8).

af. After changing from hydraulic fluid, Item 154, Appendix D, to hydraulic fluid, Item 155, Appendix D, hydraulic fluid sample should be taken to determine flashpoint. Before taking samples, perform the following:

(1) Remove APU exhaust plug (PARA 1-6-5).

(2) Start up APU 5 times (Appropriate Flight Manual).

(3) Using backup system, cycle controls for 5 minutes.

(4) If antiflap cam was secured in flight position with tiedown straps, cut and remove tiedown straps.

(5) With rotor turning, cycle controls through short ranges.

(6) Take fluid samples from reservoirs for flashpoint check.

(7) Install APU exhaust plug (PARA 1-6-5).

ag. Close main rotor pylon sliding cover.

7-4-12 NO. 1 AND NO. 2 HYDRAULIC PUMP MODULES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-16
- PARA 7-4-17
- PARA 7-4-18
- PARA 7-4-65

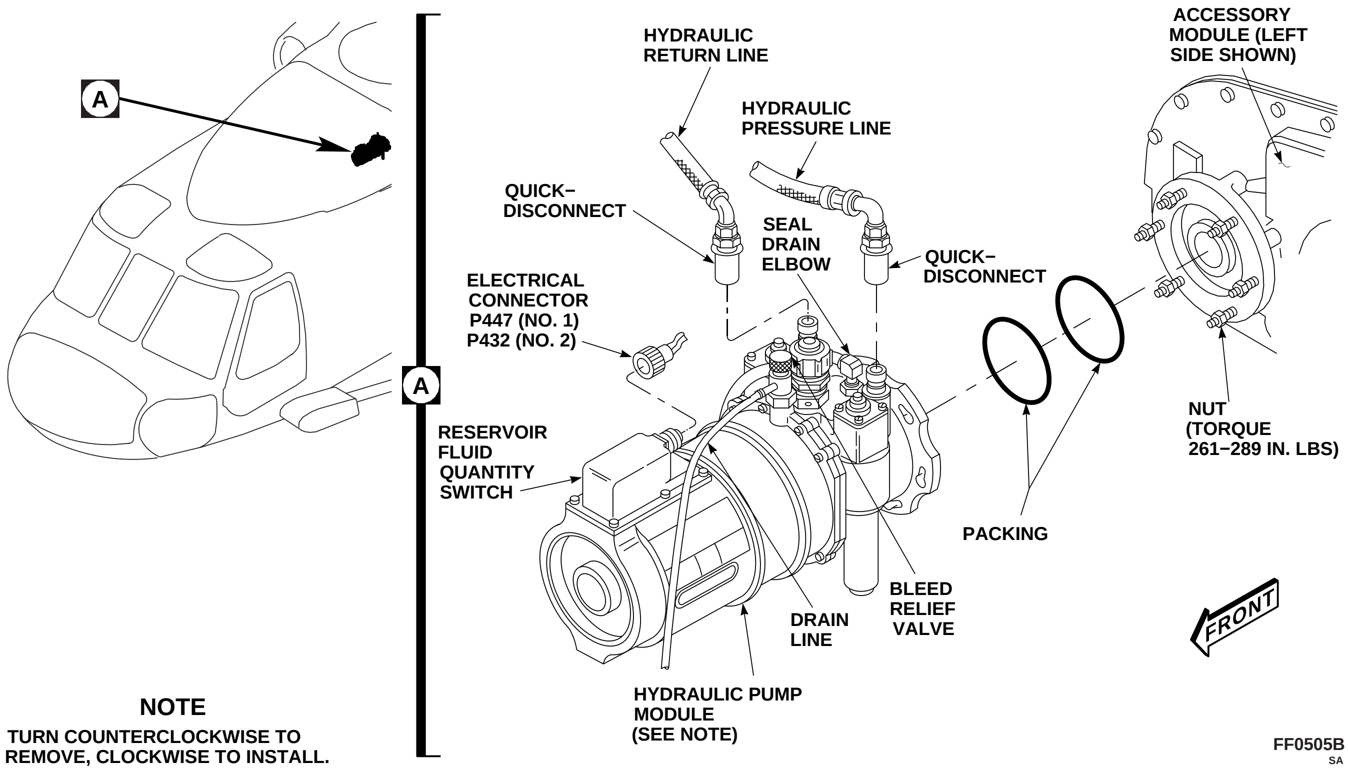


FIGURE 7-4-12-1. REPLACE NO. 1 AND NO. 2 HYDRAULIC PUMP MODULES

7-4-12.1 Replace No. 1 and No. 2 Hydraulic Pump Modules.

NOTE

Replacement of No. 1 and No. 2 hydraulic pump modules is the same, except as noted.

7-4-12.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect hydraulic pressure and hydraulic return lines at quick-disconnects (Figure 7-4-12-1, Detail A).

d. If installed, disconnect drain line from tee at bottom of hydraulic pump module.

e. Disconnect drain line from seal drain elbow.

f. Disconnect electrical connector, P447 (No. 1), or P432 (No. 2), from reservoir fluid quantity switch.

g. Install protective caps and plugs on electrical connectors.

h. Disconnect drain line from bleed relief valve.

i. Loosen nuts securing hydraulic pump module to accessory module.

j. Turn hydraulic pump module counterclockwise until nuts will go through large holes in slots.

k. Remove hydraulic pump module.

l. Remove packings from hydraulic pump module. Discard packings.

m. Turn knurled head and push down bleed relief valve on top of removed hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container of at least 2 quarts.

n. If required, teardown/buildup hydraulic pump module (PARA 7-4-12.1.2).

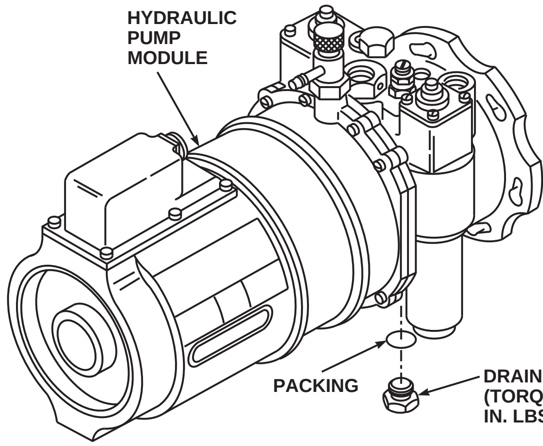


FIGURE 7-4-12-2. TEARDOWN/BUILDUP HYDRAULIC PUMP MODULE

7-4-12.1.2 Hydraulic Pump Module Teardown/Buildup.

NOTE

If replacement pump module is not equipped, fittings shall be removed from old pump module or obtained from stock and installed on replacement pump module.

a. Remove the following:

NOTE

Install protective plugs as items are removed.

(1) Hydraulic pump module pressure and return quick-disconnects (PARA 7-4-17).

(2) Seal drain elbow (PARA 7-4-18).

(3) External hydraulic power quick-disconnect (PARA 7-4-16).

b. Unscrew and remove drain plug from pump module. Discard packing. Install protective plug (Figure 7-4-12-2).

c. Install the following:

NOTE

Remove protective plugs as items are installed.

(1) Hydraulic pump module pressure and return quick-disconnects (PARA 7-4-17).

(2) Seal drain elbow (PARA 7-4-18).

(3) External hydraulic power quick-disconnect (PARA 7-4-16).

d. Coat packing and threads of drain plug with hydraulic fluid, Item 154 or Item 155, Appendix D.

e. Remove protective plug. Install packing on drain plug and install in replacement pump module. **TORQUE DRAIN PLUG TO 119 -**

131 INCH-POUNDS. Using lockwire, Item 197, Appendix D, lockwire drain plug.

7-4-12.1.3 Install.

a. Thoroughly clean accessory module mounting flange, studs, washers, nuts, pump packing grooves, and accessory module seal recess with machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe cleaned areas with machinery towel dampened with isopropyl alcohol, Item 72, Appendix D.

b. Coat new packings with lubricating oil, Item 205, Item 206, or Item 208, Appendix D. Install packings on hydraulic pump module (Figure 7-4-12-1, Detail A). Wipe off excess lubricating oil.

c. Coat spline shaft with lubricating oil, Item 206, Appendix D, to lubricate spline teeth.

NOTE

Install hydraulic pump module while primer is wet.

d. Apply thin coat of epoxy primer from epoxy primer coating kit, Item 135, Appendix D, or primer, Item 258, Appendix D, to accessory module or pump flange interface. Do not prime packing recess. Coat washers and nut contact interfaces with epoxy primer.

e. Support pump module to prevent damage to splined shaft. Position pump module on mounting flange studs. Turn pump module clockwise until studs are seated in small end of keyhole slots on pump flange.

f. Tighten nuts. **TORQUE NUTS TO 261 - 289 INCH-POUNDS, IN CRISSCROSS SEQUENCE.**

g. Apply epoxy primer from epoxy primer coating kit, Item 135, Appendix D, or primer, Item 258, Appendix D, to all exposed washers, nuts, and studs securing hydraulic pump module to accessory module. Allow to dry.

h. Seal area around nuts, washers, studs, and seam between accessory module and hydraulic pump module with sealing compound, Item 292, Appendix D.

i. Apply epoxy primer from epoxy primer coating kit, Item 135, Appendix D, or primer, Item 258, Appendix D, over sealing compound. Allow to dry.

j. Apply one coat of polyurethane paint, Item 224, Appendix D, over epoxy primer.

k. Connect pressure and return lines to quick-disconnects (Figure 7-4-12-1, Detail A). Verify the following for each line:

CAUTION

To prevent total loss of hydraulic pressure and damage to hydraulic components, make sure quick-disconnect couplings are fully connected.

(1) Sleeve on quick-disconnect (female end) contacts snapring.

(2) Locking rings are no longer visible.

(3) Two alignment balls (male end) are seated on two alignment detents on quick-disconnect housing (female end).

l. If removed, connect drain line to tee at bottom of hydraulic pump module.

m. Connect drain line to seal drain elbow.

n. Connect drain line to bleed relief valve.

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Connect electrical connector, P447 (No. 1), or P432 (No. 2), to reservoir fluid quantity switch.

r. Test replacement pump module reservoir low-level switch for proper operation by filling pump module reservoir with hydraulic fluid until ¼-inch of white indicator band is visible in sight glass (PARA 1-3-8).

s. Connect external electrical power (PARA 1-6-16), or run APU (Appropriate Flight Manual). On upper console, place GENERATORS, APU TEST switch to RESET.

t. On upper console, place HYD LEAK TEST switch to RESET. Check that RSVR LOW light is on for pump module being tested.

u. Complete servicing of hydraulic pump module (PARA 1-3-8).

v. On upper console, place HYD LEAK TEST switch to RESET. On caution/advisory panel, check that all three RSVR LOW lights are off.

w. On upper console, place HYD LEAK TEST switch to TEST. On caution/advisory panel, check that caution/advisory panel #1 RSVR LOW, #2 RSVR LOW and BACKUP RSVR LOW lights are on.

x. Connect external hydraulic power to backup hydraulic system (PARA 1-6-15).

y. Bleed hydraulic pump module (PARA 7-4-65).

NOTE

If pressure does not come up after bleeding, there is a possibility hydraulic fluid is foaming. Shut down and service hydraulic pump module again (PARA 1-3-8). Repeat operational check.

z. Perform operational check of hydraulic system (PARA 7-2-1).

aa. Make sure area is clean and free of foreign material.

ab. Close and latch main rotor pylon sliding cover.

7-4-13 BACKUP HYDRAULIC PUMP MODULE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Crowfoot Attachment
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Cleaning Compound, Item 95, Appendix D
- Grease, Item 150, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-16
- PARA 7-4-17
- PARA 7-4-18
- PARA 7-4-65

7-4-13.1 Replace Backup Hydraulic Pump Module.

7-4-13.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Remove bolts, washers, nuts, inner and outer struts, and spacers (if installed) from lower clamp and inner and outer supports (Figure 7-4-13-1, Sheet 1, Detail A).

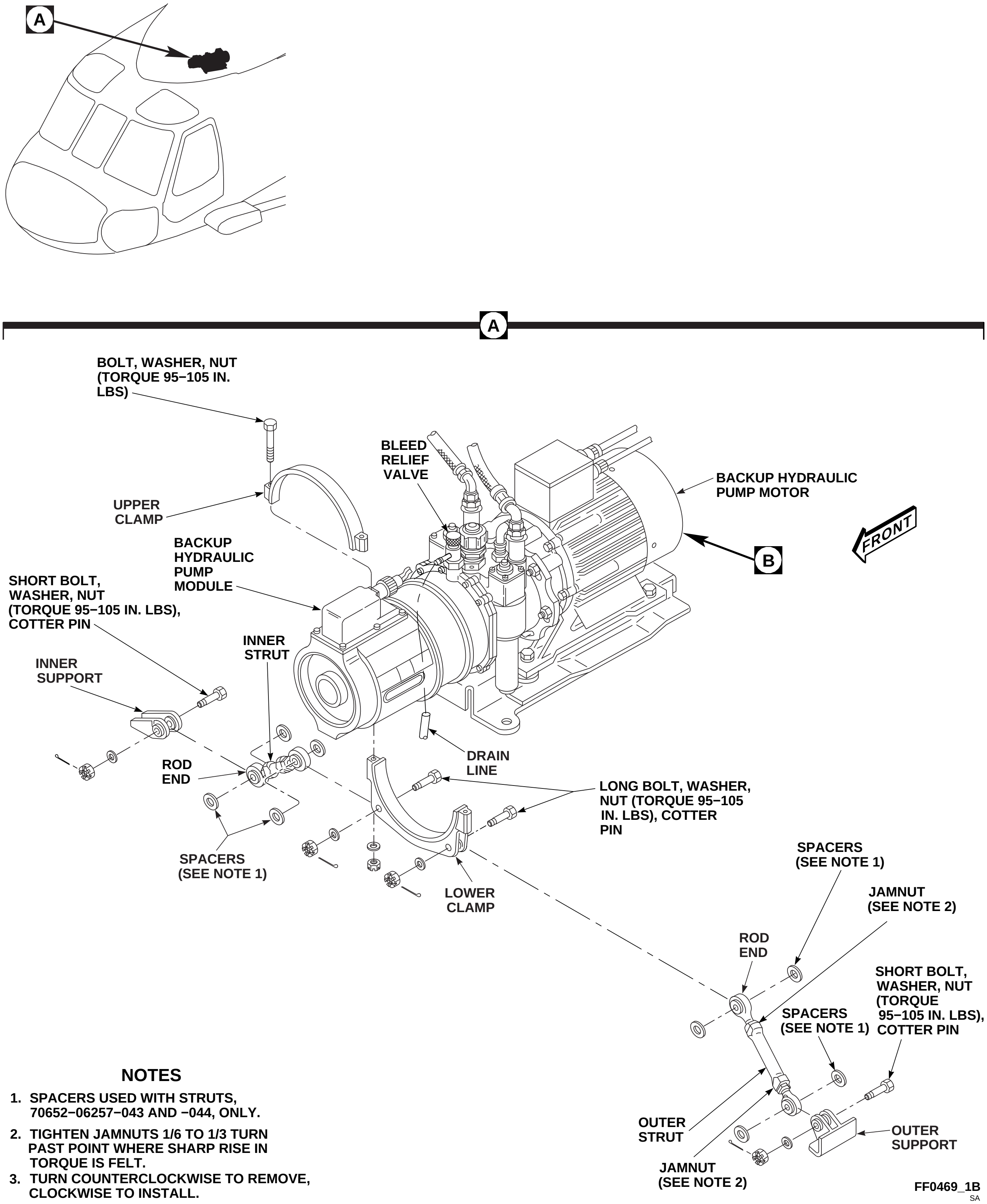
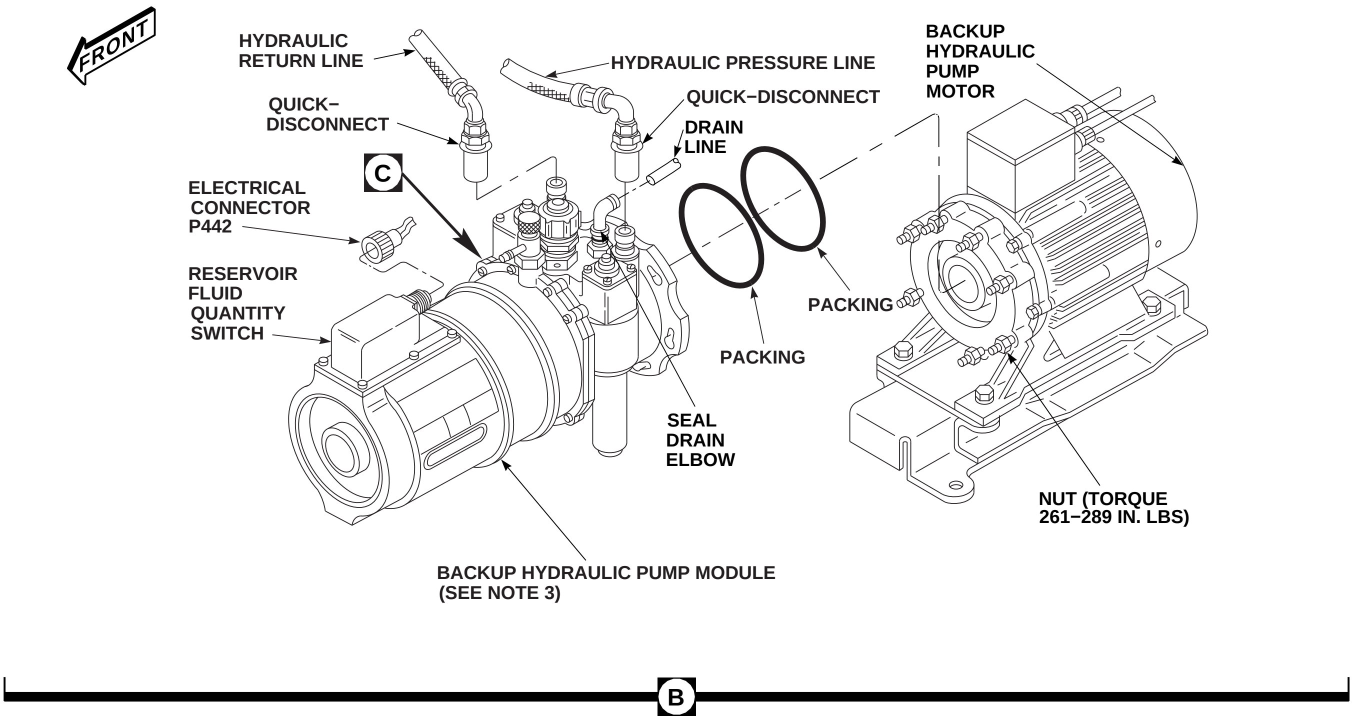


FIGURE 7-4-13-1. REPLACE BACKUP HYDRAULIC PUMP MODULE (SHEET 1 OF 2)



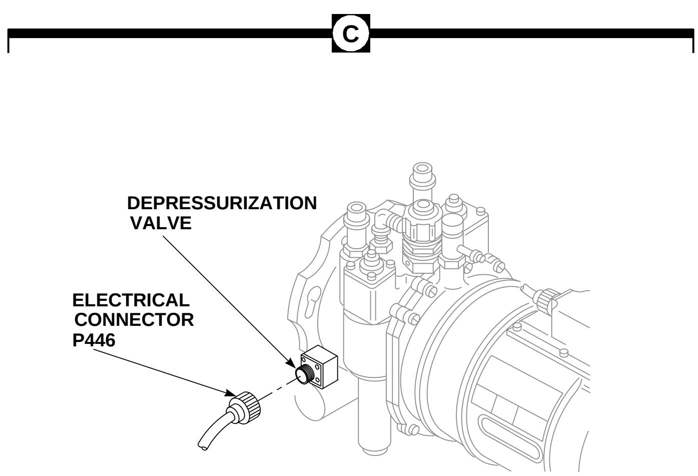


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FIGURE 7-4-13-1. REPLACE BACKUP HYDRAULIC PUMP MODULE (SHEET 2 OF 2)

e. Remove bolts, washers, and nuts securing upper clamp to lower clamp. Remove clamps from backup hydraulic pump module.

f. Disconnect hydraulic pressure and hydraulic return lines at quick-disconnects (Figure 7-4-13-1, Sheet 2, Detail B).

g. If installed, disconnect drain line from tee at bottom of backup hydraulic pump module.

h. Disconnect drain line from seal drain elbow.

i. Disconnect electrical connector, P442, from reservoir fluid quantity switch.

j. Install protective caps and plugs on electrical connectors.

k. Disconnect drain line from bleed relief valve (Figure 7-4-13-1, Sheet 1, Detail A).

l. Disconnect electrical connector, P446, from depressurization valve (Figure 7-4-13-1, Sheet 2, Detail C).

m. Install protective caps and plugs on electrical connectors.

n. Loosen nuts securing hydraulic pump module to backup hydraulic pump motor (Figure 7-4-13-1, Sheet 2, Detail B).

o. Turn backup hydraulic pump module counterclockwise until nuts will go through large holes in slots.

p. Remove backup hydraulic pump module.

q. Remove packings from backup hydraulic pump module. Discard packings.

r. Turn knurled head and push down bleed relief valve, on top of removed hydraulic pump module. Drain overflow hydraulic pump module fluid from bleed relief valve into a container of at least 2-quarts (Figure 7-4-13-1, Sheet 1, Detail A).

s. If required, teardown/buildup backup hydraulic pump module (PARA 7-4-13.1.2).

7-4-13.1.2 Backup Hydraulic Pump Module Teardown/Buildup.

NOTE

If replacement pump module is not equipped, fittings shall be removed from old pump module or obtained from stock and installed on replacement pump module.

a. Remove the following:

NOTE

Install protective plugs as items are removed.

(1) Hydraulic pump module pressure and return quick-disconnects (PARA 7-4-17).

(2) Seal drain elbow (PARA 7-4-18).

(3) External hydraulic power quick-disconnect (PARA 7-4-16).

b. Unscrew and remove drain plug from pump module. Discard packing. Install protective plug (Figure 7-4-13-2).

c. Install the following:

NOTE

Remove protective plugs as items are installed.

(1) Hydraulic pump module pressure and return quick-disconnects (PARA 7-4-17).

(2) Seal drain elbow (PARA 7-4-18).

(3) External hydraulic power quick-disconnect (PARA 7-4-16).

d. Coat packing and threads of drain plug with hydraulic fluid, Item 154 or Item 155, Appendix D.

e. Remove protective plug. Install packing on drain plug and install in replacement pump module. **TORQUE DRAIN PLUG TO 119 - 131 INCH-POUNDS.**

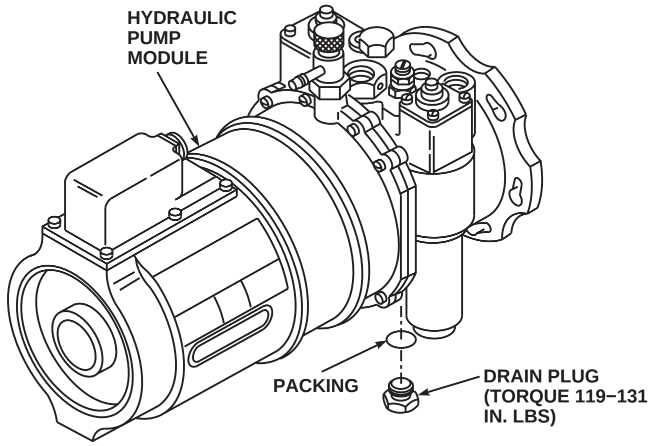


FIGURE 7-4-13-2. TEARDOWN/BUILDUP BACKUP HYDRAULIC PUMP MODULE

f. Using lockwire, Item 197, Appendix D, lockwire drain plug.

7-4-13.1.3 Install.

NOTE

Damage to backup hydraulic pump motor and backup hydraulic pump module will result if packing recesses are not clean during assembly. Make sure packing recesses are clean to allow packings to seal backup hydraulic pump motor and backup hydraulic pump module properly.

a. Inspect and clean recess in backup hydraulic pump motor with cleaning compound, Item 95, Appendix D. Wipe dry with machinery towels, Item 211, Appendix D.

b. Coat packings with lubricating oil, Item 205, Item 206, or Item 208, Appendix D.

c. Place packings on backup hydraulic pump module (Figure 7-4-13-1, Sheet 2, Detail B).

d. Coat spline shaft with enough grease, Item 150, Appendix D, to fill spline teeth.

e. Support reservoir end of backup hydraulic pump module, to prevent damage to splined shaft. Place backup hydraulic pump module on mounting flange studs.

f. Turn backup hydraulic pump module clockwise until studs are seated in small end of keyhole slots on pump flange.

NOTE

Use concentric torque values when torquing nuts.

g. **TORQUE NUTS TO 261 - 289 INCH-POUNDS, IN CRISSCROSS SEQUENCE.** Use crowfoot attachment to torque lower nuts.

h. Connect pressure and return lines to quick-disconnects on backup hydraulic pump module. Verify the following for each line:

CAUTION

To prevent total loss of hydraulic pressure and damage to hydraulic components, make sure quick-disconnect couplings are fully connected.

(1) Sleeve on quick-disconnect (female end) contacts snapring.

(2) Locking rings are no longer visible.

(3) Two alignment balls (male end) are seated on two alignment detents on quick-disconnect housing (female end).

i. If removed, connect drain line to tee at bottom of backup hydraulic pump module.

j. Connect drain line to seal drain elbow.

k. Connect drain line to bleed relief valve (Figure 7-4-13-1, Sheet 1, Detail A).

l. Remove protective caps and plugs from electrical connectors.

m. Inspect and treat electrical connectors (PARA 1-7-20).

n. Connect electrical connector, P442, to reservoir fluid quantity switch (Figure 7-4-13-1, Sheet 2, Detail B).

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Connect electrical connector, P446, to depressurization valve (Figure 7-4-13-1, Sheet 2, Detail C).

NOTE

Some upper clamps have a decal installed stating: TORQUE TO 50 INCH-POUNDS. If present, the decal is to be removed prior to installation of upper clamp.

r. Check upper clamp for decal stating: TORQUE TO 50 INCH-POUNDS. If present, remove decal.

s. Install upper and lower clamps on backup hydraulic pump module reservoir between two ribs. Secure clamps together with bolts, washers, and nuts. Leave clamps loose so they can be turned if necessary (Figure 7-4-13-1, Sheet 1, Detail A).

t. If new inner strut or rod end is being installed, coat threads of rod end with antiseize compound, Item 74, Appendix D, then set length, rod end bolthole center to rod end bolthole center, to 3 5/8-inches. Do not lockwire now.

u. If new outer strut or rod end is being installed, coat threads of rod end with antiseize compound, Item 74, Appendix D, then set length, rod end bolthole center to rod end bolthole center, to 6 1/8-inches. Do not lockwire now.

NOTE

If struts, 70652-06257-043 and -044 are used, install spacers between rod ends and supports/clamp clevises.

v. Install large rod end of inner strut in inner side of lower clamp and install long bolt (bolthead to face rear), washer, spacers, if required, and nut. Do not torque now.

w. Install spacers, if required, and small rod end of inner strut in inner side of inner support and install short bolt (bolthead to face rear), washer, and nut. Do not torque now.

x. Install spacers, if required, and large end of outer strut in outer side of lower clamp and install long bolt (bolthead to face rear), washer, and nut. Do not torque now.

y. Install spacers, if required, and small rod end of outer strut in outer side of outer support and install short bolt (bolthead to face rear), washer, and nut. Do not torque now.

z. Slide clamp between ribs on backup hydraulic pump module and adjust both inner and outer strut lengths until attaching bolts at inner and outer supports may be installed without being forced.

aa. **TORQUE CLAMP BOLTS TO 95 - 105 INCH-POUNDS.**

ab. **TIGHTEN STRUT JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

ac. Using lockwire, Item 197, Appendix D, lockwire jamnuts to tube.

ad. Check that bolts securing strut are not binding.

ae. **TORQUE NUTS SECURING LONG AND SHORT BOLTS TO 95 - 105 INCH-POUNDS.** Install cotter pins.

af. Service backup hydraulic pump module (PARA 1-3-8).

ag. Bleed backup hydraulic pump module using bleed relief valve (PARA 7-4-65).

NOTE

If pressure does not come up after bleeding, there is a possibility hydraulic fluid is foaming. Shut down and service backup hydraulic pump module again (PARA 1-3-8). Repeat operational check.

ah. Perform operational check of hydraulic system (PARA 7-2-1).

ai. Make sure area is clean and free of foreign material.

aj. Close and latch main rotor pylon sliding cover.

7-4-14 HYDRAULIC PUMP MODULE FILTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Socket Extension, 3⁄8-inch
- Wrench

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- Appendix Flight Manual
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-65

7-4-14.1 Replace Hydraulic Pump Module Filter.

NOTE

- Replacement of the No. 1, No. 2, and backup hydraulic pump module filters is the same.
- There are two filters in each hydraulic pump module.

7-4-14.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under bleed relief valve drain line.

e. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-14-1, Detail A).

f. Place machinery towels, Item 211, Appendix D, under filter bowl to catch any hydraulic fluid that might spill.

g. Using wrench on flats of filter bowl, unscrew filter bowl from hydraulic pump module.

h. Remove filter and discard (Figure 7-4-14-1, Detail B).

i. Using hydraulic fluid, Item 154 or Item 155, Appendix D, clean filter bowl.

j. Reset filter indicator pushbutton before installing filter as follows:

(1) Push indicator pushbutton down (Figure 7-4-14-1, Detail A).

(2) While holding indicator pushbutton down, push up reset plunger inside cavity, using clean ⅜-inch socket extension.

7-4-14.1.2 Install.

NOTE

Make sure indicator pushbutton has been reset before installing new filter.

a. Place clean filter into filter bowl (Figure 7-4-14-1, Detail B).

b. Make sure hydraulic pump module filter bowl mounting base is clean.

c. Fill filter bowl with clean, fresh hydraulic fluid, Item 154 or Item 155, Appendix D.

CAUTION

Damage to the hydraulic pump module and filter bowl will result if filter bowl is overtightened. When installing filter bowl into pump, do not overtighten.

d. Install filter bowl into pump (Figure 7-4-14-1, Detail A). **TIGHTEN ONLY ENOUGH TO SEAT BOWL.**

e. Service hydraulic pump module (PARA 1-3-8).

f. Bleed hydraulic pump module (PARA 7-4-65).

NOTE

If external hydraulic power is in use, increase hydraulic pressure to 3,000 psi. If backup hydraulic pump module is used, service backup hydraulic pump module reservoir, if required, prior to placing BACKUP HYD PUMP switch to ON.

g. Connect external electrical power (PARA 1-6-16), or run APU (Appropriate Flight Manual).

h. Turn on all electrical power (PARA 1-6-16).

i. On caution/advisory panel, make sure #1 RSVR LOW, #2 RSVR LOW, or BACKUP PUMP RSVR LOW light is off.

j. If any caution lights are on and oil level window on hydraulic pump module shows full, perform hydraulic leak test (PARA 7-2-1).

k. Inspect filter bowl for leaks. **NONE ALLOWED.** If filter bowl leaks, replace bowl.

l. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

m. Make sure area is clean and free of foreign material.

n. Close main rotor pylon sliding cover.

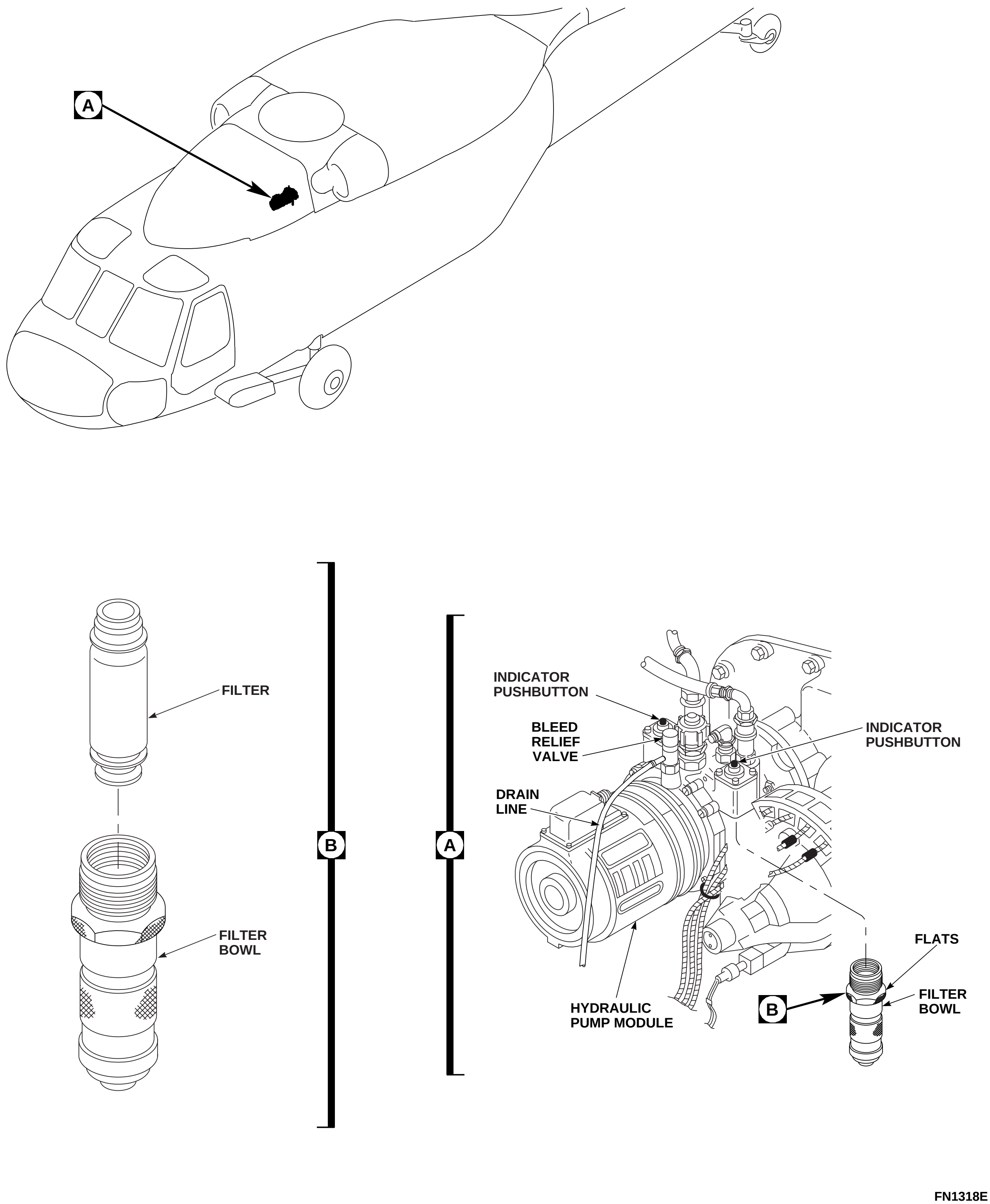


FIGURE 7-4-14-1. REPLACE HYDRAULIC PUMP MODULE FILTER

7-4-15 HYDRAULIC PUMP MODULE FILTER HOUSING PACKING AND RETAINER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- C-Clamp, 6-inch opening
- Container, 2-Quart
- Filter Tool, 62617-T201
- Pusher Tube, Locally-Made, Figure H-172, Appendix H

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Sealing Compound, Item 276, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-12
- PARA 7-4-14
- PARA 7-4-23

7-4-15.1 Replace Hydraulic Pump Module Filter Housing Packing and Retainer.

NOTE

- Replacement of the No. 1, No. 2, and backup hydraulic pump module filter housing packing and retainers is the same.
- There are two filter housings in each hydraulic pump module.

7-4-15.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Remove hydraulic pump module (PARA 7-4-12).

c. Position 2-quart container under drain line under bleed relief valve.

d. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain

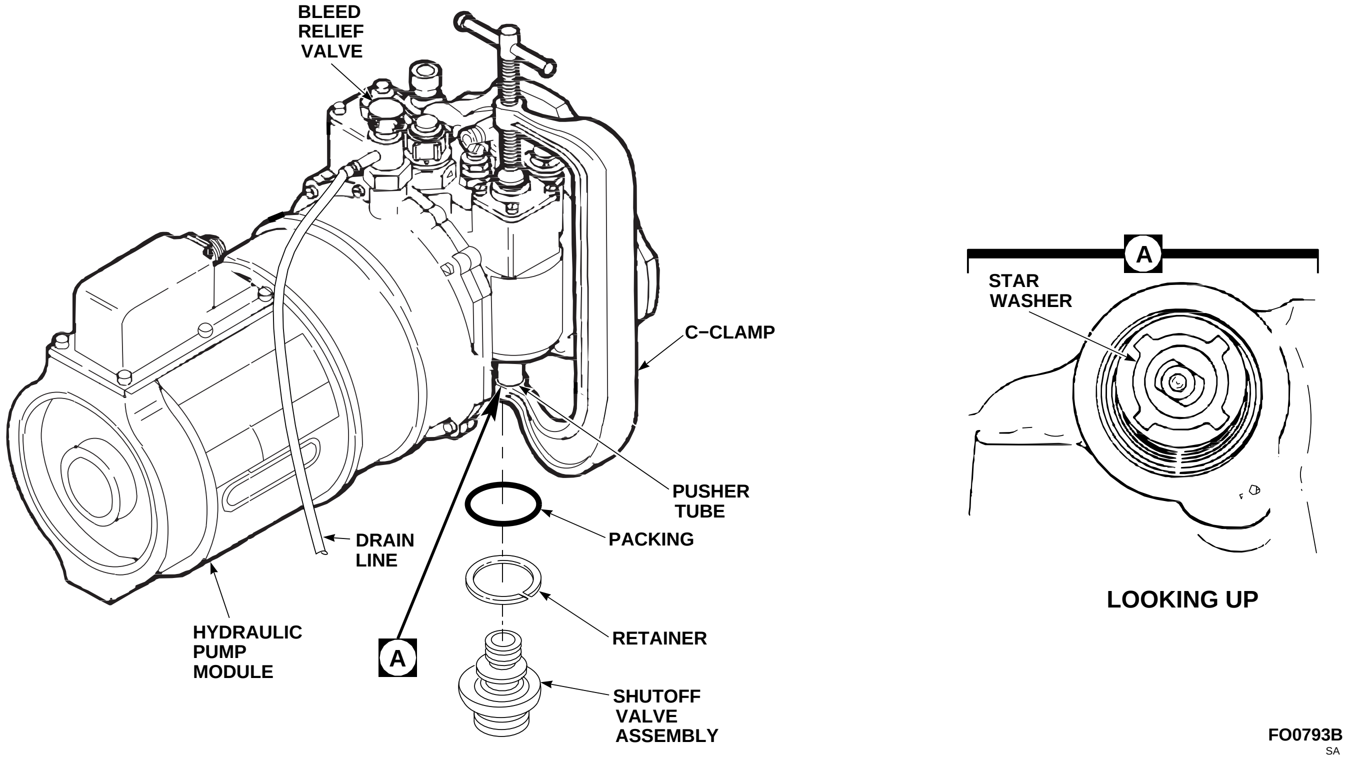


FIGURE 7-4-15-1. REPLACE HYDRAULIC PUMP MODULE FILTER HOUSING PACKING AND RETAINER

overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-15-1).

e. Remove filter from filter housing that is leaking (PARA 7-4-14).

f. Locally make pusher tube (Figure H-172, Appendix H).

g. Using C-clamp with 6-inch opening and pusher tube, depress star washer in filter cavity (Figure 7-4-15-1 and Detail A).

h. Remove packing and retainer from filter housing and discard. Remove C-clamp and pusher tube.

7-4-15.1.2. Install.

a. Coat new packing and retainer with hydraulic fluid, Item 154 or Item 155, Appendix D, and install them in filter housing (Figure 7-4-15-1).

b. If necessary, install new filter (PARA 7-4-14).

c. Install hydraulic pump module (PARA 7-4-12). Check for leaks.

d. If leakage is detected at filter bowl, perform the following:

(1) Remove hydraulic pump module (PARA 7-4-12).

(2) Remove filter from leaking bowl (PARA 7-4-14).

(3) Remove differential pressure indicator from leaking bowl (PARA 7-4-23).

NOTE

Shutoff valve assembly has left-hand threads.

(4) Remove shutoff valve assembly from hydraulic pump module, using filter tool.

(5) Remove and replace filter housing packing and retainer.

(6) Apply sealing compound, Item 276, Appendix D, to threads of shutoff valve assembly.

(7) Apply hydraulic fluid, Item 154 or Item 155, Appendix D, to filter bowl mating area below threads of shutoff valve assembly.

(8) Using filter tool, install shutoff valve assembly in hydraulic pump module.

(9) Install filter (PARA 7-4-14).

(10) Install differential pressure indicator (PARA 7-4-23).

(11) Install hydraulic pump module (PARA 7-4-12). Repeat leak check.

7-4-16 EXTERNAL HYDRAULIC POWER QUICK-DISCONNECT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-65

7-4-16.1 Replace External Hydraulic Power Quick-Disconnect.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module external hydraulic power quick-disconnect is the same.

7-4-16.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under drain line at left main wheel.

e. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-16-1, Detail A).

f. Remove dust cap from quick-disconnect and let it hang by chain.

g. Unscrew and remove external hydraulic quick-disconnect from hydraulic pump module. Remove and discard packing.

7-4-16.1.2 Install.

a. On new quick-disconnect, perform the following:

(1) Disconnect quick-disconnect as necessary to remove perforated circular plate. Remove plate.

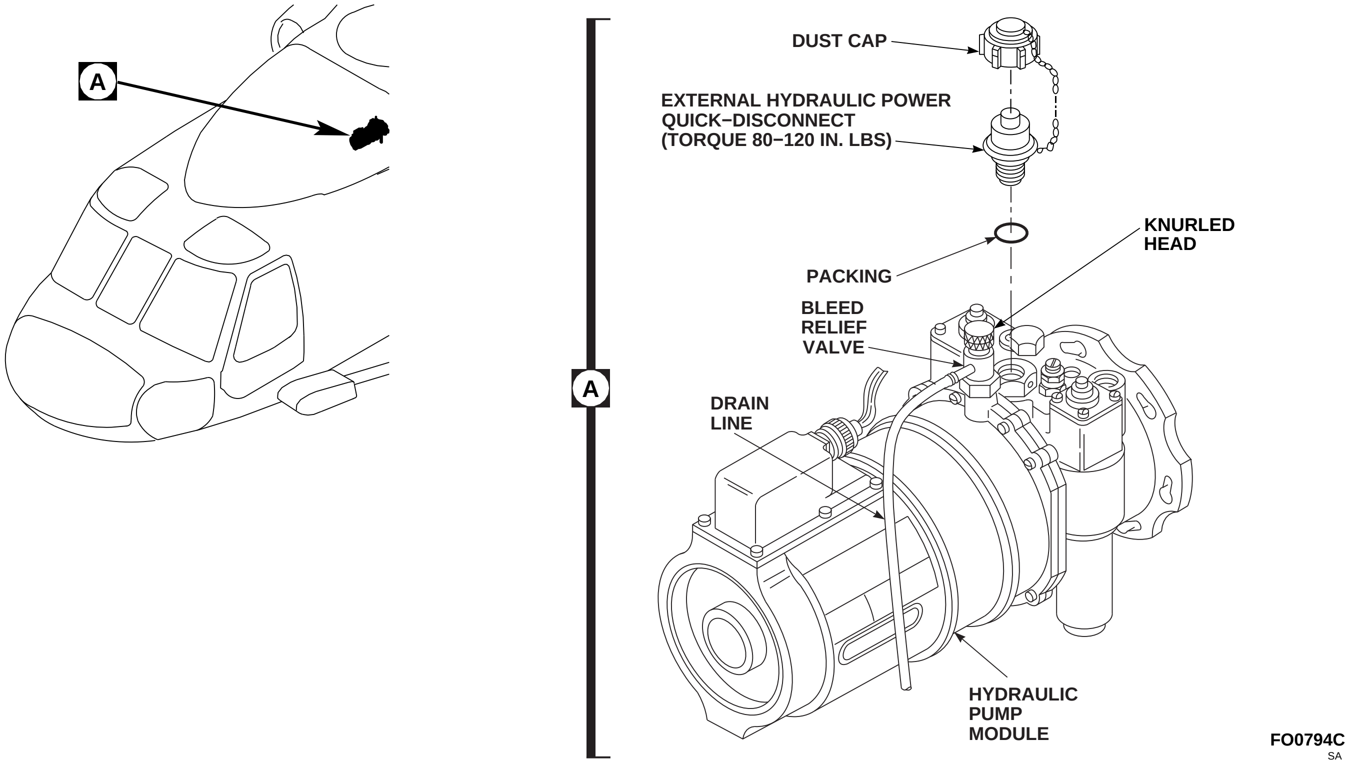


FIGURE 7-4-16-1. REPLACE EXTERNAL HYDRAULIC POWER QUICK-DISCONNECT

(2) Assemble quick-disconnect.

(3) Connect lanyard to cap anti-slip lock on quick-disconnect.

b. Coat packing and threads of quick-disconnect with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Install external hydraulic power quick-disconnect and packing into hydraulic pump module. **TORQUE DISCONNECT HALF TO 80 - 120 INCH-POUNDS** (Figure 7-4-16-1, Detail A).

d. Install dust cap on quick-disconnect.

e. Service hydraulic pump module (PARA 1-3-8).

f. Bleed hydraulic pump module (PARA 7-4-65).

g. Make sure area is clean and free of foreign material.

h. Close main rotor pylon sliding cover.

7-4-16.2 Replace External Hydraulic Power Quick-Disconnect Port Fitting.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module external hydraulic power quick-disconnect port fitting is the same.

7-4-16.2.1 Remove.

a. Remove external hydraulic power quick-disconnect (PARA 7-4-16.1.1).

b. Remove port fitting and packing from hydraulic pump module (Figure 7-4-16-2). Discard packing.

7-4-16.2.2 Install.

a. Coat packing and threads of port fitting with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install packing on port fitting and install in hydraulic pump module. **TORQUE PORT**

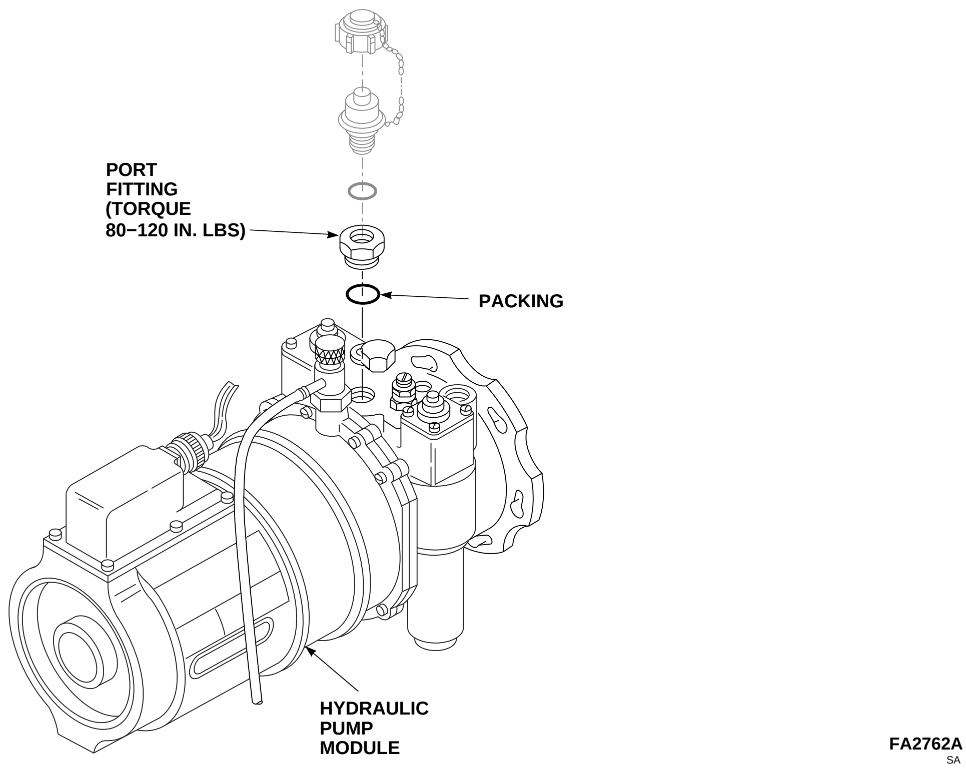


FIGURE 7-4-16-2. REPLACE EXTERNAL HYDRAULIC POWER QUICK-DISCONNECT PORT FITTING

FITTING TO 80 - 120 INCH-POUNDS (Figure 7-4-16-2). Lockwire, Item 197, Appendix D, port fitting.

c. Install external hydraulic power quick-disconnect (PARA 7-4-16.1.2).

7-4-17 HYDRAULIC PUMP MODULE PRESSURE AND RETURN QUICK-DISCONNECTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-65

7-4-17.1 Replace Hydraulic Pump Module Pressure and Return Quick-Disconnects.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module pressure and return quick-disconnects is the same.

7-4-17.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under drain line at left main wheel.

e. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-17-1, Detail A).

f. Disconnect quick-disconnect (female end) and hydraulic line from quick-disconnect (male end) as an assembly (Figure 7-4-17-1, Detail B).

g. Unscrew and remove quick-disconnect (male end) from pump boss. Remove packings from

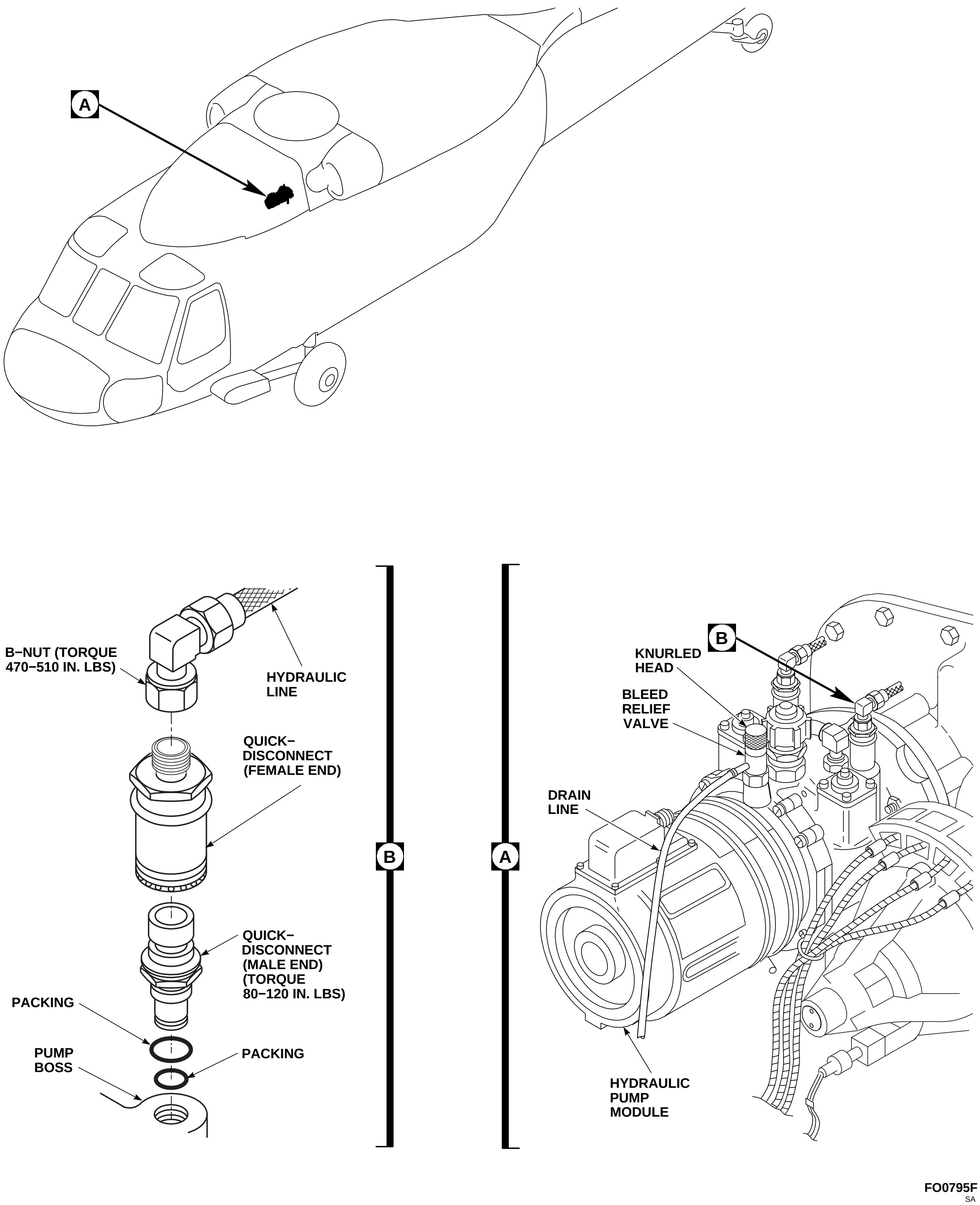


FIGURE 7-4-17-1. REPLACE HYDRAULIC PUMP MODULE PRESSURE AND RETURN QUICK-DISCONNECTS quick-disconnect and pump boss. Discard packings.

h. Remove quick-disconnect (female end) from hydraulic line.

7-4-17.1.2 Inspect.

Inspect pressure and return quick-disconnects for corrosion and evidence of wear on locking balls. **NONE ALLOWED.** If corrosion or wear is found, replace quick-disconnect.

7-4-17.1.3 Install.

a. Coat all packings and quick-disconnect threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install packings on quick-disconnect and pump boss (Figure 7-4-17-1, Detail B).

c. Install quick-disconnect (male end) and packings into pump boss. **TORQUE QUICK-DISCONNECTS TO 80 - 120 INCH-POUNDS.** Lockwire, Item 197, Appendix D, quick-disconnect.

d. Connect hydraulic line to quick-disconnect (female end). **TORQUE B-NUT TO 470 - 510 INCH-POUNDS.**

CAUTION

To prevent total loss of hydraulic pressure and damage to hydraulic components, make sure quick-disconnect couplings are fully connected.

e. Connect quick-disconnects, making sure of the following:

(1) Sleeve on quick-disconnect (female end) contacts snapring.

(2) Locking rings are no longer visible.

(3) Two alignment balls (male end) are seated on the two alignment detents of quick-disconnect housing (female end).

f. Service hydraulic pump module (PARA 1-3-8).

g. Bleed hydraulic pump module (PARA 7-4-65).

h. Connect external electrical power and external hydraulic power to system in which component was replaced (PARA 1-6-15 and PARA 1-6-16), or ground run helicopter and check for leaks (Appropriate Flight Manual).

i. Make sure area is clean and free of foreign material.

j. Close main rotor pylon sliding cover.

7-4-18 HYDRAULIC PUMP MODULE SEAL DRAIN ELBOW.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16

7-4-18.1 Replace Hydraulic Pump Module Seal Drain Elbow.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module seal drain elbows is the same.

7-4-18.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect drain line from seal drain elbow (Figure 7-4-18-1, Detail A).

e. Loosen jamnut and remove seal drain elbow and packing from hydraulic pump module. Discard packing.

f. Remove jamnut from seal drain elbow.

7-4-18.1.2 Install.

a. Install jamnut on replacement seal drain elbow (Figure 7-4-18-1, Detail A).

b. Coat packing and threads of seal drain elbow with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Install packing on seal drain elbow. Install seal drain elbow in hydraulic pump module and position elbow toward drain line. **TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**

d. Connect drain line on seal drain elbow.

e. Make sure area is clean and free of foreign material.

f. Close main rotor pylon sliding cover.

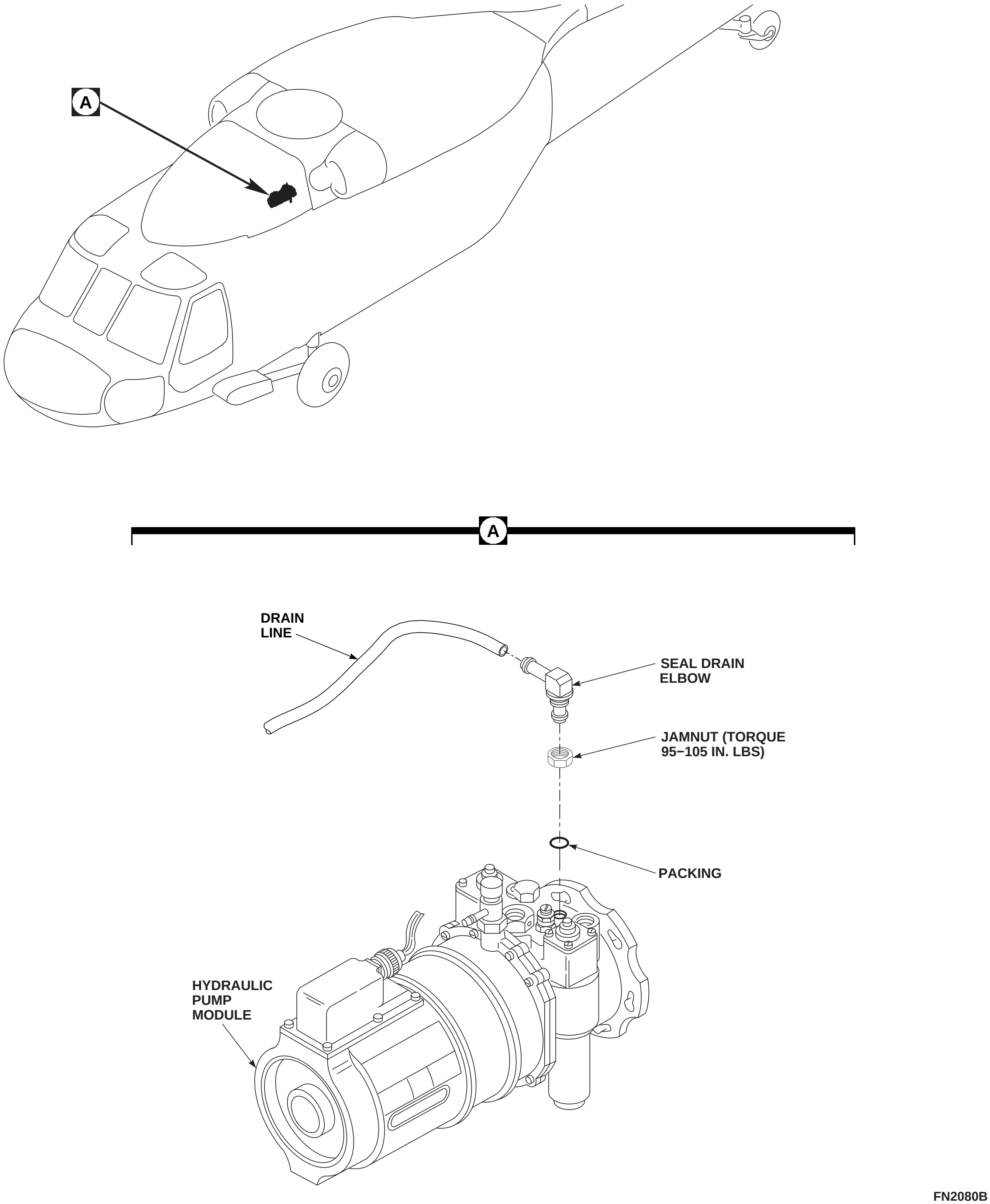


FIGURE 7-4-18-1. REPLACE HYDRAULIC PUMP MODULE SEAL DRAIN ELBOW

7-4-19 HYDRAULIC PUMP MODULE BLEED RELIEF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16

7-4-19.1 Replace Hydraulic Pump Module Bleed Relief Valve.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module bleed relief valves is the same.

7-4-19.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under drain line at left main wheel.

e. Turn knurled head on bleed relief valve, on top of hydraulic pump module, and push bleed relief valve down. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-19-1, Detail A).

f. Disconnect drain line from bleed relief valve.

g. Unscrew and remove bleed relief valve and packing from top of hydraulic pump module. Discard packing.

7-4-19.1.2 Install.

a. Coat packing and threads of bleed relief valve with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install bleed relief valve and packing in hydraulic pump module (Figure 7-4-19-1, Detail A). **TORQUE VALVE TO 80 - 100**

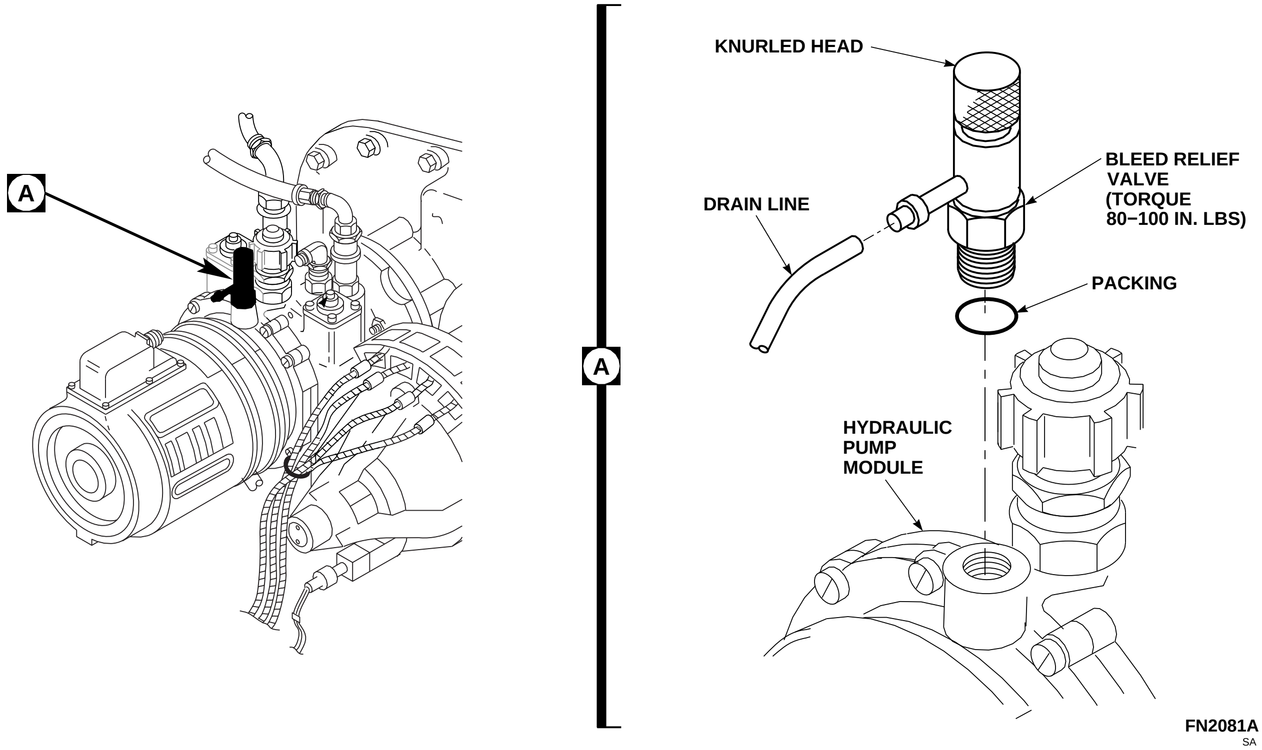


FIGURE 7-4-19-1. REPLACE HYDRAULIC PUMP MODULE BLEED RELIEF VALVE

INCH-POUNDS. Lockwire, Item 197, Appendix D, valve.

c. Connect drain line to bleed relief valve.

d. Service hydraulic pump module (PARA 1-3-8).

e. Check for leaks.

f. Make sure area is clean and free of foreign material.

g. Close and latch main rotor pylon sliding cover.

7-4-20 HYDRAULIC PUMP MODULE SIGHT GLASS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 63, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-3-8

7-4-20.1 Replace Hydraulic Pump Module Sight Glass.

7-4-20.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Position 2-quart container under drain line at left main wheel.

c. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-20-1, Detail A).

d. Remove screw and washer from sight glass. Remove sight glass from hydraulic pump module (Figure 7-4-20-1, Detail B).

7-4-20.1.2 Inspect/Repair.

a. Inspect gasket on sight glass for damage that may cause leakage. **NONE ALLOWED.** If damage is found, replace gasket as follows:

(1) Using nonmetallic scraper, remove gasket from sight glass (Figure 7-4-20-1, Detail B). Discard gasket.

(2) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from sight glass.

(3) Apply a light coat of adhesive, Item 63, Appendix D, on gasket. Place gasket in groove of sight glass and allow to air dry for 30 minutes.

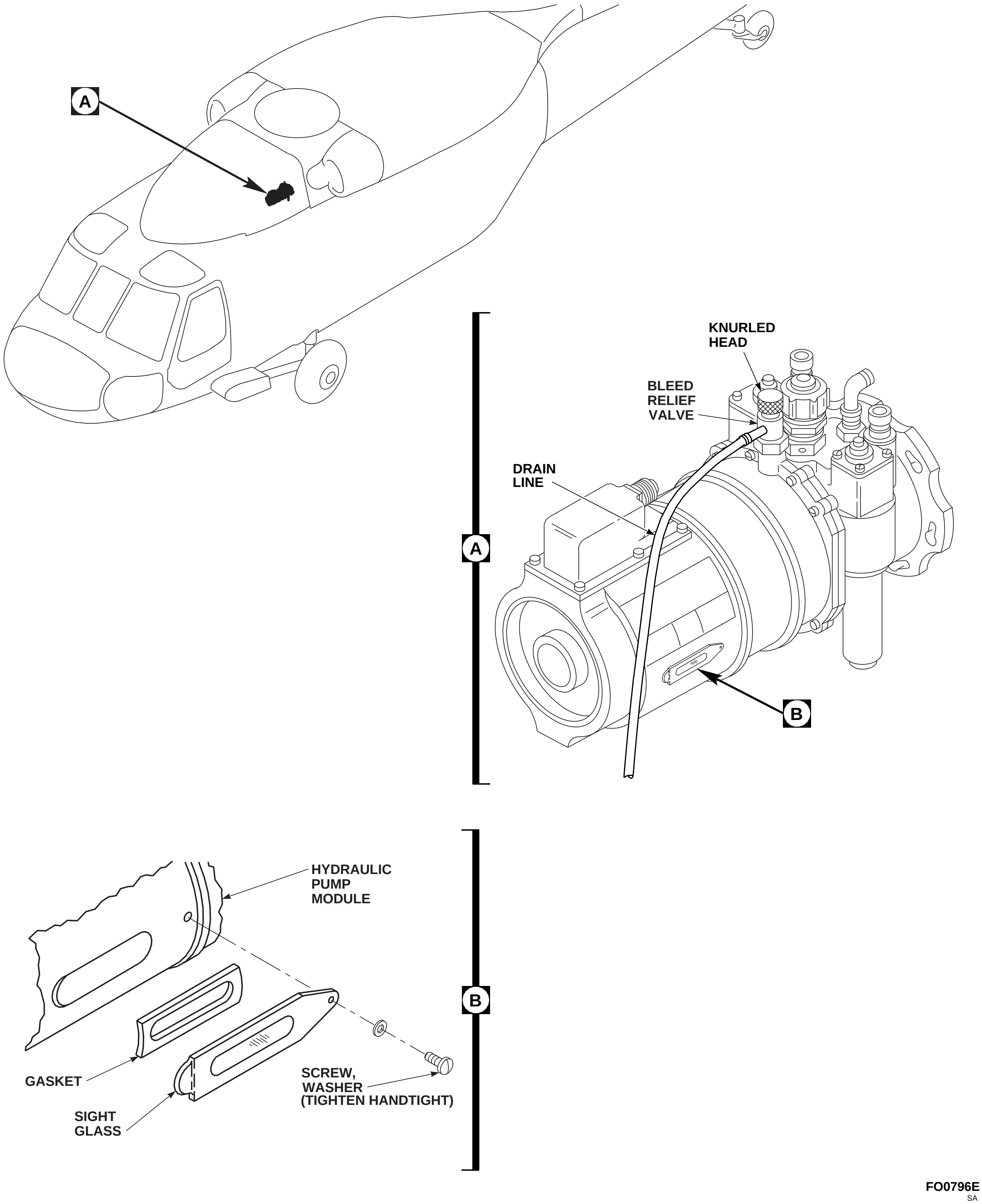


FIGURE 7-4-20-1. REPLACE HYDRAULIC PUMP MODULE SIGHT GLASS

7-4-20.1.3 Install.

a. Install sight glass in recess of hydraulic pump module using screw and washer (Figure 7-4-20-1, Detail B). **TIGHTEN SCREW HANDTIGHT.**

b. Service hydraulic pump module (PARA 1-3-8).

c. Make sure area is clean and free of foreign material.

d. Close main rotor pylon sliding cover.

7-4-21 HYDRAULIC PUMP MODULE TEMPERATURE INDICATORS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Identification Marker, Item 156, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 245, Appendix D
- Temp Indicator Card, Item 323, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 7-3-3

7-4-21.1 Replace Hydraulic Pump Module Temperature Indicators.

NOTE

- There are two temperature indicators located on each hydraulic pump module, one on each side. Replacement for both temperature indicators is the same.
- Two different types of temperature indicators are available. Either type may be used.
- Replace temperature indicator if loose or contaminated.

a. Use nonmetallic scraper to remove temperature indicator (Figure 7-4-21-1).

b. Clean area where temperature indicator is attached using machinery towel, Item 211, Appendix D, dampened in methyl ethyl ketone, Item 217, Appendix D.

c. Position identification marker, Item 156, Appendix D, or temp indicator card, Item 323, Appendix D, on module.

d. Apply pressure sensitive adhesive tape, Item 245, Appendix D, over temperature indicator with at least 1/4-inch of tape overlapping all sides of indicator.

e. Ground run system for 15 minutes (Appropriate Flight Manual).

f. Check temperature indicators on both sides of hydraulic pump module:

(1) **ON TEMP INDICATOR CARD, ITEM 323, APPENDIX D, IF DOTS UP TO AND INCLUDING 275°F (135°C) INDICATE BLACK, INSPECT**

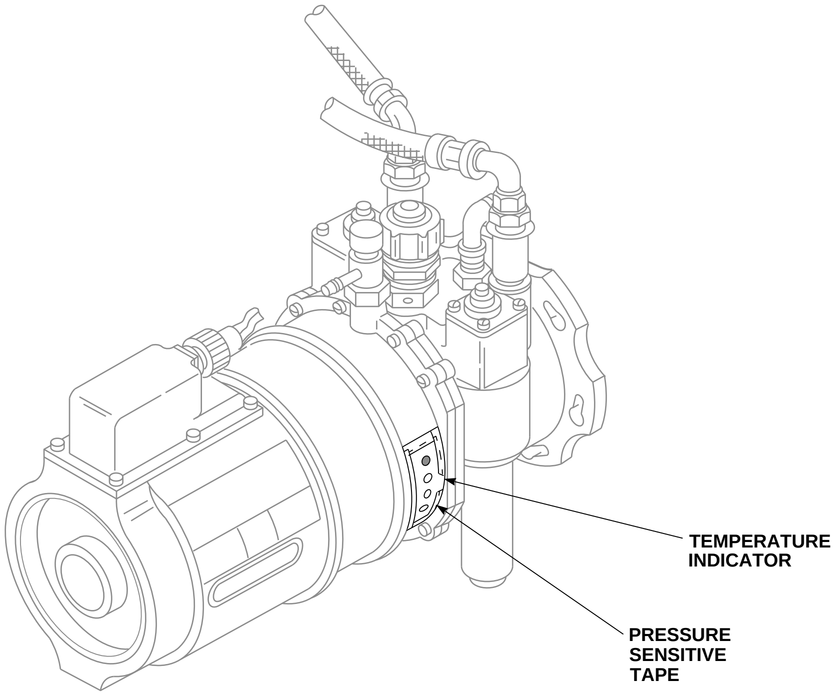


FIGURE 7-4-21-1. REPLACE HYDRAULIC PUMP MODULE TEMPERATURE INDICATORS

HYDRAULIC PUMP MODULE (PARA 7-3-3).

(2) **ON IDENTIFICATION MARKER, ITEM 156, APPENDIX D, IF DOTS UP TO AND INCLUDING 270°F (132°C) INDICATE BLACK, INSPECT HYDRAULIC PUMP MODULE** (PARA 7-3-3).

7-4-22 HYDRAULIC PUMP MODULE FLUID IDENT AND LEVEL INDICATOR PLATE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

7-4-22.1 Replace Hydraulic Pump Module Fluid Ident and Level Indicator Plate.

NOTE

There are two plates 180° apart on the hydraulic module. Replacement of both plates is the same.

7-4-22.1.1. Remove.

a. Using nonmetallic scraper, remove fluid ident and level indicator plate from hydraulic pump module (Figure 7-4-22-1).

b. Clean area on hydraulic pump module where fluid ident and level indicator plate was bonded using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

7-4-22.1.2. Install.

a. Remove backing from fluid ident and level indicator plate and position plate on hydraulic pump module (Figure 7-4-22-1).

b. Press down firmly to make sure fluid ident and level indicator plate is in full contact with hydraulic pump module.

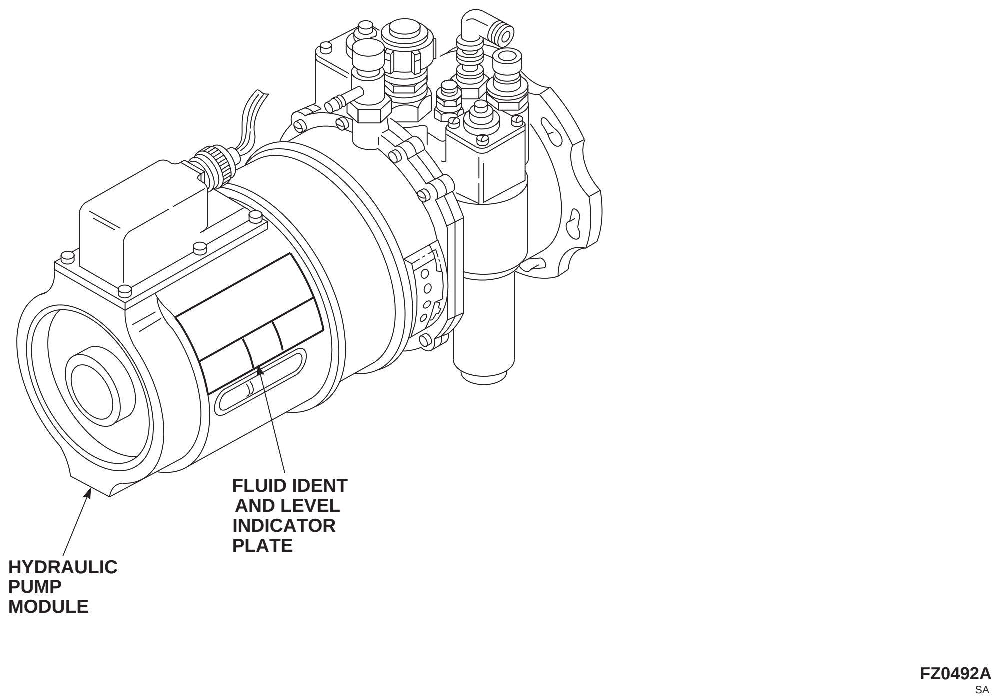


FIGURE 7-4-22-1. REPLACE HYDRAULIC PUMP MODULE FLUID IDENT AND LEVEL INDICATOR PLATE

7-4-23 HYDRAULIC PUMP MODULE DIFFERENTIAL PRESSURE INDICATOR AND SHUTOFF ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Shutoff Assembly Tool, 62617-T201
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Sealing Compound, Item 285, Appendix D
- Sealing Compound, Item 286, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-14
- PARA 7-4-65

7-4-23.1 Replace Hydraulic Pump Module Differential Pressure Indicator.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module differential pressure indicators is the same.

7-4-23.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under drain line at left main wheel.

e. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-23-1).

f. Remove retaining band and dust cap from differential pressure indicator on hydraulic pump module, 70652-02300-049 and 70652-02300-050.

g. Remove differential pressure indicator from hydraulic pump module. Discard retainer and packings.

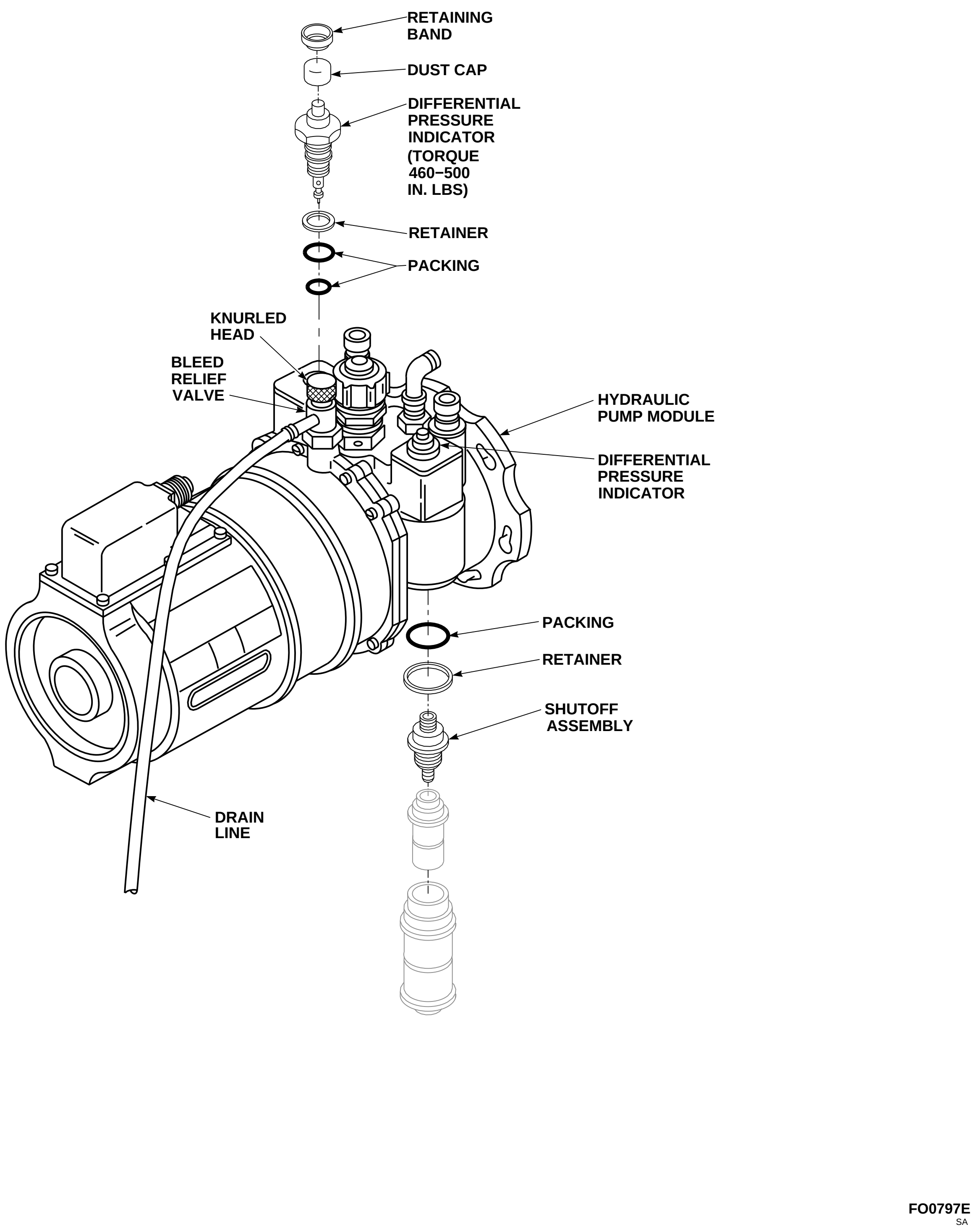


FIGURE 7-4-23-1. REPLACE HYDRAULIC PUMP MODULE DIFFERENTIAL PRESSURE INDICATOR AND SHUTOFF ASSEMBLY

7-4-23.1.2 Install.

a. Coat new retainer and packings with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install retainer and packings on differential pressure indicator (Figure 7-4-23-1).

c. Install differential pressure indicator in hydraulic pump module. **TORQUE DIFFERENTIAL PRESSURE INDICATOR TO 460 - 500 INCH-POUNDS.**

d. If required, remove hydraulic pump module filter and reset differential pressure indicator button (PARA 7-4-14).

e. Install dust cap and retaining band on differential pressure indicator of hydraulic pump module, 70652-02300-049 and 70652-02300-050.

f. Service hydraulic pump module (PARA 1-3-8).

g. Bleed hydraulic pump module (PARA 7-4-65).

h. Make sure area is clean and free of foreign material.

i. Close main rotor pylon sliding cover.

7-4-23.2 Replace Hydraulic Pump Module Shutoff Assembly.

NOTE

Replacement of the No. 1, No. 2, and backup hydraulic pump module shutoff assemblies is the same.

7-4-23.2.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 2-quart container under drain line at left main wheel.

e. Turn knurled head and push down bleed relief valve on top of hydraulic pump module. Drain overflow hydraulic fluid from bleed relief valve into container (Figure 7-4-23-1).

f. Remove hydraulic pump module filter (PARA 7-4-14).

g. Using shutoff assembly tool, remove shutoff assembly from hydraulic pump module. Discard retainer and packing (Figure 7-4-23-1).

7-4-23.2.2 Install.

a. Coat new retainer and packing with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install retainer and packing on shutoff assembly (Figure 7-4-23-1).

c. Coat shutoff assembly threads with sealing compound, Item 285 or Item 286, Appendix D.

NOTE

Shutoff assembly has left-hand threads.

d. Install shutoff assembly on hydraulic pump module.

e. Install hydraulic pump module filter (PARA 7-4-14).

f. Service hydraulic pump module (PARA 1-3-8).

g. Bleed hydraulic pump module (PARA 7-4-65).

h. Make sure area is clean and free of foreign material.

i. Close main rotor pylon sliding cover.

7-4-24 NO. 1 TRANSFER MODULE MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 291, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-26
- PARA 7-4-65
- PARA 7-5-7

7-4-24.1 Replace No. 1 Transfer Module Manifold.

7-4-24.1.1 Remove.

a. Remove No. 1 transfer module (PARA 7-4-26).

b. Use machinery towels, Item 211, Appendix D, to catch hydraulic fluid as tubing is disconnected.

NOTE

Cap or plug all tubes, ports, and fittings as parts are removed and disconnected.

c. Disconnect utility module pressure and return tubes from adapters (Figure 7-4-24-1, Detail A).

d. Disconnect 1st stage tail rotor servo pressure and return tubes from adapters.

e. Disconnect reservoir fill and ground test return tube from adapter.

f. Remove sealing compound from base of No. 1 transfer module manifold and helicopter fuselage using nonmetallic scraper.

NOTE

Bolts securing No. 1 transfer module manifold are different lengths and must be tagged to indicate location for installation.

g. Remove bolts, washers, and No. 1 transfer module manifold. Tag bolts when removed to indicate manifold location.

h. Check shims on fuselage for looseness, delamination, or damage. **NONE ALLOWED.** Replace damaged shims (PARA 7-5-7).

i. Clean sealing compound from bolts using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

7-4-24.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

NOTE

Remove caps and plugs from tubes, ports, and fittings as parts are installed and connected.

b. Apply sealing compound, Item 291, Appendix D, to bottom of No. 1 transfer module manifold. Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

c. Position inboard mounting boss of No. 1 transfer module manifold under primary servo boss on flight controls deck. Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

NOTE

Bolts securing No. 1 transfer module manifold are different lengths. Make sure to keep track of hardware when tags are removed.

d. Remove tags and install washers on bolts (Figure 7-4-24-1, Detail A). Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts. Do not get sealing compound on threads.

NOTE

Make sure bolts are installed in same location they were removed from.

e. Install bolts in No. 1 transfer module manifold. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts to manifold.

f. Apply a bead of sealing compound, Item 291, Appendix D, to bottom edge of No. 1 transfer module manifold and helicopter fuselage.

g. Connect 1st stage tail rotor servo pressure and return tubes to adapters. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

h. Connect utility module pressure and return tubes to adapters. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

i. Connect reservoir fill and ground test return tube to adapter. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

j. Install No. 1 transfer module (PARA 7-4-26).

k. Perform operational check of No. 1 hydraulic system (PARA 7-2-1).

l. Bleed No. 1, No. 2, and backup hydraulic systems after maintenance (PARA 7-4-65).

m. Make sure area is clean and free of foreign material.

n. Close main rotor pylon sliding cover.

o. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

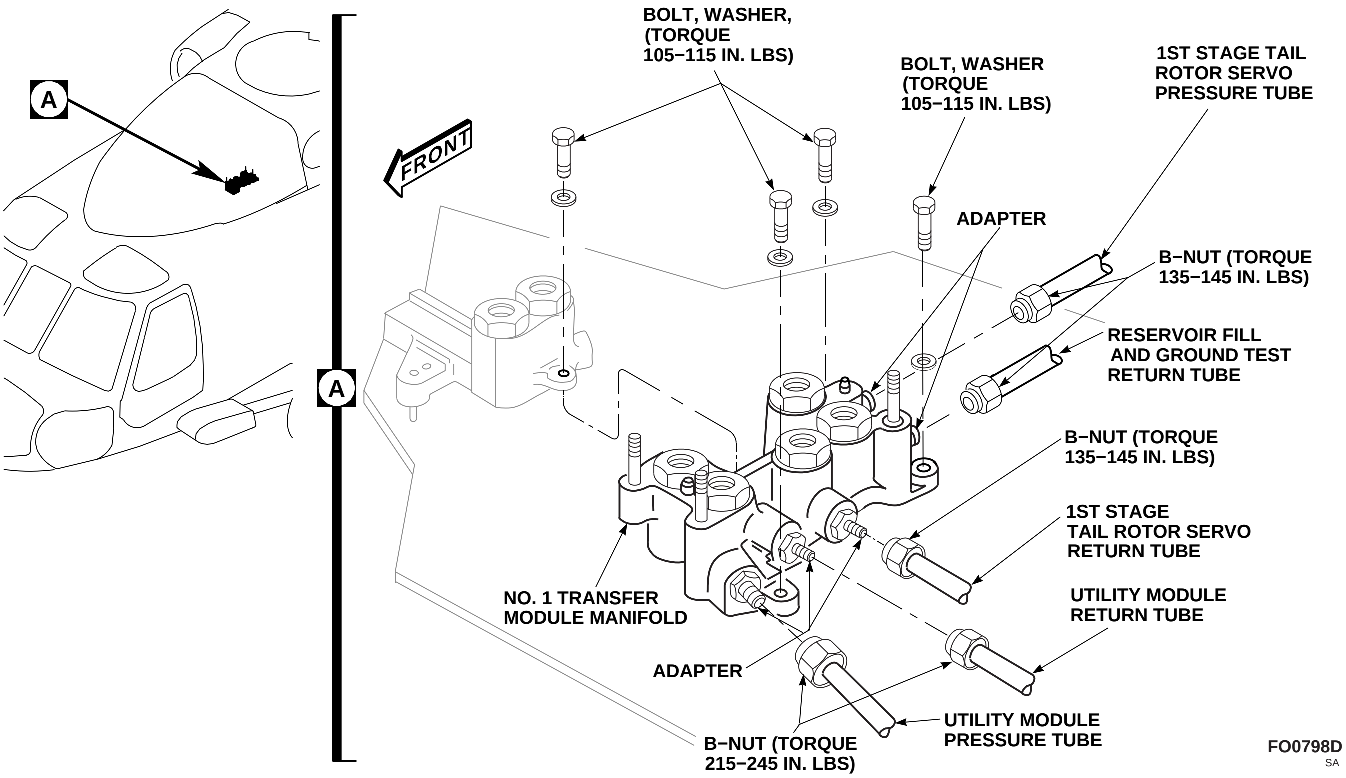


FIGURE 7-4-24-1. REPLACE NO. 1 TRANSFER MODULE MANIFOLD

7-4-25 NO. 1 PRIMARY SERVO MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cleaning Compound, Item 95, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 291, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-26
- PARA 7-4-65
- PARA 7-5-7
- PARA 11-4-87

7-4-25.1 Replace No. 1 Primary Servo Manifold.

7-4-25.1.1 Remove.

NOTE

Cap or plug all ports and fittings as parts are removed.

a. Remove all three primary servos (PARA 11-4-87).

b. Remove No. 1 transfer module (PARA 7-4-26).

c. Using nonmetallic scraper, remove sealing compound from base of No. 1 primary servo manifold and helicopter fuselage.

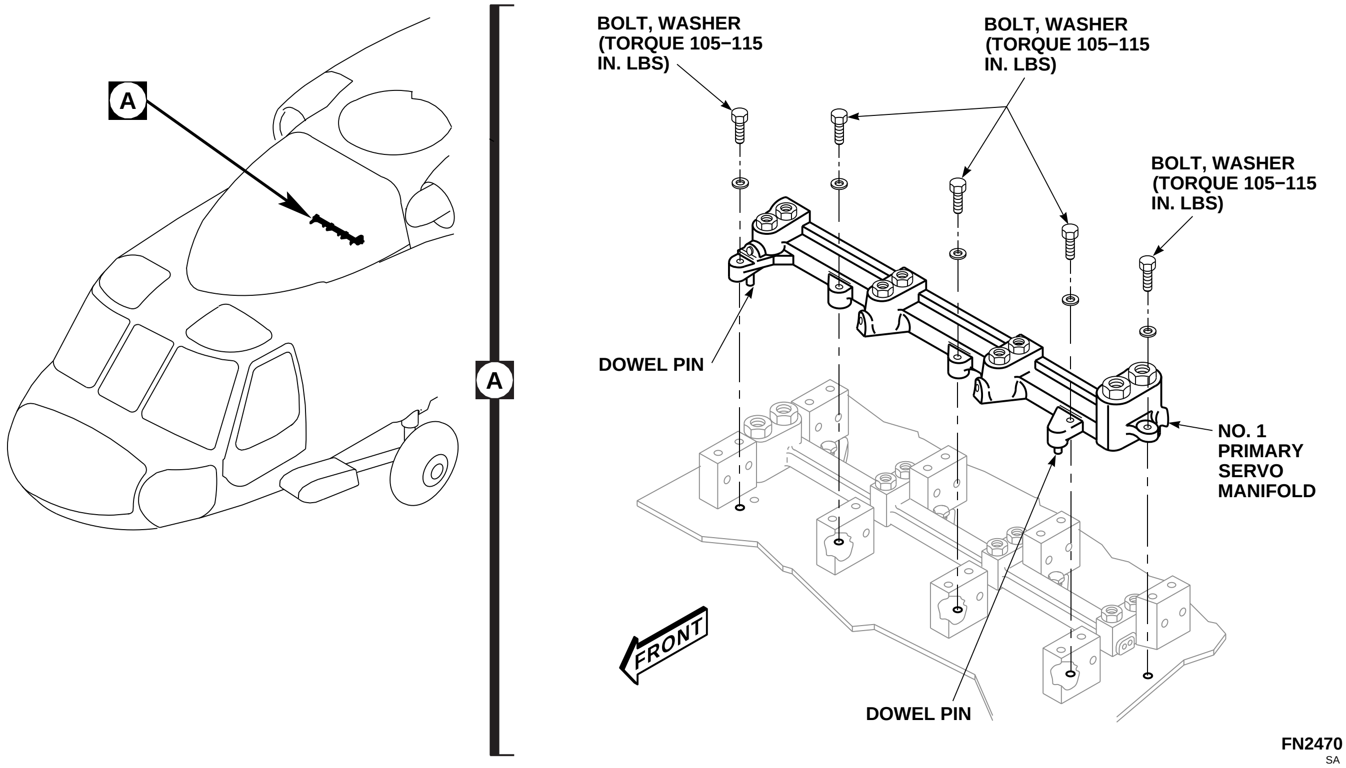


FIGURE 7-4-25-1. REPLACE NO. 1 PRIMARY SERVO MANIFOLD

NOTE

Bolts securing No. 1 primary servo manifold are different lengths and must be tagged to indicate manifold location.

d. Remove bolts, washers, and No. 1 primary servo manifold (Figure 7-4-25-1, Detail A). Tag bolts as removed.

e. Check shims on fuselage for looseness, delamination, or damage. **NONE ALLOWED.** If shim is loose, delaminated, or damaged, replace shim (PARA 7-5-7).

f. Clean sealing compound from bolts using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

7-4-25.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Apply sealing compound, Item 291, Appendix D, to bottom edge of No. 1 primary servo manifold. Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

c. Line up dowel pins with holes and position No. 1 primary servo manifold on helicopter fuselage (Figure 7-4-25-1, Detail A). Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

NOTE

Bolts securing No. 1 primary servo manifold are different lengths. Make sure to keep track of hardware when tags are removed.

d. Remove tags and install washers on bolts. Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts. Do not get sealing compound on threads.

NOTE

Make sure bolts are installed in same location from which they were removed.

e. Install bolts in No. 1 primary servo manifold. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts to servo manifold.

f. Apply sealing compound, Item 291, Appendix D, to edge of No. 1 primary servo manifold and helicopter fuselage.

g. Clean top surfaces of female couplings with machinery towels, Item 211, Appendix D. Flush with cleaning compound, Item 95, Appendix D.

NOTE

Remove caps and plugs from ports and fittings as parts are installed.

h. Install No. 1 transfer module (PARA 7-4-26).

i. Install all three primary servos (PARA 11-4-87).

j. Bleed No. 1 hydraulic system after maintenance (PARA 7-4-65).

k. Perform operational check of No. 1 hydraulic system (PARA 7-2-1).

l. Make sure area is clean and free of foreign material.

m. Close and latch main rotor pylon sliding cover.

n. Perform maintenance test flight (Appropriate MTF).

7-4-26 NO. 1 TRANSFER MODULE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Barrier Material, Item 79A, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cleaning Compound, Item 95, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, Item 212, Appendix D
- Masking Tape, Item 213, Appendix D
- Masking Tape, Item 214, Appendix D
- Masking Tape, Item 215, Appendix D
- Protective Caps and Plugs
- Protective Cover
- Tags

References

- Appendix D
- Appropriate MTF
- ASTM E1417
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-27
- PARA 7-4-28
- PARA 7-4-65

7-4-26.1 Replace No. 1 Transfer Module.

7-4-26.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 1-quart container to catch hydraulic fluid.

NOTE

Cap or plug all ports and fittings as parts are removed.

e. Disconnect pressure and return hoses from No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

f. Tag and disconnect electrical connector, P421, from pressure switch, and electrical connectors, P441 and P417, from solenoid shutoff valves (Figure 7-4-26-1, Detail A).

g. Install protective caps and plugs on electrical connectors.

h. Remove nuts, washers, bolt, and countersunk washer securing No. 1 transfer module to No. 1 transfer module manifold.

i. Use machinery towels, Item 211, Appendix D, to soak up hydraulic fluid during removal of No. 1 transfer module.

j. Lift module straight up off studs and remove from manifold.

k. Install protective cover over exposed self-sealing couplings to avoid contamination.

l. If replacing No. 1 transfer module with new module, perform the following:

(1) Remove No. 1 transfer module pressure switch (PARA 7-4-27).

(2) Remove No. 1 transfer module hoses and fittings (PARA 7-4-28).

7-4-26.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. If new module is being installed, perform the following:

(1) Install No. 1 transfer module pressure switch (PARA 7-4-27).

(2) Install No. 1 transfer module hoses and fittings (PARA 7-4-28).

NOTE

Remove caps and plugs from ports and fitting as parts are installed.

c. Flush No. 1 transfer module manifold surface, and No. 1 transfer module disconnect couplings with cleaning compound, Item 95, Appendix D, to prevent self-sealing coupling damage due to dust, dirt, etc.

d. Position No. 1 transfer module on studs, and lower module until it rests on self-sealing couplings on manifold (Figure 7-4-26-1, Detail A). Carefully press on No. 1 transfer module to seat couplings.

e. Secure No. 1 transfer module as follows:

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

(1) Secure No. 1 transfer module with nuts, washers, bolt, and countersunk washer.

(2) **TIGHTEN NUTS AND BOLT FINGER-TIGHT.**

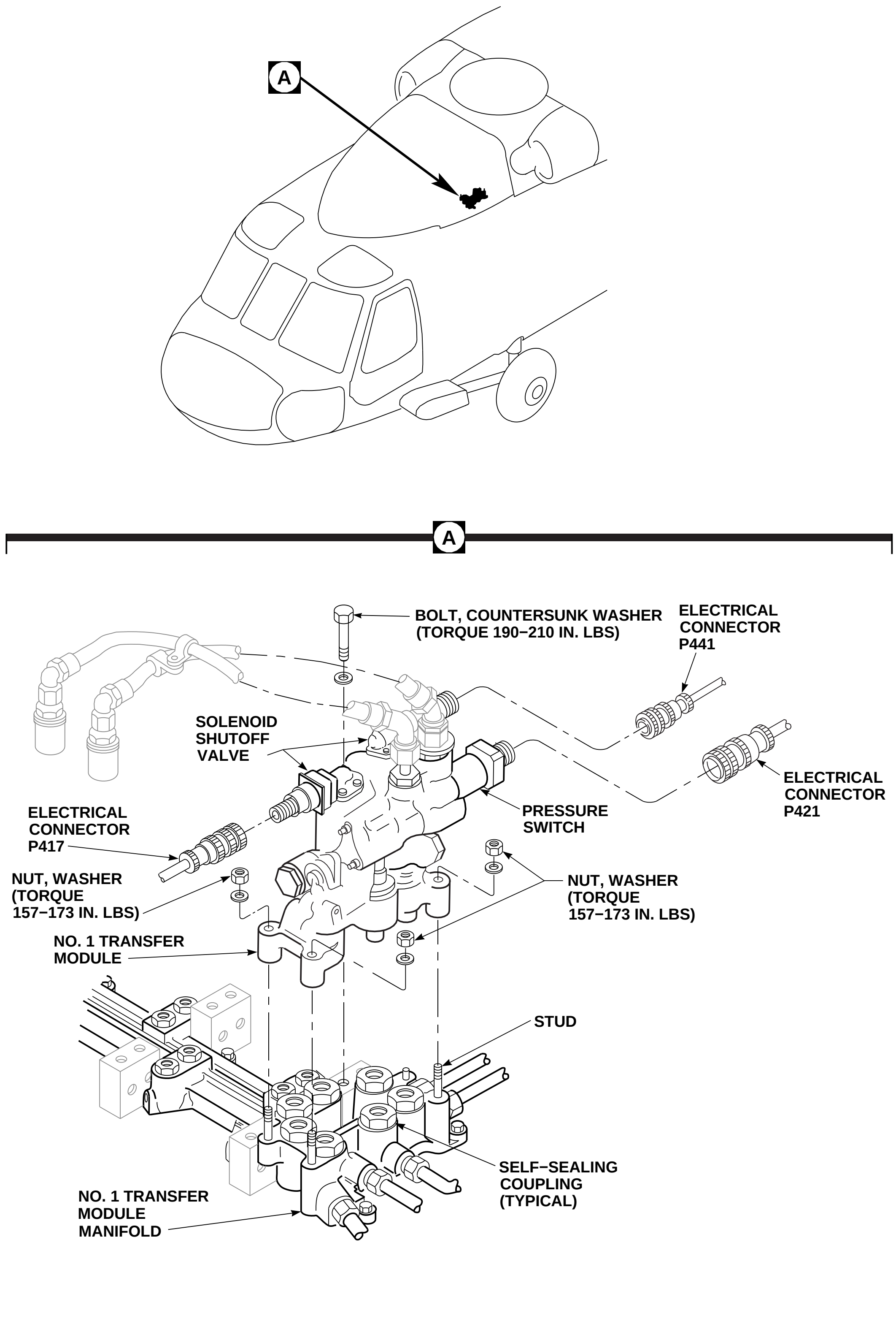


FIGURE 7-4-26-1. REPLACE NO. 1 TRANSFER MODULE

NOTE

Tighten three nuts and one bolt evenly to avoid cocking module.

(3) **TIGHTEN EACH NUT ONE TURN AT A TIME, STARTING WITH ONE NUT, THEN DIAGONALLY OPPOSITE BOLT, AND THEN REMAINING NUT AND ITS DIAGONALLY OPPOSITE NUT.**

(4) **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

(5) **TORQUE BOLTS TO 190 - 210 INCH-POUNDS.**

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connector, P421, to pressure switch, and electrical connectors, P441 and P417, to solenoid shutoff valves. Remove tags.

i. Connect pressure and return hoses to No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

j. Bleed No. 1 and backup hydraulic systems (PARA 7-4-65).

k. Perform clearance check (PARA 7-4-26.1.3).

l. Perform operational check of No. 1 and backup hydraulic systems (PARA 7-2-1).

m. Make sure area is clean and free of foreign material.

n. Close main rotor pylon sliding cover.

o. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

7-4-26.1.3 Clearance Check.

a. Fold down boot to gain access to area between mixer lever and No. 1 transfer module (Figure 7-4-26-2, Detail A).

b. Visually inspect mixer lever for evidence of contact with front face of No. 1 transfer module (Figure 7-4-26-2, Detail B and Detail C). **NONE ALLOWED.** Proceed as follows:

(1) If evidence of contact is found, proceed to step c.

(2) If no evidence of contact is found, measure clearance between mixer lever and No. 1 transfer module using feeler gage. **CLEARANCE MUST NOT BE LESS THAN 0.030-INCH.** If clearance is at least 0.030-inch, inspection is complete. Unfold boot to normal position (Figure 7-4-26-2, Detail A). If clearance is less than 0.030-inch, proceed to step c.

c. If necessary, blend/repair mixer lever and/or No. 1 transfer module as follows:

(1) Using barrier material, Item 79A, Appendix D, and suitable masking tape, Item 212 through Item 215, Appendix D, mask off surrounding areas where blending will occur to prevent contamination from entering components.

(2) Using appropriate tools, blend mixer lever where contact has occurred or gap was measured (Figure 7-4-26-2, Detail C). Blend to remove any sharp edges on No. 1 transfer module to obtain minimum clearance of 0.030-inch between mixer lever and No. 1 transfer module (Figure 7-4-26-2, Detail B and Detail C).

(3) Turn on all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

NOTE

Clearance measurements must be taken at all stick and pedal positions.

(4) Place collective stick approximately mid-position, then slowly cycle pedals from left to right and back again while measuring for a minimum of 0.030-inch clearance between mixer lever and No. 1 transfer module. Repeat this procedure

with collective stick slightly above and slightly below mid-positions. Note locations of contact or insufficient clearance.

NOTE

Front vertical face of No. 1 transfer module may be blended to a maximum depth of 0.060-inch.

(5) If clearance is met, inspection is complete, proceed to step c. (6). If minimum clearance of 0.030-inch is not met, repeat step c. (2) through step c. (4) until clearance is met. If necessary, blend front vertical face of No. 1 transfer module to a maximum depth of 0.060-inch.

(6) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect blended area (ASTM E1417).

(7) Wipe off remaining penetrant with cheesecloth, Item 90, Appendix D.

(8) Remove barrier material and masking tape from masked areas.

(9) Touchup blended area with corrosion resistant coating, Item 118, Appendix D.

(10) Touchup paint (PARA 2-6-5).

(11) Unfold boot to normal position.

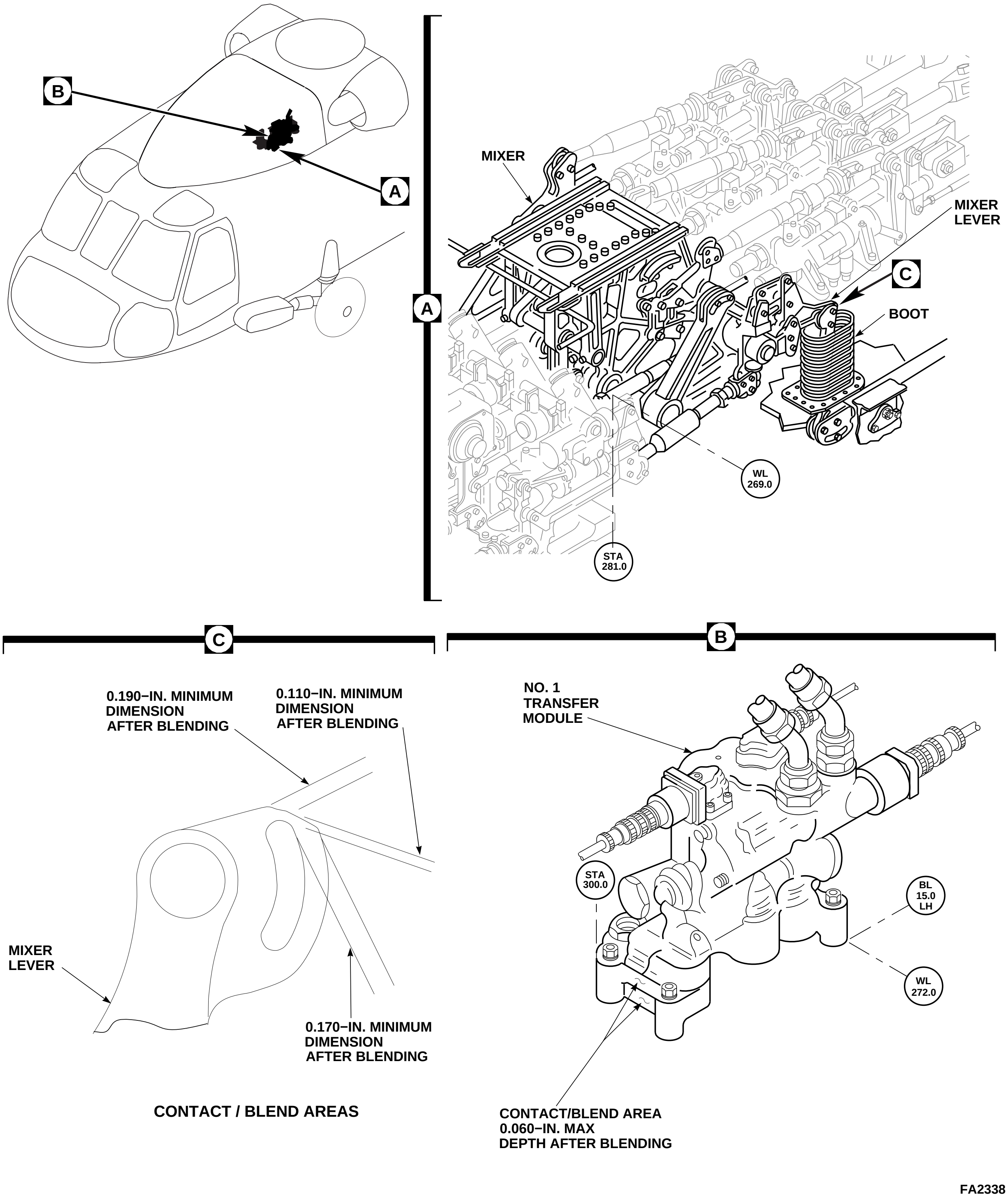


FIGURE 7-4-26-2. CLEARANCE CHECK/REPAIR

7-4-27 NO. 1 TRANSFER MODULE PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-27.1 Replace No. 1 Transfer Module Pressure Switch.

7-4-27.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

d. Disconnect electrical connector, P421, from pressure switch (Figure 7-4-27-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

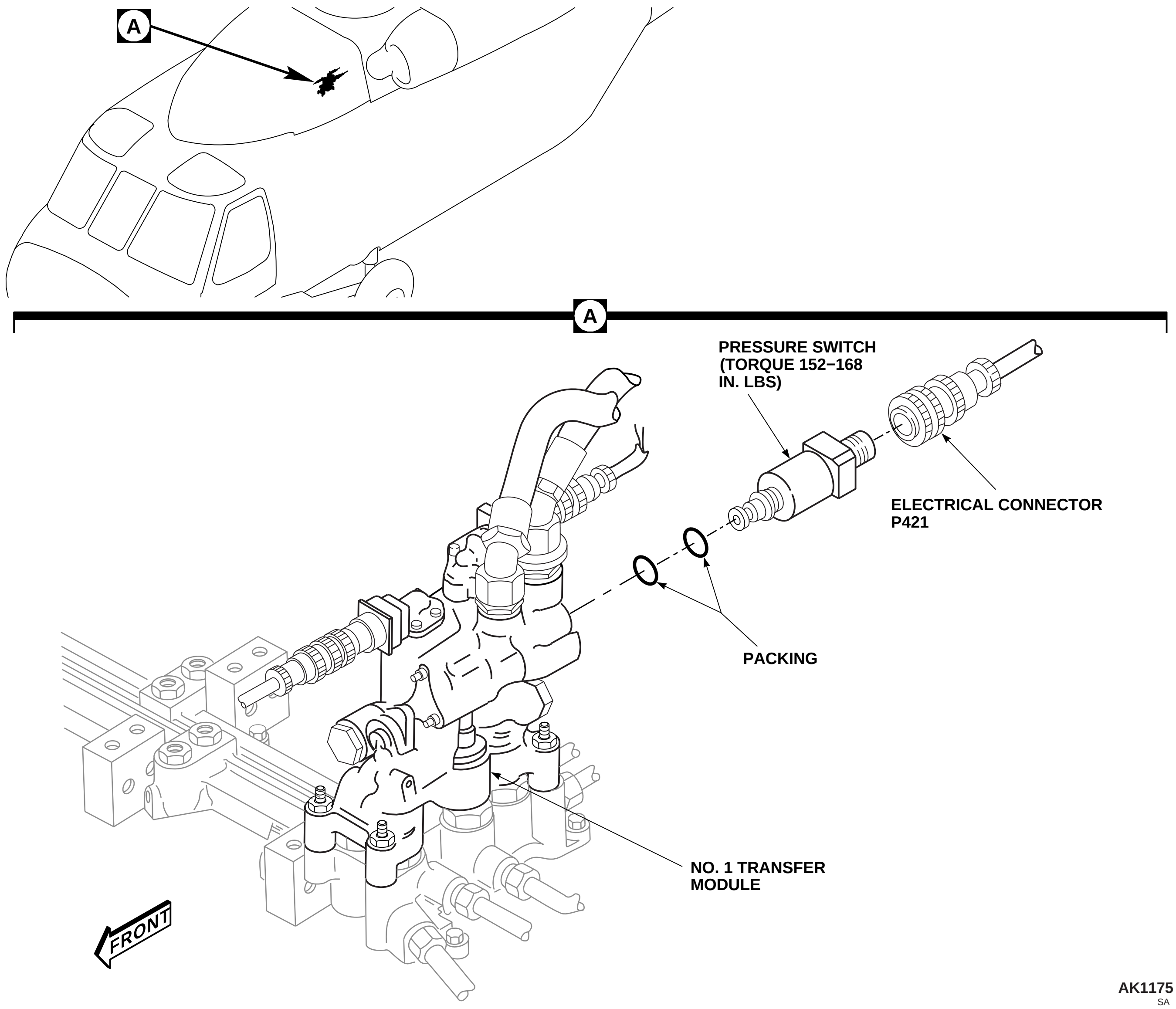


FIGURE 7-4-27-1. REPLACE NO. 1 TRANSFER MODULE PRESSURE SWITCH

f. Position 1-quart container to catch hydraulic fluid.

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter No. 1 transfer module. Make sure replacement pressure switch is ready for immediate installation to prevent fluid loss and keep dirt out.

g. Remove pressure switch and packings from No. 1 transfer module. Discard packings.

7-4-27.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packings and packing seats with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

c. Install packings on pressure switch. Install pressure switch in No. 1 transfer module

(Figure 7-4-27-1, Detail A). **TORQUE PRESSURE SWITCH TO 152 - 168 INCH-POUNDS.** Lockwire, Item 196, Appendix D, pressure switch.

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P421, to pressure switch.

g. Connect pressure and return hoses to No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

h. Bleed hydraulic system (PARA 7-4-65).

i. Perform operational check of No. 1 and backup hydraulic systems to check for proper operation of #1 HYD PUMP capsule on caution/advisory panel (PARA 7-2-1).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

7-4-28 NO. 1 TRANSFER MODULE HOSES AND FITTINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-28.1 Replace No. 1 Transfer Module Hoses and Fittings.

7-4-28.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

NOTE

Plug or cap hoses, ports, and fittings during removal.

d. Disconnect pressure and return hoses from unions on top of No. 1 transfer module (Figure 7-4-28-1, Detail A).

NOTE

If replacing only one hose, leave clamp in position on opposite hose to mark clamp location.

e. Remove screw, washers, and nut securing clamps to hose assemblies. Remove clamps one at a time from hose assemblies using opposite hose to mark clamp location on replacement hose assemblies (Figure 7-4-28-1, Detail B).

f. Remove quick-disconnect halves from No. 1 transfer module pressure and return hoses.

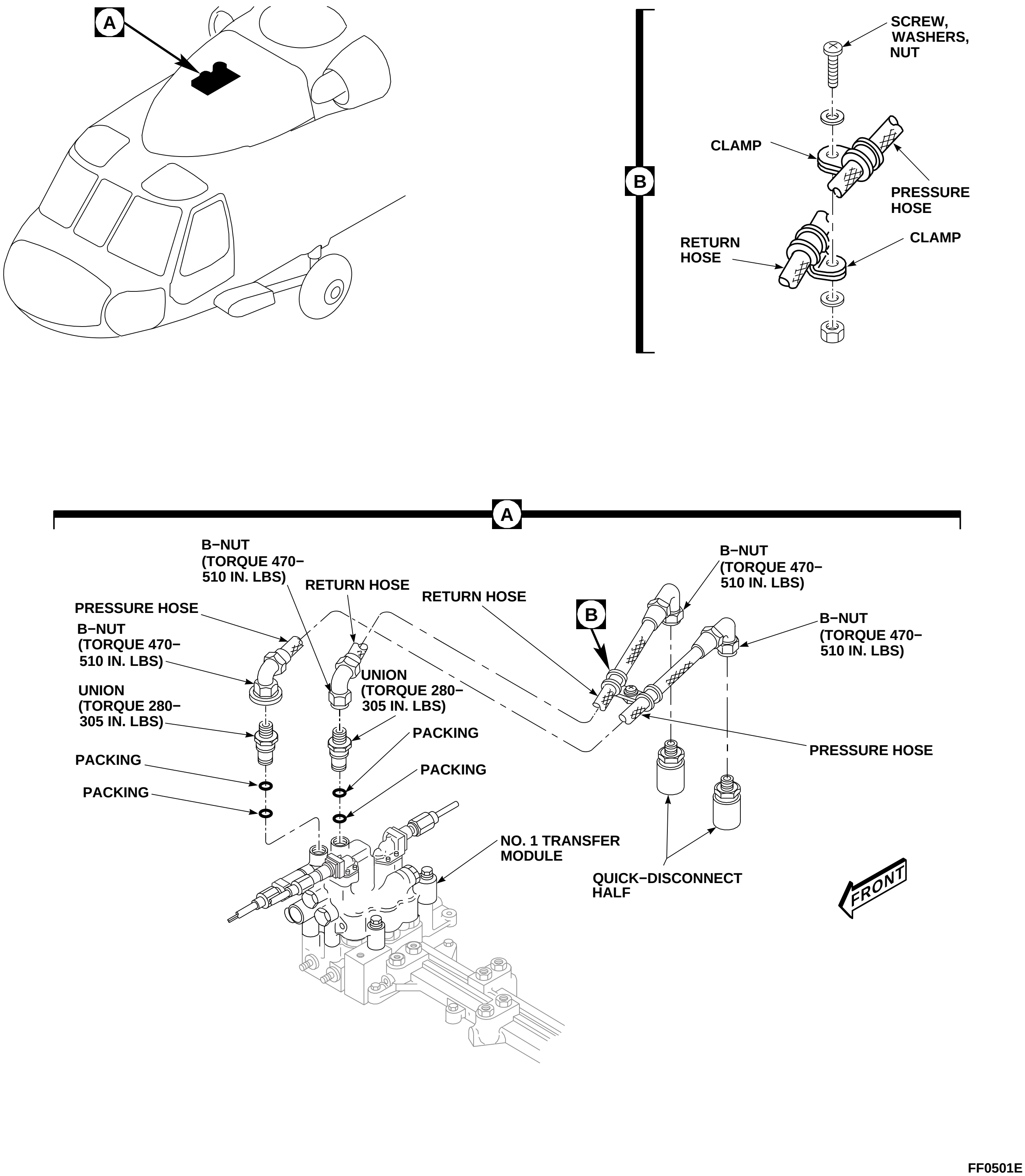


FIGURE 7-4-28-1. REPLACE NO. 1 TRANSFER MODULE HOSES AND FITTINGS

g. Remove two unions and packings from No. 1 transfer module. Discard packings (Figure 7-4-28-1, Detail A).

7-4-28.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to No. 1 transfer module will result if hoses and fittings are not clean prior to installation. Make sure hoses and fittings are clean inside and out before installing.

NOTE

Remove caps and plugs from hoses, ports, and fittings during installation.

b. Before installation blow all hoses and fittings clear with dry compressed air.

c. Coat packings with hydraulic fluid, Item 154 or Item 155, Appendix D.

d. Install packings on unions and install unions in top ports of No. 1 transfer module (Figure 7-4-28-1, Detail A). **TORQUE UNIONS TO 280 - 305 INCH-POUNDS.**

e. Install quick-disconnect halves on pressure and return hoses. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

NOTE

Hydraulic pressure hose interference washer prevents it from being installed in module return port.

f. Fill hoses with hydraulic fluid, Item 154 or Item 155, Appendix D, and connect pressure and return hoses to unions on top of module. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

CAUTION

To prevent hydraulic flexible hose from chafing, make sure hose radius is oriented toward front and clamps are centralized.

g. Position and connect clamps on replacement hoses with screw, washers, and nut (Figure 7-4-28-1, Detail B).

h. Connect pressure and return hoses to No. 1 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

i. Bleed hydraulic system (PARA 7-4-65).

j. Perform operational check of No. 1 and backup hydraulic systems to check for proper operation of #1 HYD PUMP capsule on caution/advisory panel (PARA 7-2-1).

k. Make sure area is clean and free of foreign material.

l. Close main rotor pylon sliding cover.

m. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

7-4-29 NO. 2 TRANSFER MODULE MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cleaning Compound, Item 95, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 291, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-31
- PARA 7-4-65
- PARA 7-5-7

7-4-29.1 Replace No. 2 Transfer Module Manifold.

7-4-29.1.1 Remove.

a. Remove No. 2 transfer module (PARA 7-4-31).

b. Use machinery towel, Item 211, Appendix D, to catch hydraulic fluid as tubing is disconnected.

NOTE

Cap or plug all tubes, ports, and fittings as parts are removed and disconnected.

c. Remove pilot boost servo pressure and return tubes from adapters (Figure 7-4-29-1, Detail A).

d. Remove backup system pressure and return tubes from adapters.

e. Remove reservoir fill and ground test return tube from adapter.

f. Remove sealing compound from base of No. 2 transfer module manifold and helicopter fuselage using nonmetallic scraper.

NOTE

Bolts securing No. 2 transfer module manifold are different lengths and must be tagged to indicate location for installation.

g. Remove bolts, washers, and remove No. 2 transfer module manifold. Tag bolts when removed to indicate manifold location.

h. Check shims on fuselage for looseness, delamination, or damage. **NONE ALLOWED.** Replace damaged shims (PARA 7-5-7).

i. Clean sealing compound from bolts using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

7-4-29.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Apply sealing compound, Item 291, Appendix D, to bottom of No. 2 transfer module manifold. Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

c. Position inboard mounting boss of No. 2 transfer module manifold under primary servo boss on flight controls deck (Figure 7-4-29-1, Detail A). Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

NOTE

Bolts securing No. 2 transfer module manifold are different lengths. Make sure to keep track of hardware when tags are removed.

d. Remove tags and install washers on bolts. Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts. Do not get sealing compound on threads.

NOTE

Make sure bolts are installed in same location they were removed from.

e. Install bolts in No. 2 transfer module manifold. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to manifold.

f. Apply a bead of sealing compound, Item 291, Appendix D, to base of No. 2 transfer module manifold and helicopter fuselage.

NOTE

Remove caps and plugs from tubes, ports, and fittings as parts are installed and connected.

g. Connect reservoir fill and ground test return tube to adapter on No. 2 transfer module manifold. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

h. Connect pilot boost servo pressure and return tubes to adapters on No. 2 transfer module manifold. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

i. Connect backup system pressure and return tubes to adapters. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

j. Clean top surfaces of female couplings with clean machinery towels, Item 211, Appendix D. Flush with cleaning compound, Item 95, Appendix D.

k. Install No. 2 transfer module (PARA 7-4-31).

l. Bleed No. 1 and No. 2 hydraulic systems (PARA 7-4-65).

m. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

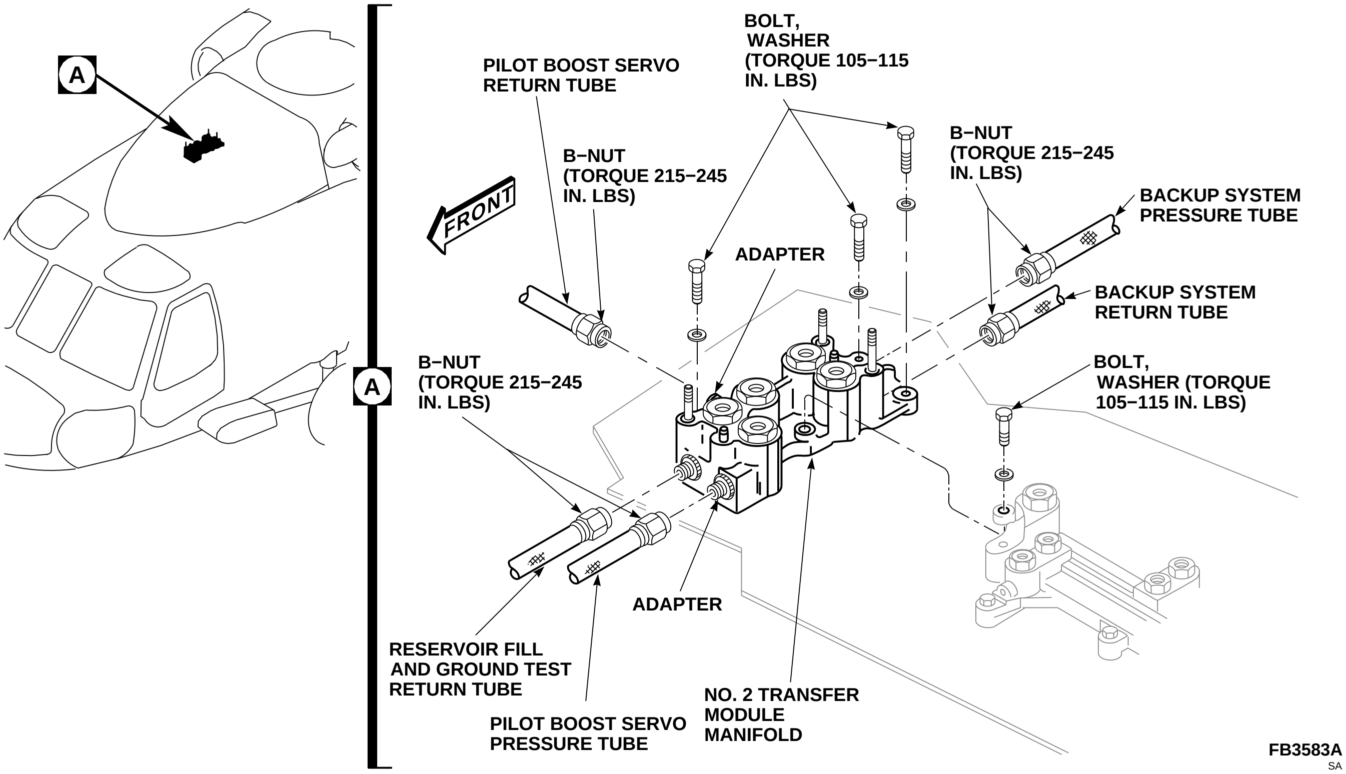


FIGURE 7-4-29-1. REPLACE NO. 2 TRANSFER MODULE MANIFOLD

n. Make sure area is clean and free of foreign material.

o. Close main rotor pylon sliding cover.

p. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

7-4-30 NO. 2 PRIMARY SERVO MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cleaning Compound, Item 95, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 291, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-31
- PARA 7-4-65
- PARA 7-5-7
- PARA 11-4-87

7-4-30.1 Replace No. 2 Primary Servo Manifold.

7-4-30.1.1 Remove.

NOTE

Cap or plug all ports and fittings as parts are removed.

a. Remove all three primary servos (PARA 11-4-87).

b. Remove No. 2 transfer module (PARA 7-4-31).

c. Remove sealing compound from base of No. 2 primary servo manifold and helicopter fuselage using nonmetallic scraper.

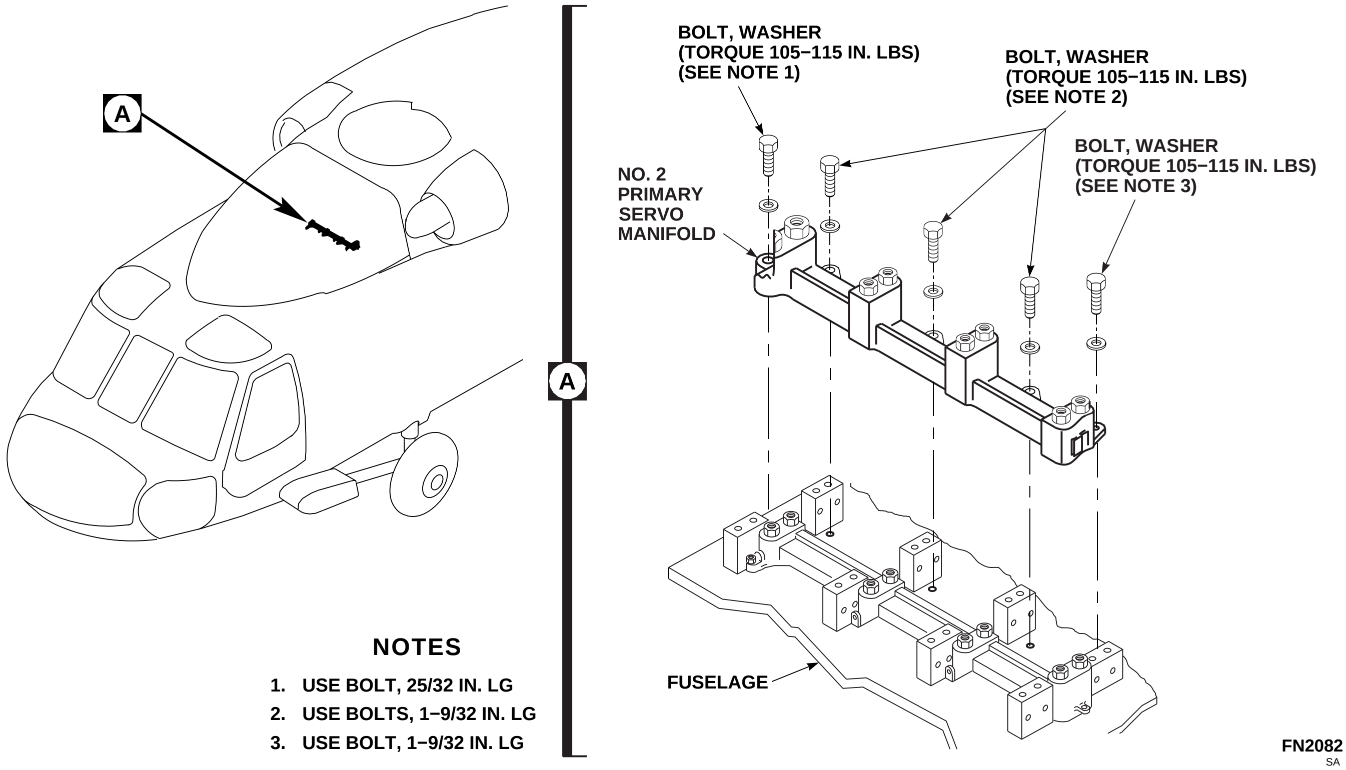


FIGURE 7-4-30-1. REPLACE NO. 2 PRIMARY SERVO MANIFOLD

NOTE

Bolts securing No. 2 primary servo manifold are different lengths and must be tagged to indicate manifold location.

d. Remove bolts, washers, and No. 2 primary servo manifold (Figure 7-4-30-1, Detail A). Tag bolts as removed.

e. Check shims on fuselage for looseness, delamination, or damage. **NONE ALLOWED.** Replace damaged shims (PARA 7-5-7).

f. Clean sealing compound from bolts using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

7-4-30.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Apply sealing compound, Item 291, Appendix D, to bottom edge of No. 2 primary servo manifold. Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

c. Line up dowel pins with holes and position No. 2 primary servo manifold on helicopter fuselage (Figure 7-4-30-1, Detail A). Clean any sealing compound from holes using machinery towel, Item 211, Appendix D.

NOTE

Bolts securing No. 2 primary servo manifold are different lengths. Make sure to keep track of hardware when tags are removed.

d. Remove tags and install washers on bolts. Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts. Do not get sealing compound on threads.

NOTE

Make sure bolts are installed in same location they were removed from.

e. Install bolts and washers in No. 2 primary servo manifold. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to servo manifold.

f. Apply sealing compound, Item 291, Appendix D, to edge of No. 2 primary servo manifold and helicopter fuselage.

g. Clean top surfaces of female couplings with machinery towels, Item 211, Appendix D. Flush with cleaning compound, Item 95, Appendix D.

NOTE

Remove caps and plugs from ports and fittings as parts are installed.

h. Install No. 2 transfer module (PARA 7-4-31).

i. Install all three primary servos (PARA 11-4-87).

j. Bleed No. 2 hydraulic system (PARA 7-4-65).

k. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

l. Make sure area is clean and free of foreign material.

m. Close main rotor pylon sliding cover.

n. Perform maintenance test flight (Appropriate MTF).

7-4-31 NO. 2 TRANSFER MODULE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cleaning Compound, Item 95, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Protective Cover

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-32
- PARA 7-4-33
- PARA 7-4-65

7-4-31.1 Replace No. 2 Transfer Module.

7-4-31.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Position 1-quart container to catch hydraulic fluid.

NOTE

Cap or plug all ports and fittings as parts are removed.

e. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

f. Disconnect electrical connector, P420, from pressure switch, and electrical connectors, P448 and P416, from solenoid shutoff valves (Figure 7-4-31-1, Detail A).

g. Install protective caps and plugs on electrical connectors.

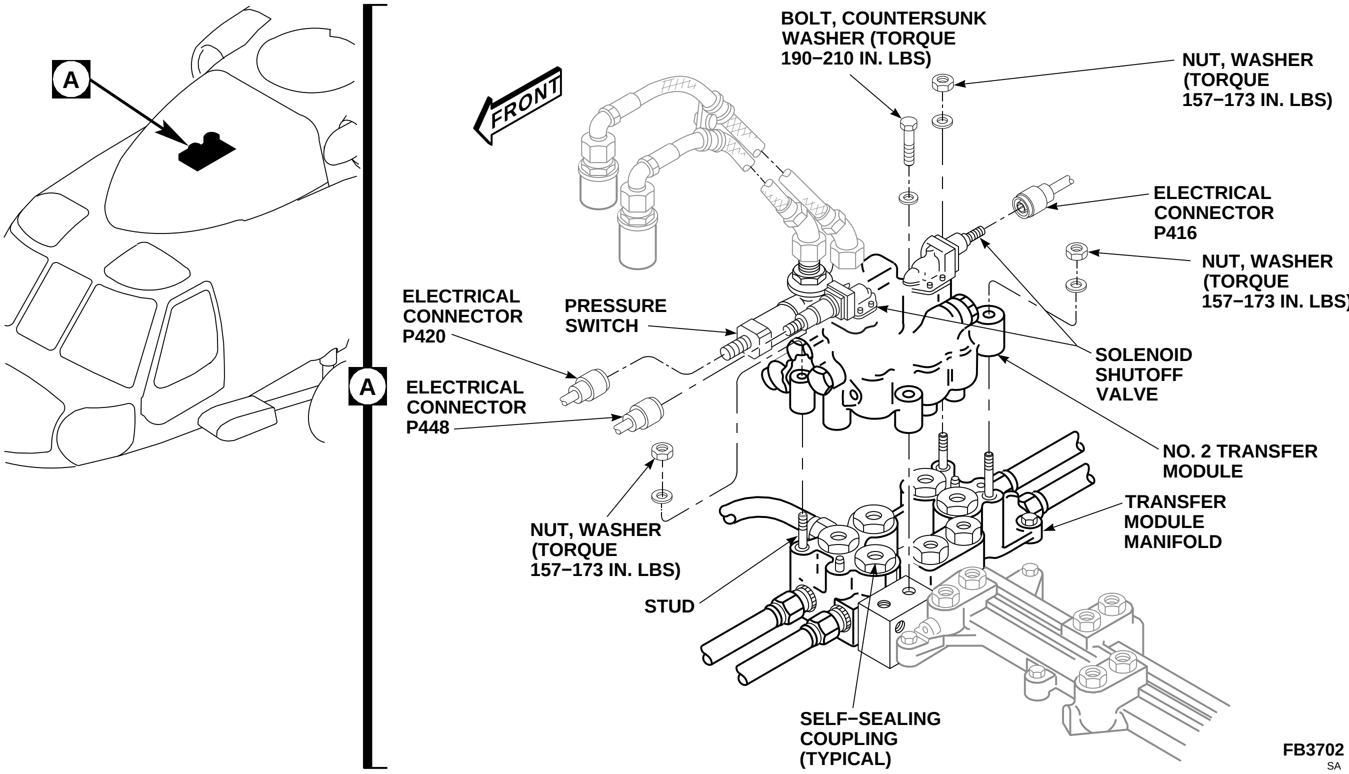


FIGURE 7-4-31-1. REPLACE NO. 2 TRANSFER MODULE

h. Remove nuts, washers, bolt, and countersunk washer holding No. 2 transfer module to transfer module manifold.

i. Use machinery towels, Item 211, Appendix D, to soak up hydraulic fluid during removal of transfer module.

j. Lift No. 2 transfer module straight up and remove from transfer module manifold.

k. Cover exposed self-sealing couplings to avoid contamination.

l. If replacing No. 2 transfer module, remove pressure switch (PARA 7-4-32).

m. If replacing No. 2 transfer module, remove pressure and return hoses and fittings (PARA 7-4-33).

7-4-31.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. If removed, install No. 2 transfer module pressure switch (PARA 7-4-32).

c. If removed, install No. 2 transfer module hoses and fittings (PARA 7-4-33).

NOTE

Remove caps and plugs from ports and fittings as parts are installed.

d. Flush manifold surface, and No. 2 transfer module disconnect couplings with cleaning compound, Item 95, Appendix D, to prevent self-sealing coupling damage due to dust, dirt, etc.

e. Position No. 2 transfer module on studs and lower module until it rests on self-sealing couplings (Figure 7-4-31-1, Detail A). Carefully press on No. 2 transfer module to seat couplings.

f. Secure No. 2 transfer module with nuts, washers, bolt, and countersunk washer. Make sure countersunk side of washer faces bolthead.

TIGHTEN NUTS AND BOLT FINGER-TIGHT. Tighten three nuts and one bolt evenly to avoid cocking module. Tighten each one turn at a time starting with one nut, then diagonally opposite bolt, and then remaining nut and its diagonally opposite nut.

g. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

h. **TORQUE BOLT TO 190 - 210 INCH-POUNDS.**

i. Remove protective caps and plugs from electrical connectors.

j. Inspect and treat electrical connectors (PARA 1-7-20).

k. Connect electrical connector, P420, to pressure switch, and electrical connectors, P448 and P416, to solenoid shutoff valves.

l. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

m. Bleed No. 2 and backup hydraulic systems (PARA 7-4-65).

n. Perform operational check of No. 2 and backup hydraulic system (PARA 7-2-1).

o. Make sure area is clean and free of foreign material.

p. Close main rotor pylon sliding cover.

q. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

7-4-32 NO. 2 TRANSFER MODULE PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-32.1 Replace No. 2 Transfer Module Pressure Switch.

7-4-32.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

d. Disconnect electrical connector, P420, from pressure switch (Figure 7-4-32-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

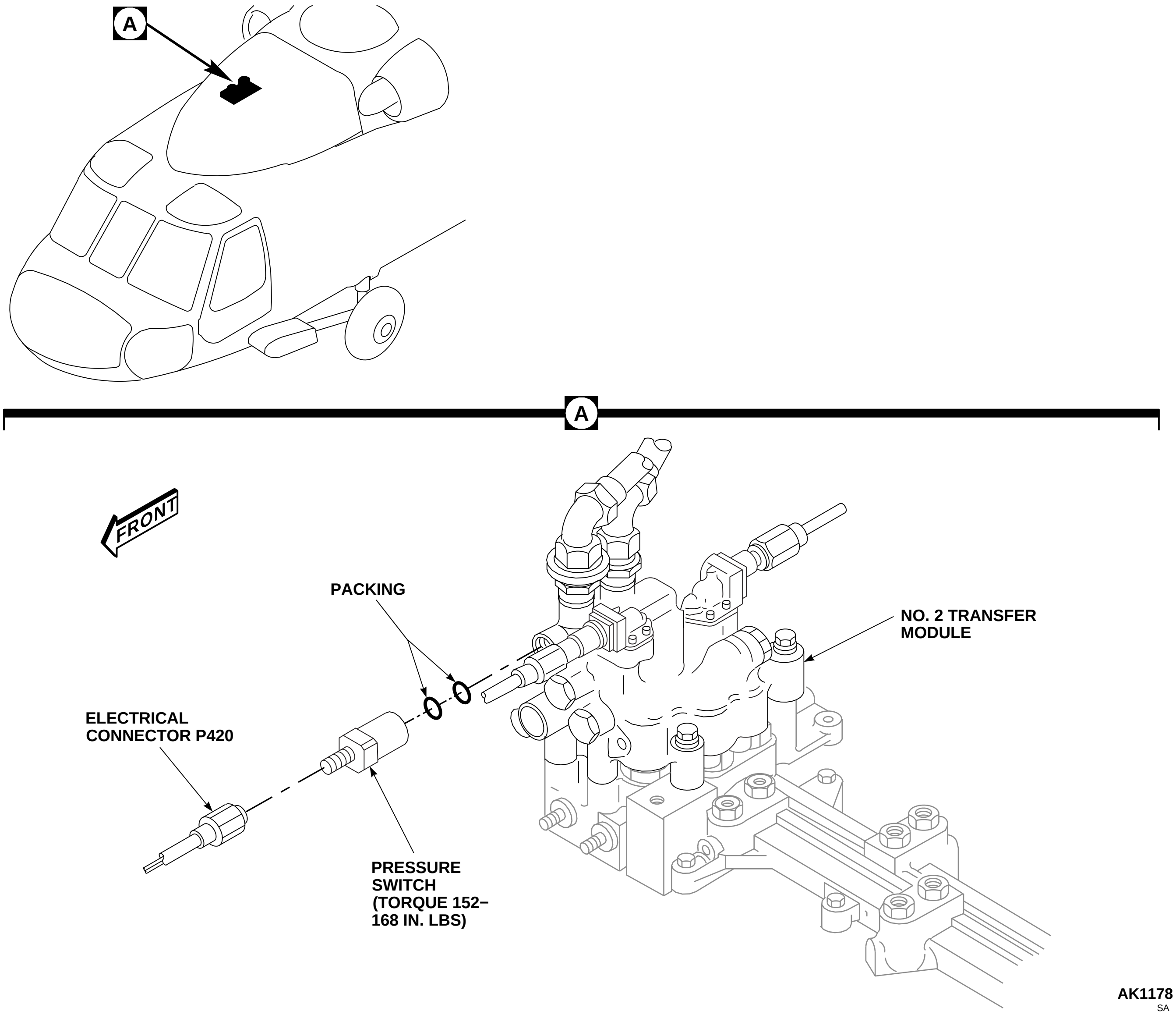


FIGURE 7-4-32-1. REPLACE NO. 2 TRANSFER MODULE PRESSURE SWITCH

f. Position 1-quart container to catch hydraulic fluid.

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter No. 2 transfer module. Make sure replacement pressure switch is ready for immediate installation to prevent fluid loss and keep dirt out.

g. Remove pressure switch and packings from No. 2 transfer module. Discard packings.

7-4-32.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packings and packing seats with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

c. Install packings on pressure switch. Install pressure switch in No. 2 transfer module

(Figure 7-4-32-1, Detail A). **TORQUE PRESSURE SWITCH TO 152 - 168 INCH-POUNDS.** Lockwire, Item 196, Appendix D, pressure switch.

d. Remove protective caps and plug from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P420, to pressure switch.

g. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

h. Bleed hydraulic system (PARA 7-4-65).

i. Perform operational check of No. 2 and backup hydraulic systems to check for proper operation of #2 HYD PUMP capsule on caution/advisory panel (PARA 7-2-1).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

7-4-33 NO. 2 TRANSFER MODULE HOSES AND FITTINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-33.1 Replace No. 2 Transfer Module Hoses and Fittings.

7-4-33.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

NOTE

Plug or cap hoses, ports, and fittings during removal.

d. Disconnect pressure and return hoses from unions on top of No. 2 transfer module (Figure 7-4-33-1, Detail A).

NOTE

If replacing just one hose, leave clamp in position on opposite hose to mark clamp location.

e. Remove screw, washers, and nut securing clamps to hose assemblies. Remove clamps one at a time from hose assemblies using opposite hose to mark clamp location on replacement hose assemblies (Figure 7-4-33-1, Detail B).

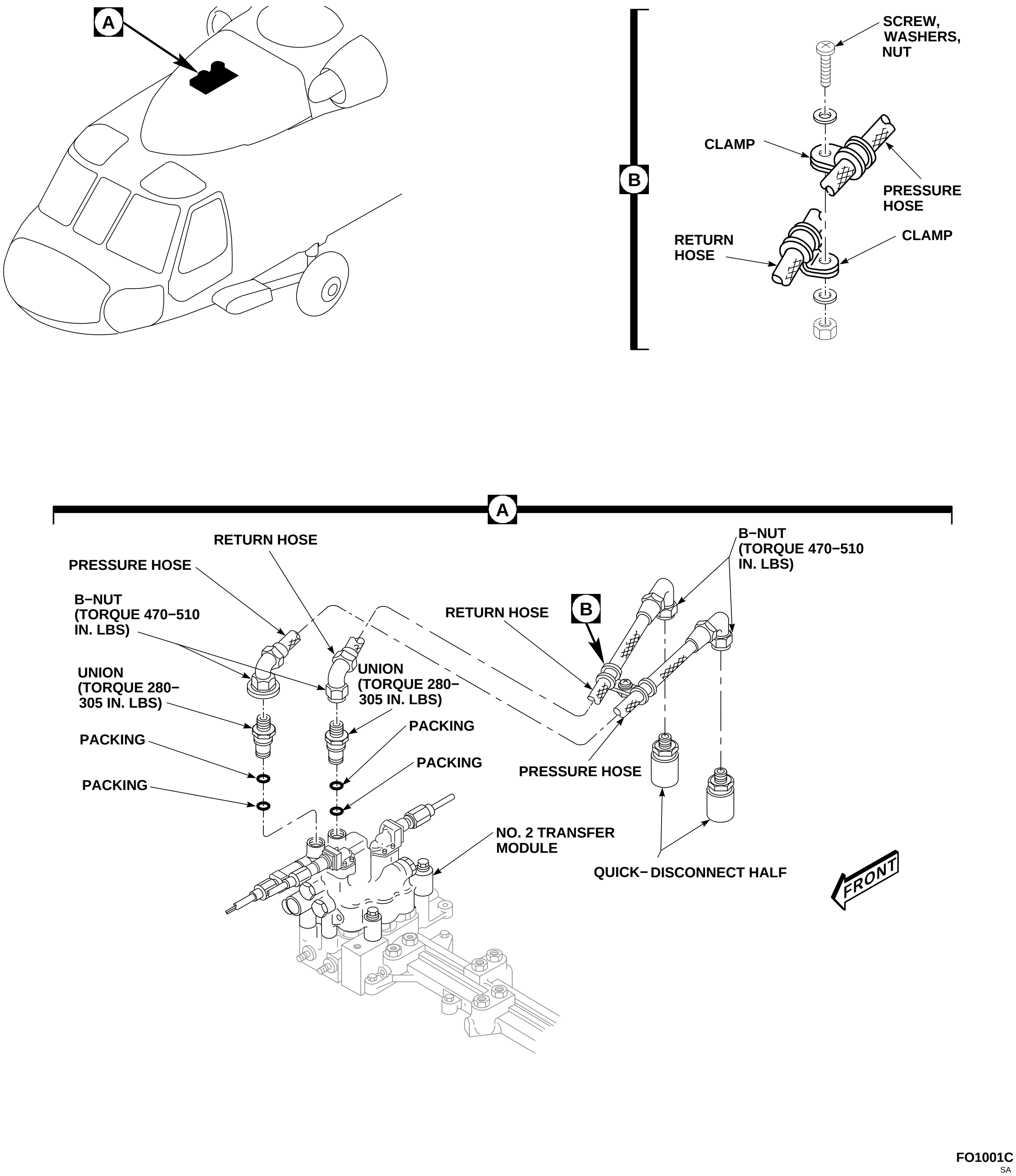


FIGURE 7-4-33-1. REPLACE NO. 2 TRANSFER MODULE HOSES AND FITTINGS

f. Remove quick-disconnect halves from No. 2 transfer module pressure and return hoses (Figure 7-4-33-1, Detail A).

g. Remove two unions and packings from No. 2 transfer module. Discard packings.

7-4-33.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to No. 2 transfer module will result if hoses and fittings are not clean prior to installation. Make sure hoses and fittings are clean inside and out before installing.

NOTE

Remove caps and plugs from hoses, ports, and fittings during installation.

b. Before installation blow all hoses and fittings clear with dry compressed air.

c. Coat packings with hydraulic fluid, Item 154 or Item 155, Appendix D.

d. Install packings on unions and install unions in top ports of No. 2 transfer module (Figure 7-4-33-1, Detail A). **TORQUE UNIONS TO 280 - 305 INCH-POUNDS.**

e. Connect quick-disconnect halves to pressure and return hoses. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

NOTE

Hydraulic pressure hose interference washer prevents it from being installed in module return port.

f. Fill hoses with hydraulic fluid, Item 154 or Item 155, Appendix D, and connect pressure and return hoses to unions on top of No. 2 transfer module. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

g. Position and connect clamps on replacement hoses using screw, washers, and nut (Figure 7-4-33-1, Detail B).

h. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

i. Bleed hydraulic system (PARA 7-4-65).

j. Perform operational check of No. 2 and backup hydraulic systems to check for proper operation of #2 HYD PUMP capsule on caution/advisory panel (PARA 7-2-1).

k. Make sure area is clean and free of foreign material.

l. Close main rotor pylon sliding cover.

m. Perform post maintenance ground run to verify proper module/manifold function and no external leakage of hydraulic fluid (Appropriate MTF).

7-4-34 PILOT-ASSIST MODULE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cleaning Compound, Item 95, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13

7-4-34.1 Replace Pilot-Assist Module.

7-4-34.1.1 Inspect.

Check for extended indicator button (Figure 7-4-34-1, Detail A). If button is extended (not flush with fitting), perform operational check of hydraulic system (PARA 7-2-1).

7-4-34.1.2 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

d. Disconnect electrical connectors as follows (Figure 7-4-34-1, Detail A):

(1) P468R from SAS shutoff valve.

(2) P469R from boost shutoff valve.

(3) P466R from trim shutoff valve.

(4) P467R from SAS pressure switch.

e. Install protective caps and plugs on electrical connectors.

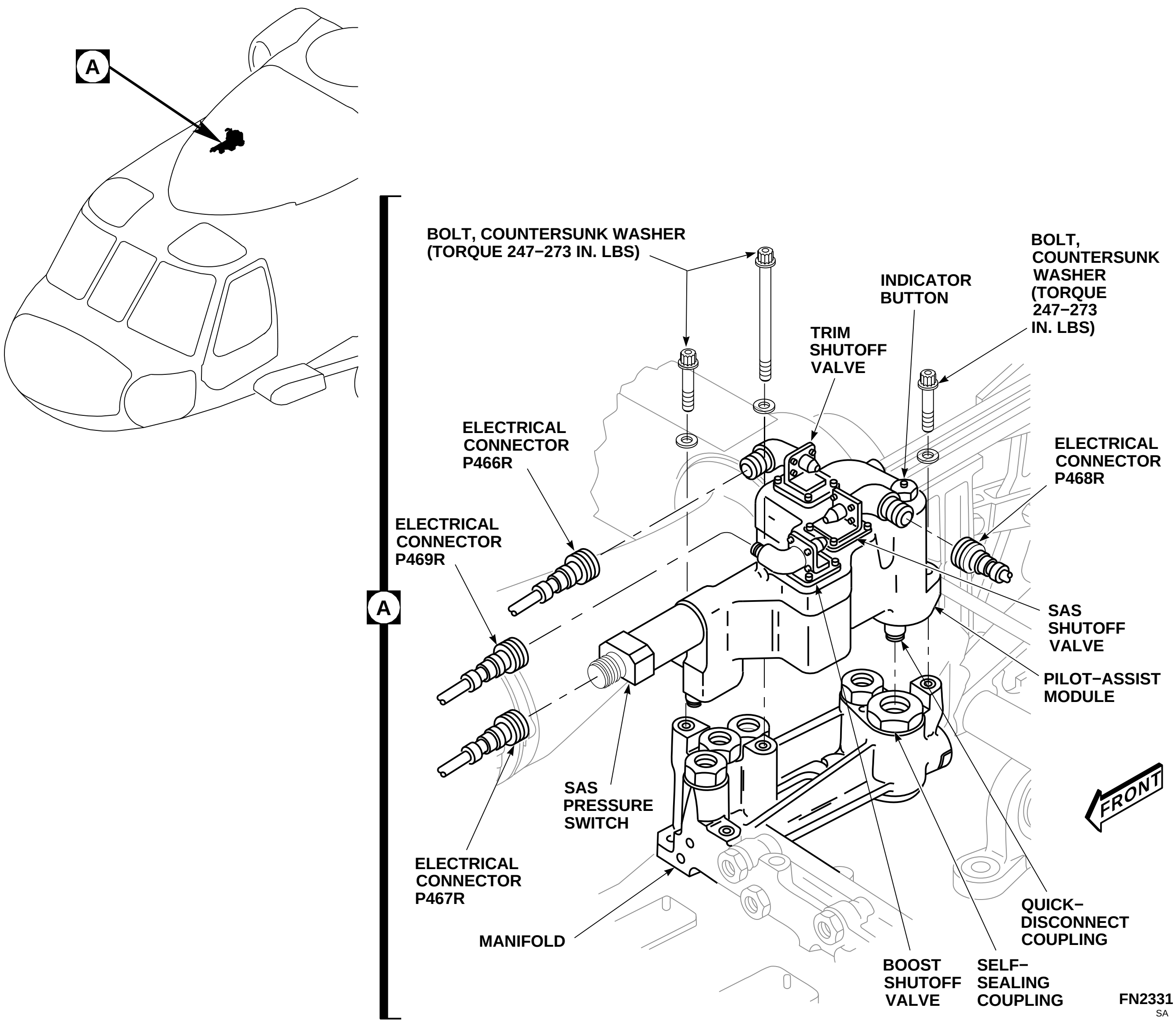


FIGURE 7-4-34-1. REPLACE PILOT-ASSIST MODULE

CAUTION

Damage to hydraulic system will result, if dirt or foreign materials enter boost servo and pilot-assist manifold. Make sure area is clean and have replacement module ready to install or cover manifold with clean machinery towel, Item 211, Appendix D, when module is removed.

f. Remove bolts and countersunk washers and remove pilot-assist module from manifold by pulling straight up on module.

7-4-34.1.3 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Clean top surfaces of self-sealing couplings with clean machinery towel, Item 211, Appendix D. Flush with cleaning compound, Item 95, Appendix D.

c. Position pilot-assist module on manifold, making sure quick-disconnect couplings on module line up with self-sealing couplings on manifold (Figure 7-4-34-1, Detail A).

d. Push straight down on pilot-assist module. Install bolts and countersunk washers. Make sure countersunk side of washer faces bolthead. Tighten three bolts down evenly, one turn at a time, to avoid cocking module and possible damage to couplings. **TORQUE BOLTS TO 247 - 273 INCH-POUNDS.**

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors as follows:

(1) P468R to SAS shutoff valve.

(2) P469R to boost shutoff valve.

(3) P466R to trim shutoff valve.

(4) P467R to SAS pressure switch.

h. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

i. Perform operational check of No. 2 and backup hydraulic systems (PARA 7-2-1).

j. Make sure area is clean and free of foreign material.

k. Close and latch main rotor pylon sliding cover.

l. Perform maintenance test flight (Appropriate MTF).

7-4-35 PILOT-ASSIST MODULE SAS PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-35.1 Replace Pilot-Assist Module SAS Pressure Switch.

7-4-35.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

d. Disconnect electrical connector, P467R, from SAS pressure switch (Figure 7-4-35-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

f. Position 1-quart container to catch hydraulic fluid.

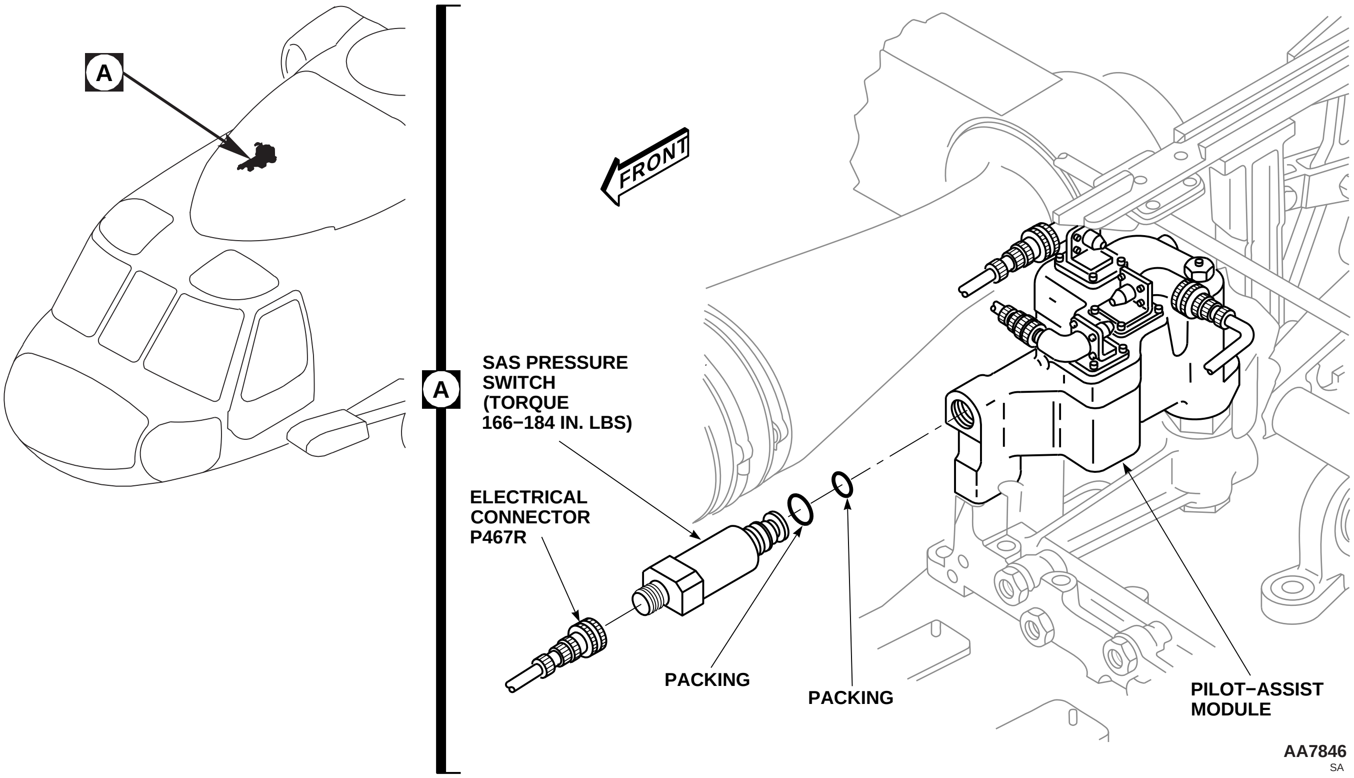


FIGURE 7-4-35-1. REPLACE PILOT-ASSIST MODULE SAS PRESSURE SWITCH

CAUTION

Damage to hydraulic system will result if dirt or foreign material enters pilot-assist module. Make sure area is clean and have replacement pressure switch ready for immediate installation to prevent fluid loss and keep dirt out.

g. Remove SAS pressure switch and packings from pilot-assist module. Discard packings.

7-4-35.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packings and packing seats with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

c. Install packings on SAS pressure switch. Install SAS pressure switch in pilot-assist module (Figure 7-4-35-1, Detail A). **TORQUE PRESSURE SWITCH TO 166 - 184 INCH-POUNDS.** Lockwire, Item 196, Appendix D, pressure switch to pilot-assist module.

d. Remove protective caps and plugs on electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P467R, to SAS pressure switch.

g. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

h. Bleed hydraulic system (PARA 7-4-65).

i. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

7-4-36 PILOT-ASSIST MODULE THERMAL RELIEF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-65

7-4-36.1 Replace Pilot-Assist Module Thermal Relief Valve.

7-4-36.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter pilot-assist module. Make sure area is clean and have replacement thermal relief valve ready for installation before old valve is removed.

d. Remove thermal relief valve and packing from pilot-assist module (Figure 7-4-36-1, Detail A). Discard packing.

7-4-36.1.2. Install.

a. Coat packing with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

b. Install packing on thermal relief valve and install valve in pilot-assist module (Figure 7-4-36-1, Detail A). **TORQUE THERMAL RELIEF VALVE TO 71 - 79 INCH-POUNDS.** Lockwire, Item 196, Appendix D, thermal relief valve to module.

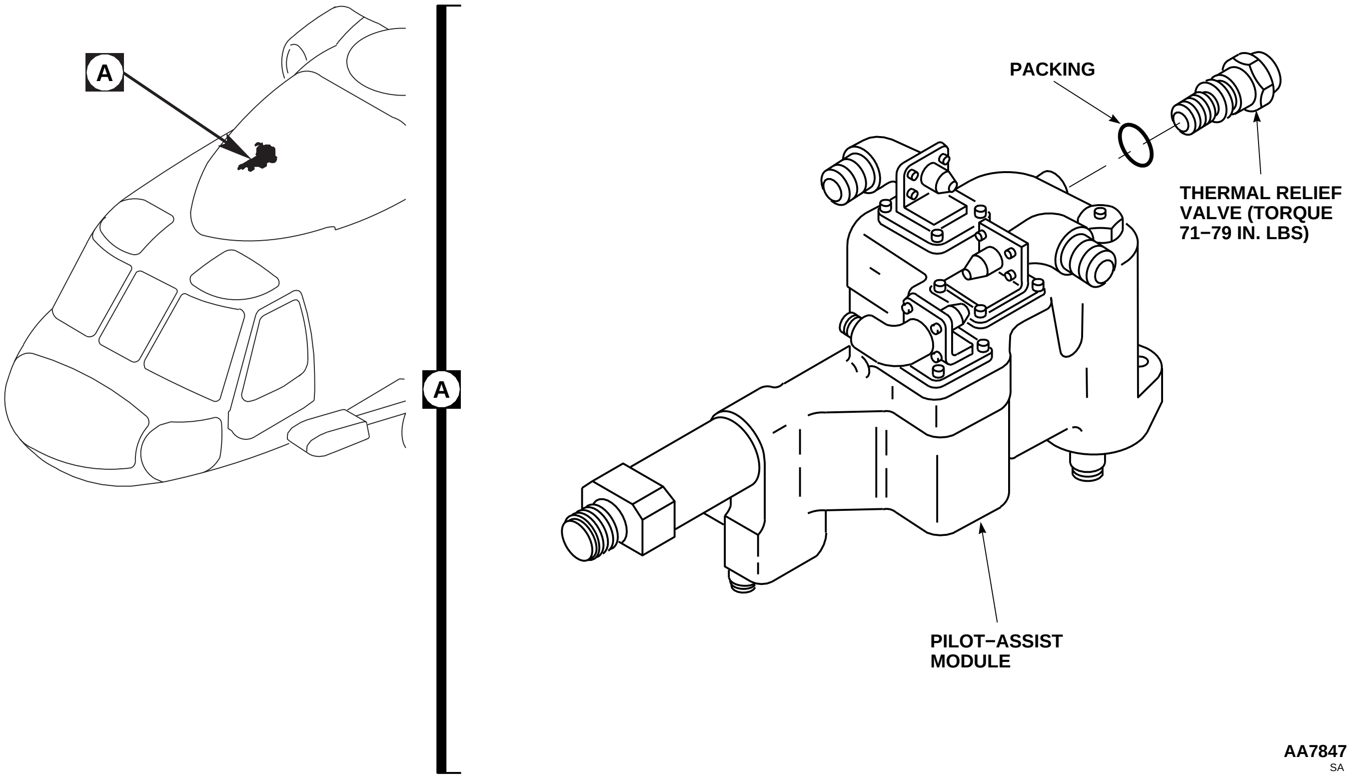


FIGURE 7-4-36-1. REPLACE PILOT-ASSIST MODULE THERMAL RELIEF VALVE

c. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

d. Bleed hydraulic system (PARA 7-4-65).

e. Perform operational check of No. 2 and backup hydraulic systems (PARA 7-2-1).

f. Make sure area is clean and free of foreign material.

g. Close main rotor pylon sliding cover.

7-4-37 BOOST SERVO AND PILOT-ASSIST MODULE MANIFOLD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Protective Cover
- Sealing Compound, Item 291, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-11
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-34
- PARA 7-4-65
- PARA 11-4-67
- PARA 11-4-68
- PARA 11-4-69
- PARA 11-4-70

7-4-37.1 Replace Boost Servo and Pilot-Assist Module Manifold.

7-4-37.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return hoses from No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

e. Remove pilot-assist servo module (PARA 7-4-34).

f. Remove collective boost servo (PARA 11-4-67).

g. Remove pitch/trim assembly (PARA 11-4-68).

h. Remove roll actuator (PARA 11-4-69).

i. Remove yaw boost servo (PARA 11-4-70).

NOTE

Cap or plug all tubes, ports, and fittings as parts are removed and disconnected.

j. Disconnect pilot boost servo pressure and return tubes from fittings on boost servo and pilot-assist manifold (Figure 7-4-37-1, Detail A).

k. Remove sealing compound from base of manifold and baseplate using nonmetallic scraper.

NOTE

Bolts securing manifold to baseplate are different lengths and must be tagged to indicate location.

l. Remove bolts and washers and remove manifold from baseplate. Tag bolts for reinstallation. Remove manifold.

m. Remove sealing compound from edges of baseplate and helicopter fuselage using nonmetallic scraper.

NOTE

- Bolts securing baseplate to helicopter fuselage are different lengths and must be tagged to indicate location.
- Brackets must be tagged to indicate location and orientation.

n. Remove bolts, washers, 70583–701743 clip, and brackets securing baseplate to helicopter fuselage and remove baseplate (Figure 7-4-37-1, Detail B). Tag bolts and brackets for reinstallation.

o. Clean remaining sealing compound from edge of boost servo and pilot-assist manifold using nonmetallic scraper and machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

p. Clean remaining sealing compound from edge of baseplate and helicopter fuselage using nonmetallic scraper and machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

q. Clean sealing compound from shank of bolts using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

r. If new manifold is to be installed, remove fittings and packings from manifold (Figure 7-4-37-1, Detail A). Discard packings.

s. Cover manifold to keep dirt out of hydraulic system.

7-4-37.1.2 Inspect.

a. **DAMAGE TO ANY FLAT SURFACE UP TO MAXIMUM DEPTH OF 0.005-INCH IS CONSIDERED NEGLIGIBLE DAMAGE.**

b. **DAMAGE WITH SHARP DENTS OR SCRATCHES MUST BE DEBURRED AND WITHIN 0.005-INCH DEPTH AFTER CLEAN UP.**

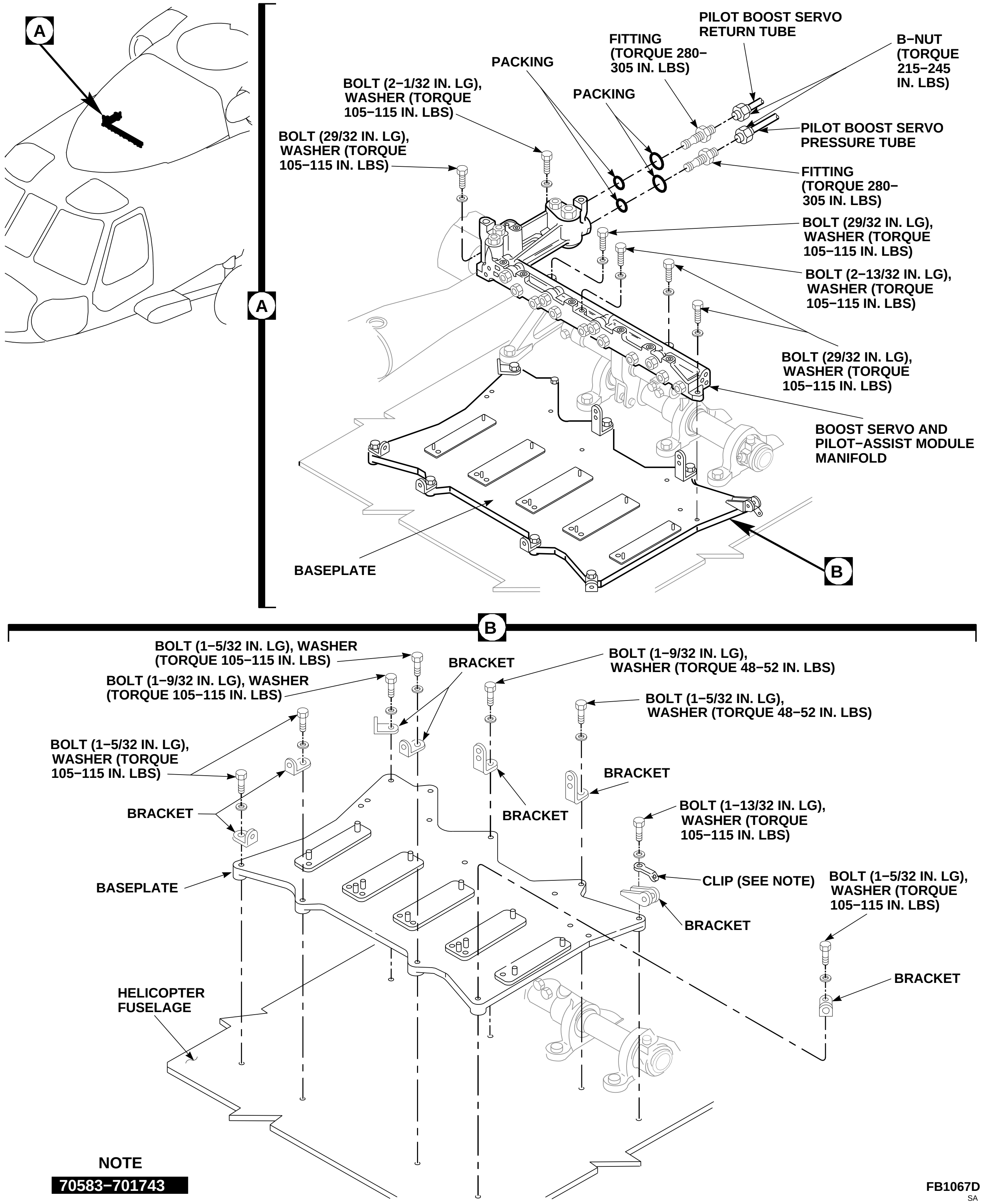


FIGURE 7-4-37-1. REPLACE BOOST SERVO AND PILOT-ASSIST MODULE MANIFOLD

c. **IF THERE WAS NO SEALING COMPOUND ON UNDERSIDE OF BASEPLATE, CLEAN AND/OR TREAT FUSELAGE AREA** (PARA 2-6-11).

7-4-37.1.3 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Remove protective cover from manifold.

NOTE

Remove caps and plugs from tubes, ports, and fittings as parts are installed and connected.

c. If removed, install fittings as follows:

(1) Coat packings and fittings with hydraulic fluid, Item 154 or Item 155, Appendix D.

(2) Install packings on fittings (Figure 7-4-37-1, Detail A).

(3) Install fittings in manifold ports. **TORQUE FITTINGS 280 - 305 INCH-POUNDS.**

d. Install baseplate as follows (Figure 7-4-37-1, Detail B):

(1) Position baseplate over mounting holes in helicopter fuselage.

NOTE

- Do not get sealing compound on threads of bolts.
- Bolts securing brackets and baseplate are different lengths. Make sure to keep track of hardware when tags are removed.
- Brackets must be installed in same location and orientation they were removed from. Make sure to keep track of location and orientation when tags are removed.

(2) Remove tags and install washers, brackets, **70583–701743** and clip on bolts. Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts.

NOTE

Bolts securing baseplate are different lengths. Remove tags from bolts as they are installed.

(3) Install bolts in front holes of baseplate. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts to baseplate.

(4) Install bolt in right rear mounting hole of baseplate. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to baseplate.

(5) Install bolt in left rear mounting hole of baseplate. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to baseplate.

(6) Install bolt in right center rear mounting hole of baseplate. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to baseplate.

(7) Install bolt in left center rear mounting hole of baseplate. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolt to baseplate.

(8) Apply a bead of sealing compound, Item 291, Appendix D, to edge of baseplate and helicopter fuselage and heads of each baseplate-to-fuselage bolt.

e. Install boost servo and pilot-assist manifold as follows (Figure 7-4-37-1, Detail A):

NOTE

Do not get sealing compound in mounting holes.

(1) Apply a light coat of sealing compound, Item 291, Appendix D, to base of manifold.

(2) Position manifold over mounting holes in baseplate.

NOTE

Bolts securing manifold are different lengths. Make sure to keep track of hardware when tags are removed.

(3) Remove tags and install washers on bolts. Apply a light coat of sealing compound, Item 291, Appendix D, to shank of bolts.

NOTE

Make sure bolts are installed in same location they were removed from.

(4) Install bolts in manifold. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts to manifold.

f. Connect pilot boost servo pressure and return tubes to fittings on manifold. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

g. Install yaw boost servo (PARA 11-4-70).

h. Install roll actuator (PARA 11-4-69).

i. Install pitch/trim assembly (PARA 11-4-68).

j. Install collective boost servo (PARA 11-4-67).

k. Install pilot-assist servo module (PARA 7-4-34).

l. Connect pressure and return hoses to No. 2 and backup hydraulic pump module quick-disconnects (PARA 7-4-12 and PARA 7-4-13).

m. Perform operational check of No. 2 and backup hydraulic systems (PARA 7-2-1).

n. Bleed No. 2 and backup hydraulic systems (PARA 7-4-65).

o. Make sure area is clean and free of foreign material.

p. Close main rotor pylon sliding cover.

q. Perform maintenance test flight (Appropriate MTF).

7-4-38 UTILITY MODULE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-13
- PARA 7-4-39
- PARA 7-4-40
- PARA 7-4-42
- PARA 7-4-65

7-4-38.1 Replace Utility Module.

7-4-38.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect electrical connector, P445, from pressure switch (Figure 7-4-38-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

f. Disconnect pressure and return hoses from backup hydraulic pump module quick-disconnects (PARA 7-4-13).

NOTE

Install caps and plugs on tubes and fittings as tubes are disconnected.

g. Use 2-quart container and machinery towel, Item 211, Appendix D, to catch hydraulic fluid.

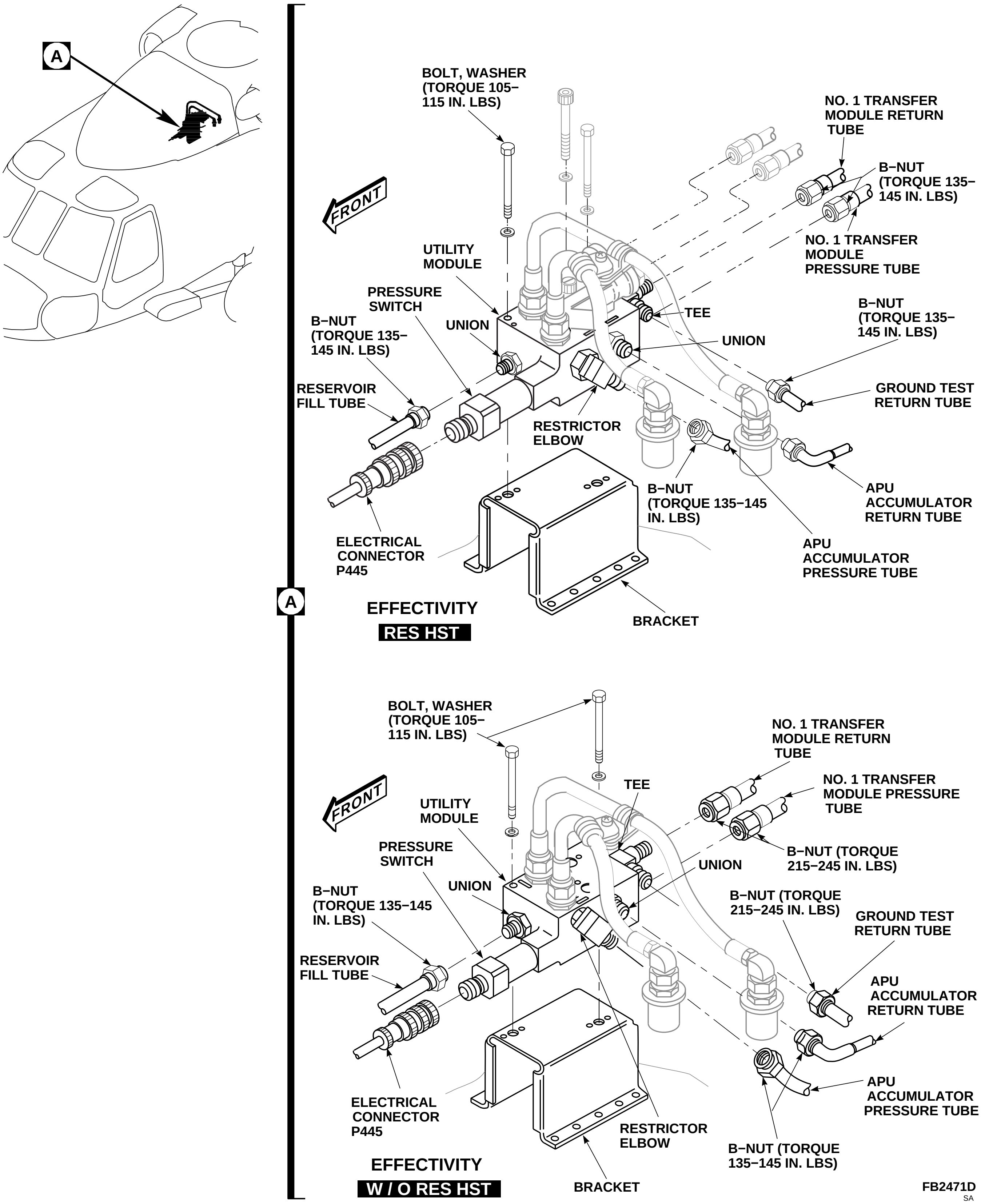


FIGURE 7-4-38-1. REPLACE UTILITY MODULE

h. RES HST Remove rescue hoist manifold (PARA 7-4-42).

i. Disconnect ground test and No. 1 transfer module return tubes from tee.

j. Disconnect No. 1 transfer module pressure tube from union.

k. Disconnect APU accumulator pressure and return tubes from restrictor elbow and union.

l. Disconnect reservoir fill tube from union.

m. If new utility module is to be installed, remove hoses and fittings and pressure switch from module (PARA 7-4-39 and PARA 7-4-40).

n. Remove W/O RES HST bolts and washers RES HST bolt and washer securing module to bracket. Remove utility module from bracket.

7-4-38.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. W/O RES HST Place utility module on bracket with pressure switch facing front. Install bolts and washers (Figure 7-4-38-1, Detail A). **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

c. RES HST Place utility module on bracket with pressure switch facing front, and secure with front bolt and washer. Do not torque bolt now (Figure 7-4-38-1, Detail A).

d. Coat all threaded connections with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D, before installation.

NOTE

Remove caps and plugs from tubes and fittings as tubes are installed.

e. Connect No. 1 transfer module pressure tube to union on rear of module. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

f. Connect APU accumulator pressure tube to restrictor elbow on side of module. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

g. Connect APU accumulator return tube to union on side of module. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

h. Connect reservoir fill tube to union on front of module. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

i. Connect No. 1 transfer module return tube to tee on rear of module. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

j. Connect ground test return tube to tee on rear of module. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

k. If removed, install hoses and fittings and pressure switch in replacement utility module (PARA 7-4-39 and PARA 7-4-40).

l. RES HST Install rescue hoist manifold (PARA 7-4-42).

m. RES HST **TORQUE FRONT BOLT TO 105 - 115 INCH-POUNDS.**

n. Connect pressure and return hoses to backup hydraulic pump module quick-disconnects (PARA 7-4-13).

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Connect electrical connector, P445, to pressure switch. Lockwire, Item 196, Appendix D, connector.

r. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

s. Connect external electrical power (PARA 1-6-16).

t. Service backup hydraulic pump module (PARA 1-3-8), if required, then place

BACKUP HYD PUMP switch on upper console to ON.

u. Move all flight controls through their ranges, stop to stop.

NOTE

Make sure APU CONTR switch is OFF prior to motoring APU to discharge accumulator pressure.

v. In cabin, at APU start valve, manually operate APU start valve lever to motor APU to discharge accumulator pressure.

w. Check that backup system recharges accumulator.

x. On upper console, place BACKUP HYD PUMP switch to OFF.

y. Perform operational check of hydraulic system (PARA 7-2-1).

z. Disconnect external electrical power (PARA 1-6-16).

aa. Make sure area is clean and free of foreign material.

ab. Close main rotor pylon sliding cover.

7-4-39 UTILITY MODULE PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-13
- PARA 7-4-65

7-4-39.1 Replace Utility Module Pressure Switch.

7-4-39.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return hoses from backup hydraulic pump module quick-disconnects (PARA 7-4-13).

e. Disconnect electrical connector, P445, from pressure switch (Figure 7-4-39-1, Detail A).

f. Install protective caps and plugs on electrical connectors.

g. Remove pressure switch from utility module. Discard packings.

7-4-39.1.2 Install.

a. Coat packings and threads of pressure switch with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install packings on replacement pressure switch (Figure 7-4-39-1, Detail A).

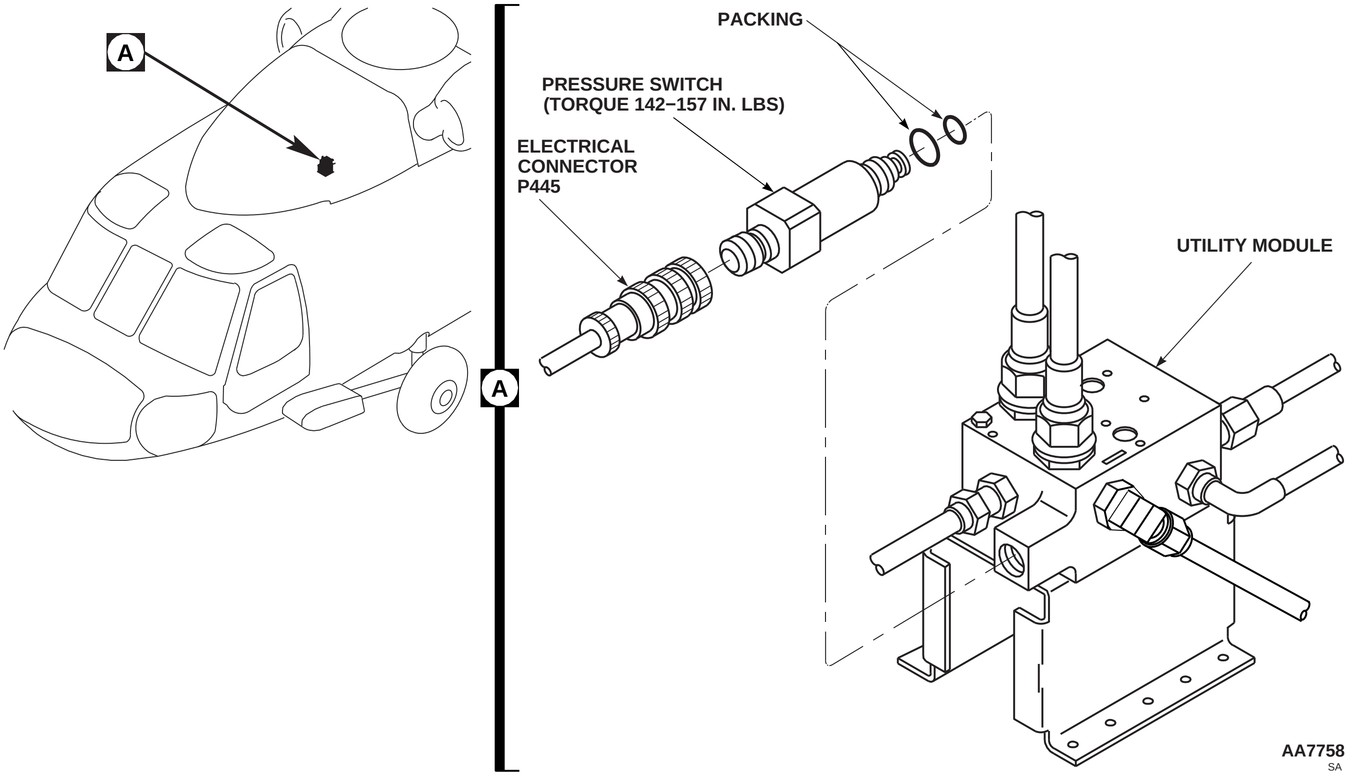


FIGURE 7-4-39-1. REPLACE UTILITY MODULE PRESSURE SWITCH

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter utility module. Make sure area is clean and replacement pressure switch is ready for installation before old switch is removed.

c. Install replacement pressure switch on utility module.

d. **TORQUE PRESSURE SWITCH TO 142 - 157 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire pressure switch.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connector, P445, to pressure switch. Using lockwire, Item 196, Appendix D, lockwire connector.

h. Connect pressure and return hoses to backup hydraulic pump module quick-disconnects (PARA 7-4-13).

i. Bleed hydraulic system (PARA 7-4-65).

j. Perform operational check of hydraulic system (PARA 7-2-1).

k. Make sure area is clean and free of foreign material.

l. Close main rotor pylon sliding cover.

7-4-40 UTILITY MODULE HOSES AND FITTINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-13
- PARA 7-4-65

7-4-40.1 Replace Utility Module Hoses and Fittings.

7-4-40.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return hoses from backup hydraulic pump module quick-disconnects (PARA 7-4-13).

NOTE

Install caps and plugs as hoses and fittings are disconnected.

e. Disconnect pressure and return hoses from unions on top of utility module (Figure 7-4-40-1, Detail A).

f. Remove quick-disconnect halves from pressure and return hoses.

NOTE

If replacing just one hose, leave clamp in position on opposite hose to mark clamp location.

g. Remove screw, washers, and nut securing clamps to hose assemblies. Remove clamps one at a time from hose assemblies using opposite hose to mark clamp location on replacement hose assemblies (Figure 7-4-40-1, Detail B).

h. Loosen jamnuts and remove tee and restrictor elbow from module. Remove retainers and packings from tee and restrictor elbow. Discard retainers and packings (Figure 7-4-40-1, Detail A).

i. Remove unions and packings from utility module. Discard packings.

7-4-40.1.2. Install.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Damage to utility module will result if hoses and fittings are not clean prior to installation. Make sure hoses and fittings are clean inside and out before installing.

NOTE

Remove plugs or caps as fittings and hoses are installed.

a. Before installation blow all hoses and fittings clear with dry compressed air.

b. Coat packings and fitting threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Install packings on unions. Install one union in right front port and one in left rear port of module (Figure 7-4-40-1, Detail A). **TORQUE UNIONS TO 155 - 165 INCH-POUNDS.**

d. Coat packings and fitting threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

e. Install packings on union and install union in front right port of module. **TORQUE UNION TO 280 - 305 INCH-POUNDS.**

f. Coat packings and fitting threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

g. Install packings on unions and install unions in top ports of utility module. **TORQUE UNIONS TO 280 - 305 INCH-POUNDS.**

h. Coat packings, retainer, and fitting threads, with hydraulic fluid, Item 154 or Item 155, Appendix D.

i. Install jamnut, retainer, and packings on tee. Install tee in front left port of module. Tighten jamnut after tee is aligned with hydraulic line. **TORQUE JAMNUT TO 280 - 305 INCH-POUNDS.**

j. Coat packing, retainer, and fitting threads, with hydraulic fluid, Item 154 or Item 155, Appendix D.

k. Install jamnut, retainer, and packings on restrictor elbow. Install elbow in right rear port of module. Tighten jamnut after elbow is aligned with hydraulic line. **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS.**

l. Coat quick-disconnect and hose threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

m. Install quick-disconnect halves on pressure and return hoses. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

NOTE

Pressure hose interference washer prevents it from being installed in module return port.

n. Fill hoses and coat threaded connections with hydraulic fluid, Item 154 or Item 155, Appendix D, if module is being installed now.

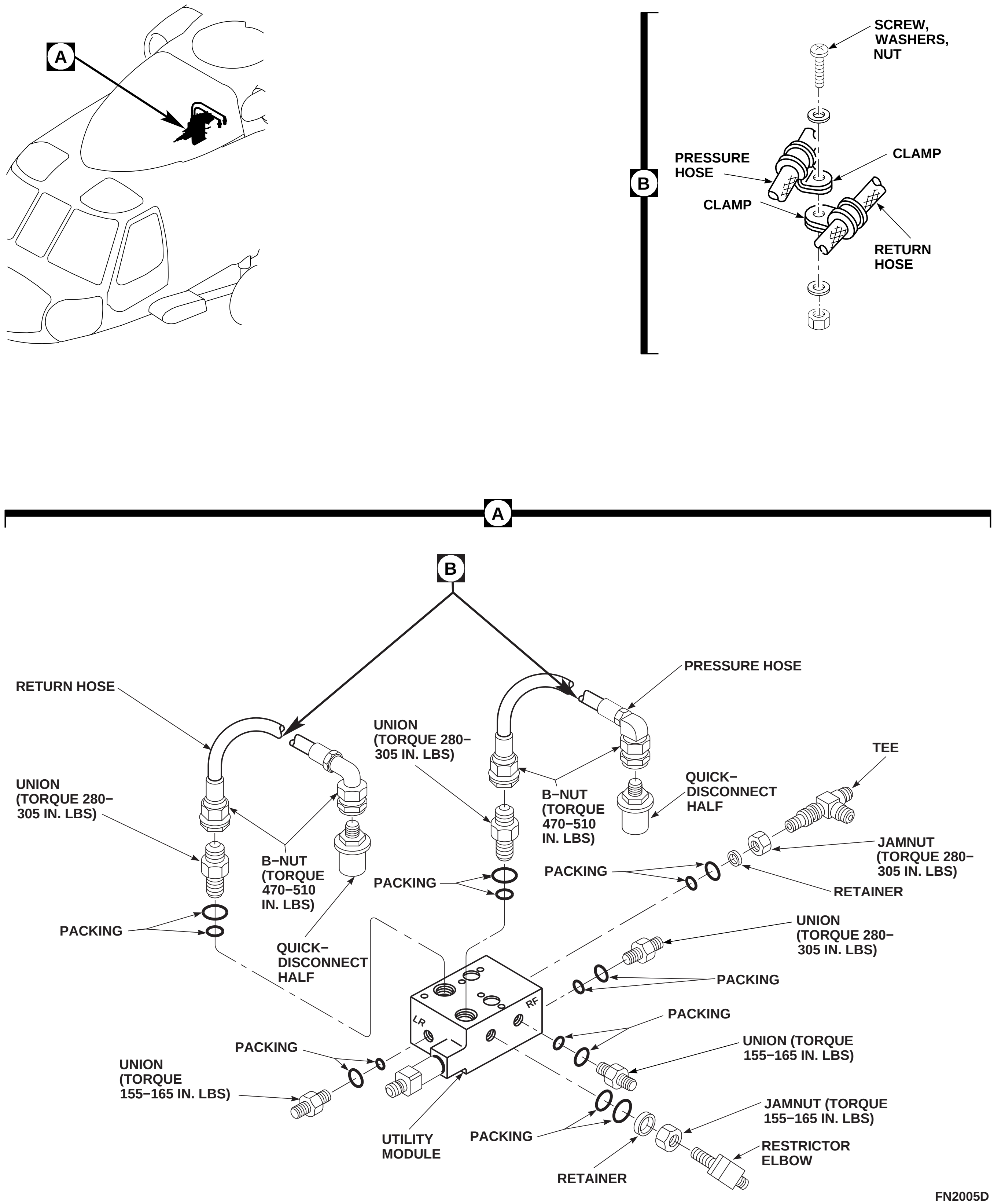


FIGURE 7-4-40-1. REPLACE UTILITY MODULE HOSES AND FITTINGS

o. Connect pressure and return hoses to unions on top of module. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

p. Connect pressure and return hoses to backup hydraulic pump module quick-disconnects (PARA 7-4-13).

q. Position and connect clamps on replacement hoses using screw, washers, and nut (Figure 7-4-40-1, Detail B).

r. Bleed hydraulic system (PARA 7-4-65).

s. Make sure area is clean and free of foreign material.

t. Close main rotor pylon sliding cover.

7-4-41 TAIL ROTOR SERVO 2ND STAGE SHUTOFF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-13
- PARA 7-4-65

7-4-41.1 Replace Tail Rotor Servo 2nd Stage Shutoff Valve.

7-4-41.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return hoses from backup hydraulic pump module quick-disconnects to isolate pressure from backup hydraulic pump module reservoir (PARA 7-4-13).

e. Disconnect electrical connector, P433, from shutoff valve (Figure 7-4-41-1, Detail A).

f. Install protective caps and plugs on electrical connectors.

g. Use 1-quart container and machinery towels, Item 211, Appendix D, to catch hydraulic fluid.

NOTE

Install caps and plugs as tubes are disconnected.

h. Disconnect pressure and return tubes from shutoff valve and tee check valve.

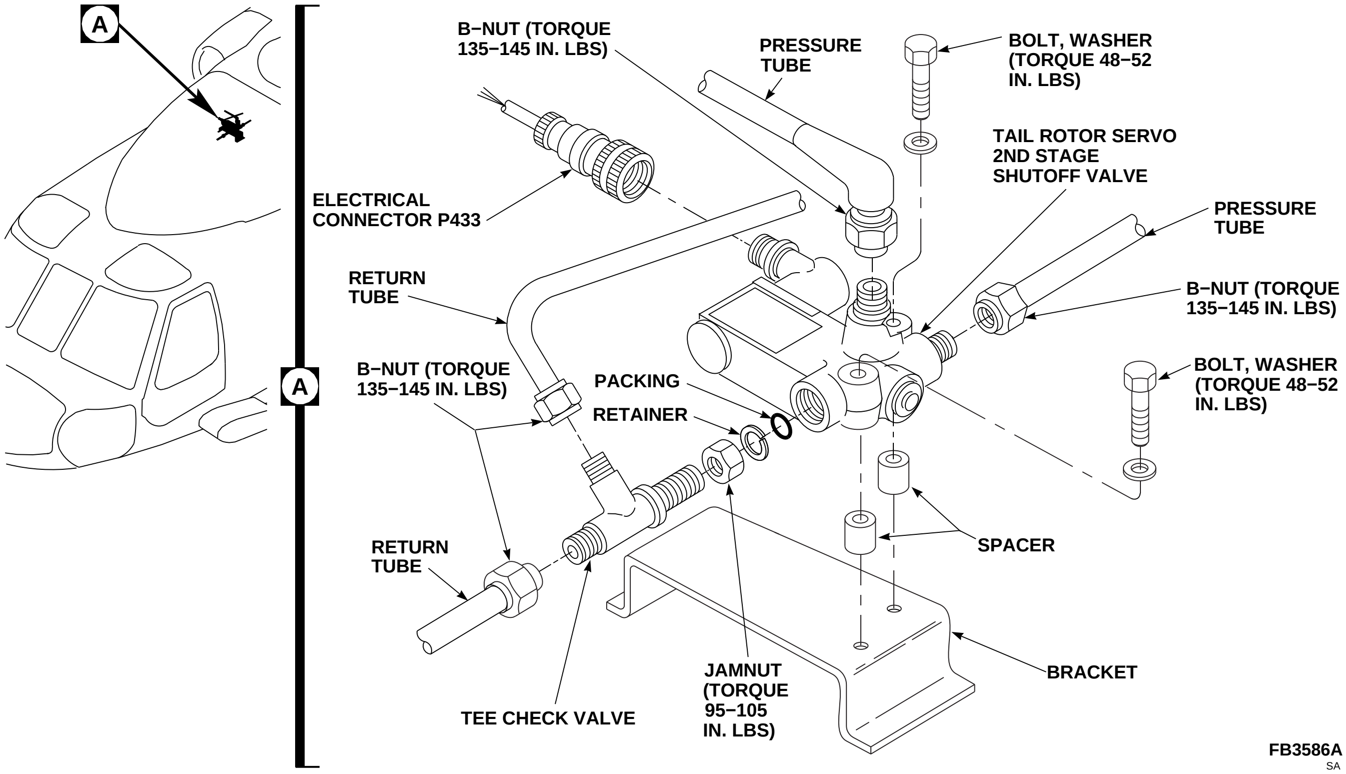


FIGURE 7-4-41-1. REPLACE TAIL ROTOR SERVO 2ND STAGE SHUTOFF VALVE

i. Loosen jamnut and remove tee check valve, retainer, and packing. Discard packing and retainer.

j. Remove bolts, washers, and spacers securing shutoff valve to bracket.

7-4-41.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Place spacers under shutoff valve and secure to bracket with bolts and washers (Figure 7-4-41-1, Detail A). **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

NOTE

Remove caps and plug as tubes and fittings are installed.

c. Coat retainer, packing, and tee threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

d. Install jamnut, retainer, and packing on tee check valve. Install tee check valve in shutoff valve. Do not tighten jamnut now.

e. Connect return tubes to tee check valve. Tighten jamnut on tee check valve. **TORQUE JAMNUT TO 95 - 105 INCH-POUNDS. TORQUE RETURN TUBE B-NUTS TO 135 - 145 INCH-POUNDS.**

f. Coat connection threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

g. Connect pressure tubes to shutoff valve. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

h. Remove protective caps and plugs on electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. Connect electrical connector, P433, to shutoff valve.

k. Connect pressure and return hoses to backup hydraulic pump module quick-disconnects (PARA 7-4-13).

l. Connect external electrical power (PARA 1-6-16), or run APU (Appropriate Flight Manual).

m. Service backup hydraulic pump module (PARA 1-3-8), if required, then place BACKUP HYD PUMP switch on upper console to ON.

n. On lower console, place TAIL SERVO switch to BACKUP. Run for 2 minutes, then return switch to NORMAL.

o. Bleed backup hydraulic system (PARA 7-4-65). Repeat step m. and step n. until all air is bled from system.

p. On upper console, place BACKUP HYD PUMP switch to OFF, and disconnect external electrical power (PARA 1-6-16), or shut down APU (Appropriate Flight Manual).

q. Perform operational check of hydraulic system (PARA 7-2-1).

r. Make sure area is clean and free of foreign material.

s. Close and latch main rotor pylon sliding cover.

7-4-42 RESCUE HOIST MANIFOLD RES HST .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-13
- PARA 7-4-38
- PARA 7-4-65

7-4-42.1 Replace Rescue Hoist Manifold.

7-4-42.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return lines at backup hydraulic pump module quick-disconnects (PARA 7-4-13).

e. Relieve hydraulic pressure and drain hydraulic fluid from utility module (PARA 7-4-38).

NOTE

Install caps and plugs on tubes and ports as tubes are disconnected and rescue hoist module is removed.

f. Disconnect rescue hoist return tube from rescue hoist manifold (Figure 7-4-42-1, Detail A).

g. Disconnect rescue hoist pressure tube from rescue hoist manifold.

h. Remove bolt and washer holding rescue hoist manifold to utility module and bracket.

i. Remove capscrews and washers securing rescue hoist manifold to utility module.

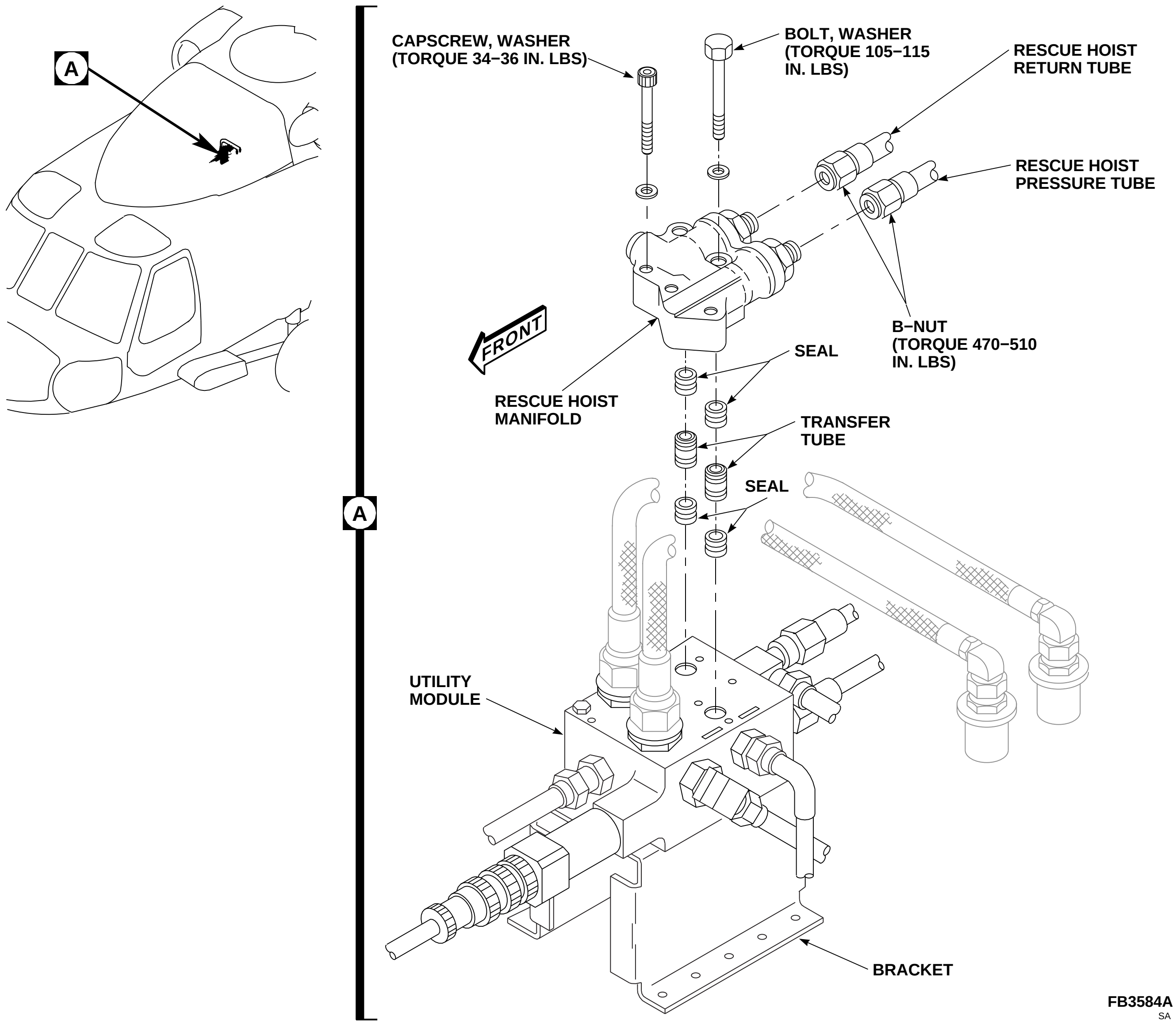


FIGURE 7-4-42-1. REPLACE RESCUE HOIST MANIFOLD

j. Carefully remove rescue hoist manifold from utility module. Remove transfer tubes and seals from manifold. Discard seals.

7-4-42.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

NOTE

Remove caps and plugs from tubes and ports as rescue hoist module is installed.

b. Lubricate all packings, seals, and fitting threads with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

CAUTION

Damage to transfer tubes will result if rescue hoist manifold is incorrectly installed on utility module. Make sure transfer tubes are positioned correctly when installing rescue hoist manifold on utility module.

c. Install seals on transfer tubes and install tubes in utility module (Figure 7-4-42-1, Detail A).

d. Carefully line up ports in rescue hoist manifold with tubes in utility module and install manifold.

e. Install capscrews and washers. **TORQUE CAPSCREWS TO 34 - 36 INCH-POUNDS.** Lockwire, Item 197, Appendix D, capscrews.

f. Install bolt and washer holding rescue hoist manifold and utility module to bracket. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.**

g. Connect rescue hoist pressure and return tubes to rescue hoist manifold. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

h. Connect pressure and return lines at backup hydraulic pump module quick-disconnects (PARA 7-4-13).

i. Bleed backup hydraulic system (PARA 7-4-65).

j. Perform operational check of hydraulic system (PARA 7-2-1).

k. Close main rotor pylon sliding cover.

7-4-43 RESCUE HOIST TUBES RES HST .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 2-Quart
- Nonmetallic Scraper, Locally-Made
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 287, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-13
- PARA 7-4-38
- PARA 7-4-44
- PARA 7-4-45
- PARA 7-4-65
- PARA 14-2-1

7-4-43.1 Replace Rescue Hoist Tubes.

7-4-43.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return lines at backup hydraulic pump module quick-disconnects (PARA 7-4-13).

e. Relieve hydraulic pressure and drain hydraulic fluid from utility module (PARA 7-4-38).

f. To remove rescue hoist tubing, perform the following:

(1) Remove screws, washers, spacer, nut, and clamps from rescue hoist tubes (Figure 7-4-43-1, Sheet 1, Detail A and Detail B).

(2) Remove screws, washers, channels, spacers, and blocks from rescue hoist tubes (Figure 7-4-43-1, Sheet 1, Detail C).

(3) Remove rescue hoist priority valve (PARA 7-4-44). Remove pressure tube section.

(4) Use 2-quart container and machinery towels, Item 211, Appendix D, to catch hydraulic fluid.

NOTE

Install caps and plugs on tubes and fittings as tubes are disconnected.

(5) Disconnect return tube from rescue hoist manifold (Figure 7-4-43-1, Sheet 2, Detail H).

(6) Loosen B-nuts at unions and remove sections of tubing (Figure 7-4-43-1, Sheet 1).

(7) Loosen rescue hoist return tube B-nut at tee. Remove tube from helicopter (Figure 7-4-43-1, Sheet 2, Detail D).

(8) Loosen rescue hoist pressure tube B-nut at elbow. Remove tube from helicopter (Figure 7-4-43-1, Sheet 2, Detail E).

g. To remove rescue hoist support hoses, perform the following:

(1) Using nonmetallic scraper, remove sealing compound from hydraulic hoses and rescue hoist structure to allow removal of hoses (Figure 7-4-43-1, Sheet 2, Detail F).

(2) Use machinery towels, Item 211, Appendix D, to catch fluid as hoses are disconnected.

NOTE

Install caps and plugs on hoses and fittings as hoses are disconnected.

(3) Disconnect rescue hoist return hose from tee (Figure 7-4-43-1, Sheet 2, Detail D).

(4) Disconnect rescue hoist pressure hose from elbow (Figure 7-4-43-1, Sheet 2, Detail E).

(5) Disconnect pressure and return hoses from unions and remove hoses (Figure 7-4-43-1, Sheet 2, Detail G). Discard packings from unions.

h. Remove jamnut, washers, and packing from elbow and remove elbow (Figure 7-4-43-1, Sheet 2, Detail E). Discard packing.

i. Remove plugstat (PARA 7-4-45).

j. Remove jamnut, washers, and packing from tee and remove tee. Discard packing (Figure 7-4-43-1, Sheet 2, Detail D).

7-4-43.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Install tee with washer and packing through flight control deck. Position tee forward (Figure 7-4-43-1, Sheet 2, Detail D).

c. Install washer and jamnut on tee. **TORQUE JAMNUT 240 - 280 INCH-POUNDS.**

d. Insert elbow with washer and packing through flight control deck. Position elbow forward (Figure 7-4-43-1, Sheet 2, Detail E).

e. Install washer and jamnut on elbow. **TORQUE JAMNUT 120 - 150 INCH-POUNDS.**

f. To install rescue hoist hoses, perform the following:

(1) Remove old sealing compound from rescue hoist support access hole. Using dry-cleaning solvent, Item 124, Appendix D, loosen up sealing compound and scrape off with nonmetallic scraper (Figure 7-4-43-1, Sheet 2, Detail F).

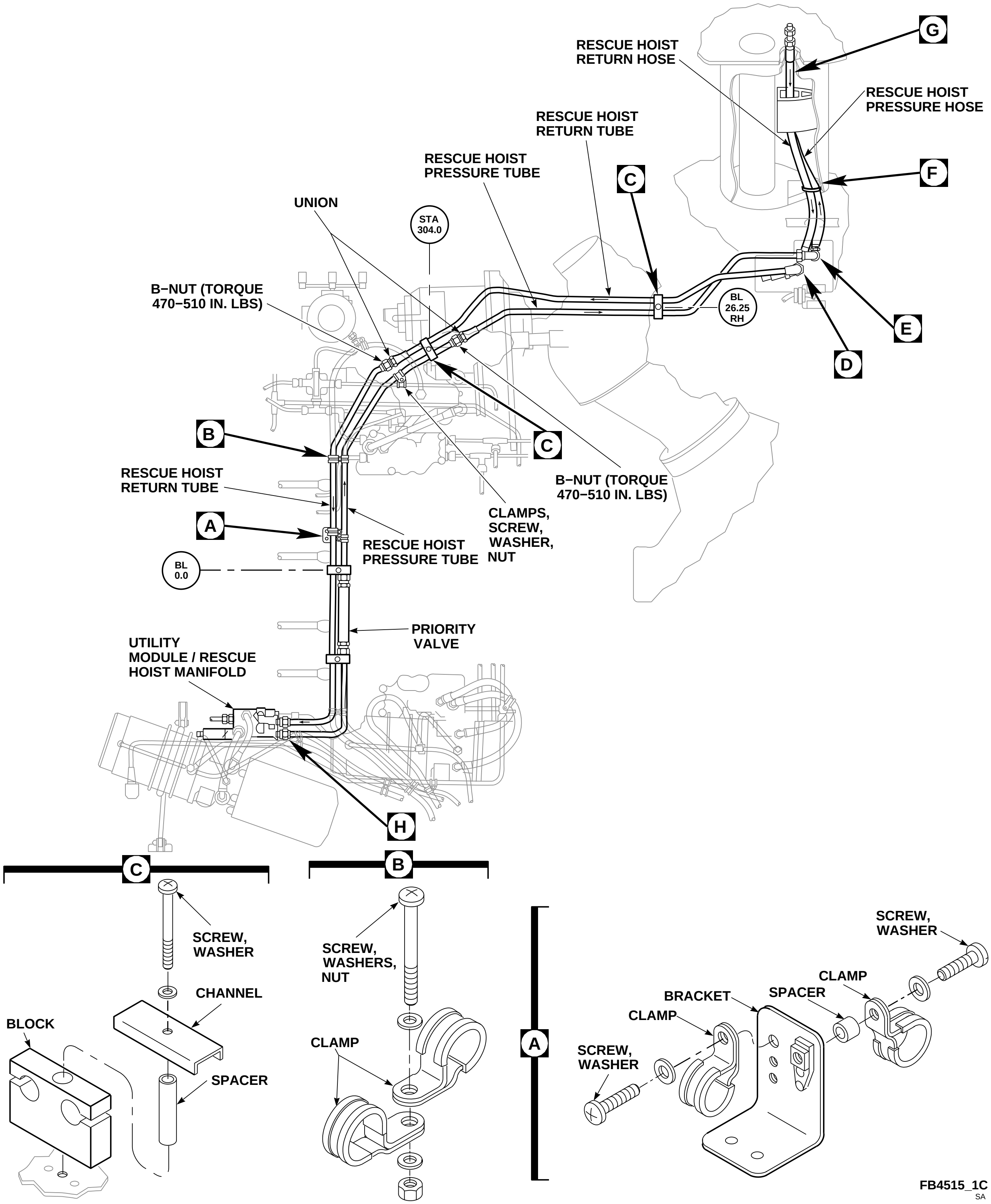


FIGURE 7-4-43-1. REPLACE RESCUE HOIST TUBES (SHEET 1 OF 2)

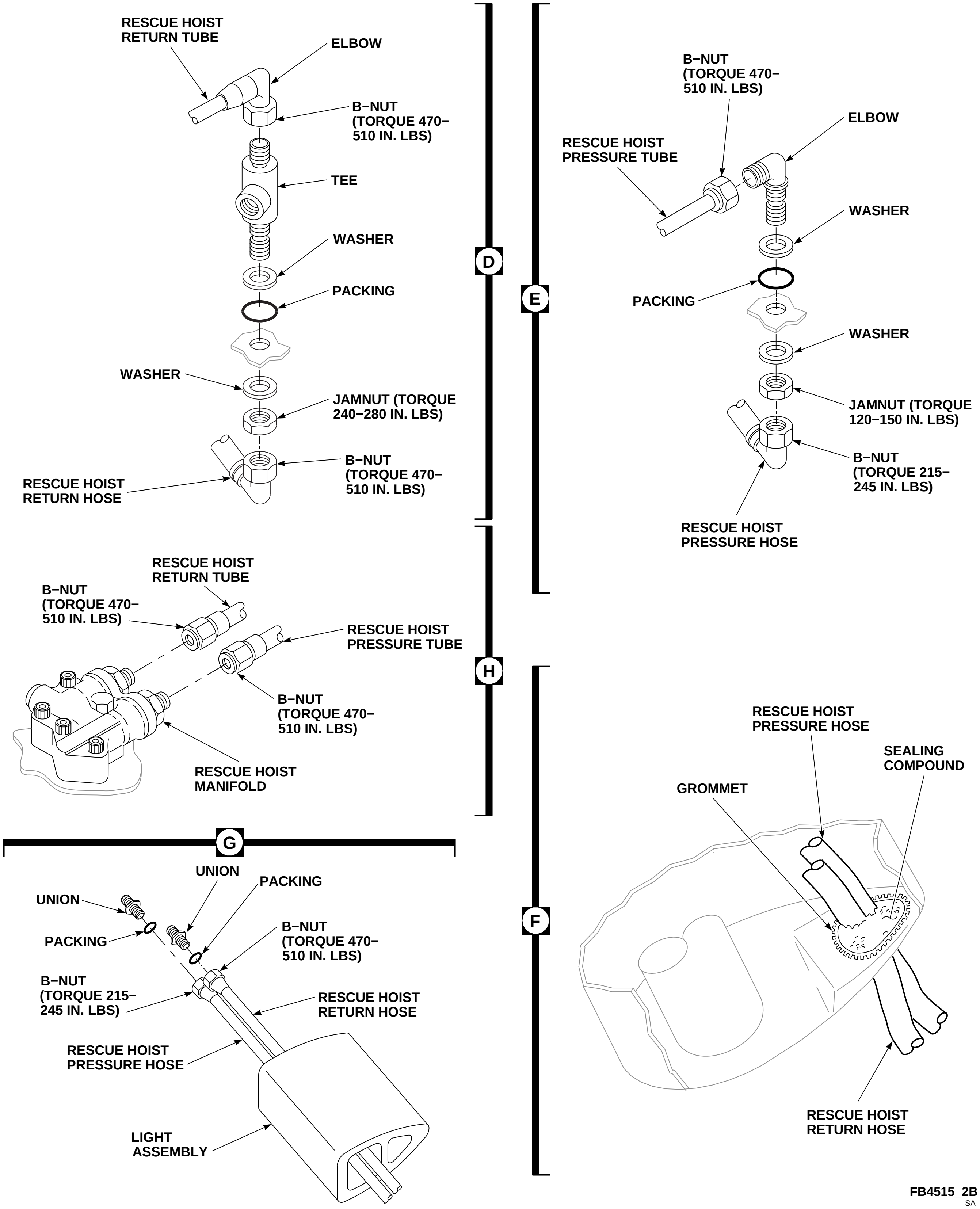


FIGURE 7-4-43-1. REPLACE RESCUE HOIST TUBES (SHEET 2 OF 2)

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Blow all hoses clear at installation with clean, dry compressed air.

(2) Install replacement hoses through access hole and thread through light assembly.

(3) Install packings on unions (Figure 7-4-43-1, Sheet 2, Detail G).

NOTE

Remove caps and plugs from hoses and fittings as hoses are connected.

(4) Connect rescue hoist pressure hose to union. **TORQUE B-NUT 215 - 245 INCH-POUNDS.**

(5) Connect rescue hoist return hose to union. **TORQUE B-NUT 470 - 510 INCH-POUNDS.**

(6) Connect rescue hoist pressure hose to elbow (Figure 7-4-43-1, Sheet 2, Detail E). **TORQUE HOSE B-NUT 215 - 245 INCH-POUNDS.**

(7) Connect rescue hoist return hose to tee (Figure 7-4-43-1, Sheet 2, Detail D). **TORQUE B-NUT 470 - 510 INCH-POUNDS.**

(8) Apply sealing compound, Item 275, Appendix D, to fill large gaps between rescue hoist hoses and edge of access hole (Figure 7-4-43-1, Sheet 2, Detail F).

(9) Apply sealing compound, Item 292, Appendix D, to rescue hoist hoses and access hole area to form a continuous flexible seal.

g. To install rescue hoist tubes, perform the following:

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Blow all tubes clear at installation with clean, dry compressed air.

(1) Lubricate threads of fittings with hydraulic fluid, Item 154 or Item 155, Appendix D, before connecting tubes.

CAUTION

Damage to hydraulic lines will result if tubes are bent, strained, or forced into place. Carefully check routing and connect ends loosely to check for proper installation before tightening.

NOTE

Remove caps and plugs from tubes and fittings as tubes are connected.

(2) Connect rescue hoist return tubes between rescue hoist manifold, union and tee (Figure 7-4-43-1, Sheet 1, and Figure 7-4-43-1, Sheet 2, Detail D and Detail H). **TORQUE B-NUTS 470 - 510 INCH-POUNDS.**

(3) Connect rescue hoist pressure tubes between elbow, union and rescue hoist priority valve (Figure 7-4-43-1, Sheet 1, and Figure 7-4-43-1, Sheet 2, Detail E and Detail H). **TORQUE B-NUTS 470 - 510 INCH-POUNDS.**

(4) Install priority valve (PARA 7-4-44) and pressure tube.

(5) Install clamps on rescue hoist tubes (Figure 7-4-43-1, Sheet 1, Detail A and Detail B).

(6) Install block and channels on rescue hoist tubes as follows (Figure 7-4-43-1, Sheet 1, Detail C):

(a) Install tubes into blocks.

NOTE

Screws must be installed while sealing compound is still wet. Do not allow sealing compound to dry on screws prior to installation.

(b) Using sealing compound, Item 287, Appendix D, coat threads of screws.

(c) Secure blocks to airframe with channels, spacers, screws, and washers.

(7) Using sealing compound, Item 292, Appendix D, seal block to deck and head of screw to channel.

h. Install plugstat (PARA 7-4-45).

i. Connect pressure and return lines to backup hydraulic pump module at quick-disconnects (PARA 7-4-13).

j. Bleed backup hydraulic system (PARA 7-4-65).

k. Make sure area is clean and free of foreign material.

l. Close main rotor pylon sliding cover.

m. Perform operational check of rescue hoist system (PARA 14-2-1).

7-4-44 RESCUE HOIST PRIORITY VALVE RES HST .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 287, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-38
- PARA 7-4-65
- PARA 14-2-1

7-4-44.1 Replace Rescue Hoist Priority Valve.

7-4-44.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor sliding pylon cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return lines at backup pump module quick-disconnects (PARA 7-4-38).

e. Remove screws, washers, channels, and spacers securing tube blocks on either end of priority valve to deck (Figure 7-4-44-1, Detail A).

f. Slide tube blocks a few inches away from priority valve.

NOTE

Install caps and plugs on tubes and fittings as tubes are disconnected.

g. Use machinery towel, Item 211, Appendix D, to catch hydraulic fluid as tubing is disconnected.

h. Disconnect rescue hoist pressure tube from fitting at rescue hoist manifold.

i. Disconnect rescue hoist pressure tubes from unions at priority valve.

j. Remove unions and packings from priority valve. Discard packings.

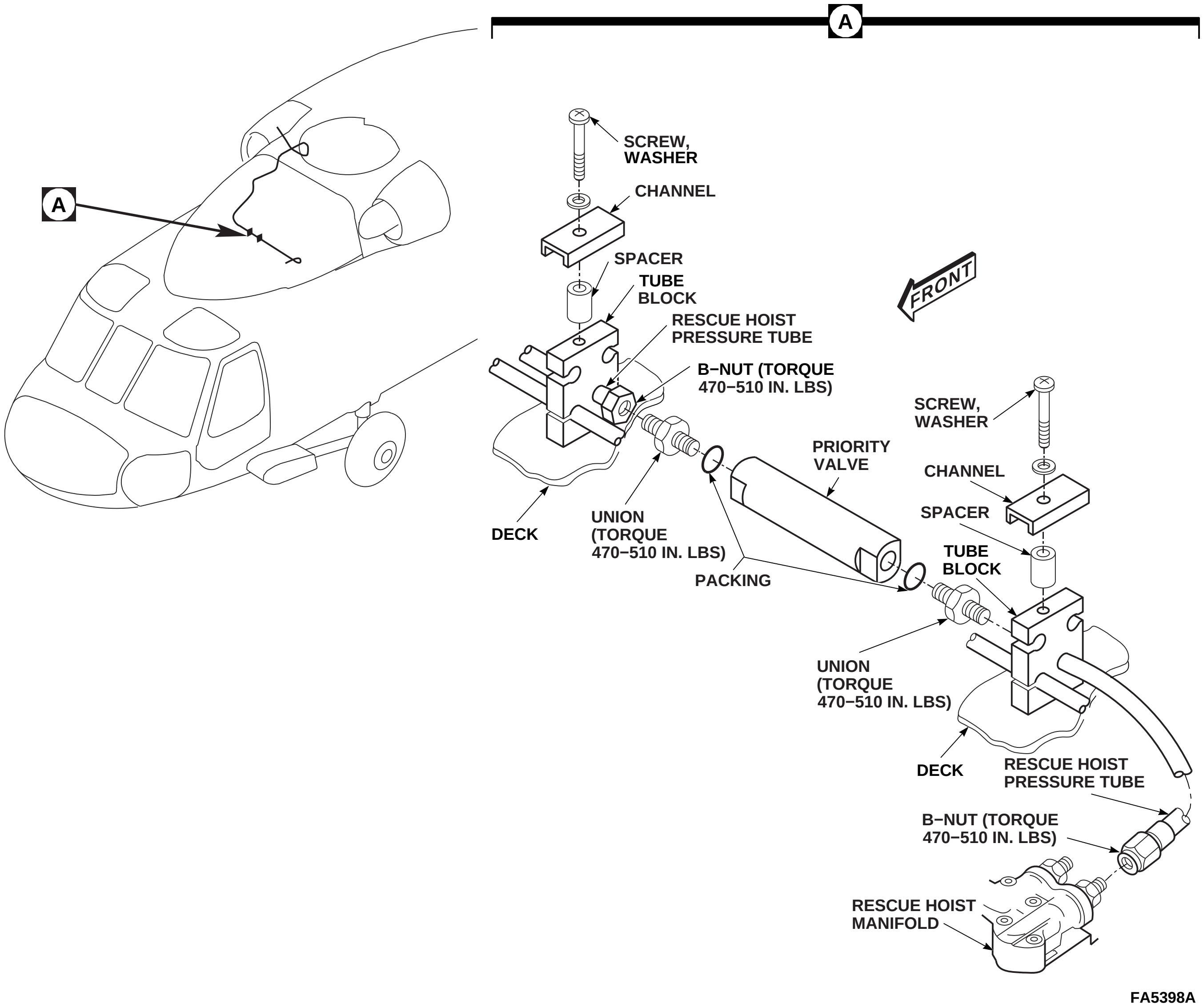


FIGURE 7-4-44-1. REPLACE RESCUE HOIST PRIORITY VALVE

7-4-44.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

NOTE

Remove caps and plug from tubes and fittings as tubes are installed.

b. Using hydraulic fluid, Item 154 or Item 155, Appendix D, lubricate packings. Install packings on unions (Figure 7-4-44-1, Detail A).

c. Install unions in rescue hoist priority valve. **TORQUE UNIONS TO 470 - 510 INCH-POUNDS.**

d. Connect rescue hoist pressure tubes to unions at priority valve. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

e. Connect pressure tube to fitting at rescue hoist manifold. **TORQUE B-NUTS TO 470 - 510 INCH-POUNDS.**

f. Slide tube blocks into position and secure as follows:

NOTE

Screws must be installed while sealing compound is still wet. Do not allow sealing compound to dry on screws prior to installation.

(1) Using sealing compound, Item 287, Appendix D, coat threads of screws.

(2) Secure tube blocks to airframe with channels, spacers, screws, and washers.

g. Using sealing compound, Item 292, Appendix D, seal tube block to deck and screwhead to channel.

h. Connect pressure and return lines to backup pump module at quick-disconnects (PARA 7-4-38).

i. Bleed backup hydraulic system (PARA 7-4-65).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

l. Perform operational check of rescue hoist system (PARA 14-2-1).

7-4-45 RESCUE HOIST PLUGSTAT (THERMAL SWITCH) RES HST .

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1
- PARA 7-4-13
- PARA 7-4-65

7-4-45.1 Replace Rescue Hoist Plugstat (Thermal Switch).

7-4-45.1.1. Remove.

NOTE

To keep loss of hydraulic fluid to a minimum have replacement plugstat (thermal switch), with packing installed, ready to install as soon as the old plugstat is removed.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Disconnect pressure and return lines from backup pump module at quick-disconnects (PARA 7-4-13).

e. Disconnect electrical connector, P967, from plugstat (Figure 7-4-45-1, Detail A).

f. Install protective caps and plugs on electrical connectors.

g. Use machinery towel, Item 211, Appendix D, to catch hydraulic fluid when plugstat is removed.

h. Remove plugstat. Discard packing.

7-4-45.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

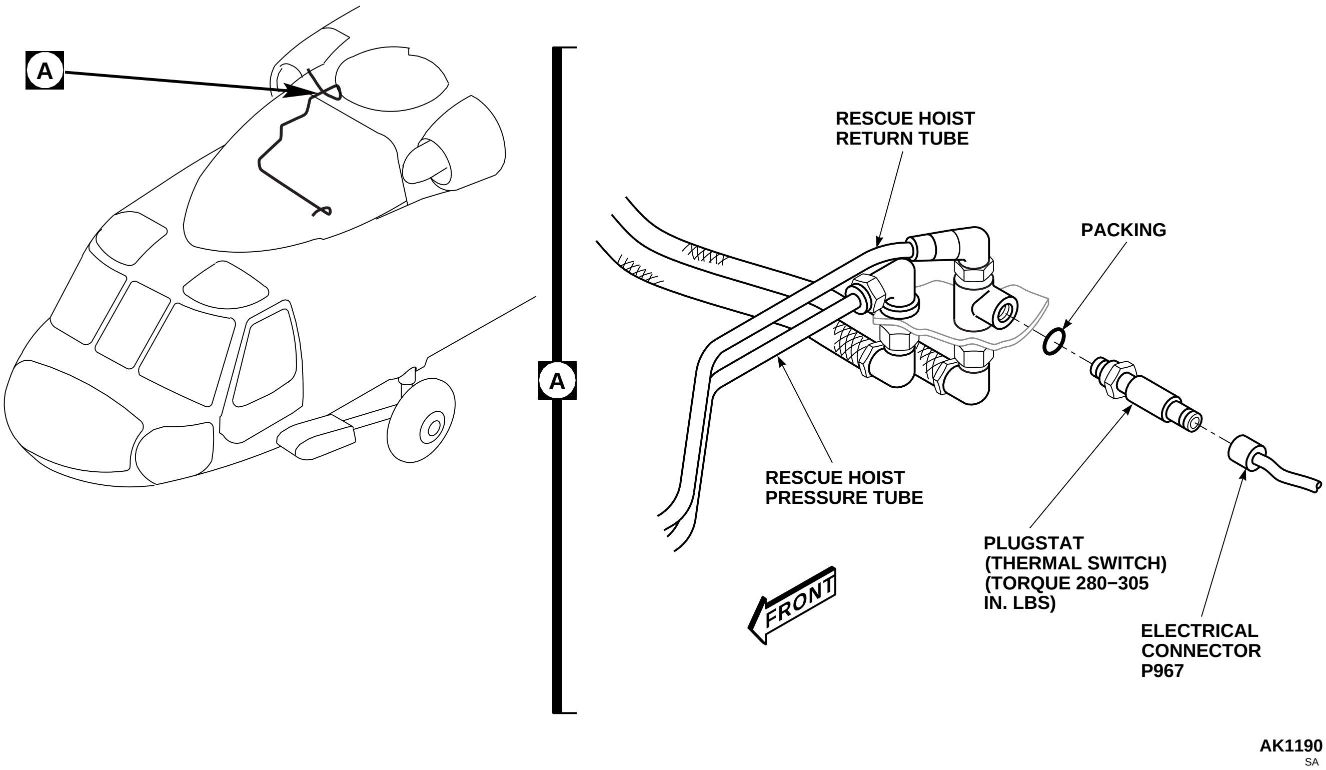


FIGURE 7-4-45-1. REPLACE RESCUE HOIST PLUGSTAT (THERMAL SWITCH)

b. Lubricate packing with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D, and install packing on replacement plugstat.

c. Install replacement plugstat and packing in tee (Figure 7-4-45-1, Detail A). **TORQUE PLUGSTAT TO 280 - 305 INCH-POUNDS.**

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P967, to plugstat.

g. Connect pressure and return lines to backup pump module at quick-disconnects (PARA 7-4-13).

h. Bleed backup hydraulic system (PARA 7-4-65).

i. Make sure area is clean and free of foreign material.

j. Close main rotor pylon sliding cover.

k. Perform operational check of rescue hoist system (PARA 7-2-1).

7-4-46 MANIFOLD SELF-SEALING COUPLINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-12
- PARA 7-4-13
- PARA 7-4-24
- PARA 7-4-25
- PARA 7-4-29
- PARA 7-4-30
- PARA 7-4-34
- PARA 7-4-37
- PARA 7-4-65
- PARA 11-4-68
- PARA 11-4-69

7-4-46.1 Replace Manifold Self-Sealing Couplings.

NOTE

Two sizes of self-sealing couplings are used in hydraulic manifolds. Replacement of both size self-sealing couplings is the same, except as noted.

7-4-46.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Remove the following component(s) as necessary to access affected manifold self-sealing coupling:

(1) No. 1 primary servo manifold (PARA 7-4-25).

(2) No. 2 primary servo manifold (PARA 7-4-30).

(3) No. 1 transfer module manifold (PARA 7-4-24).

(4) No. 2 transfer module manifold (PARA 7-4-29).

(5) Pilot-assist module (PARA 7-4-34).

(6) Pitch/trim assembly (PARA 11-4-68).

(7) Roll trim actuator (PARA 11-4-69).

(8) Boost servo (PARA 7-4-37).

d. Disconnect pressure and return lines at quick-disconnects of the following component(s), as required:

(1) No. 1/No. 2 hydraulic pump module (PARA 7-4-12).

(2) Backup hydraulic pump module (PARA 7-4-13).

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter manifold. Make sure area is clean and self-sealing coupling is ready for installation before old self-sealing coupling is removed.

e. Using cleaning cloth, Item 102, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, clean self-sealing coupling and surrounding area.

f. Remove self-sealing coupling and seal from manifold (Figure 7-4-46-1). Discard seal.

7-4-46.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Using hydraulic fluid, Item 154 or Item 155, Appendix D, coat seals.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

c. Using dry, clean, compressed air, clean large self-sealing coupling. Flush with hydraulic fluid, Item 154 or Item 155, Appendix D.

d. Install large self-sealing coupling as follows (Figure 7-4-46-1):

(1) Install large self-sealing coupling and seal in manifold.

(2) **TORQUE LARGE SELF-SEALING COUPLING TO 285 - 315 INCH-POUNDS.**

(3) Using lockwire, Item 196, Appendix D, lockwire coupling.

e. Install small self-sealing coupling as follows:

(1) Install small self-sealing coupling and seal in manifold.

(2) **TORQUE SMALL SELF-SEALING COUPLING TO 143- 157 INCH-POUNDS.**

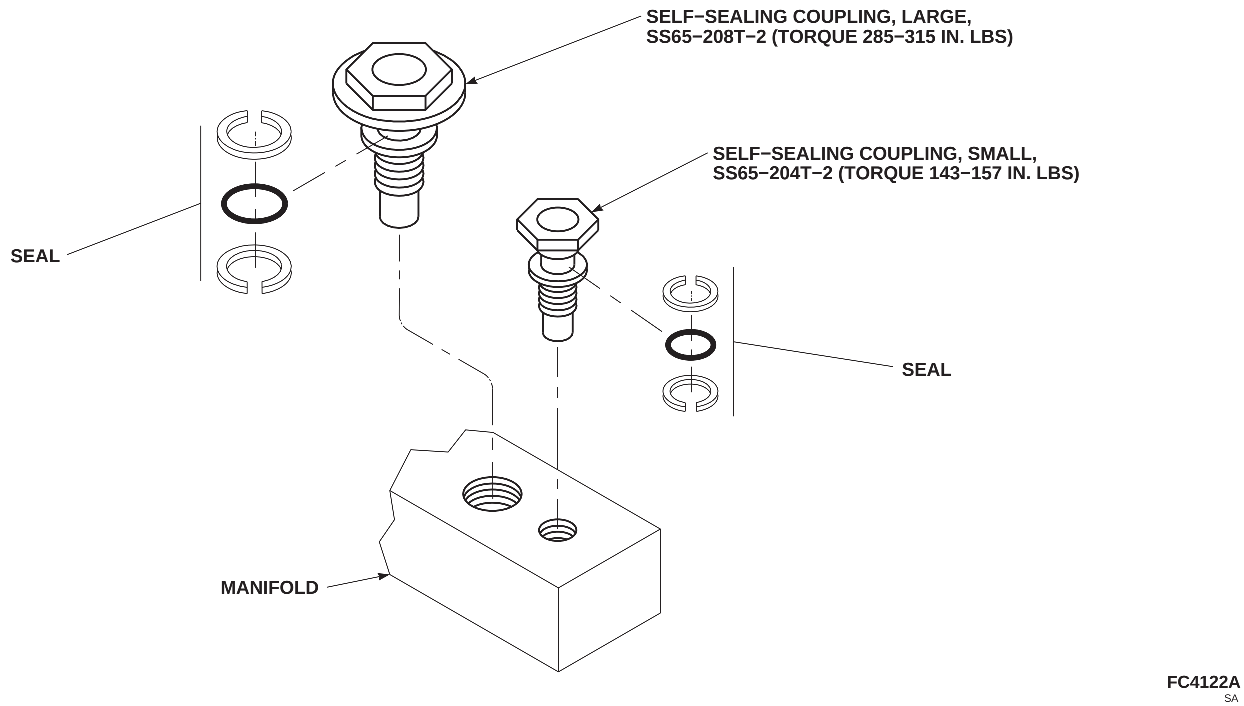


FIGURE 7-4-46-1. REPLACE MANIFOLD SELF-SEALING COUPLINGS

(3) Using lockwire, Item 196, Appendix D, lockwire coupling.

f. Install the following previously removed component(s):

(1) No. 1 primary servo manifold (PARA 7-4-25).

(2) No. 2 primary servo manifold (PARA 7-4-30).

(3) No. 1 transfer module manifold (PARA 7-4-24).

(4) No. 2 transfer module manifold (PARA 7-4-29).

(5) Pilot-assist module (PARA 7-4-34).

(6) Pitch/trim assembly (PARA 11-4-68).

(7) Roll trim actuator (PARA 11-4-69).

(8) Boost servo (PARA 7-4-37).

g. Connect pressure and return lines at quick-disconnects of the following component(s), as required:

(1) No. 1/No. 2 hydraulic pump module (PARA 7-4-12).

(2) Backup hydraulic pump module (PARA 7-4-13).

h. Service hydraulic pump module reservoirs (PARA 1-3-8).

i. Bleed hydraulic system (PARA 7-4-65).

j. Perform operational check of No. 1, No. 2, or backup hydraulic system as necessary to check self-sealing coupling for leakage (PARA 7-2-1).

k. Make sure area is clean and free of foreign material.

l. Close main rotor pylon sliding cover.

7-4-47 HYDRAULIC SELF-SEALING COUPLINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Padded Pliers
- Padded Vise
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Protective Cover

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-26
- PARA 7-4-31
- PARA 7-4-34
- PARA 7-4-65
- PARA 11-4-67
- PARA 11-4-68
- PARA 11-4-69
- PARA 11-4-70
- PARA 11-4-87

7-4-47.1 Replace Hydraulic Self-Sealing Couplings.

NOTE

Two sizes of self-sealing couplings are used on modules and servos. Procedures are for both sizes.

7-4-47.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Remove the following components as required to access hydraulic self-sealing coupling:

(1) No. 1 transfer module (PARA 7-4-26).

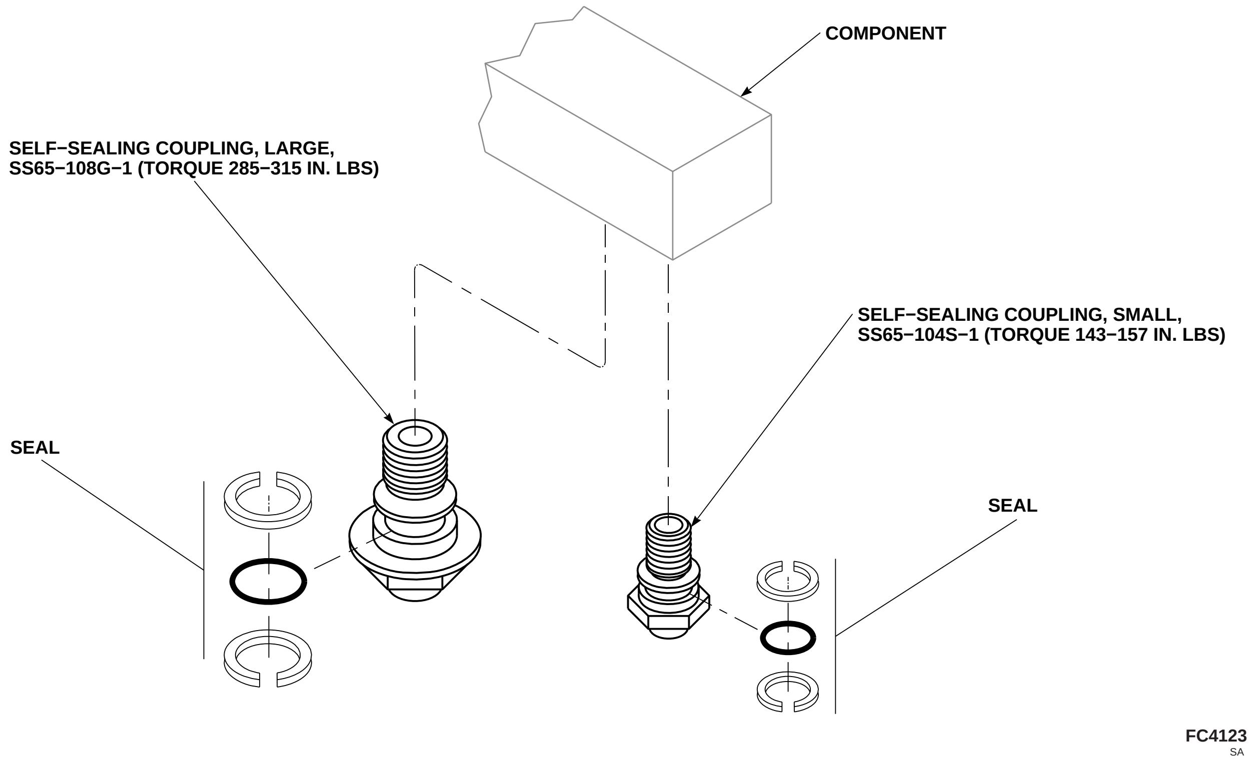


FIGURE 7-4-47-1. REPLACE HYDRAULIC SELF-SEALING COUPLINGS

(2) No. 2 transfer module (PARA 7-4-31).

(3) Pilot-assist module (PARA 7-4-34).

(4) Yaw boost servo (PARA 11-4-70).

(5) Pitch/trim assembly (PARA 11-4-68).

(6) Collective boost servo (PARA 11-4-67).

(7) Roll actuator (PARA 11-4-69).

(8) Aft, forward, or lateral primary servo (PARA 11-4-87).

c. Cover component manifold to prevent possible system contamination.

CAUTION

Damage to hydraulic system will result if dirt or foreign materials enter self-sealing coupling or component. Make sure area is clean and self-sealing coupling is ready for installation before old self-sealing coupling is removed.

d. Using cleaning cloth, Item 102, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, clean self-sealing coupling and surrounding area.

CAUTION

Transfer modules are composed of three separate housings. When removing self-sealing couplings, properly secure only the housing that contains the defective coupling with a padded vise or padded pliers. Failure to secure the housing properly may cause internal valve malfunction and/or leaking between the three housings due to torquing of the whole assembly.

e. Remove self-sealing coupling and seal from component. Discard seal (Figure 7-4-47-1).

7-4-47.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat seals using hydraulic fluid, Item 154 or Item 155, Appendix D

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

c. Using dry, clean compressed air, clean large self-sealing coupling. Flush coupling using hydraulic fluid, Item 154 or Item 155, Appendix D.

CAUTION

Transfer modules are composed of three separate housings. When installing self-sealing couplings, properly secure only the housing that contains the defective coupling with a padded vise or padded pliers. Failure to secure the housing properly may cause internal valve malfunction and/or leaking between the three housings due to torquing of the whole assembly.

d. Install large self-sealing coupling as follows (Figure 7-4-47-1):

(1) Install large self-sealing coupling and seal in manifold.

(2) **TORQUE LARGE SELF-SEALING COUPLING TO 285 - 315 INCH-POUNDS.**

(3) Using lockwire, Item 196, Appendix D, lockwire coupling.

CAUTION

Transfer modules are composed of three separate housings. When installing self-sealing couplings, properly secure only the housing that contains the defective coupling with a padded vise or padded pliers. Failure to secure the housing properly may cause internal valve malfunction and/or leaking between the three housings due to torquing of the whole assembly.

e. Install small self-sealing coupling as follows:

(1) Install small self-sealing coupling and seal in manifold.

(2) **TORQUE SMALL SELF-SEALING COUPLING TO 143 - 157 INCH-POUNDS.**

(3) Using lockwire, Item 196, Appendix D, lockwire coupling.

f. Install the following previously removed component(s):

(1) No. 1 transfer module (PARA 7-4-26).

(2) No. 2 transfer module (PARA 7-4-31).

(3) Pilot-assist module (PARA 7-4-34).

(4) Yaw boost servo (PARA 11-4-70).

(5) Pitch/trim assembly (PARA 11-4-68).

(6) Collective boost servo (PARA 11-4-67).

(7) Roll actuator (PARA 11-4-69).

(8) Aft, forward, or lateral primary servo (PARA 11-4-87).

g. Bleed hydraulic system (PARA 7-4-65).

h. Perform operational check of No. 1, No. 2, or backup hydraulic system as necessary to check self-sealing couplings for leakage (PARA 7-2-1).

7-4-48 TAIL ROTOR SERVO HYDRAULIC CONNECTOR SELF-SEALING COUPLINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 11-4-116

7-4-48.1 Replace Tail Rotor Servo Hydraulic Connector Self-Sealing Couplings.

NOTE

Replacement of all tail rotor servo hydraulic connector couplings is the same.

7-4-48.1.1. Remove.

a. Remove tail rotor servo hydraulic connector (PARA 11-4-116).

CAUTION

Damage to hydraulic system will result if dirt or foreign material is allowed to enter system. Make sure area is clean and have replacement self-sealing coupling ready for immediate installation before old self-sealing coupling is removed.

b. Clean self-sealing coupling and surrounding area using cleaning cloth, Item 102, Appendix

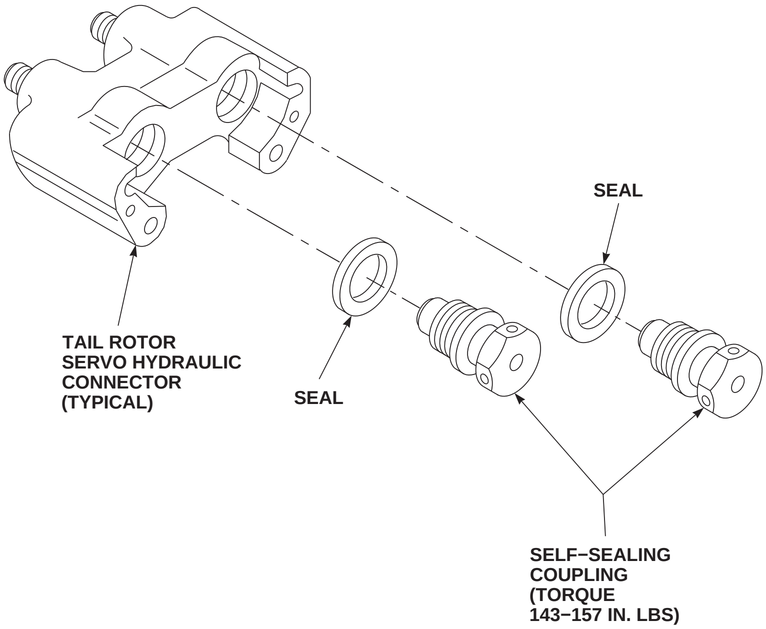


FIGURE 7-4-48-1. REPLACE TAIL ROTOR SERVO HYDRAULIC CONNECTOR SELF-SEALING COUPLINGS

D, dampened with denatured alcohol, Item 70, Appendix D.

c. Remove self-sealing coupling from tail rotor servo hydraulic connector (Figure 7-4-48-1).

d. Remove seal from self-sealing coupling. Discard seal.

7-4-48.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Lightly coat seal with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on self-sealing coupling.

c. Lightly coat threads of self-sealing coupling with hydraulic fluid, Item 154 or Item 155, Appendix D, and install self-sealing coupling in tail rotor servo connector (Figure 7-4-48-1). **TORQUE SELF-SEALING COUPLING TO 143 - 157 INCH-POUNDS.** Lockwire, Item 196, Appendix D, connectors in pairs.

d. Install tail rotor servo hydraulic connector (PARA 11-4-116).

e. Perform operational check of No. 1 or backup hydraulic systems (PARA 7-2-1).

7-4-49 HYDRAULIC SYSTEM RIGID TUBING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Borescope
- Hydraulic Repairer's Toolkit
- Light Source, Strong
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-2-1
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-2-1
- PARA 7-4-65
- PARA 7-5-8
- PARA 11-4-112

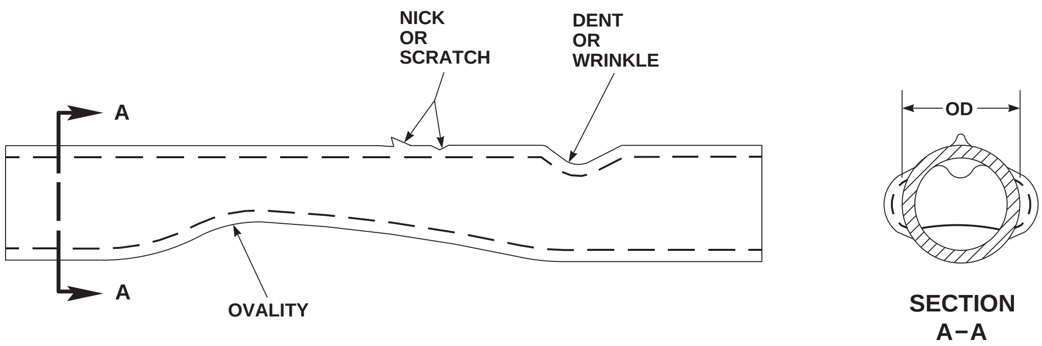
7-4-49.1 Replace Hydraulic System Rigid Tubing.

NOTE

Replacement of rigid tubing and fittings from any section of helicopter is performed in generally the same manner.

7-4-49.1.1 Inspect.

a. Inspect tubing for nicks, scratches, dents, wrinkles, and out-of-roundness (ovality). **LIMITS STATED IN FIGURE 7-4-49-1 MUST NOT BE EXCEEDED.** If limits are exceeded, repair or replace tubing (PARA 7-5-8).



TUBES − INCHES		MAXIMUM DEPTH−INCHES		
NOMINAL OD	WALL THICKNESS	NICK / SCRATCH	DENT / WRINKLE	OVALITY
0.375 (3/8)	0.020	0.0008	0.003	0.018
0.250 (1/4)	0.016	0.0005	0.002	0.010
0.250 (1/4)	0.020	0.0008	0.002	0.010

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FIGURE 7-4-49-1. INSPECT HYDRAULIC SYSTEM RIGID TUBING

CAUTION

Damage to equipment may result if chafing or fretting occurs. Chafing or fretting reduces ability of metal tubing to withstand internal pressures and vibration. Make sure tubing is not chafing or fretting.

b. Using 10X magnifying glass and strong light source or borescope, visually inspect tubing for chafing or fretting. **NONE ALLOWED.** If chafing or fretting is found, replace tubing (PARA 7-5-8).

c. Check tubing for adequate clearance from adjacent components and other tubing as follows:

(1) Make sure minimum clearance of 1/16-inch is maintained for supported tubing where no motion is possible.

(2) Make sure minimum clearance of 1/4-inch is maintained where relative motion of adjacent components may exist.

d. Check tubing for loose or missing clamps. **NONE ALLOWED.** If loose or missing clamps are evident, tighten or install clamps as required.

7-4-49.1.2 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Open other covers and access panels as necessary to access hydraulic lines or fittings that need replacing (PARA 1-2-1).

NOTE

- Tubing is clamped to each other or to structure for support. In some places, clamps are screwed to brackets. In some places, spacers are used with clamps to support tubing and keep it in line.
- Clamps are installed to tubing they support. Make sure correct size clamp is used when replacing tubing.

d. When removing section of tubing, perform the following (Figure 7-4-49-2, Sheet 1, Sheet 2, Sheet 3, Sheet 4, Sheet 5, Sheet 6, Sheet 7, Sheet 8, Sheet 9, Sheet 10, Sheet 11, and Sheet 12):

NOTE

Pump module reservoirs are spring-loaded to keep 25 psi on system.

(1) Disconnect system pump module quick-disconnects or bleed fluid from reservoir before opening hydraulic system tubes.

(2) Remove screws, washers, and nuts securing clamps together.

(3) Leave clamps on tubing. Clamp on tube not being replaced will mark location of clamp for new tube.

(4) Where spacer is used, screw remaining clamp (on tube not being replaced) and spacer back in place. This will keep spacer in location and ready for replacement tubing.

(5) Have machinery towels, Item 211, Appendix D, to catch any spilled fluid at each end of tubing when fitting nuts are disconnected.

(6) Disconnect fitting nuts from each end of tubing being removed. Use assistant to help guide tubing as it is being removed from helicopter.

(7) Install protective caps and plugs on tubes, fittings, and ports to prevent contamination.

(8) If required, remove and save clamps from tubing removed.

CAUTION

Damage to tubing will result if bent or forced when removing couplings, elbows, or unions. Remove clamps and tubing assemblies, if necessary, to allow removal of tubing.

e. Remove nuts and washers from couplings, elbows, or unions, and remove from airframe.

f. Using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, remove sealing compound from airframe, couplings, fittings, elbows, unions, and nuts.

g. Remove screws and washers, and separate block supports (Figure 7-4-49-2, Sheet 6, Detail S).

h. Remove bolts, washers, and nuts to remove male and female self-sealing couplings from tail cone and pylon bulkheads (Figure 7-4-49-2, Sheet 8, Detail Y and Detail Z, and Sheet 10, Detail AC).

7-4-49.1.3 Inspect.

a. Inspect tubing for nicks, scratches, dents, wrinkles, and out-of-roundness (ovality) beyond limits. **LIMITS STATED IN FIGURE 7-4-49-1 MUST NOT BE EXCEEDED.** If limits are exceeded, repair or replace tubes as required (PARA 7-5-8).

CAUTION

Damage to equipment may result if chafing or fretting occurs. Chafing or fretting reduces ability of metal tubing to withstand internal pressures and vibration. Make sure tubing is not chafing or fretting.

b. Using 10X magnifying glass and strong light source or borescope, visually inspect tubing for

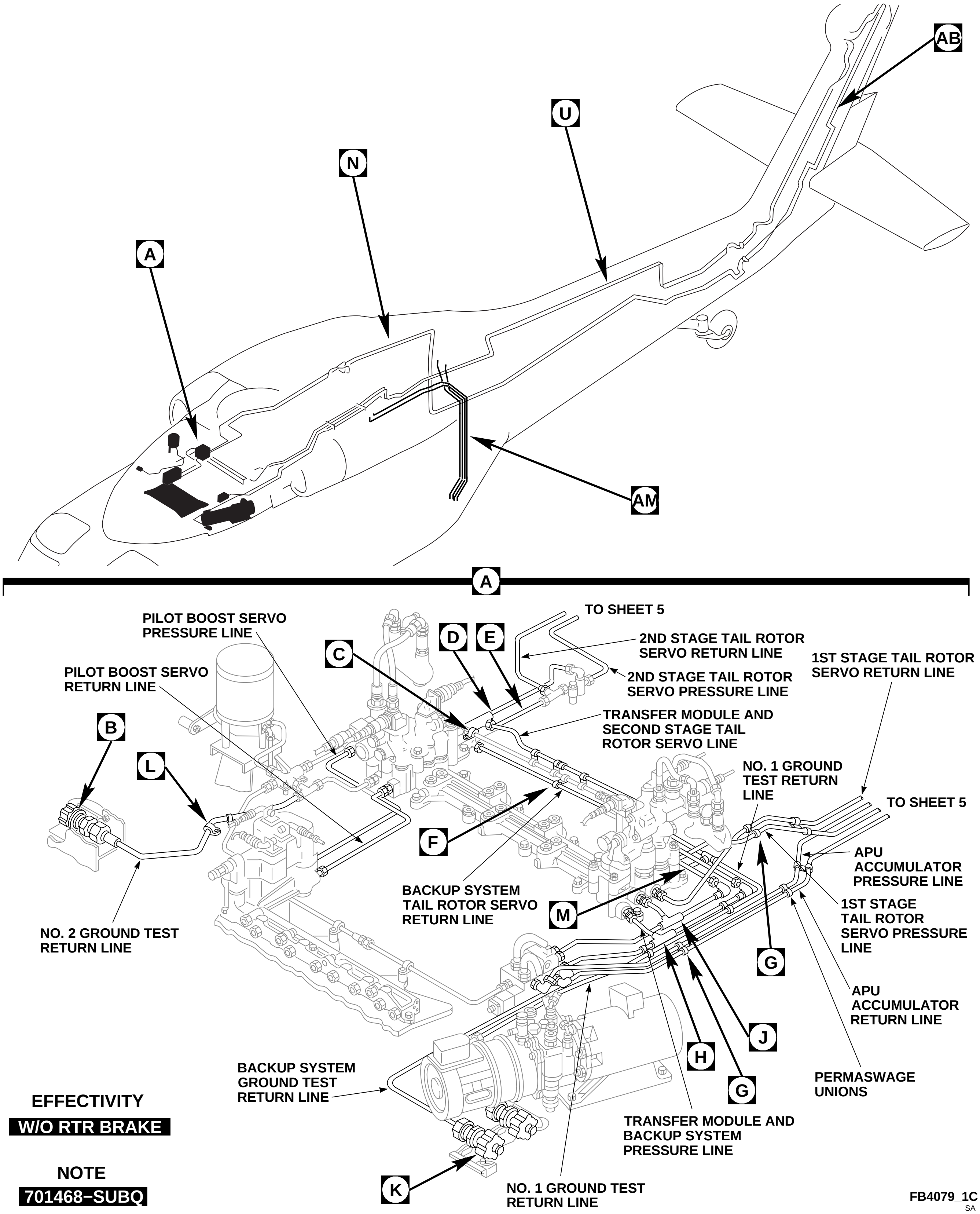
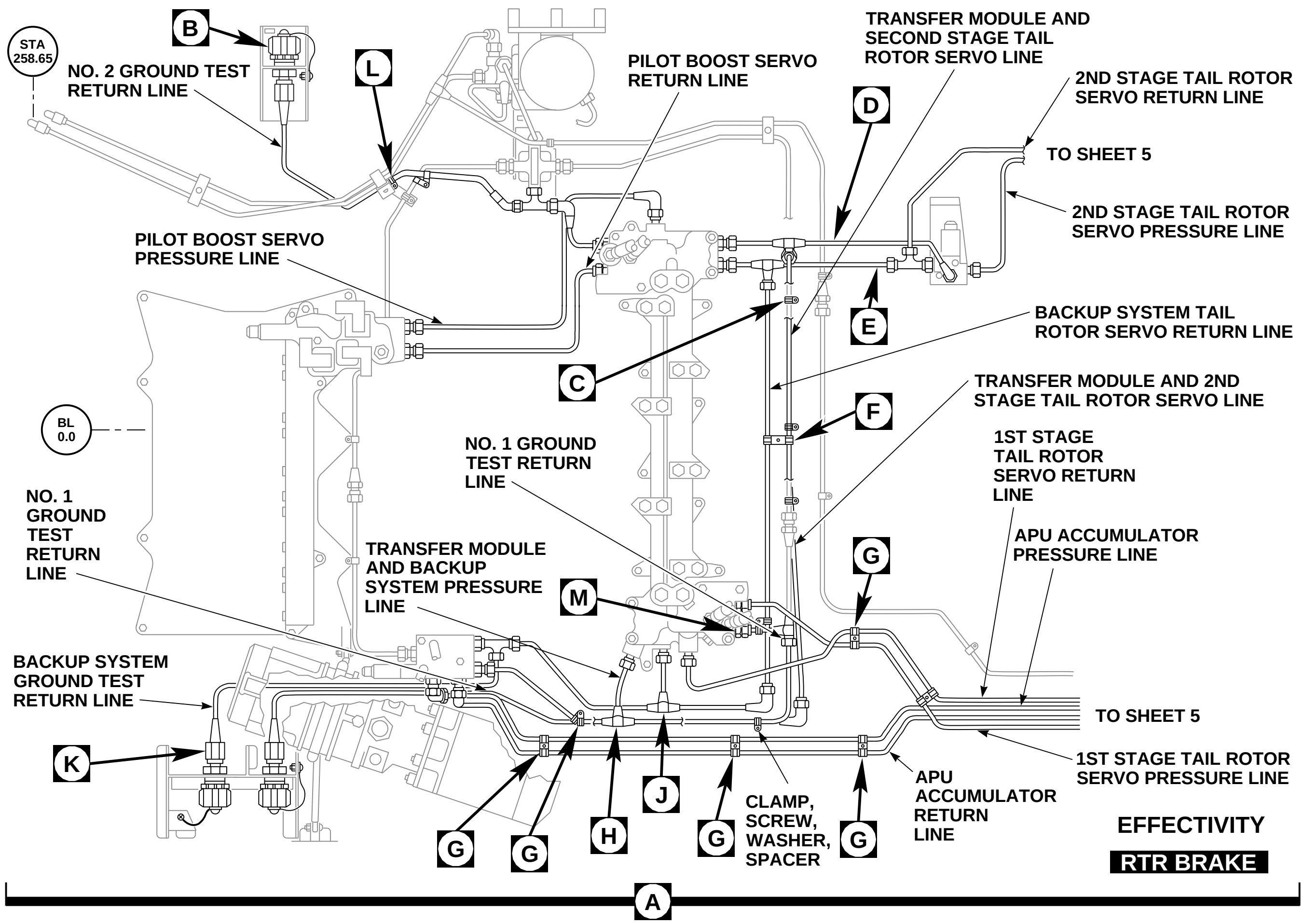


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 1 OF 12)



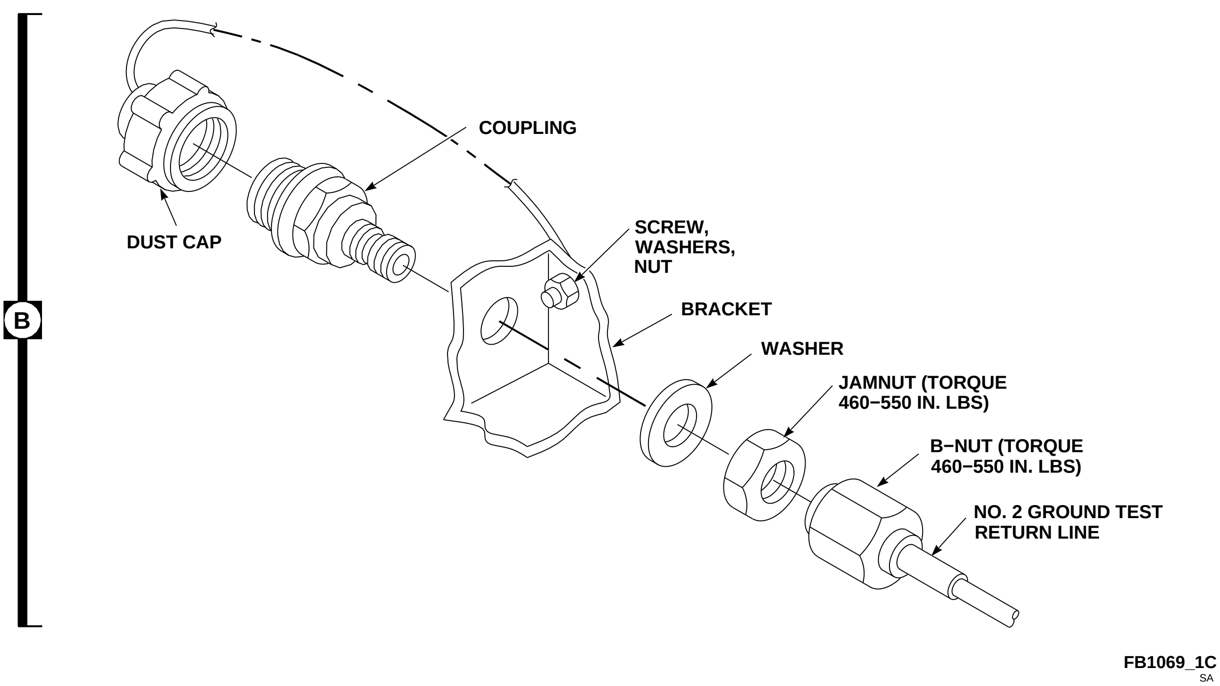


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 2 OF 12)

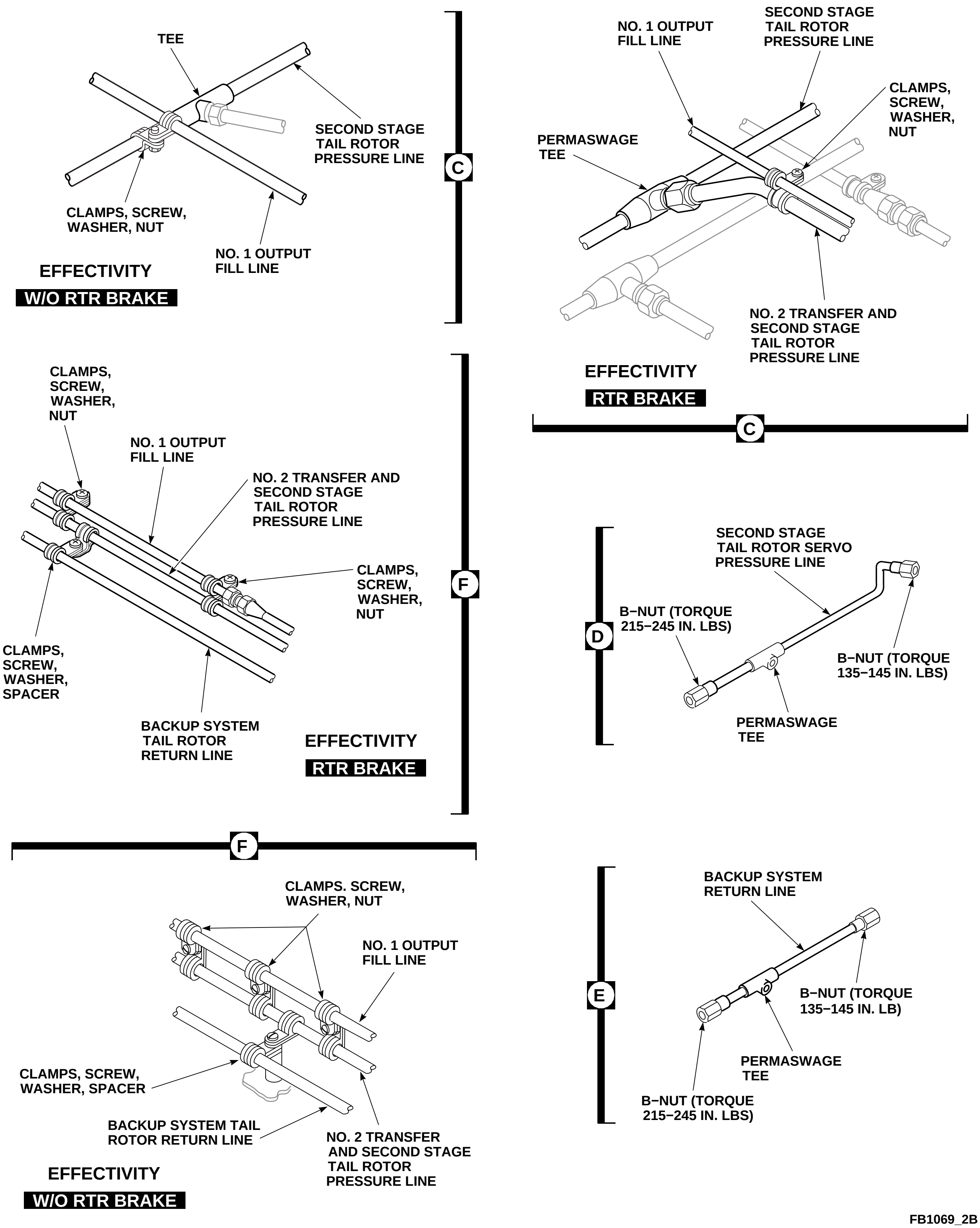


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 3 OF 12)

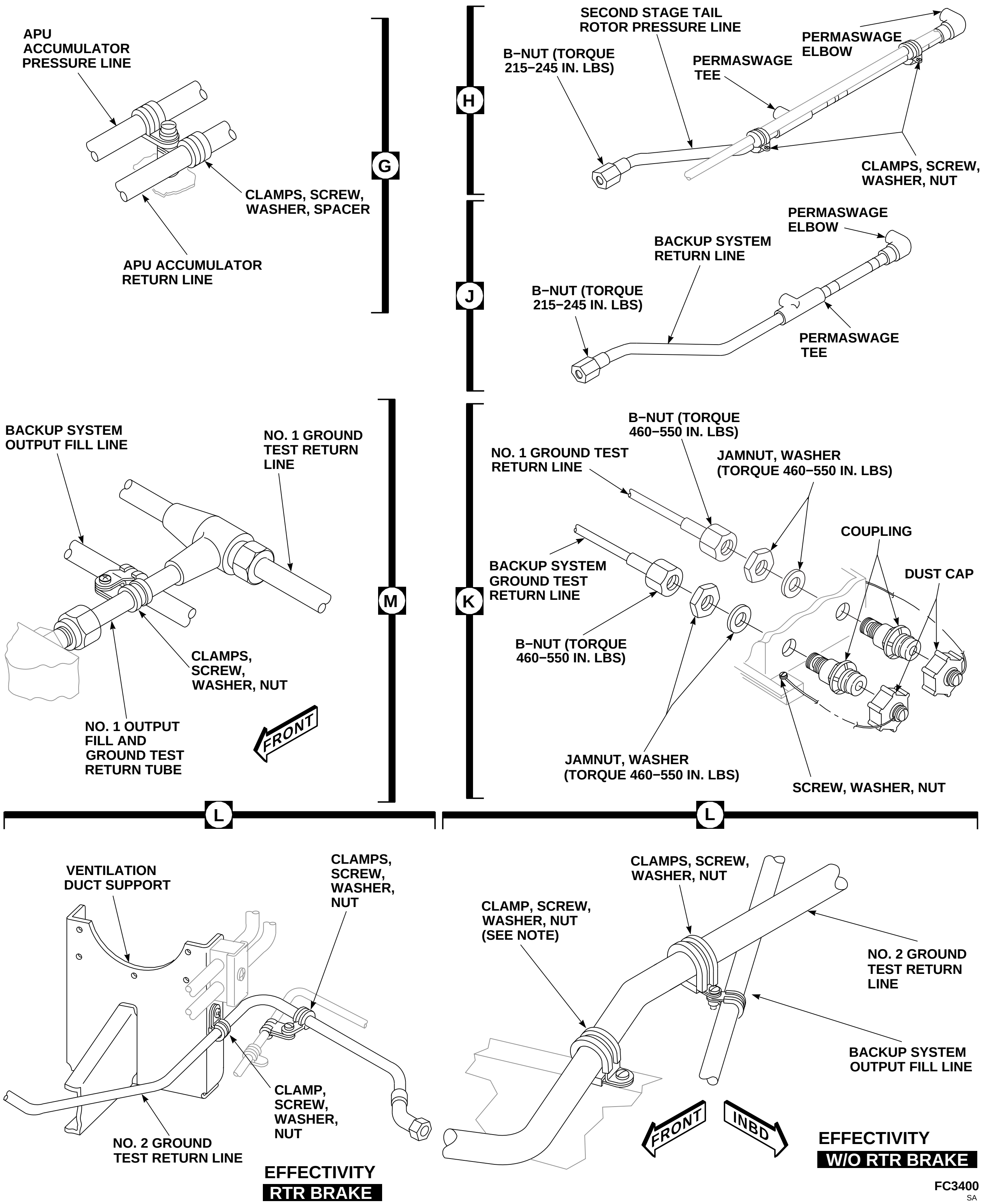


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 4 OF 12)

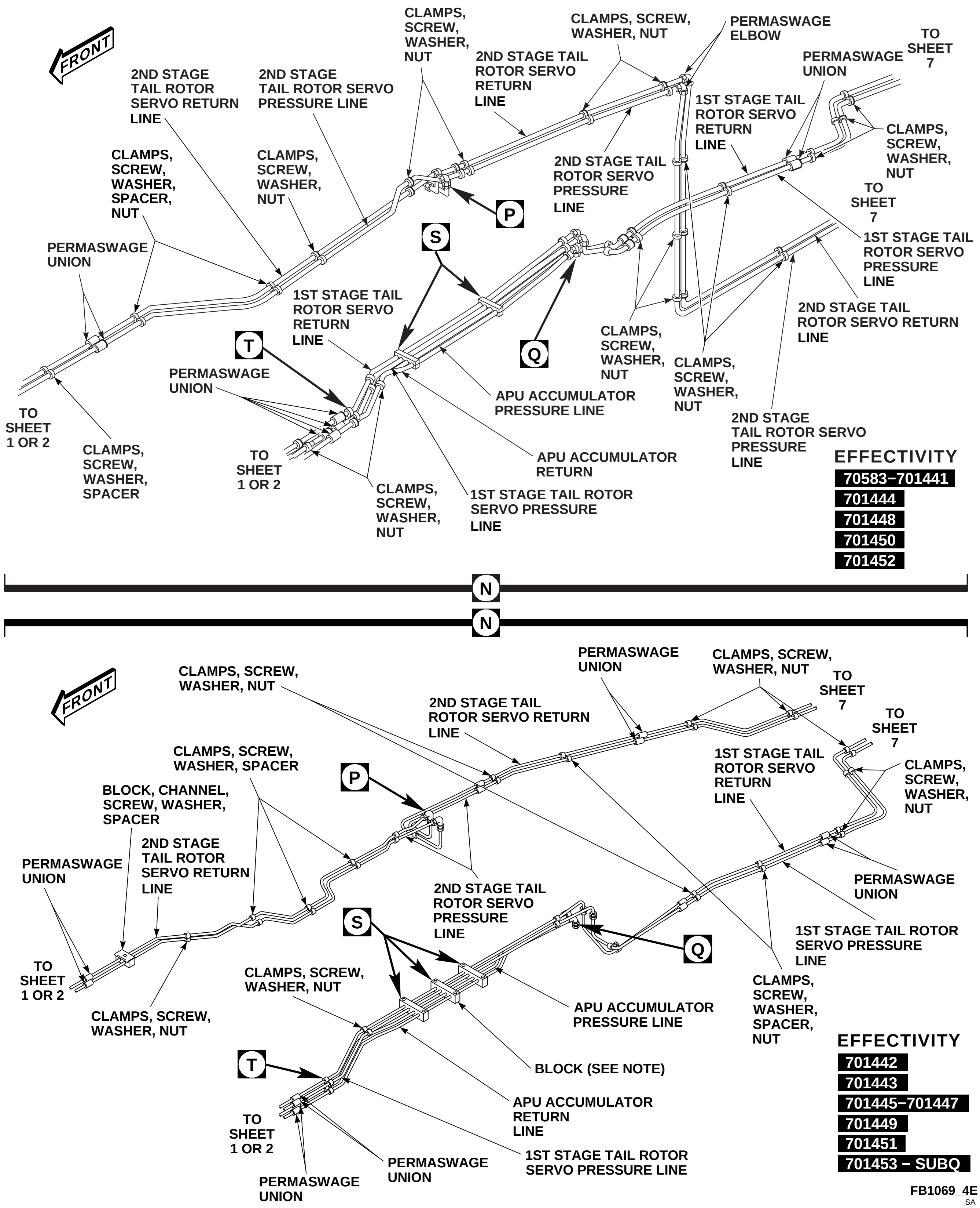


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 5 OF 12)

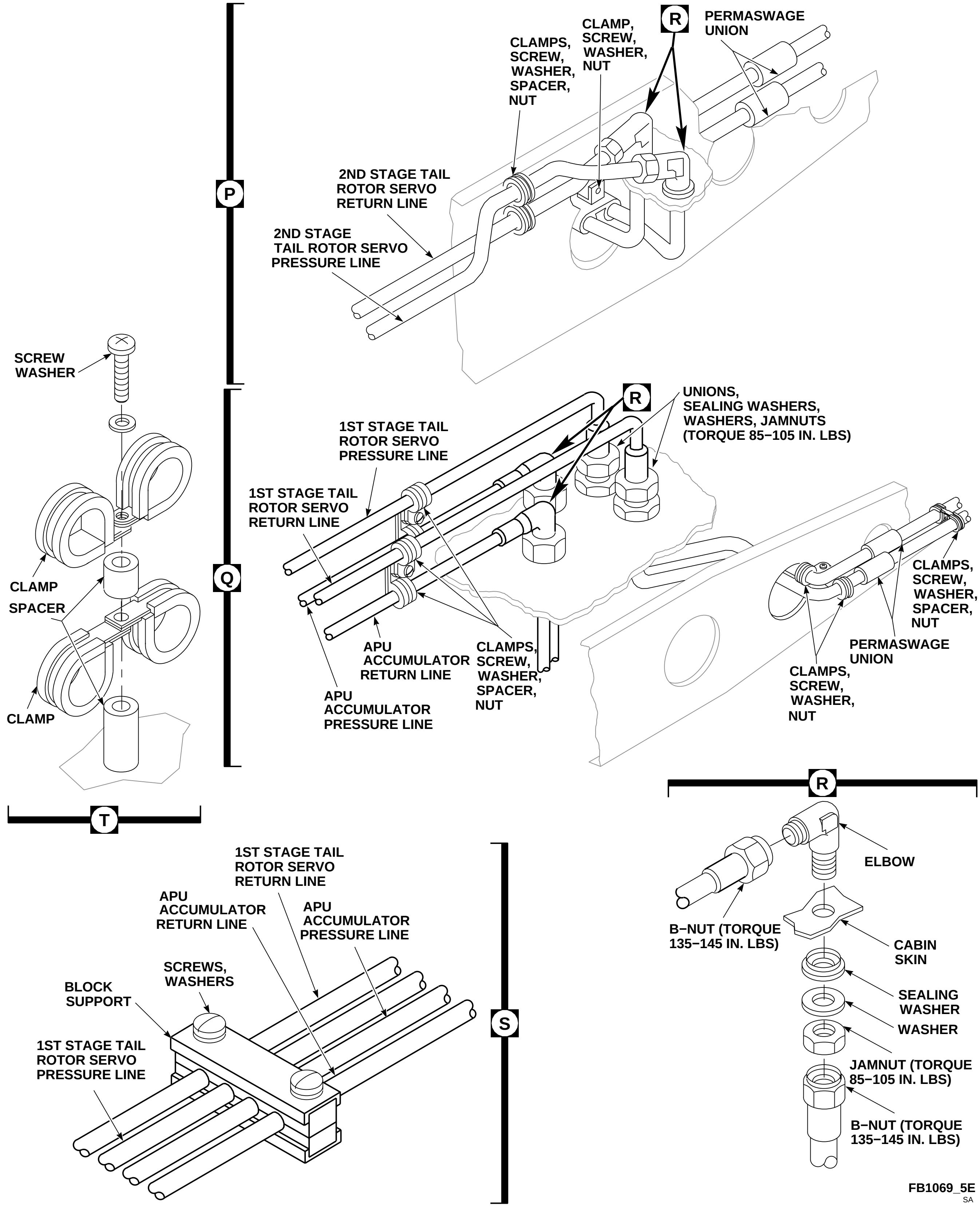


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 6 OF 12)

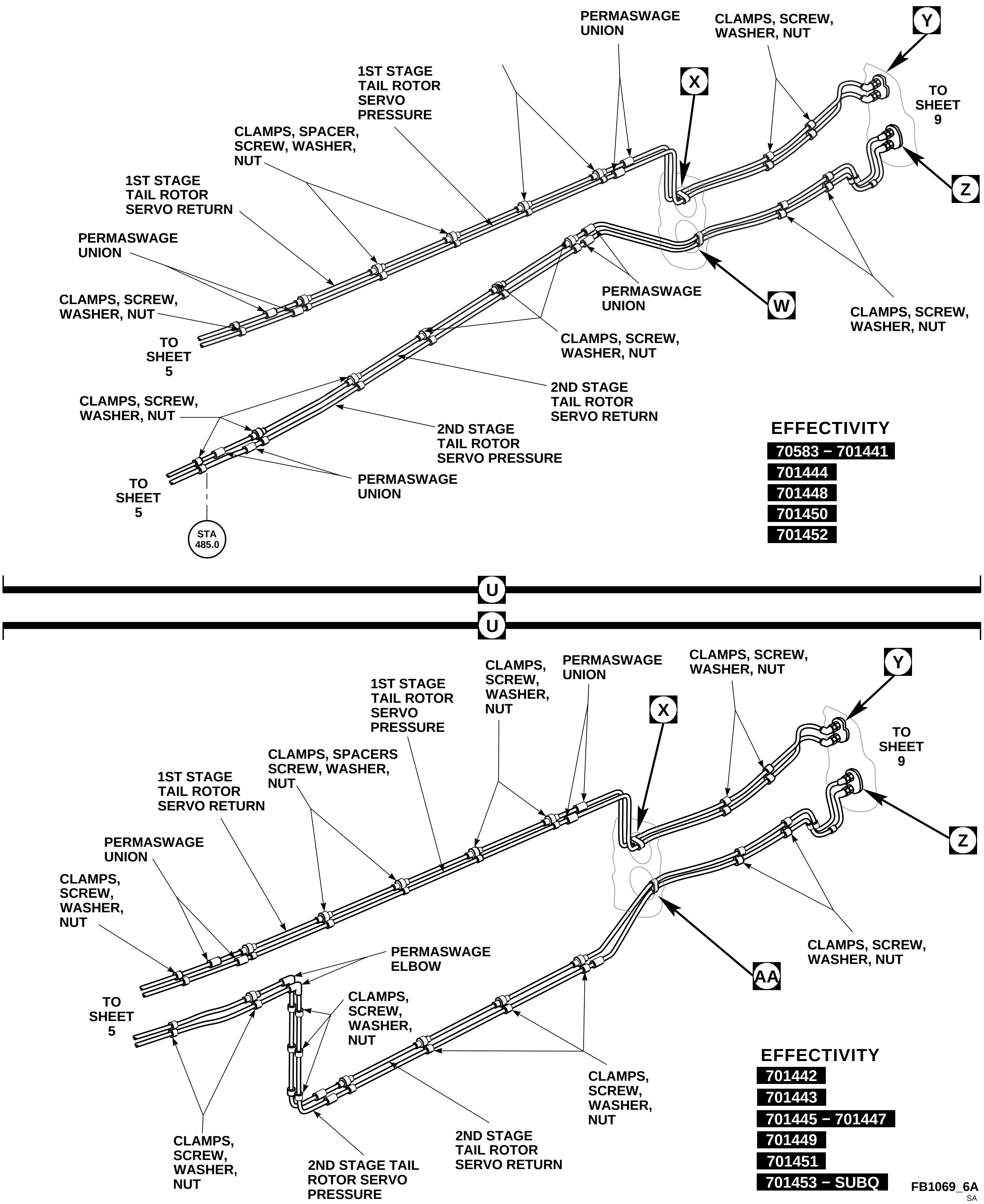


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 7 OF 12)

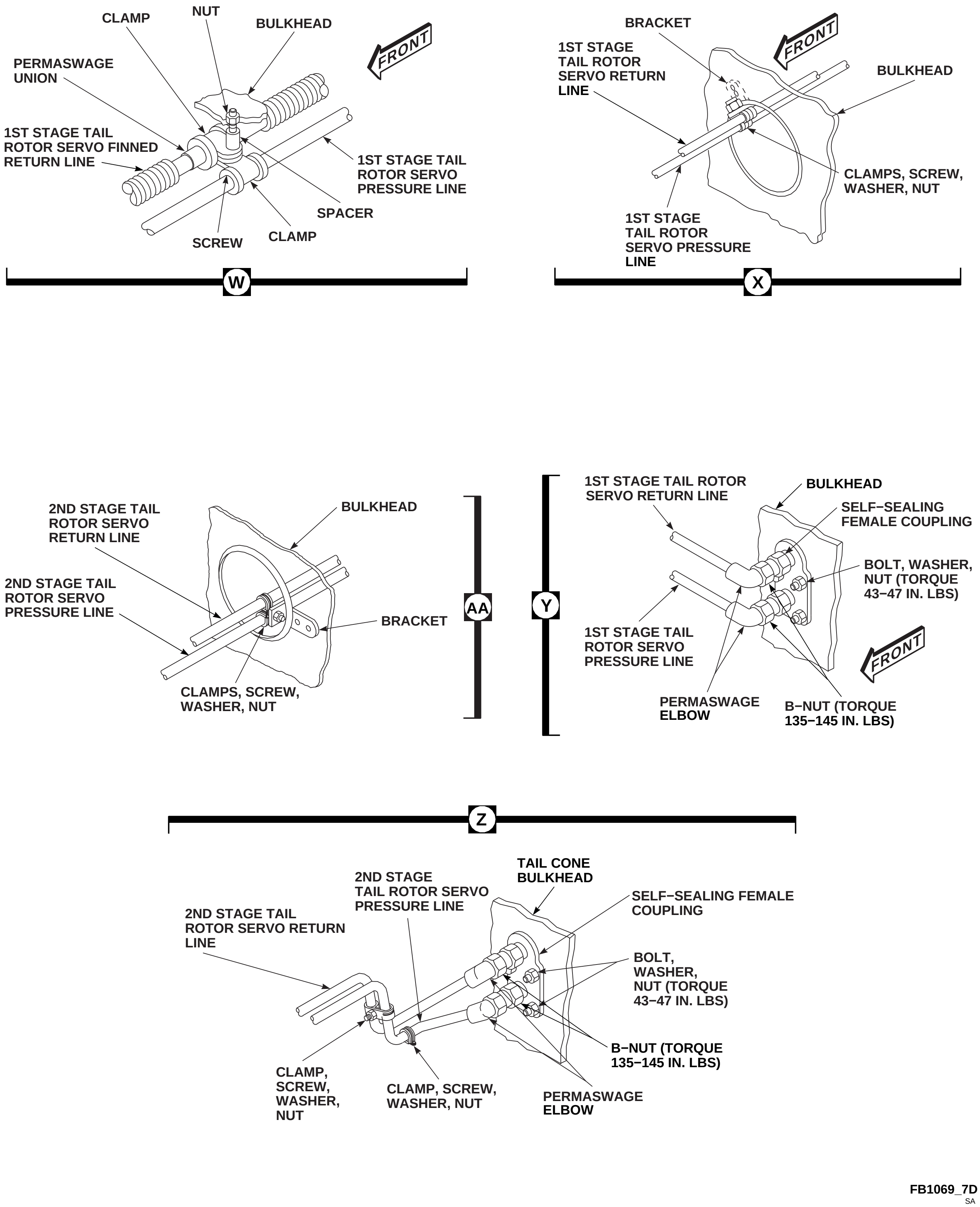


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 8 OF 12)

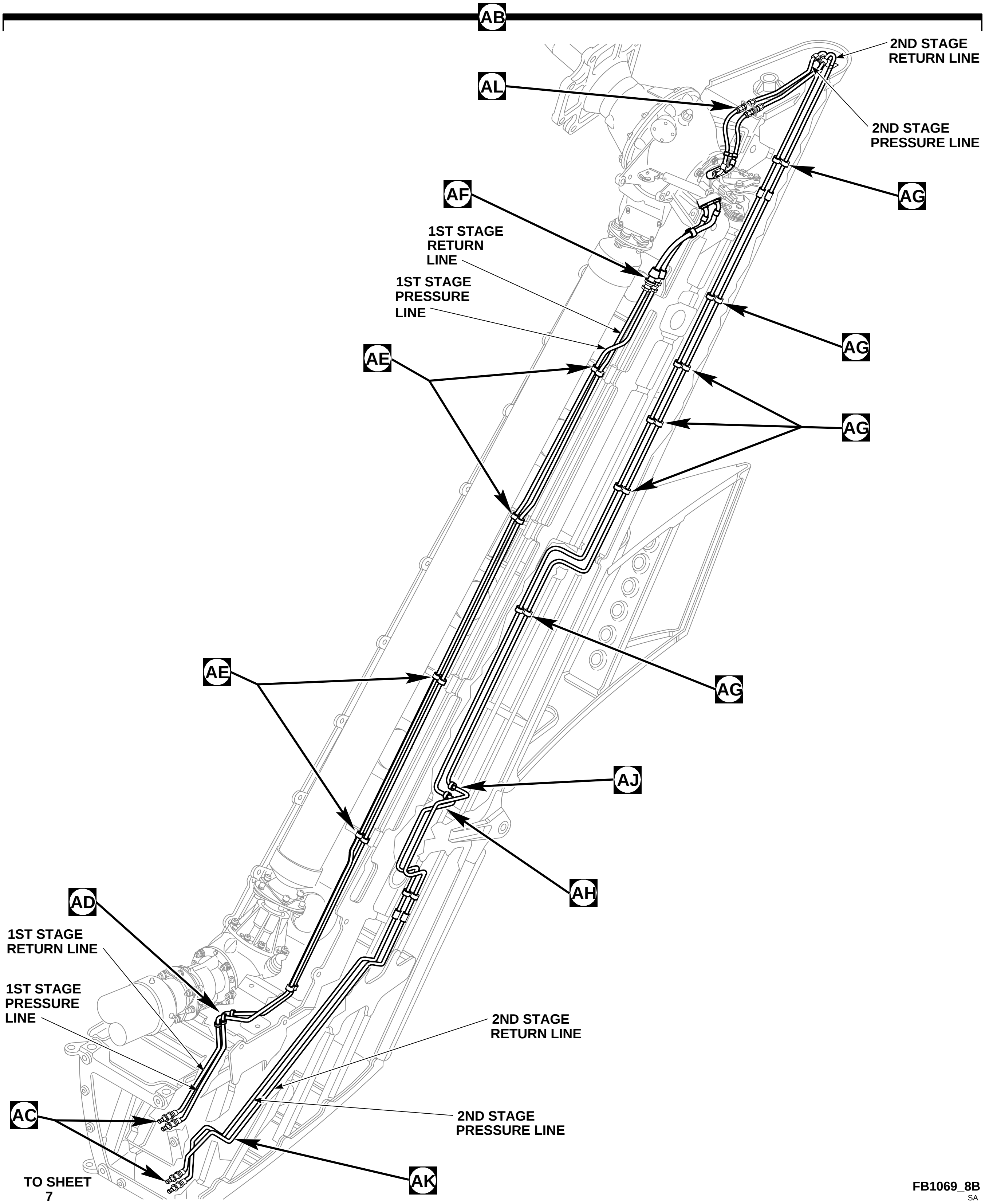


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 9 OF 12)

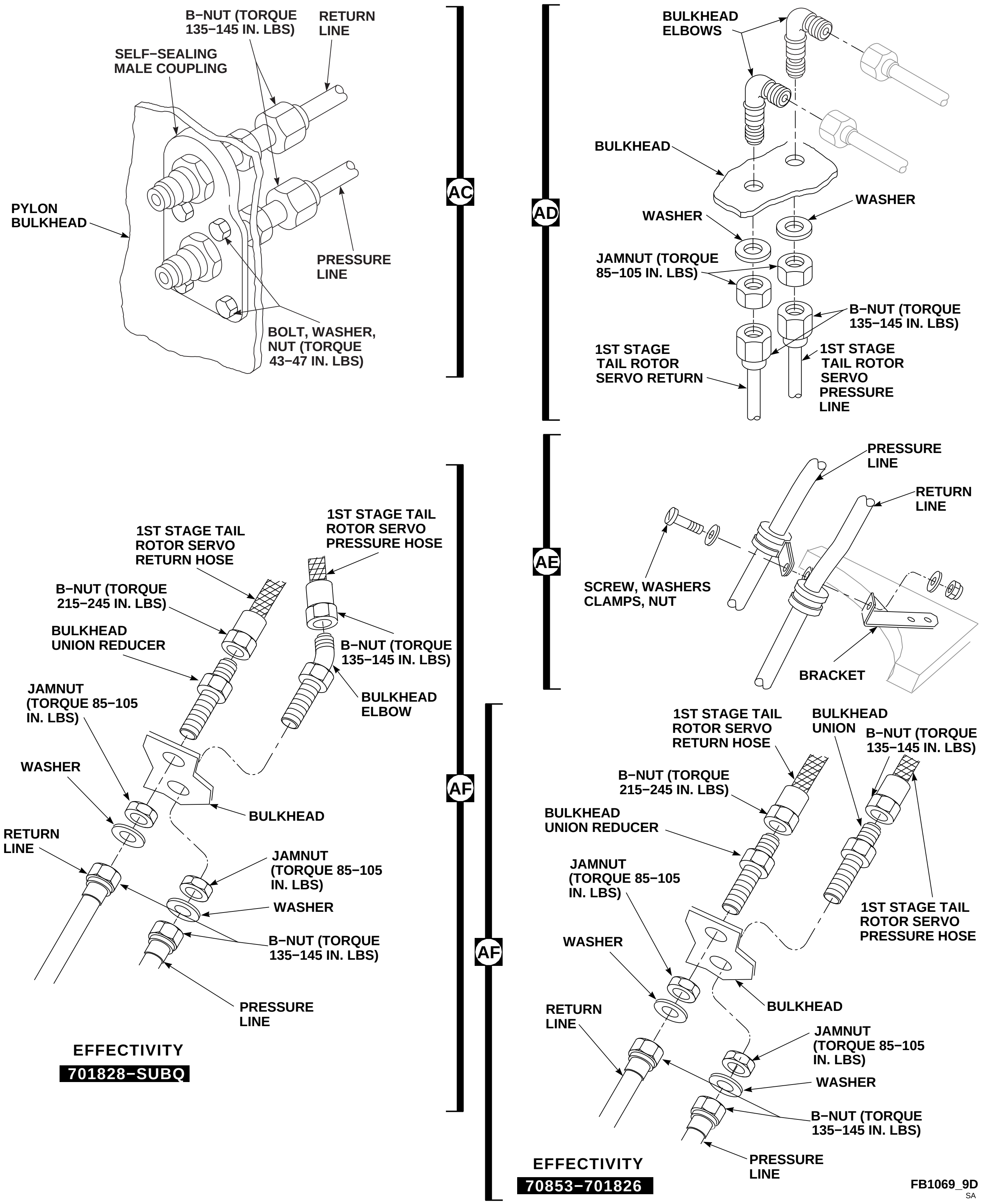


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 10 OF 12)

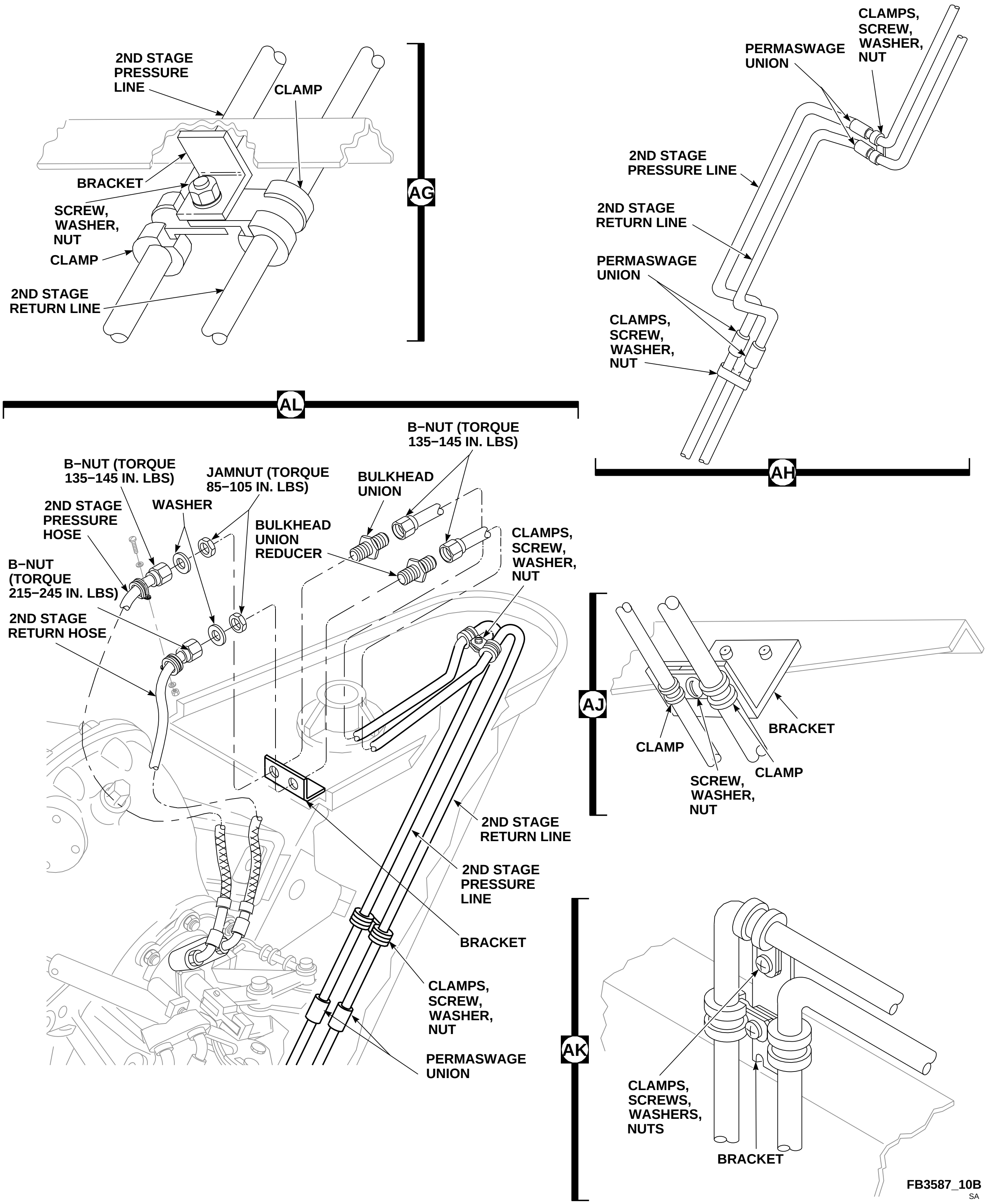


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 11 OF 12)

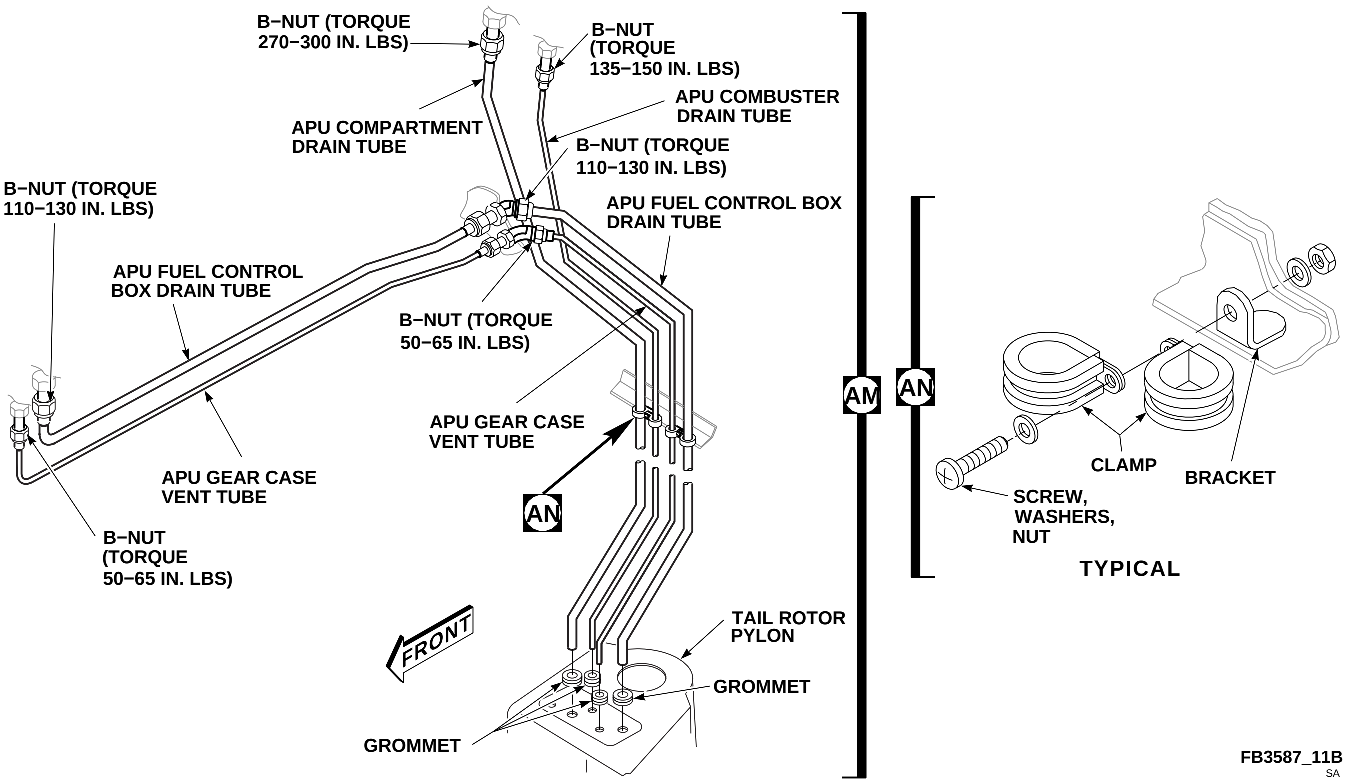


FIGURE 7-4-49-2. REPLACE HYDRAULIC SYSTEM RIGID TUBING (SHEET 12 OF 12)

chafing or fretting. **NONE ALLOWED.** If chafing or fretting is found, replace tubing (PARA 7-5-8).

c. Check identification tapes for condition. **IDENTIFICATION TAPES MUST BE PRESENT AND LEGIBLE.** If missing or unreadable, replace tapes as required.

d. Inspect grommets for cracks and wear. **NONE ALLOWED.** If cracked or worn, replace grommets as required.

7-4-49.1.4 Install.

NOTE

- Remove protective caps and plugs from tubes, fittings, and ports immediately before installation.
- Make sure replacement line is clear and clean.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Before connecting nuts to fittings, coat all threads using hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Using machinery towel, Item 211, Appendix D, dampened in acetone, Item 25, Appendix D, clean all areas of cabin skin and bulkheads that elbows or unions pass through. Using sealing compound, Item 292, Appendix D, seal cleaned areas.

d. Install elbows of 1st and 2nd stage tail rotor servo pressure and return lines through cabin skin with sealing washers, washers, and nuts. **TORQUE JAMNUTS TO 85 - 105 INCH-POUNDS. TORQUE B-NUTS ON 1ST AND 2ND STAGE TAIL ROTOR SERVO PRESSURE HOSES TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 6, Detail P, Detail Q, and Detail R).

e. Install bulkhead elbows of 1st stage tail rotor servo pressure and return lines to bulkhead with washers and jamnuts. **TORQUE JAMNUTS TO 85 - 105 INCH-POUNDS.**

TORQUE B-NUTS ON 1ST AND 2ND STAGE TAIL ROTOR SERVO PRESSURE HOSES TO 135 - 145 INCH-POUNDS (Figure 7-4-49-2, Sheet 10, Detail AD).

f. **70583–701826** Install bulkhead union of 1st stage tail rotor servo pressure hose to bulkhead with washer and jamnut. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS. TORQUE B-NUT ON 1ST STAGE TAIL ROTOR PRESSURE HOSE TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 10, Detail AF).

g. **701828–SUBQ** Install bulkhead elbow of 1st stage tail rotor servo pressure hose to bulkhead with washer and jamnut. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS. TORQUE B-NUT ON 1ST STAGE TAIL ROTOR PRESSURE HOSE TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 10, Detail AF).

h. Install bulkhead union of 1st and 2nd stage tail rotor servo pressure hose to bracket on bulkhead with washer and jamnut. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS. TORQUE B-NUT ON 1ST AND 2ND STAGE TAIL ROTOR SERVO PRESSURE HOSES TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 11, Detail AL).

i. Install bulkhead union reducers of 1st and 2nd stage tail rotor servo return hoses to bulkhead with washers and jamnuts. **TORQUE JAMNUTS TO 85 - 105 INCH-POUNDS. TORQUE B-NUTS ON 1ST AND 2ND STAGE TAIL ROTOR SERVO RETURN HOSES TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 10, Detail AF and Sheet 11, Detail AL).

j. Install male self-sealing coupling in pylon bulkhead with return fitting up. Install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.** Secure pressure and return tubes to coupling. **TORQUE TUBE B-NUTS TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 10, Detail AC).

k. Install female self-sealing coupling in tail cone bulkhead with return fitting up. Install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.** Secure tubes to coupling. **TORQUE TUBE B-NUTS TO 135 - 145 INCH-POUNDS** (Figure 7-4-49-2, Sheet 8, Detail Y and Detail Z).

NOTE

If new clamps are used, make sure they are correct size for tubing.

l. Install clamps on replacement tubing and connect to other support clamps with screw, washer, and nut (Figure 7-4-49-2, Sheet 3, Detail F, and Sheet 6, Detail T).

m. Using sealing compound, Item 292, Appendix D, seal areas where clamp hardware goes through cabin skin.

n. Where spacer is used, remove screw and washer from clamp and spacer. Position clamp of replacement tube on clamp and spacer and install screw and washer (Figure 7-4-49-2, Sheet 3, Detail F and Sheet 4, Detail G).

o. Where lines go through bulkhead holes, secure clamps to brackets (Figure 7-4-49-2, Sheet 8, Detail X and Detail AA).

p. Put block supports on tubing. Make sure block holes match tube sizes. Install screws and washers (Figure 7-4-49-2, Sheet 6, Detail S).

CAUTION

Damage to equipment may result if chafing or fretting occurs. Chafing or fretting reduces ability of metal tubing to withstand internal pressures and vibration. Make sure tubing is not chafing or fretting.

q. Check tubing for adequate clearance from adjacent components and other tubing as follows:

(1) Make sure minimum clearance of 1/16-inch is maintained for supported tubing where no motion is possible.

(2) Make sure minimum clearance of ¼-inch is maintained where relative motion of adjacent components may exist.

7-4-49.1.5 Test.

a. Disconnect tail rotor servo input pushrod from tail rotor servo input arm (PARA 11-4-112).

b. Connect external hydraulic power to backup system (PARA 1-6-15), connect external electrical power (PARA 1-6-16), or run APU.

c. If required, service backup hydraulic pump module reservoir (PARA 1-3-8) prior to placing BACKUP HYD PUMP switch on upper console to ON to pressurize No. 1 and No. 2 hydraulic systems.

d. On lower console, place TAIL RTR SERVO switch to NORMAL.

WARNING

Injury to personnel will result if caution is not used around control linkages. Make sure all personnel are clear of control linkages while moving control pedals.

e. Hold right pedal to position servo input pushrod out of way as required. Do not move cyclic or collective sticks.

f. Change tail rotor servo input arm to full extended position. Release arm. **SERVO MUST CHANGE BACK TO MIDSTROKE POSITION.**

g. If servo does not center, check for crossed first stage pressure and return lines. Correct crossed line condition. Recheck servo for correct hydraulic line installation.

h. On lower console, place TAIL RTR SERVO switch to BACKUP.

i. On caution/advisory panel, check that NO. 1 TAIL RTR SERVO and NO. 2 TAIL RTR SERVO lights are on.

j. Change tail rotor servo input arm to full extended position. Release arm. **SERVO MUST CHANGE BACK TO MIDSTROKE POSITION.**

k. If servo does not center, check for crossed second stage pressure and return lines. Correct crossed line condition. Recheck servo for correct hydraulic line installation.

l. Disconnect external hydraulic power (PARA 1-6-15), external electrical power (PARA 1-6-16), or shut down APU (Appropriate Flight Manual).

m. Connect tail rotor servo input pushrod to tail rotor servo input arm (PARA 11-4-112).

n. Bleed flight controls hydraulic system (PARA 7-4-65).

o. If APU drain lines were replaced (Figure 7-4-49-2, Sheet 12, Detail AM), pressure-test APU drain lines for leakage at 15 psig for 5 minutes. **NONE ALLOWED.** If leakage is found, make sure all connections are properly tightened.

p. Perform operational check of No. 1, No. 2, and backup hydraulic systems (PARA 7-2-1).

q. Make sure area is clean and free of foreign material.

r. Install any access panels or covers removed to replace tubing.

s. Close main rotor pylon sliding cover.

7-4-50 HYDRAULIC SYSTEM FLEX HOSES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Hydraulic Repairer's Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 165, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 7-4-26
- PARA 7-4-31
- PARA 7-4-38
- PARA 7-4-49
- PARA 7-4-65

7-4-50.1 Replace Hydraulic System Flex Hoses.

NOTE

Refer to PARA 7-4-26, PARA 7-4-31, or PARA 7-4-38 for hydraulic module pump-to-transfer module hose maintenance.

7-4-50.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Remove tail gear box fairings (PARA 2-4-137).

NOTE

Pump module reservoirs are spring-loaded to keep 25 psi on system.

c. To prevent fluid loss, uncouple pump module quick-disconnects on affected system, or drain fluid from reservoir before opening hydraulic system (PARA 1-3-8).

d. Use machinery towels, Item 211, Appendix D, and 1-quart container to catch fluid as hoses are disconnected.

e. Remove screws, washers, nuts, and disconnect clamps from pressure and return hoses (Figure 7-4-50-1, Detail A and Detail B).

NOTE

Leave clamp on opposite hose to mark location of clamp when replacing hose.

f. Remove clamp from hose being replaced.

g. Disconnect 1st stage pressure and return hoses at tail rotor servo connector. Drain hoses into 1-quart container. Disconnect hoses from bulkhead union reducer 70583-701826 and bulkhead union 701828-SUBQ and bulkhead elbow. Cap fittings (Figure 7-4-50-1, Detail B).

h. Disconnect 2nd stage pressure and return hoses from bulkhead union and bulkhead union reducer. Drain hoses into 1-quart container. Disconnect hoses at tail rotor servo connector. Cap fittings (Figure 7-4-50-1, Detail A).

7-4-50.1.2 Inspect.

a. Check hose identification tapes for condition. Replace missing or unreadable tapes, as required.

b. Check pressure hoses for heat-shrinkable insulation sleeving around metal identification tag. If missing, replace heat-shrinkable insulation sleeving, Item 165, Appendix D.

c. Check hose B-nuts, threads, end fittings, and hose exterior for damage. Repair or replace damaged hoses, as required.

7-4-50.1.3 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Remove caps and plugs from hoses, ports, and fittings during installation.

b. Using clean, dry, compressed air, blow dry replacement hose clear.

c. Coat threads of fittings with hydraulic fluid, Item 154 or Item 155, Appendix D, before connecting hoses.

CAUTION

Damage to flex hoses will result if kinked or twisted when installed. Do not force hose connections or routing which will strain, kink, or twist hose. Connect hoses loosely at each end, making sure fittings match squarely before tightening nuts.

d. Connect 90° elbow of 2nd stage return hose to tail rotor servo connector return fitting (Figure 7-4-50-1, Detail A). Connect straight end to

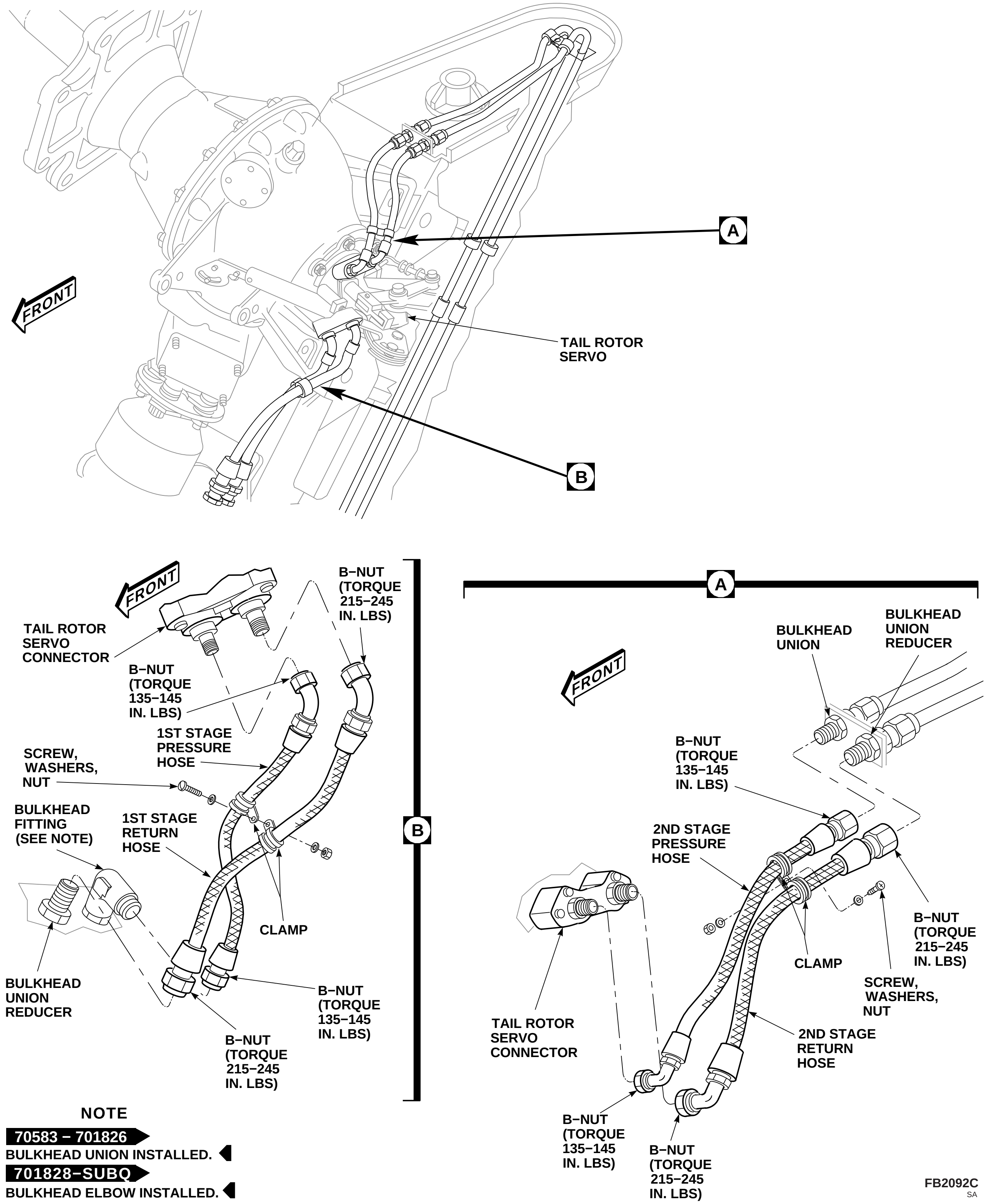


FIGURE 7-4-50-1. REPLACE HYDRAULIC SYSTEM FLEX HOSES

2nd stage return bulkhead union reducer. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

e. Connect 90° elbow of 2nd stage pressure hose to tail rotor servo connector pressure fitting. Connect straight end to pressure bulkhead union. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

f. Connect 45° elbow of 1st stage pressure hose to tail rotor servo 1st stage connector pressure fittings (Figure 7-4-50-1, Detail B). **70583–701826** Connect straight end of 1st stage pressure hose to bulkhead union. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

g. **701828–SUBQ** Connect straight end of 1st stage pressure hose to bulkhead elbow. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

h. Connect straight end of 1st stage return hose to 1st stage bulkhead union reducer. Connect 45° hose end to tail rotor servo connector return fitting. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

i. Check hose installation and test tail rotor servo for proper operation (PARA 7-4-49).

j. Place clamp on replacement hose.

k. Connect clamps using screws, washers, and nuts (Figure 7-4-50-1, Detail A and Detail B).

l. Service backup hydraulic pump module reservoir (PARA 1-3-8), if required.

m. Bleed 1st and backup stage hydraulic systems (PARA 7-4-65).

n. Make sure area is clean and free of foreign material.

o. Install tail gear box fairings (PARA 2-4-137).

7-4-51 HYDRAULIC SYSTEM DRAIN LINES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container
- Hydraulic Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69

7-4-51.1 Replace Hydraulic System Drain Lines.

NOTE

Replacement of all drain lines is the same, except for location of clamps and use of spacers.

7-4-51.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. When removing a drain line, perform the following:

(1) Remove screws, washers, and nuts securing clamps together (Figure 7-4-51-1, Sheet 1, Detail A, Detail B, and Detail C).

(2) Leave clamps on tubing. Clamp on tube not being replaced will serve to mark location of clamp for new tube.

(3) Where a spacer is used, install remaining clamp (on tube not being replaced) and spacer back in place. This will keep spacer in location and ready for replacement tubing (Figure 7-4-51-1, Sheet 1, Detail B, and Sheet 2, Detail J).

(4) Place a container and machinery towel, Item 211, Appendix D, under mating area to catch any fluid that might spill when tubing is removed from fitting.

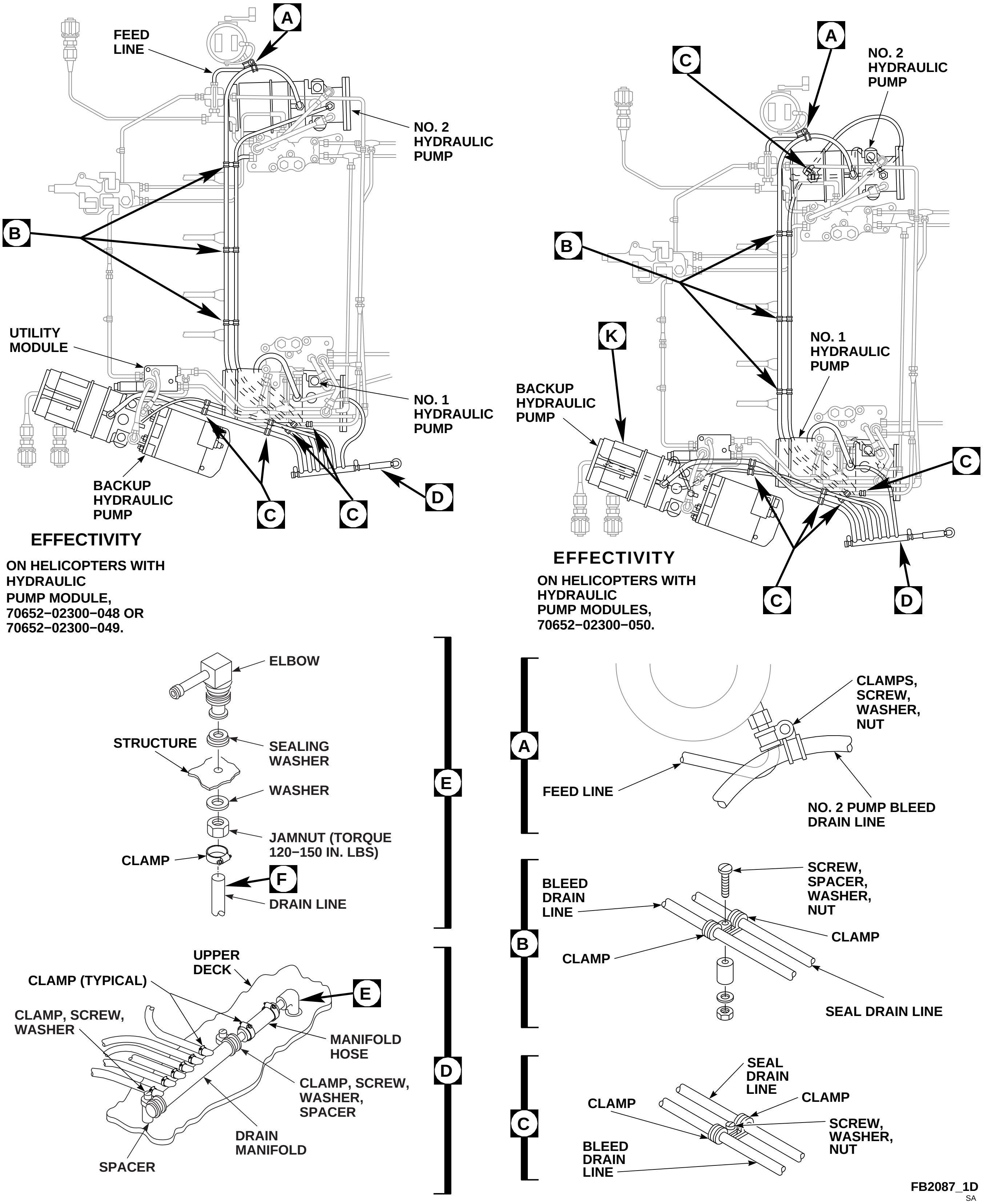


FIGURE 7-4-51-1. REPLACE HYDRAULIC SYSTEM DRAIN LINES (SHEET 1 OF 2)

NOTE

Tube may be bonded to drain manifold. If necessary, cut tube from drain manifold.

d. Disconnect clamps. Pull tube from hydraulic pump module bleed relief valves, tees, or seal drain elbows. Pull tube from drain manifold fitting on left side of helicopter (Figure 7-4-51-1, Sheet 1, Detail D, and Sheet 2, Detail K).

e. Inside cabin, remove soundproofing panels (PARA 2-4-69) as necessary, to get to drain tube running down left side of cabin.

f. Disconnect drain lines from both elbows and remove drain lines (Figure 7-4-51-1, Sheet 1, Detail E and Sheet 2, Detail G).

g. Remove clamps where required. Remove tubing from bulkhead union and from left side of cabin (Figure 7-4-51-1, Sheet 2, Detail H).

h. Remove clamp and remove manifold hose from elbow. Remove block support, screws, washers, and spacer from drain manifold. Remove drain manifold and manifold hose from upper deck (Figure 7-4-51-1, Sheet 1, Detail D).

i. If necessary, remove manifold hose from drain manifold as required.

j. Remove jamnuts, washers, elbows, and bulkhead union from structure (Figure 7-4-51-1, Sheet 1, Detail E, and Sheet 2, Detail G and Detail H).

7-4-51.1.2 Inspect.

Check drain line identification tapes for condition. **IDENTIFICATION TAPES MUST BE PRESENT AND LEGIBLE.** If missing or unreadable, replace tapes as required.

7-4-51.1.3 Install.

a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean fittings and area around fittings. Wipe cleaned areas using machinery towel dampened with isopropyl alcohol, Item 72, Appendix D.

b. Assemble sealing washer onto threaded end of elbow. Install elbow in upper deck from top side, point tube fitting toward front, and install washer and jamnut (Figure 7-4-51-1, Sheet 1, Detail E). **TORQUE JAMNUT TO 120 - 150 INCH-POUNDS.**

c. Install elbow in cabin ceiling structure (Figure 7-4-51-1, Sheet 2, Detail G). Install washer and jamnut. **TORQUE JAMNUT TO 120 - 150 INCH-POUNDS.**

d. Install bulkhead union in cabin ceiling structure. Install washer and jamnut. **TORQUE JAMNUT TO 120 - 150 INCH-POUNDS** (Figure 7-4-51-1, Sheet 2, Detail H).

e. In cabin ceiling, install drain tube between elbows with clamps (Figure 7-4-51-1, Sheet 1, Detail E and Sheet 2, Detail G).

f. Install tubing between elbow and bulkhead union with clamps (Figure 7-4-51-1, Sheet 2, Detail G and Detail H).

g. Secure tubing down side of structure with clamps, screws, washers, nuts, and spacers (Figure 7-4-51-1, Sheet 2, Detail J).

h. Using sealing compound, Item 292, Appendix D, seal areas where fittings go through cabin ceiling.

i. Using machinery towel, Item 211, Appendix D, and paint remover, Item 230, Appendix D, clean old adhesive off drain manifold tube fittings.

NOTE

Clamping of hoses to manifold is authorized to replace bonding.

j. Install manifold hose on drain manifold and elbow with clamps. Secure drain manifold to upper deck with block supports, clamp, screw, washers, and spacers (Figure 7-4-51-1, Sheet 1, Detail D).

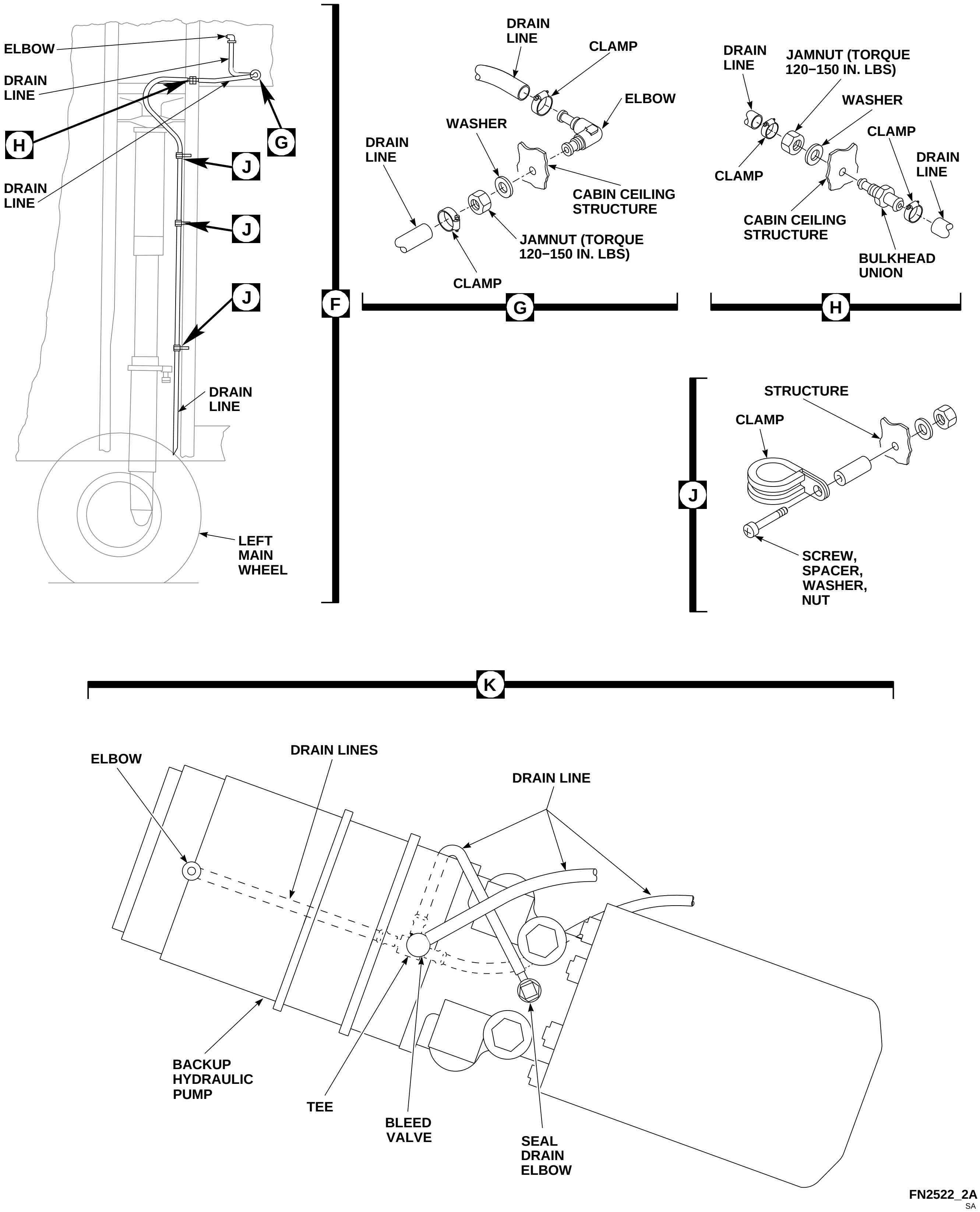


FIGURE 7-4-51-1. REPLACE HYDRAULIC SYSTEM DRAIN LINES (SHEET 2 OF 2)

NOTE

Position drain manifold so tubing fittings extend inboard and up. Position so drain lines will slope in direction of flow.

k. Install drain tubing on drain manifold fittings with clamps.

l. To seal areas where hardware passes through cabin ceiling, perform the following:

(1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean areas. Wipe cleaned areas with machinery towel dampened with isopropyl alcohol, Item 72, Appendix D.

(2) Install clamps with nuts, washers, and spacers.

(3) Using sealing compound, Item 292, Appendix D, seal area around nuts, washers, and spacers.

m. Install drain tubes on hydraulic pump module bleed valves, tees, or seal drain elbows. Route lines as shown (Figure 7-4-51-1, Sheet 2, Detail K). Install clamps.

n. Make sure area is clean and free of foreign material.

o. Close main rotor pylon sliding cover.

p. Install soundproofing panels (PARA 2-4-69).

7-4-52 APU ACCUMULATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Container, 1-Pint
- Nonmetallic Scraper, Locally-Made
- Rivet Gun
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Adhesive, Item 37, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 7-2-1
- PARA 7-4-53
- PARA 7-4-61
- PARA 7-4-65
- PARA 13-5-3
- PARA 13-5-6

- PARA 16-6-1
- PARA 16-6-2

7-4-52.1 Replace APU Accumulator.

7-4-52.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. If winterization kit is installed, remove second accumulator (PARA 16-6-2).

c. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

d. Remove screws from front and top panels covering fuel system components on top of fuel cell. Remove panels.

e. On upper console, place BACKUP HYD PUMP switch to OFF.

f. On upper console, make sure APU CONTR switch is OFF.

g. Disconnect electrical connector, P290, from APU start valve (Figure 7-4-52-1, Sheet 1, Detail A).

h. Install protective caps and plugs on electrical connectors.

i. Install APU exhaust plug (PARA 1-6-5).

j. Manually operate APU start valve lever to motor APU and release hydraulic charge.

k. Remove accumulator servicing valve cap on pressure gage.

l. Loosen jamnut on servicing valve and slowly release nitrogen charge. Make sure all nitrogen pressure is depleted.

m. Disconnect nitrogen gas line from accumulator. Cap and plug open line and fitting.

n. Use 1-pint container and machinery towels, Item 211, Appendix D, to catch spilled fluid when lines are disconnected.

o. Disconnect handpump pressure and return lines from overhead tee fittings. Cap and plug lines and fittings.

p. Open APU compartment access doors.

q. Disconnect pressure and return hoses from APU start valve (PARA 7-4-61).

r. Remove jamnuts and washers from APU start valve at boot (Figure 7-4-52-1, Sheet 2, Detail B).

s. Loosen T-bolts on strap hangers (Figure 7-4-52-1, Sheet 1, Detail A).

t. Instruct assistant to support accumulator.

u. Release T-bolts on strap hangers.

v. Carefully remove accumulator.

w. Check boot for cracks, splits, holes, or tears. **NONE ALLOWED.** If damaged, remove boot as follows (Figure 7-4-52-1, Sheet 2, Detail B):

(1) Remove screws, washers, and sealing washers securing boot to APU deck. Remove boot.

(2) Remove sealing compound from APU deck using nonmetallic scraper. Remove any remaining sealing compound with

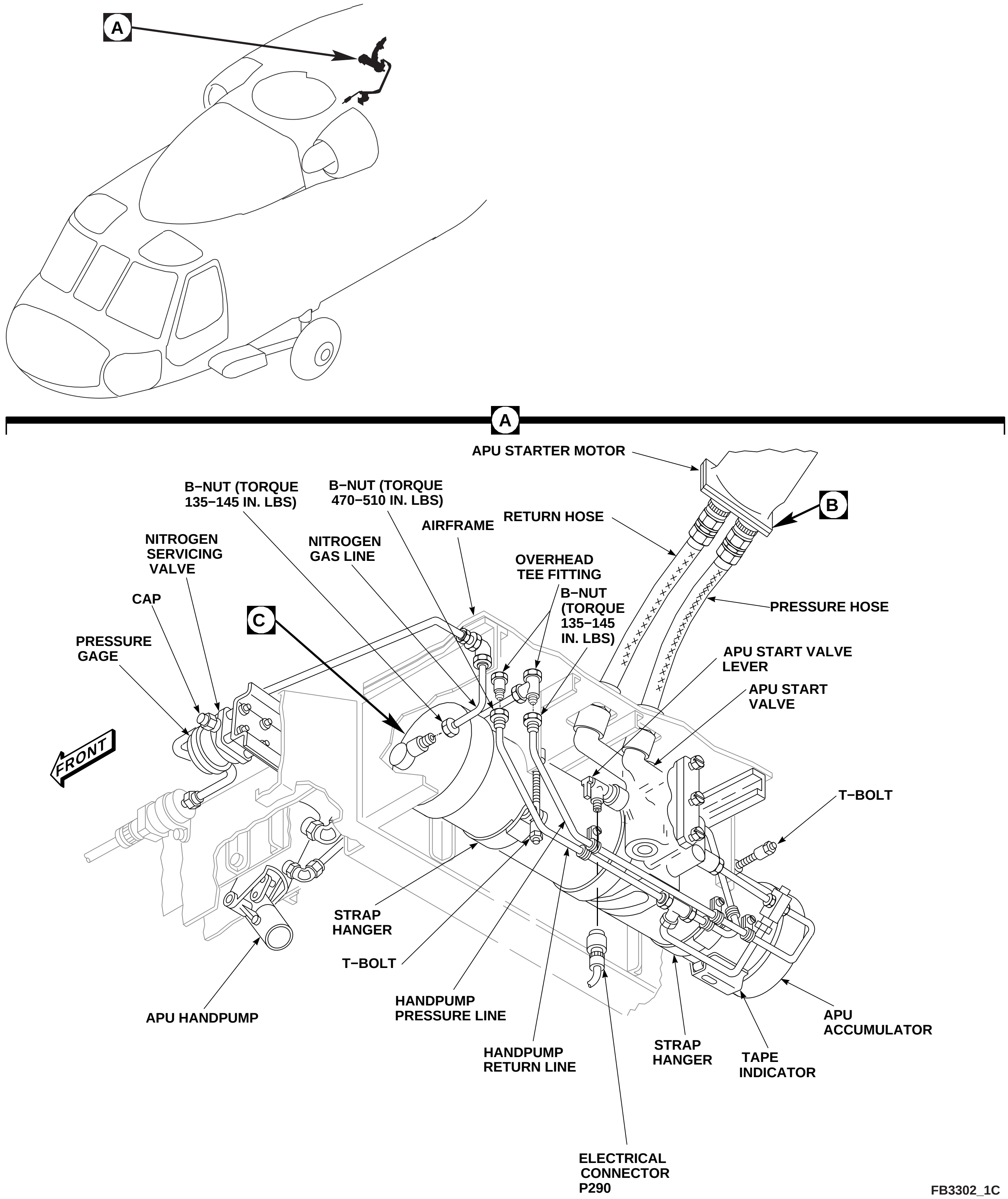


FIGURE 7-4-52-1. REPLACE APU ACCUMULATOR (SHEET 1 OF 2)

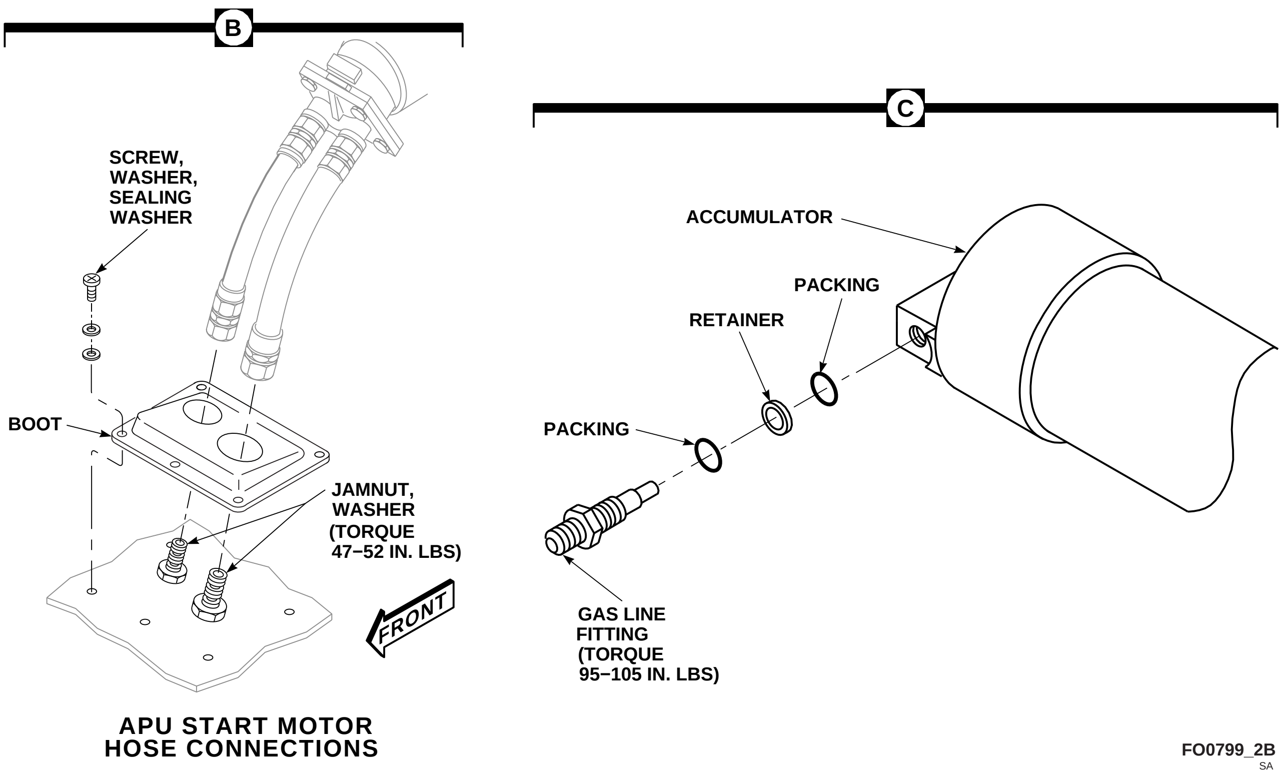


FIGURE 7-4-52-1. REPLACE APU ACCUMULATOR (SHEET 2 OF 2)

machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

x. Check accumulator strap hangers for damage. **NONE ALLOWED.** If damaged, remove strap hangers as follows (Figure 7-4-52-1, Sheet 1, Detail A):

NOTE

Removal of left and right strap hangers from accumulator is the same, except where noted.

(1) On right side, remove pin, washers, and cotter pin from front side of strap hanger.

(2) Remove rivets from strap hanger.

y. If required, replace or rebond rubber edging on strap hanger as follows:

(1) Remove rubber edging using nonmetallic scraper.

(2) Clean adhesive from strap hanger with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Remove remaining adhesive using nonmetallic scraper.

(3) Roughen up surface of rubber edging using abrasive cloth, Item 2, Appendix D. Clean with machinery towel, Item 211, Appendix D.

(4) Apply a thin even coat of adhesive, Item 37, Appendix D, to strap hanger and mating surface of rubber edging. Allow adhesive to air-dry for about 10 minutes or until tacky.

(5) Press seal firmly in place. Make sure all surfaces are in good contact. Allow seal to cure for 24 hours at 68°F (20°C).

7-4-52.1.2. APU Accumulator Buildup.

NOTE

Replace attaching parts only if new APU accumulator is to be installed.

a. Remove APU start valve (PARA 7-4-53).

b. Remove gas line fitting, retainer, and packings from accumulator. Discard packings (Figure 7-4-52-1, Sheet 2, Detail C).

NOTE

Use only hydraulic fluid, Item 155, Appendix D, when preserving accumulator.

c. Preserve removed accumulator and depreserve replacement accumulator.

d. Place cleaning cloth, Item 102, Appendix D, around hose of air supply. Apply air pressure to pressure (PRESS) port of replacement accumulator until piston indicator shows 100%. Remove air supply.

e. Install APU start valve (PARA 7-4-53).

f. Install gas line fitting, packings, and retainer in replacement accumulator. **TORQUE GAS LINE FITTING TO 95 - 105 INCH-POUNDS.**

7-4-52.1.3. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. If required, install replacement boot as follows (Figure 7-4-52-1, Sheet 2, Detail B):

(1) Apply sealing compound, Item 292, Appendix D, to outside edge of boot.

(2) Install boot on APU deck with raised end forward.

(3) Secure boot to APU deck with screws, washers, and sealing washers wet with primer, Item 258, Appendix D.

c. If required, install replacement strap hangers as follows (Figure 7-4-52-1, Sheet 1, Detail A):

NOTE

Installation of left and right strap hangers are the same, except as noted.

(1) On right side, install pin and washers through front side of strap hanger. Install cotter pin.

(2) Using rivet gun, rivet strap hanger to bracket.

d. Instruct assistant to lift accumulator into place. Line up and place APU start valve fittings through bulkhead.

e. Place strap hangers around accumulator and fasten T-bolts. Do not tighten nuts.

f. In APU compartment, install washers and jamnuts on APU start valve fittings (Figure 7-4-52-1, Sheet 2, Detail B). **TORQUE JAMNUTS TO 47 - 52 INCH-POUNDS.**

g. Connect pressure and return hoses to APU start valve (PARA 7-4-61).

h. **TIGHTEN ACCUMULATOR CLAMP T-BOLT NUTS** (Figure 7-4-52-2, Sheet 1, Detail A).

i. If required, install second winterization kit (PARA 16-6-1).

j. Coat threads of lines and overhead tee fittings with hydraulic fluid, Item 154 or Item 155, Appendix D.

k. Remove caps and plugs and connect handpump pressure and return lines to overhead tee fittings (Figure 7-4-52-1, Sheet 1, Detail A). **TORQUE PRESSURE LINE B-NUT TO 135 - 145 INCH-POUNDS. TORQUE RETURN LINE B-NUT TO 470 - 510 INCH-POUNDS.**

l. Remove cap and plug and connect nitrogen gas line to accumulator. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

m. Remove protective caps and plugs from electrical connectors.

n. Inspect and treat electrical connectors (PARA 1-7-20).

o. Connect electrical connector, P290, to APU start valve. Lockwire, Item 196, Appendix D, connector to start valve.

p. Keep servicing valve open.

q. Connect external electrical power (PARA 1-6-16).

r. On upper console, place BACKUP HYD PUMP switch to OFF.

s. Service APU accumulator (PARA 1-3-10).

t. Bleed hydraulic system (PARA 7-4-65).

u. On copilot's circuit breaker panel, pull out BACKUP PUMP PWR circuit breaker.

v. Remove APU exhaust plug (PARA 1-6-5).

w. Manually operate APU start valve to motor APU and release hydraulic charge.

x. On caution/advisory panel, check that APU ACCUM LOW light is on.

NOTE

The pressures and tape indicator readings following are based on a 70°F (21°C) temperature day. Use the Temperature/Precharge Chart for correct pressure.

y. Check pressure gage for 1450 psi pressure.

z. Check accumulator tape indicator. Indicator should read 0.

WARNING

Injury to personnel will result if APU accumulator handpump handle is not installed correctly. Make sure pump handle is firmly locked into handle receptacle before operating handpump.

aa. Operate APU accumulator handpump until a small pressure increase is noticed on pressure gage. This will show handpump is working.

CAUTION

Whenever the No. 1 AC generator is inoperative and the BACKUP PUMP PWR circuit breaker is out for any reason, AC electrical power must be shut off before resetting BACKUP PUMP PWR circuit breaker. Otherwise it is possible to damage the current limiters.

NOTE

Backup system should recharge accumulator in no more than 90 seconds. If it takes longer, perform operational check of hydraulic system (PARA 7-2-1).

ab. On copilot's circuit breaker panel, push in BACKUP PUMP PWR circuit breaker. On caution/advisory panel, BACK-UP PUMP ON light should go on and backup hydraulic pump module will pressurize accumulator.

ac. On caution/advisory panel, check that APU ACCUM LOW light goes off, and then BACKUP PUMP ON light to go off when backup pump shuts off.

ad. Disconnect external electrical power (PARA 1-6-16).

ae. Check accumulator tape indicator. Indicator should read 63% to 86%.

af. Check pressure gage. It should read between 2700 to 3100 psi.

ag. Start and operate APU (Appropriate Flight Manual).

ah. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

ai. Make sure fuel cell area is clean and free of foreign material.

aj. Place front and top panels over fuel system components and install screws.

ak. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

al. Make sure area is clean and free of foreign material.

am. Close APU compartment access doors.

an. Install APU exhaust plug (PARA 1-6-5).

ao. Service backup hydraulic pump module (PARA 1-3-8).

7-4-53 APU START VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 7-4-52

7-4-53.1 Replace APU Start Valve.

7-4-53.1.1 Remove.

a. Remove APU accumulator (PARA 7-4-52).

b. Remove bolts, washers, and APU start valve from accumulator. Transfer tubes will stay with accumulator. Remove packings and backup retainers from transfer tubes. Discard packings (Figure 7-4-53-1).

NOTE

Be careful not to tip accumulator towards open ports with APU start valve removed. This could cause possible loss of hydraulic fluid.

c. If new APU start valve is to be installed, remove check valve restrictor, union, and packings from APU start valve. Discard packings.

7-4-53.1.2 Install.

a. Coat packings and threads of parts with hydraulic fluid, Item 154 or Item 155, Appendix D, before installing.

b. If required, install check valve restrictor and packing into pressure port (PR) of replacement APU start valve (Figure 7-4-53-1). **TORQUE CHECK VALVE RESTRICTOR TO 280 - 305 INCH-POUNDS.**

c. If required, install union and packing into return port (RET) of replacement APU start valve. **TORQUE UNION TO 550 - 600 INCH-POUNDS.**

d. Install packings and backup retainers on transfer tubes.

e. Install APU start valve with bolts and washers. **TORQUE BOLTS TO 309 - 341 INCH-POUNDS.** Lockwire, Item 194, Appendix D, bolts.

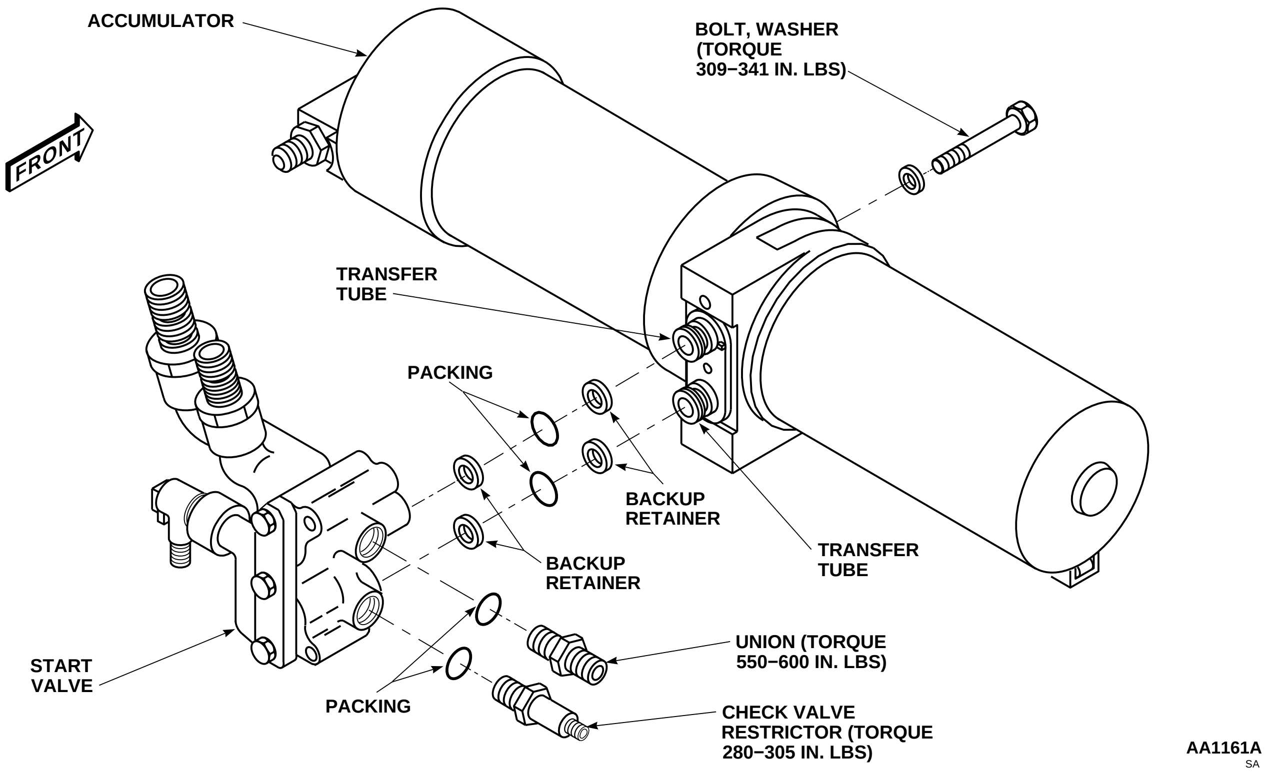


FIGURE 7-4-53-1. REPLACE APU START VALVE

f. Install APU accumulator (PARA 7-4-52).

7-4-54 APU ACCUMULATOR HANDPUMP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Pint
- Drift
- Retaining Ring Pliers
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 150, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

7-4-54.1 Replace APU Accumulator Handpump.

7-4-54.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Use 1-pint container and machinery towels, Item 211, Appendix D, to catch spilled fluid when lines are disconnected.

e. Disconnect pressure and return lines from APU accumulator handpump elbows. Cap and plug all open lines and fittings (Figure 7-4-54-1, Detail A).

f. Remove bolts, washers, and spacers, and remove APU accumulator handpump from mounting brackets.

g. If new APU accumulator handpump is to be installed, remove elbows, jamnuts, retainers, and packings from handpump. Discard packings and retainers.

h. Preserve APU accumulator handpump. Use only hydraulic fluid, Item 155, Appendix D, when preserving handpump.

7-4-54.1.2 Disassemble.

a. Using retaining ring pliers, remove retaining rings from link pivot pins (Figure 7-4-54-2).

b. Using retaining ring pliers, remove retaining rings from socket pivot pin.

c. Using drift, press out link and socket pivot pins. Remove socket.

d. Remove remaining link pin and attaching hardware as required.

e. Clean parts with dry-cleaning solvent, Item 124, Appendix D, and machinery towels, Item 211, Appendix D. Wipe parts dry.

7-4-54.1.3 Assemble.

a. Pack linkage bushings with grease, Item 150, Appendix D.

b. Line up holes in APU handpump, links, and socket. Install link pivot pins and retaining rings (Figure 7-4-54-2).

c. Install socket pivot pin and retaining rings.

7-4-54.1.4 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Depreserve APU accumulator handpump. Use only hydraulic fluid, Item 155, Appendix D, when depreserving handpump.

c. Coat packings, retainers, and threads of jamnut and elbow with hydraulic fluid, Item 154 or Item 155, Appendix D.

d. If removed, install elbow, jamnut, retainer, and packing in SUCTION port. Face elbow to rear. Do not tighten jamnut (Figure 7-4-54-1, Detail A).

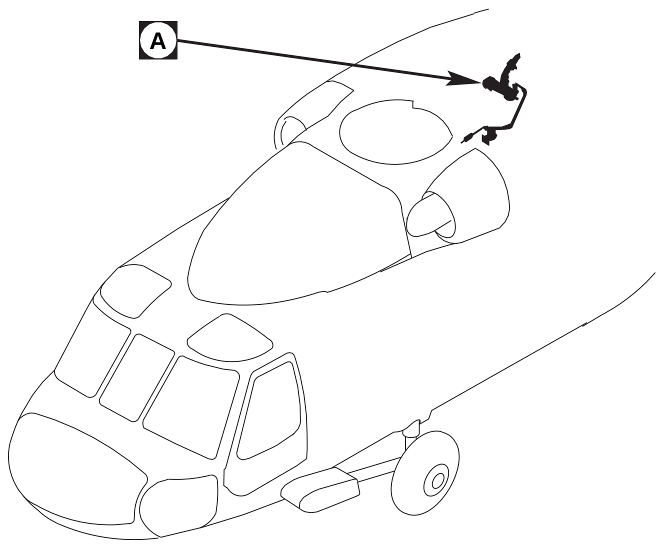
e. Coat packings, retainers, and threads of jamnut and elbow with hydraulic fluid, Item 154 or Item 155, Appendix D.

f. If removed, install elbow, jamnut, retainer, and packing in PRESSURE port. Face elbow to rear left side. Do not tighten jamnut.

g. Fill APU accumulator handpump with hydraulic fluid, Item 154 or Item 155, Appendix D.

h. Place APU accumulator handpump between mounting brackets. Install bolts, washers, and spacers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

i. Connect pressure and return lines to APU accumulator handpump elbows. Leave pressure line slightly loose. **TORQUE RETURN LINE JAMNUT TO 280 - 305 INCH-POUNDS. TORQUE RETURN LINE B-NUT TO 470 - 510 INCH-POUNDS. TORQUE PRESSURE LINE JAMNUT TO 95 - 105 INCH-POUNDS.**



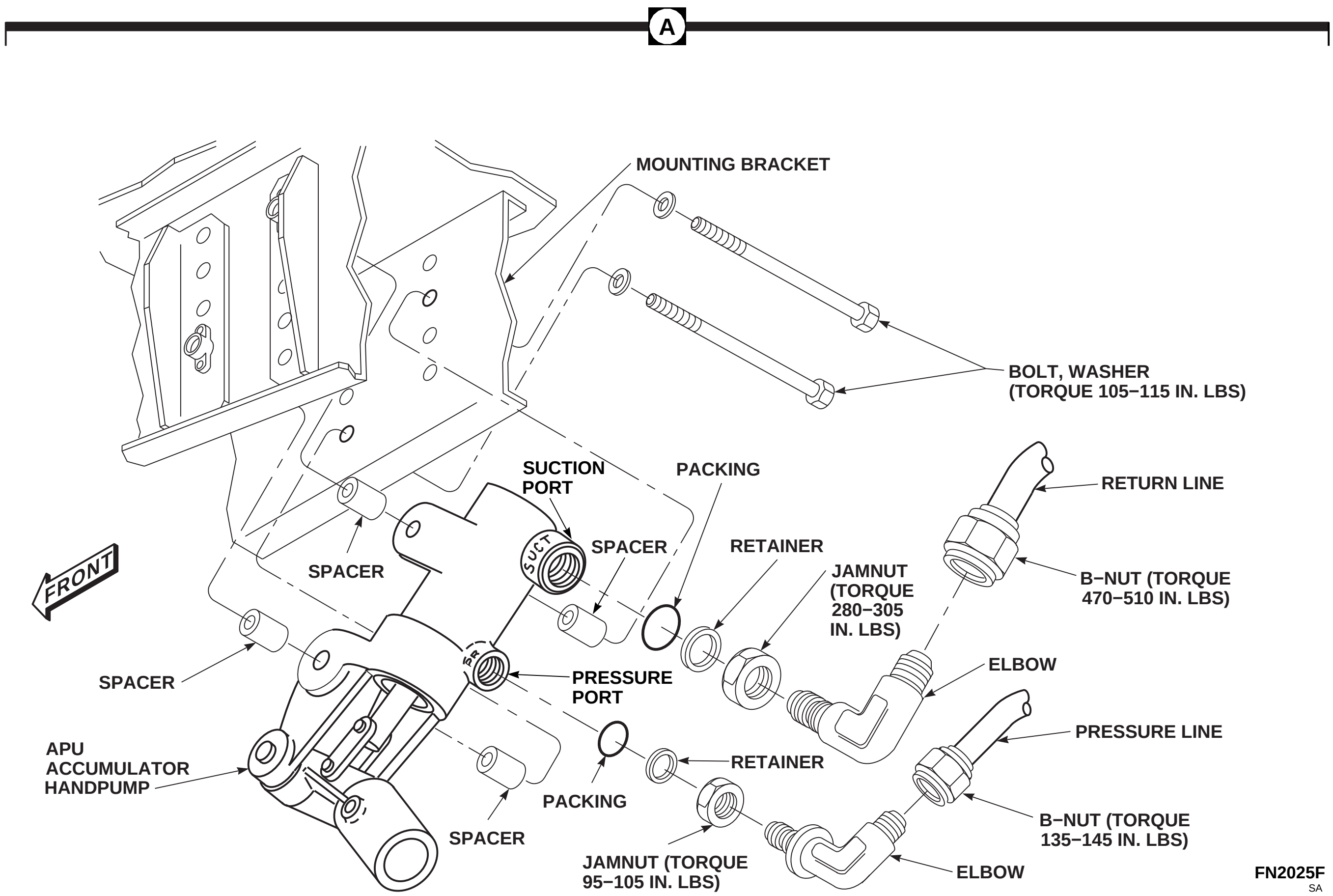


FIGURE 7-4-54-1. REPLACE APU ACCUMULATOR HANDPUMP

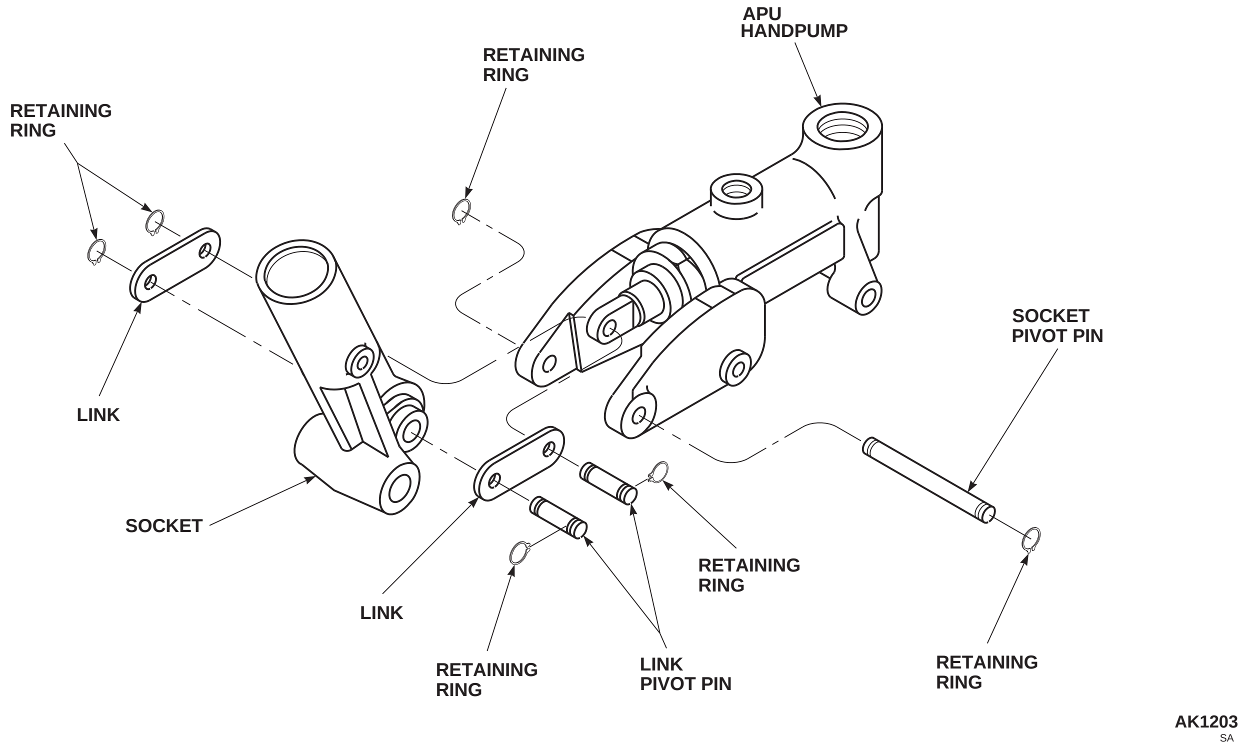


FIGURE 7-4-54-2. REPAIR APU ACCUMULATOR HANDPUMP

WARNING

Injury to personnel will result if APU accumulator handpump is not installed correctly. Make sure APU accumulator handpump handle is firmly locked into handle receptacle before operating handpump.

NOTE

APU accumulator handpump handle is in left stowage compartment behind rear crew compartment.

j. Place handle into handpump and operate handpump by moving handle back and forth.

k. Place machinery towels, Item 211, Appendix D, underneath line to catch bleed-off hydraulic fluid. Pump until no more air bubbles come from fitting. **TORQUE PRESSURE LINE B-NUT TO 135 - 145 INCH-POUNDS.**

l. Check APU accumulator handpump for proper operation as follows:

(1) Turn off all electrical power (PARA 1-6-16).

(2) On upper console, make sure APU CONTR switch is set to OFF.

(3) On copilot's circuit breaker panel, pull out BACKUP PUMP PWR, .5-amp, circuit breaker.

(4) Manually operate APU start valve lever quickly, only allowing a small percent of charge to be released.

(5) Recharge APU accumulator by moving handle back and forth on handpump. Use long, slow strokes to build up pressure.

m. Make sure area is clean and free of foreign material.

n. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

o. On copilot's circuit breaker panel, push in BACKUP PUMP PWR, .5-amp, circuit breaker.

7-4-55 HYDRAULIC REFILL HANDPUMP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16

7-4-55.1 Replace Hydraulic Refill Handpump.

7-4-55.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. RTR BRAKE Disconnect rotor brake hose from hydraulic refill handpump. Use machinery towels, Item 211, Appendix D, to catch hydraulic fluid (Figure 7-4-55-1).

d. Disconnect pressure line from hydraulic refill handpump. Use machinery towels, Item 211, Appendix D, to catch hydraulic fluid.

e. Remove bolts and washers securing hydraulic refill handpump to support bracket.

f. Drain any hydraulic fluid that may be trapped inside hydraulic refill handpump.

NOTE

If hydraulic refill handpump is not to be installed now, plug ports and cap lines.

g. If new hydraulic refill handpump is to be installed, remove unions and packings from old hydraulic refill handpump. Discard packings.

h. If new hydraulic refill handpump is to be installed, preserve removed hydraulic refill handpump. Use only hydraulic fluid, Item 155, Appendix D, when preserving refill handpump.

7-4-55.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

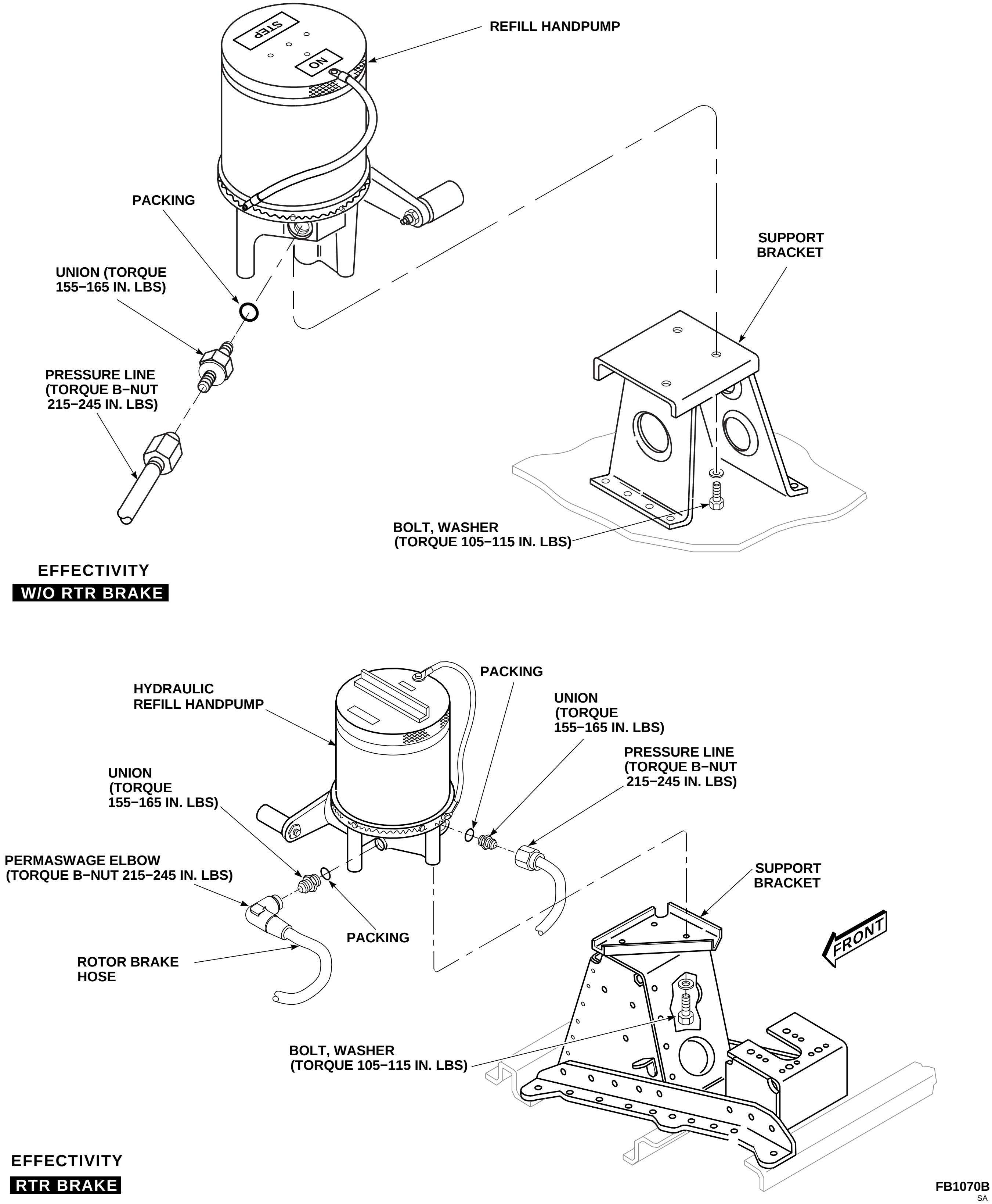


FIGURE 7-4-55-1. REPLACE HYDRAULIC REFILL HANDPUMP

b. If necessary, depreserve handpump.

c. Lubricate packings with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

d. If required, install packings on unions and install unions in hydraulic refill handpump. **TORQUE UNIONS TO 155 - 165 INCH-POUNDS** (Figure 7-4-55-1).

e. Before installing hydraulic refill handpump, open top and fill handpump reservoir with hydraulic fluid of the same type used in hydraulic systems. Crank handle until fluid comes out pressure port freely. Use machinery towels, Item 211, Appendix D, to catch fluid.

f. Place hydraulic refill handpump on support bracket. Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** If drilled-head bolts are installed, lockwire, Item 196, Appendix D, bolts.

g. Connect pressure line to union in hydraulic refill handpump. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

h. RTR BRAKE Connect rotor brake hose to union in hydraulic refill handpump. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

i. Top off handpump reservoir with hydraulic fluid. Check operation by draining some fluid from system hydraulic pump module reservoir and servicing low reservoir (PARA 1-3-8).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

7-4-56 HYDRAULIC REFILL HANDPUMP FILTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16

7-4-56.1 Replace Hydraulic Refill Handpump Filter.

7-4-56.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. Unscrew and remove filter bowl, filter, and packings as an assembly from hydraulic refill handpump (Figure 7-4-56-1, Detail A).

d. Remove filter. Discard packings from filter bowl.

e. Clean filter bowl with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

7-4-56.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packings with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

c. Place filter and packings in filter bowl (Figure 7-4-56-1, Detail A).

d. Install filter bowl and packings on hydraulic refill handpump. Lockwire, Item 194, Appendix D, filter bowl.

e. Make sure area is clean and free of foreign material.

f. Close main rotor pylon sliding cover.

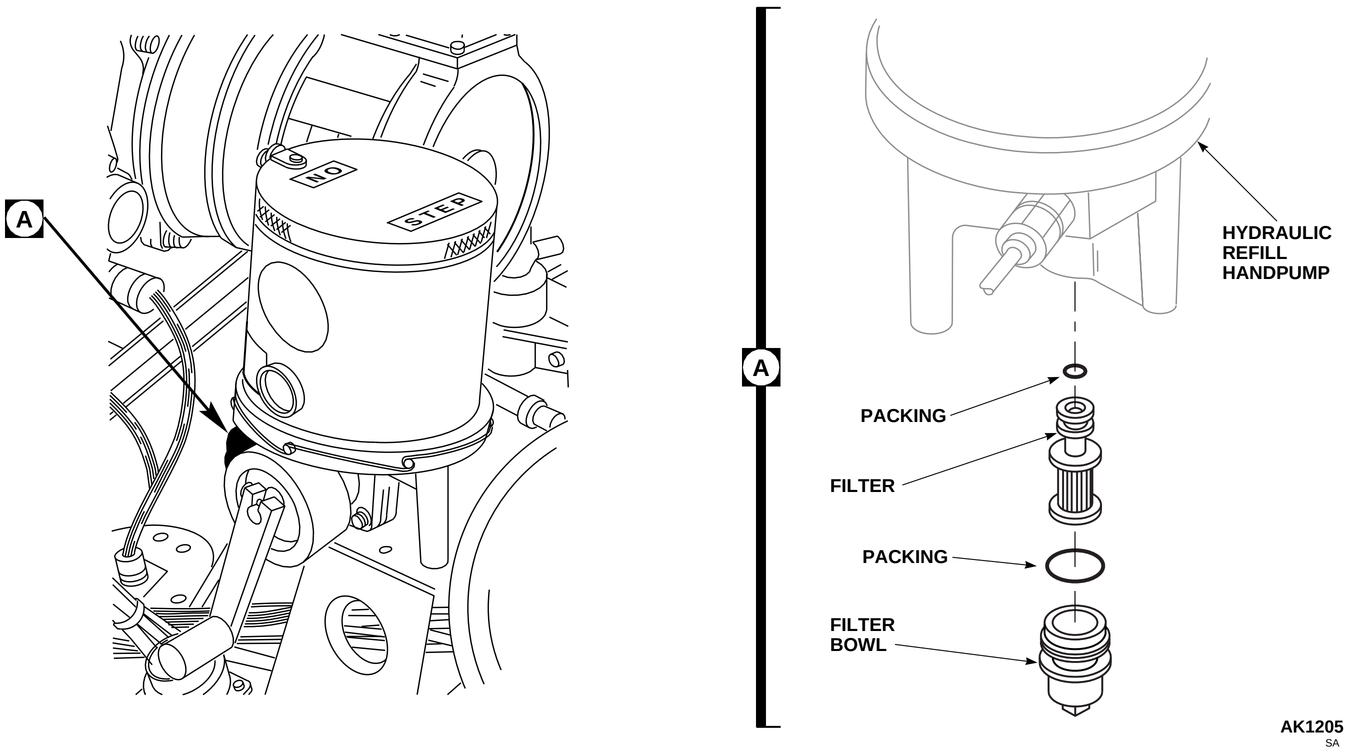


FIGURE 7-4-56-1. REPLACE HYDRAULIC REFILL HANDPUMP FILTER

7-4-57 APU START CHECK VALVE RESTRICTOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 7-4-65
- PARA 13-5-3
- PARA 13-5-6

7-4-57.1 Replace APU Start Check Valve Restrictor.

7-4-57.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

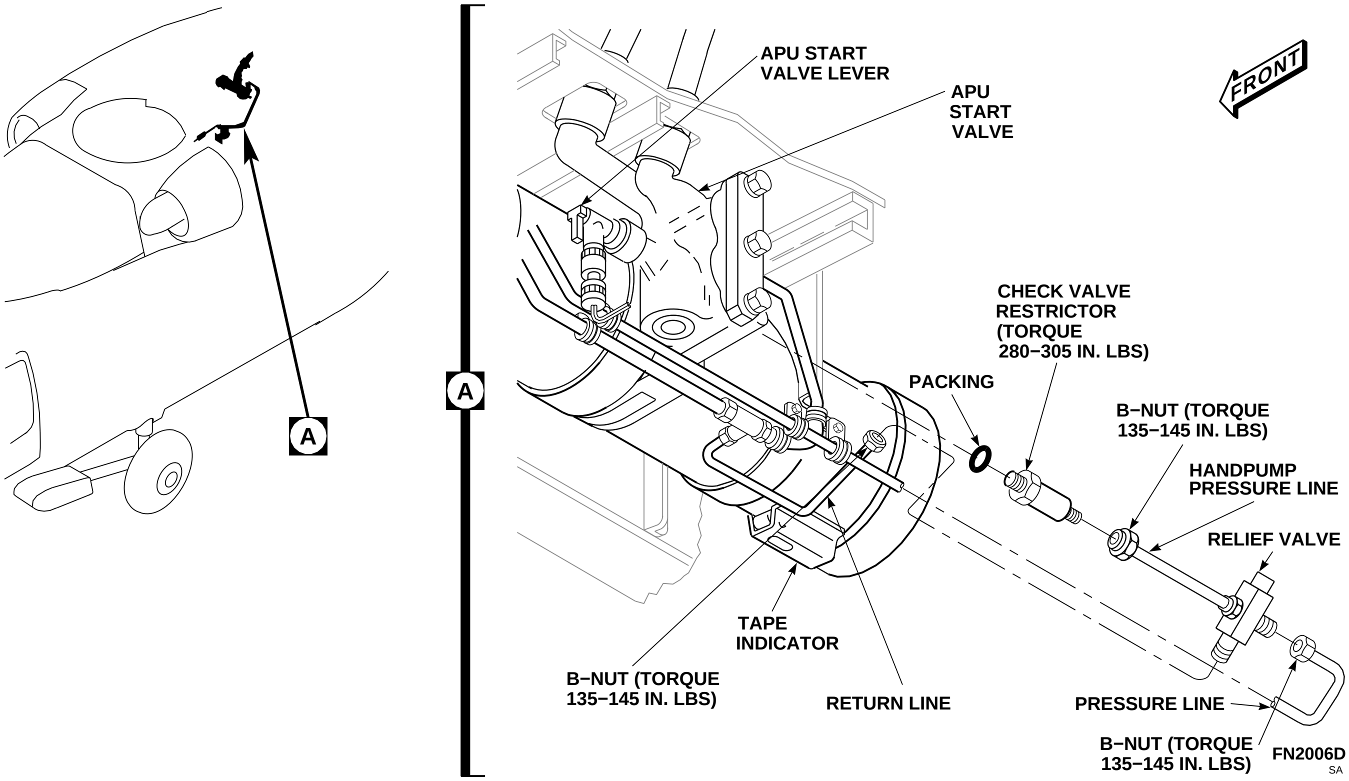


FIGURE 7-4-57-1. REPLACE APU START CHECK VALVE RESTRICTOR

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. Remove screws from front and top panels enclosing fuel system component on top of fuel cell. Remove panels.

f. Remove APU exhaust plug (PARA 1-6-5).

g. Manually operate APU start valve lever to motor APU and release hydraulic charge. Tape indicator on accumulator will show 0.

h. Disconnect pressure and return lines from relief valve (Figure 7-4-57-1, Detail A).

i. Disconnect handpump pressure line from check valve restrictor.

j. Unscrew and remove check valve restrictor and packing. Discard packing.

k. Plug port if check valve restrictor is not to be installed now.

7-4-57.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packing and threads of check valve restrictor with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. If installed, remove plug from port. Install check valve restrictor and packing into APU start valve (Figure 7-4-57-1, Detail A). **TORQUE CHECK VALVE RESTRICTOR TO 280 - 305 INCH-POUNDS.**

d. Connect handpump pressure line to check valve restrictor. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

e. Connect pressure and return lines to relief valve. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

f. Service APU accumulator (PARA 1-3-10).

g. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

h. Bleed hydraulic system (PARA 7-4-65).

i. Make sure fuel cell area is clean and free of foreign material.

j. Place front and top panels over fuel system components and install screws.

k. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

l. Install APU exhaust plug (PARA 1-6-5).

7-4-58 APU ACCUMULATOR PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 196, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 7-4-65
- PARA 13-5-3
- PARA 13-5-6

7-4-58.1 Replace APU Accumulator Pressure Switch.

7-4-58.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. Remove APU exhaust plug (PARA 1-6-5).

f. Manually operate APU start valve lever to motor APU and release hydraulic charge.

g. Remove accumulator servicing valve cap on pressure gage (Figure 7-4-58-1, Detail B).

h. Loosen jamnut on servicing valve and slowly release nitrogen charge.

i. Make sure tape indicator on accumulator, and pressure gage, each show 0 (Figure 7-4-58-1, and Detail A).

j. Disconnect nitrogen line from pressure switch. Plug nitrogen line (Figure 7-4-58-1, Detail B).

k. Disconnect electrical connector, P302, from pressure switch.

l. Install protective caps and plugs on electrical connectors.

m. Remove screw, washer, clamp, and pressure switch from frame and remove pressure switch from clamp.

7-4-58.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Install pressure switch into clamp, and install clamp and pressure switch on frame with screw and washer (Figure 7-4-58-1, Detail B).

c. Connect nitrogen line to pressure switch. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector, P302, to pressure switch. Lockwire, Item 196, Appendix D, connector to pressure switch.

g. Service APU accumulator (PARA 1-3-10).

h. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

i. Bleed hydraulic system (PARA 7-4-65).

j. Make sure area is clean and free of foreign material.

k. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

l. Install APU exhaust plug (PARA 1-6-5).

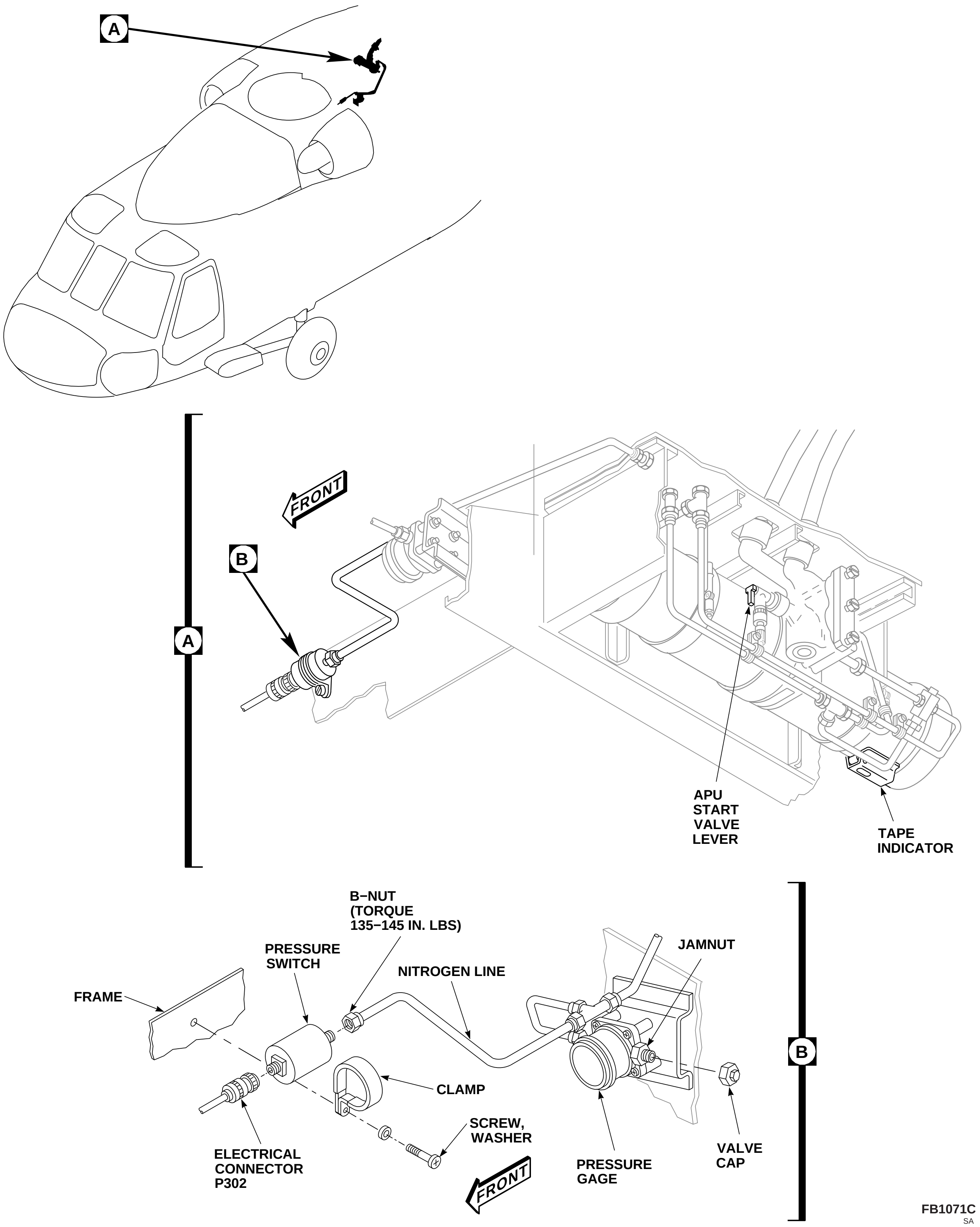


FIGURE 7-4-58-1. REPLACE APU ACCUMULATOR PRESSURE SWITCH

7-4-59 APU ACCUMULATOR PRESSURE GAGE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

7-4-59.1 Replace APU Accumulator Pressure Gage.

7-4-59.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger’s seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. Remove APU exhaust plug (PARA 1-6-5).

f. Manually operate APU start valve lever to motor APU and release hydraulic charge (Figure 7-4-59-1, Detail A).

g. Remove accumulator servicing valve cap on pressure gage (Figure 7-4-59-1, Detail C).

h. Loosen jamnut on servicing valve and slowly release nitrogen charge.

i. Make sure tape indicator on accumulator, and pressure gage, each show 0 (Figure 7-4-59-1, Detail A and Detail B).

j. Disconnect nitrogen line between tee and pressure gage (Figure 7-4-59-1, Detail B). Plug nitrogen line.

k. Remove screws, washers, and spacers attaching pressure gage to airframe. Remove pressure gage.

l. If new pressure gage is to be installed, remove union and packing from pressure gage. Discard packing. Cap and plug ports and fittings.

7-4-59.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packing with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Remove caps and plugs from ports and fittings. If new pressure gage is to be installed, install union and packing into pressure gage (Figure 7-4-59-1, Detail B). **TORQUE UNION TO 95 - 105 INCH-POUNDS.**

d. Install pressure gage on airframe and secure with screws, washers, and spacers.

e. Remove plug and connect nitrogen line to union. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

f. Service APU accumulator (PARA 1-3-10).

g. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

h. Make sure area is clean and free of foreign material.

i. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger’s seats (PARA 2-4-45).

j. Install APU exhaust plug (PARA 1-6-5).

7-4-59.2 Replace APU Accumulator Pressure Gage Servicing Valve.

7-4-59.2.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger’s seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

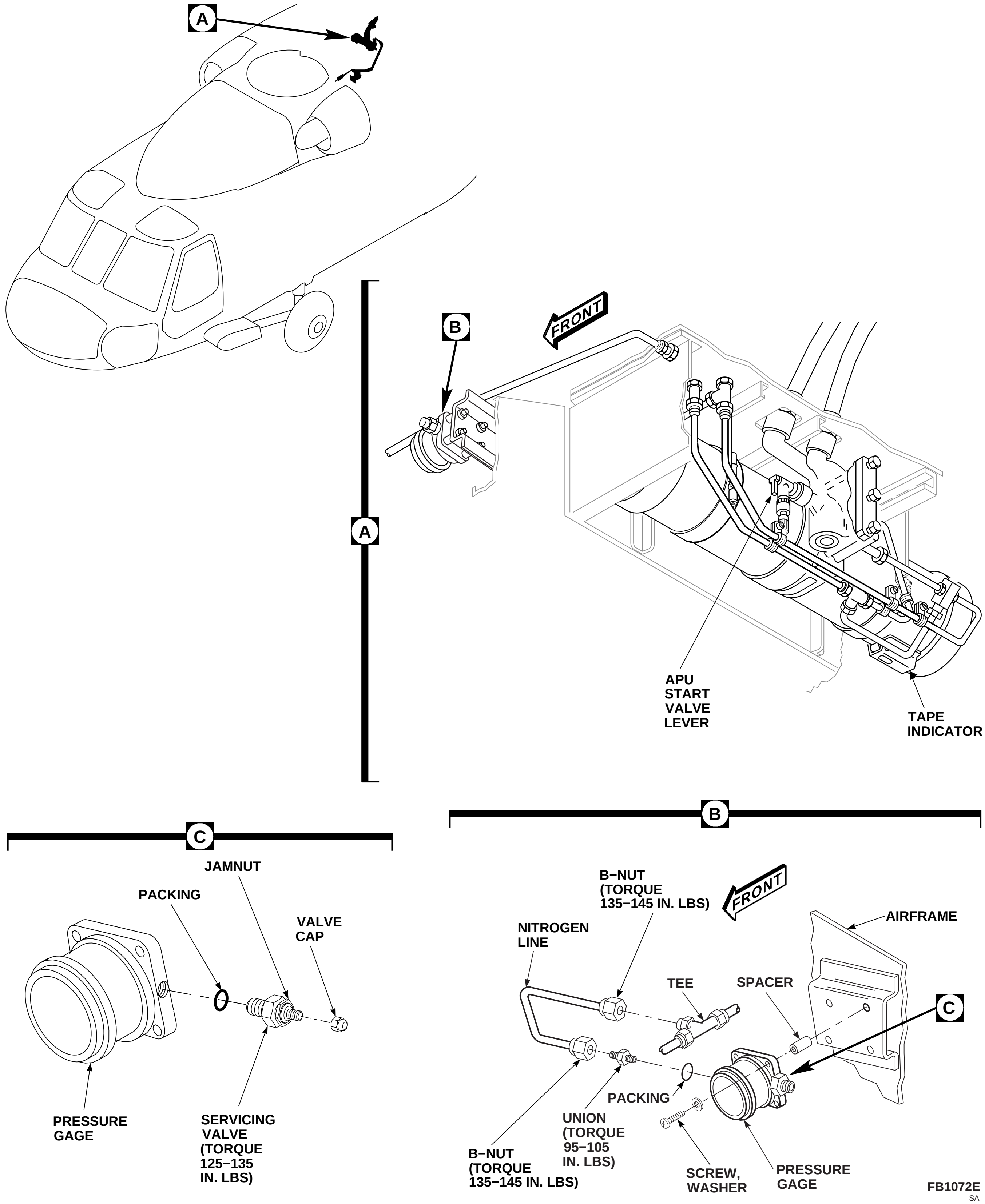


FIGURE 7-4-59-1. REPLACE APU ACCUMULATOR PRESSURE GAGE

d. Remove APU exhaust plug (PARA 1-6-5).

e. Manually operate APU start valve lever to motor APU and release hydraulic charge(Figure 7-4-59-1, Detail A).

f. Remove accumulator servicing valve cap on pressure gage (Figure 7-4-59-1, Detail C).

g. Loosen jamnut on servicing valve and slowly release nitrogen charge.

h. Make sure tape indicator on accumulator and pressure gage each show 0 (Figure 7-4-59-1, Detail A and Detail B).

i. Remove servicing valve and packing from pressure gage (Figure 7-4-59-1, Detail C). Discard packing.

7-4-59.2.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat packing with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. Install packing on servicing valve (Figure 7-4-59-1, Detail C).

d. Install servicing valve in pressure gage. **TORQUE SERVICING VALVE TO 125 - 135 INCH-POUNDS.**

e. Lockwire, Item 197, Appendix D, servicing valve to pressure gage.

f. Service APU accumulator (PARA 1-3-10).

g. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

h. Make sure area is clean and free of foreign material.

i. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger’s seats (PARA 2-4-45).

j. Install APU exhaust plug (PARA 1-6-5).

7-4-60 APU START TEE CHECK VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-5
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 13-5-3
- PARA 13-5-6

7-4-60.1 Replace APU Start Tee Check Valve.

7-4-60.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. Remove APU exhaust plug (PARA 1-6-5).

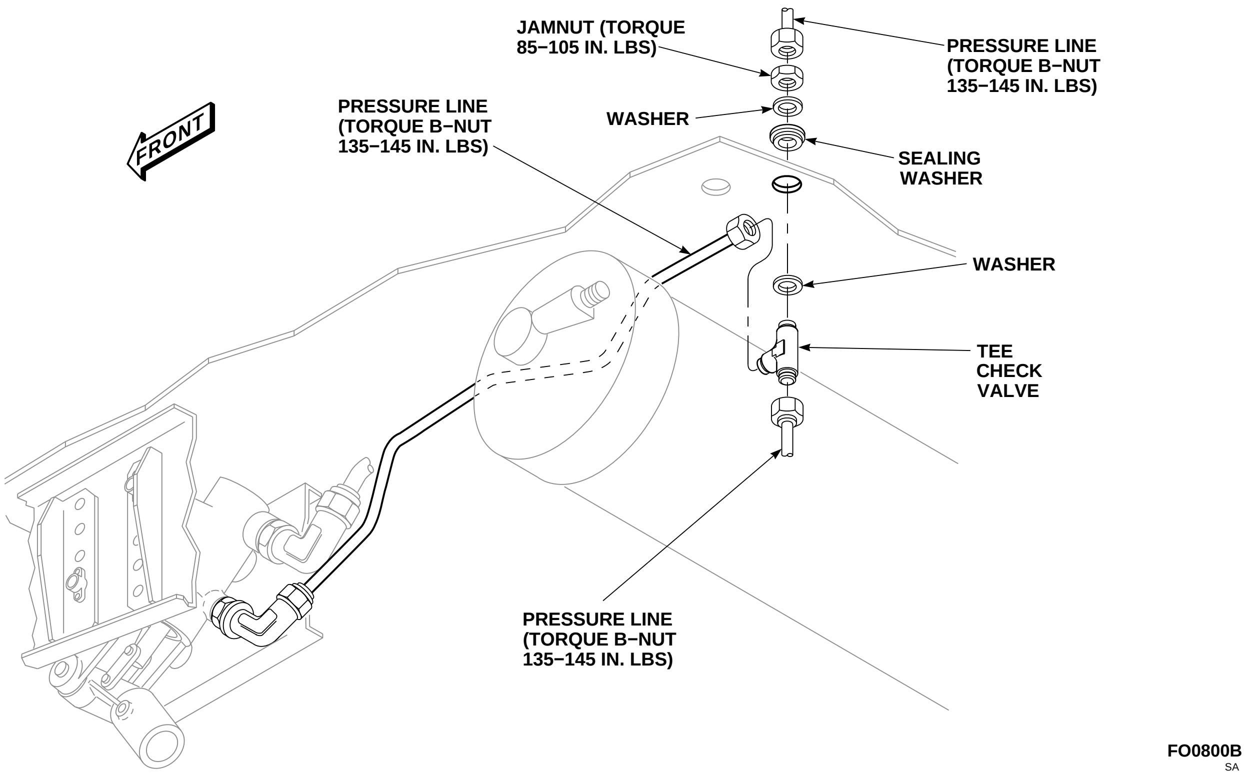


FIGURE 7-4-60-1. REPLACE APU START TEE CHECK VALVE

f. Manually operate APU start valve lever to motor APU and release hydraulic charge.

g. Open APU compartment access doors.

h. In APU compartment, disconnect pressure line from tee check valve (Figure 7-4-60-1).

i. From inside cabin, disconnect pressure lines from tee check valve.

j. In APU compartment, remove jamnut, washer, and sealing washer. Discard sealing washer.

k. Remove tee check valve.

l. Cap or plug lines if tee check valve is not to be installed now.

7-4-60.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat threads of tee check valve with hydraulic fluid, Item 154 or Item 155, Appendix D.

c. From inside cabin, have assistant hold tee check valve in place (Figure 7-4-60-1). In APU compartment, install washer, sealing washer, and jamnut. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS.**

d. In APU compartment, connect pressure line to tee check valve. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

e. In cabin, connect pressure lines to tee check valve. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**

f. Seal area around tee check valve and helicopter skin with sealing compound, Item 292, Appendix D.

g. Service APU accumulator (PARA 1-3-10).

h. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

i. Make sure area is clean and free of foreign material.

j. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

k. Make sure area is clean and free of foreign material.

l. Close APU compartment access doors.

m. Install APU exhaust plug (PARA 1-6-5).

7-4-61 APU START HYDRAULIC FLEX HOSES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Pint
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-15
- PARA 1-6-16
- PARA 4-2-8
- PARA 7-4-65

7-4-61.1 Replace APU Start Hydraulic Flex Hoses.

7-4-61.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open APU compartment access doors.

c. Use 1-pint container and machinery towels, Item 211, Appendix D, around APU start valve fittings to catch fluid when flex hoses are disconnected.

NOTE

Plug or cap fittings as flex hoses are removed.

d. Disconnect and remove pressure and return flex hoses (Figure 7-4-61-1, Detail A).

7-4-61.1.2. Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Coat threads of APU start valve fittings and APU start motor with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

NOTE

Remove plugs or caps from APU start valve fittings and flex hoses as flex hoses are installed.

c. Connect pressure and return flex hoses to APU start valve fittings (Figure 7-4-61-1, Detail A).

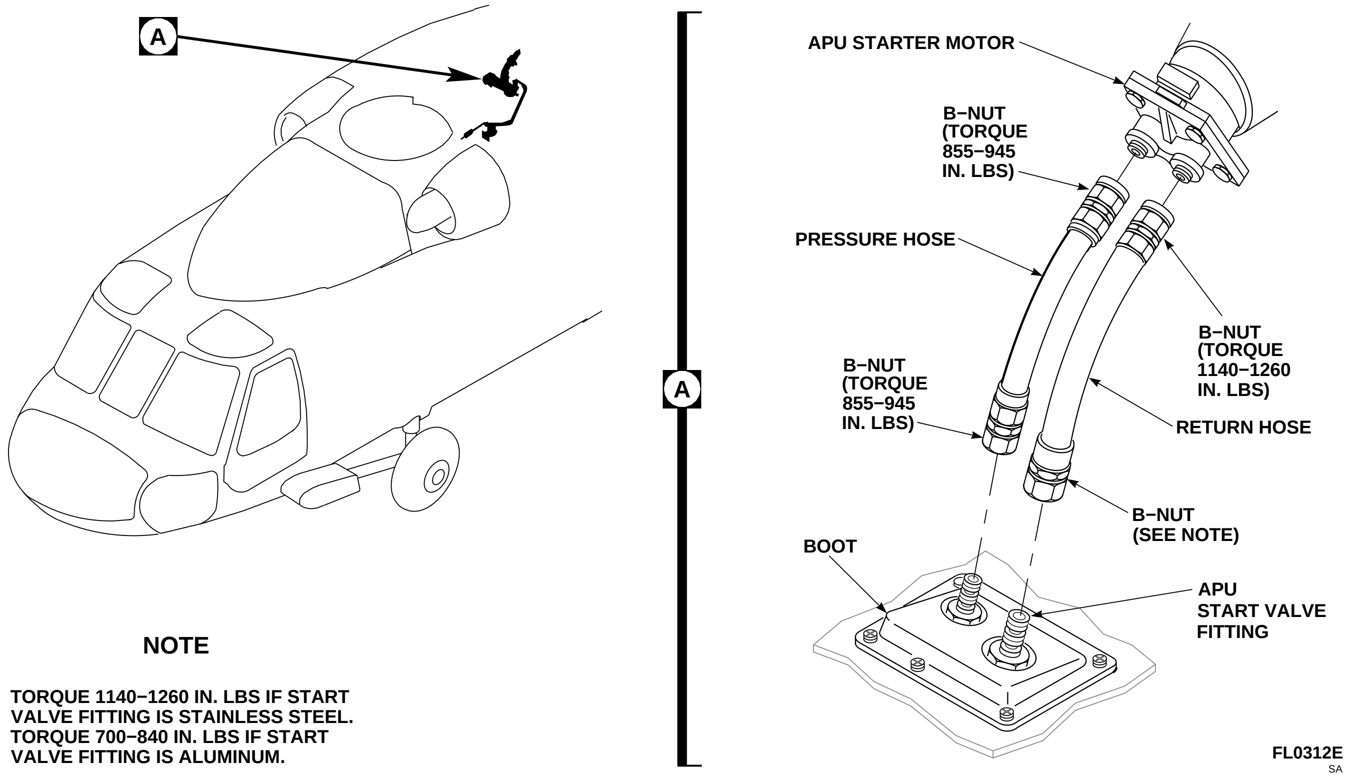


FIGURE 7-4-61-1. REPLACE APU START HYDRAULIC FLEX HOSES

d. Fill flex hoses with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D.

NOTE

Hold APU start valve fittings with wrench while tightening flex hoses on APU start valve fittings.

e. Connect flex hoses to APU start motor as follows:

(1) **TORQUE PRESSURE HOSE B-NUTS TO 855 - 945 INCH-POUNDS.**

(2) **IF APU START VALVE FITTING IS STAINLESS STEEL, TORQUE B-NUTS AT BOTH ENDS OF RETURN FLEX HOSE TO 1140 - 1260 INCH-POUNDS.**

(3) **IF APU START VALVE FITTING IS ALUMINUM, TORQUE RETURN FLEX HOSE B-NUT AT APU START VALVE TO 700 - 840 INCH-POUNDS. TORQUE RETURN FLEX HOSE B-NUT AT APU START MOTOR TO 1140 - 1260 INCH-POUNDS.**

f. Service APU accumulator (PARA 1-3-10).

g. Bleed hydraulic system (PARA 7-4-65).

h. Perform operational check of engine start and ignition system (PARA 4-2-8).

i. Make sure area is clean and free of foreign material.

j. Close APU compartment access doors.

7-4-62 APU START HYDRAULIC RIGID TUBING LINES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 7-2-1
- PARA 7-4-65
- PARA 13-5-3
- PARA 13-5-6

7-4-62.1 Replace APU Start Hydraulic Rigid Tubing Lines.

7-4-62.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. On upper console, make sure APU CONTR switch is OFF.

c. Gain access to fuel cell compartment as follows:

(1) Remove rear passenger's seats (PARA 2-4-45).

(2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).

(3) Remove cargo netting from stowage compartments.

(4) If air conditioning is installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

d. Manually operate APU start valve lever to motor APU and release hydraulic charge.

e. Remove accumulator servicing valve cap on pressure gage (Figure 7-4-62-1, Sheet 1, Detail A).

f. Loosen jamnut on servicing valve and slowly release nitrogen charge.

g. Make sure tape indicator on accumulator and pressure gage each show 0.

NOTE

Plug or cap hoses, ports, and fittings during removal.

h. Disconnect lines at both ends, remove screws, spacers, washers, nuts, and clamps as required. Carefully remove line from fittings (Figure 7-4-62-1, Sheet 1, Detail A and Detail B and Sheet 2, Detail C).

7-4-62.1.2 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

NOTE

Remove caps and plugs from hoses, ports, and fittings during installation.

b. Before installation blow all hoses and fittings clear with dry compressed air.

c. Coat all threads with hydraulic fluid, Item 154 or Item 155, Appendix D, before installation.

d. Connect lines loosely between fittings. Install clamps with screws, spacers, washers, and nuts as required (Figure 7-4-62-1, Sheet 1, Detail A and Detail B and Sheet 2, Detail C). **TORQUE ¼-INCH B-NUTS TO 135 - 145 INCH-POUNDS. TORQUE ½-INCH B-NUTS TO 470 - 510 INCH-POUNDS.**

e. Seal nitrogen gas line elbow going through helicopter structure with sealing compound, Item 292, Appendix D.

f. Service APU accumulator (PARA 1-3-10).

g. Check system for leaks with leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D. **NONE ALLOWED.** Repair or replace all damaged components and repeat leak check as necessary.

h. Bleed hydraulic system (PARA 7-4-65).

i. Perform operational check of hydraulic system (PARA 7-2-1).

j. Make sure area is clean and free of foreign material.

k. Close fuel cell compartment as follows:

(1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

(2) Install cargo netting.

(3) Install soundproofing panels (PARA 2-4-69).

(4) Install rear passenger's seats (PARA 2-4-45).

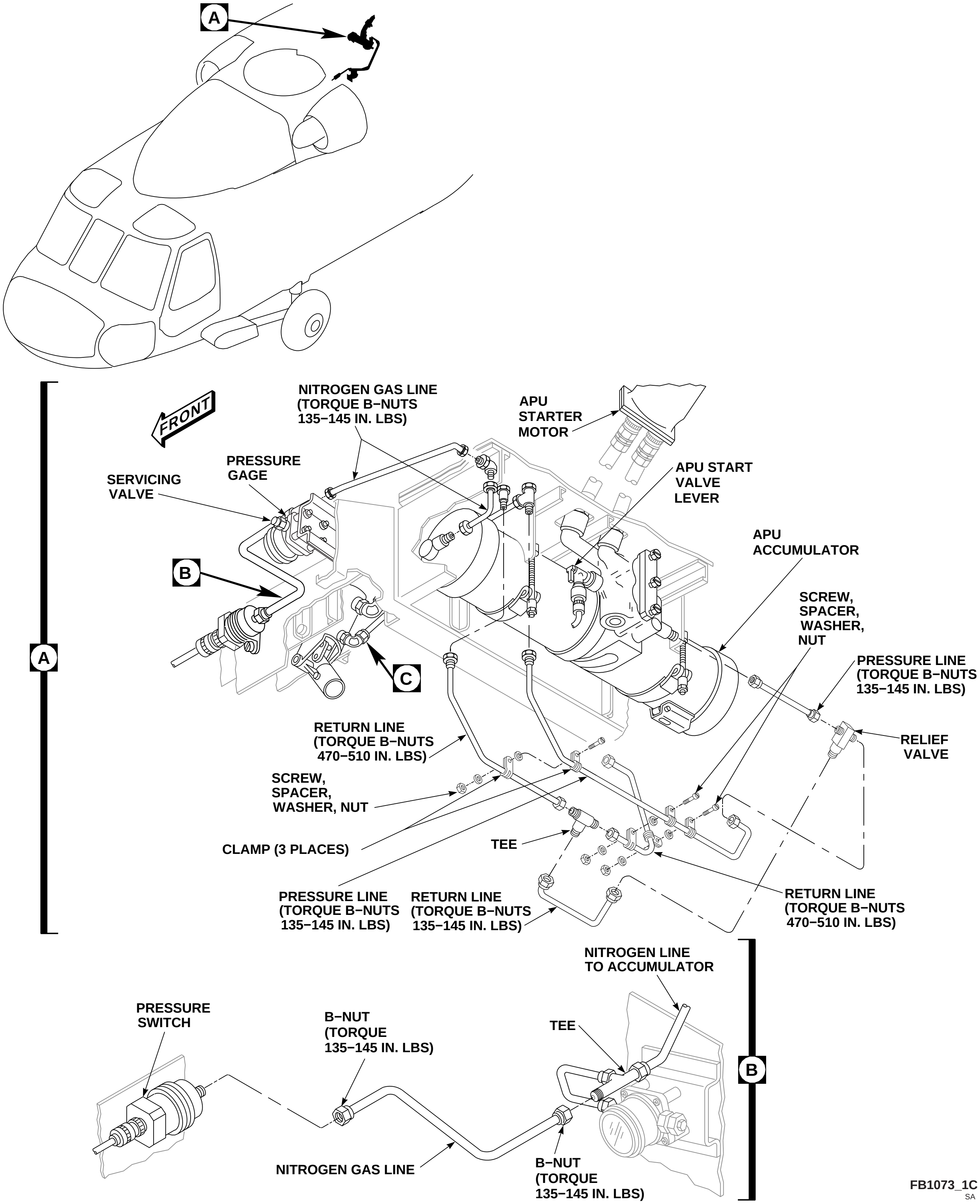


FIGURE 7-4-62-1. REPLACE APU START HYDRAULIC RIGID TUBING LINES (SHEET 1 OF 2)

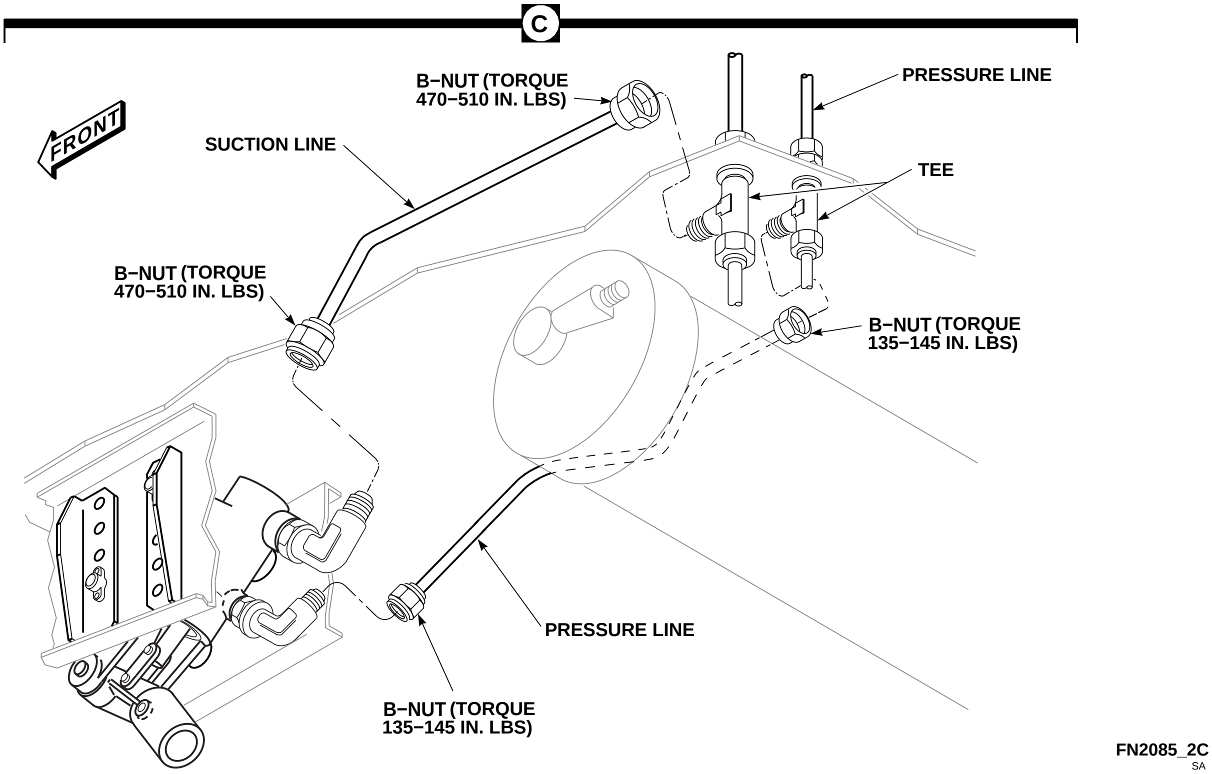


FIGURE 7-4-62-1. REPLACE APU START HYDRAULIC RIGID TUBING LINES (SHEET 2 OF 2)

7-4-63 HYDRAULIC SYSTEM SELECTOR VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 6-4-68
- PARA 7-4-65
- PARA 7-5-2

7-4-63.1 Replace Hydraulic System Selector Valve.

7-4-63.1.1 Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. If main rotor blades are installed, manually turn main rotor head to allow overhead clearance when removing selector valve.

e. Use machinery towels, Item 211, Appendix D, to catch hydraulic fluid as line is disconnected.

NOTE

As parts are removed and disconnected, cap or plug all lines and fittings.

f. Disconnect input line at selector valve bottom elbow. Plug input line quickly so fluid in hydraulic refill handpump will not drain (Figure 7-4-63-1, Sheet 1, Detail A).

g. Disconnect No. 1 output line from check valve, and No. 3 output line from reducer. Disconnect No. 2 output and ground test lines from tee check valve.

h. RTR BRAKE Disconnect No. 4 output line from elbow.

i. Remove screws and washers, and remove selector valve and spacers from bracket.

7-4-63.1.2 Disassemble.

a. **W/O RTR BRAKE** Remove reducer, tee check valve, check valve, and bleeder plug from selector valve. Remove jamnut, packings, and retainers from fittings as required. Discard packings (Figure 7-4-63-1, Sheet 2, Detail B).

b. **RTR BRAKE** Remove reducer, tee check valve, check valve, and elbow from selector valve. Remove jamnuts, packings, and retainers from fittings as required. Discard packings (Figure 7-4-63-1, Sheet 2, Detail C).

c. Remove elbow, jamnut, retainer, and packing from bottom of selector valve. Discard packing.

d. If internal parts of selector valve will be replaced, repair selector valve (PARA 7-5-2).

7-4-63.1.3 Assemble.

a. Using hydraulic fluid, Item 154 or Item 155, Appendix D, coat all threads and packings.

b. Install jamnut, retainer, and packing on elbow, and install elbow in bottom of replacement selector valve (Figure 7-4-63-1, Sheet 2, Detail B or Detail C).

c. Position selector valve on bracket on helicopter and line up elbow to input line (Figure 7-4-63-1, Sheet 1, Detail A). Hold elbow. **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS.** (Figure 7-4-63-1, Sheet 2, Detail B). Remove selector valve from helicopter.

d. Install check valve, and packing in No. 1 port of selector valve. **TORQUE CHECK VALVE TO 155 - 165 INCH-POUNDS.**

e. Install reducer and packing in No. 3 port of selector valve. **TORQUE REDUCER TO 155 - 165 INCH-POUNDS.**

f. Install jamnut, retainer, and packing on tee check valve. Install tee in No. 2 port of selector valve. Do not tighten jamnut.

g. **W/O RTR BRAKE** Install packing on bleeder plug. Install bleeder plug in No. 4 port of selector valve. **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS** (Figure 7-4-63-1, Sheet 2, Detail B).

h. **RTR BRAKE** Install jamnut, retainer, and packing on elbow and install elbow in No. 4 port of selector valve. Position elbow facing up. **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS** (Figure 7-4-63-1, Sheet 2, Detail C).

7-4-63.1.4 Install.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Put selector valve on bracket with No. 4 port facing outboard (Figure 7-4-63-1, Sheet 1, Detail A). Place spacers under valve. Install screws and washers. **TORQUE SCREWS TO 48 - 52 INCH-POUNDS.**

NOTE

As parts are installed and connected, remove caps and plugs from lines and fittings.

c. Connect input line to selector valve bottom elbow. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**

d. Connect No. 1 output line to check valve. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

e. Connect No. 3 output line to reducer. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**

f. Connect No. 2 output and ground test lines to tee check valve. **TORQUE TEE JAMNUT TO 155 - 165 INCH-POUNDS** (Figure 7-4-63-1, Sheet 2, Detail B or Detail C). **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS** (Figure 7-4-63-1, Sheet 1, Detail A).

g. **RTR BRAKE** Connect No. 4 output line to elbow. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**

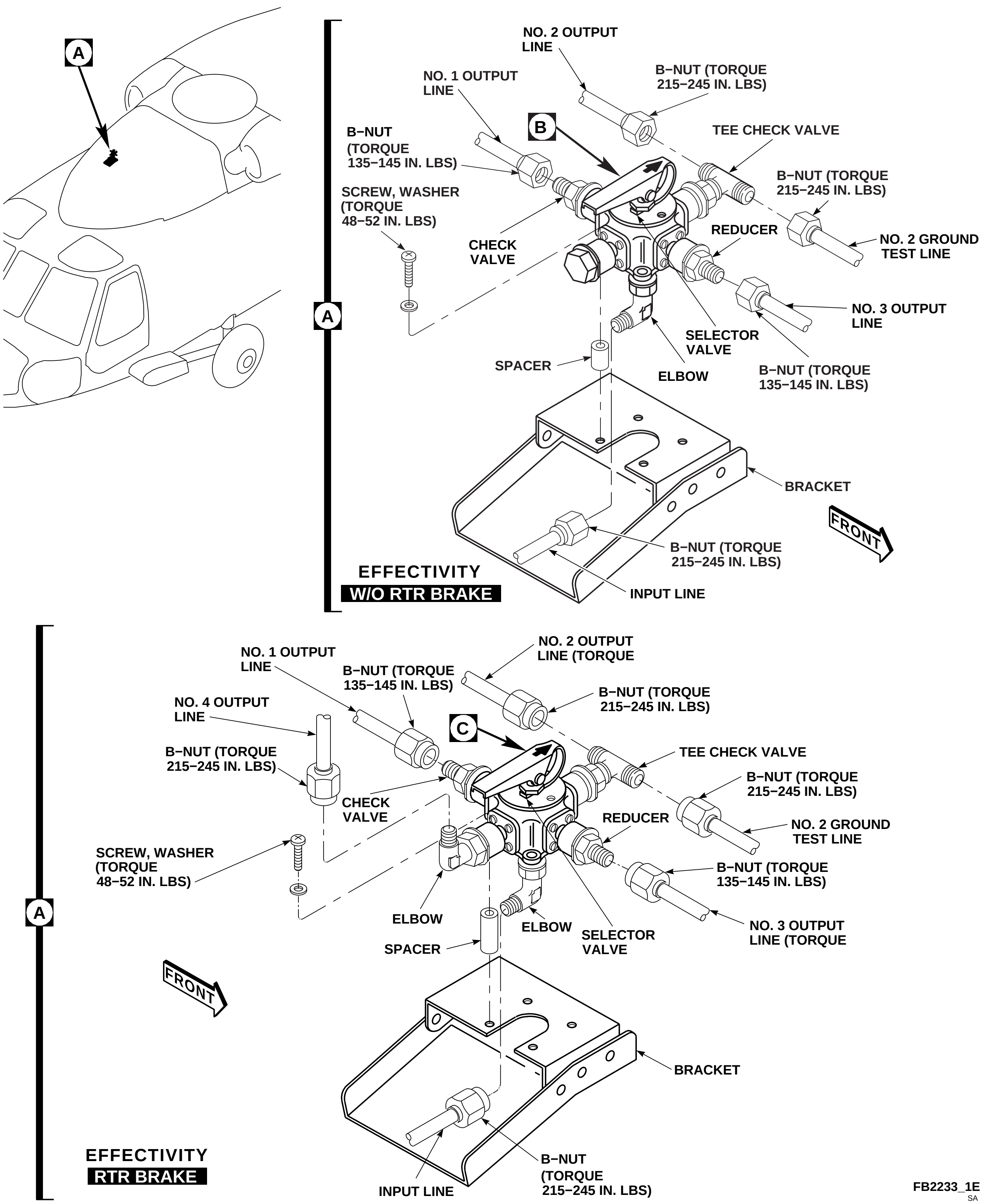


FIGURE 7-4-63-1. REPLACE HYDRAULIC SYSTEM SELECTOR VALVE (SHEET 1 OF 2)

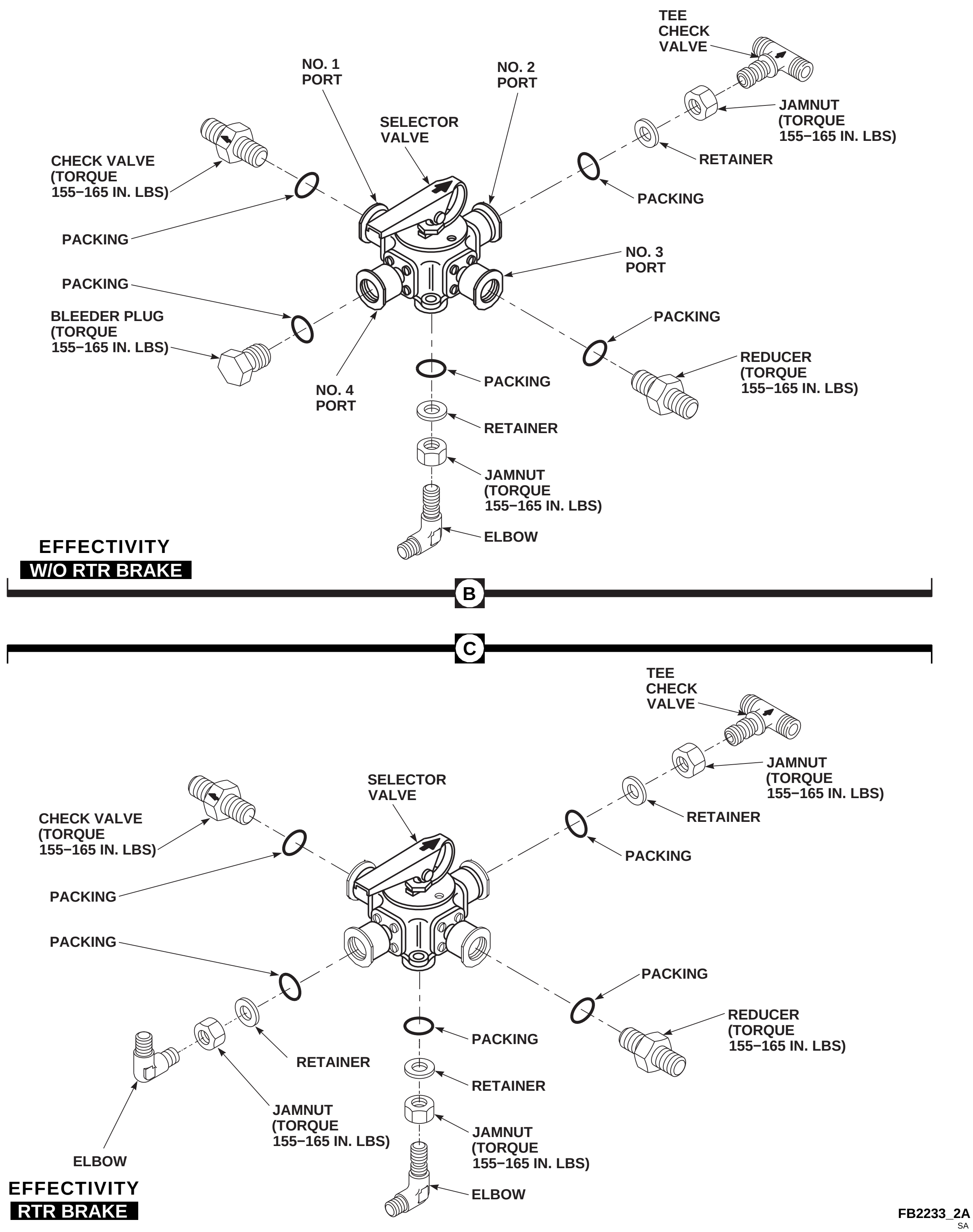


FIGURE 7-4-63-1. REPLACE HYDRAULIC SYSTEM SELECTOR VALVE (SHEET 2 OF 2)

h. Bleed hydraulic system (PARA 6-4-68 and PARA 7-4-65).

i. Check operation of replacement selector valve by servicing each reservoir (PARA 1-3-8).

j. Make sure area is clean and free of foreign material.

k. Close main rotor pylon sliding cover.

7-4-64 HYDRAULIC SYSTEM SELECTOR VALVE FILL LINES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16

7-4-64.1 Replace Hydraulic System Selector Valve Fill Lines.

7-4-64.1.1. Remove.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Open main rotor pylon sliding cover.

c. On upper console, place BACKUP HYD PUMP switch to OFF.

d. Use machinery towel, Item 211, Appendix D, to catch any fluid in line being replaced. Remove component as necessary to access lines and clamps.

NOTE

Install caps and plugs in tubes and fittings after line removal.

e. Loosen B-nuts at both ends of line. Remove clamps or tube block. Remove line (Figure 7-4-64-1, Sheet 1, Detail A, Detail B, and Detail C or Figure 7-4-64-1, Sheet 2, Detail A, Detail B, Detail C, Detail D, and Detail E).

7-4-64.1.2. Install.

NOTE

Remove caps and plugs from tubes and fittings prior to installation.

a. Coat threads of fitting with hydraulic fluid, Item 154 or hydraulic fluid, Item 155, Appendix D, before installing lines.

b. Install replacement line between fittings. Install clamps or tube block (Figure 7-4-64-1, Sheet 1, Detail A, Detail B, and Detail C or Figure 7-4-64-1, Sheet 2, Detail A, Detail B, Detail C,

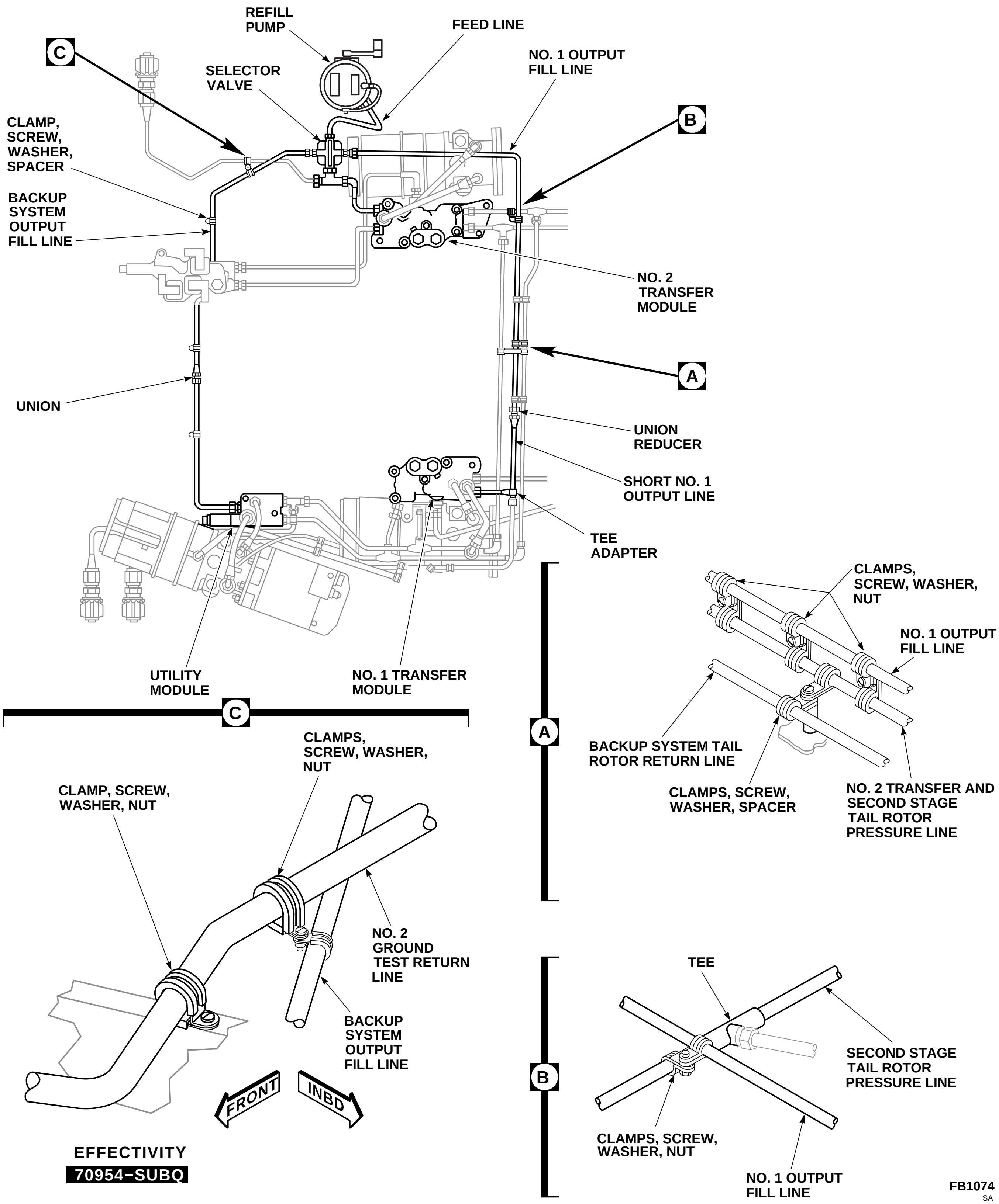


FIGURE 7-4-64-1. REPLACE HYDRAULIC SYSTEM SELECTOR VALVE FILL LINES W/O RTR BRAKE (SHEET 1 OF 2)

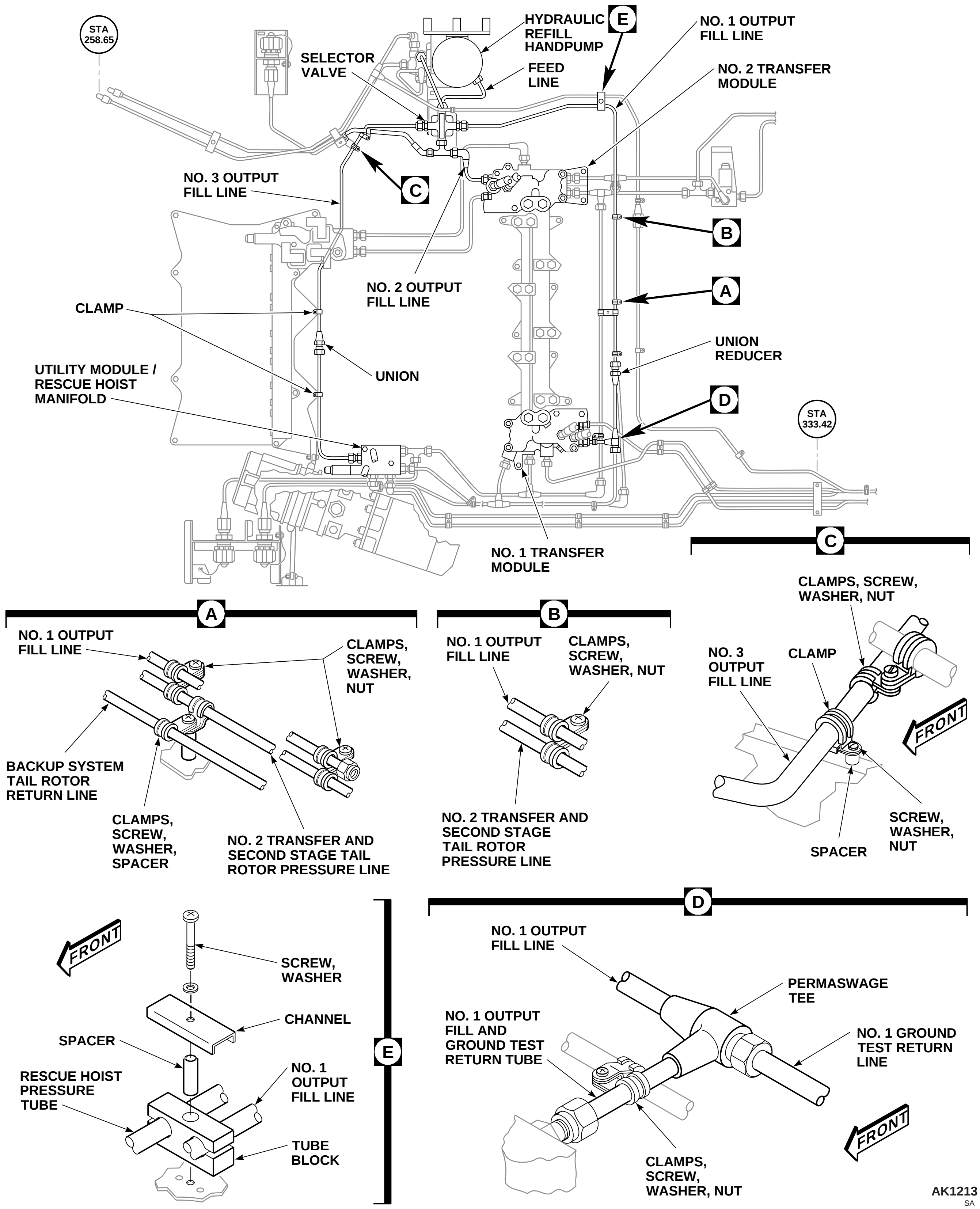


FIGURE 7-4-64-1. REPLACE HYDRAULIC SYSTEM SELECTOR VALVE FILL LINES RTR BRAKE (SHEET 2 OF 2)

Detail D, and Detail E). **TORQUE 1/4-INCH B-NUTS TO 135 - 145 INCH-POUNDS. TORQUE 3/8-INCH B-NUTS TO 215 - 245 INCH-POUNDS.**

c. Seal areas where clamp hardware goes through helicopter skin with sealing compound, Item 292, Appendix D.

d. Replace any disconnected component.

e. Check operation of system by refilling reservoir of system being worked on (PARA 1-3-8).

f. Make sure area is clean and free of foreign material.

g. Close main rotor pylon sliding cover.

7-4-65 BLEED HYDRAULIC SYSTEM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container, 1-Quart
- Plastic Bag

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Machinery Towel, Item 211, Appendix D
- Tiedown Straps, Item 338, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-12
- PARA 11-4-72

7-4-65.1 General.

NOTE

Hydraulic system must be bled after replacing any servos, modules, manifolds, tubing, hoses, quick-disconnects, or any other parts that makes it necessary to open system.

a. Bleed hydraulic system with system operating using external hydraulic power or backup hydraulic system (PARA 1-6-15).

b. To make sure all air is out of system, run system for at least 15 minutes.

CAUTION

Do not use unseparated gas-pressurized hydraulic service equipment to fill hydraulic handpump reservoir. This type of equipment pumps air/gas into reservoir. Air/gas trapped in hydraulic system can cause stiffened/sluggish flight control movement during certain maneuvers.

c. Keep hydraulic pump module reservoirs filled at all times.

7-4-65.2 Bleeding After Replacing Any Servo, Module, Manifold, Tubing, Hose, or Disconnect.

NOTE

- Use backup hydraulic pump module to operate hydraulic systems after replacing any component, except No. 1 and No. 2 hydraulic pump module.
- If No. 1 or No. 2 hydraulic pump module is replaced, refer to PARA 7-4-65.3.

a. Open main rotor pylon sliding cover.

b. Connect external hydraulic power or use backup hydraulic pump module (PARA 1-6-15).

c. Connect external electrical power (PARA 1-6-16), or run APU.

d. If main rotor blades are installed, manually turn main rotor head to position so they will not interfere with bleeding procedure.

CAUTION

Air will enter system if handpump reservoir is not maintained with hydraulic fluid during bleeding. Backup pump module reservoir fluid level must be kept in green area. If reservoir piston goes from green into rear red area, service handpump reservoir (PARA 1-3-8).

e. Position 1-quart container under drain tube next to left main wheel.

f. If external hydraulic power is being used, increase hydraulic pressure to 3000 psi. If backup hydraulic pump module is used, service backup hydraulic pump module reservoir (PARA 1-3-8), if required, prior to placing BACKUP HYD PUMP switch to ON.

g. On collective stick grip, place 1ST STAGE, 2ND STAGE SVO OFF switch in center position (both primary servo stages ON).

h. On lower console MISC SW panel, place TAIL SERVO switch to NORMAL.

i. On AUTO FLIGHT CONTROL panel, engage SAS 1, SAS 2, TRIM, and BOOST switches.

CAUTION

Damage to antiflap cam and antiflap stop attaching bolt will result if cam is not secured in flight position when cycling flight controls with main rotor blades removed. Secure antiflap cam in flight position prior to cycling flight controls.

j. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and secure them to spindle horn with tiedown straps, Item 338, Appendix D.

k. Cycle flight controls, slowly, stop-to-stop to purge air trapped in lines as follows:

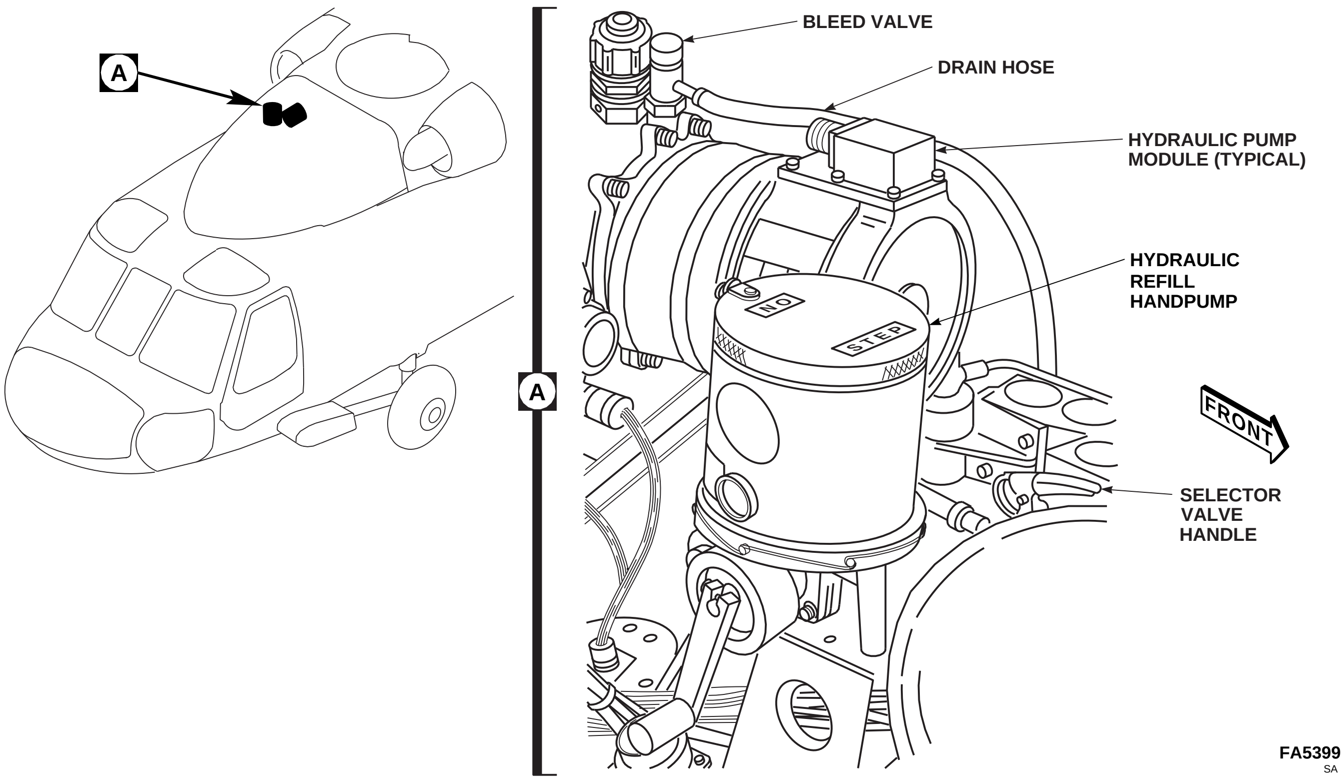


FIGURE 7-4-65-1. BLEEDING AFTER REPLACING ANY SERVO, MODULE, MANIFOLD, TUBING, HOSE, OR DISCONNECT

NOTE

Keep watching fluid level in backup hydraulic pump reservoir while cycling controls.

(1) Perform 20 cycles, low collective, forward left cyclic, and right pedal to high collective, aft right cyclic, and left pedal.

(2) Perform 20 cycles low collective, aft right cyclic, and right pedal to high collective, forward left cyclic, and left pedal.

l. Push down and turn bleed relief valve to open. Keep open until air-free hydraulic fluid flows from valve. Release bleed relief valve (Figure 7-4-65-1, Detail A).

m. Service hydraulic pump module (PARA 1-3-8).

n. Turn off all electrical switches.

o. Disconnect external electrical power (PARA 1-6-16), or shut down APU.

p. If installed, remove tiedown straps from anti-flap cam levers.

q. Remove drained fluid and container from under fuselage.

r. Make sure area is clean and free of foreign material.

s. Close main rotor pylon sliding cover.

7-4-65.3 Hydraulic Pump Module Bleeding.

NOTE

A new hydraulic pump module may have air trapped in it. This air might not bleed out using reservoir bleed relief valve.

a. If hydraulic pump module does not develop pressure after running for 6 seconds, bleed air from pump as follows:

(1) Push down and turn bleed relief valve to open and deplete reservoir static pressure.

(2) Remove both pressure and return quick-disconnect male halves from pump module. Store quick-disconnects in a plastic bag to keep them clean (PARA 7-4-12).

CAUTION

Damage to hydraulic system will result if dirt or foreign materials are allowed to enter pump module. Make sure area is clean and protected from contaminates when bleeding pump module.

(3) Use machinery towels, Item 211, Appendix D, to soak up spilled hydraulic fluid.

(4) Fill both ports with hydraulic fluid, Item 154 or Item 155, Appendix D.

(5) Manually turn rotor head, keeping both ports filled until no more air bubbles come from pressure port.

(6) Install quick-disconnects in hydraulic pump module (PARA 7-4-12).

b. Service hydraulic pump module (PARA 1-3-8).

7-4-65.4 Bleeding System After Maintenance.

a. Open main rotor pylon sliding cover.

b. Operate backup hydraulic pump module.

c. Connect external electrical power (PARA 1-6-16), or run APU (Appropriate Flight Manual).

d. On collective stick grip, place 1ST STAGE, 2ND STAGE SVO OFF switch in center position (both primary servo stages on).

e. On lower console, place TAIL SERVO switch on MISC SW panel to NORMAL.

f. On AUTO FLIGHT CONTROL panel, engage SAS 1, SAS 2, TRIM, and BOOST switches.

CAUTION

Damage to antiflap cam and antiflap stop attaching bolt will result if cam is not secured in flight position when cycling flight controls with main rotor blades removed. Secure antiflap cam in flight position prior to cycling flight controls.

g. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and secure them to spindle horn with tiedown straps, Item 338, Appendix D.

h. Cycle flight controls slowly, stop-to-stop to purge air trapped in lines as follows:

(1) Perform 20 cycles low collective, forward left cyclic, and right pedal to high collective, aft right cyclic, and left pedal.

(2) Perform 20 cycles low collective, aft right cyclic, and right pedal to high collective, forward left cyclic, and left pedal.

i. On lower console, place TAIL SERVO switch on MISC SW panel to BACKUP.

j. Perform 20 cycles low collective right pedal to high collective left pedal.

k. Turn off backup hydraulic pump module.

l. Service backup hydraulic pump module (PARA 1-3-8).

m. Connect external hydraulic power to No. 1 hydraulic system. Loop hoses over engine cowling and main gear box fairing.

NOTE

With pilot assist functions unpowered, cockpit control forces will be increased.

n. Perform 20 cycles low collective, forward left cyclic, right pedal to high collective, aft right cyclic, left pedal.

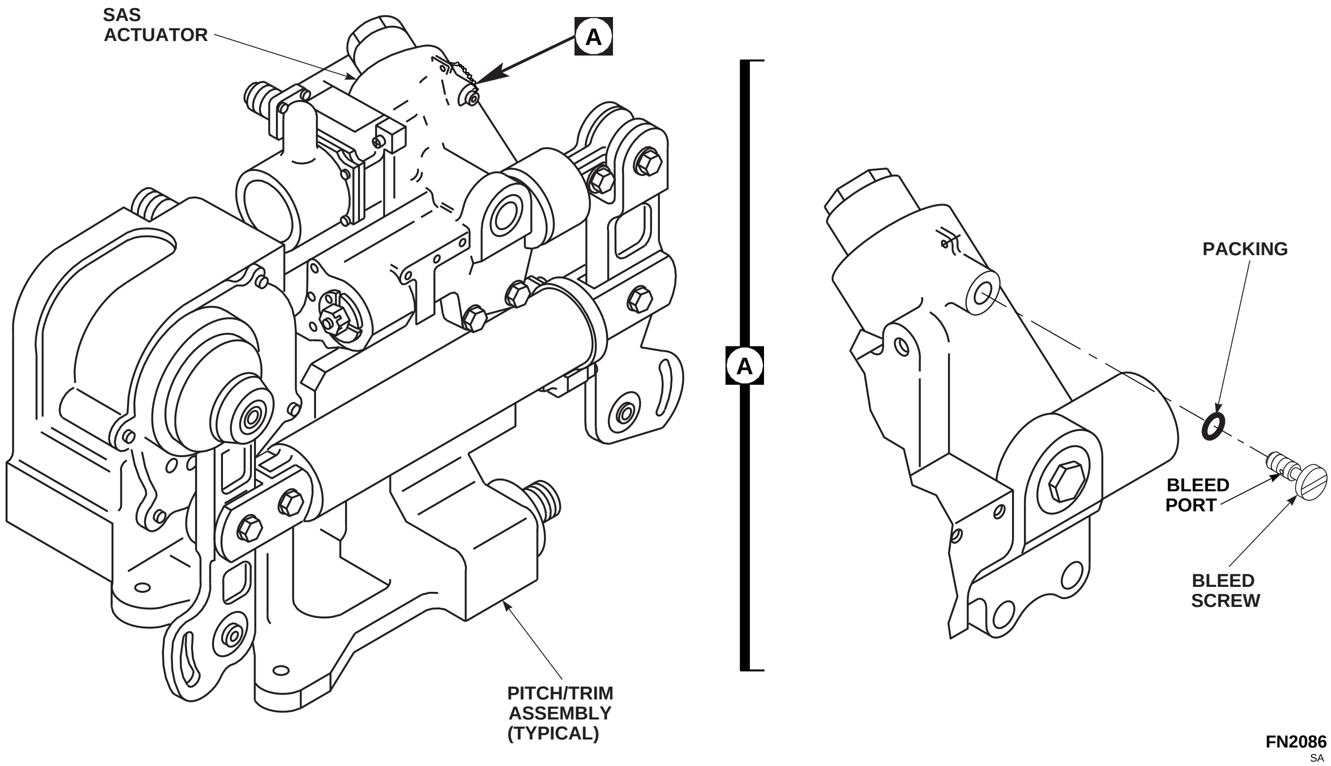


FIGURE 7-4-65-2. BLEEDING SYSTEM AFTER MAINTENANCE

o. Disconnect external hydraulic power from No. 1 hydraulic system.

p. Service No. 1 hydraulic pump module (PARA 1-3-8).

q. Connect external hydraulic power to No. 2 hydraulic system. Loop hoses over engine cowling and main gear box fairing.

r. Perform 20 cycles low collective, forward left cyclic, right pedal to high collective, aft right cyclic, left pedal.

NOTE

- Use machinery towels, Item 211, Appendix D, to catch fluid while bleeding actuators.
- Back out SAS actuator bleed screws just far enough to uncover bleed port drilled in side of screw (Figure 7-4-65-2, Detail A).

s. With No. 2 system still pressurized, bleed each of three SAS actuators at bleed screw.

t. After tightening bleed screw, hold system pressure long enough to see that no hydraulic fluid leaks past bleed screw packing. If bleed screw leaks, replace packing and discard (PARA 11-4-72).

u. Using lockwire, Item 196, Appendix D, lockwire bleed screw.

v. Disconnect external hydraulic power from No. 2 hydraulic system.

w. Service No. 2 hydraulic pump module (PARA 1-3-8).

x. Turn off all electrical switches.

y. Disconnect external electrical power (PARA 1-6-16), or shut down APU (Appropriate Flight Manual).

z. If installed, remove tiedown straps from antiflap cam levers.

aa. Make sure area is clean and free of foreign material.

ab. Close main rotor pylon sliding cover.

7-4-66 FLUSH HYDRAULIC SYSTEM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Container
- Hydraulic Fluid Contamination Analysis Kit
- Portable Hydraulic Test Stand
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- NAS 1638
- PARA 1-3-8
- PARA 1-3-10
- PARA 1-6-16
- PARA 2-4-137
- PARA 7-4-14
- PARA 11-4-90
- PARA 11-4-91
- PARA 11-4-92

7-4-66.1 Flush Hydraulic System.

NOTE

- Flushing is a decontamination process in which contaminated hydraulic fluid is removed from helicopter to maximum extent possible and then discarded.
- Flushing is a means to effectively decontaminate a system found to contain water, other foreign fluids, or fluid that is chemically unacceptable. Contamination of this type cannot be remedied by recirculation cleaning.
- Severe cases of particulate contamination, such as those resulting from major component failures, may be more easily corrected by flushing than by recirculation cleaning.
- Portable hydraulic test stand must be maintained to test at NAS 1638, Class 7 (or cleaner) cleanliness level before it can be used on helicopter.
- A contamination analysis must be performed on test stand before connecting it to helicopter.

a. Open controls/accessories access.

b. Unscrew lid from hydraulic refill handpump. Take fluid sample from handpump. Screw lid on handpump tightly.

c. Analyze fluid sample using hydraulic fluid contamination analysis kit.

d. If contamination level is acceptable, proceed to step e. If contamination level is not acceptable, continue flushing procedures as follows:

(1) Position container to catch hydraulic fluid. Empty as required between steps.

(2) Rotate refill selector valve handle to OUT 3 position (Figure 7-4-66-1, Sheet 1, Detail A).

(3) Crank pump handle and pump refill handpump dry (Figure 7-4-66-1, Sheet 1, Detail B).

(4) Press in and rotate bleed relief valve on No. 1 pump module. Allow reservoir to drain.

(5) Press in and rotate bleed relief valve on No. 2 pump module. Allow reservoir to drain.

(6) Press in and rotate bleed relief valve on backup pump module. Allow reservoir to drain.

(7) Close bleed relief valves.

(8) Replace filter element in hydraulic refill handpump (PARA 7-4-14).

NOTE

Performing of step d. (9) through step d. (14) will purge refill handpump to transfer and utility modules.

(9) Unscrew lid from refill handpump.

(10) Fill reservoir to top with clean hydraulic fluid, Item 155, Appendix D.

(11) Make sure lid is clean. **SCREW LID TIGHTLY ONTO REFILL HANDPUMP.**

(12) Crank pump handle and fill each pump module until each reservoir piston moves ½-inch.

(13) Continue to fill each pump module until hydraulic fluid flows from reservoir overboard drain.

(14) Repeat step d. (9) through step d. (13) five times.

CAUTION

During installation of filter elements, make sure filter bowls are filled with clean hydraulic fluid.

e. Replace filter elements in No. 1, No. 2, and backup pump modules (PARA 7-4-14).

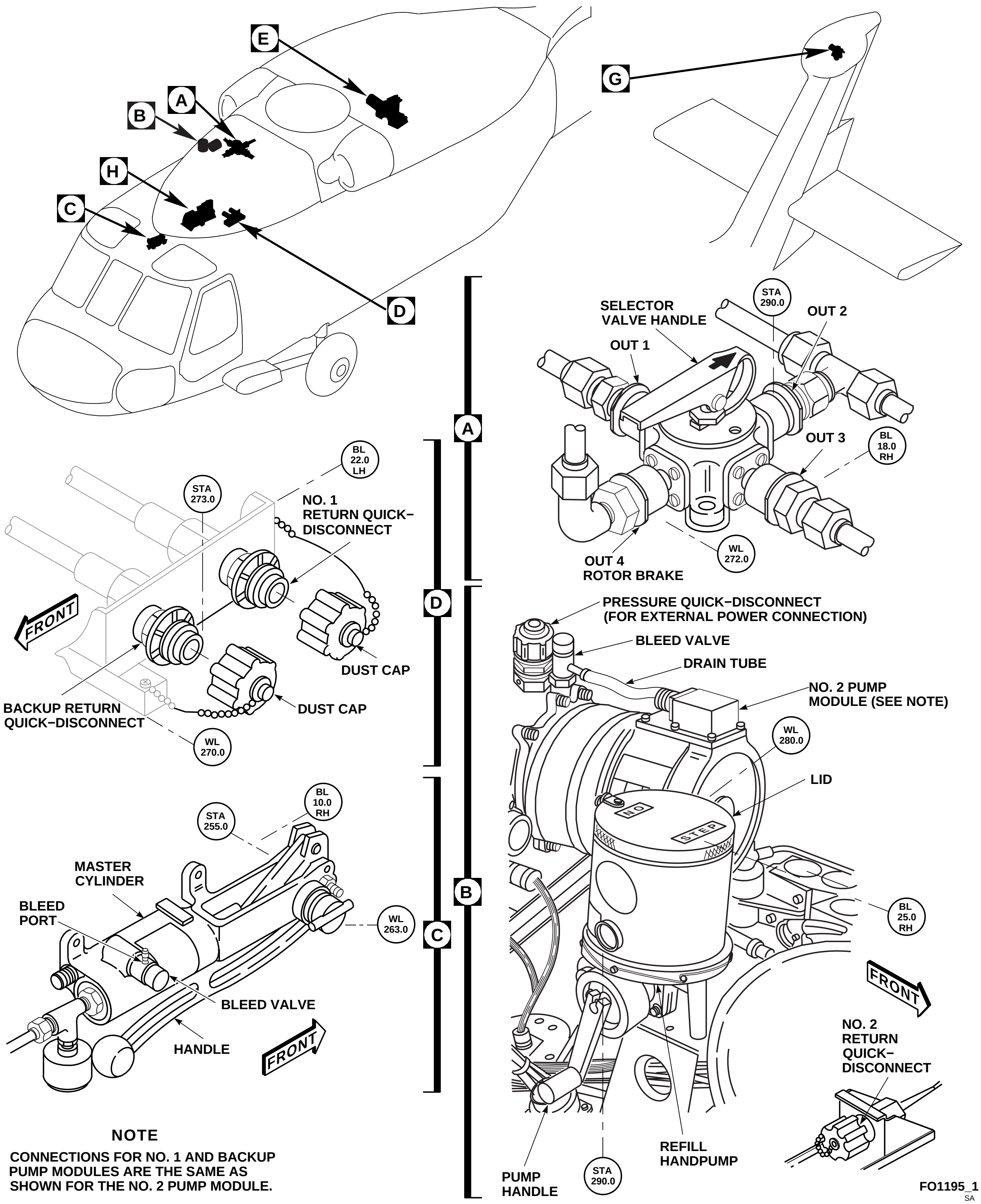


FIGURE 7-4-66-1. FLUSH HYDRAULIC SYSTEM (SHEET 1 OF 3)

f. Connect test stand to pressure quick-disconnect on backup pump module.

g. Connect drain line to backup return quick-disconnect (Figure 7-4-66-1, Sheet 1, Detail D).

h. Place open end of drain line into container.

i. Release nitrogen pressure from auxiliary power unit (APU) accumulator (PARA 1-3-10).

j. Disconnect return line from APU handpump (Figure 7-4-66-1, Sheet 2, Detail E).

k. Connect bleed line to handpump return line. Place open end of bleed line into container.

l. Position APU start valve lever to OPEN (Figure 7-4-66-1, Sheet 2, Detail F).

m. Slowly increase test stand pressure to 500 psi.

n. Fill accumulator until piston position indicator reads 100% (Figure 7-4-66-1, Sheet 2, Detail E).

o. Watch for air-free flow of hydraulic fluid from bleed line; then CLOSE APU start valve lever (Figure 7-4-66-1, Sheet 2, Detail F).

p. Disconnect bleed line from APU return line (Figure 7-4-66-1, Sheet 2, Detail E).

q. Connect return line to handpump. **TORQUE NUT TO 365 - 515 INCH-POUNDS.**

r. Reduce test stand pressure to 0 psi.

s. Connect nitrogen pressure to accumulator (PARA 1-3-10).

t. Hold APU start valve lever in OPEN position (Figure 7-4-66-1, Sheet 2, Detail F).

CAUTION

The following step will motor APU, fill accumulator reservoir, and bottom out piston.

u. Slowly increase nitrogen pressure until piston bottoms out and piston position indicator reads 0 psi (Figure 7-4-66-1, Sheet 2, Detail E).

v. Service accumulator with nitrogen for outside ambient temperature; then remove nitrogen source (PARA 1-3-10).

CAUTION

If blades are in folded position and hydraulic power is applied, flight controls must not be moved with control rods connected or damage to flight control system will occur.

w. If blades are folded, perform the following:

(1) Remove aft control rod (PARA 11-4-91).

(2) Disconnect forward control rod from servo (PARA 11-4-90).

(3) Disconnect lateral control rod from servo (PARA 11-4-92).

(4) Using lockwire, Item 196, Appendix D, secure control rods in up position to allow free travel of servos without damage to control rods.

x. Disconnect test stand from backup pump module.

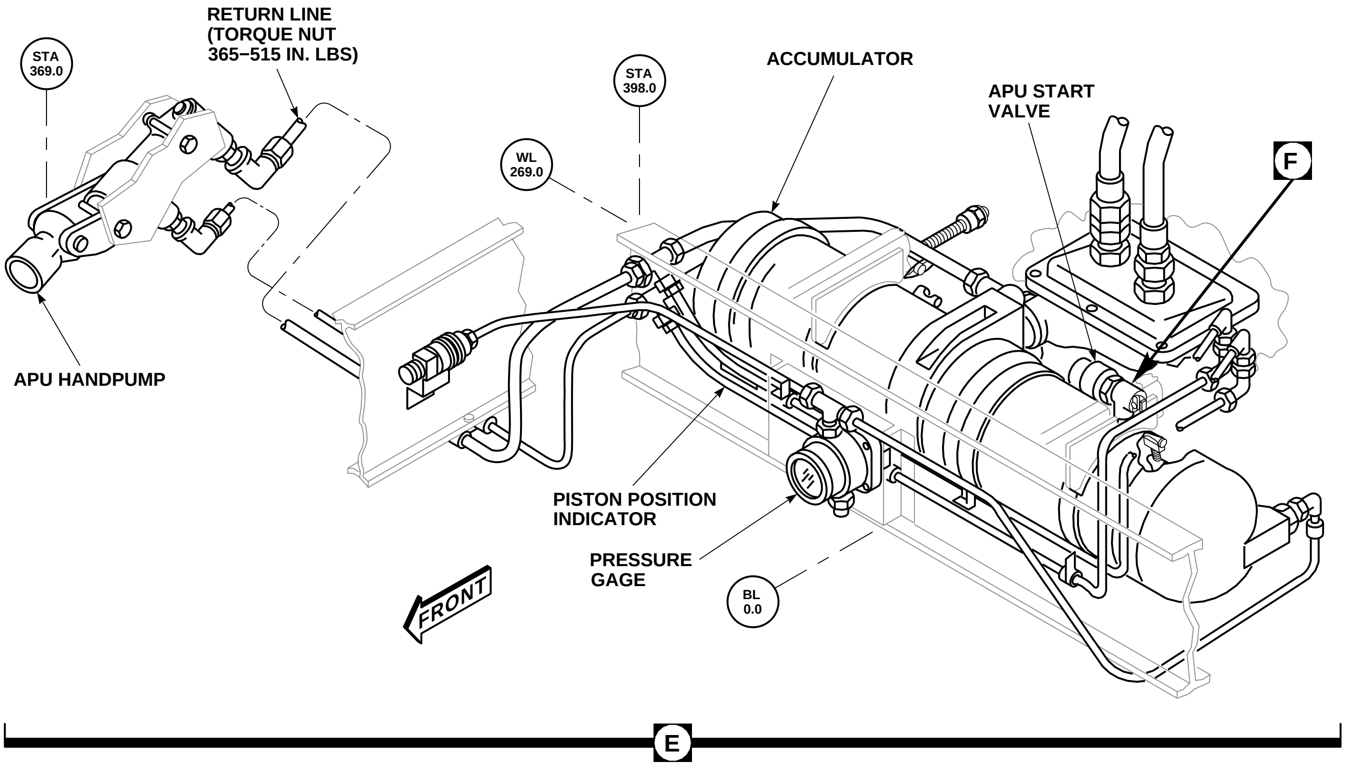
y. Connect test stand to pressure quick-disconnect on No. 1 pump module (Figure 7-4-66-1, Sheet 1, Detail B).

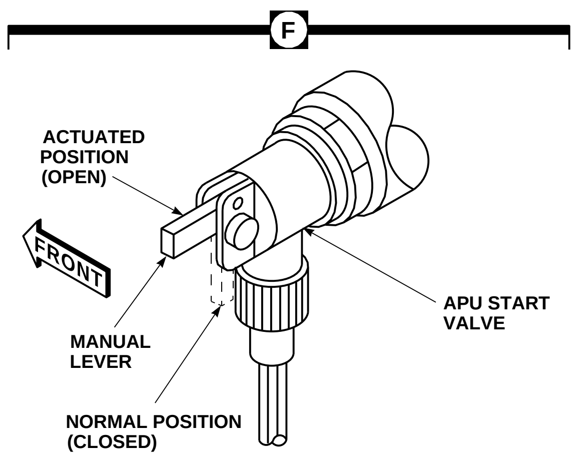
z. Connect drain line to No. 1 return quick-disconnect (Figure 7-4-66-1, Sheet 1, Detail D).

aa. Place open end of line into container.

ab. Turn on all electrical power (PARA 1-6-16).

ac. On pilot's and copilot's collective stick grip, SERVO switch must be at center position (both 1ST and 2ND primary servo stages on).





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SA

FIGURE 7-4-66-1. FLUSH HYDRAULIC SYSTEM (SHEET 2 OF 3) ad. On lower console, place TAIL SERVO switch on MISC SW panel to NORMAL.

ae. On AUTO FLIGHT CONTROL panel, engage SAS 1, SAS 2, TRIM, BOOST, and POWER ON RESET switches.

CAUTION

- To prevent cavitation and possible failure of test stand, make sure clean hydraulic fluid is added to test stand when fluid level becomes low (about ½ full).
- When hydraulic system is pressurized, periodically monitor indicator button on pump module filters and test stand filters.
- Make sure return line from backup return quick-disconnect is still in container.

af. Slowly increase test stand pressure to 3,000 psi.

ag. Slowly move flight controls through 30 cycles as shown in Table 7-4-66-1. Controls must move smoothly through entire sequence.

Table 7-4-66-1. Repeat Cycle 30 Times

Lo Side	Hi Side
Lo Coll/Aft Cyc	Hi Coll/Fwd Cyc
Lo Coll/Fwd Cyc	Hi Coll/Aft Cyc
Lo Coll/L Cyc	Hi Coll/R Cyc
Lo Coll/R Cyc	Hi Coll/L Cyc
Lo Coll/R Pedal	Hi Coll/L Pedal
Lo Coll/L Pedal	Hi Coll/R Pedal

ah. If necessary, reduce hydraulic pressure to 0 psi and perform the following:

(1) Service test stand, as required.

(2) Discard contaminated hydraulic fluid or replace container.

(3) Replace filter element on No. 1 pump module if indicator pin is extended to indicate a clogged filter (PARA 7-4-14).

ai. On lower console, place TAIL SERVO switch on MISC SW panel to BACKUP.

aj. Increase hydraulic pressure to 3,000 psi.

ak. Remove tail gear box fairing (PARA 2-4-137).

al. Move collective stick to midposition and hold.

am. Slowly cycle yaw pedals through 14 cycles.

NOTE

During last cycling of yaw pedals, loosening of nuts on each tail rotor servo manifold return hose will remove air from return lines.

an. Slightly loosen nuts on 1st and 2nd stage tail rotor servo manifold return hoses (Figure 7-4-66-1, Sheet 3, Detail G).

ao. Slowly cycle yaw pedals through one cycle.

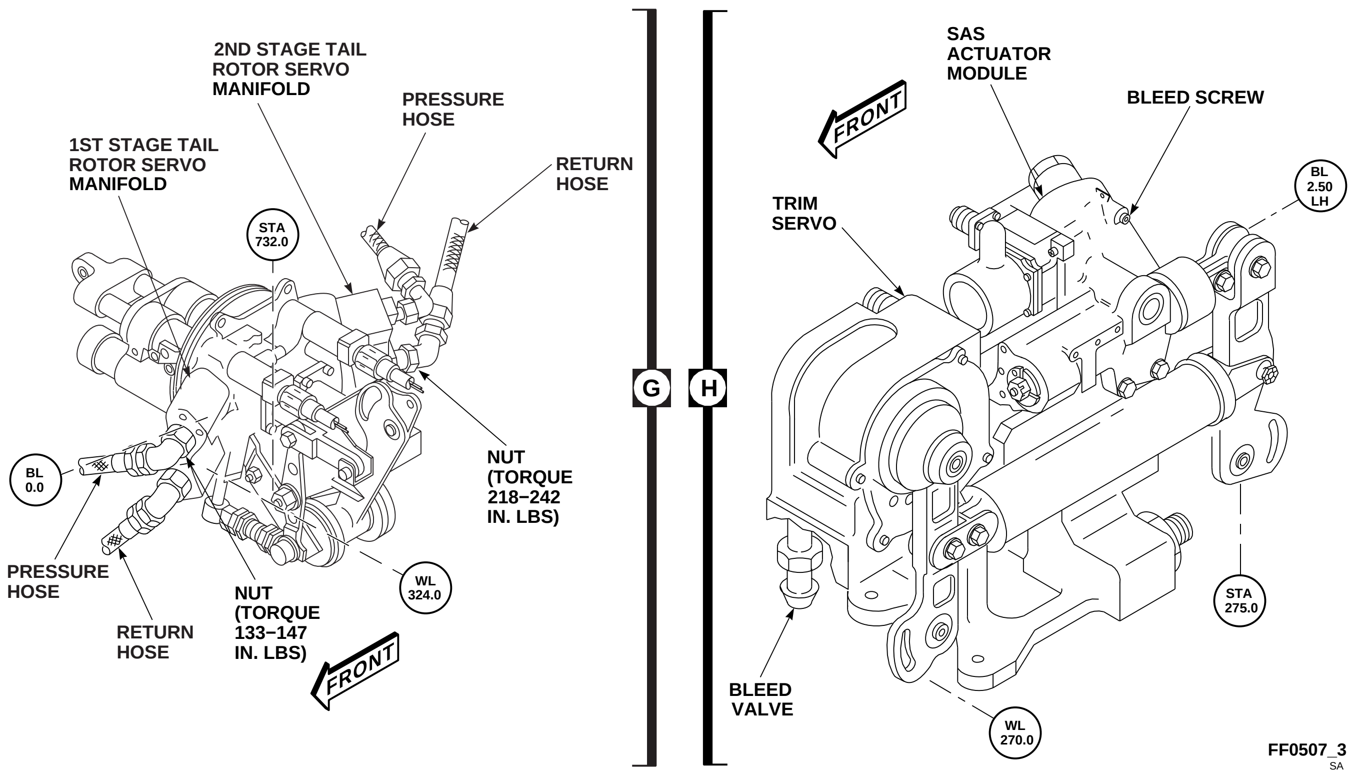


FIGURE 7-4-66-1. FLUSH HYDRAULIC SYSTEM (SHEET 3 OF 3)

ap. **TORQUE NUTS ON RETURN HOSES TO 218 - 242 INCH-POUNDS.**

aq. Move collective stick down.

ar. Reduce hydraulic pressure to 0 psi.

as. Connect test stand to pressure quick-disconnect on No. 2 pump module (Figure 7-4-66-1, Sheet 1, Detail B).

at. Connect drain line to No. 2 return quick-disconnect and place drain line open end in container.

au. Repeat steps af. through ar.

av. Connect test stand to pressure quick-disconnect on backup pump module.

aw. Connect drain line to backup return quick-disconnect (Figure 7-4-66-1, Sheet 1, Detail D) and place open end in container.

ax. Repeat step af. through step ar.

ay. Bleed SAS actuator modules as follows (Figure 7-4-66-1, Sheet 3, Detail H):

(1) Turn on all electrical power (PARA 1-6-16).

(2) On lower console, place BACKUP HYD PMP switch on MISC SW panel to ON.

(3) On AUTO FLIGHT CONTROL panel, engage SAS 1, SAS 2, and BOOST switches.

(4) Position container to catch hydraulic fluid. Empty as required between steps.

(5) Slowly loosen bleed screw until hydraulic fluid and/or air bubbles flow from bleed port.

(6) Watch for an air-free flow of hydraulic fluid from bleed port.

(7) **TIGHTEN BLEED SCREW.**

(8) Using lockwire, Item 196, Appendix D, lockwire bleed screw to SAS actuator module.

az. Repeat step ay. for each SAS actuator module.

ba. Repeat SAS and BOOST ON procedure (step ay.) and listen for ″click″ or ″clicks″ of locking pins retracting in SAS actuator modules.

bb. If all three clicks are completed within an elapsed time of one second, helicopter is ready for flight.

bc. If clicks take longer than one second, repeat SAS and BOOST ON procedure (step ay.) and locate defective SAS actuator. If SAS actuator is identified as a problem area, replace only that assembly.

bd. Repeat SAS and BOOST ON procedure (step ay.) until ″click″ or ″clicks″ is verified within one second.

be. Bleed off about 20 cc of hydraulic fluid from pitch and roll trim servo bleed relief valves.

bf. Unscrew lid from hydraulic refill handpump. Take fluid sample from handpump. **SCREW LID TIGHT ONTO REFILL HANDPUMP.**

bg. Using hydraulic fluid contamination analysis kit, analyze fluid sample.

bh. If contamination level is not acceptable, repeat flushing procedures.

bi. If contamination level is acceptable, perform the following:

(1) Replace filter element in hydraulic refill handpump (PARA 7-4-14).

(2) Replace filter elements in No. 1, No. 2, and backup pump modules (PARA 7-4-14).

bj. On lower console, place TAIL SERVO switch on MISC SW panel to NORMAL.

bk. On AUTO FLIGHT CONTROL panel, disengage SAS 1, SAS 2, TRIM, BOOST, and POWER ON RESET switches.

bl. Turn off all electrical power (PARA 1-6-16).

bm. Disconnect test stand from helicopter.

bn. Disconnect return line from backup pump module and return quick-disconnect.

bo. Make sure dust caps are installed on all quick-disconnects.

bp. Service No. 1, No. 2, and backup systems with hydraulic fluid (PARA 1-3-8).

bq. If required, perform the following:

(1) Install aft control rod (PARA 11-4-91).

(2) Connect forward control rod to servo (PARA 11-4-90).

(3) Connect lateral control rod to servo (PARA 11-4-92).

br. Make sure area is clean and free of foreign material.

bs. Close controls/accessories access.

bt. Install tail gear box fairing (PARA 2-4-137).

SECTION 7-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

7-5-1 BLEED-AIR SHUTOFF VALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Rubber Mallet
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise

Materials/Parts

- Adhesive, Item 53, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 193, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

7-5-1.1 Clean/Repair Bleed-Air Shutoff Valve (BENCH).

7-5-1.1.1 Disassemble.

a. Remove screws, washers, and manifold assembly from housing. If necessary, tap lightly with rubber mallet to separate pieces (Figure 7-5-1-1).

b. Remove flange from housing.

c. Remove self-locking nut and shim from shouldered shaft. It may be necessary to bend shim back away from nut to remove nut.

d. Remove directional sleeve and valve coupling as an assembly from housing. Separate valve coupling from directional sleeve.

e. Remove spring from housing.

f. Remove nut, shim, valve seal, and shouldered shaft from housing. It may be necessary to bend shim back away from nut to remove nut.

g. Remove cylinder guide shaft and valve seal from shouldered shaft.

h. Using dry-cleaning solvent, Item 124, Appendix D, and machinery towels, Item 211, Appendix D, clean all parts and housing.

i. Check seals on valve coupling for damage. If seals are damaged, perform the following:

(1) Remove seals and plate spacer from valve coupling. Discard seals.

(2) Break new seals at one place only, by gently bending each using index finger and thumb of each hand.

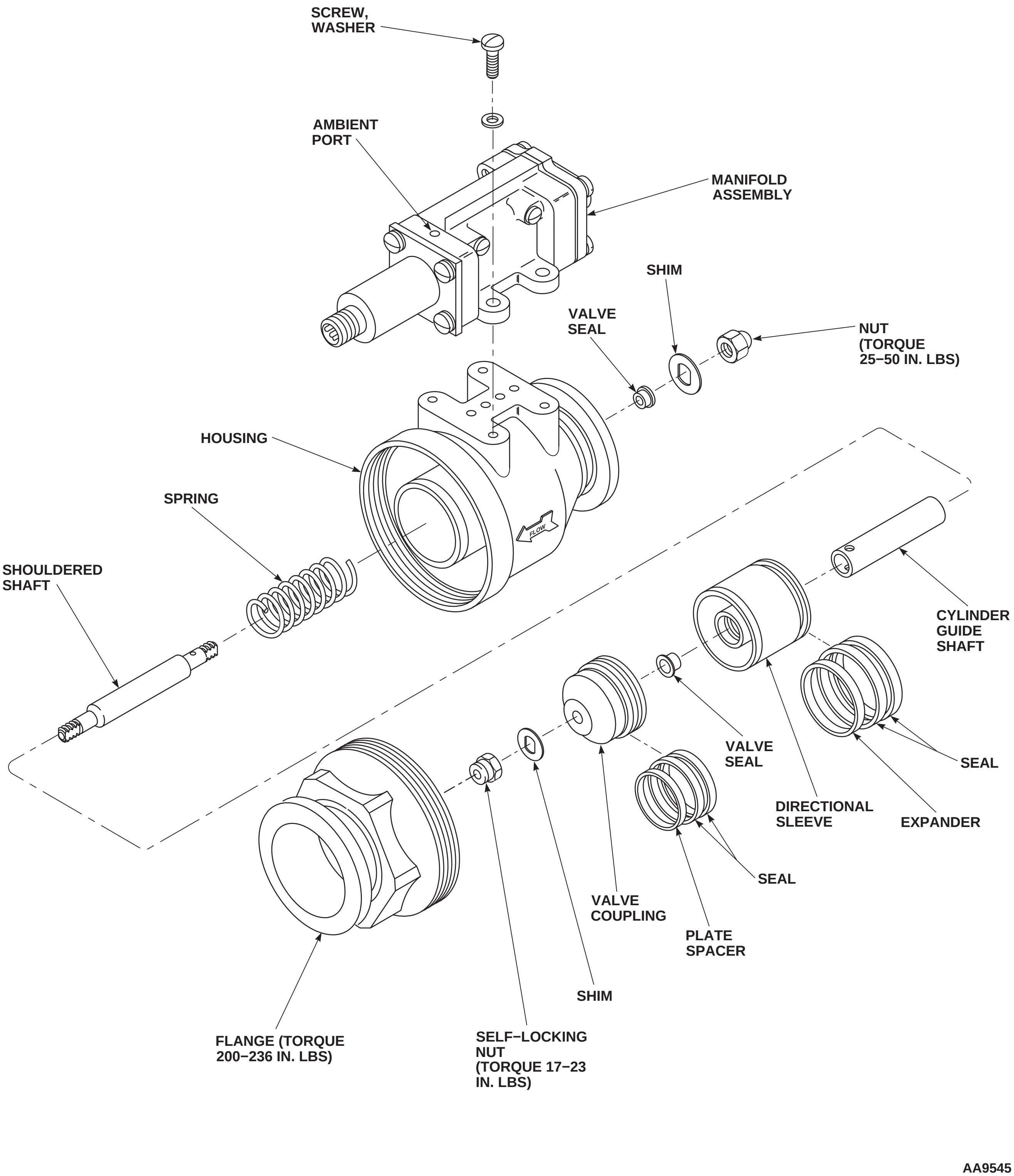


FIGURE 7-5-1-1. CLEAN/REPAIR BLEED-AIR SHUTOFF VALVE (BENCH)

(3) Install plate spacer and seals on valve coupling and orient broken gaps of seals 180° apart to minimize leakage.

j. Check seals on directional sleeve for damage. If seals are damaged, perform the following:

(1) Remove seals and expander from directional sleeve. Discard seals.

(2) Break new seals at one place only, by gently bending each using index finger and thumb of each hand.

(3) Install expander and seals on directional sleeve and orient broken gaps of seals 180° apart to minimize leakage.

k. Using dry-cleaning solvent, Item 124, Appendix D, clean and flush all ports and passages of housing and manifold. Flush out ambient port in manifold by spraying dry-cleaning solvent in and then dumping out.

7-5-1.1.2 Assemble.

a. Using adhesive, Item 53, Appendix D, coat inlet end of shouldered shaft. Insert shaft into housing.

b. Install valve seal, shim, and nut onto shaft. Hold opposite end of shaft in vise to keep shaft from moving and tighten nut. **TORQUE NUT TO 25 - 50 INCH-POUNDS.** Secure nut in place by bending one side of shim against nut and other side against shaft (Figure 7-5-1-1).

c. Using adhesive, Item 53, Appendix D, coat outlet end of shouldered shaft.

d. Install cylinder guide shaft onto shouldered shaft. Install spring on cylinder guide shaft.

e. Install valve seal into directional sleeve.

f. Compress seals on valve coupling and insert valve coupling into directional sleeve.

g. Compress seals on directional sleeve and insert directional sleeve and valve coupling assembly into housing.

h. Install shim and self-locking nut on shouldered shaft. **TORQUE SELF-LOCKING NUT TO 17 - 23 INCH-POUNDS.** Secure self-locking nut in place by bending one side of shim against nut.

NOTE

Do not allow sealant within ⅛-inch of any air passage.

i. Apply adhesive, Item 53, Appendix D, to threads of flange and install in housing. **TORQUE FLANGE TO 200 - 236 INCH-POUNDS.**

j. Apply adhesive, Item 53, Appendix D, to mating surface of housing and manifold.

k. Install manifold assembly on housing with screws and washers. Using lockwire, Item 193, Appendix D, lockwire screws.

7-5-2 SELECTOR VALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Hydraulic Test Stand
- Plug
- Protection, Eye, Skin, and Breathing
- Spring Scale

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 7-4-63

7-5-2.1 Repair Selector Valve (BENCH).

7-5-2.1.1 Disassemble.

a. Remove screws, and remove port fitting and packing from selector valve body. Discard packing. Remove spring from port fitting (Figure 7-5-2-1).

b. Pull shear seal from selector valve body, and remove backup rings and packing from shear seal. Discard packing.

c. Repeat step a. and step b. for other ports.

NOTE

- While removing nut, hold detent down, as it is under spring pressure from detent ball and spring.
- To prevent losing ball, be ready to catch it when removing detent disc.

d. Loosen nut securing handle and detent disc to rotor.

e. Remove clevis pin, washer, and spacers. Remove nut, and remove handle from poppet valve and rotor.

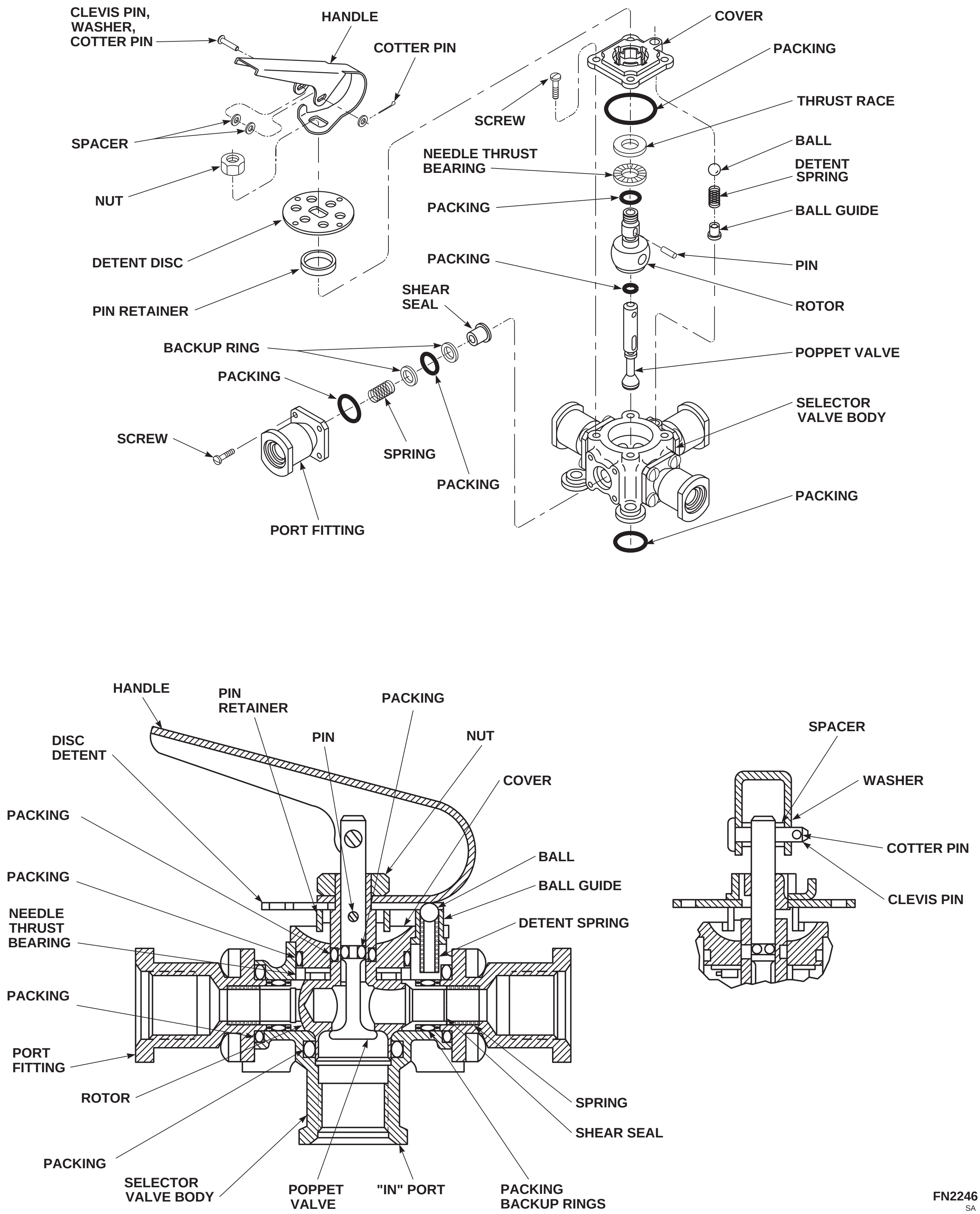


FIGURE 7-5-2-1. REPAIR SELECTOR VALVE (BENCH)

f. Remove detent disc, ball, pin retainer, and pin from rotor shaft.

g. Remove screws and cover from valve body, and at the same time, remove detent spring from selector valve body.

h. Remove ball guide and packing from cover. Discard packing.

i. Pull rotor, needle thrust bearing, thrust race, and poppet valve from selector valve body. Remove thrust bearing and thrust race from rotor. Remove packing from rotor. Discard packing.

j. Push poppet valve from rotor. Remove packing from poppet. Discard packing.

k. Remove packing from selector valve body. Discard packing.

7-5-2.1.2 Inspect.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

a. Clean all metal parts in dry-cleaning solvent, Item 124, Appendix D. Blow dry using compressed air.

b. Inspect selector valve body for cracks, internal scratches, and damaged threads. **NONE ALLOWED.** If crack, internal scratch, or damaged thread is found, replace selector valve body.

c. Inspect selector valve body for scoring and corrosion. **NONE ALLOWED.** If necessary, repair using abrasive cloth, Item 5, Appendix D, as follows:

(1) **BLEND SCORING AND CORROSION UP TO 0.010-INCH DEEP.** If damage exceeds limit, replace selector valve body.

(2) Polish repair smooth using abrasive cloth, Item 10, Appendix D.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to reworked area.

(4) Touch up paint (PARA 2-6-5).

d. Inspect detent spring as follows:

(1) Measure free length. **FREE LENGTH MUST BE 0.594-INCH.** If spring does not meet limit, replace detent spring.

(2) Using spring scale, measure spring loaded to 2.7 - 3.3 pounds. **LENGTH MUST BE 0.439-INCH.** If spring does not meet limit, replace detent spring.

(3) Using spring scale, measure spring loaded to 3.47 - 4.25 pounds. **LENGTH MUST BE 0.394-INCH.** If spring does not meet limit, replace detent spring.

(4) Compress spring to 0.375-inch. Release spring. Measure free length for loss. **NONE ALLOWED.** If spring does not meet limit, replace detent spring.

e. Inspect springs for corrosion, bends, or breaks. **NONE ALLOWED.** If corroded, bent, or broken, replace springs.

f. Inspect port fittings for cracks, corrosion, or damaged threads. **NONE ALLOWED.** If crack, corrosion, or damaged thread is found, replace port fittings.

g. Inspect all metal parts for scratches, scoring, rough edges, out-of-round, and corrosion. **NONE ALLOWED.** If scratched, scored, rough-edged, out-of-round, or corroded, replace part.

7-5-2.1.3 Assemble.

a. Coat all parts using hydraulic fluid, Item 154 or Item 155, Appendix D, as they are assembled.

b. Put new packing in the ″IN″ port of selector valve body (Figure 7-5-2-1).

c. Put new packing on rotor.

d. Install new packing on poppet valve. Install poppet in rotor through inlet port.

e. Install rotor in selector valve body.

f. Install needle thrust bearing and thrust race on rotor.

g. Put new packing on cover. Install ball guide into cover, with flange under cover flange.

h. Install cover over rotor, with ball guide over spring hole in selector valve body, and secure with four screws. Using lockwire, Item 197, Appendix D, lockwire screws.

i. Install pin through rotor and poppet valve. Install pin retainer over pin and in cover.

j. Put spring in ball guide in cover. Put ball on spring and install detent disc on rotor. Keep pressure on detent disc to prevent ball from popping out of spring guide.

k. Install handle over poppet valve and onto rotor shaft and at the same time install nut over poppet valve. Secure handle and detent disc to rotor with nut.

l. Install clevis pin through handle and poppet valve with spacers on each side of poppet valve. Install washer and cotter pin. Using lockwire, Item 197, Appendix D, lockwire nut.

m. Install packing and backup rings on shear seal. Install shear seal in port of valve body.

n. Install packing on port fitting. Install spring in port fitting. Secure port fitting to body with screws. Using lockwire, Item 197, Appendix D, lockwire screws.

o. Repeat step m. and step n. for other ports.

7-5-2.2 Test Selector Valve (BENCH).

a. Plug three output ports of selector valve.

b. Connect pressure line from hydraulic test stand to inlet port of selector valve. Connect a hose to outlet port with open end to drain.

c. Using hydraulic fluid, Item 154 or Item 155, Appendix D, set hydraulic test stand pressure to 40 psi.

d. Apply 40 psi hydraulic pressure to selector valve.

e. Check handle as follows:

(1) Using spring scale, push down on handle, and check for the following:

(a) **6 - 10 POUNDS OF FORCE SHOULD UNSEAT POPPET VALVE AND HYDRAULIC FLUID SHOULD FLOW FROM OUTPUT PORT.**

(b) Three pounds or less of force should seat poppet valve and shut off flow of hydraulic fluid.

(2) If handle does not pass either test, replace handle.

f. Check operating torque as follows:

(1) Secure spring scale to handle, about ¾-inch from end.

(2) Pull handle to next detent.

(3) **READING ON SCALE MUST NOT EXCEED SIX POUNDS, INCLUDING PULLING OUT OF DETENT.** If reading on scale exceeds limit, repeat PARA 7-5-2.1.

g. **EXTERNAL LEAKAGE MUST NOT EXCEED ONE DROP PER 25 CYCLES BETWEEN POSITIONS.** If external leakage exceeds limit, repeat PARA 7-5-2.1, or replace selector valve (PARA 7-4-63).

h. Internal leakage must not exceed the following:

(1) **2 CC PER MINUTE WITH HANDLE DEPRESSED AND VALVE CYCLED TO EACH OUTLET POSITION.**

(2) **2 CC PER MINUTE WITH HANDLE RELEASED AND VALVE CYCLED TO EACH MIDPOSITION.**

(3) **2 CC PER MINUTE WITH HANDLE RELEASED AND VALVE CYCLED TO EACH OUTLET POSITION.**

(4) If selector valve does not pass internal leakage test, repeat PARA 7-5-2.1, or replace selector valve (PARA 7-4-63).

7-5-3 HYDRAULIC PUMP MODULE LOW LEVEL SWITCH CONNECTOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Heat Gun
- Hydraulic Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 164, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 169, Appendix D
- Lockwire, Item 197, Appendix D
- Sealing Compound, Item 292, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 7-4-12

7-5-3.1 Replace Hydraulic Pump Module Low Level Switch Connector (BENCH).

7-5-3.1.1 Remove.

NOTE

Leave wires attached to electrical connector.

a. Remove screws and electrical connector from switch cover (Figure 7-5-3-1, Detail A).

b. Remove screws and washers securing switch cover to reservoir housing. Pull back and remove switch cover.

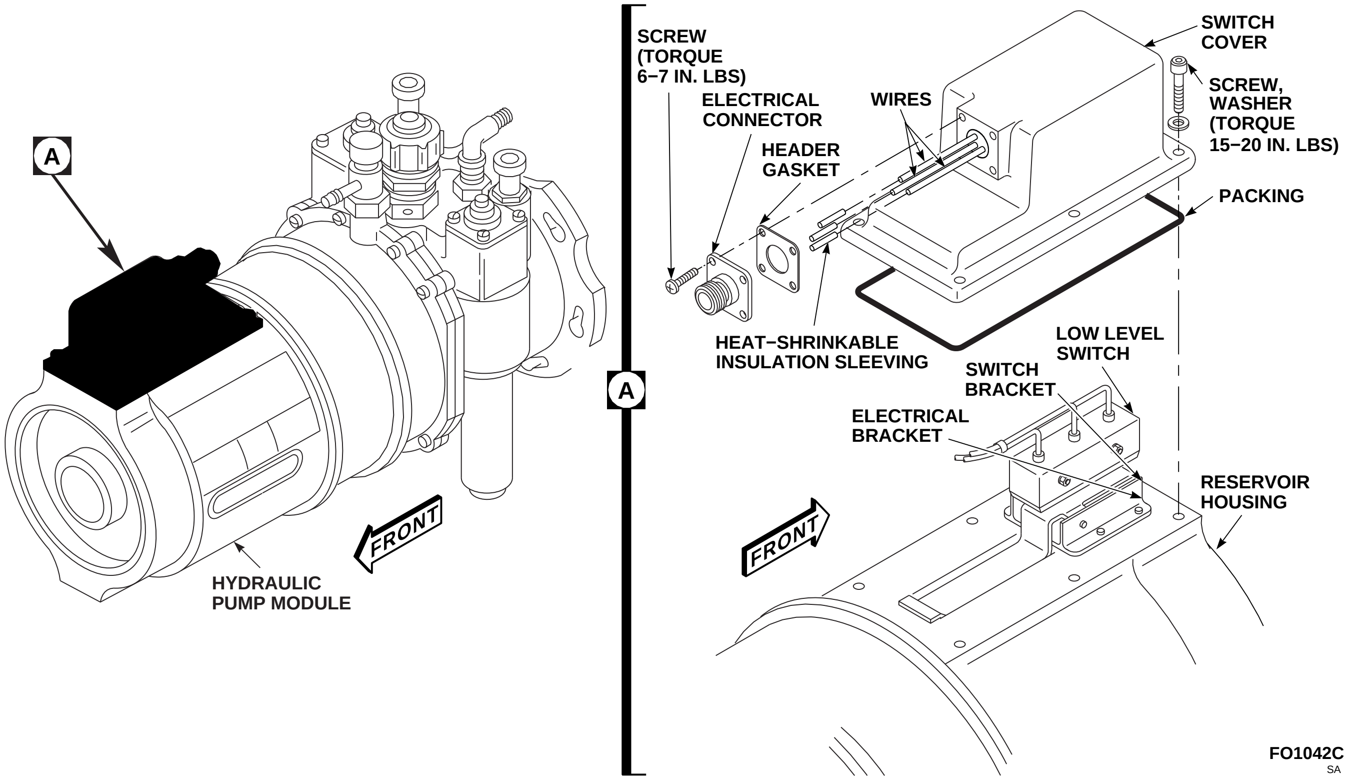


FIGURE 7-5-3-1. REPLACE HYDRAULIC PUMP MODULE LOW LEVEL SWITCH CONNECTOR (BENCH)

c. Inspect low level switch and switch bracket for looseness. **NONE ALLOWED.** If loose, replace hydraulic pump module (PARA 7-4-12).

NOTE

- Do not remove heat-shrink tubing from end nearest switch.
- Be careful not to damage wires under heat-shrink tubing.

d. Remove insulation sleeving from each wire at electrical connector end.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin, eyes, or clothing.

e. Tag wires. Using soldering iron, unsolder wires from electrical connector.

f. Remove and discard header gasket.

7-5-3.1.2 Inspect.

a. Inspect electrical connector for corrosion. **NONE ALLOWED.** If corrosion is found, replace electrical connector.

b. Discard packing from switch cover.

c. Inspect switch cover for corrosion. **NONE ALLOWED IN SCREWHOLES.** If corrosion is present in screwholes, replace switch cover. If corrosion is present on remaining area of switch cover, perform the following:

(1) Using abrasive paper, Item 15, Appendix D, clean corrosion from switch cover.

(2) Using acid brush, Item 84, Appendix D, touch up switch cover using corrosion resistant coating, Item 118, Appendix D.

7-5-3.1.3 Install.

a. On each wire, strip insulation ⅜-inch from wire end.

b. Install ¾-inch length of small diameter heat-shrinkable insulation sleeving, Item 164, Appendix D, over individual wires.

c. Install 1-inch length of large diameter heat-shrinkable insulation sleeving, Item 169, Appendix D, over wires.

d. Position switch cover and header gasket on reservoir housing (Figure 7-5-3-1, Detail A). Allow for access to insulation sleeving.

e. Position electrical connector on back of reservoir housing.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin, eyes, or clothing.

f. Using soldering iron, and solder, Item 313, Appendix D, solder wires to electrical connector. Untag wires.

g. Slide small diameter heat-shrinkable insulation sleeving, Item 164, Appendix D, over electrical connector pins. Make sure tubing contacts base of electrical connector. Apply heat using heat gun.

h. Slide large diameter heat-shrinkable insulation sleeving, Item 169, Appendix D, against base of electrical connector. Apply heat using heat gun.

i. Using sealing compound, Item 292, Appendix D, fill back end of electrical connector, completely covering pins. Using sealing compound, Item 292, Appendix D, seal end of heat-shrinkable insulation sleeving, Item 164, Appendix D.

j. Position header gasket on switch cover. Install electrical connector to switch cover with new screws and washers. **TORQUE SCREWS TO 6 - 7 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire screws.

NOTE

- Make sure wire bundle is completely inside switch cover.
- Make sure wire bundle does not prevent operation of switch lever.

k. Install new packing on switch cover. Using new screws and washers, install switch cover on reservoir housing. **TORQUE SCREWS TO 15 - 20 INCH-POUNDS.**

7-5-4 HYDRAULIC PUMP MODULE LOW LEVEL SWITCH (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Container, 2-Quart
- Continuity Lamp
- Dial Indicator
- Electrical Repairer's Toolkit
- Feeler Gage, 0.032-inch
- Heat Gun
- Hydraulic Repairer's Toolkit
- Hydraulic Test Stand
- Protection, Eye and Skin
- Soldering Iron
- Switch Adjustment Tool, Locally-Made, Figure H-217, Appendix H
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Paper, Item 15, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 164, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 169, Appendix D
- Lockwire, Item 197, Appendix D
- Sealing Compound, Item 292, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- Appendix H
- PARA 7-2-1

7-5-4.1 Replace/Adjust Hydraulic Pump Module Low Level Switch (BENCH).

7-5-4.1.1 Remove.

NOTE

Leave wires attached to connector.

a. Remove screws and electrical connector from switch cover (Figure 7-5-4-1, Detail A).

b. Remove screws and washers securing switch cover to reservoir housing. Pull back and remove switch cover. Discard packing.

c. Inspect switch cover for corrosion. **NONE ALLOWED IN SCREWHOLES.** Replace switch cover if corrosion is present in screwholes.

d. Remove corrosion on remaining area of switch cover as follows:

(1) Clean corrosion from switch cover with abrasive paper, Item 15, Appendix D.

(2) Using an acid brush, Item 84, Appendix D, touch up switch cover with corrosion resistant coating, Item 118, Appendix D.

NOTE

- Do not remove heat-shrink tubing from the end nearest the switch.
- Be careful not to damage wires under heat-shrink tubing.

e. Remove insulation sleeving from each wire at connector end.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Tag and using soldering iron, unsolder wires from electrical connector. Discard header gasket.

g. Inspect connector for corrosion. **NONE ALLOWED.** Replace connector if corrosion is found.

h. Remove capscrews, washers, and nuts securing low level switch to switch bracket. Loosen, but do not remove, all screws attaching switch bracket to hydraulic pump module reservoir.

7-5-4.1.2 Install.

a. On each switch lead wire, strip the insulation 3/8-inch from wire end.

b. Install 3/4-inch length of small diameter heat-shrinkable insulation sleeving, Item 164, Appendix D, over individual wires.

c. Install 1-inch length of large diameter heat-shrinkable insulation sleeving, Item 169, Appendix D, over wires.

d. Install low level switch on switch bracket with capscrews, washers, and nuts. Tighten, but do not torque, capscrews (Figure 7-5-4-1, Detail A).

e. Position switch cover and header gasket on reservoir housing. Allow for access to insulation sleeving. Pass wires through connector opening.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection, while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Using soldering iron, solder, Item 313, Appendix D, wires to electrical connector. Untag wires.

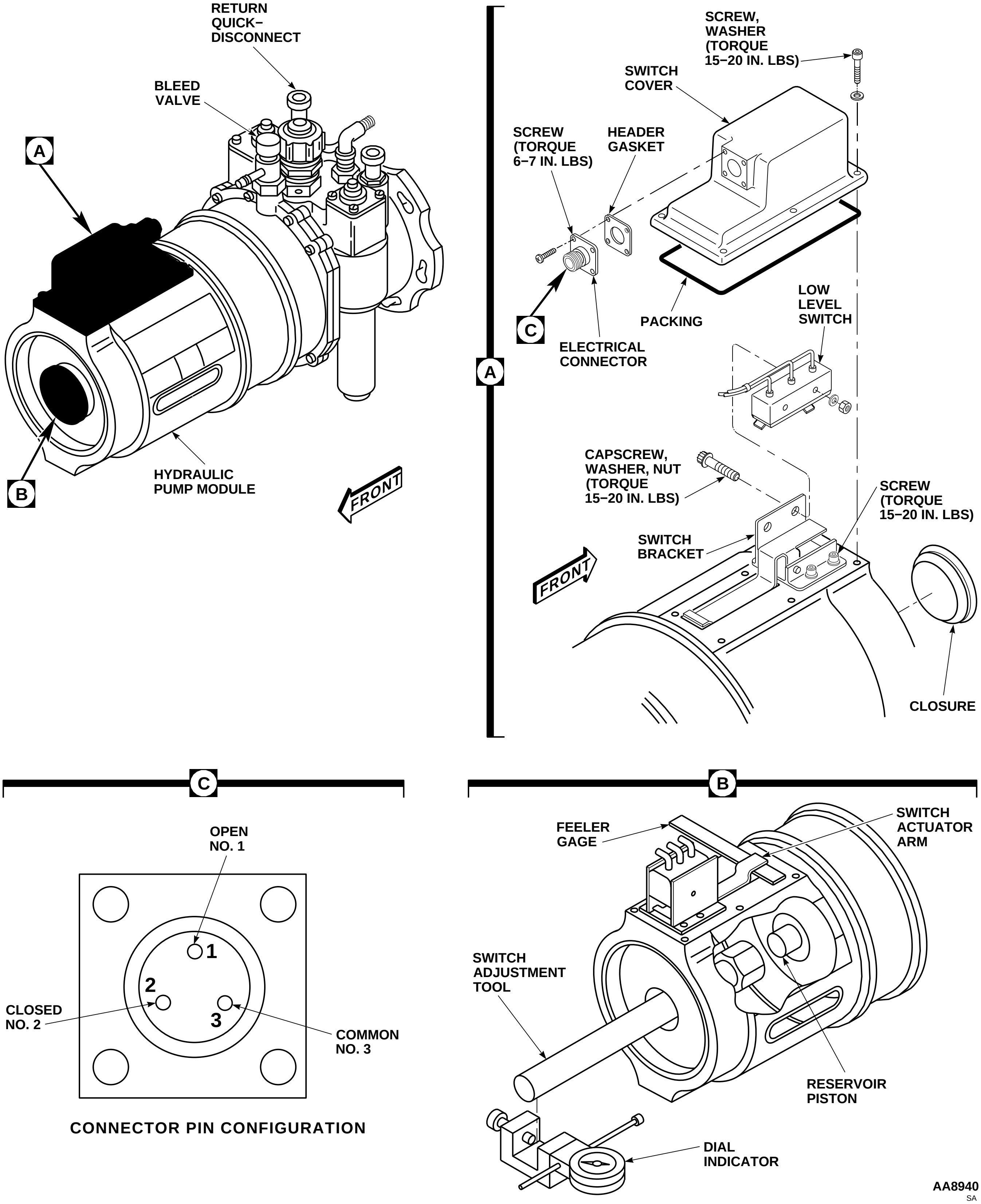


FIGURE 7-5-4-1. REPLACE/ADJUST HYDRAULIC PUMP MODULE LOW LEVEL SWITCH (BENCH)

g. Slide small diameter heat-shrinkable insulation sleeving, Item 164, Appendix D, over electrical connector pins. Make sure tubing contacts base of connector. Apply heat, using heat gun.

h. Slide large diameter heat-shrinkable insulation sleeving, Item 169, Appendix D, against base of electrical connector. Apply heat, using heat gun.

i. Fill back end of electrical connector with sealing compound, Item 292, Appendix D, completely covering pins. Seal end of heat-shrinkable insulation sleeving, Item 164, Appendix D, with sealing compound, Item 292, Appendix D.

j. Position header gasket on switch cover. Install electrical connector to switch cover with new screws and washers. **TORQUE SCREWS TO 6 - 7 INCH-POUNDS.** Lockwire, Item 197, Appendix D, screws.

7-5-4.1.3 Adjust.

a. Locally make switch adjustment tool (Figure H-217, Appendix H).

b. Remove closure from endplate on hydraulic pump module (Figure 7-5-4-1, Detail A).

c. Insert switch adjustment tool through opening in hydraulic pump module endplate. Thread switch adjustment tool onto reservoir piston (Figure 7-5-4-1, Detail B).

d. Install dial indicator on switch adjustment tool.

e. Push reservoir piston all the way in (extreme empty position) with switch adjustment tool. Adjust dial indicator to zero.

f. Connect hydraulic test stand to return quick-disconnect on hydraulic pump module.

g. Refill the hydraulic pump module, using hydraulic test stand. Pull reservoir piston all the way out (full position) with switch adjustment tool and hold.

h. Position 0.032-inch feeler gage between hydraulic pump module reservoir and switch actuator arm.

i. Install continuity lamp to low level switch connector with one lead to Open No. 1 and one lead to Common No. 3 (Figure 7-5-4-1, Detail C).

j. Tighten two screws to secure switch bracket to hydraulic pump module.

k. Adjust position of switch bracket so continuity lamp is on.

l. Tighten remaining switch bracket screws. **TORQUE ALL SWITCH BRACKET SCREWS TO 15 - 20 INCH-POUNDS.**

m. Remove feeler gage.

n. Turn knurled knob and push down bleed valve to drain hydraulic fluid into a 2-quart container. Drain until reservoir piston is 1.14 to 1.20-inches from the extreme empty position.

o. Adjust low level switch position or switch bracket so continuity lamp is off. **TORQUE LOW LEVEL SWITCH CAPSCREWS TO 15 - 20 INCH-POUNDS. TORQUE SWITCH BRACKET SCREWS TO 15 - 20 INCH-POUNDS.**

p. Turn knurled knob and push down bleed valve to drain hydraulic fluid. As reservoir empties, observe that switch reenergizes as reservoir piston passes 1.14 to 1.20-inches towards extreme empty position.

q. Refill the hydraulic pump module, using hydraulic test stand.

r. Disconnect hydraulic test stand from hydraulic pump module.

s. Remove switch adjustment tool and dial indicator from hydraulic pump module.

t. Install closure as follows:

(1) Check closure for damage. If damaged, replace closure.

(2) Make sure closure and receiving bore are clean and free of oil.

(3) Apply sealing compound, Item 292, Appendix D, to closure and install closure on hydraulic pump module endplate.

NOTE

- Make sure wire bundle is completely inside switch cover.
- Make sure wire bundle does not prevent operation of switch lever.

u. Install new packing on switch cover. Install switch cover to hydraulic pump module with screws and washers. **TORQUE SCREWS TO 15 - 20 INCH-POUNDS.**

v. Perform operational test of hydraulic pump module low level switch (PARA 7-2-1).

7-5-5 HYDRAULIC PUMP MODULE SHAFT SEALS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Seal Puller
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D
- PARA 7-2-1

7-5-5.1 Replace Hydraulic Pump Module Shaft Seals (BENCH).

7-5-5.1.1. Remove.

a. Remove retaining ring, and remove external drive shaft and spring from internal drive shaft and bearing subassembly (Figure 7-5-5-1).

b. Remove capscrews, washers, seal and retainer assembly, and seal assembly from hydraulic pump module. Remove and discard packing from seal and retainer assembly.

c. On seal and retainer assembly, separate plain encased seal from seal retainer using seal puller (Figure 7-5-5-1, Detail A).

d. Remove step pin, retainer, and packings from seal assembly (Figure 7-5-5-1). Discard packings.

e. Disassemble seal assembly by separating mating face seal, nose seal, flat washer, and wavy washer (Figure 7-5-5-1, Detail B).

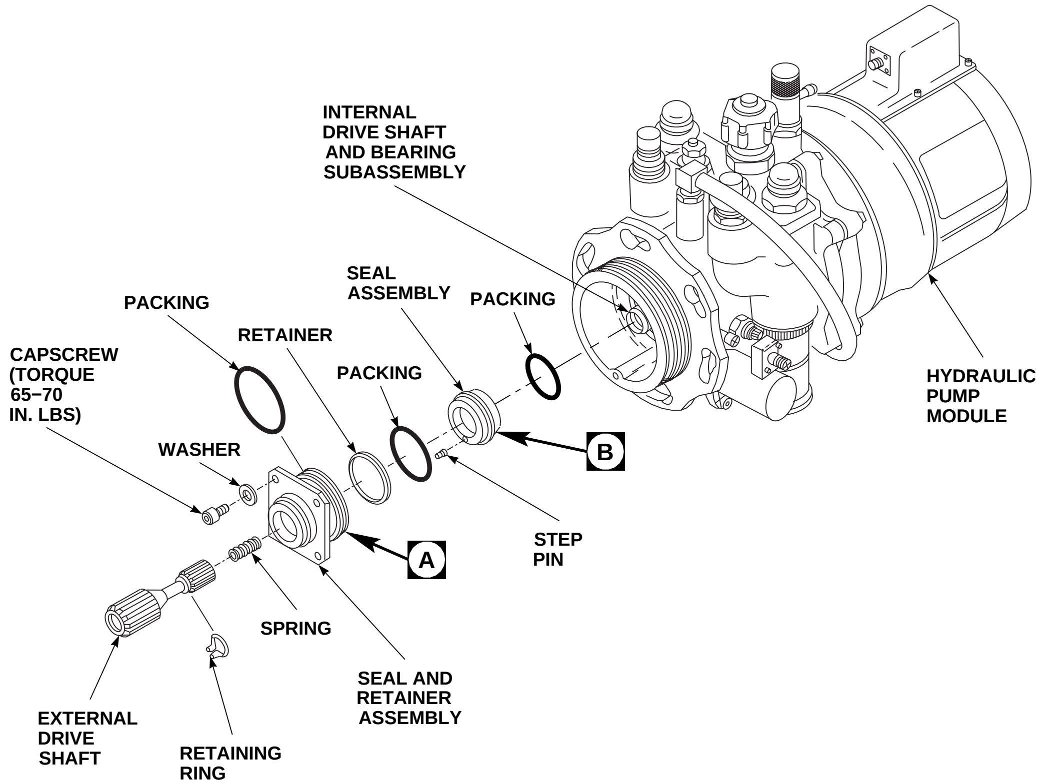
7-5-5.1.2. Install.

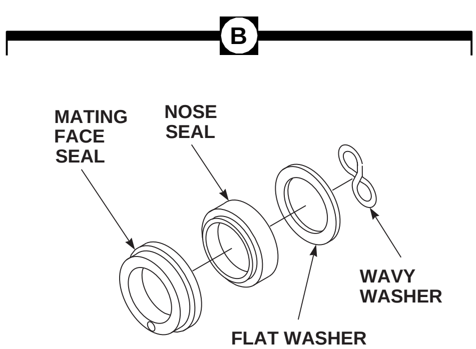
a. Soak all seals and packings in hydraulic fluid, Item 154 or Item 155, Appendix D, for not less than one hour prior to installation. Wipe excess fluid from parts prior to assembly.

NOTE

Seal assembly is comprised of wavy washer, flat washer, and nose seal (Figure 7-5-5-1, Detail B). Packing is to be installed in nose seal.

b. Install packing in seal assembly (Figure 7-5-5-1).





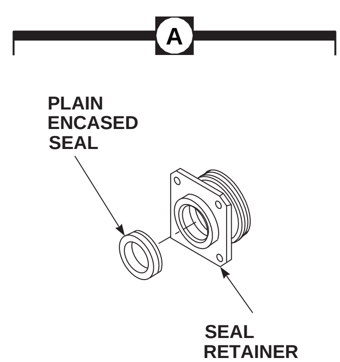


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FIGURE 7-5-5-1. REPLACE HYDRAULIC PUMP MODULE SHAFT SEALS (BENCH)

NOTE

Refer to Figure 7-5-5-1, Detail B, for proper stackup of wavy washer, flat washer, and nose seal.

c. Install seal assembly on internal drive shaft and bearing subassembly by installing wavy washer, flat washer, and nose seal (Figure 7-5-5-1).

d. Install packing and retainer on seal assembly.

e. Install mating face seal on internal drive shaft and bearing subassembly.

f. Install step pin in mating face seal.

g. Install plain encased seal in seal retainer (Figure 7-5-5-1, Detail A).

h. Install packing on seal and retainer assembly (Figure 7-5-5-1).

i. Install seal and retainer assembly on internal drive shaft and bearing subassembly. Rotate seal and retainer assembly to engage the notched groove with the step pin of mating face seal.

j. Attach seal and retainer assembly to hydraulic pump module with capscrews and washers. **TORQUE CAPSCREWS TO 65 - 70 INCH-POUNDS.**

k. Install retaining ring on neck of external drive shaft.

l. Install spring and external drive shaft into internal drive shaft and bearing subassembly.

m. Press external drive shaft in to expose locking groove in internal drive shaft.

n. Insert retaining ring in locking groove of internal drive shaft. Align tines of ring with space where teeth have been removed on internal drive shaft.

o. Perform maintenance operational check (PARA 7-2-1).

7-5-6 HYDRAULIC PUMP MODULE ELECTRICAL SOLENOID (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

7-5-6.1 Replace Hydraulic Pump Module Electrical Solenoid (BENCH).

7-5-6.1.1. Remove.

a. Remove bolts securing electrical solenoid to hydraulic pump module (Figure 7-5-6-1).

b. Pull electrical solenoid from hydraulic pump module. Remove and discard packings and retainers.

7-5-6.1.2. Install.

NOTE

Position packings and retainers on electrical solenoid as shown in Figure 7-5-6-1, Detail A.

a. Install packings and retainers on electrical solenoid (Figure 7-5-6-1).

b. Install electrical solenoid into hydraulic pump module and install bolts. **TORQUE BOLTS TO 40 - 45 INCH-POUNDS.**

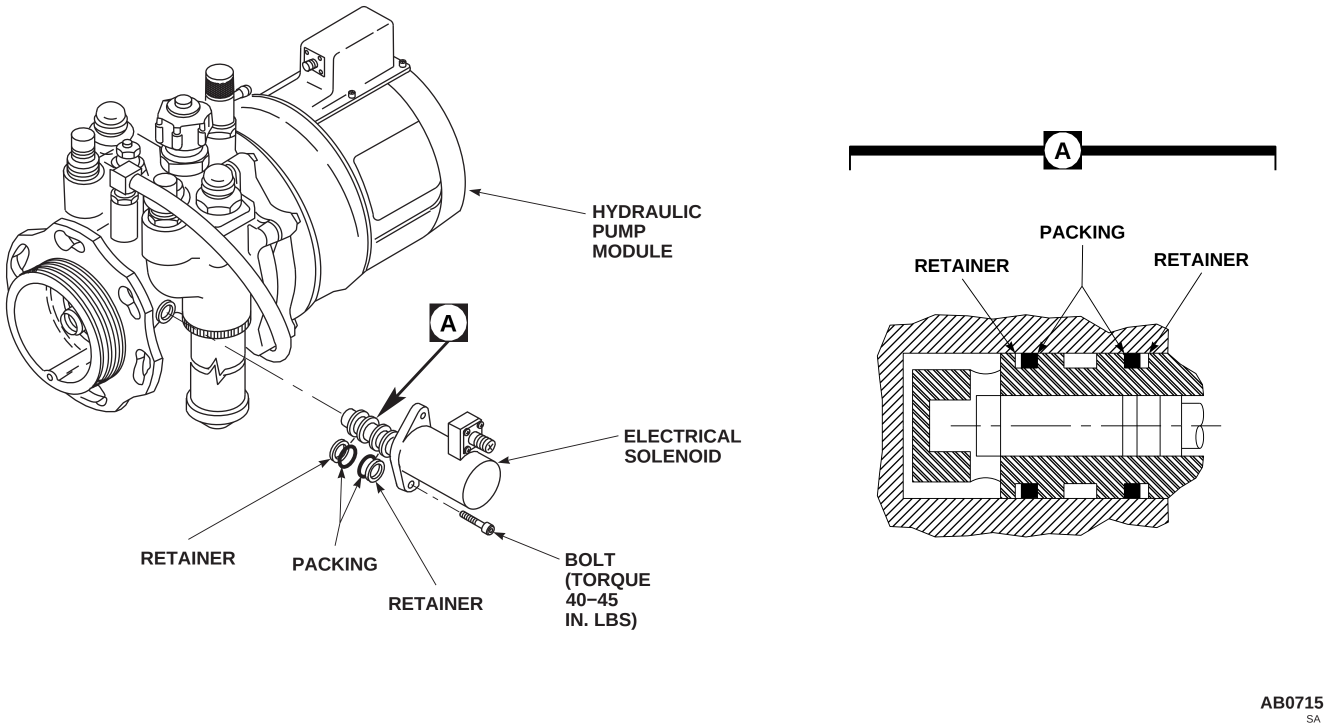


FIGURE 7-5-6-1. REPLACE HYDRAULIC PUMP MODULE ELECTRICAL SOLENOID (BENCH)

7-5-7 MANIFOLD SHIMS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Alignment Toolkit, 70652-02150-041T046
- Heat Gun
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 32, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 291, Appendix D
- Shims, 70652-02151-101
- Shims, 70652-02151-102
- Shims, 70652-02151-103

References

- Appendix D
- PARA 7-4-24
- PARA 7-4-25
- PARA 7-4-29
- PARA 7-4-30

7-5-7.1 Replace Manifold Shims (BENCH).

a. Remove the following components as required:

(1) No. 1 transfer module manifold (PARA 7-4-24).

(2) No. 1 primary servo manifold (PARA 7-4-25).

(3) No. 2 transfer module manifold (PARA 7-4-29).

(4) No. 2 primary servo manifold (PARA 7-4-30).

NOTE

The following procedure requires alignment toolkit.

b. Remove all old shims from fuselage using nonmetallic scraper.

c. Clean remaining sealing compound from helicopter fuselage and manifolds, using a nonmetallic scraper and machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

NOTE

- Use only the barrel-nuts supplied with the alignment toolkit.
- The four locations which use the two hole shims need to be measured only once.

d. Mount tool plate from alignment kit on servo beams, with locating pins and barrel-nuts (Figure 7-5-7-1, Detail B).

e. Install gage on tool plate above one of the shim locations (Figure 7-5-7-1, Detail A).

f. Extend gage probe until gage indicator reads 3.091-inches, and then lock gage probe. This dimension locates plane to which fuselage will be shimmed.

g. Reset gage indicator to zero.

h. Extend gage probe down to top of fuselage deck. Record gage indicator reading.

i. Repeat step e. through step h. at all 22 shim locations and record dimension for each location.

j. Peel laminated shims to recorded dimension. Remove an additional 0.005-inch from shim for bonding adhesive thickness.

k. Bond shims at their respective positions on fuselage deck, using adhesive, Item 32, Appendix D. Press down to squeeze out excessive adhesive. Remove any adhesive from mounting holes.

l. Repeat gage indicator readings at all 22 shim locations to make sure shim installation creates a level plane required for manifold installation. Gage indicator reading to be -0.005-inch to +0.005-inch.

m. If gage indicator readings are not within the range specified in step l., do the following:

(1) If the gage indicator reading is too high, remove required laminations.

(2) If the gage indicator reading is too low or laminations cannot be removed as specified in step m. (1), replace shim and repeat steps f. through step k.

n. Remove gage, locating pins, tool plate, and barrel-nuts.

o. Allow adhesive to cure for 24 hours at 70° - 80°F (21° - 27°C) or, using heat gun, for two hours at 140° - 160°F (60° - 71°C).

p. Apply a thin layer of sealing compound, Item 291, Appendix D to shim surfaces. Make sure holes are free of sealing compound.

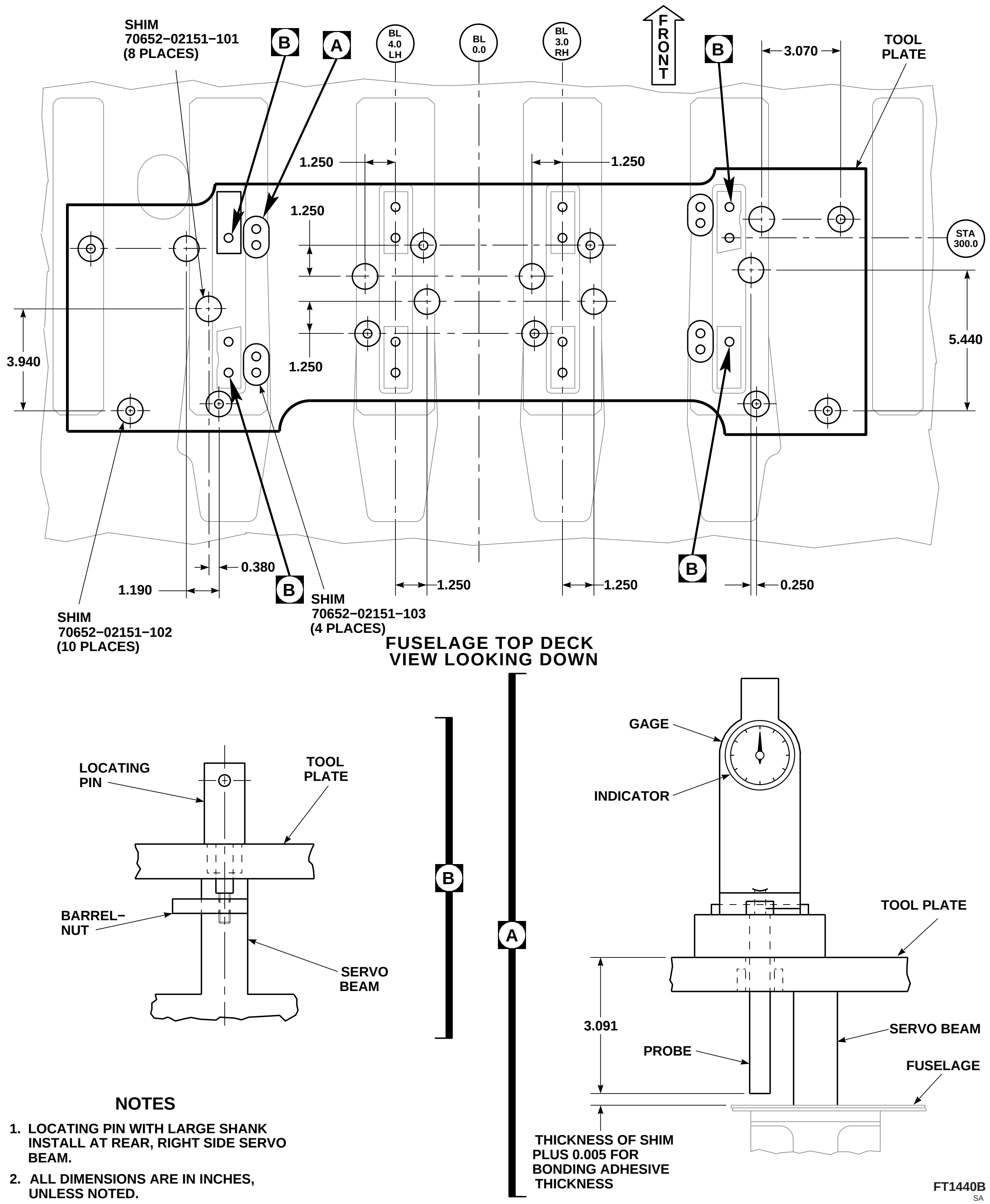
q. If removed, install the following components:

(1) No. 1 transfer module manifold (PARA 7-4-24).

(2) No. 1 primary servo manifold (PARA 7-4-25).

(3) No. 2 transfer module manifold (PARA 7-4-29).

(4) No. 2 primary servo manifold (PARA 7-4-30).



NOTES

1. LOCATING PIN WITH LARGE SHANK INSTALL AT REAR, RIGHT SIDE SERVO BEAM.
2. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FIGURE 7-5-7-1. REPLACE MANIFOLD SHIMS (BENCH)

7-5-8 HYDRAULIC SYSTEM RIGID TUBING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench
- Cutting Tool, D9872
- Deburring Tool, D9851
- Felt Tip Pen
- Filter, D10004-104
- Hydraulic Repairer's Toolkit
- Inspection Gage, D9892
- Marking Pen
- Marking Tool, D9862S
- Pliers
- Portable Hydraulic Pump, D10004
- Protection, Eye
- Ruler
- Screwdriver
- Spanner Wrench
- Swaging Tool, D10001

Materials/Parts

- Acetone, Item 25, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lubricating Oil, Item 209, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

7-5-8.1 Repair Hydraulic System Rigid Tubing (BENCH).

7-5-8.1.1 General.

NOTE

Repair may be done on or off helicopter.

a. Repair to rigid tubing is done by replacing fittings and by splicing sections of tubing, using Permaswage Repair Procedure (PARA 7-5-8.1.11).

b. Basic repair method is to mechanically swage Permaswage fitting onto tube by a hydraulically-operated tool. Fittings are permanent unions, elbows, tees, crosses, flared, flareless and separable sleeves; and combinations of permanent and threaded fittings.

7-5-8.1.2 Portable Hydraulic Pump, D10004.

Portable hydraulic pump provides pressure required to operate swaging tool. Hydraulic fluid, Item 154 or Item 155, Appendix D, is fed to tool through a ¼-inch quick-disconnect high pressure hose. A swaging pressure of 5,500 psi, applied to all tubing sizes and materials, may be done by either manual operation of handpump or by using air-to-hydraulic fluid pressure from a 60 - 100 psi shop air source. A remote control air switch is provided to activate air-to-hydraulic fluid valve. As a precaution against tool damage, swaging pressure level is kept from reaching more than 5,500 psi by a pressure relief valve. After swaging is finished, pressure is reduced to zero by turning a manual four-way selector valve to exhaust (Figure 7-5-8-1, Sheet 1 and Sheet 2).

7-5-8.1.3 Calibrate Portable Hydraulic Pump, D10004.

a. These instructions apply to calibration adjustments of hydraulic pressure relief valve and air pressure regulator (Figure 7-5-8-1, Sheet 2).

b. Adjust hydraulic pressure relief valve as follows:

(1) Loosen setscrew on hydraulic pressure relief valve.

(2) Set selector valve to either hydraulic output (one on each side).

(3) Manually pump output pressure until gage shows relief valve has actuated, and record pressure move.

(4) Turn selector valve to exhaust.

(5) Using screwdriver, turn adjustment screw counterclockwise to lower relief point, or clockwise to raise relief point.

(6) Repeat above procedure until valve relieves at 5,250 - 5,750 psig.

c. Adjust air pressure regulator as follows:

(1) Connect shop air line.

(2) Set selector valve to either hydraulic output (one on each side).

(3) Pull up on red snapring.

(4) Press air switch and adjust cap until regulator functions at 5,500 psi on output pressure gage.

7-5-8.1.4 Air Supply.

NOTE

Portable hydraulic pump air filter should be cleaned every 30 days.

a. Clean air filter as follows:

(1) Remove regulator sediment bowl.

(2) Remove mounting bolt.

(3) Slide flow diverter and filter off mounting bolt.

(4) Wash filter in dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(5) Using clean, dry, compressed air, blow dry filter.

(6) Install filter and flow diverter on mounting bolt.

(7) Install mounting bolt in regulator. **TIGHTEN MOUNTING BOLT HANDTIGHT.**

(8) Install sediment bowl.

7-5-8.1.5 Hydraulic Fluid Filter.

NOTE

Change hydraulic fluid filter every 1,000 operating hours or once per year, whichever comes first.

a. Change hydraulic fluid filter as follows:

(1) Tip portable hydraulic pump on its side.

(2) Remove three retaining screws and cover.

(3) Remove plug and filter.

(4) Install new filter, D10004-104, and reinstall plug.

(5) Install cover and secure with three retaining screws.

7-5-8.1.6 Basic Swaging Tool, D10001.

a. Swaging tool is used to make permanent tubing joints by swaging (squeezing) a Permaswage fitting onto tubing. Swaging tool covers a range of tubing sizes as shown in Table 7-5-8-1:

Table 7-5-8-1. Swaging Tool, D10001

Part No.	Assembly
D10001-52	Basic Swage Tool Assembly
D10001-56-3	Die Block Assembly (3⁄16-in.)
D10001-56-4	Die Block Assembly (1⁄4-in.)
D10001-56-5	Die Block Assembly (5⁄16-in.)
D10001-56-6	Die Block Assembly (3⁄8-in.)

b. Operation of Swaging Tool. Hydraulic pressure supplied by portable hydraulic power supply causes die segments contained within swaging tool to do swaging. Swaging tool covers a range of tubing sizes and types of fittings by changing die block assemblies and/or fitting locators. Die block assemblies are in sets of upper and lower die blocks, dies, and locators. Lower die block is retained on basic swage tool assembly to assure automatic retraction and consistent repeatability. Upper die block assembly is removable for easy loading (Figure 7-5-8-1, Sheet 3).

NOTE

Die block assemblies listed in Table 7-5-8-1 are used for standard Permaswage union, elbow, tee, and cross, and are supplied in matched sets of one upper die block assembly and one lower die block assembly.

c. Basic swage tool assembly, D10001-52, contains an actuating piston and a lock-type latch which insures upper die block retention during swage cycle.

d. Upper die block assembly. Upper die block assembly consists of a die-holder, die-retainers, and die segments which move from force applied through lower die block assembly.

e. Lower die block assembly. Lower die block assembly consists of a die-holder, die-retainers, die segments, and fitting locator. Lower die block assembly is held on top of swaging tool piston. Movement of piston brings lower dies in contact with upper dies, and continuation of this movement closes dies, thus swaging fitting. Fitting locator is attached to one side of lower die block assembly and positions fitting in swaging dies. During swaging operation, fitting locator keeps fitting from moving. Three fitting locators are used:

(1) Standard Fitting Locator, D10001-56-X (X = dash size), used with standard union, elbow, tee, and cross fittings.

(2) Separable Fitting Locator, used with separable sleeves and unions (flared and flareless), consists of Locator Adapter, D10050-1, -2, -3 (small, medium, and large), and Locator Insert Assembly, D10051-3 through -6 sizes. Locator adapter is attached in place of standard fitting locator. Locator insert assembly of appropriate size is inserted into locator adapter in one direction for separable sleeves and turned 180° for separable unions (Figure 7-5-8-1, Sheet 4).

(3) No fitting locator is used when installing special fittings such as reducers, bulkhead fittings, and 90° adapter.

7-5-8.1.7 Die Block Inspection and Lubrication.

a. Die block should be inspected periodically for foreign matter between die segments. If foreign matter is found, wash die segments in drycleaning solvent, Item 124, Appendix D, and dry thoroughly. After die segments are thoroughly dry, lubricate them using general purpose lubricating oil, Item 209, Appendix D, to prevent rusting.

b. Adjust end cover as follows:

(1) Install die block assemblies, D10001-6, on swaging tool.

(2) Install end cover into body with spanner wrench (do not try to bottom).

(3) Adjust until upper die block slides over Permaswage fitting without forcing it.

(4) Back off end cover until setscrew on side of body lines up with notch in end cover. **TIGHTEN SETSCREW.**

7-5-8.1.8 Cutting Tool, D9872.

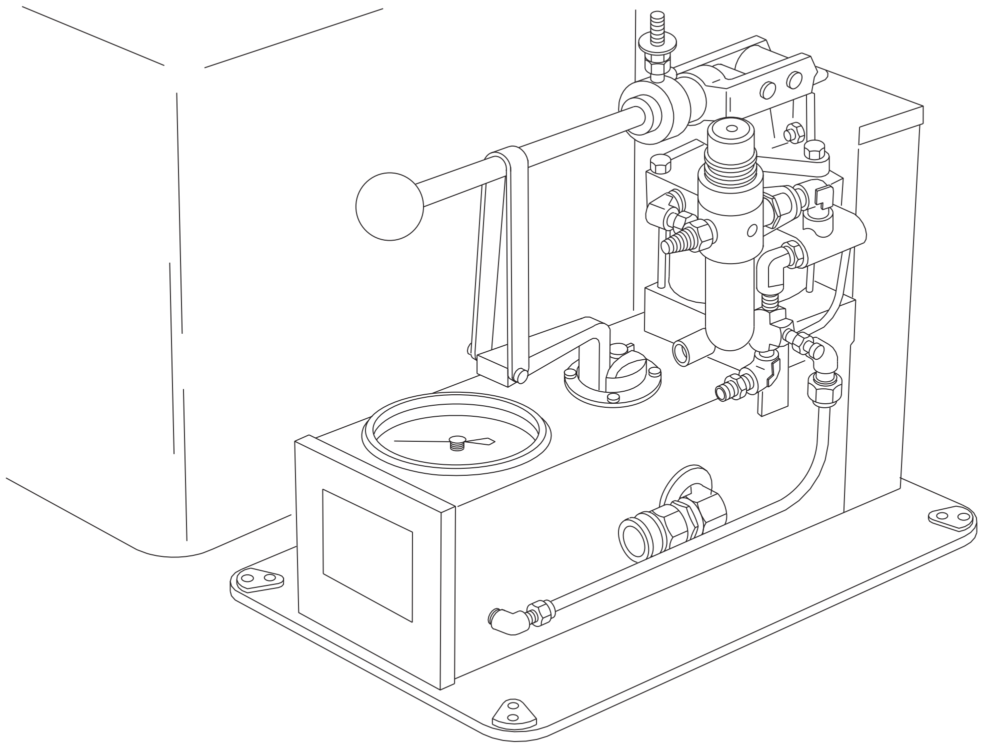
a. To ensure ease of operation, lubricate ratchet, rollers, and cutter wheel regularly, using lubricating oil, Item 209, Appendix D (Figure 7-5-8-1, Sheet 5).

b. When cutter wheel does not cut tubing easily, replace as follows:

(1) **TIGHTEN DRIVE SCREW UNTIL CUTTER PIVOT PIN BECOMES ACCESSIBLE.**

(2) Remove cutter pivot pin and slide cutter wheel out.

(3) Slide new cutter wheel into place and secure with pivot pin.



PORTABLE HYDRAULIC PUMP, D10004

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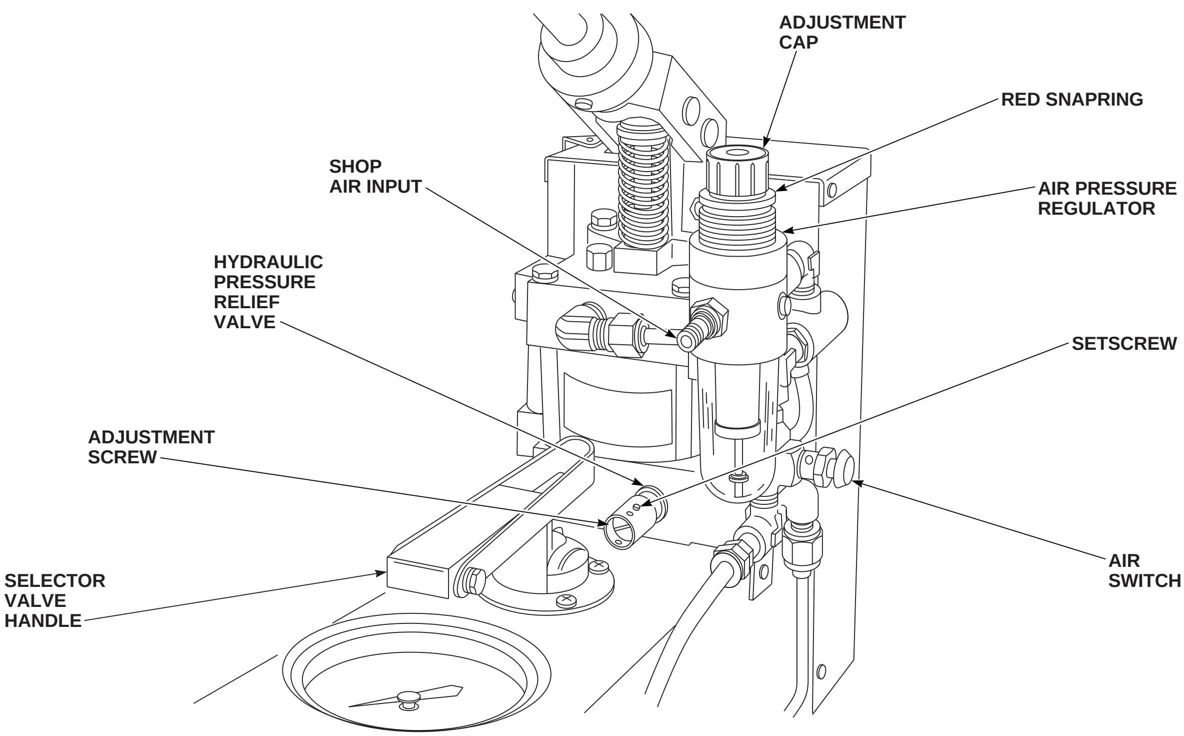
FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 1 OF 17)



PORTABLE HYDRAULIC PUMP, D10004

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 2 OF 17)

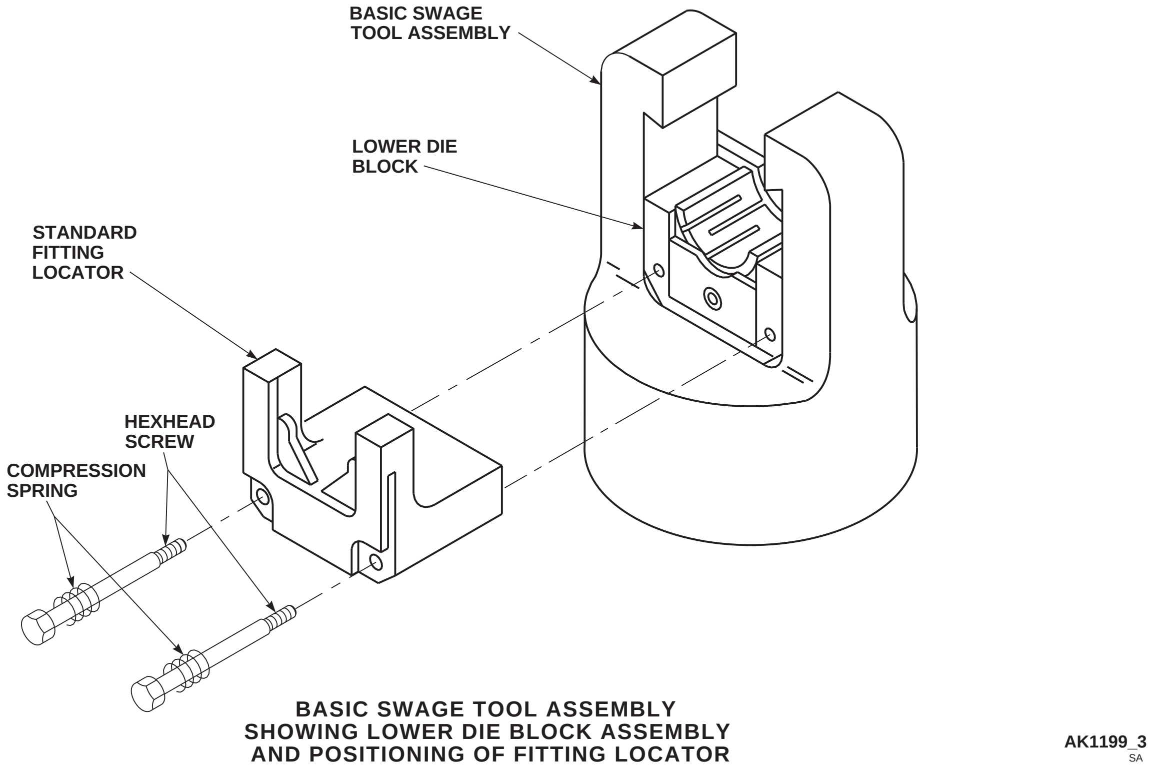


FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 3 OF 17)

7-5-8.1.9 Deburring Tool, D9851.

A special deburring tool is provided to remove burrs from inside diameter (ID) of tubing. Tool consists of a plunger-activated stem subassembly, having an elastic plug inserted into tube end up to point where cutter contacts burr. Releasing plunger causes elastic plug of stem subassembly to expand, thus creating an interference fit with ID of tube. Turning knurled tool body by hand operates cutter and deburrs, while elastic plug prevents chips from deburring operation from entering tubing. After trapping chips, plug wipes tube ID clean as it is withdrawn. Tool has a set of stem subassemblies that match tube sizes (Figure 7-5-8-1, Sheet 9).

7-5-8.1.10 Marking Tool, D9862S.

Making a tube insertion mark with marking tool may be done by positioning lip stop of marking tool over end of tube to be swaged. Use correct tube size slot as a marking guide, making sure total width of mark is made on tube. Tube insertion length and 0.300-inch marking band width are part of marking tool (Figure 7-5-8-1, Sheet 10).

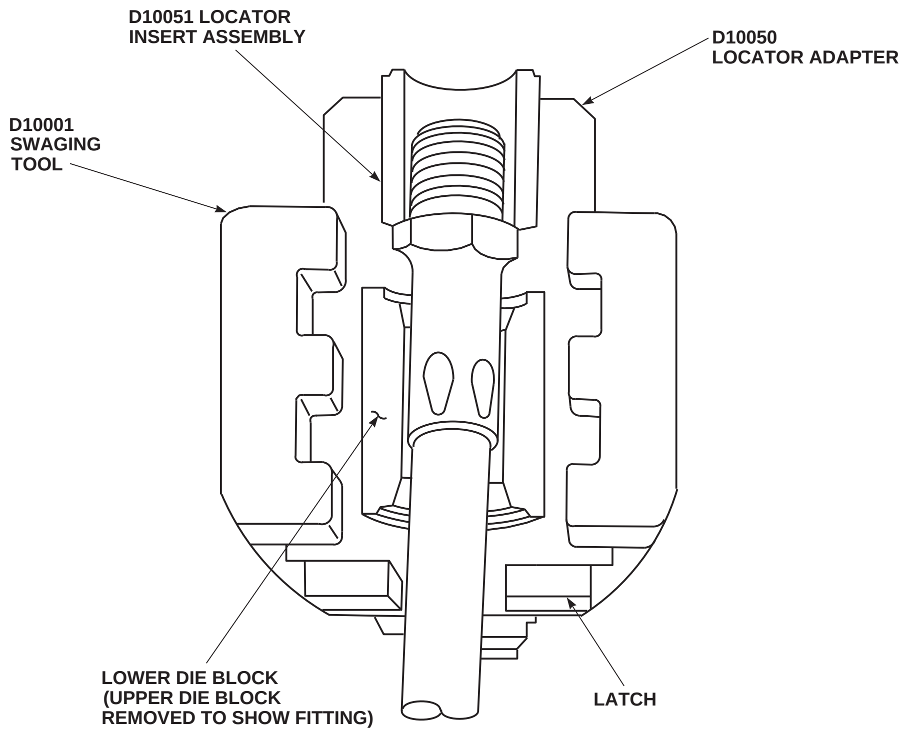
7-5-8.1.11 Permaswage Repair Procedure.

a. Permaswage procedures for tubing repair are cutting, deburring, tube end preparation, swaging, and gaging operations.

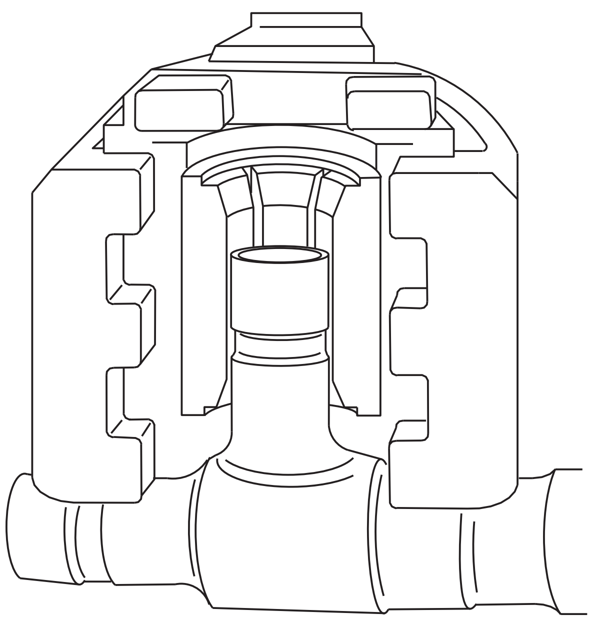
CAUTION

Permaswage fittings are manufactured using 21-6-9 corrosion-resistant steel and must be used only with those tubing materials specified in Table 7-5-8-2. Use of 21-6-9 Permaswage fittings with tubing material other than specified, will result in unacceptable, unsafe repair, and must not be used.

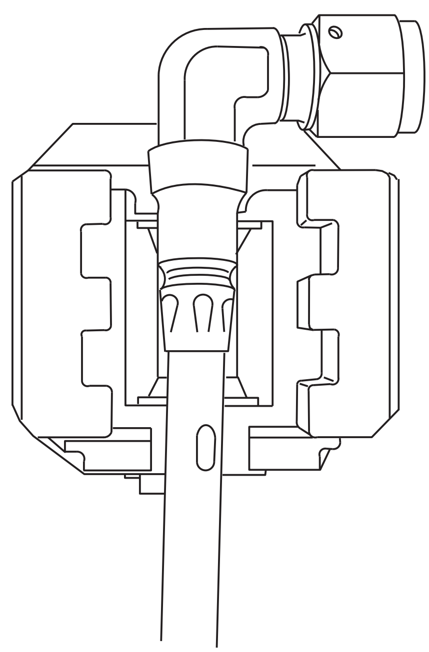
b. This repair procedure must be used only for tubing assemblies not more than material, line size, operating pressure requirements of Table 7-5-8-2. This repair method must not be used with annealed corrosion-resistant steel or 5052 aluminum tubing.



SEPARABLE FITTING LOCATOR ON SWAGING TOOL



TEE REDUCER IN SWAGING TOOL,
NO FITTING LOCATOR



ADAPTER 90° FITTING IN
SWAGING TOOL, NO FITTING LOCATOR

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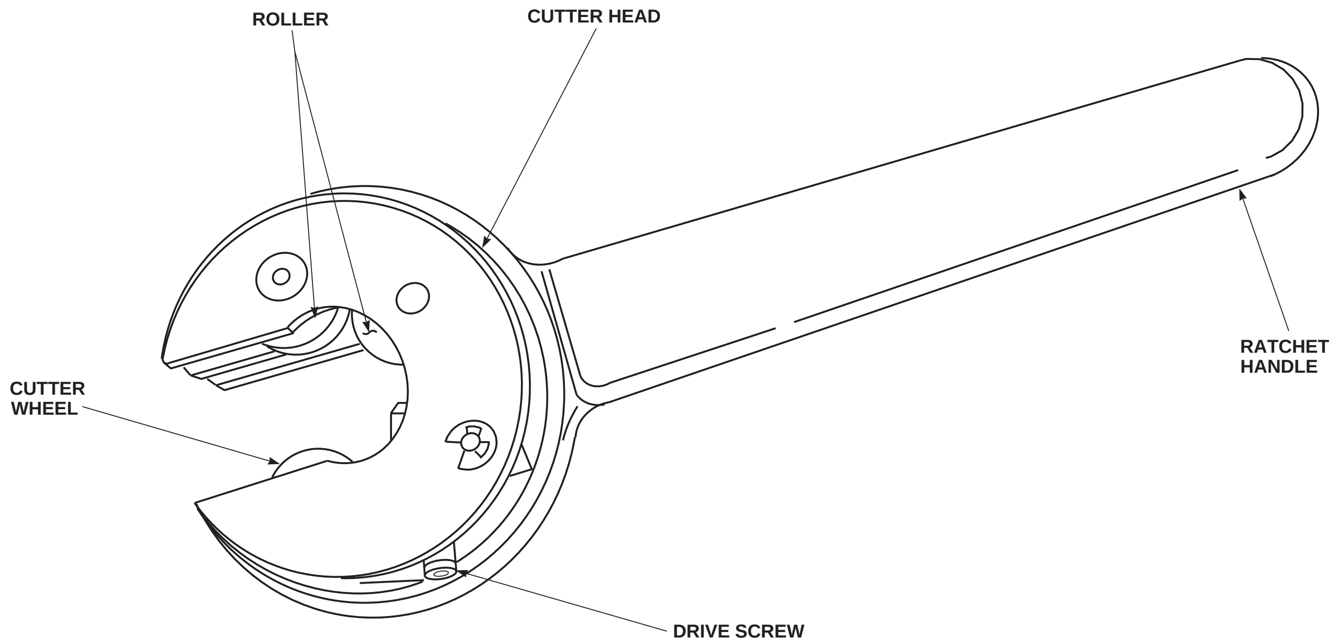
FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 4 OF 17)



CUTTING TOOL D9872

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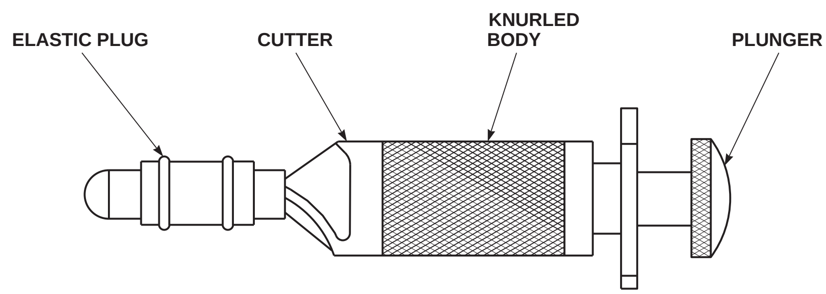
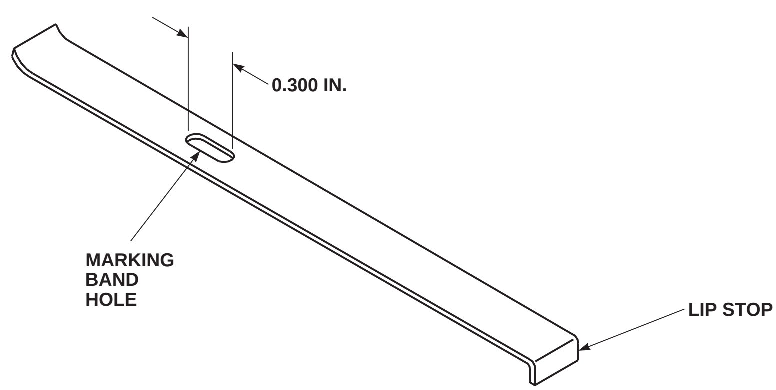
FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 5 OF 17)



DEBURRING TOOL D9851



MARKING TOOL D9862S

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 6 OF 17)

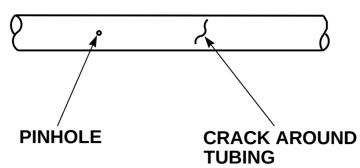
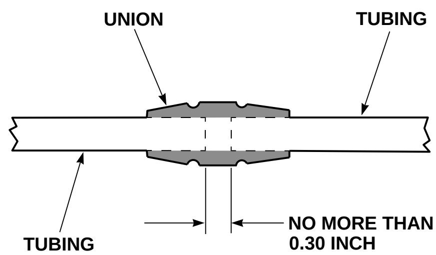
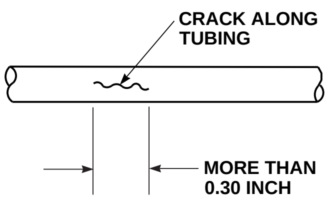
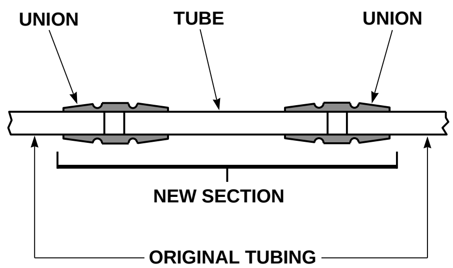
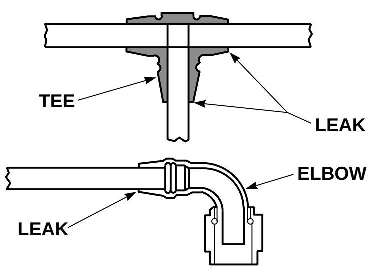
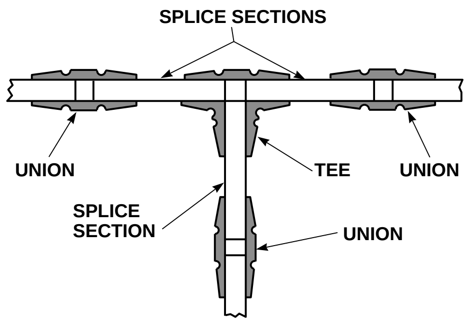
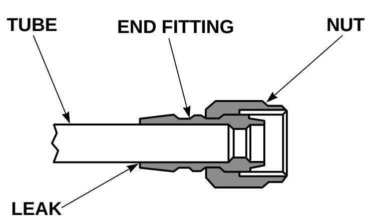
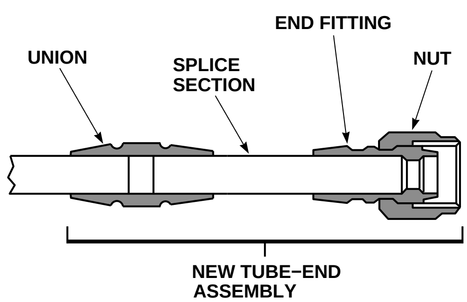
TYPE OF DAMAGE	REPAIR METHOD	
1. PINHOLE LEAK OR CRACK AROUND TUBING 		A. MAKE ONE OR TWO CUTS AS NECESSARY SO DAMAGED SECTION CAN BE REMOVED. IF TWO CUTS ARE REQUIRED, DISTANCE BETWEEN THEM SHALL NOT BE MORE THAN 0.03 INCH. IF THIS MEASUREMENT IS MORE, GO TO REPAIR METHOD 2. B. SWAGE TUBE UNION IN TUBE SECTION BEING REPAIRED.
2. CRACK ALONG TUBING (CRACK MORE THAN 0.30 INCH LONG) 		A. MAKE TWO CUTS SO DAMAGED SECTION CAN BE REMOVED. B. REMOVE DAMAGED SECTION AND MAKE REPLACEMENT TUBE SAME LENGTH. C. SWAGE REPLACEMENT SECTION INTO TUBING BEING REPAIRED USING TWO UNIONS.
3. LEAKING TEE OR ELBOW (PERMANENT TUBE CONNECTION TYPE) 		A. CUT OUT LEAKING TEE OR ELBOW. B. MAKE TUBING SECTIONS FOR EACH BRANCH SAME LENGTH. C. SWAGE SPLICE SECTIONS TO TEE OR ELBOW. D. CONNECT EACH SPLICE SECTION TO TUBING UNDER REPAIR USING A TUBE TO TUBE UNION.
4. LEAKING END FITTING 		A. CUT TUBING TO REMOVE LEAKING FITTING. B. MAKE UP SPLICE SECTION. C. SWAGE APPROPRIATE END FITTING TO END OF SPLICE SECTION. D. CONNECT NEW END FITTING TO MATING EQUIPMENT CONNECTION. TORQUING NUT AS REQUIRED. E. SWAGE ONE UNION CONNECTING NEW TUBE−END ASSEMBLY TO TUBE SECTION BEING REPAIRED.

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 7 OF 17)

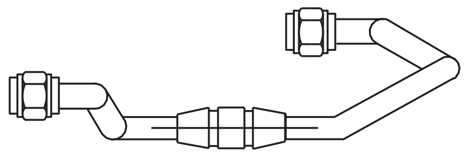
TYPE OF DAMAGE	REPAIR METHOD	
5. ADDITIONAL INFORMATION TO USE WHEN MAKING ANY OF FOUR TYPES OF REPAIRS. (1 THRU 4 ABOVE.)	 A. BEFORE CUTTING TUBE, DRAW A LINE PARALLEL TO TUBE RUN ACROSS SEC− TION TO BE CUT, USING MARKING PEN AND A RULER. B. CUT TUBING. C. IF A TUBE END IS TO BE REPLACED, MAKE SURE LINE IS PLACED IN SAME LOCATION ON NEW TUBE AS TUBE SECTION THAT HAS BEEN REMOVED. **CAUTION** DO NOT FORGET TO PLACE TUBE INSERTION MARK ON ALL TUBE ENDS. D. DRAW A LINE ACROSS FITTING.	E. INSTALL TUBE RUN AND LOCATE FITTING. FINGERTIGHTEN ANY END FITTINGS. (ONE END OF FITTING MAY BE SWAGED ON BENCH, IF POSSIBLE.) F. PLACE SWAGE TOOL ON FIRST END BEING SWAGED, LINING UP LINE ON TUBE END BEING SWAGED WITH LINE ON FITTING. G. REPEAT THIS STEP WITH OTHER ENDS TO BE SWAGED. H. TIGHTEN ANY END FITTINGS TO THEIR PROPER TORQUE VALUE.

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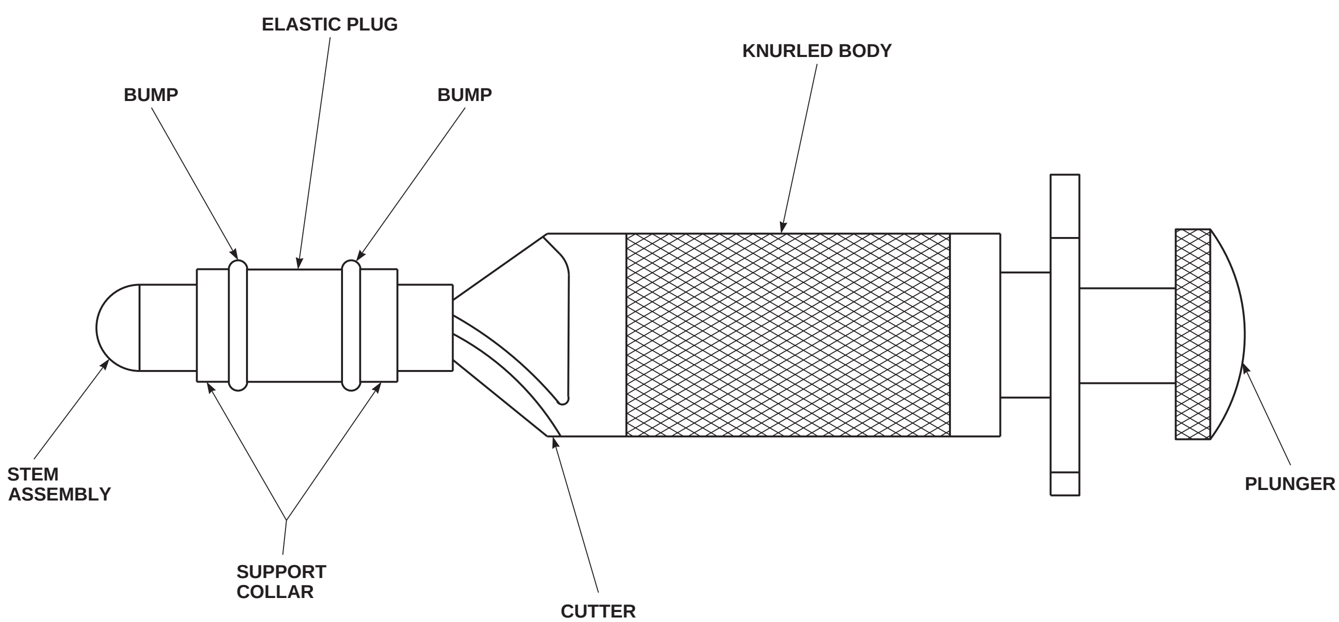
FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 8 OF 17)



DEBURRING TOOL D9851

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 9 OF 17)

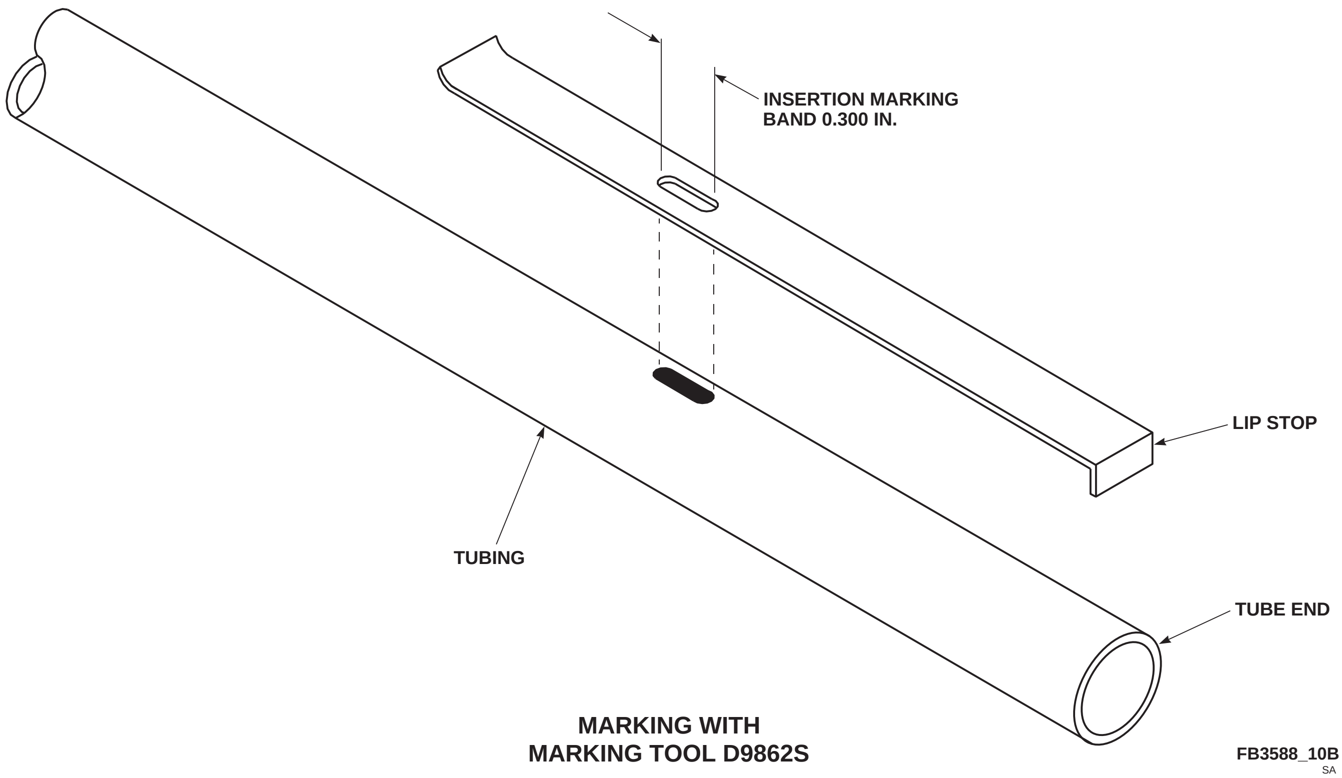


FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 10 OF 17)

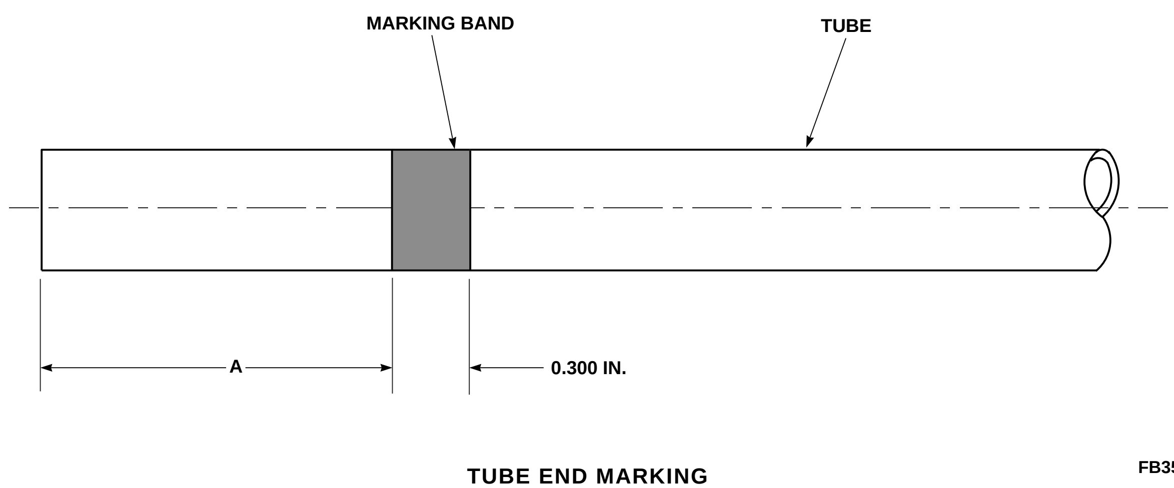


FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 11 OF 17)

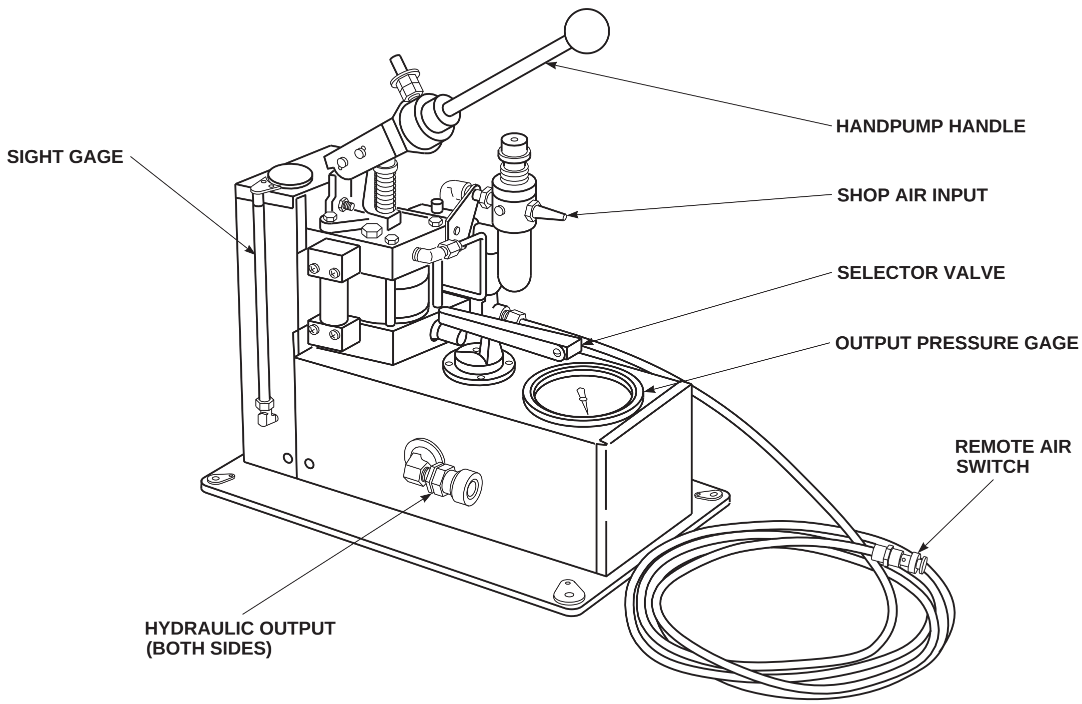
c. Check damage to tube or fitting and determine which method of repair must be used (Figure 7-5-8-1, Sheet 7 and Sheet 8).

7-5-8.1.12 Tube Cutting.

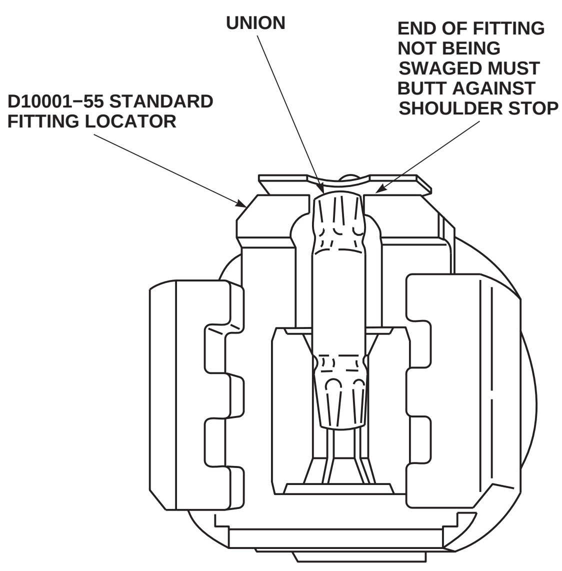
NOTE

Before performing the following procedure, make sure ratchet is operating freely and cutter wheel is clear of cutter head opening.

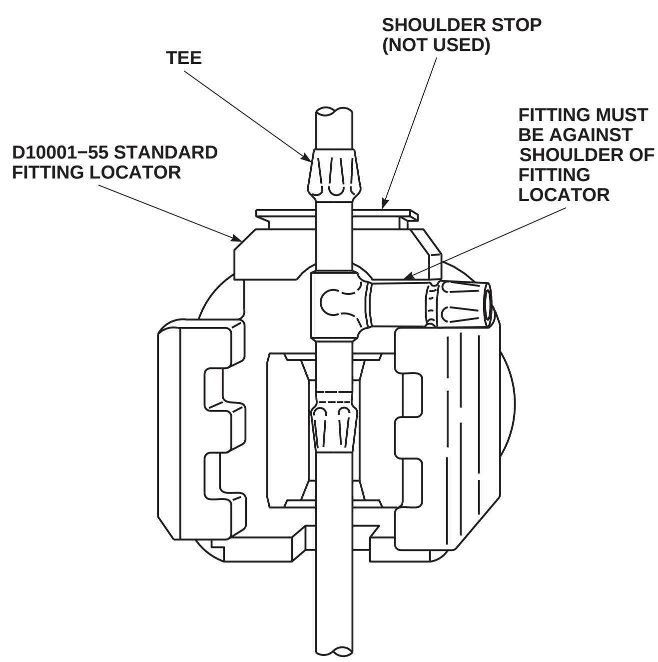
a. Turn cutter head to accept tubing, and locate in cutting position. When properly located, tubing will be centered on two rollers and cutting blade (Figure 7-5-8-1, Sheet 5).



PORTABLE HYDRAULIC PUMP (READY FOR OPERATION)



STANDARD UNION SWAGING TOOL



SHAPED FITTING IN SWAGING TOOL

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 12 OF 17)

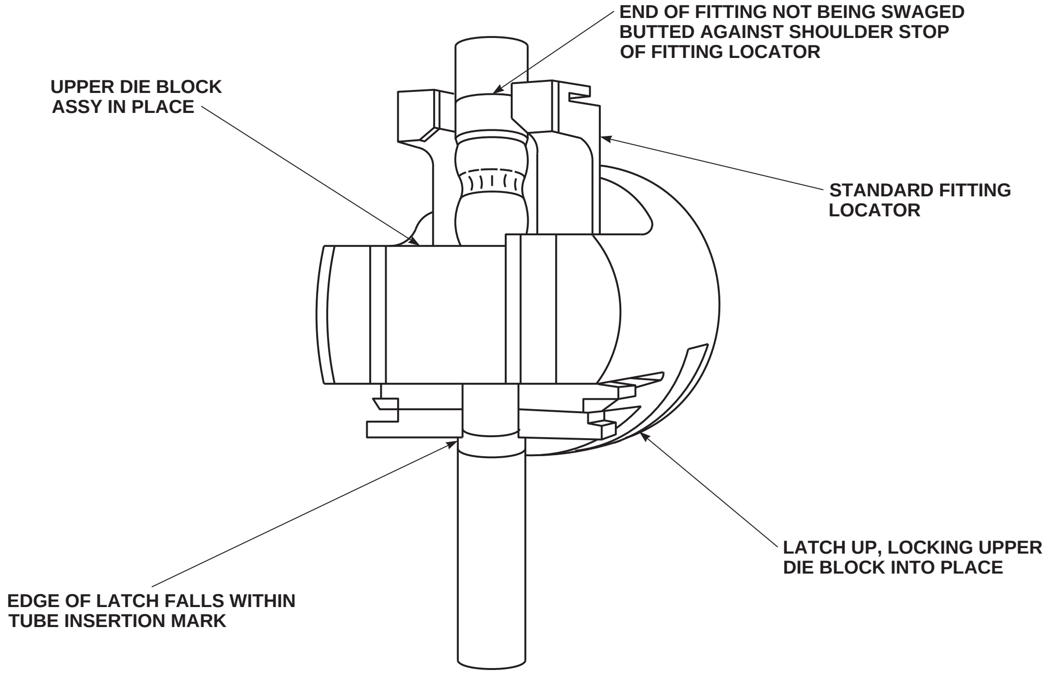
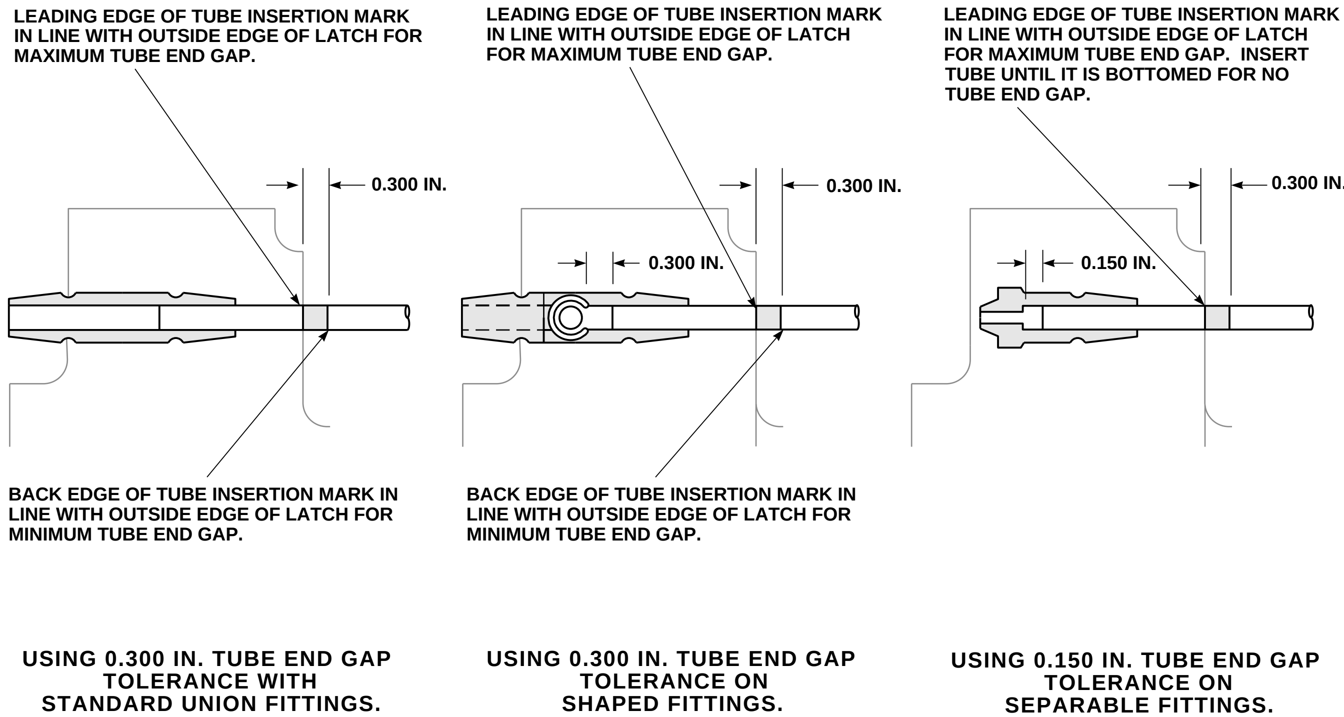


FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 13 OF 17)



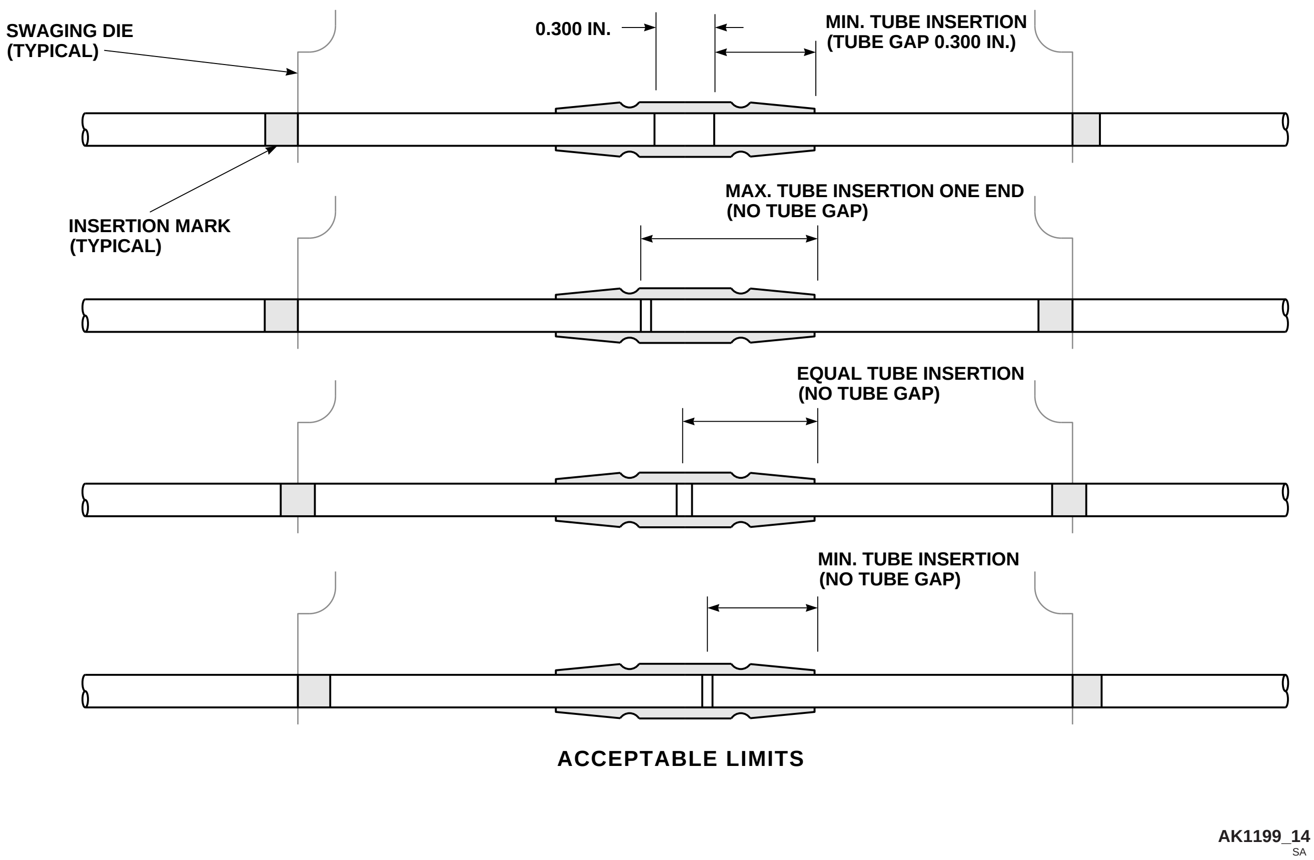
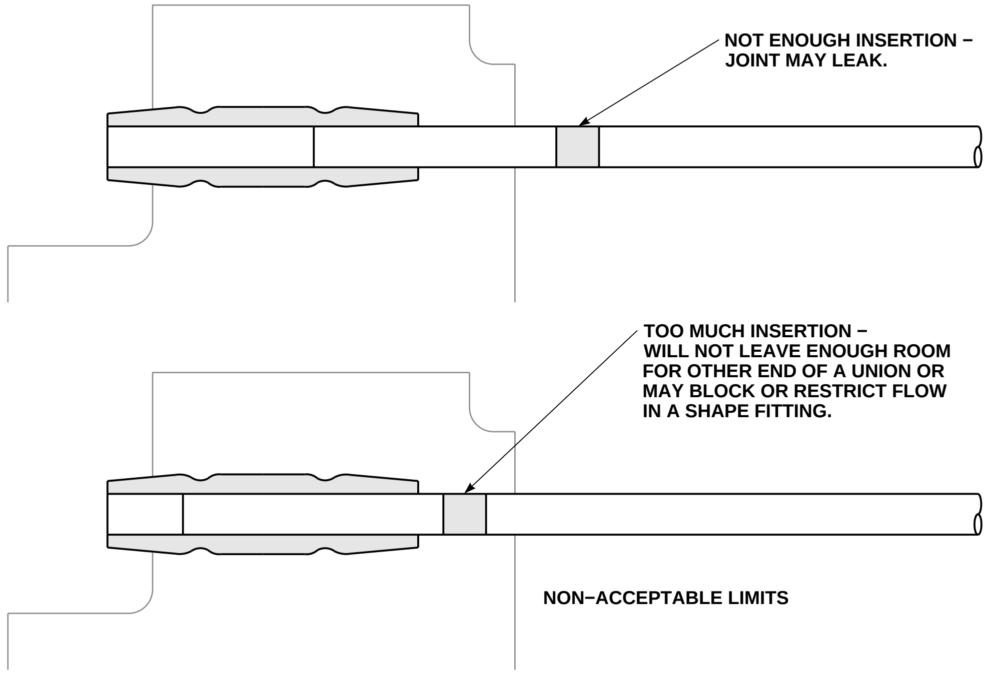


FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 14 OF 17)



TUBE INSERTION LIMITS WHEN
USING STANDARD UNION FITTINGS

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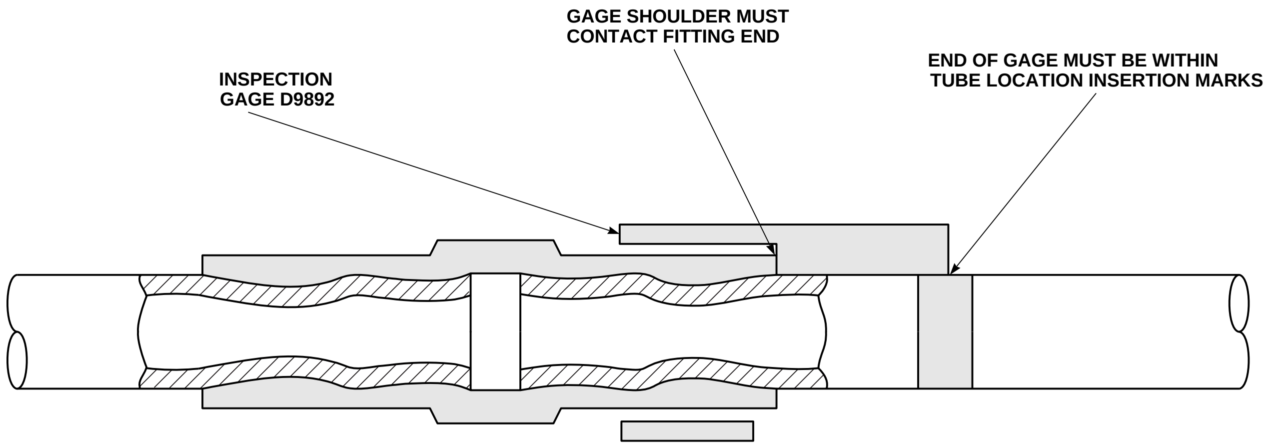
FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 15 OF 17)



CHECKING SWAGED JOINT WITH
INSPECTION GAGE D9892

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FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 16 OF 17)

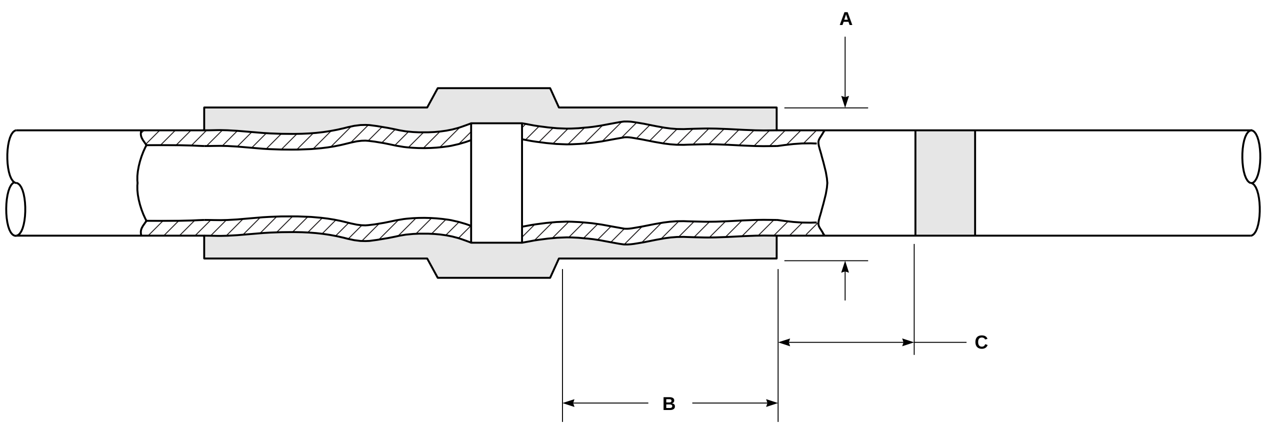

FIGURE 7-5-8-1. REPAIR HYDRAULIC SYSTEM RIGID TUBING (BENCH) (SHEET 17 OF 17)

Table 7-5-8-2. Permaswage Repair - Tube Material and Maximum Operating Pressure Limitations (PSI)

Tubing Material	Permaswage Fitting Material, 21-6-9 (MMS-229)			
	3/16-inch	1/4-inch	5/16-inch	3/8-inch
Cres*	(0.016)	(0.016)	(0.016)	(0.016)
21-6-9	3000	3000	3000	3000
AM-350	3000	3000	3000	3000
304 1/8 H	3000	3000	3000	3000
304L 1/8 H	3000	3000	3000	3000
321 1/8 H	3000	3000	3000	3000
Titanium*	(0.016)	(0.016)	(0.016)	(0.019)
3AL-2.5V Cold Red.	3000	3000	3000	3000
3AL-2.5V Ann.	3000	3000	3000	3000

Table 7-5-8-2. Permaswage Repair - Tube Material and Maximum Operating Pressure Limitations (PSI) (Cont)

Tubing Material	Permaswage Fitting Material, 21-6-9 (MMS-229)			
6AL-4V Cold Red.	3000	3000	3000	3000
Al. Al.*	(0.020)	(0.020)	(0.020)	(0.020)
6061-T6	3000	3000	3000	3000

*Minimum Tube Wall Thickness.

For Maximum Tube Wall Thickness see Table 7-5-8-4.

Table 7-5-8-3. Stem Subassembly Selection Chart

Deburring Tool, D9851

Tube Outside Diameter (Inch)	Tube Wall Thicknes (Inch)	Stem Subassembly Required
1/4	0.016 - 0.028	D9851-13-04
	0.028 - 0.050	D9851-13-03
5/16	0.016 - 0.035	D9851-13-05
3/8	0.016 - 0.035	D9851-13-06
	0.035 - 0.058	D9851-13-07

CAUTION

Damage to cutting tool and tubing will result if cutter wheel is overtightened. In soft tubing, a large burr will result. In hard tubing, cutter wheel will dull, wear, or break. Do not overtighten cutter wheel when cutting tubes. Side forces will break cutter wheel. Do not put side force on cutter handle.

b. **TIGHTEN DRIVE SCREW UNTIL LIGHT CONTACT IS MADE BY CUTTER WHEEL ON TUBE. THEN TIGHTEN SCREW AN ADDITIONAL 1/8 - 1/4 TURN. TIGHTEN SCREW WITH EITHER HEX KEY PROVIDED WITH KIT, ALLEN WRENCH, OR SCREWDRIVER.**

c. Swing ratchet handle back and forth through available clearance until it turns easily.

d. **TIGHTEN DRIVE SCREW AN ADDITIONAL 1/8 - 1/4 TURN AND CONTINUE CUTTING OPERATION.** Readjust drive screw as required to hold cutter pressure until cut has been completed. Cut must be within ½-inch square to tube centerline.

7-5-8.1.13 Tube Deburring.

a. Refer to Table 7-5-8-3 and select deburring tool and stem subassembly required for size tube to be deburred. Proper stem subassembly selection is important. Not using proper size plug will result either in chips entering tube or difficult plug insertion, with resulting plug damage (Figure 7-5-8-1, Sheet 9).

b. Assemble deburring tool as follows:

(1) Check sliding collar on end of elastic plug for free movement on stem. Lubricate using lubricating oil, Item 209, Appendix D, if required.

(2) Engage threads and insert stem subassembly into cutter end of deburring tool. **WITH PLUNGER PUSHED IN, INSTALL STEM INTO PLUNGER UNTIL BOTTOMED AND TIGHTEN FINGERTIGHT.**

(3) Check assembled tool for proper operation by pushing and releasing plunger and noting effect on elastic plug. With plunger pushed in, plug outside diameter (OD) must be reduced to same diameter as metal support collars on either end of plug. With plunger released, two distinct bumps around plug should appear on surface of plug and be higher than OD of metal support collars.

(4) Inspect elastic plug for wear or other damage. A worn or damaged plug could allow chips from deburring operation to enter tubing.

c. Make sure tube end is deburred for required squareness and perform the following:

(1) Make sure outer surface of elastic plug is clean and free of chips or other dirt. Clean, if required, and lightly lubricate using hydraulic fluid, Item 154 or Item 155, Appendix D.

(2) Grasp tool in one hand with two fingers on collar and thumb on plunger. Depress plunger with thumb and insert elastic plug into tube opening until cutter is about 1/8-inch from tube end.

(3) If plug fit is tight due to a large burr on inside diameter of tube, slowly turn plunger end of tool while gently pushing into tube end.

(4) When tool is properly positioned, release plunger, allowing elastic plug to expand and seal tube.

CAUTION

Too much deburring will cause too deep a chamfer on tube inside diameter. Chamfer width should generally not be more than one-half wall thickness of tubing.

(5) Holding tube end with one hand, turn knurled body of deburring tool clockwise while applying slight pressure to cutter. Continue to turn until cutter turns smoothly. Relax pressure and turn cutter several more times to produce a smooth surface on resulting chamfer.

CAUTION

Damage to hydraulic system will occur if deburring debris is allowed to enter tube. Plunger must not be pushed in when withdrawing tool from tube.

(6) Without pushing plunger in, ease tool from tube until first bulge of plug is exposed. Wipe off tube end and plug, and inspect tube end to see if it has been properly deburred.

NOTE

Deburring forms a slight chamfer on inner edges of cut tube. Chamfer must be over ½ of tube wall thickness at an angle no greater than 45°.

(7) If tube end appears satisfactory, pull tool completely out of tube, taking care not to operate plunger. If tube end has not been completely deburred, push tool back into tube, again being careful not to operate plunger, and repeat deburring procedure.

(8) Inspect tube end and tube inside diameter for loose chips or other debris. If required, wipe down tube using clean machinery towels, Item 211, Appendix D.

(9) Thoroughly wipe deburring tool to remove all traces of chips and dirt, particularly from elastic plug.

CAUTION

Damage to elastic plug portion of deburring tool will result if contact with hydraulic fluid occurs.

d. If elastic plug has contacted hydraulic fluid, it should be thoroughly cleaned using methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D, and immediately dried.

e. Remove stem subassembly from deburring tool by pushing in plunger and unscrewing stem from plunger.

f. Return stem subassembly and cutter body to fitted tool kit for protective storage. This is necessary to prolong life of stem subassembly.

g. Wear or damage may require replacing cutter head. Replace cutter as follows:

(1) Remove stem subassembly if one is installed on tool.

(2) Remove worn or damaged cutter by grasping with a pair of pliers and unscrewing.

(3) **INSTALL NEW CUTTER ON TOOL BODY BY ENGAGING THREADS AND TURNING UNTIL FINGER-TIGHT. DO NOT OVERTIGHTEN, AS THREAD DAMAGE MAY RESULT.**

h. Deburr and square tube end must be swaged to within ½° of square to tube centerline.

i. Mark tube ends with a tube insertion band or mark, using felt tip pen (provided with kit), ink, or other marking per one of following procedures:

(1) Tube insertion mark using Permaswage marking tool may be done by placing lip stop of marking tool over end of tube to be swaged. Use correct tube size slot as a marking guide, making sure total width of mark is made on tube. Tube insertion length and 0.300-inch marking band width are part of marking tool (Figure 7-5-8-1, Sheet 10).

(2) Using Table 7-5-8-4, determine tube insertion band location. This method is used when Permaswage marking tool is not available (Figure 7-5-8-1, Sheet 11).

(3) If required, either of above procedures may be done with permanent marking such as paint, dye, tape, or other permanent marking. Paint must be removed from dimension ″A″ section of tubing before swaging.

7-5-8.1.14 Swaging Tool Operating Instructions.

a. Make sure reservoir is full by checking level of sight gage.

NOTE

Make sure hydraulic power supply is serviced with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Connect hydraulic pressure line from hydraulic power supply to fitting on base of swaging tool (Figure 7-5-8-1, Sheet 12).

c. Set selector valve on hydraulic power supply to EXHAUST. (This relieves all pressure in line and allows enough room for upper die block to be inserted over lower die block).

Table 7-5-8-4. Tube Insertion Band Location

Tube Diameter (Inch)	Dash Size	Dimension ″A″ (+0.010-0.000-Inch)
1⁄4	(-4)	1.475
5⁄16	(-5)	1.475
3⁄8	(-6)	1.475

d. To keep air from being trapped in hose and swaging tool, install a set of die blocks in tool and position tool below level of hydraulic power supply. Then cycle hydraulic power supply to 2,000 psi, followed by EXHAUST. Two or three cycles may be required.

e. Connect air line from 60 - 100 psi shop air source to shop air input.

NOTE

When you see water in air pressure regulator, press valve core on bottom of sediment bowl to drain.

f. Select correct upper and lower die block assemblies for size of fitting to be swaged. If either die block is too small, upper die block will not load. If either is too large, fitting will not swage properly.

g. Insert lower die block (with correct fitting locator depending on type of fitting) into swaging tool. When properly inserted, fitting locator is on other end of tool from latch.

h. Place swaging tool on fitting to be swaged, with fitting over lower die block, so end not being swaged is butted up against shoulder stop of fitting locator.

NOTE

Upper die block assembly may also be inserted from top of tool-head.

i. Insert upper die block assembly into swaging tool from latch side of tool (Figure 7-5-8-1, Sheet 13).

CAUTION

Injury to personnel will result if swaging tool malfunctions. Make sure latch is in place and securely locks upper die block assembly.

j. Push latch up to lock upper die block assembly into position over lower die block assembly.

k. Position face of swaging tool against location mark on tube. Outer edge of latch must fall within location mark (Figure 7-5-8-1, Sheet 14 and Sheet 15).

NOTE

Not enough or too much tube insertion in Permaswage fitting is cause for rejection of joint.

l. For maximum tube end gap, leading edge of tube insertion mark must be in line with outside edge of swaging tool latch.

m. For minimum tube end gap when using union and shaped fittings, back edge of tube insertion mark must be in line with outside edge of latch.

n. For minimum tube end gap using separable fittings only, insert tube until it is bottomed.

o. On hydraulic pump, set selector valve toward hydraulic output being used (Figure 7-5-8-1, Sheet 12).

WARNING

Injury to personnel will result if swaging tool malfunctions. Care must be taken to make sure swaging tool is properly installed and operated. Improper installation and operation will result in sudden expulsion of high pressure fluid.

NOTE

Not enough pressure will result in improperly swaged joint. Too much pressure will overstress swaging tool, fitting and tube.

p. Actuate swaging tool with 5,250 - 5,750 psi hydraulic pressure in line from portable hydraulic pump. Perform this by manually pumping handpump on hydraulic power supply until gage shows 5,500 psi, or automatically by using shop-air connection and pressing local or remote air switch until gage shows 5,500 psi.

NOTE

Portable hydraulic pump has a pressure relief valve which should be set for 5,250 - 5,750 psig. If different reading is noticed, calibrate hydraulic pump.

q. Turn selector valve to EXHAUST to relieve pressure completely.

r. Slide latch to OPEN.

s. Remove upper die block assembly by sliding out from end of swaging tool, or up through slots in tool head.

NOTE

If upper die block assembly is difficult to remove, make sure pressure has been completely relieved.

t. Remove swaging tool from swaged joint.

u. Inspect swaged end with appropriate size D9892 go/no-go inspection gage (Figure 7-5-8-1, Sheet 16).

v. If inspection gage does not fit as required, re-swage fitting by repeating swaging procedure.

NOTE

When repositioning swaging tool on a partially swaged fitting end, opposite end must be carefully located against shoulder stop of fitting locator, or swage length could be exceeded.

w. Swage other end of fitting (if required) by repeating swaging procedure.

7-5-8.1.15 Inspection/Checking with Inspection Gage, D9892.

a. Inspection/checking of swaged joints is performed visually and with inspection gage of correct dash (-) size for swaged fitting. Proper use of correct inspection gage insures:

(1) Length of tube-to-fitting engagement.

(2) Outside diameter of swaged fitting.

(3) Length of swaged portion of fitting.

b. Visually inspect for cracks in fitting or tube.

c. Place inspection gage of correct size on swaged fitting (Figure 7-5-8-1, Sheet 16).

NOTE

Use of die segments produces longitudinal ridges (″flash″ marks) on swaged fitting.

d. Put gage between ″flash″ marks. Gage must fit over all swaged segments between ″flash″ marks.

NOTE

Gage shoulder must contact end of fitting, and end of gage must fall within tube insertion mark on tubing.

e. If fitting does not meet requirements of inspection gage, corrective action must be taken as indicated in Table 7-5-8-5, Troubleshooting Guide for Inspection/Checking.

7-5-8.1.16 Inspection/Checking Without Inspection Gage.

a. Visually inspect for cracks in fitting or tube.

NOTE

Outside diameter (Dimension ″A″ in Table 7-5-8-6) must be measured in area between long ridges (″flash″ marks) produced by swaging action of die segments.

b. In the absence of proper inspection gage, swaged fitting must meet dimensional requirements of Table 7-5-8-6 (Figure 7-5-8-1, Sheet 17).

Table 7-5-8-5. Troubleshooting Guide for Inspection Checking

Indication	Possible Problem	Corrective Action
Fitting does not swage at all.	Swaging with less than 5,250 - 5,750 psi to tool.	Check hydraulic power supply and fittings to tool.
	Improper size die blocks (too large).	Install proper die block assemblies.
Gage will not fit over swaged fitting outside diameter.	Swaging with less than 5,250 - 5,750 psi.	Reswage.
	Attempting to read over flash marks.	Turn gage off flash marks.
	Die segments and walls of die blocks must be clean.	Carefully remove die segments from die blocks and: 1) Clean between die sectors. 2) Clean and lubricate die block side surfaces and those surfaces in contact with die segments.
Gage shoulder does not contact end of fitting.	End of fitting must be in contact with fitting locator shoulder stops on swaging tool.	Reswage. Locator shoulder stops may be damaged; if so, they should be replaced.
Gage finger does not fall within tube insertion mark on tubing.	Tube insertion mark did not coincide with face of tool (outside edge of safety latch) during swaging of both ends.	Cut out fitting.

Table 7-5-8-6. Fitting Dimensions After Swaging

Tube Diameter (Inch)	Dash Size	Maximum Swaged Dia. ″A″ Dimension (Inch)	Minimum Swaged Length ″B″ Dimension (Inch)	Tube Location ″C″ Dimension (Inches)
3⁄16	(-3)	0.2465	0.340	1.010 to 0.710
1⁄4	(-4)	0.315	0.460	0.865 to 0.565
5⁄16	(-5)	0.3815	0.500	0.820 to 0.520
3⁄8	(-6)	0.447	0.530	0.784 to 0.484

7-5-9 APU ACCUMULATOR START VALVE FITTINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D

7-5-9.1 Replace APU Accumulator Start Valve Fittings (BENCH).

7-5-9.1.1. Remove.

a. Remove return fitting and packing from start valve (Figure 7-5-9-1). Discard packing.

b. Remove pressure fitting and packing from start valve. Discard packing.

7-5-9.1.2. Install.

a. Coat packings and threads of fittings with hydraulic fluid, Item 154 or Item 155, Appendix D, before installing.

b. Install packing on pressure fitting and install fitting in start valve (Figure 7-5-9-1). **TORQUE PRESSURE FITTING TO 500 - 600 INCH-POUNDS.**

c. Install packing on return fitting and install fitting in start valve. **TORQUE RETURN FITTING TO 700 - 750 INCH-POUNDS.**

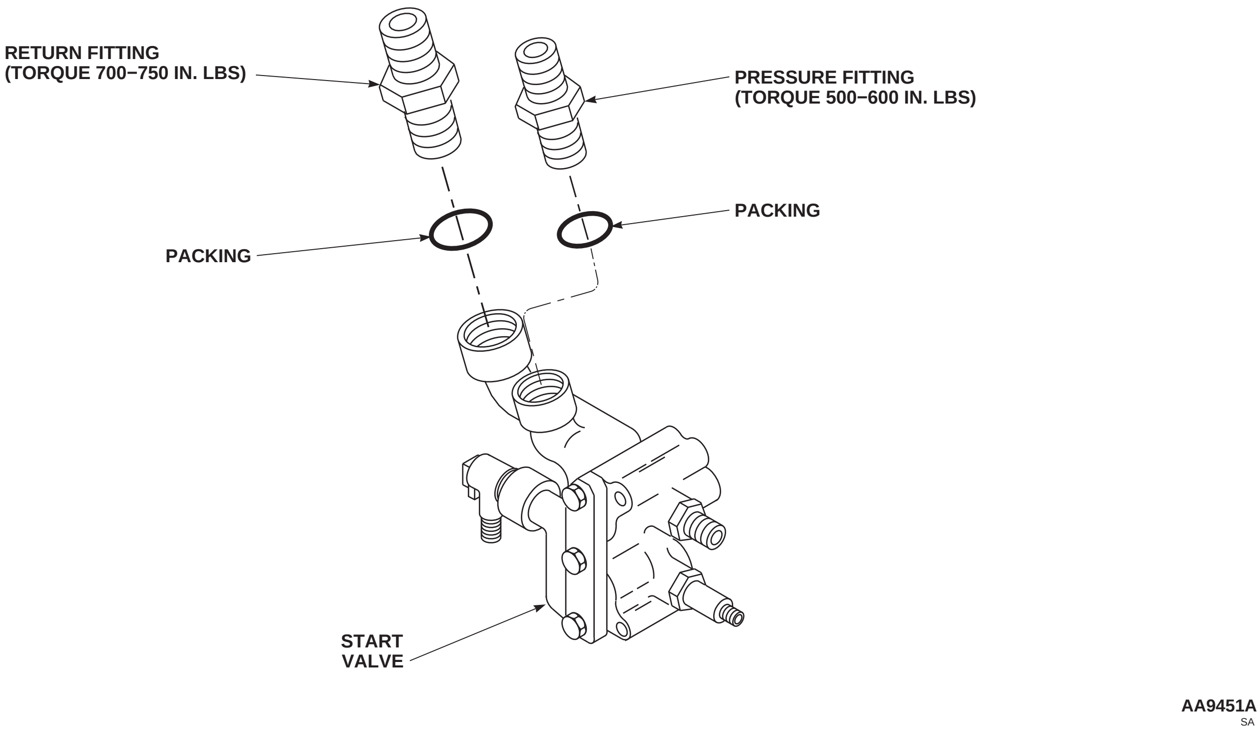

FIGURE 7-5-9-1. REPLACE APU ACCUMULATOR START VALVE FITTINGS (BENCH)

7-5-10 PRESSURE REDUCER INDICATOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D

7-5-10.1 Replace Pressure Reducer Indicator (BENCH).

7-5-10.1.1. Remove.

Remove pressure reducer indicator, packings, and seal from pilot assist module (Figure 7-5-10-1, Detail A). Discard packings and seal.

7-5-10.1.2. Install.

a. Coat packings and seal with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on pressure reducer indicator (Figure 7-5-10-1, Detail A).

b. Install pressure reducer indicator in pilot assist module. **TORQUE PRESSURE REDUCER INDICATOR TO 238 - 262 INCH-POUNDS.**

c. Lockwire, Item 196, Appendix D, pressure reducer indicator.

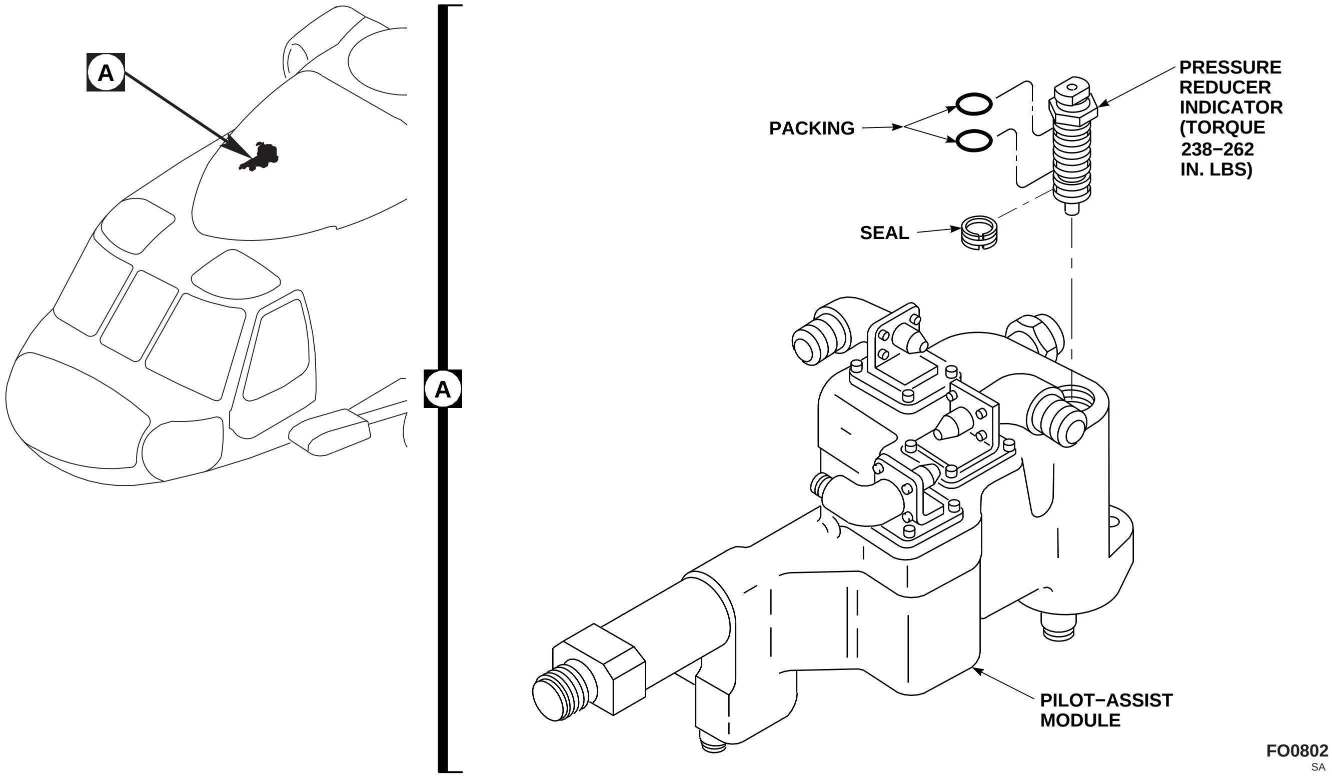

FIGURE 7-5-10-1. REPLACE PRESSURE REDUCER INDICATOR (BENCH)